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(54) **MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT AND METHODS THEREFOR**

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(57) **ABSTRACT**

Disclosed is a method of controlling a modular battery powered handheld surgical instrument. The surgical instrument including a battery, a user input sensor, a controller, a radio frequency (RF) drive circuit, an ultrasonic transducer, ultrasonic transducer drive circuit, and an end effector. The end effector including an electrode electrically coupled to RF drive circuit, an ultrasonic blade acoustically coupled to the ultrasonic transducer, and a sensor to measure tissue parameters. The method includes applying an RF current drive signal to the electrode by the RF drive circuit; applying an ultrasonic drive signal to the ultrasonic transducer by the ultrasonic transducer drive circuit to acoustically excite the ultrasonic blade; controlling intensity, wave shape, and/or

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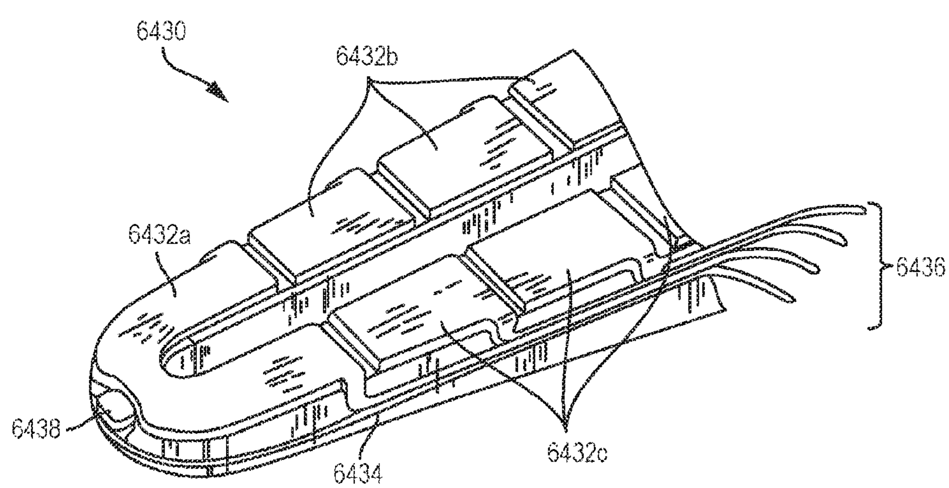

frequency of the RF current drive signal and the ultrasonic drive signal on a sensed measure of a tissue or user parameter.

17 Claims, 128 Drawing Sheets

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(58) **Field of Classification Search**

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See application file for complete search history.

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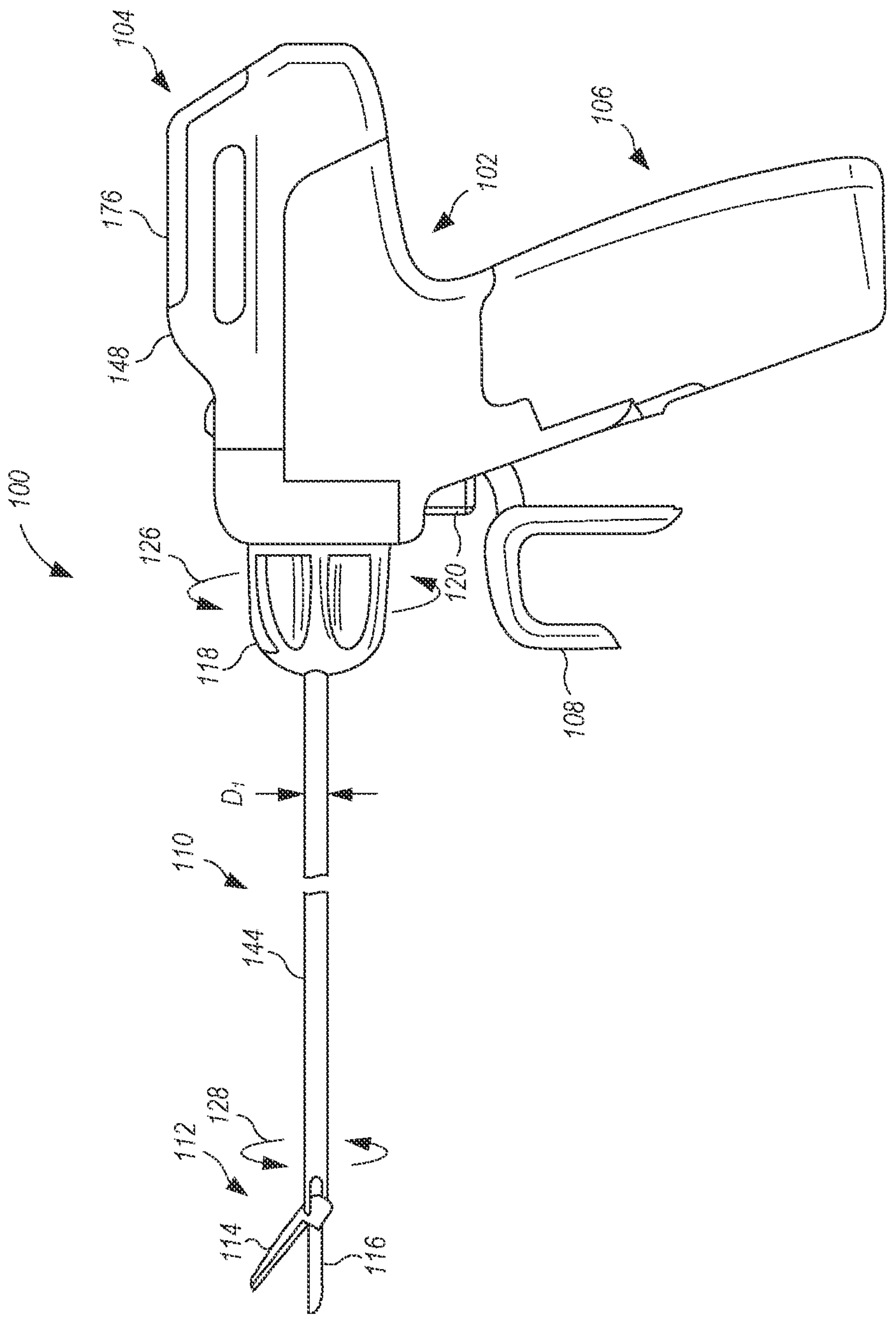

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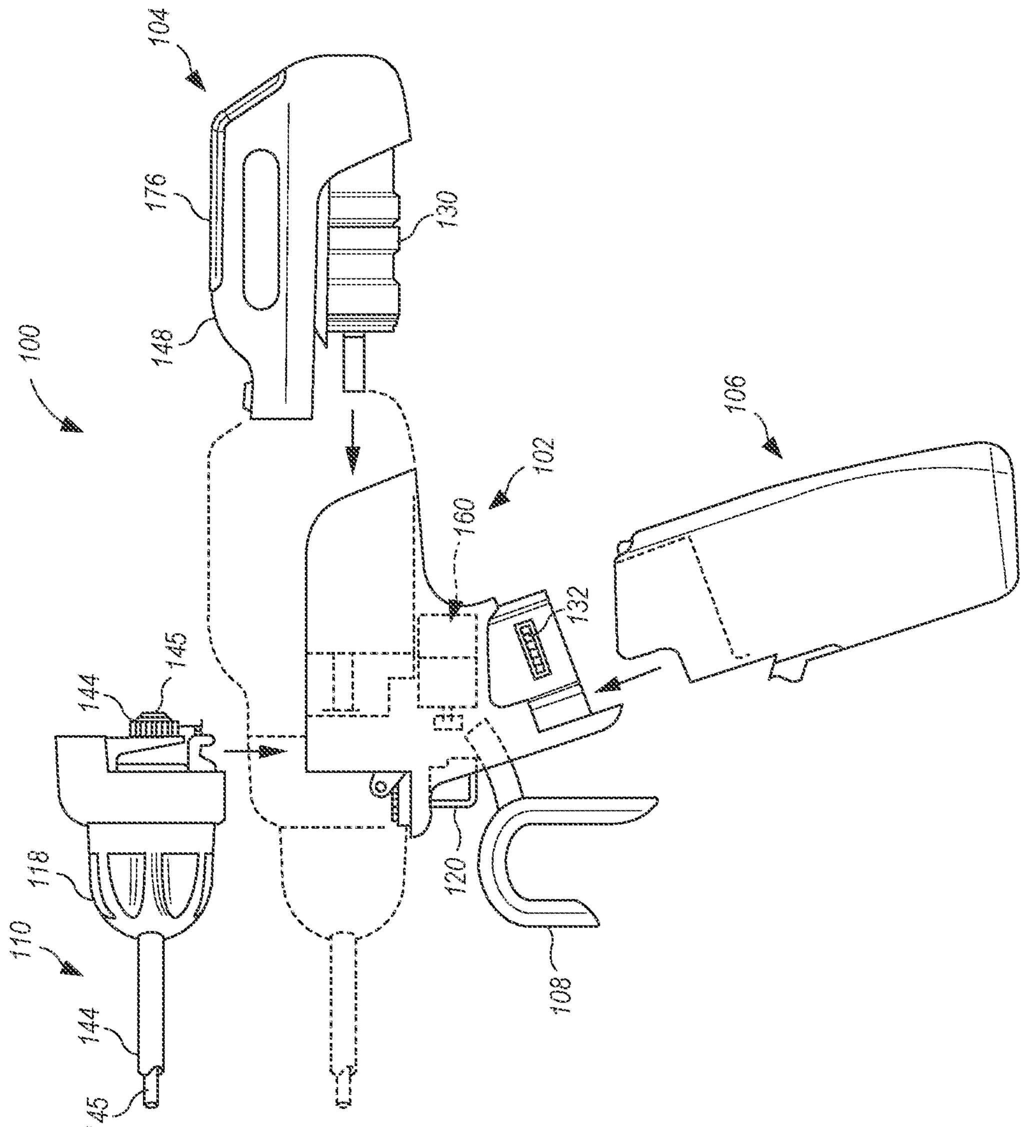
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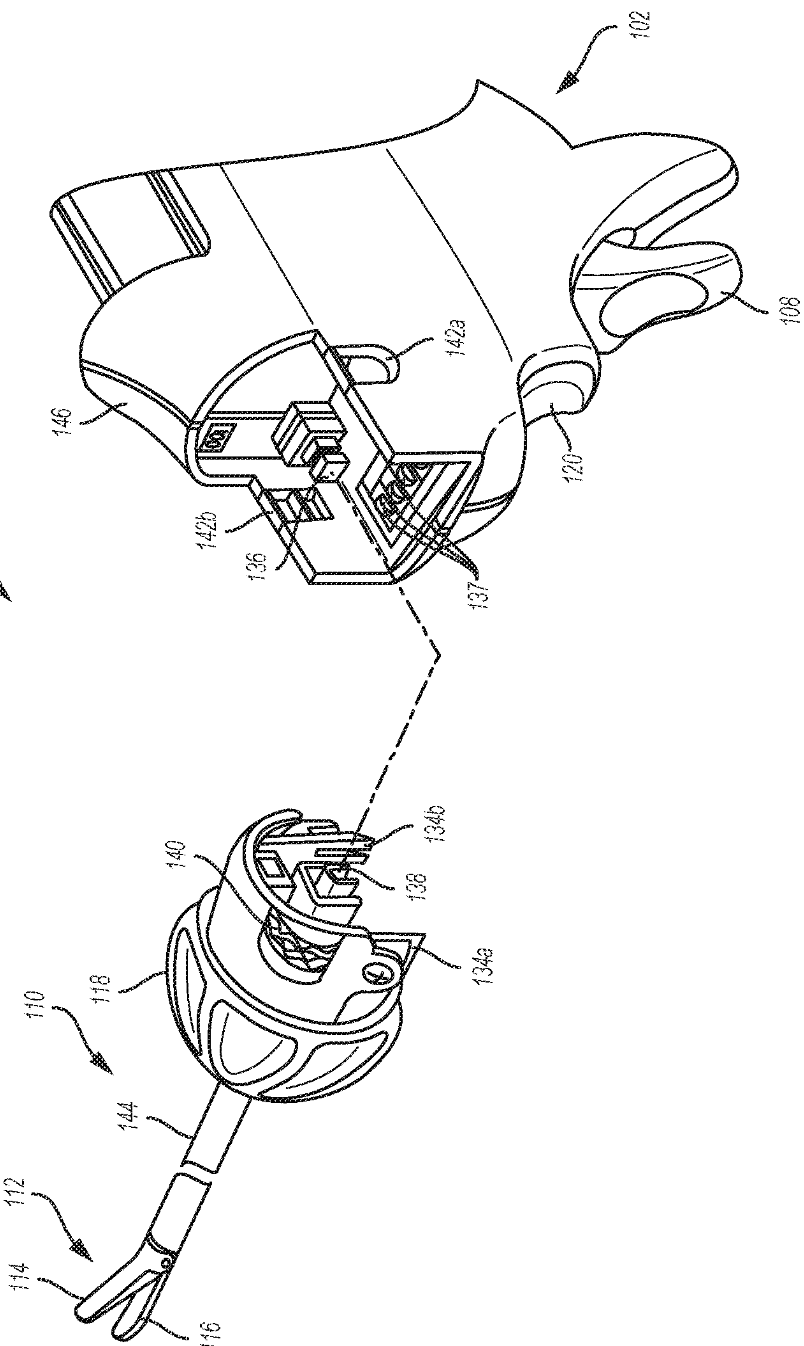
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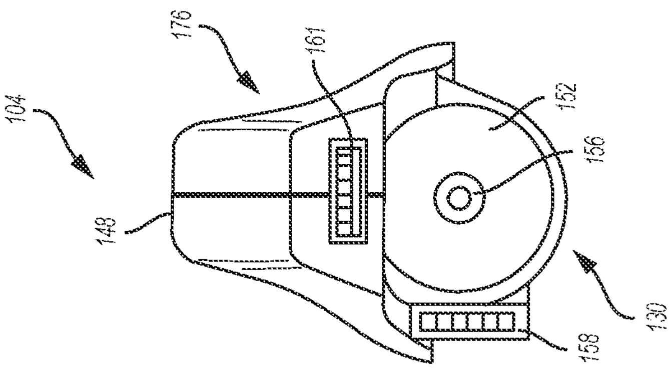
FIG. 3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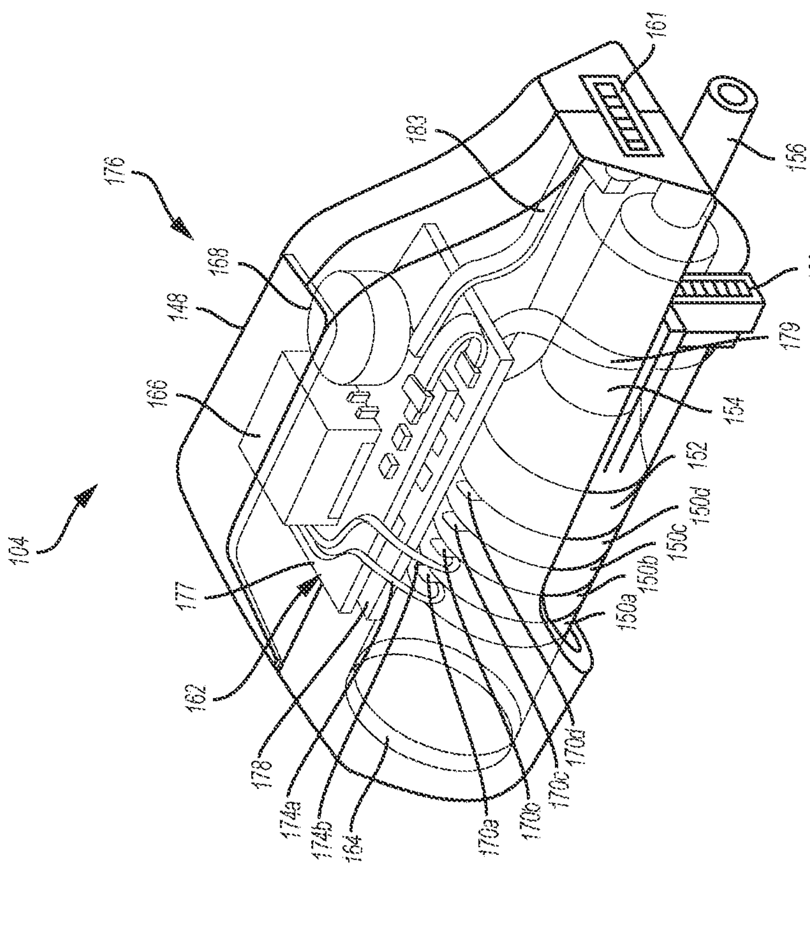
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FIG. 4

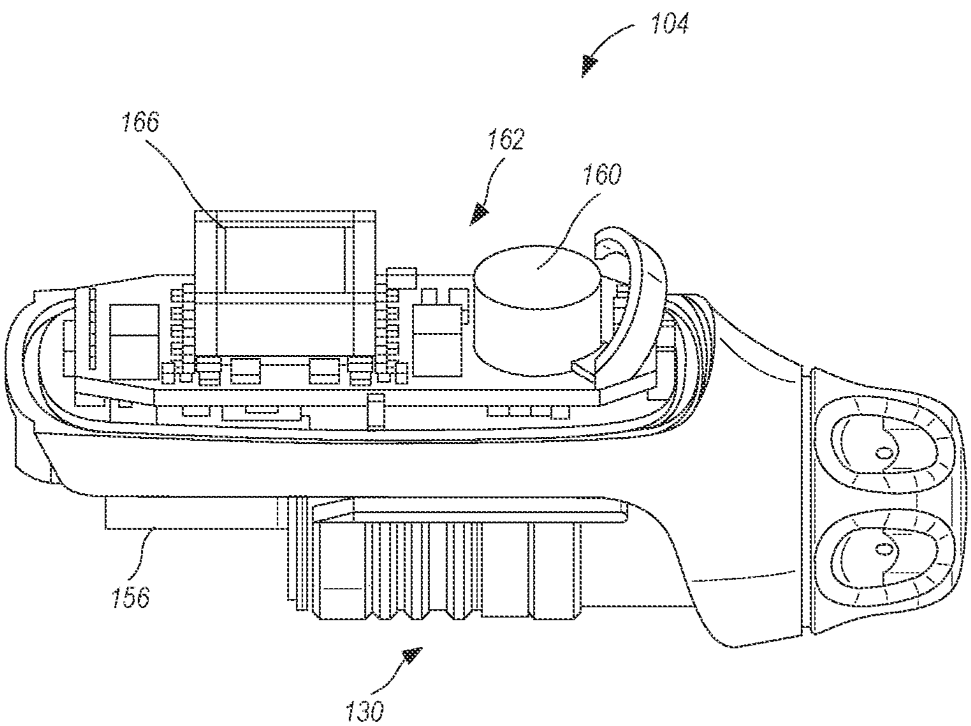
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FIG. 6

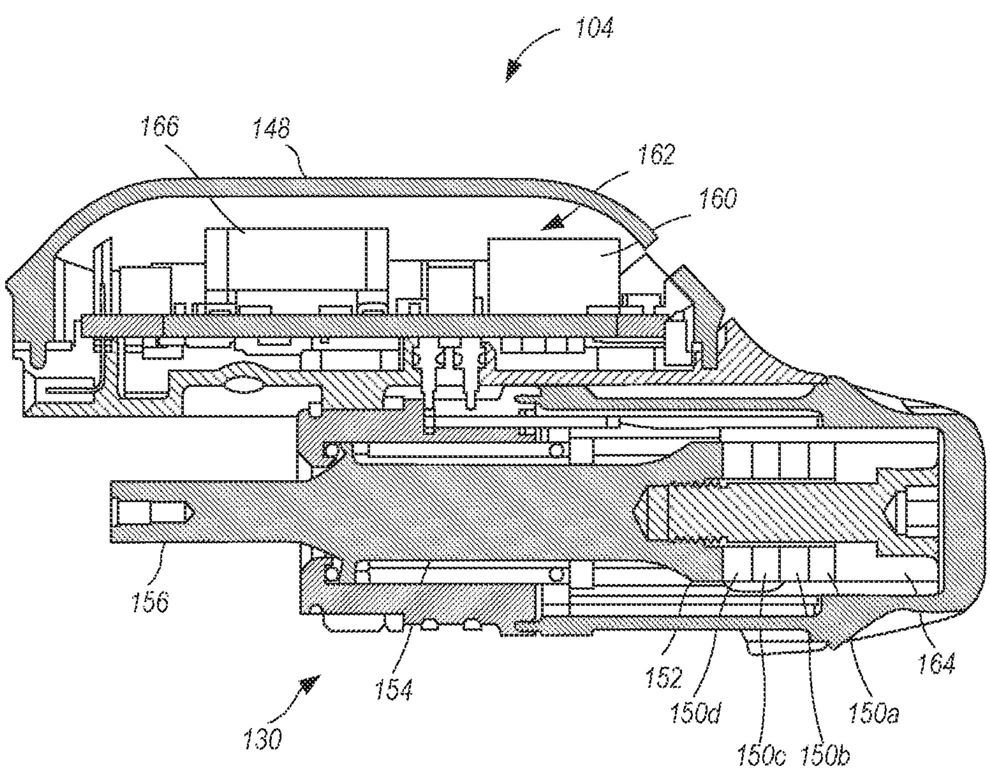
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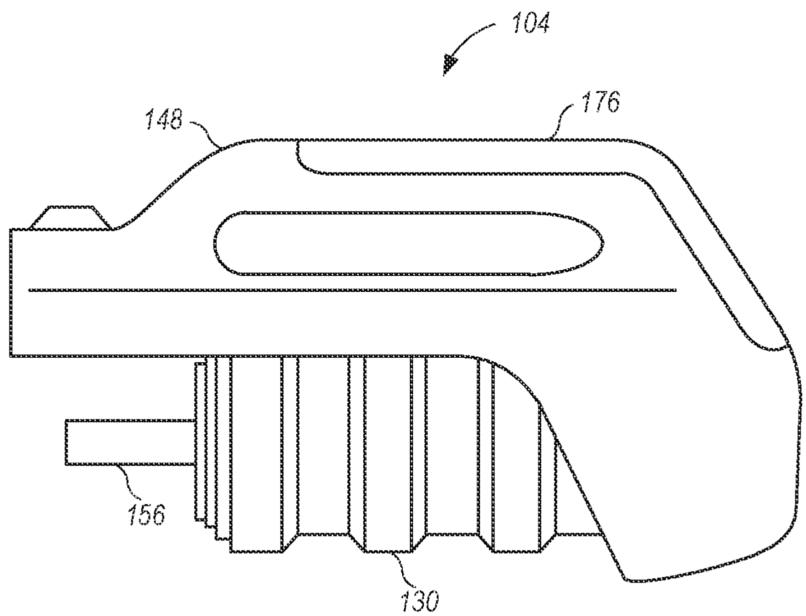
FIG. 7



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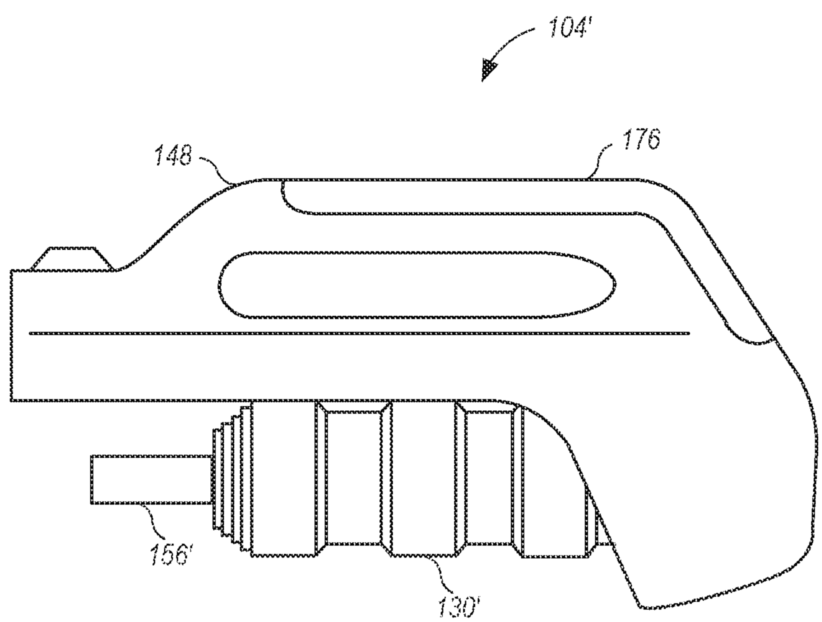

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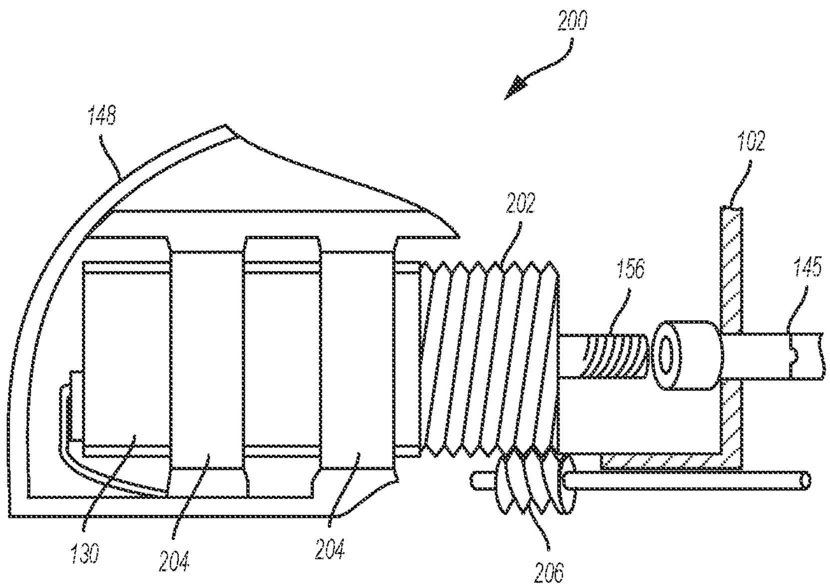

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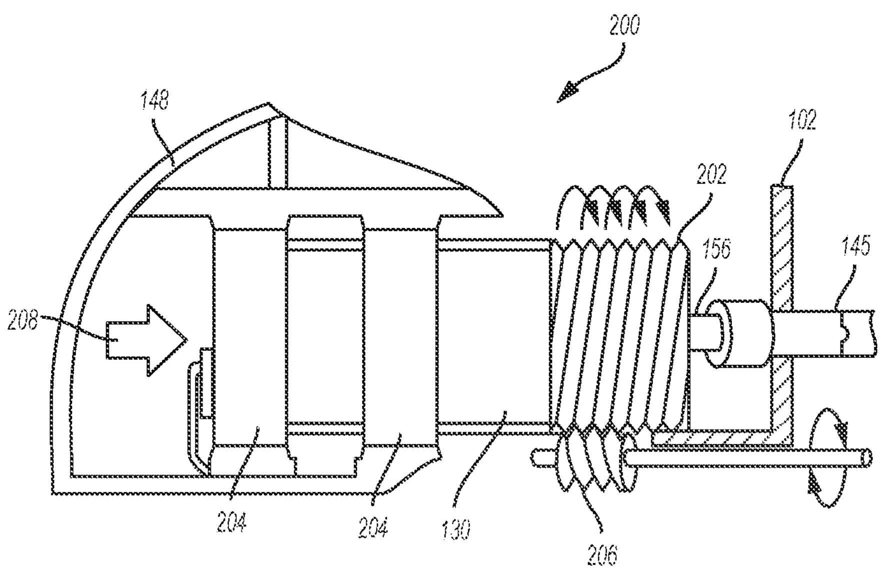


FIG. 10B

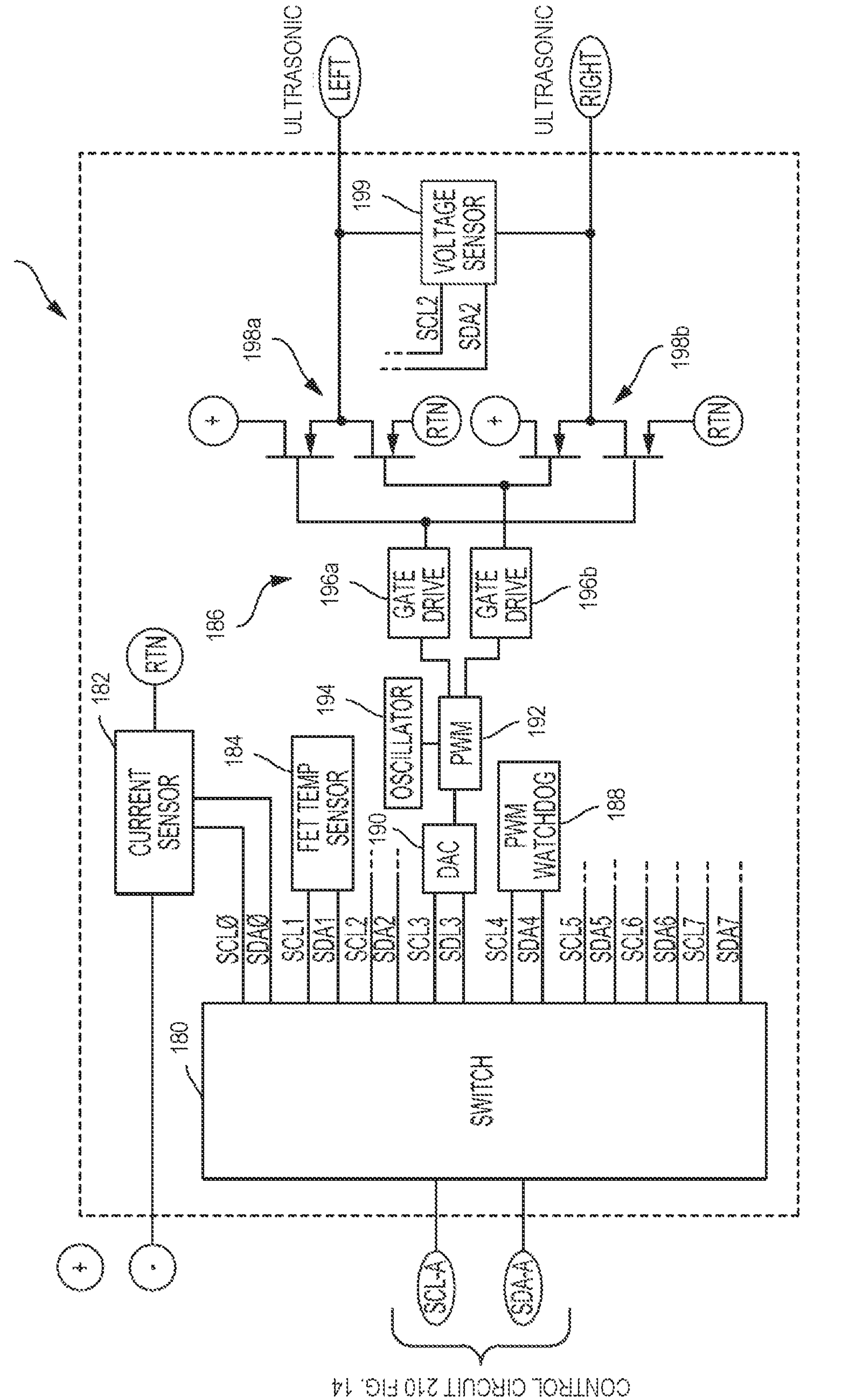
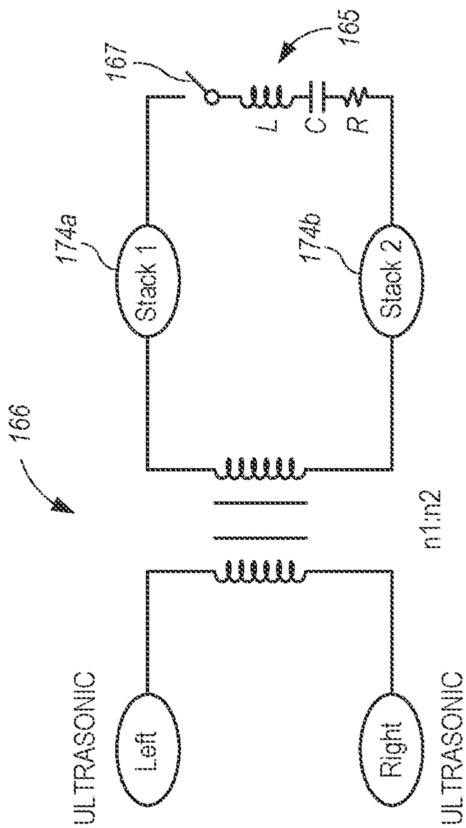
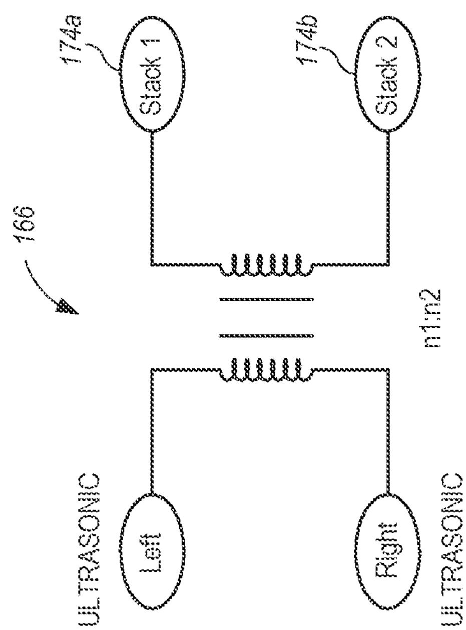

177
ULTRASONIC
LEFT
ULTRASONIC
RIGHT
199
VOLTAGE SENSOR
SCL2
SDA2
198a
198b
RTN
186
196a
GATE DRIVE
GATE DRIVE
196b
182
CURRENT SENSOR
184
FET TEMP SENSOR
194
OSCILLATOR
PWM
192
190
DAC
PWM WATCHDOG
188
SCLØ
SDAØ
SCL1
SDA1
SCL2
SDA2
SCL3
SDL3
SCL4
SDA4
SCL5
SDA5
SCL6
SDA6
SCL7
SDA7
180
SWITCH
SCL-A
SDA-A
CONTROL CIRCUIT 210 FIG. 14

FIG. 11



167
165
L
C
R
174a
Stack 1
174b
Stack 2
166
n1:n2
ULTRASONIC
Left
Right
ULTRASONIC

FIG. 13



174a
Stack 1
174b
Stack 2
166
n1:n2
ULTRASONIC
Left
Right
ULTRASONIC

FIG. 12

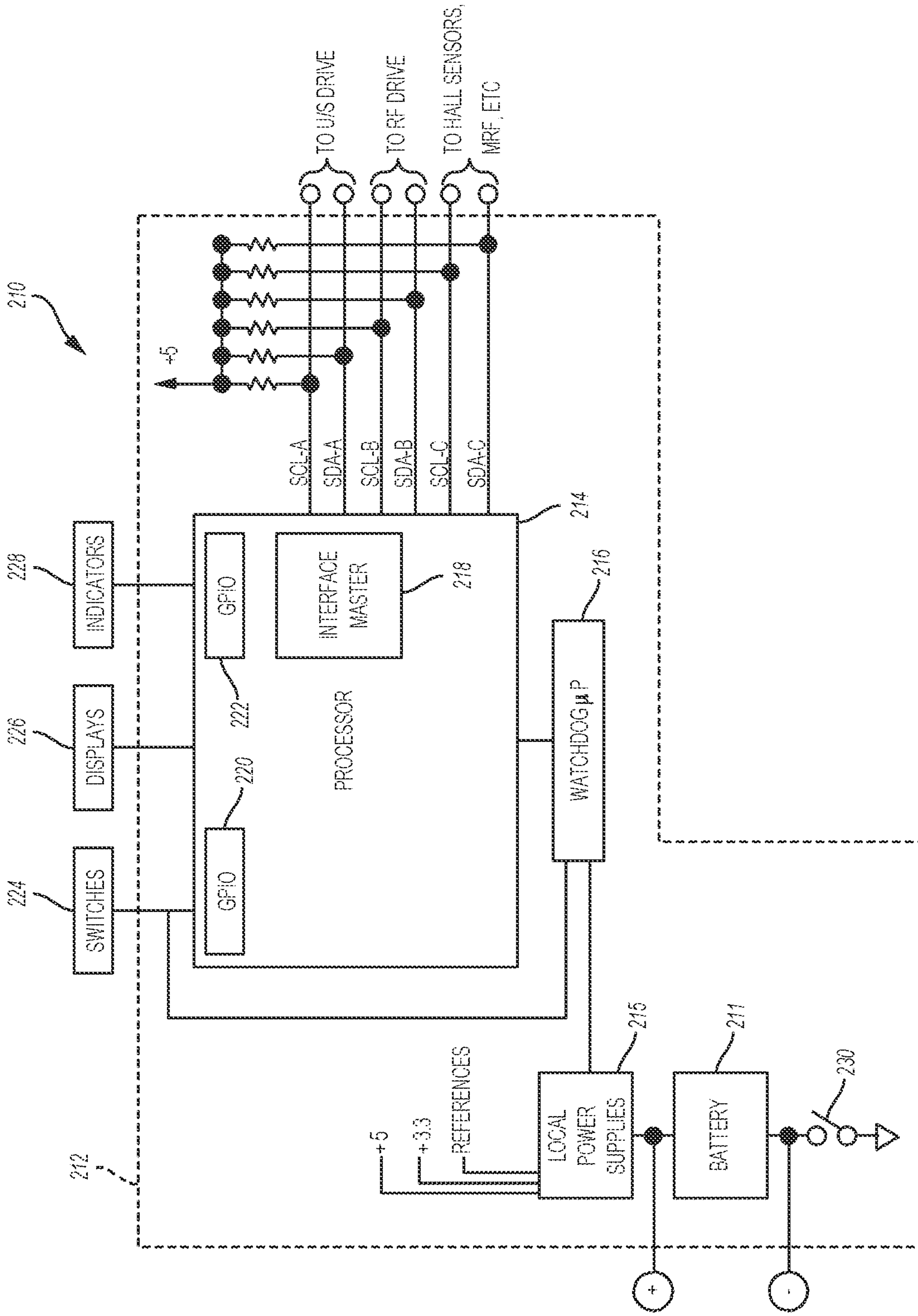

210
212
TO U/S DRIVE
TO RF DRIVE
TO HALL SENSORS,
MRF, ETC
+5
SCL-A
SDA-A
SCL-B
SDA-B
SCL-C
SDA-C
214
228
INDICATORS
226
DISPLAYS
224
SWITCHES
GPIO
222
GPIO
220
PROCESSOR
INTERFACE
MASTER
218
216
WATCHDOG µP
215
LOCAL
POWER
SUPPLIES
+5
+3.3
REFERENCES
211
BATTERY
230
+
-

FIG. 14

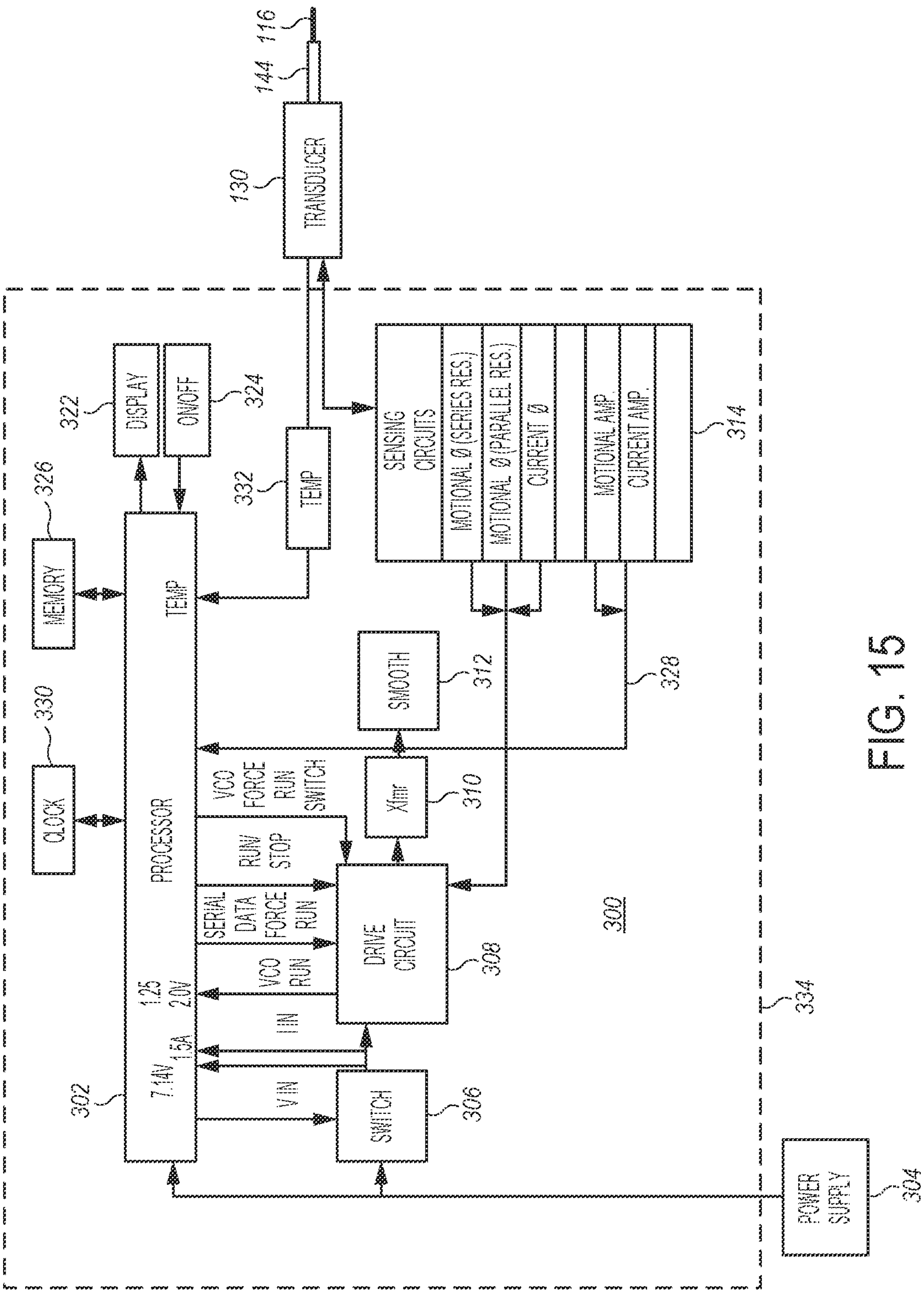

TRANSDUCER
DISPLAY
ON/OFF
MEMORY
TEMP
SENSING CIRCUITS
MOTIONAL Ø (SERIES RES.)
MOTIONAL Ø (PARALLEL RES.)
CURRENT Ø
MOTIONAL AMP
CURRENT AMP.
SMOOTH
CLOCK
PROCESSOR
TEMP
VCO FORCE RUN SWITCH
Xfmr
RUN/ STOP
SERIAL DATA FORCE RUN
VCO RUN
DRIVE CIRCUIT
1.25 2.0V
IIN
7.14V
1.5A
V IN
SWITCH
POWER SUPPLY
116
144
130
322
324
326
332
314
312
328
330
310
300
308
334
302
306
304

FIG. 15

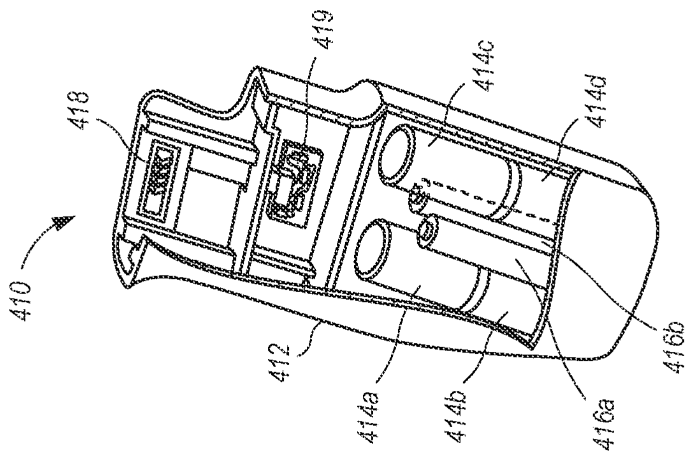

410
418
419
414c
414d
412
414a
414b
416a
416b

FIG. 17

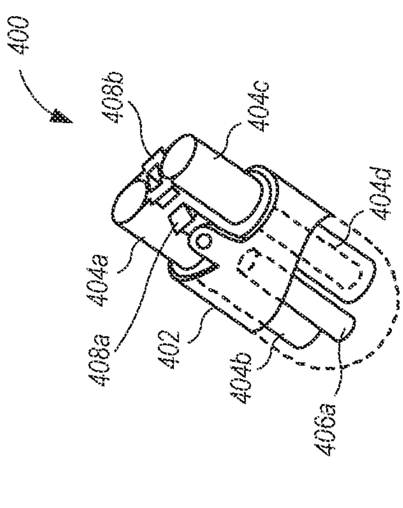

400
408b
404c
404d
404a
408a
402
404b
406a

FIG. 16

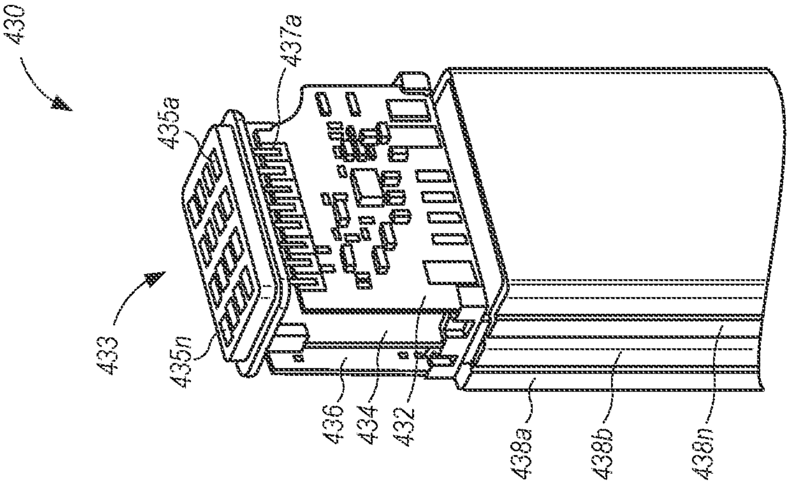

430
433
435a
435n
437a
436
434
432
438a
438b
438n

FIG. 19

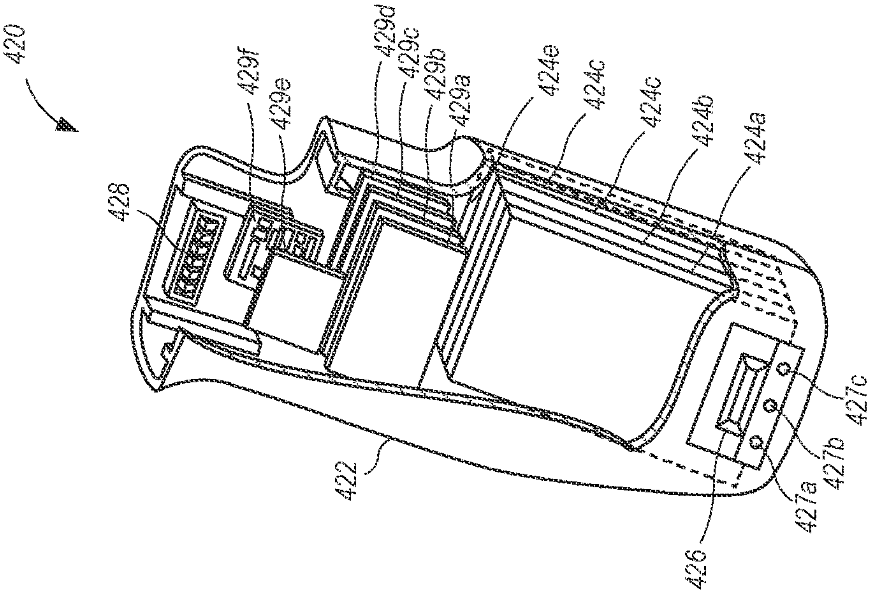

420
428
429f
429e
429d
429c
429b
429a
424e
424c
424c
424b
424a
422
426
427a
427b
427c

FIG. 18

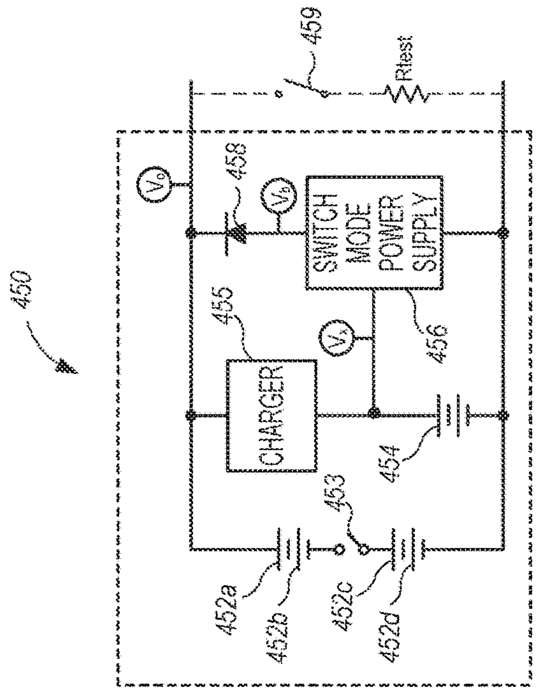


FIG. 21

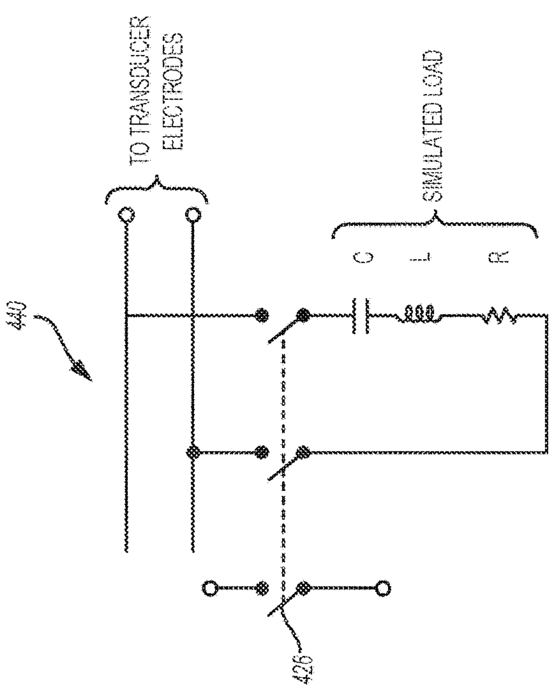


FIG. 20

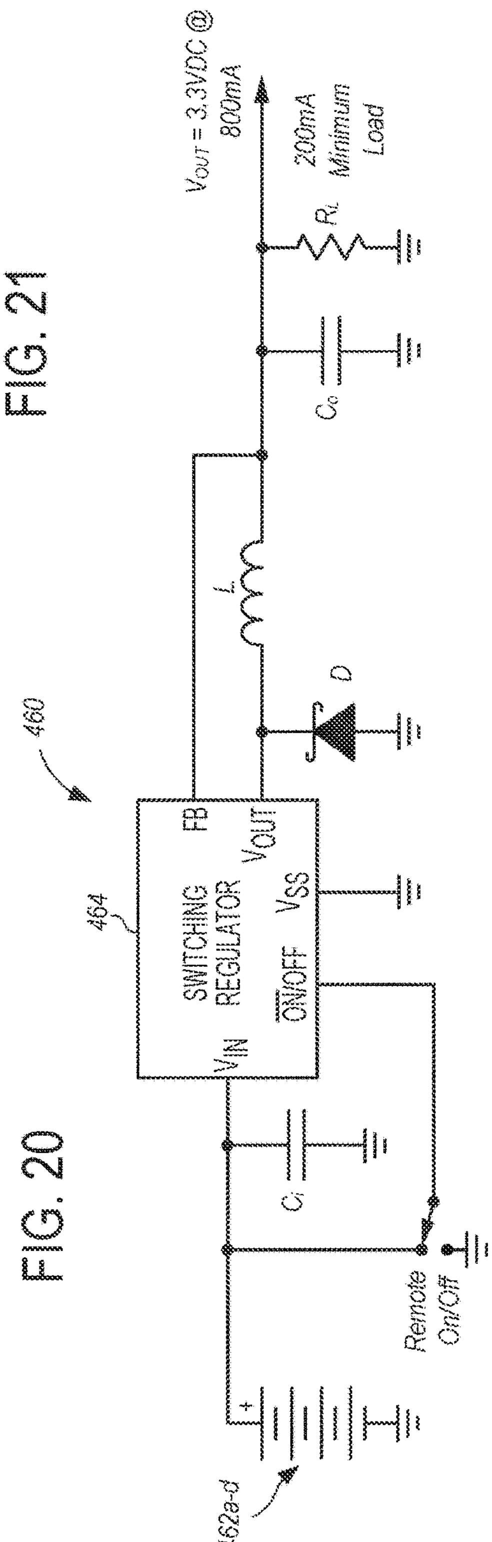


FIG. 22

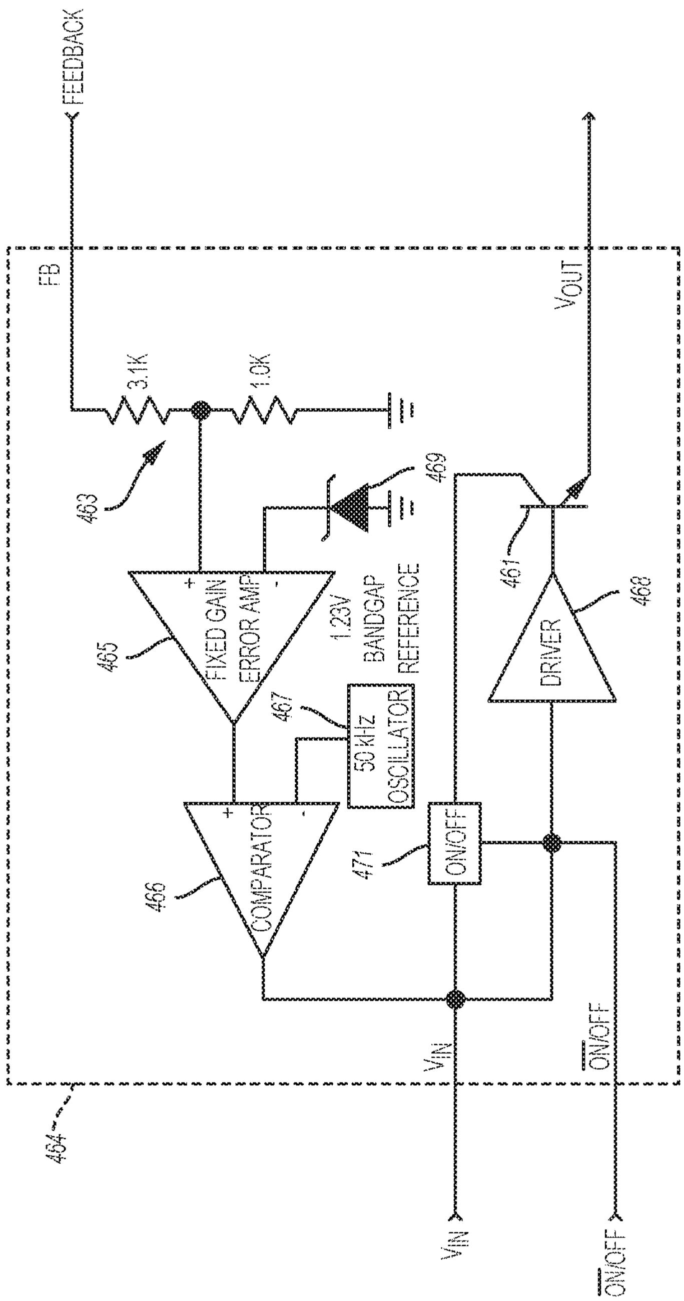

FEEDBACK
FB
3.1K
1.0K
463
FIXED GAIN ERROR AMP
465
469
1.23V BANDGAP REFERENCE
467
50 kHz OSCILLATOR
COMPARATOR
466
ON/OFF
471
DRIVER
468
461
VOUT
VIN
ON/OFF
464
VIN
ON/OFF

FIG. 23

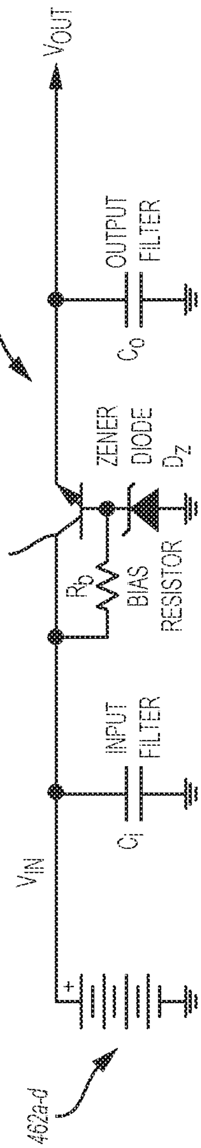

462a-d
VIN
Ci
INPUT FILTER
Rb
BIAS RESISTOR
472
ZENER DIODE Dz
470
Co
OUTPUT FILTER
VOUT

FIG. 24

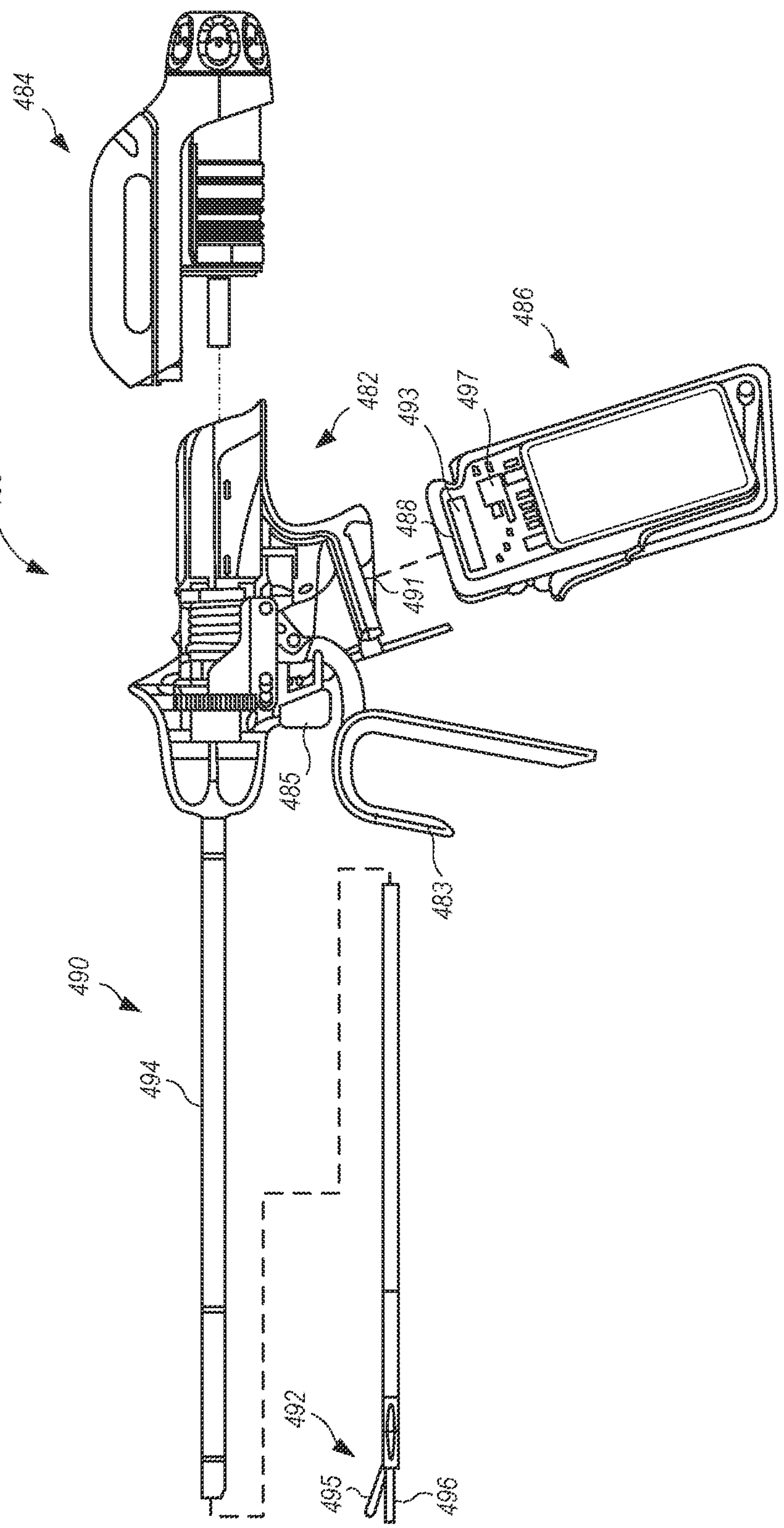


FIG. 25

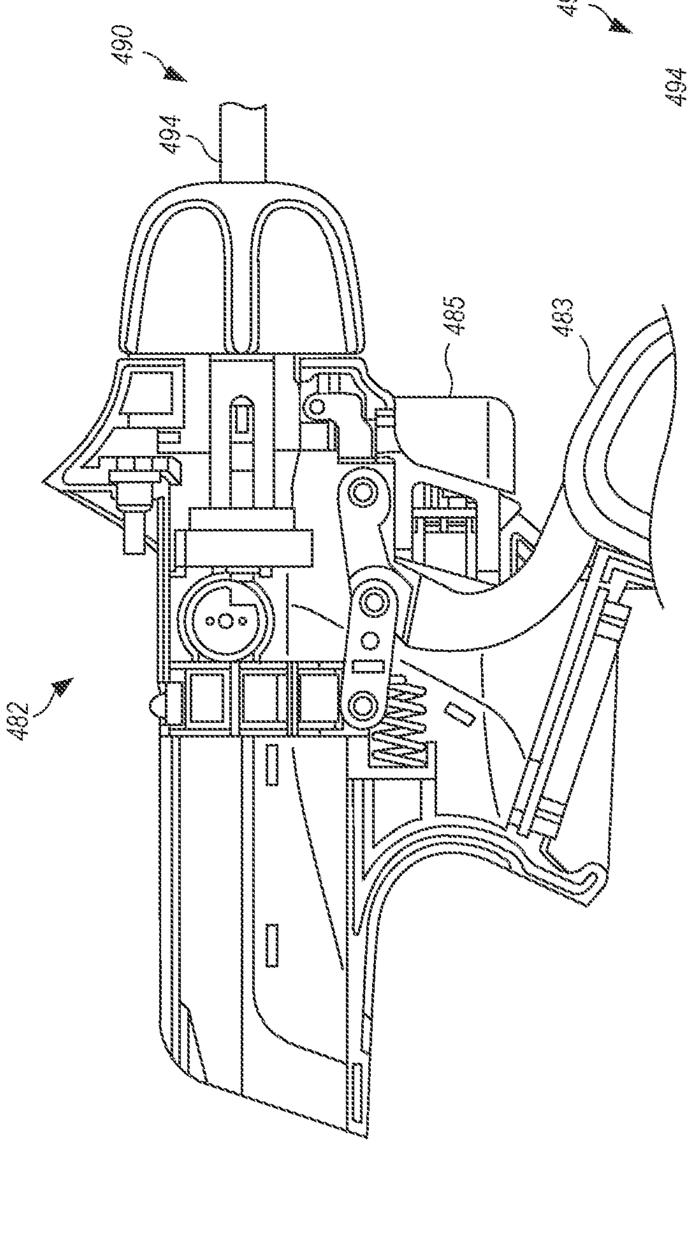


FIG. 26

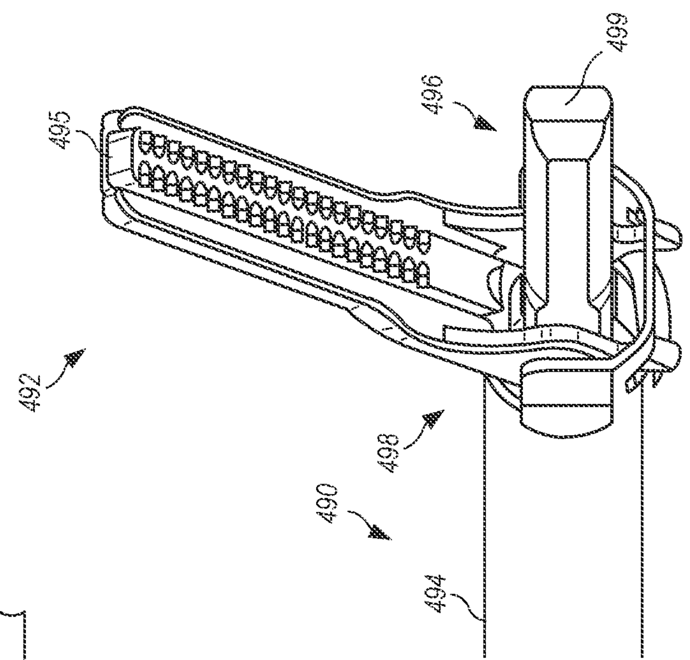


FIG. 27

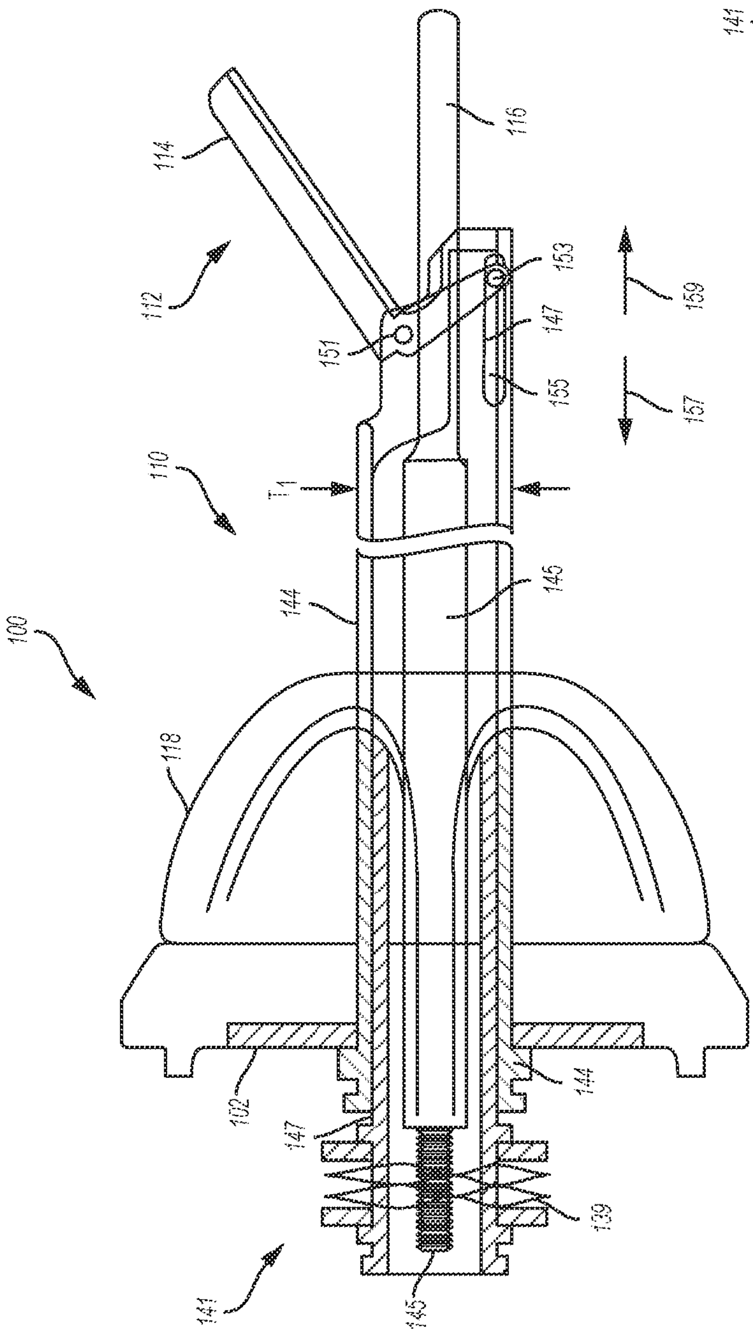

100
110
112
114
116
118
102
144
145
147
151
153
155
157
159
T1
141
139

FIG. 28

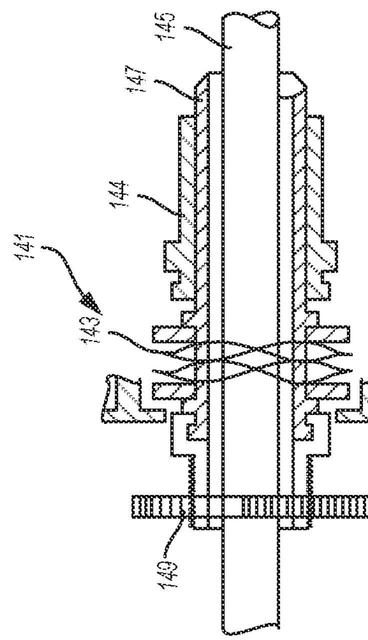
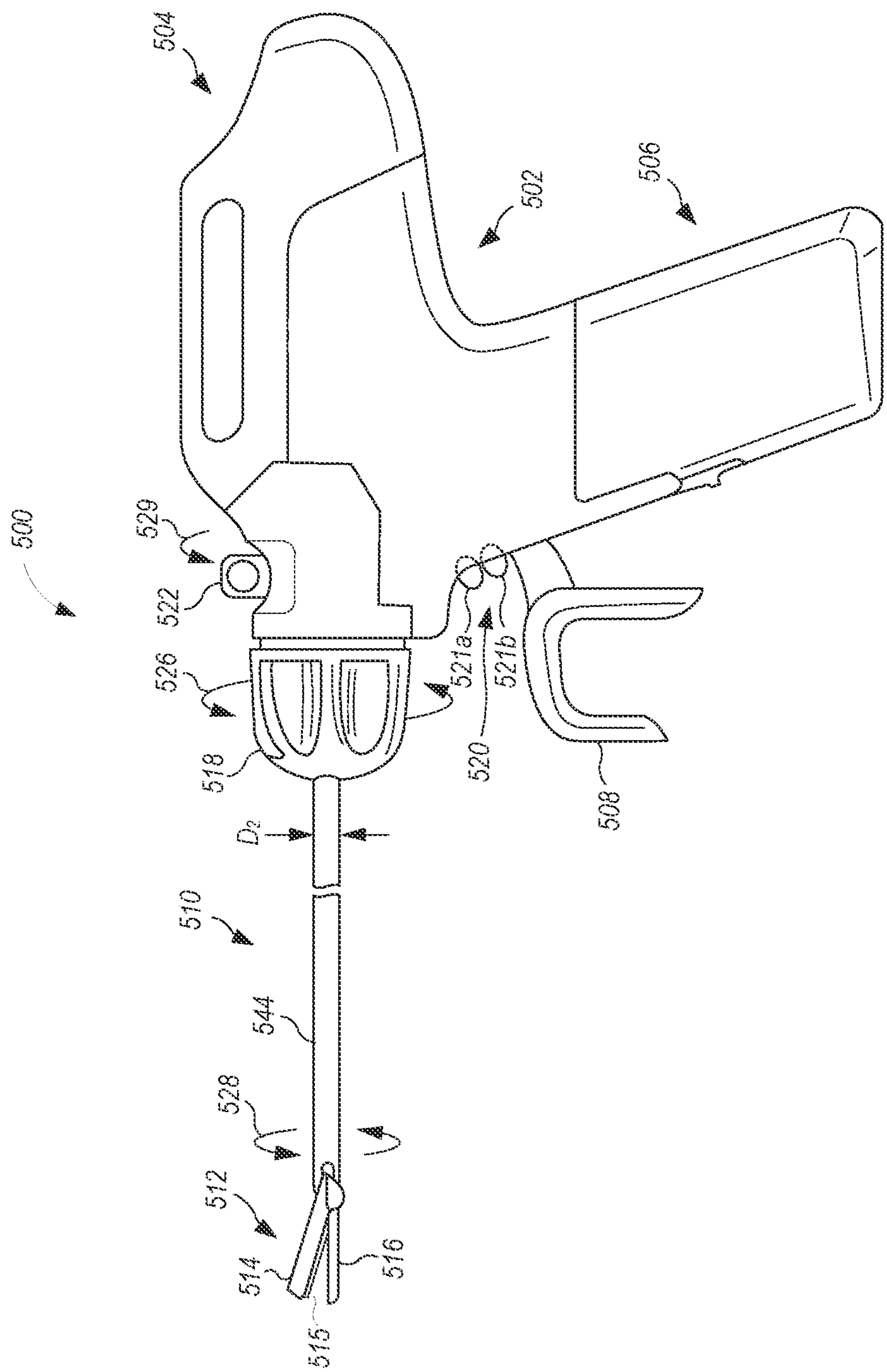

141
143
144
145
147
149

FIG. 29



504
502
506
500
529
522
526
518
521a
521b
520
508
D_2
510
544
528
512
514
515
516

FIG. 30

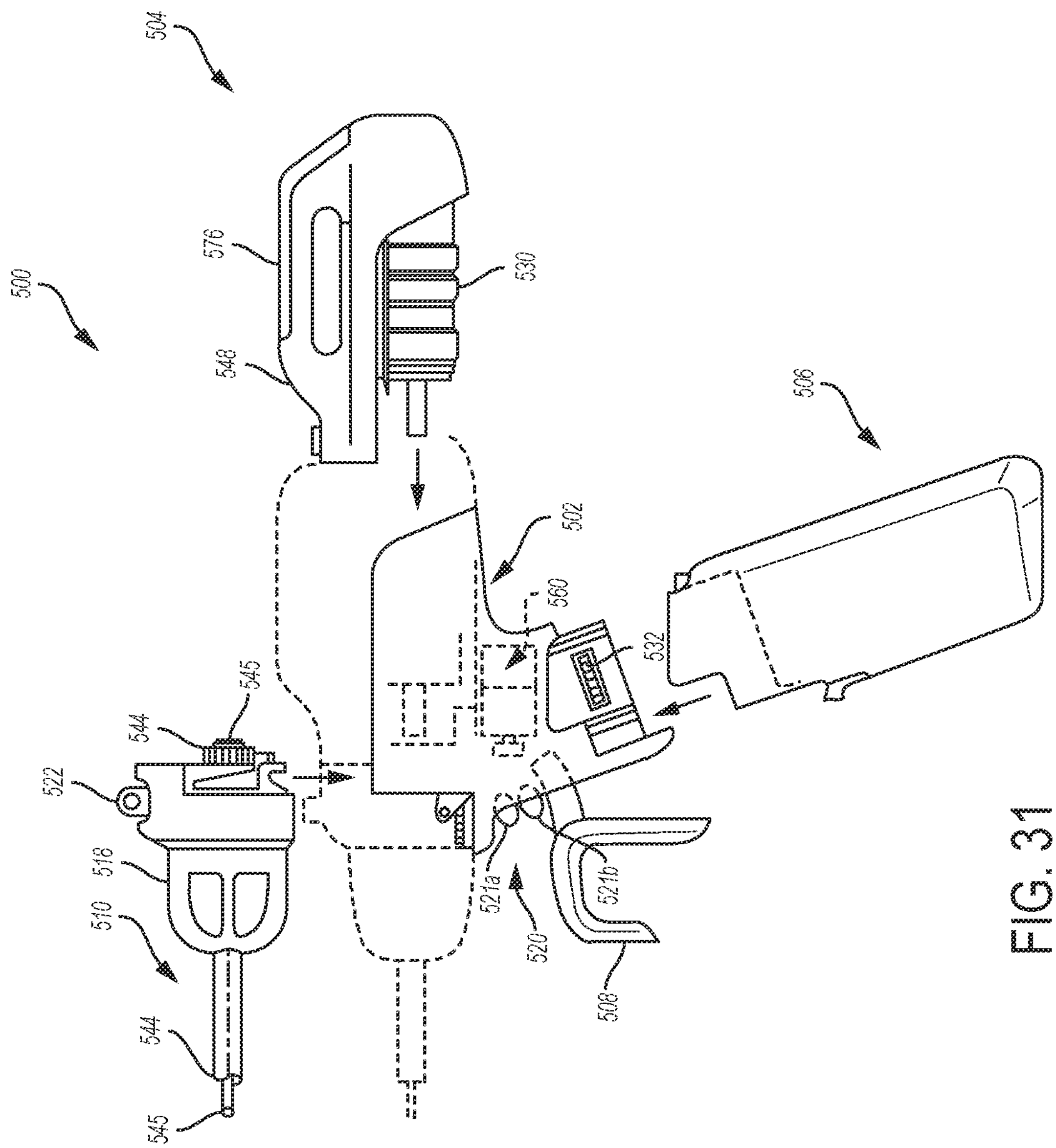

504
500
576
530
548
506
502
560
532
545
544
522
516
510
521a
521b
520
508
544
545

FIG. 31

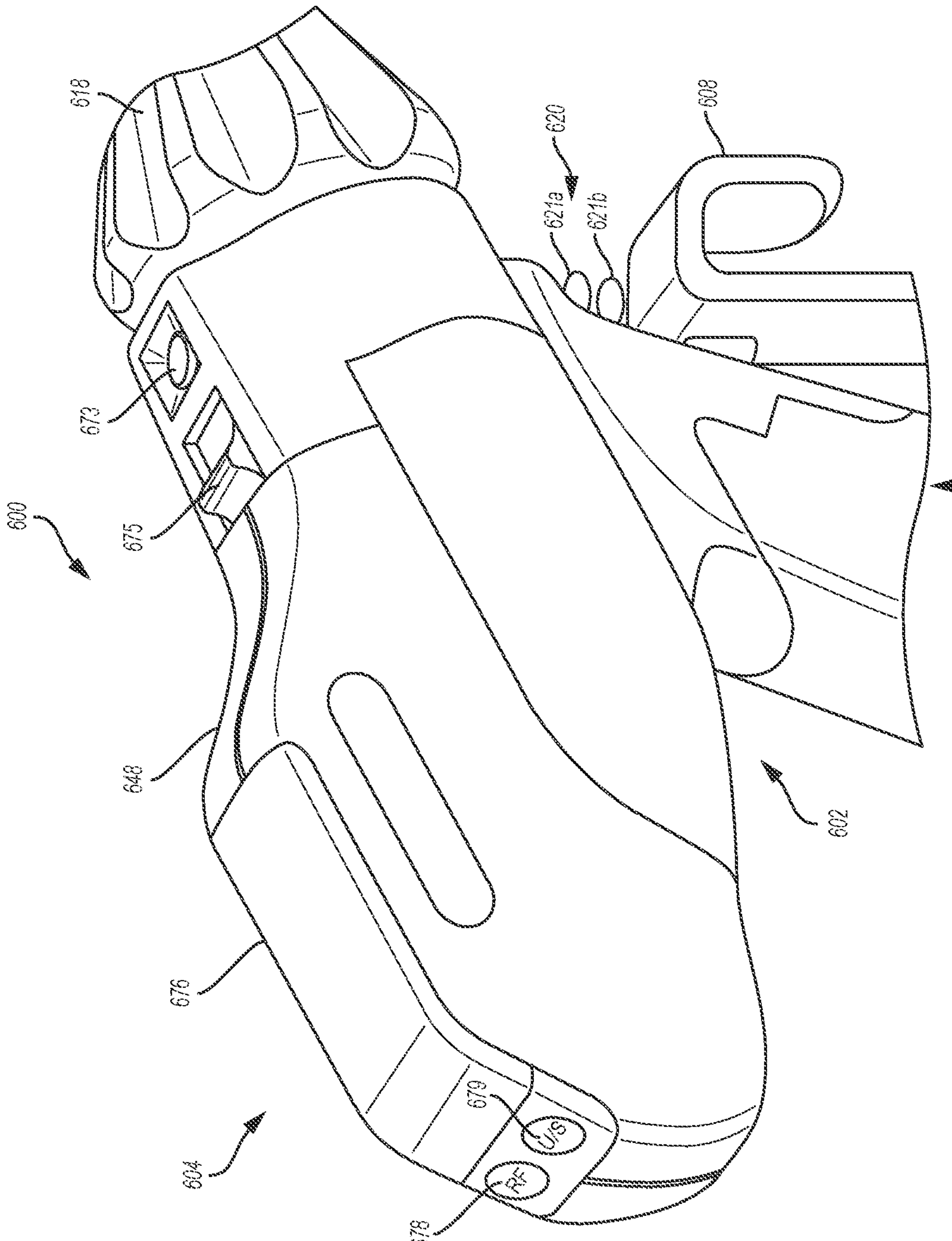

618
608
620
621a
621b
673
600
675
606
648
602
676
679
U/S
RF
604
678

FIG. 32

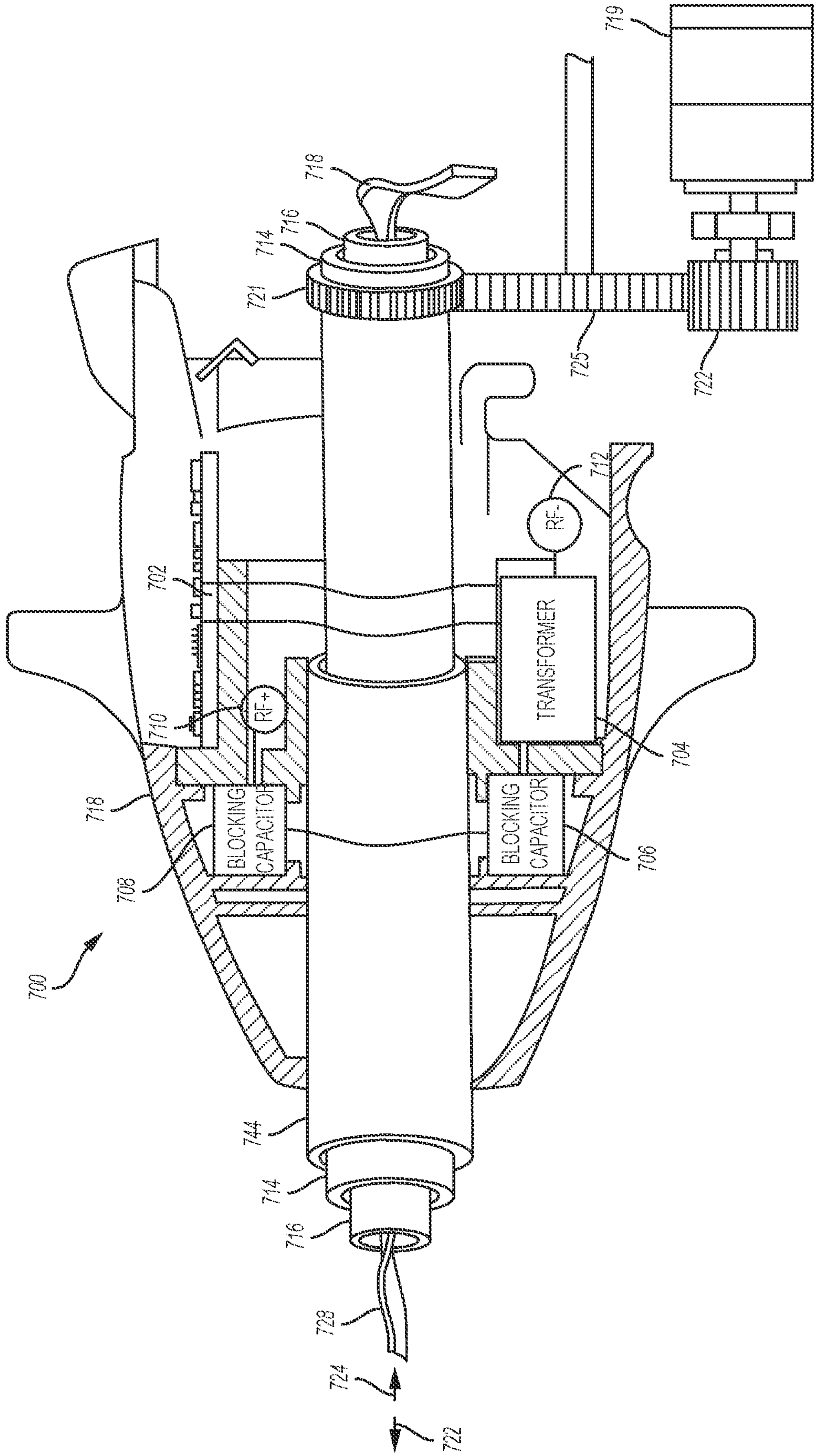

700
702
704
706
708
710
712
714
716
718
719
721
722
724
725
728
744
RF+
RF-
TRANSFORMER
BLOCKING CAPACITOR
BLOCKING CAPACITOR

FIG. 33

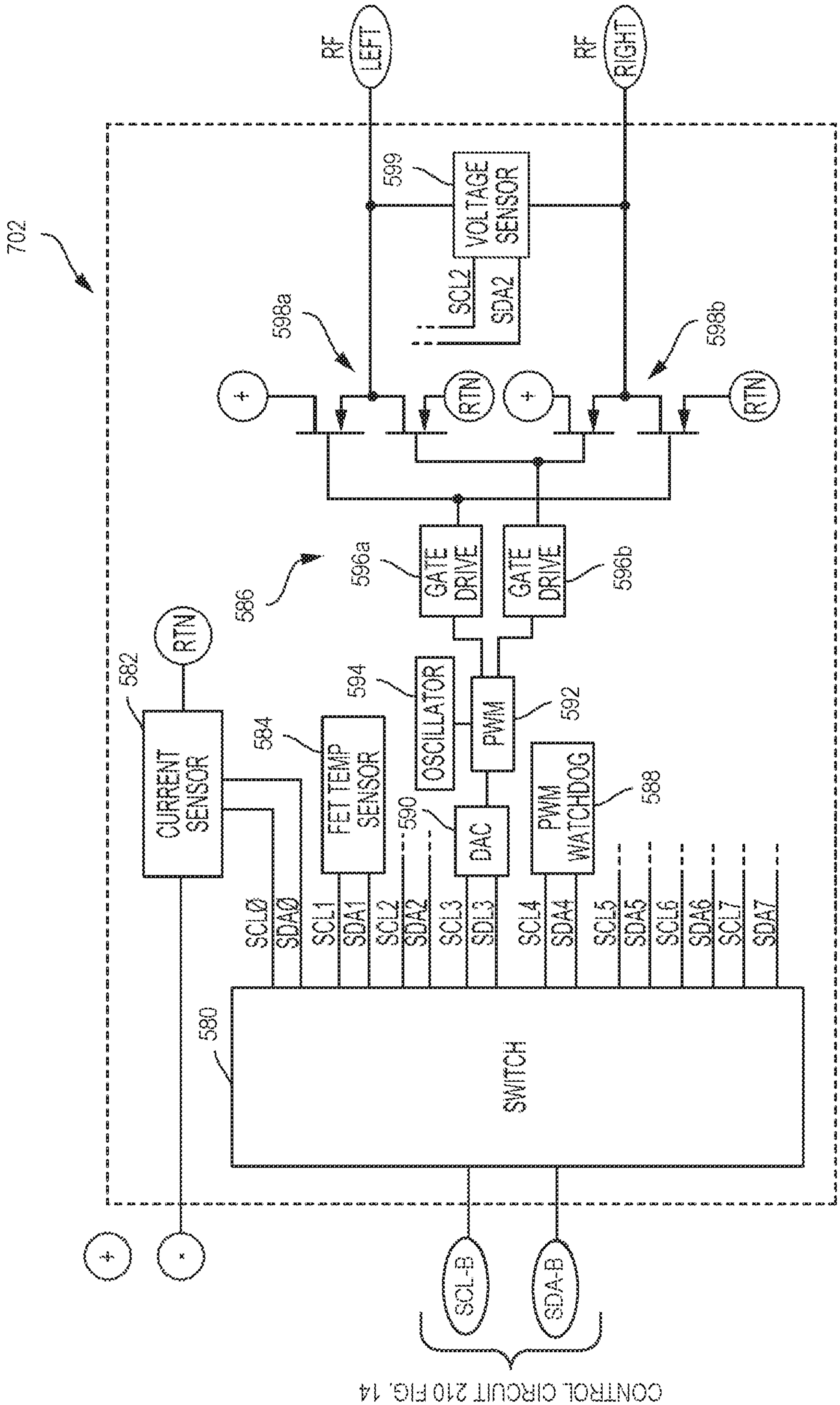

702
RF LEFT
RF RIGHT
599
VOLTAGE SENSOR
SCL2
SDA2
598a
598b
RTN
586
596a
GATE DRIVE
596b
GATE DRIVE
582
CURRENT SENSOR
584
FET TEMP SENSOR
594
OSCILLATOR
PWM
592
590
DAC
PWM WATCHDOG
588
SCLØ
SDAØ
SCL1
SDA1
SCL2
SDA2
SCL3
SDL3
SCL4
SDA4
SCL5
SDA5
SCL6
SDA6
SCL7
SDA7
580
SWITCH
SCL-B
SDA-B
CONTROL CIRCUIT 210 FIG. 14

FIG. 34

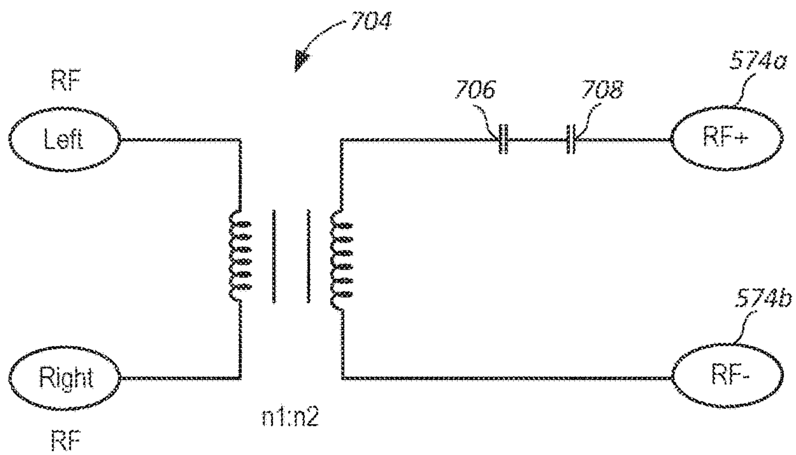

704
RF
Left
Right
RF
n1:n2
706
708
574a
RF+
574b
RF-

FIG. 35

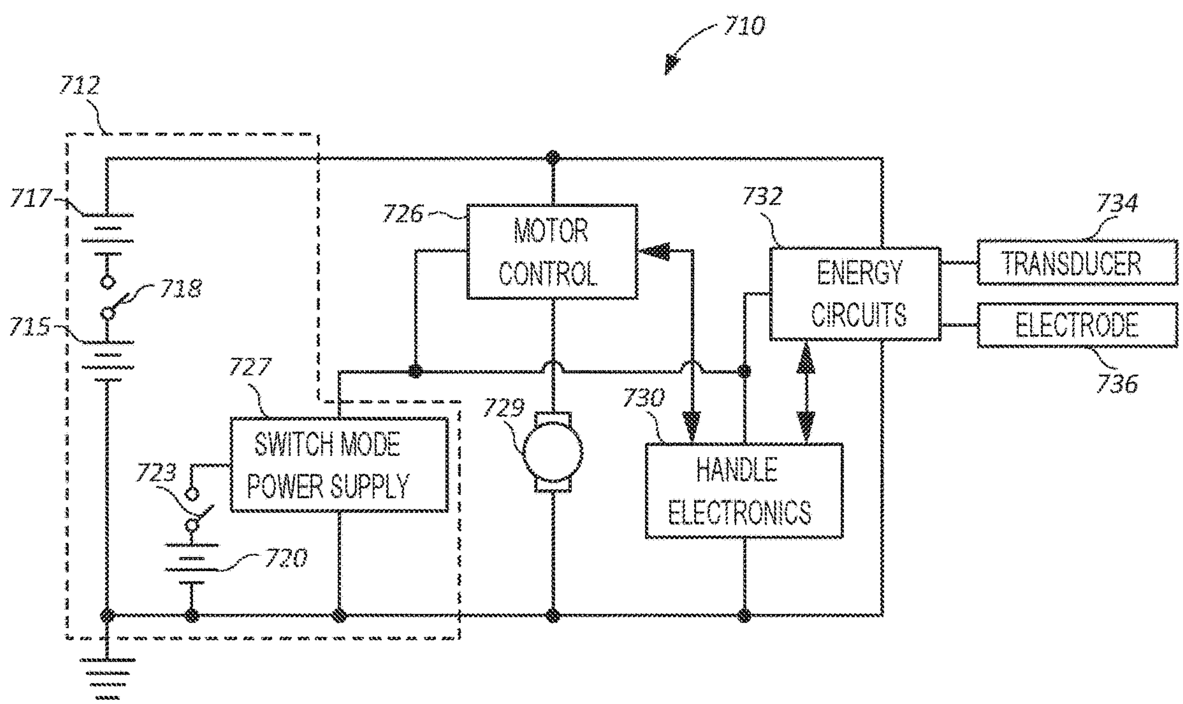
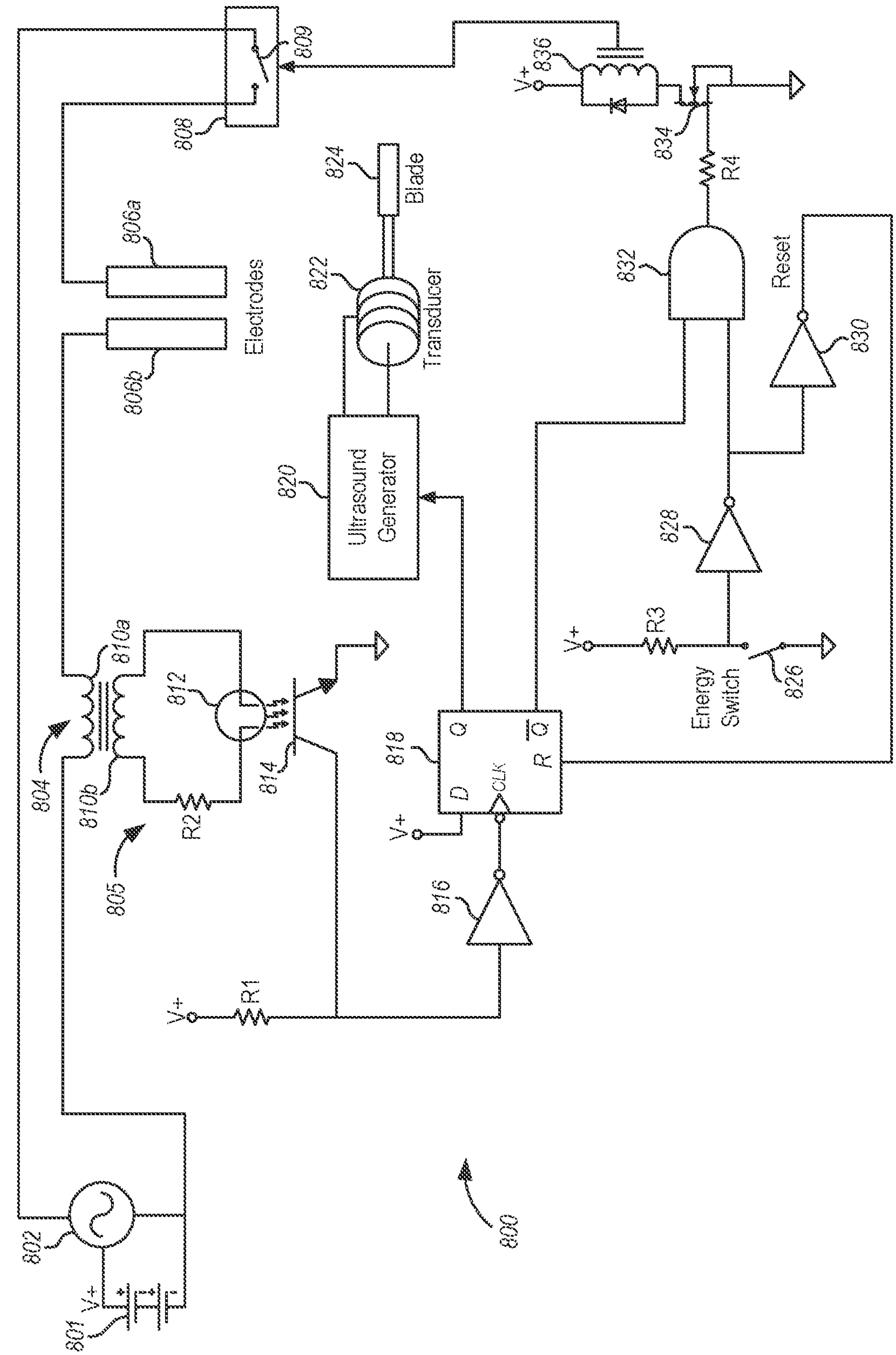

710
712
717
718
715
727
SWITCH MODE
POWER SUPPLY
723
720
726
MOTOR
CONTROL
729
730
HANDLE
ELECTRONICS
732
ENERGY
CIRCUITS
734
TRANSDUCER
ELECTRODE
736

FIG. 36



800
801
V+
802
804
805
806a
806b
Electrodes
808
809
810a
810b
812
814
R1
R2
816
818
D
Q
CLK
R
$\overline{Q}$
820
Ultrasound Generator
822
Transducer
824
Blade
826
Energy Switch
R3
828
830
Reset
832
834
R4
836

FIG. 37

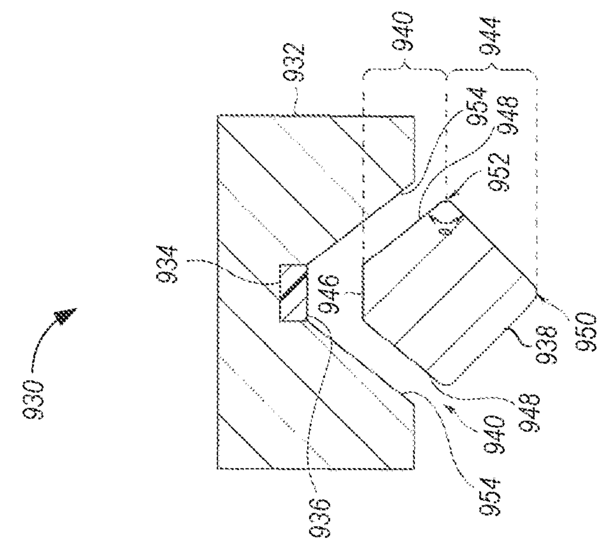

930
932
934
936
938
940
944
946
948
950
952
954

FIG. 39

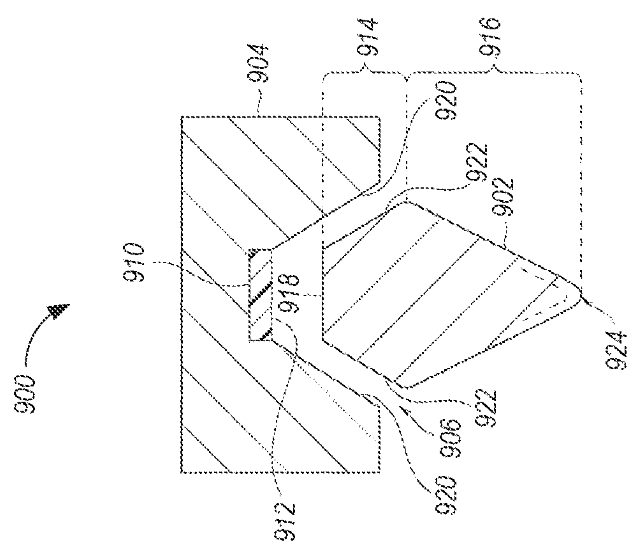

900
902
904
906
910
912
914
916
918
920
922
924

FIG. 38

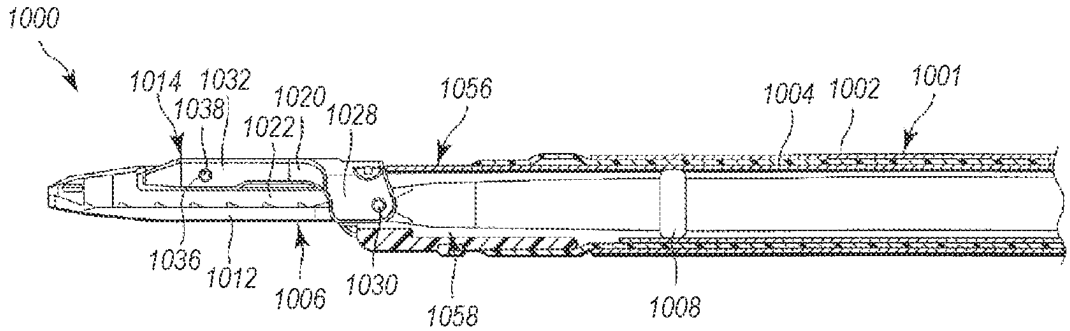


FIG. 40

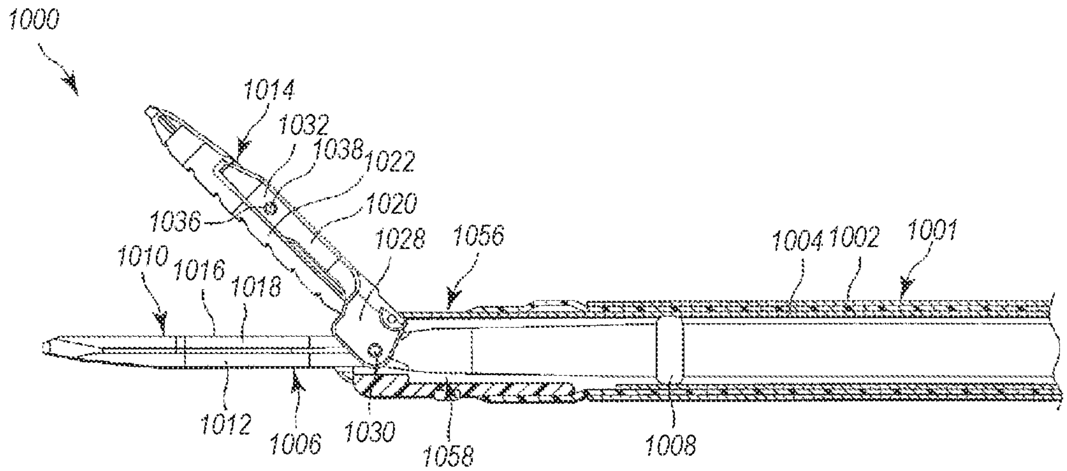


FIG. 41

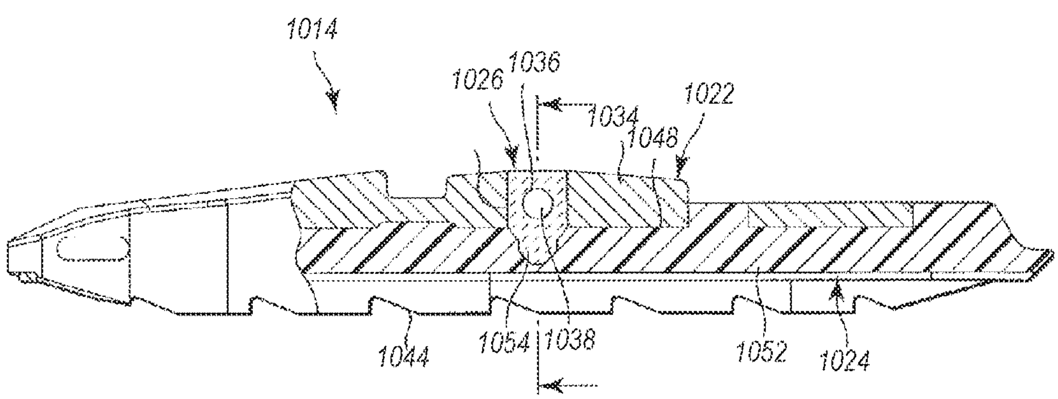


FIG. 42

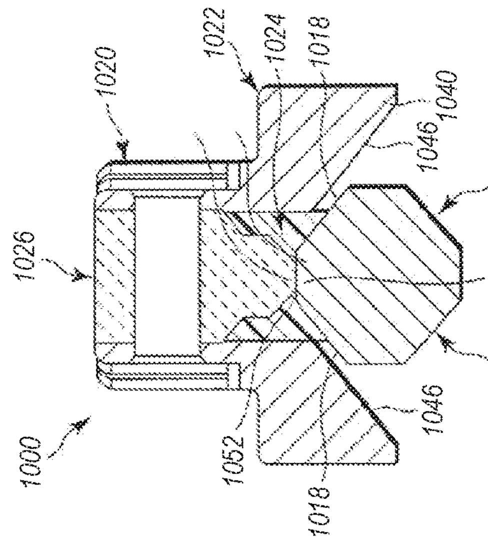

1000
1026
1020
1022
1024
1018
1040
1046
1006
1010
1012
1046
1018
1052

FIG. 44

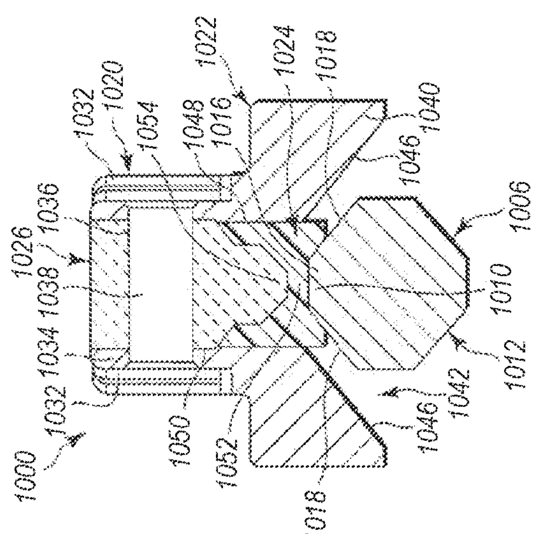

1000
1032
1034
1038
1026
1036
1032
1020
1054
1048
1016
1022
1024
1018
1040
1046
1006
1010
1012
1042
1046
1018
1052
1050

FIG. 43

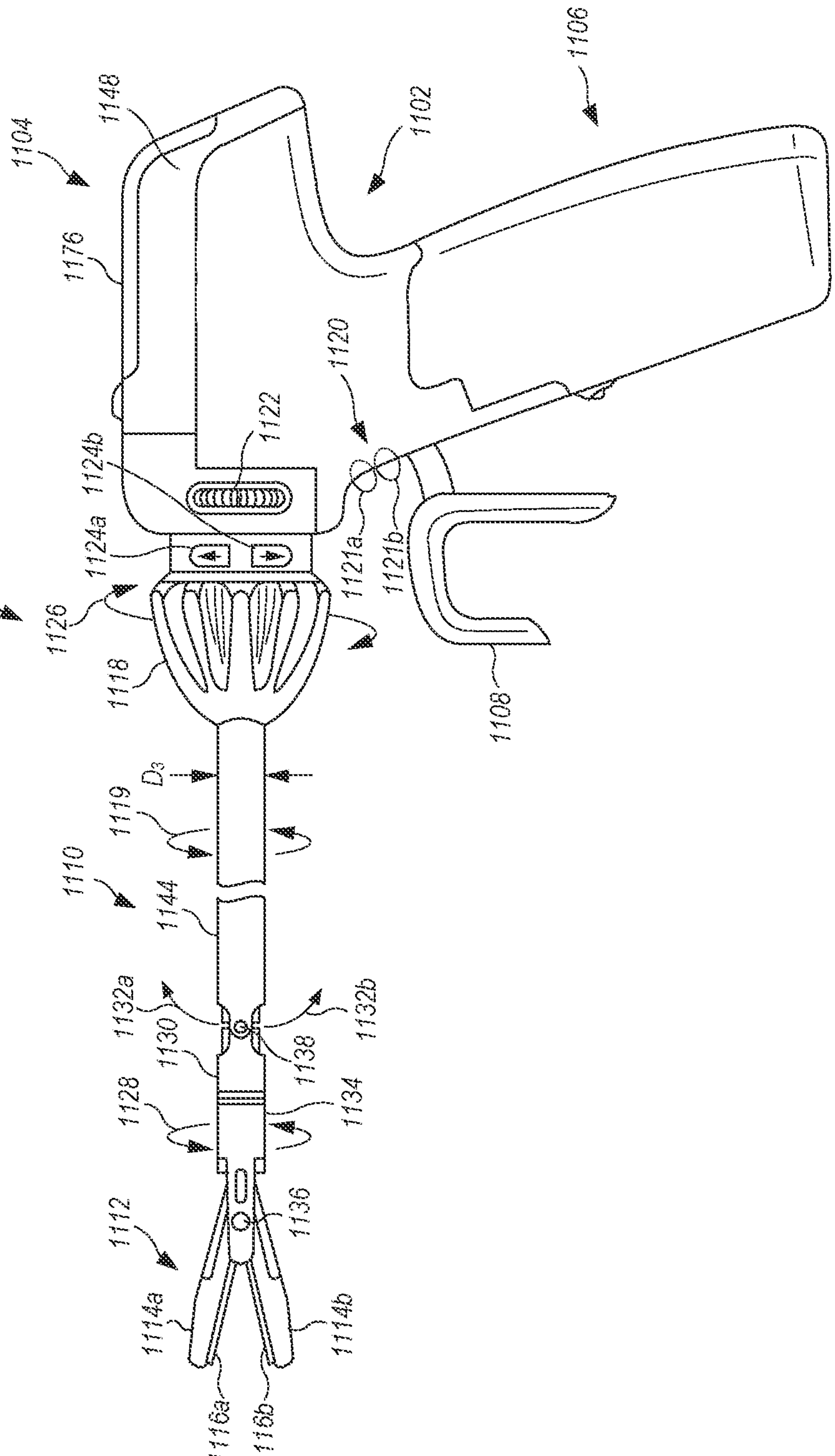

1106
1102
1148
1104
1176
1120
1122
1124b
1124a
1100
1121a
1121b
1126
1118
1108
D_3
1119
1110
1144
1132a
1132b
1130
1138
1128
1134
1136
1112
1114a
1114b
1116a
1116b

FIG. 45

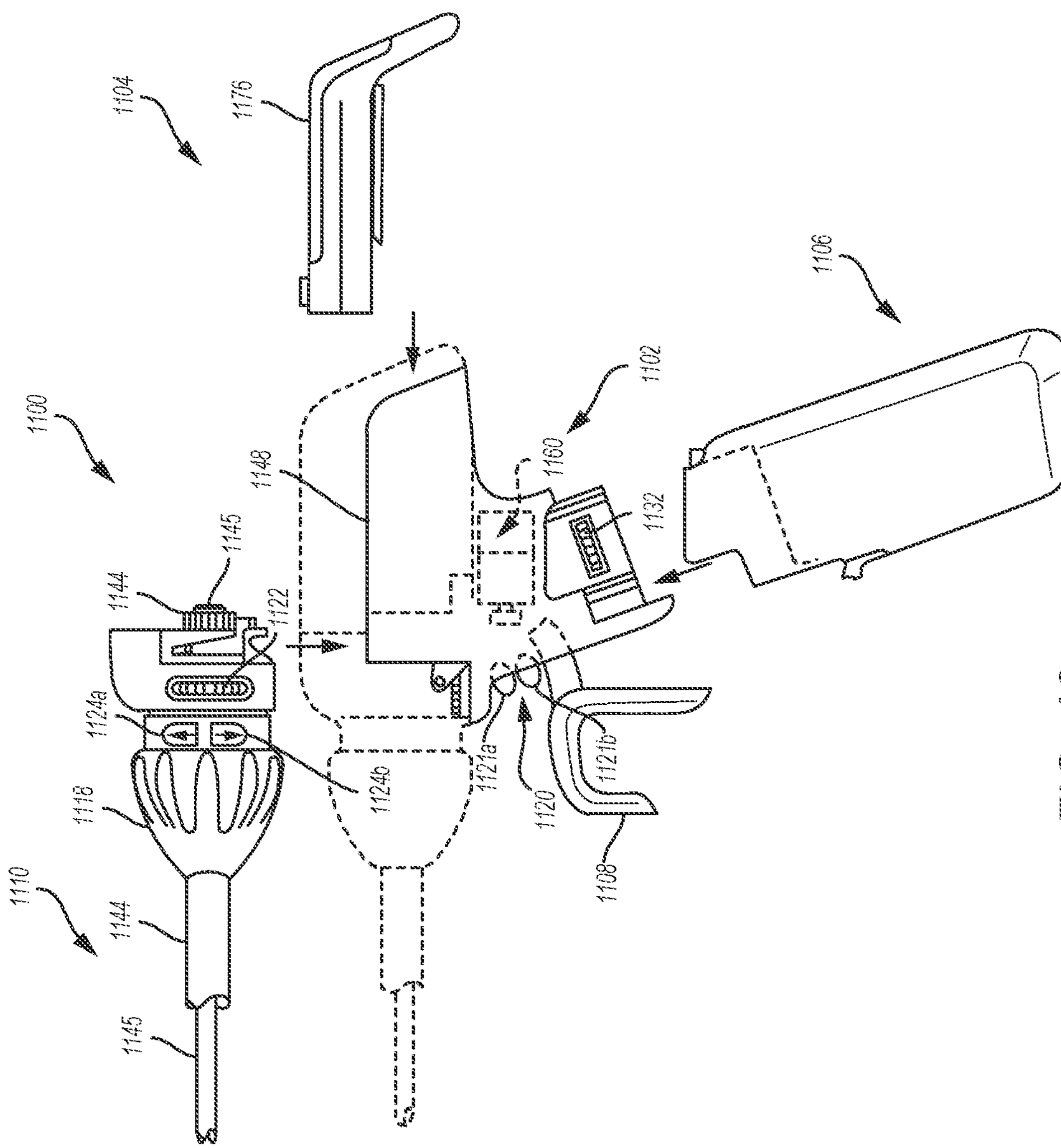

1100
1104
1176
1106
1102
1148
1160
1132
1145
1144
1122
1124a
1124b
1118
1110
1144
1145
1121a
1120
1121b
1108

FIG. 46

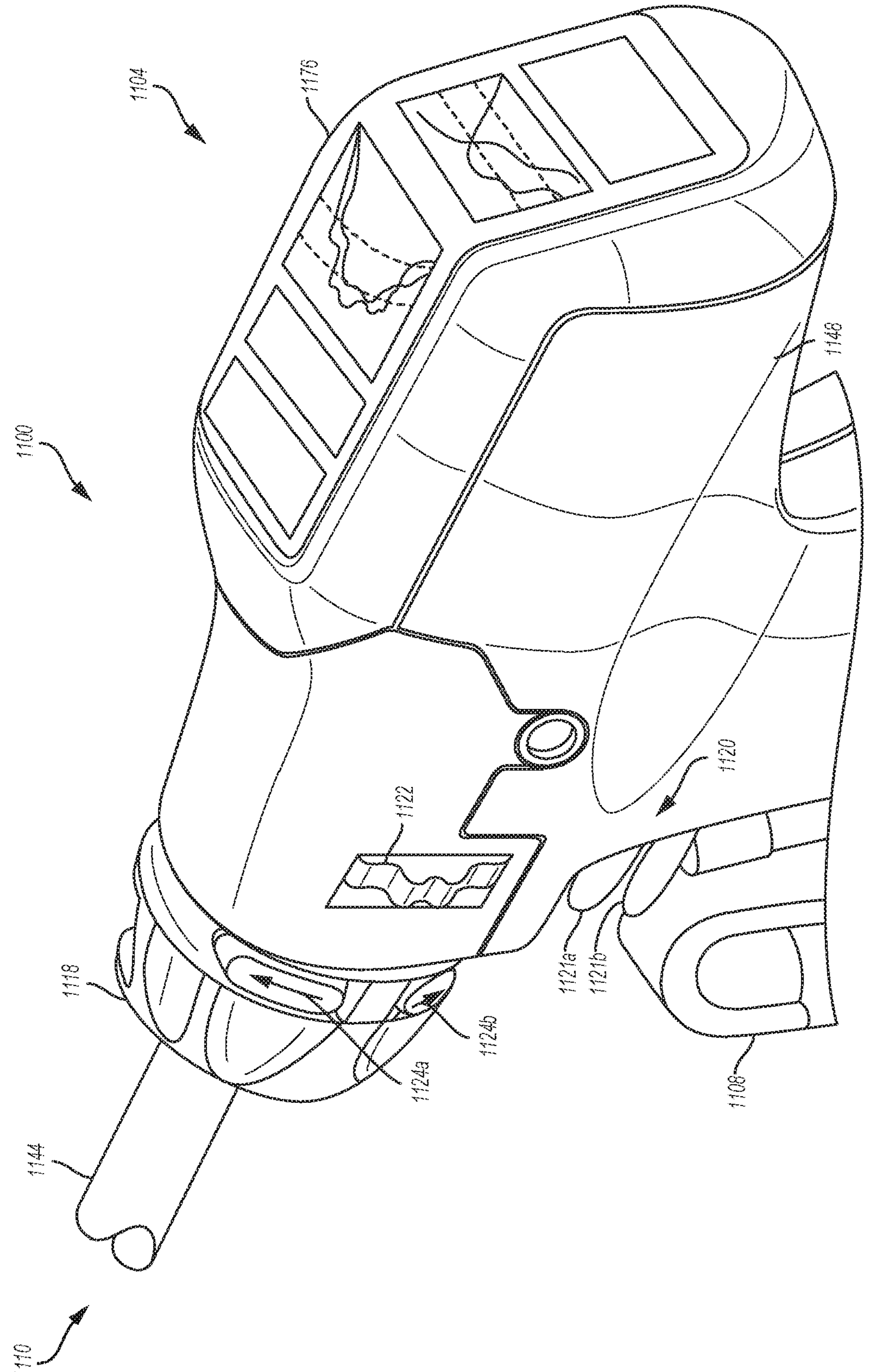

1104
1175
1100
1148
1102
1120
1122
1118
1124a
1124b
1121a
1121b
1108
1144
1110

FIG. 47

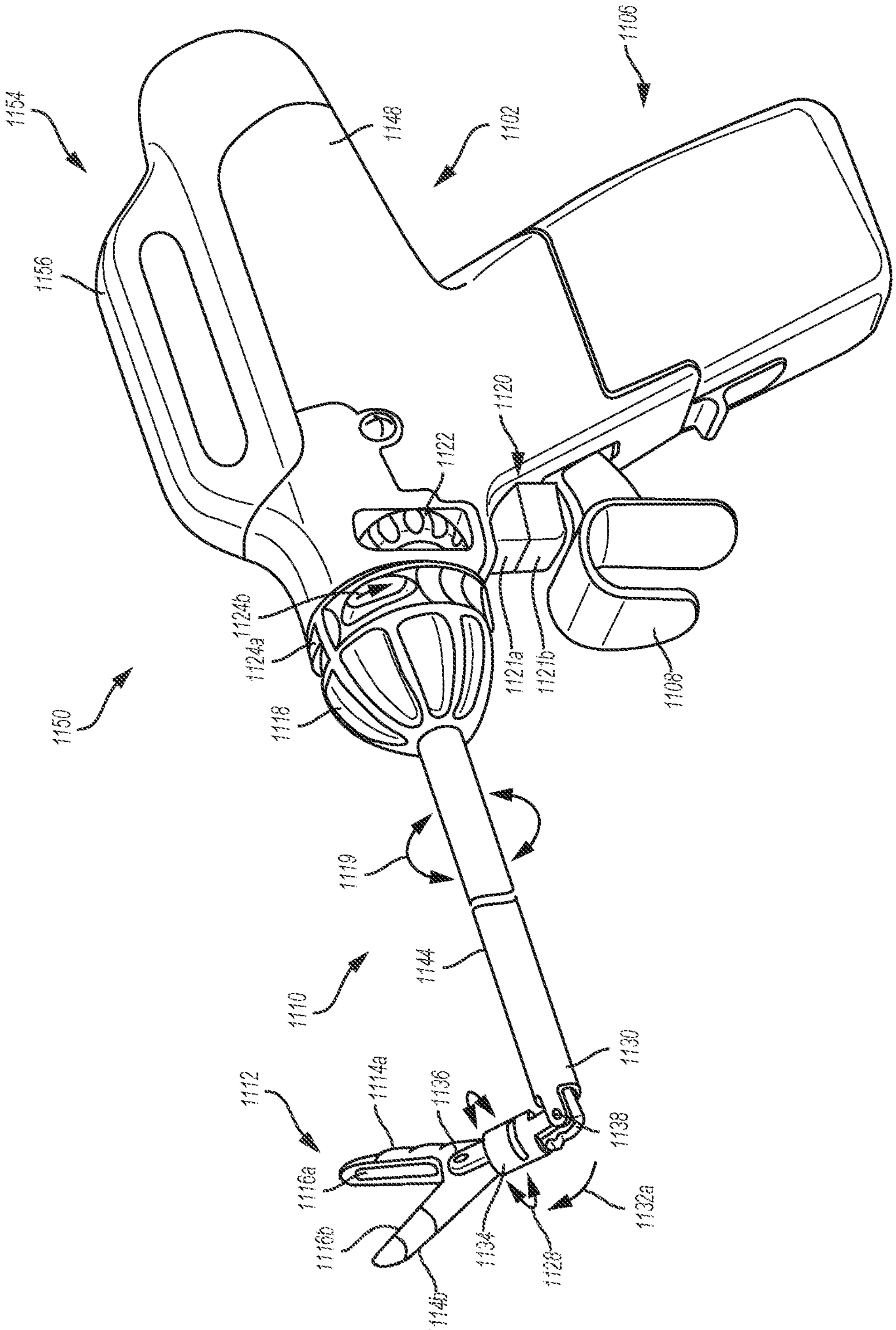

1106
1154
1148
1102
1156
1120
1122
1124b
1124a
1121a
1121b
1108
1150
1118
1119
1144
1110
1130
1112
1114a
1136
1138
1116a
1116b
1134
1128
1132a
1114b

FIG. 48

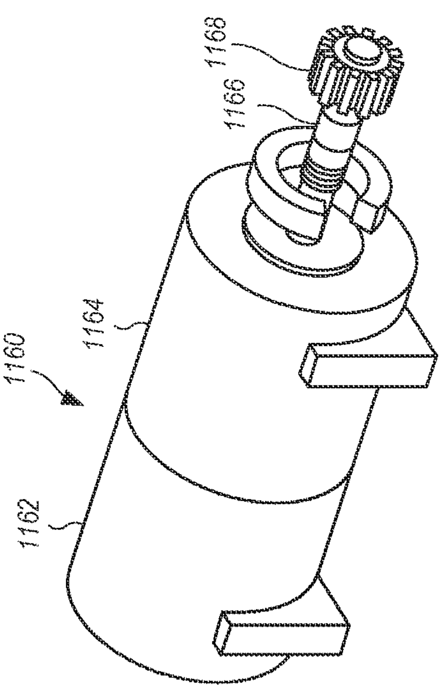

1160
1162
1164
1166
1168

FIG. 49

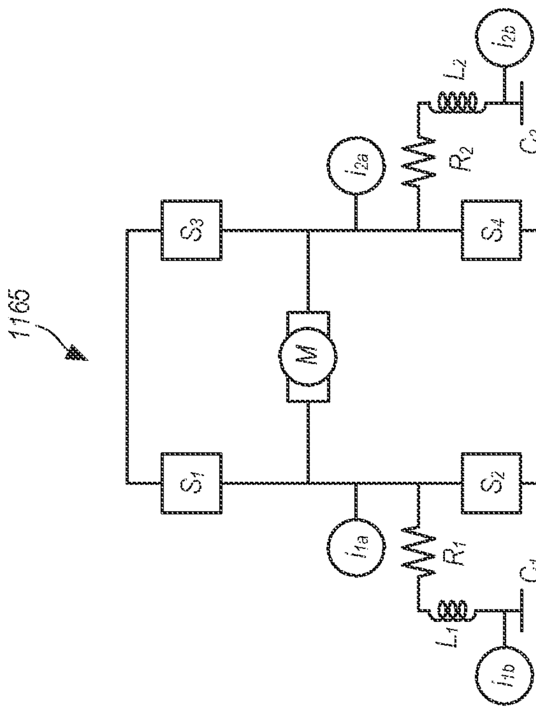

1165
S_1
S_2
S_3
S_4
M
i_{1a}
i_{1b}
i_{2a}
i_{2b}
R_1
R_2
L_1
L_2
C_1
C_2

FIG. 50

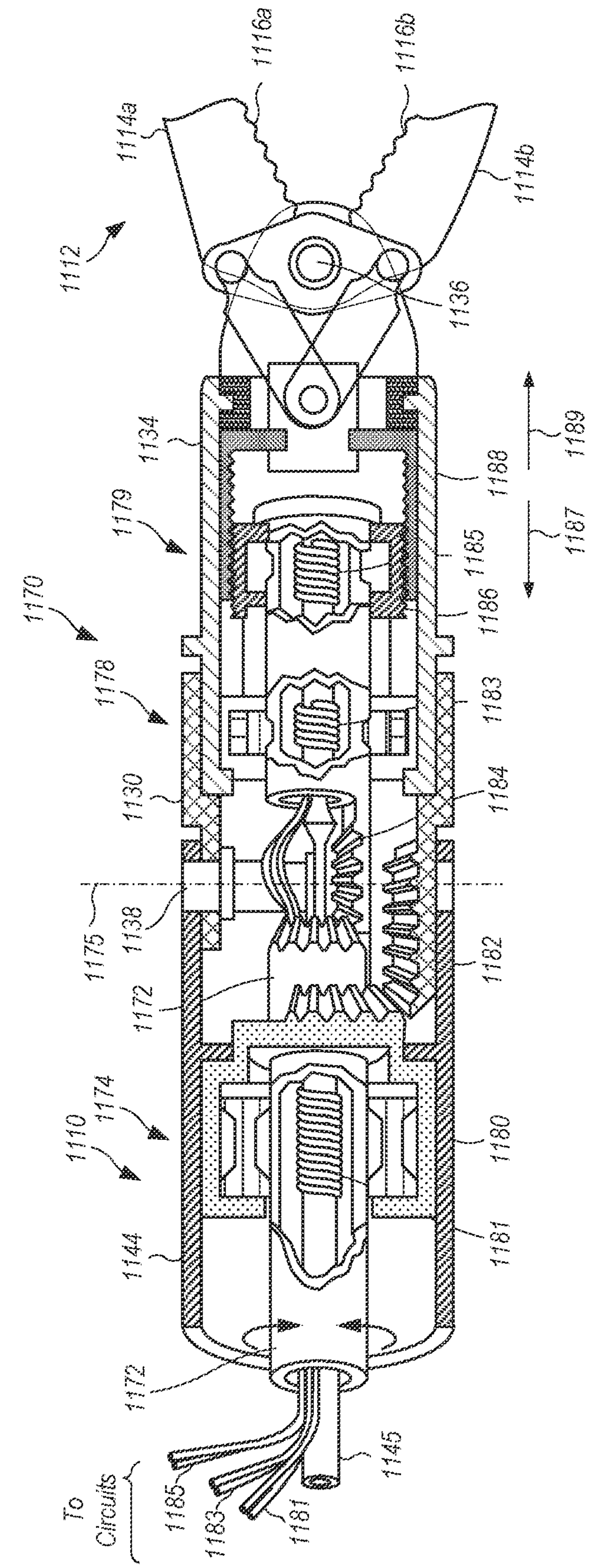

1116a
1116b
1114a
1114b
1112
1136
1134
1188
1189
1179
1187
1185
1170
1186
1178
1183
1130
1184
1175
1138
1182
1172
1110
1174
1180
1144
1181
1172
1145
To Circuits
1185
1183
1181

FIG. 51

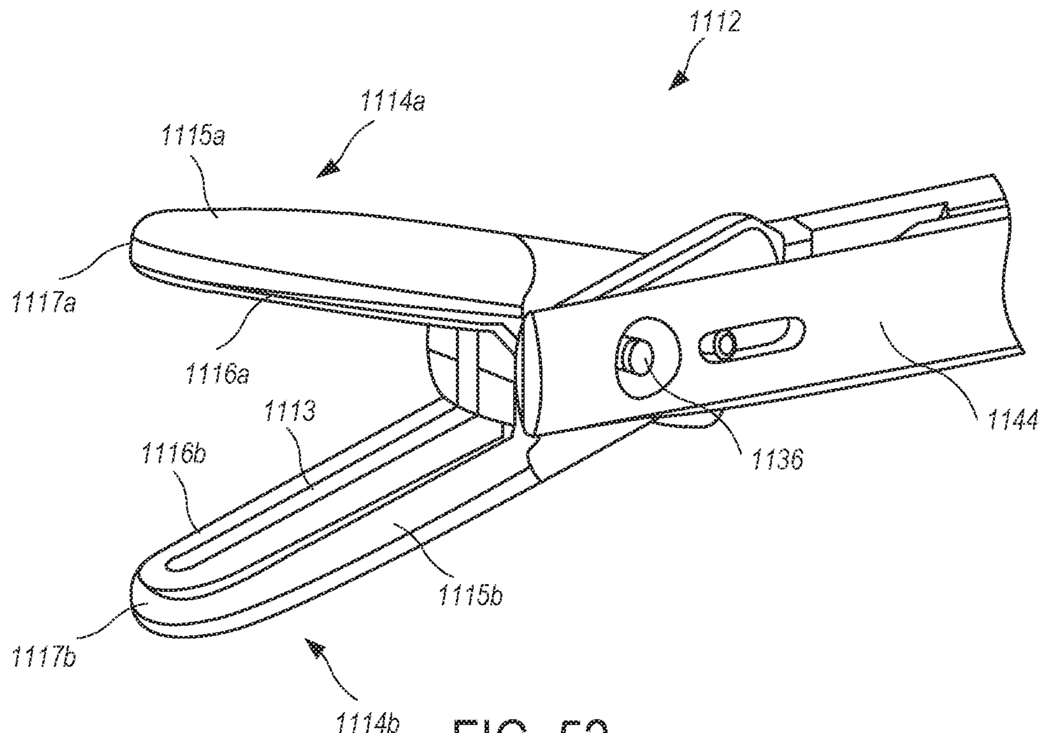

1112
1114a
1115a
1117a
1116a
1113
1116b
1115b
1117b
1114b
1136
1144

FIG. 52

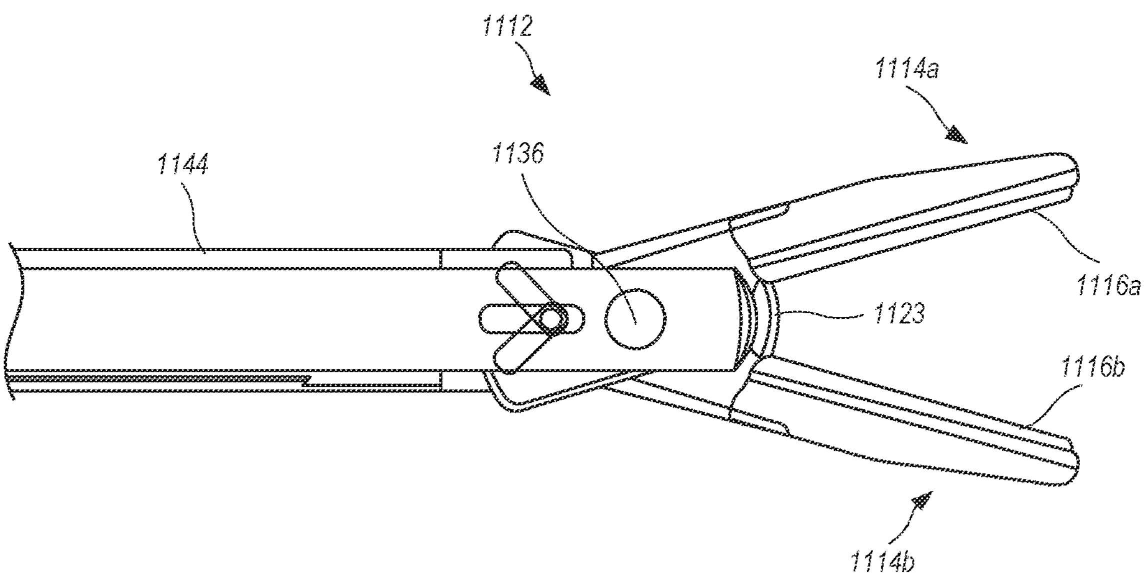

1112
1114a
1144
1136
1116a
1123
1116b
1114b

FIG. 53

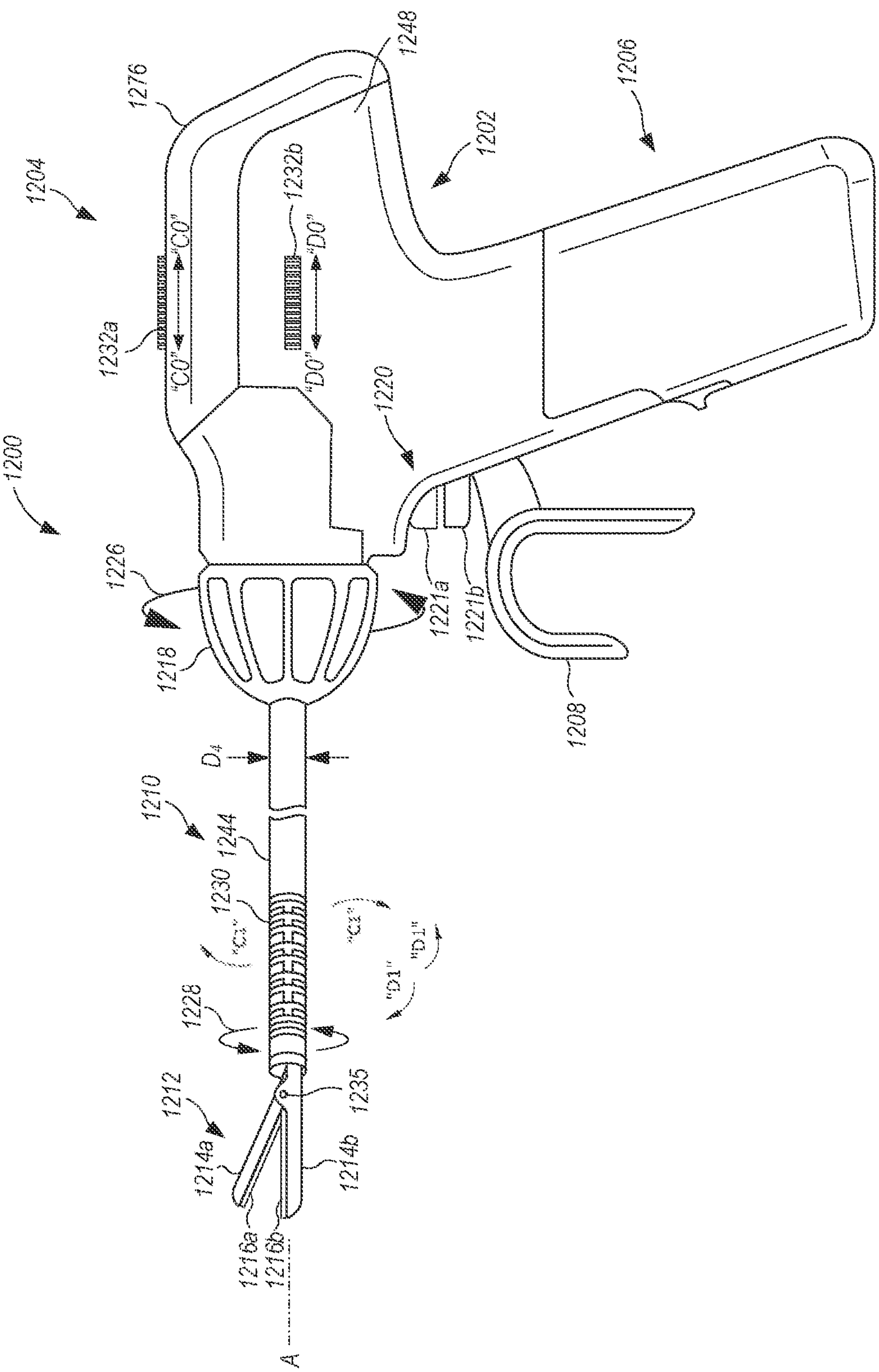

1200
1202
1204
1206
1208
1210
1212
1214a
1214b
1216a
1216b
1218
1220
1221a
1221b
1226
1228
1230
1232a
1232b
1235
1244
1248
1276
"C0"
"C0"
"D0"
"D0"
"C1"
"C1"
"D1"
"D1"
D_4
A

FIG. 54

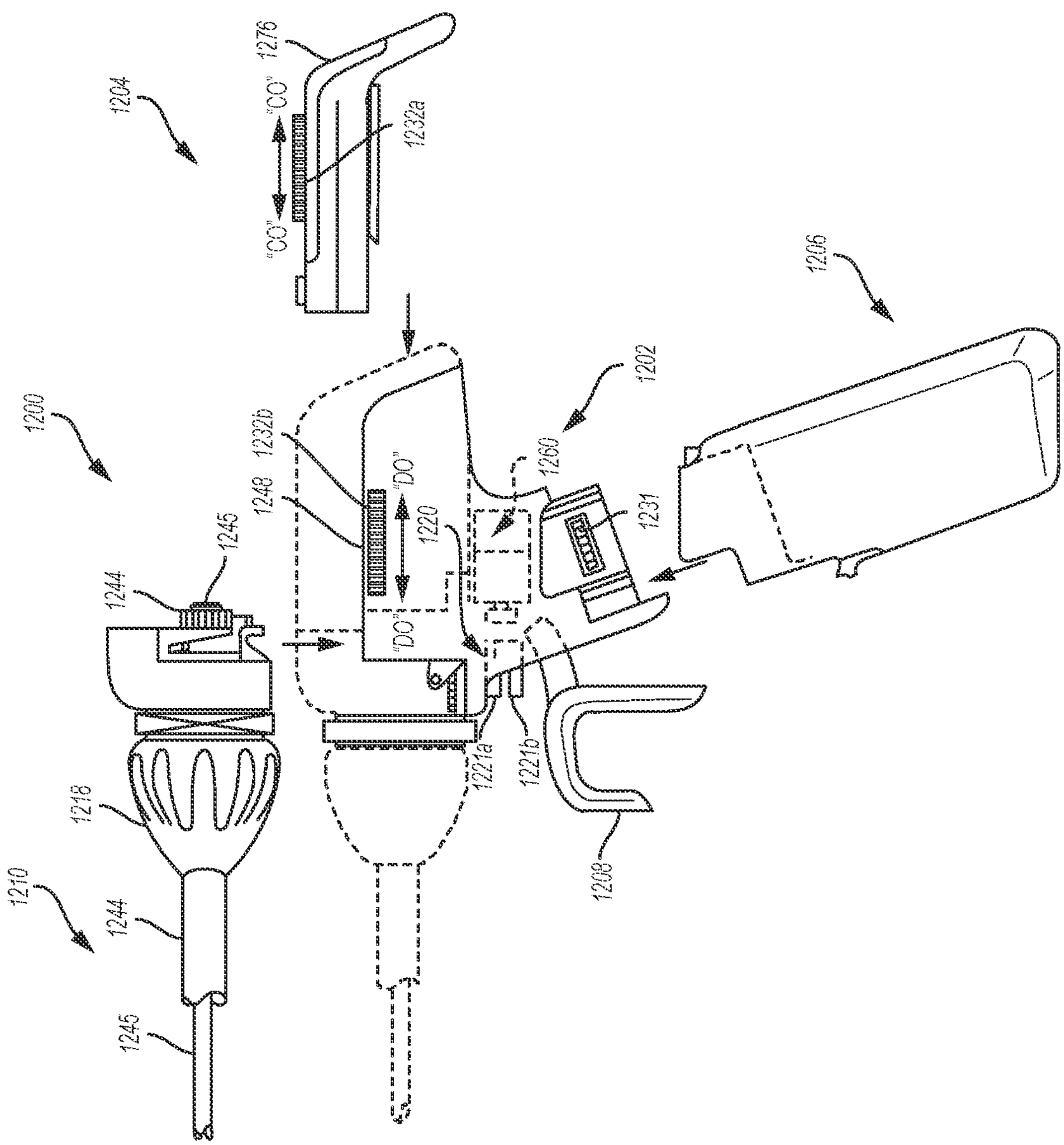

1204
1276
"CO"
"CO"
1232a
1206
1200
1202
1232b
1248
"DO"
"DO"
1220
1260
1231
1245
1244
1218
1210
1244
1245
1221a
1221b
1208

FIG. 55

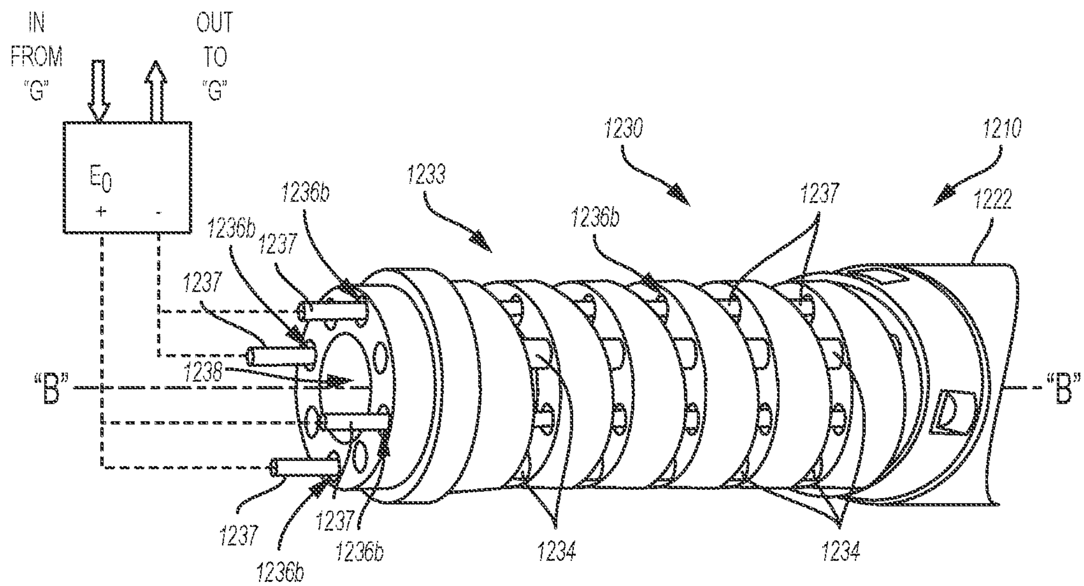

IN
FROM
"G"
OUT
TO
"G"
E_0
+
-
1230
1210
1233
1236b
1237
1222
1238
"B"
1234

FIG. 56

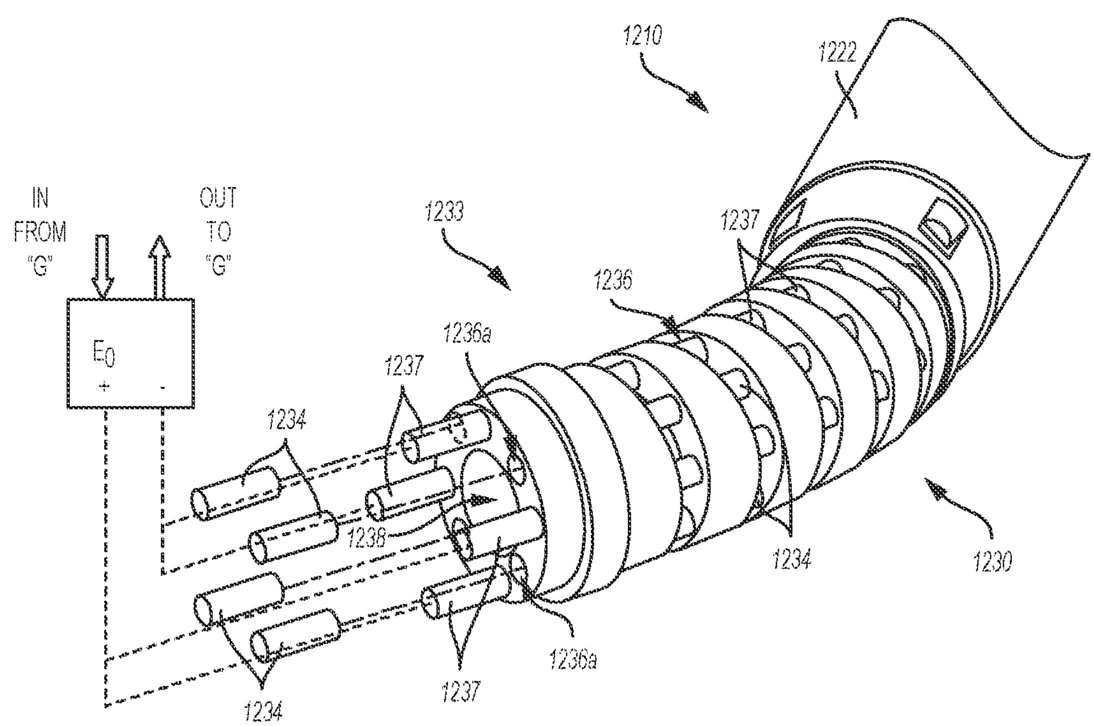

1210
1222
IN
FROM
"G"
OUT
TO
"G"
E_0
+
-
1233
1237
1236
1236a
1234
1238
1230

FIG. 57

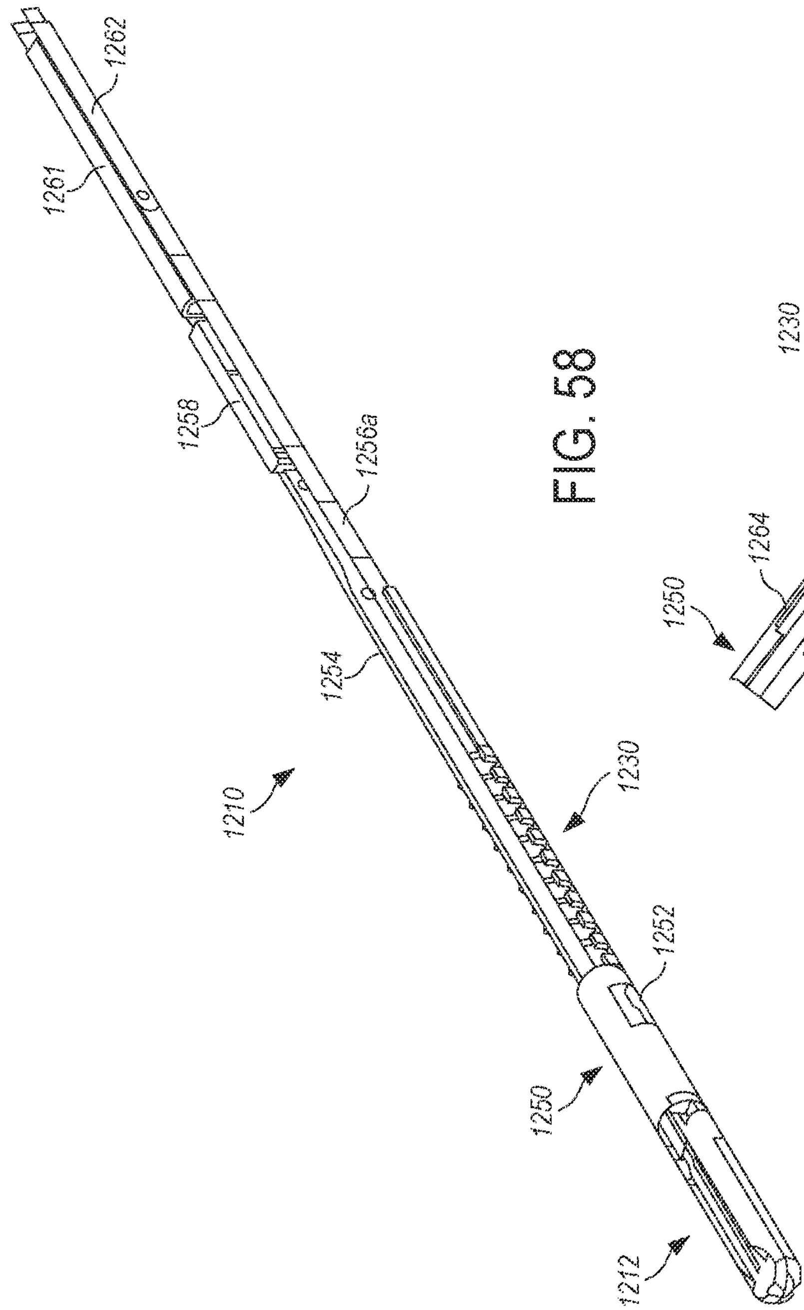

1262
1261
1258
1256a
1254
1210
1230
1252
1250
1212

FIG. 58

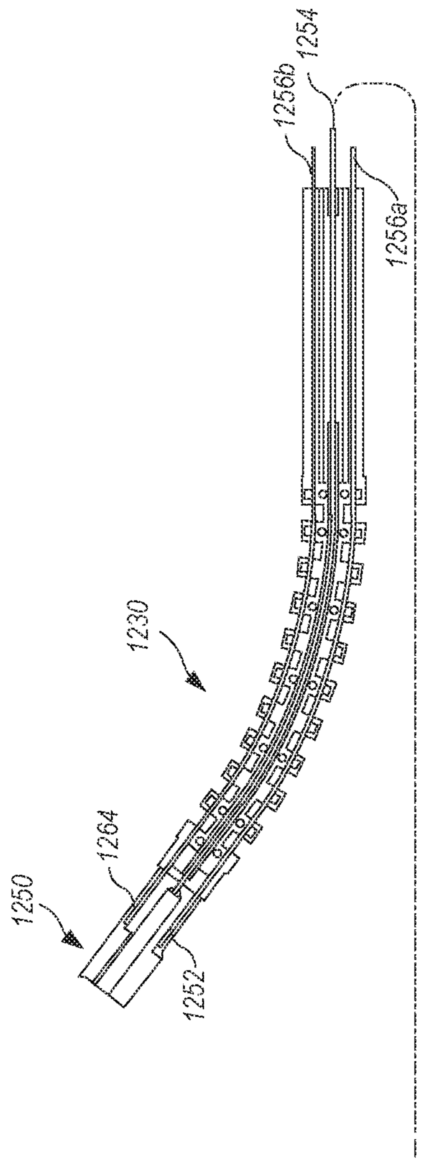
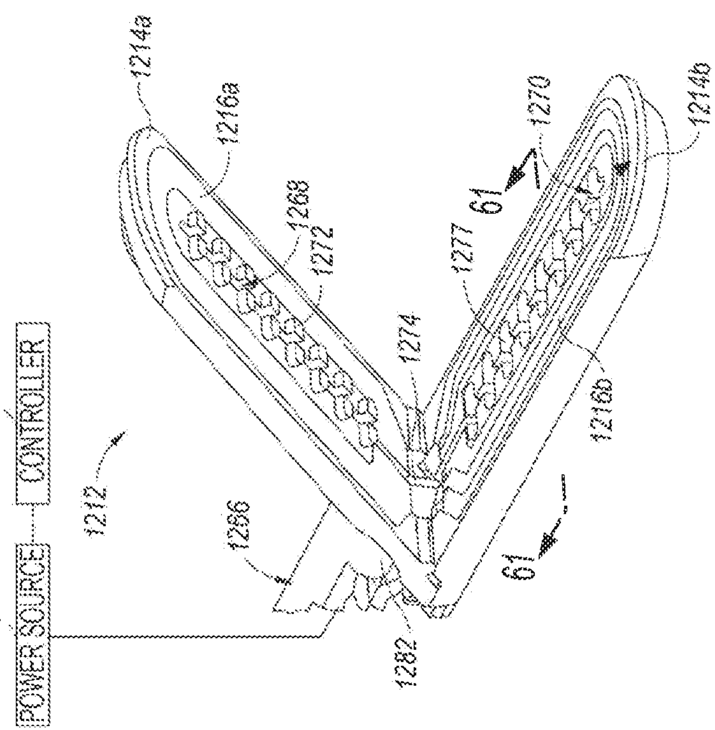
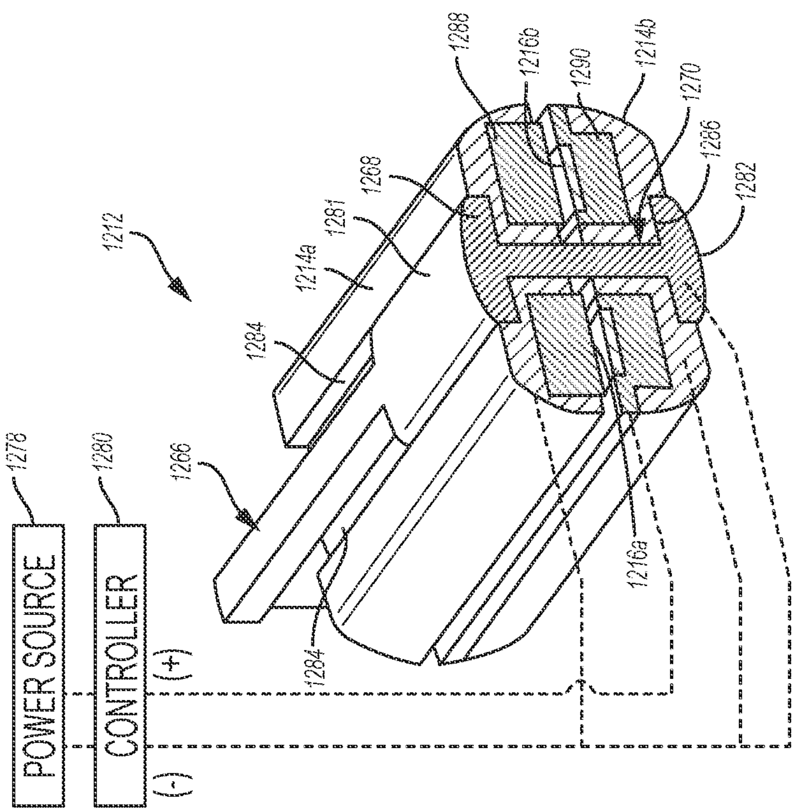

1256b
1254
1256a
1230
1264
1250
1252

FIG. 59



1278
POWER SOURCE
1280
CONTROLLER
1212
1286
1282
1214a
1216a
1268
1272
1274
1277
61
1270
1214b
1216b

FIG. 60



1278
POWER SOURCE
1280
CONTROLLER
(+)
(-)
1212
1266
1284
1214a
1281
1268
1288
1216b
1290
1214b
1270
1286
1282
1216a

FIG. 61

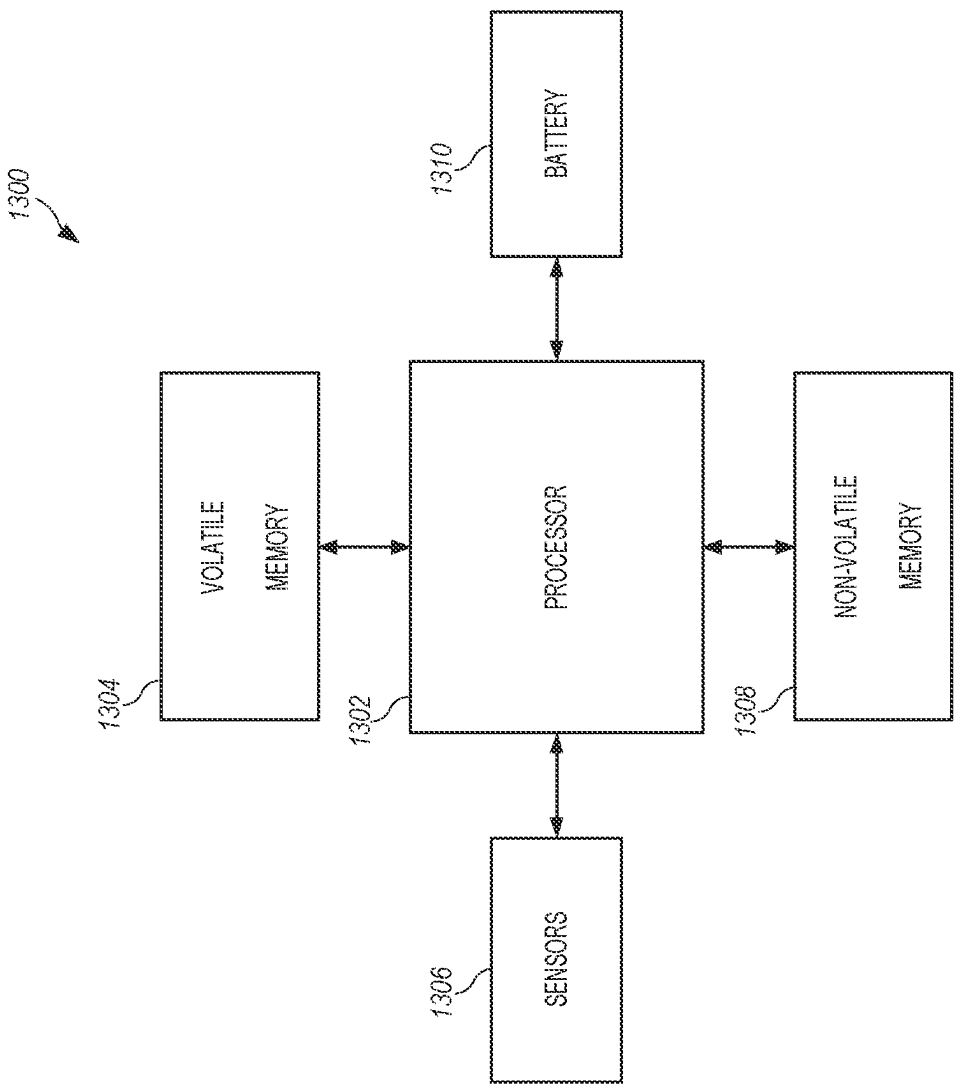

1300
1310
BATTERY
VOLATILE MEMORY
PROCESSOR
NON-VOLATILE MEMORY
1304
1302
1308
1306
SENSORS

FIG. 62

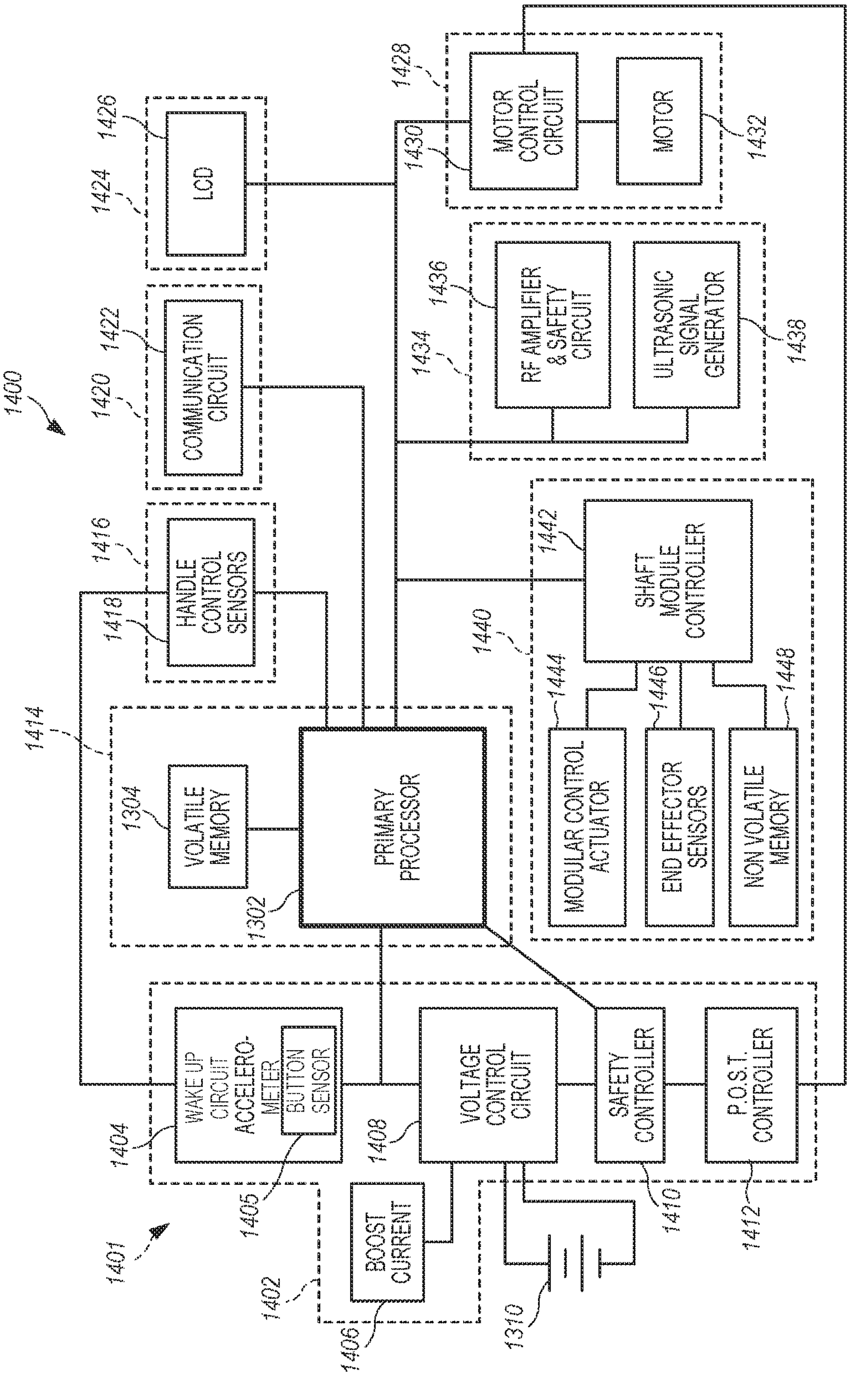

1400
1401
1402
1404
1405
1406
1408
1410
1412
1414
1416
1418
1420
1422
1424
1426
1428
1430
1432
1434
1436
1438
1440
1442
1444
1446
1448
1302
1304
1310
WAKE UP CIRCUIT ACCELERO-METER
BUTTON SENSOR
BOOST CURRENT
VOLTAGE CONTROL CIRCUIT
SAFETY CONTROLLER
P.O.S.T. CONTROLLER
VOLATILE MEMORY
PRIMARY PROCESSOR
HANDLE CONTROL SENSORS
COMMUNICATION CIRCUIT
LCD
MOTOR CONTROL CIRCUIT
MOTOR
RF AMPLIFIER & SAFETY CIRCUIT
ULTRASONIC SIGNAL GENERATOR
SHAFT MODULE CONTROLLER
MODULAR CONTROL ACTUATOR
END EFFECTOR SENSORS
NON VOLATILE MEMORY

FIG. 63

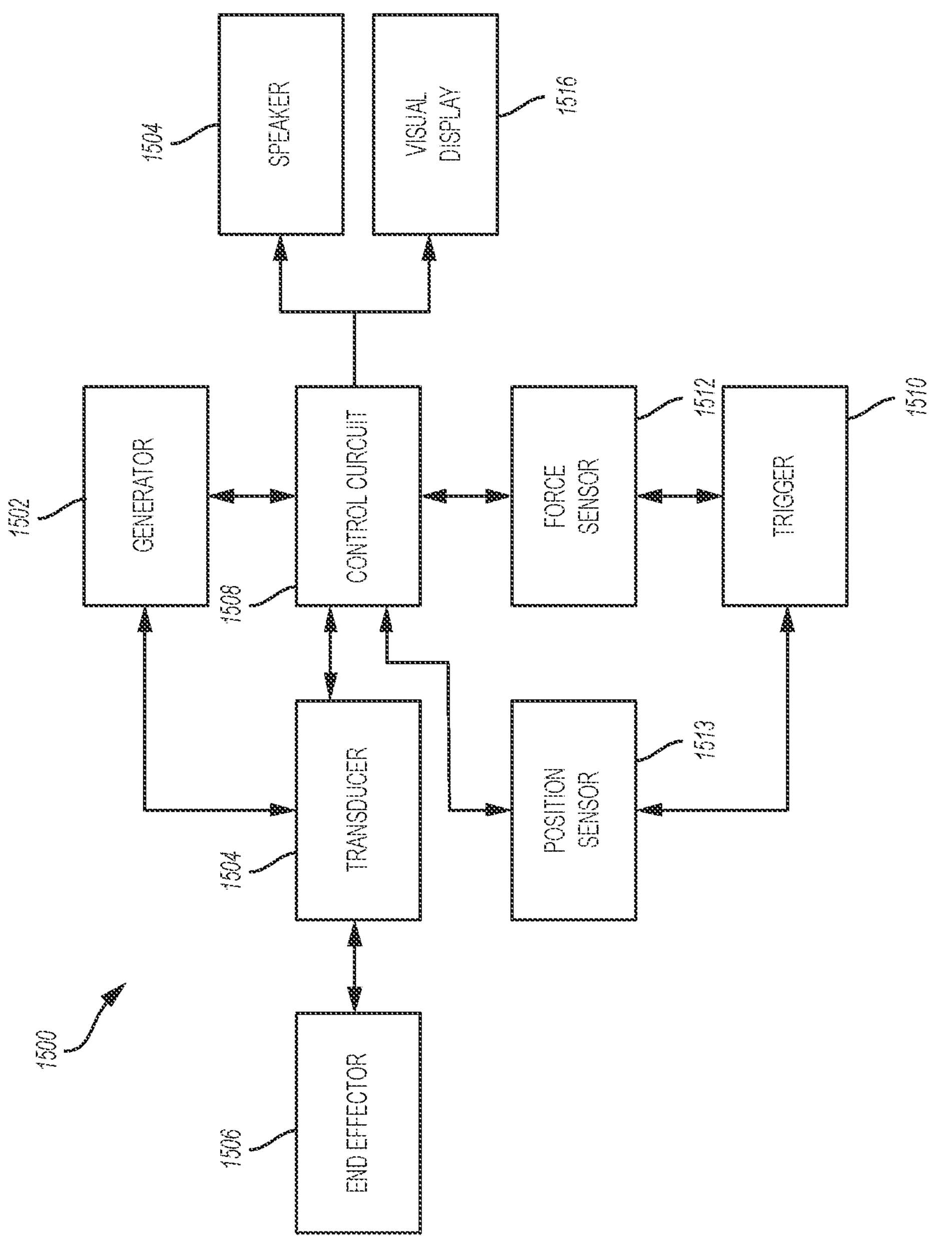

1500
1502
GENERATOR
1508
CONTROL CURCUIT
FORCE SENSOR
1512
TRIGGER
1510
1504
SPEAKER
VISUAL DISPLAY
1516
1504
TRANSDUCER
POSITION SENSOR
1513
1506
END EFFECTOR

FIG. 64

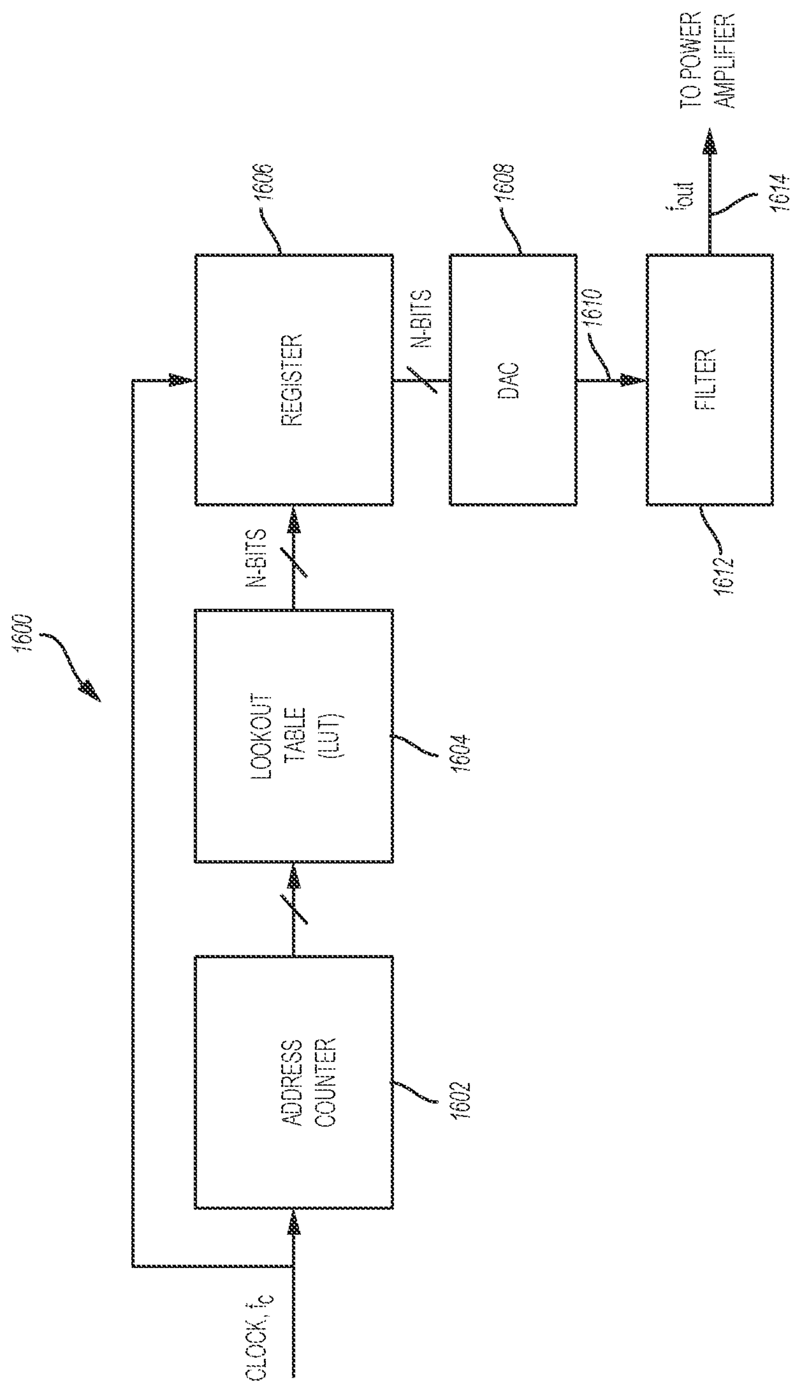

1600
CLOCK, f_c
ADDRESS COUNTER
1602
LOOKOUT TABLE (LUT)
1604
N-BITS
REGISTER
1606
N-BITS
DAC
1608
1610
FILTER
1612
f_{out}
1614
TO POWER AMPLIFIER

FIG. 65

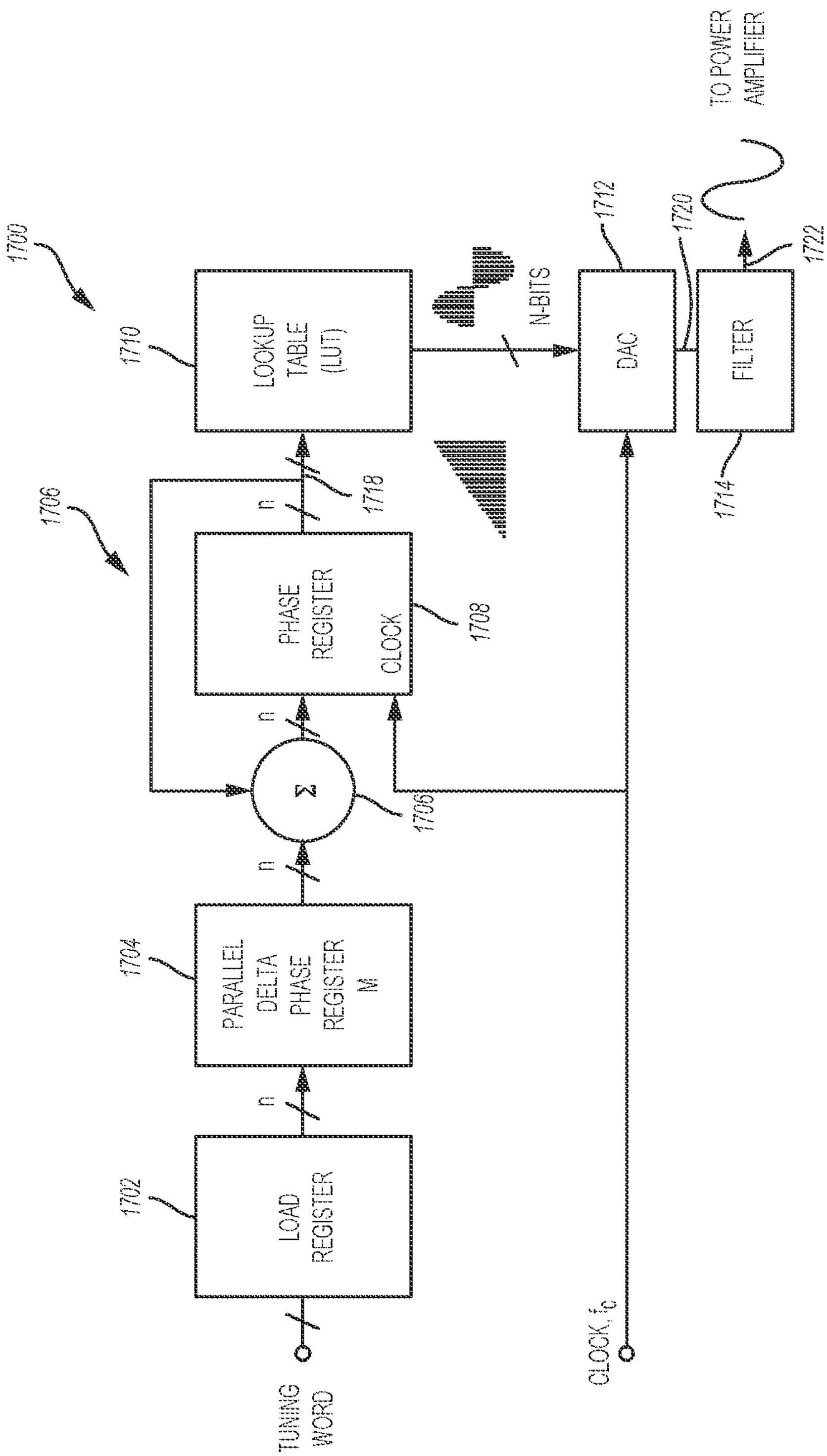

1700
1706
TUNING WORD
LOAD REGISTER
1702
n
PARALLEL DELTA PHASE REGISTER M
1704
n
Σ
1705
n
PHASE REGISTER
CLOCK
1708
n
1718
LOOKUP TABLE (LUT)
1710
N-BITS
DAC
1712
1720
FILTER
1714
1722
TO POWER AMPLIFIER
CLOCK, f_c

FIG. 66

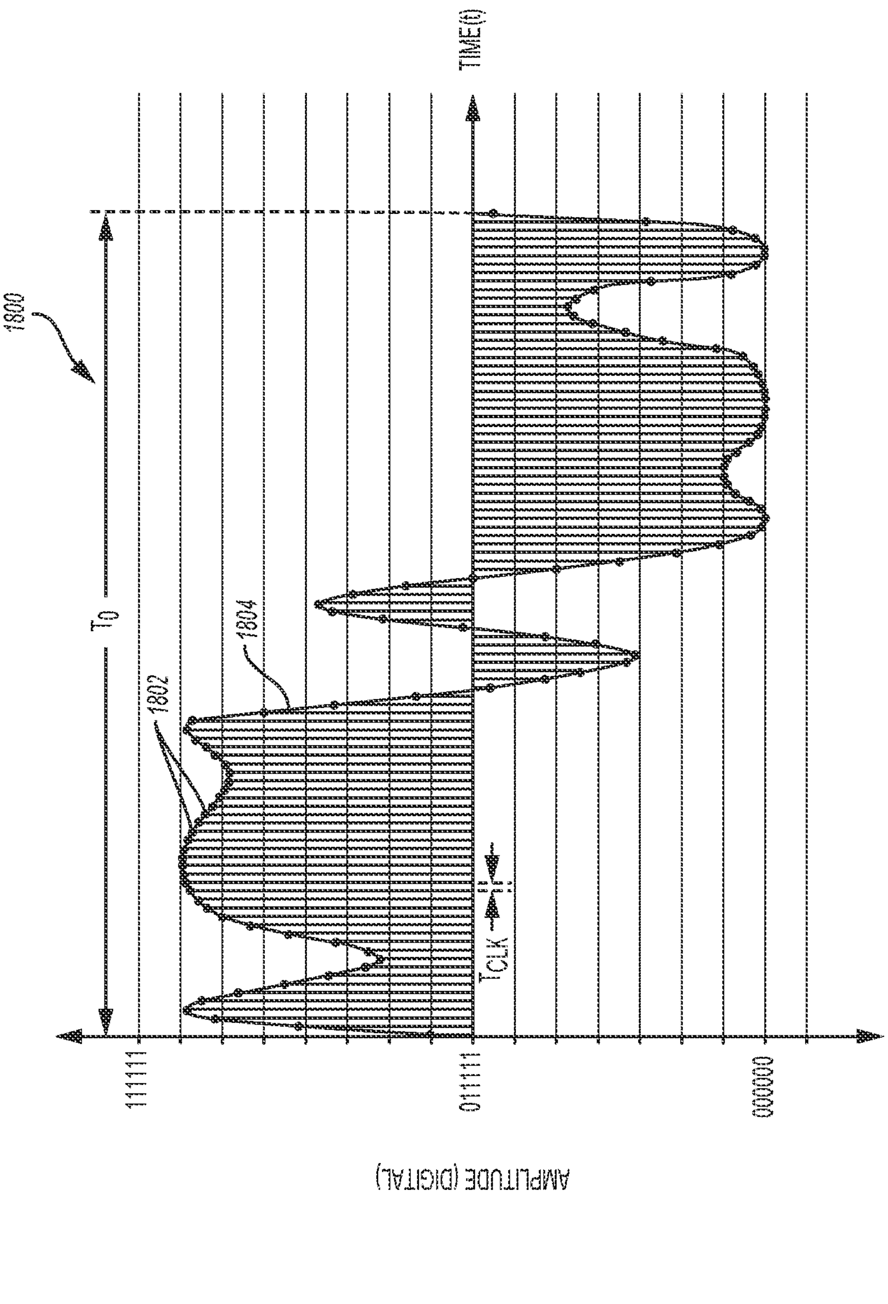

1800
1802
1804
T_0
T_{CLK}
TIME(t)
111111
011111
000000
AMPLITUDE (DIGITAL)

FIG. 67

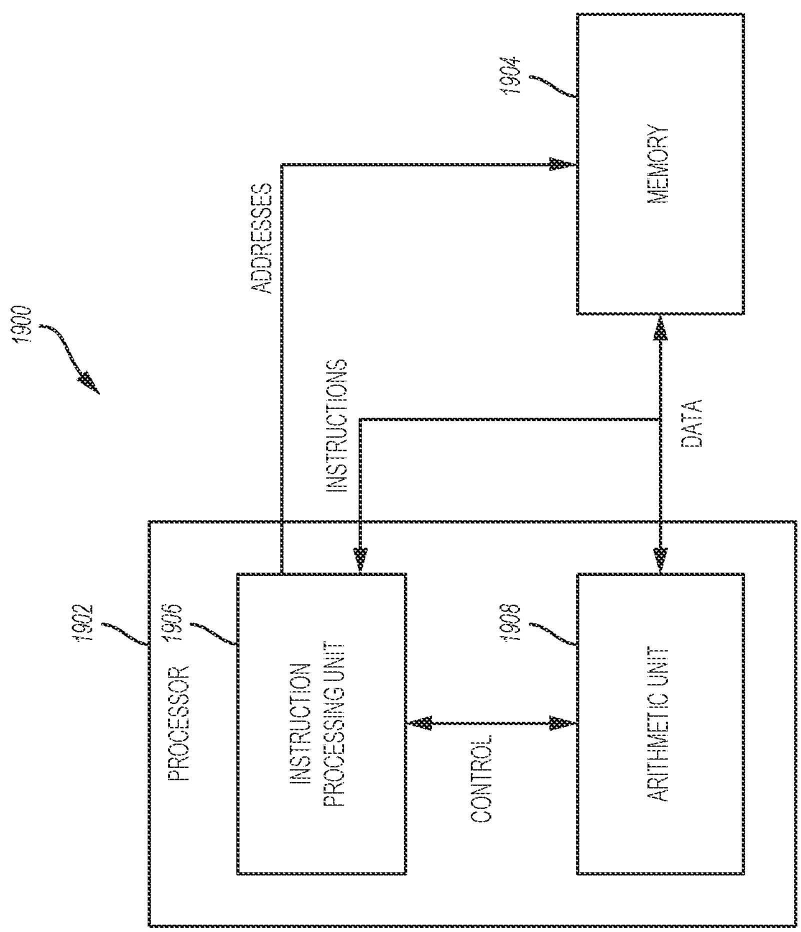

1900
1902
PROCESSOR
1906
INSTRUCTION
PROCESSING UNIT
1908
ARITHMETIC UNIT
CONTROL
ADDRESSES
INSTRUCTIONS
DATA
1904
MEMORY

FIG. 68A

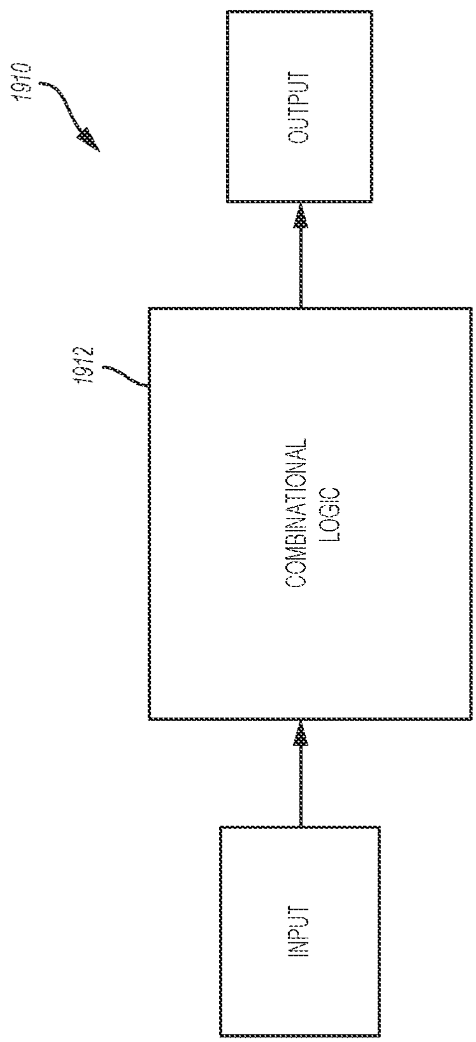


FIG. 68B

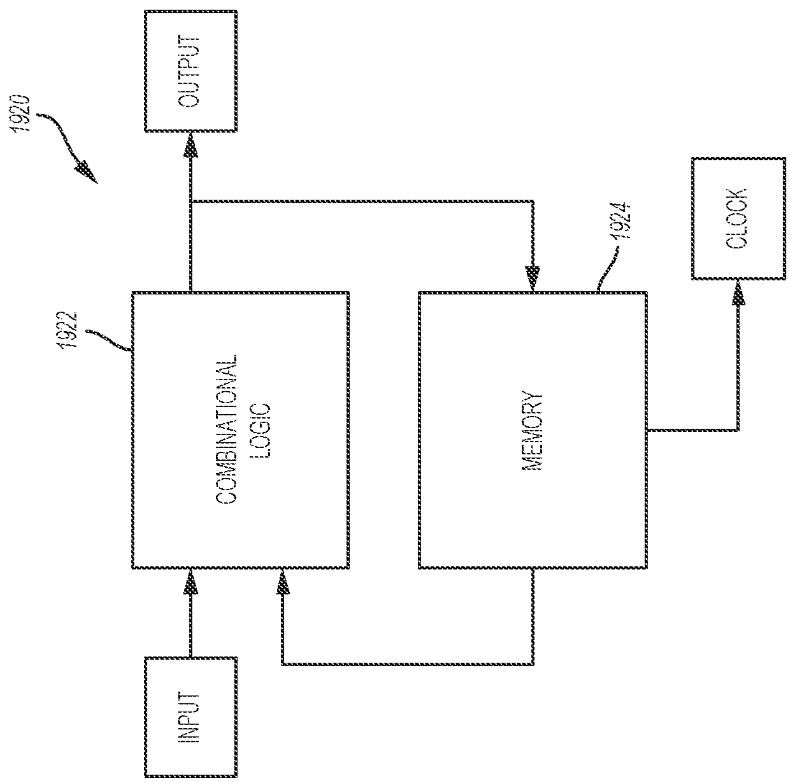

1920
OUTPUT
1922
COMBINATIONAL LOGIC
1924
MEMORY
CLOCK
INPUT

FIG. 68C

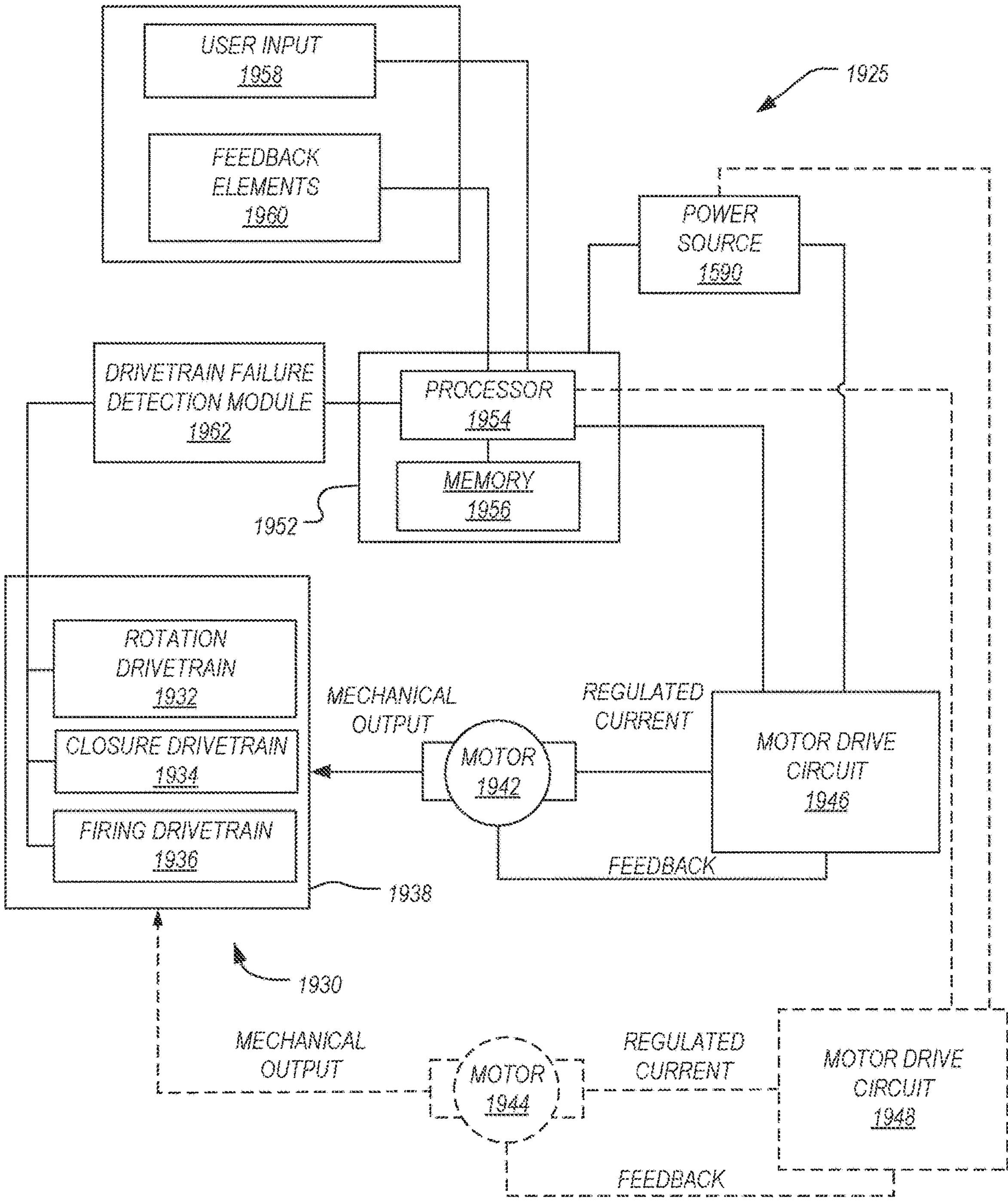

USER INPUT
1958
FEEDBACK ELEMENTS
1960
1925
POWER SOURCE
1590
DRIVETRAIN FAILURE DETECTION MODULE
1962
PROCESSOR
1954
MEMORY
1956
1952
ROTATION DRIVETRAIN
1932
CLOSURE DRIVETRAIN
1934
FIRING DRIVETRAIN
1936
MECHANICAL OUTPUT
MOTOR
1942
REGULATED CURRENT
MOTOR DRIVE CIRCUIT
1946
FEEDBACK
1938
1930
MECHANICAL OUTPUT
MOTOR
1944
REGULATED CURRENT
MOTOR DRIVE CIRCUIT
1948
FEEDBACK

FIG. 69

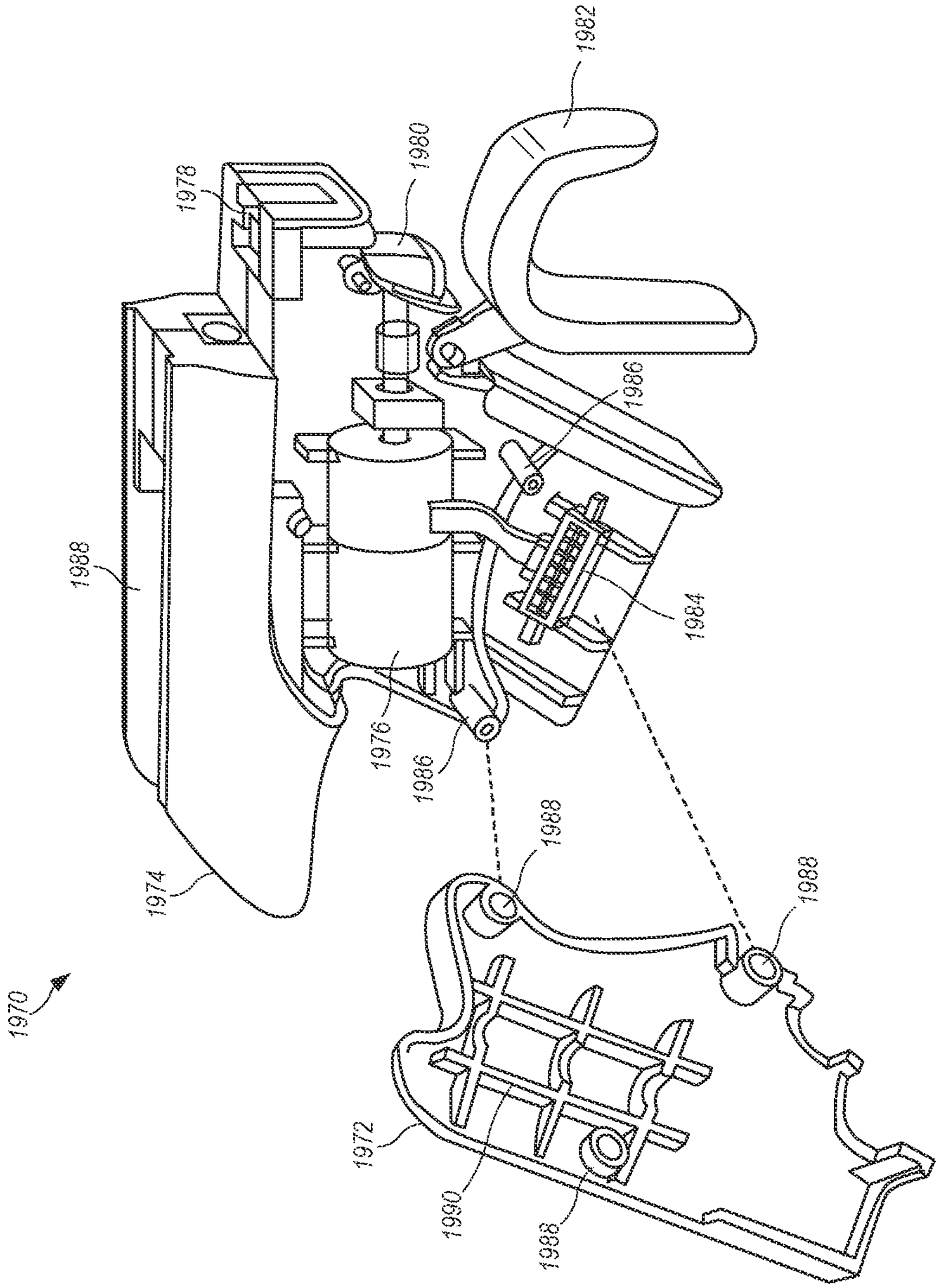

1970
1972
1974
1976
1978
1980
1982
1984
1986
1986
1988
1988
1988
1988
1990

FIG. 70

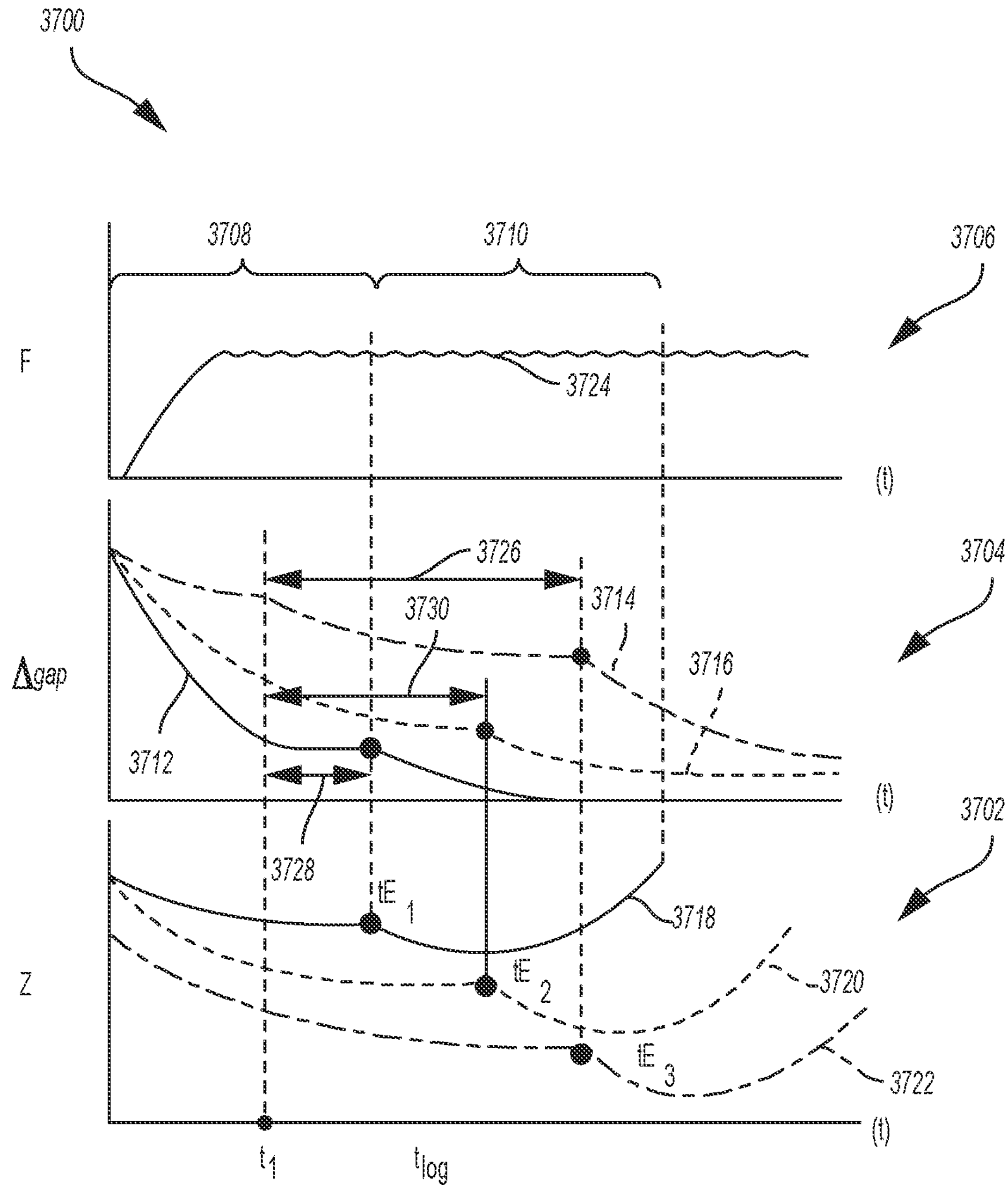

3700
3708
3710
3706
F
3724
(t)
3726
3704
3714
3730
3716
Δgap
3712
(t)
3702
3728
tE 1
3718
Z
tE 2
3720
tE 3
3722
(t)
t1
tlog

FIG. 71

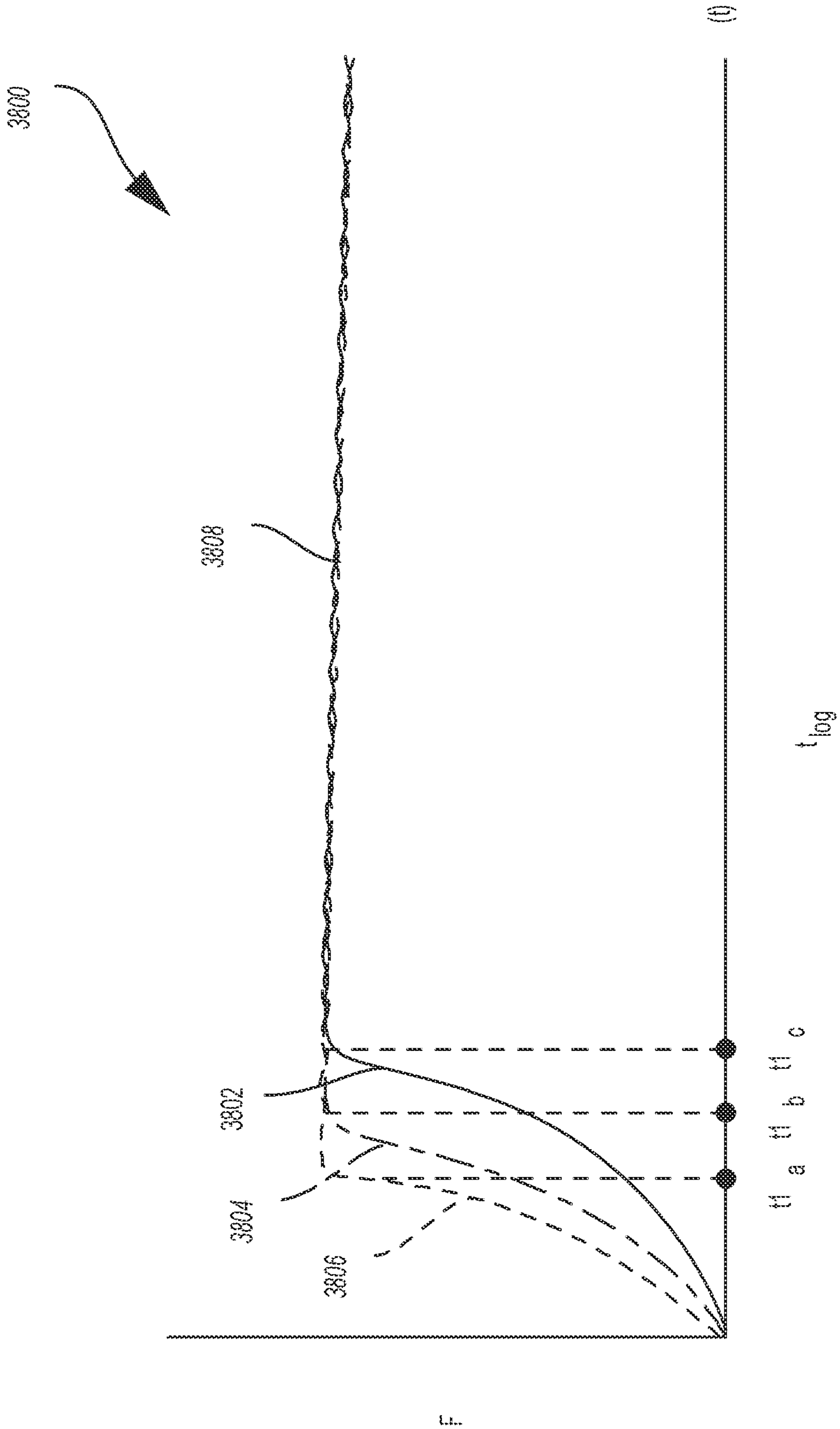


FIG. 72

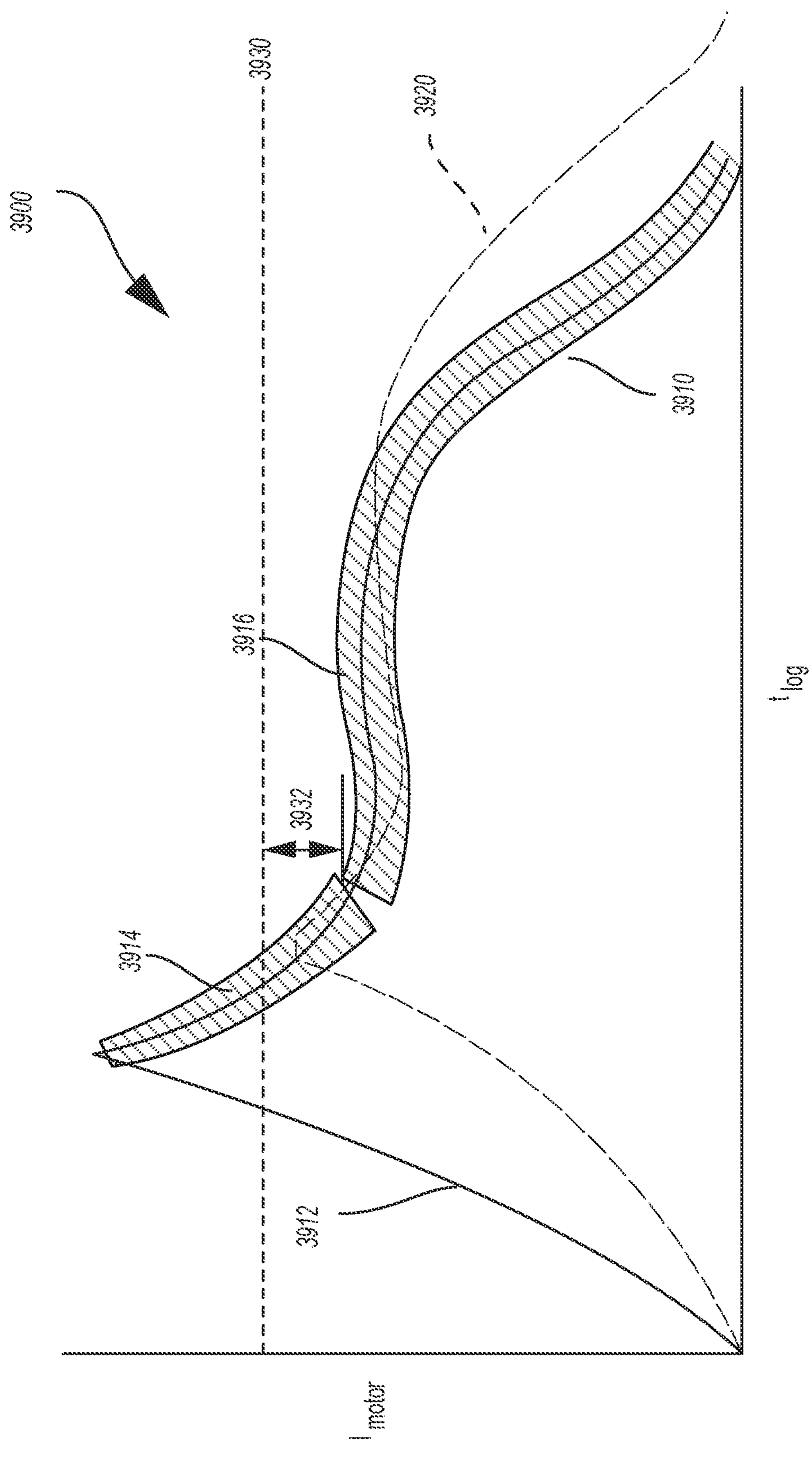


FIG. 73

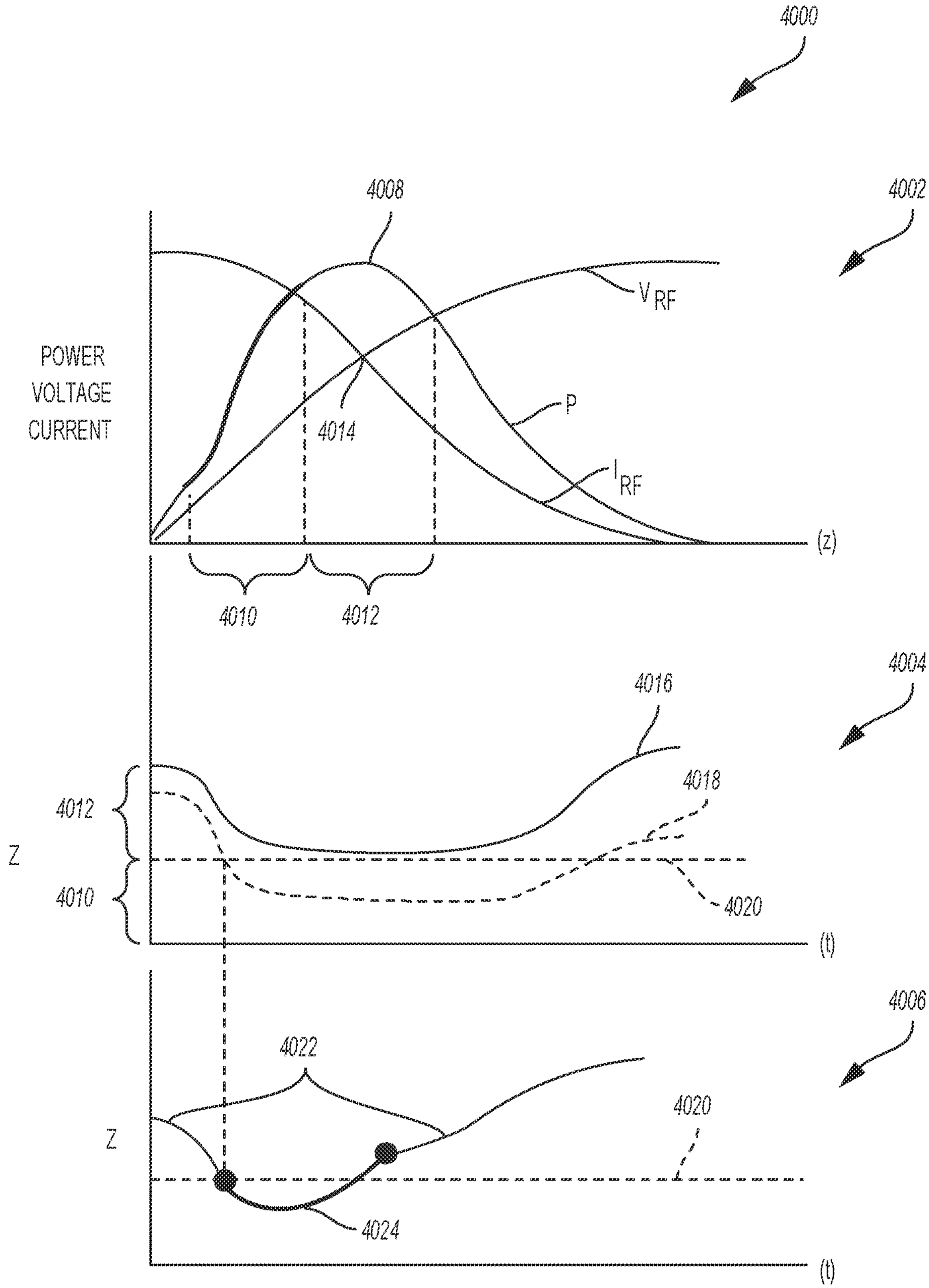

4000
4002
4008
V RF
POWER
VOLTAGE
CURRENT
P
4014
I RF
(z)
4010
4012
4004
4016
4018
4012
Z
4010
4020
(t)
4006
4022
4020
Z
4024
(t)

FIG. 74

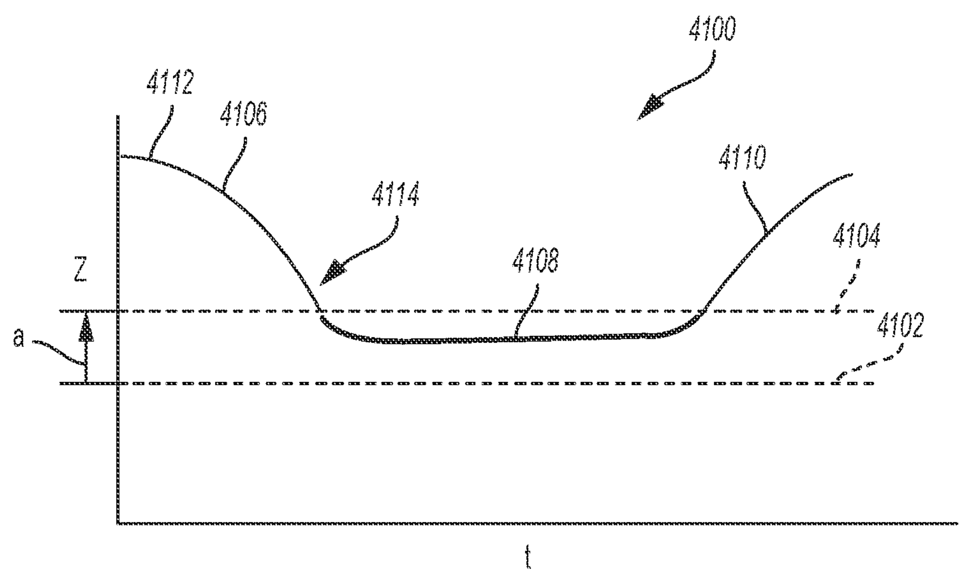


FIG. 75

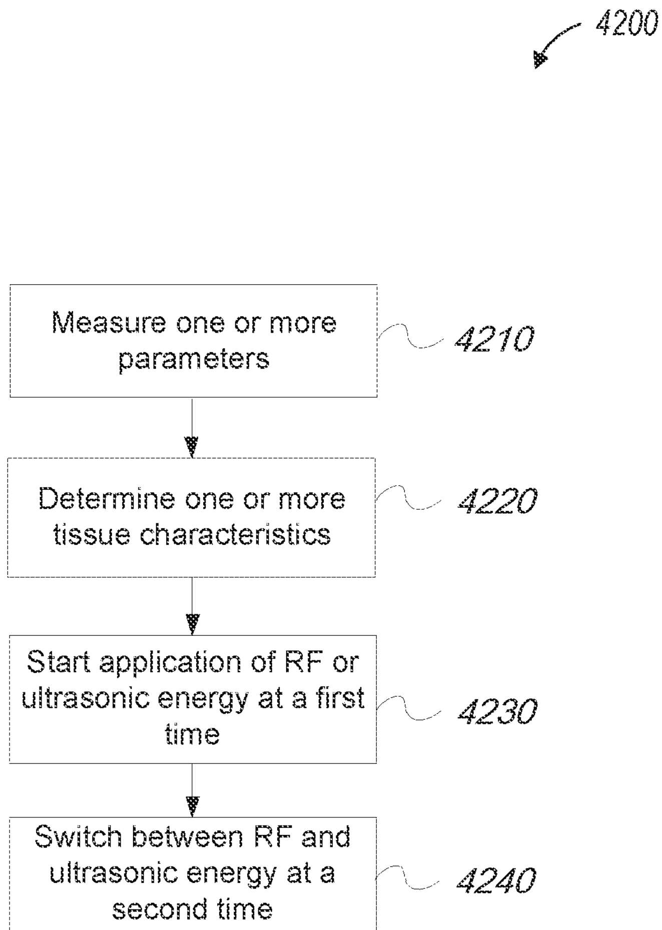

4200
Measure one or more parameters
4210
Determine one or more tissue characteristics
4220
Start application of RF or ultrasonic energy at a first time
4230
Switch between RF and ultrasonic energy at a second time
4240

FIG. 76

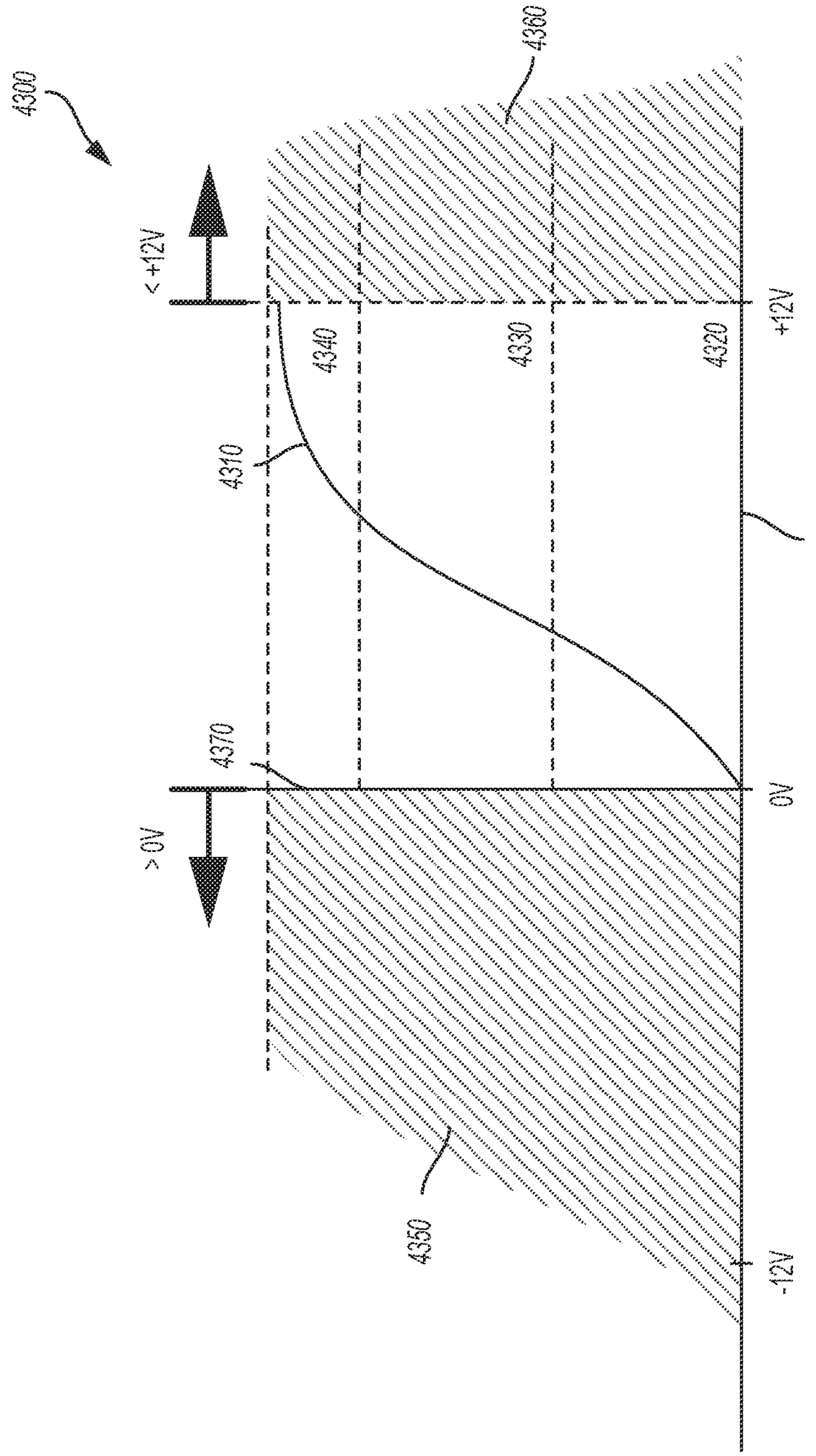


FIG. 77

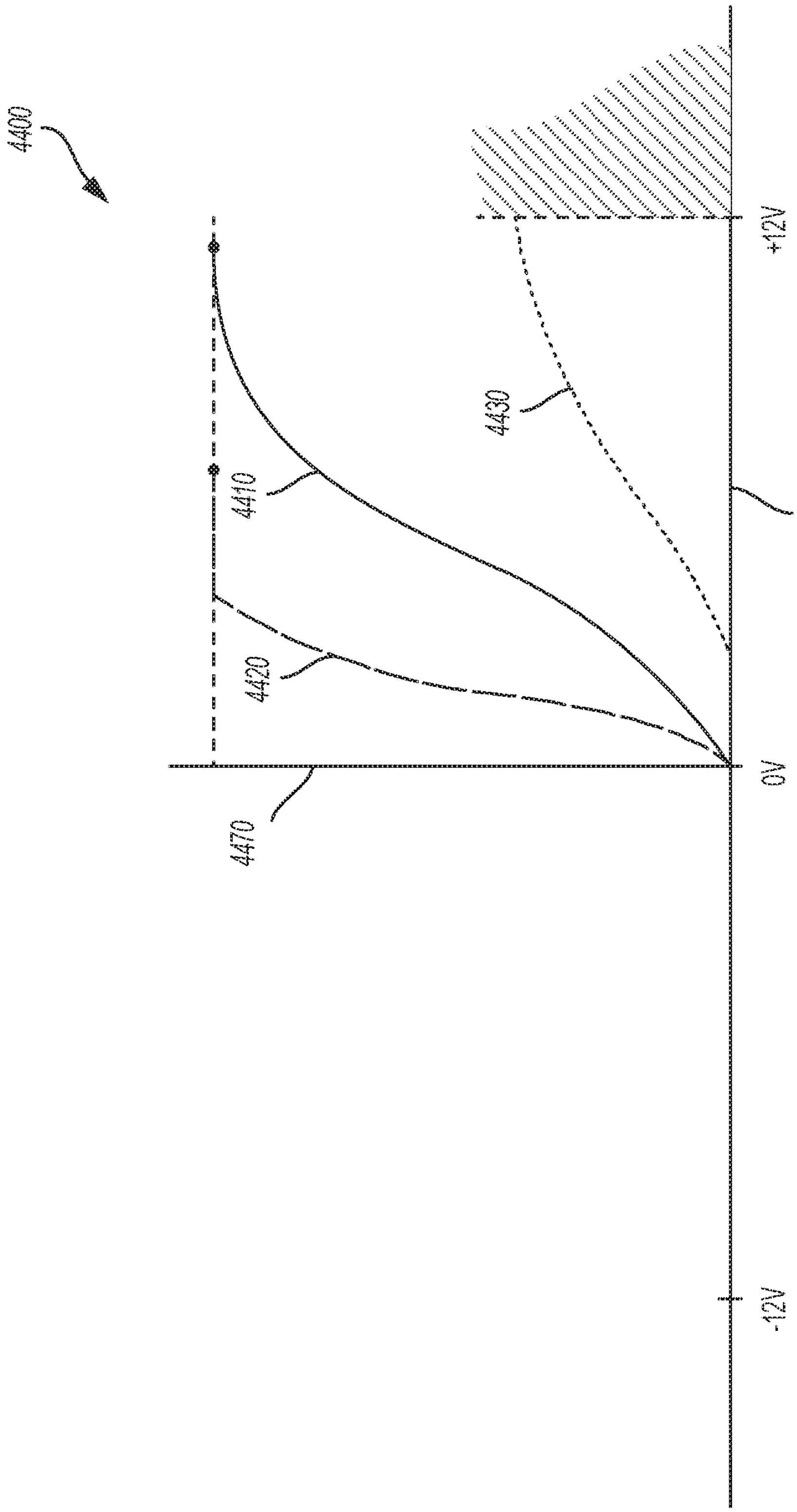


FIG. 78

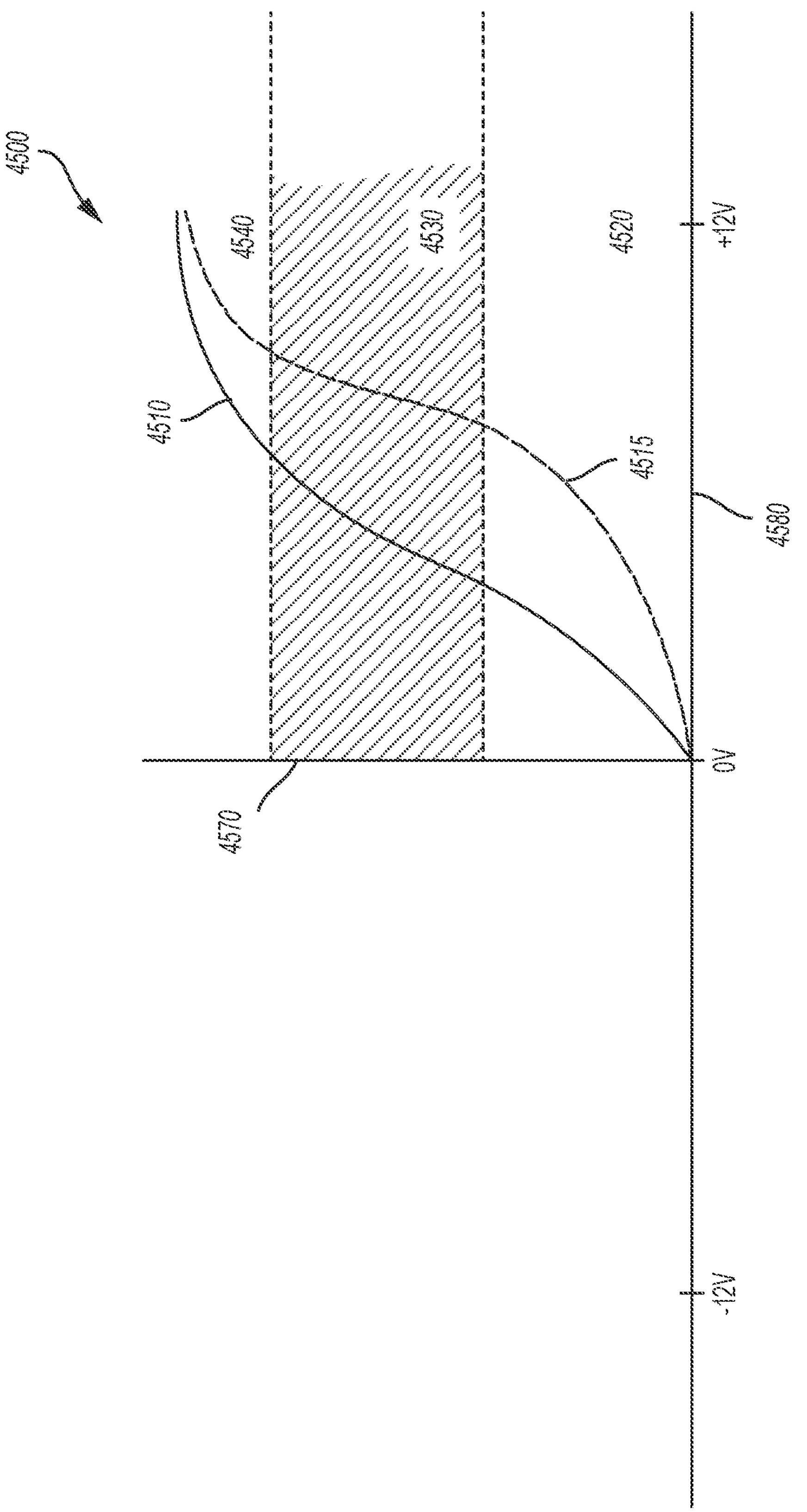


FIG. 79

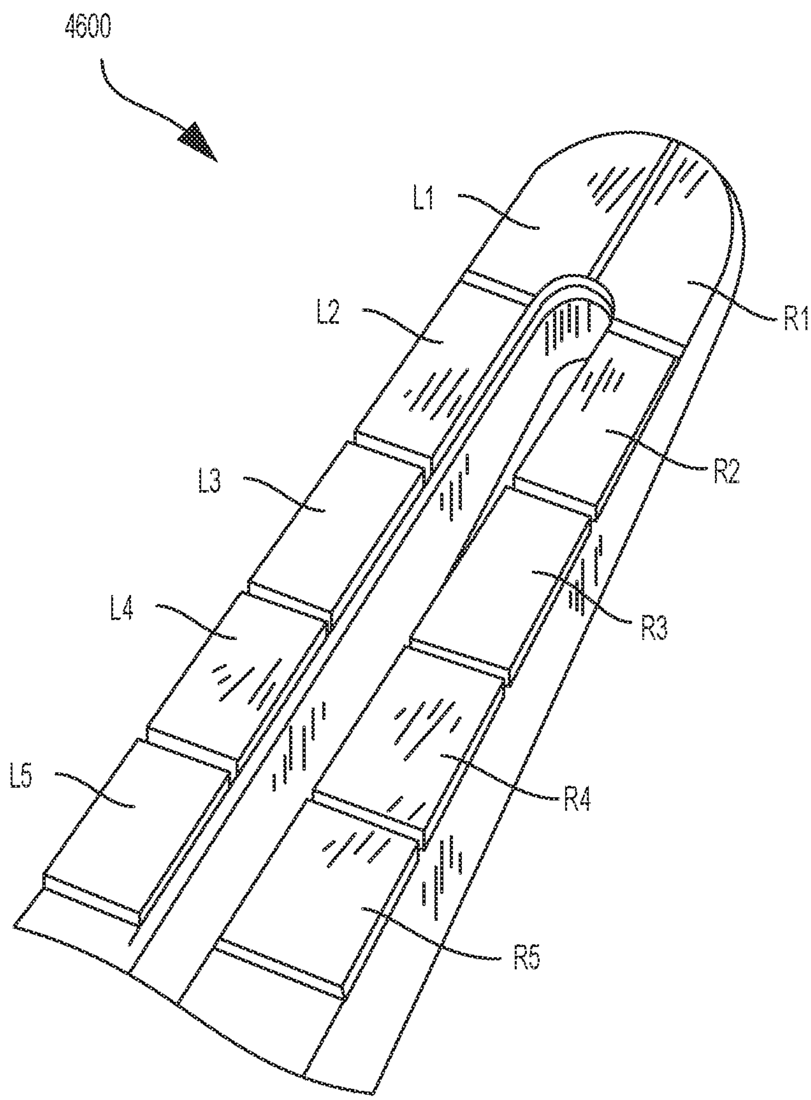


FIG. 80

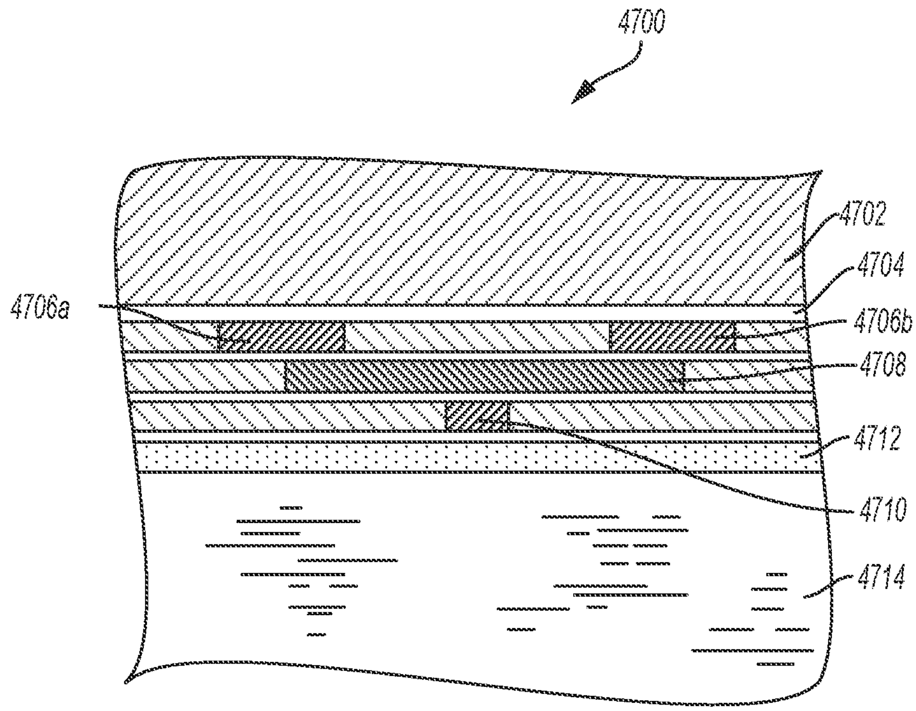


FIG. 81

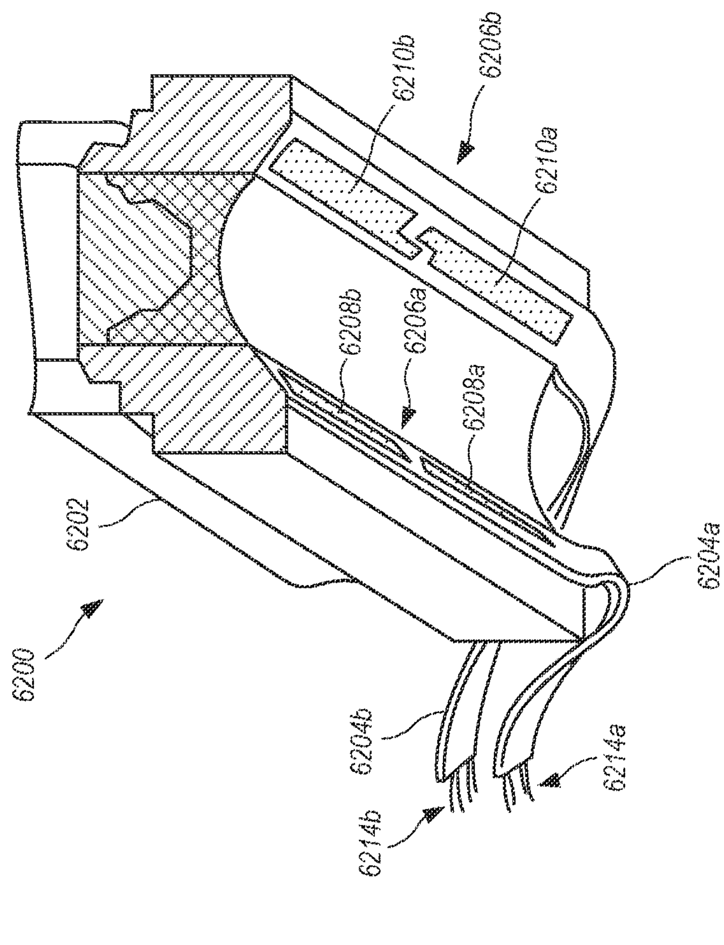

6200
6202
6210b
6206b
6210a
6208b
6206a
6208a
6204a
6204b
6214a
6214b

FIG. 82

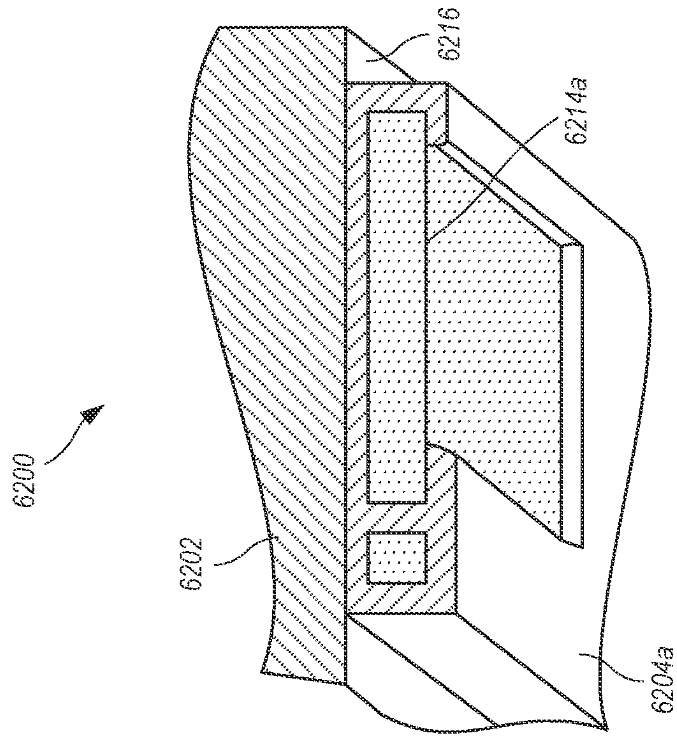

6216
6214a
6200
6202
6204a

FIG. 83

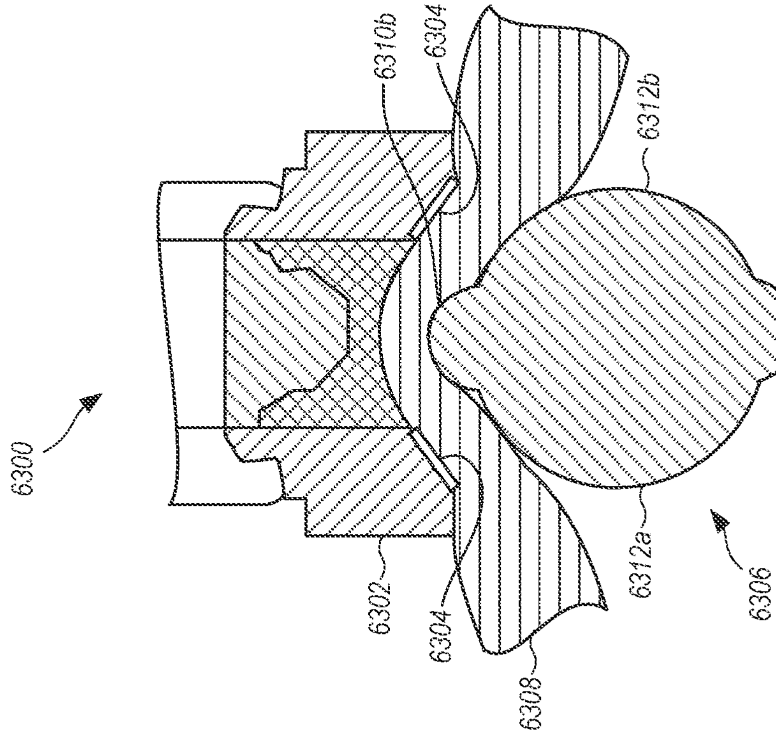


FIG. 84B

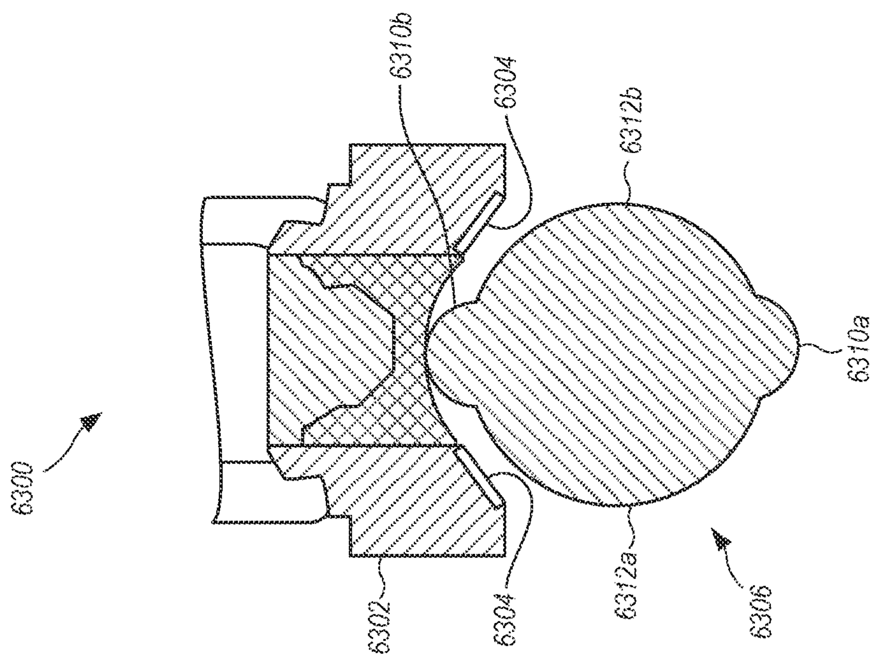


FIG. 84A

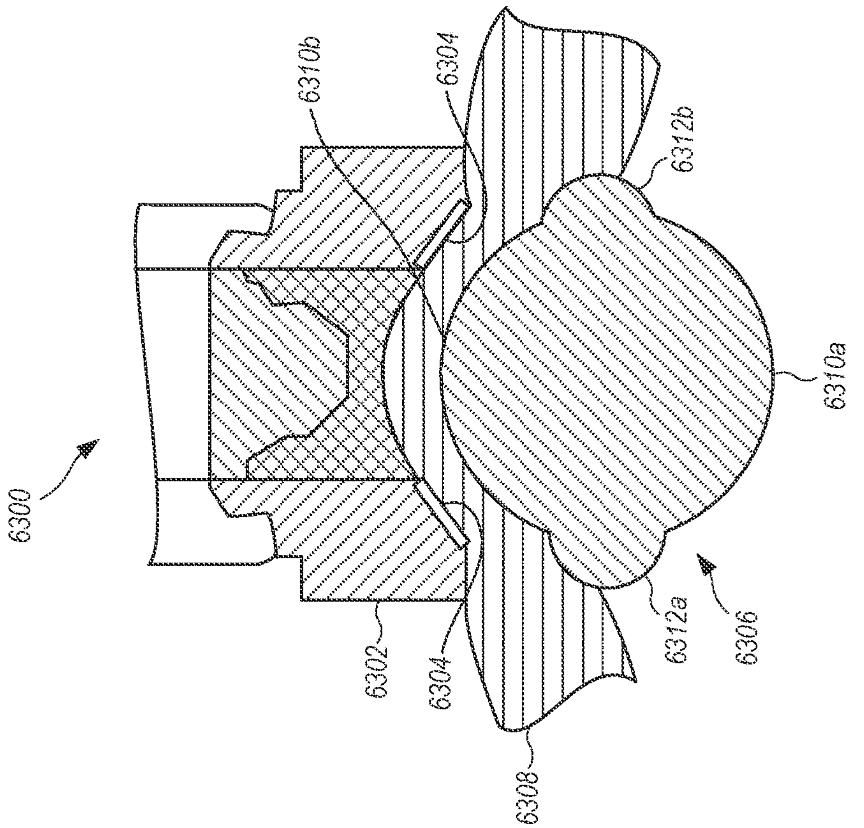


FIG. 85B

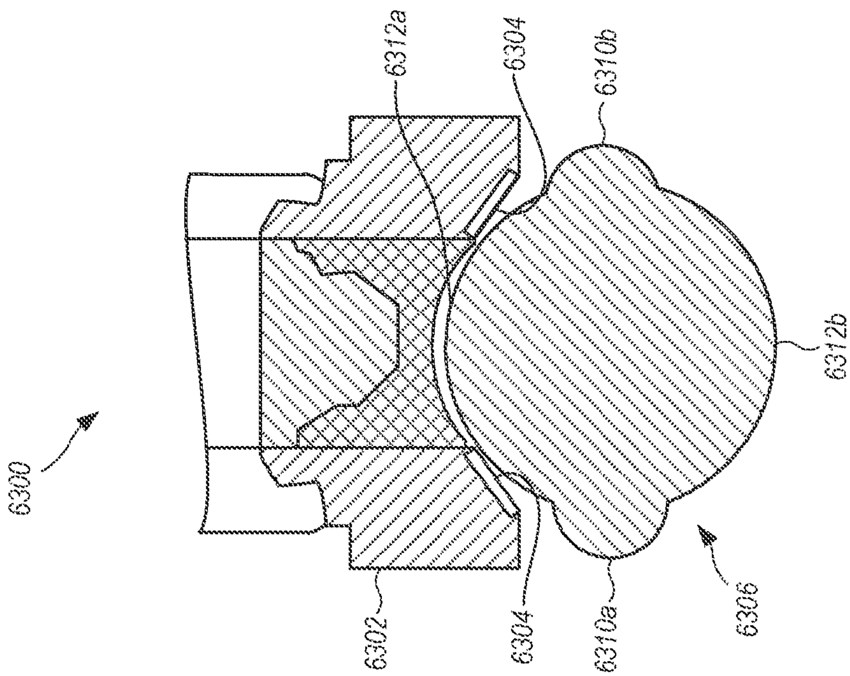


FIG. 85A

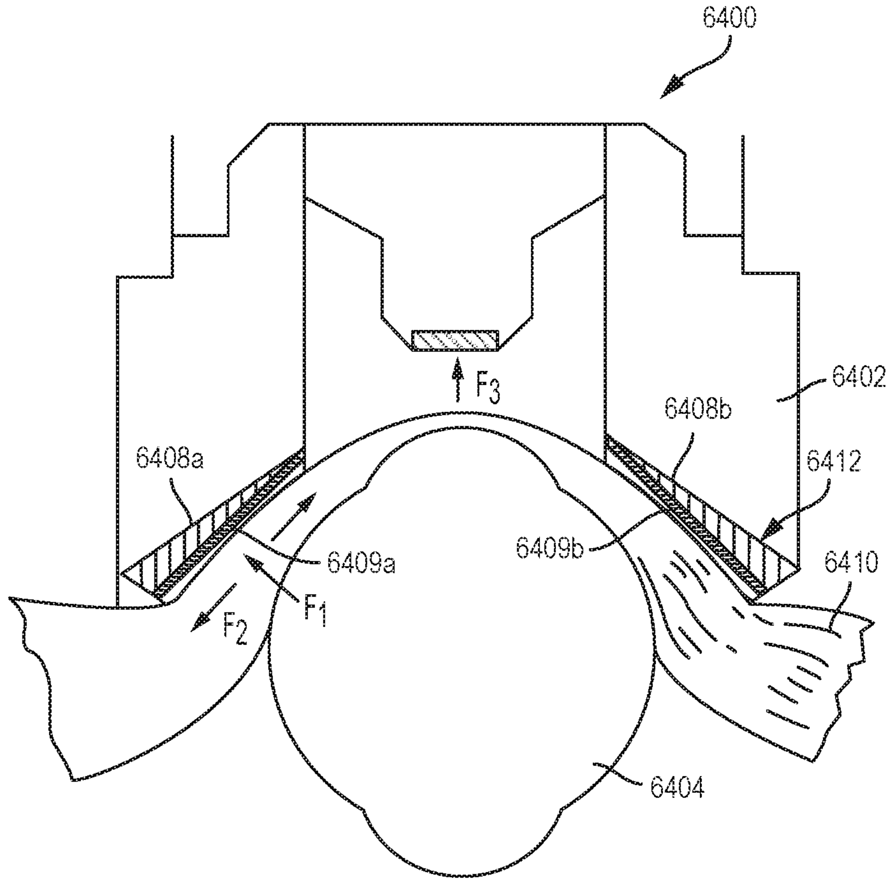

6400
6402
6408b
6412
6408a
6409a
6409b
6410
F_3
F_1
F_2
6404

FIG. 86

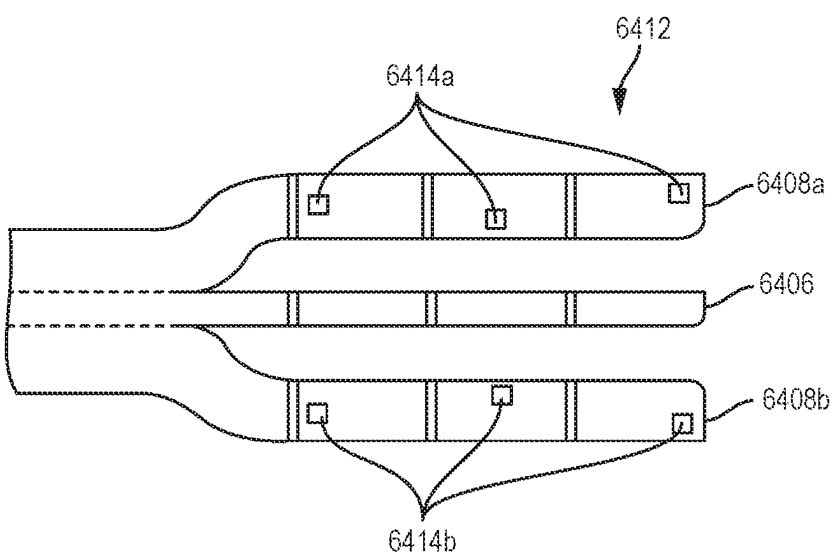

6412
6414a
6408a
6406
6408b
6414b

FIG. 87

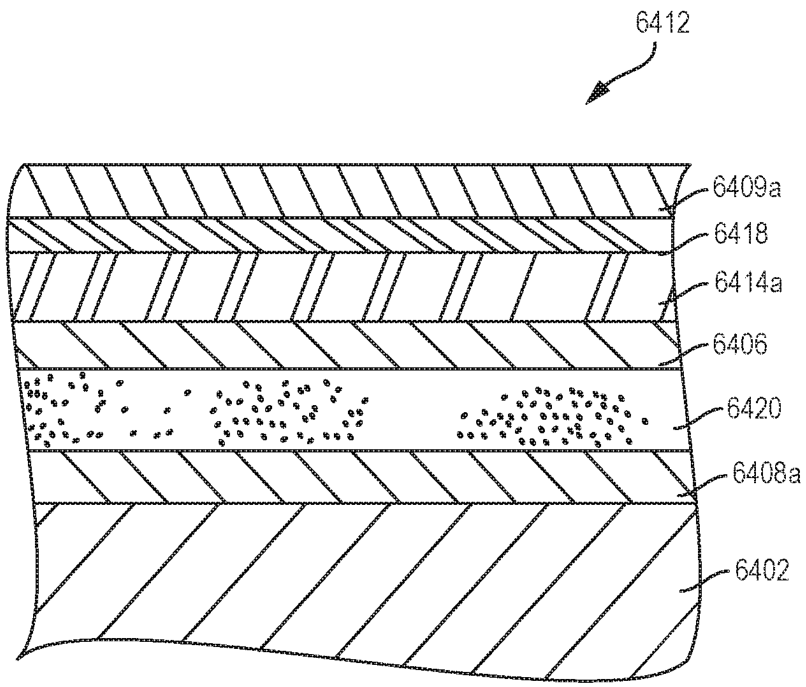

6412
6409a
6418
6414a
6406
6420
6408a
6402

FIG. 88

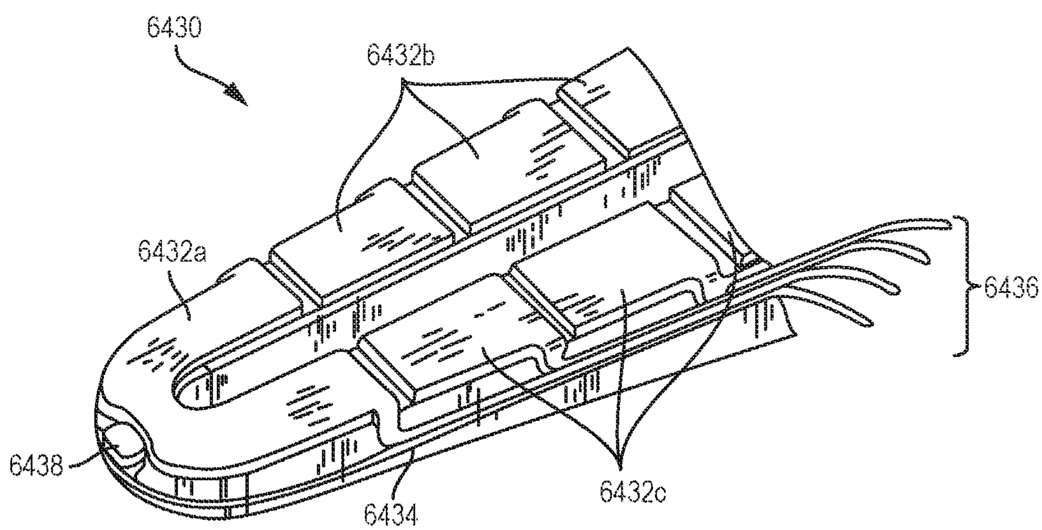

6430
6432b
6432a
6436
6438
6434
6432c

FIG. 89

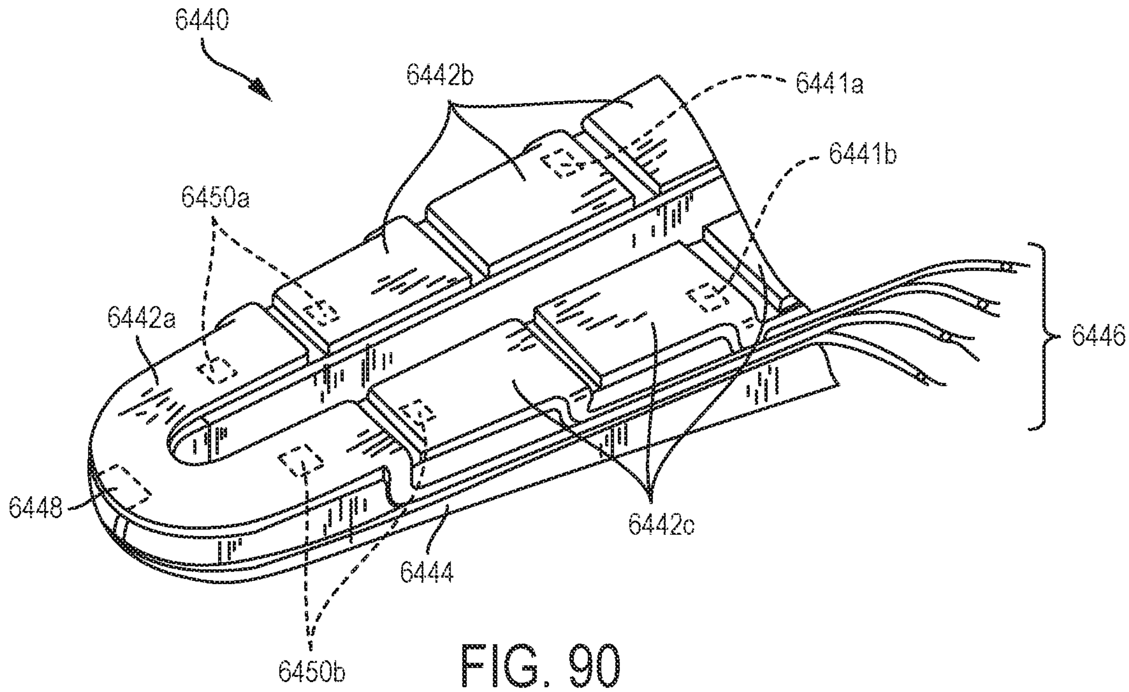

6440
6442b
6441a
6441b
6450a
6442a
6446
6448
6442c
6444
6450b

FIG. 90

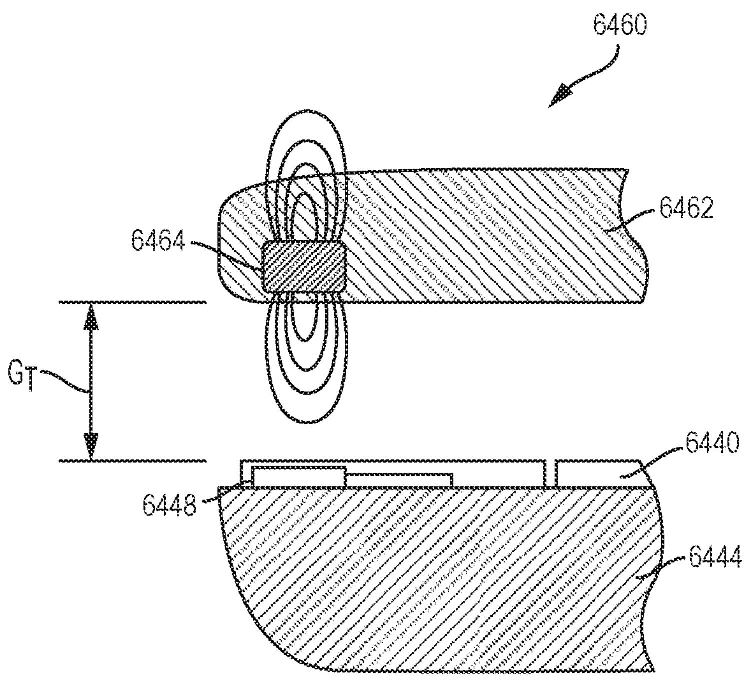
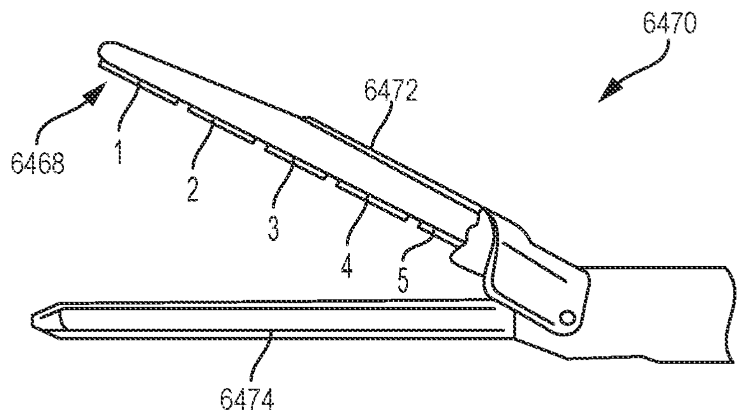
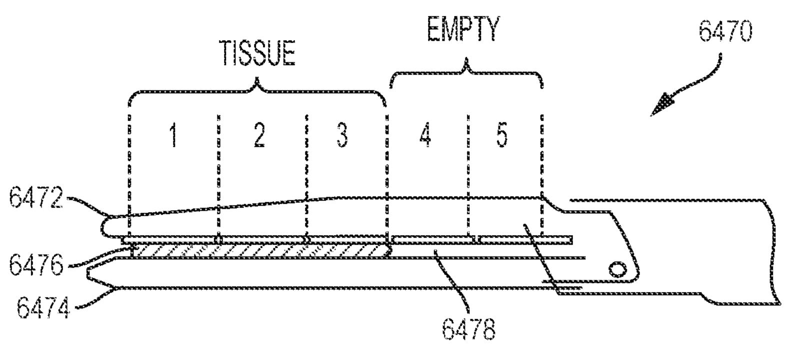

6460
6462
6464
G_T
6440
6448
6444

FIG. 91



6470
6472
6468
1
2
3
4
5
6474

FIG. 92



TISSUE
EMPTY
6470
1
2
3
4
5
6472
6476
6474
6478

FIG. 93

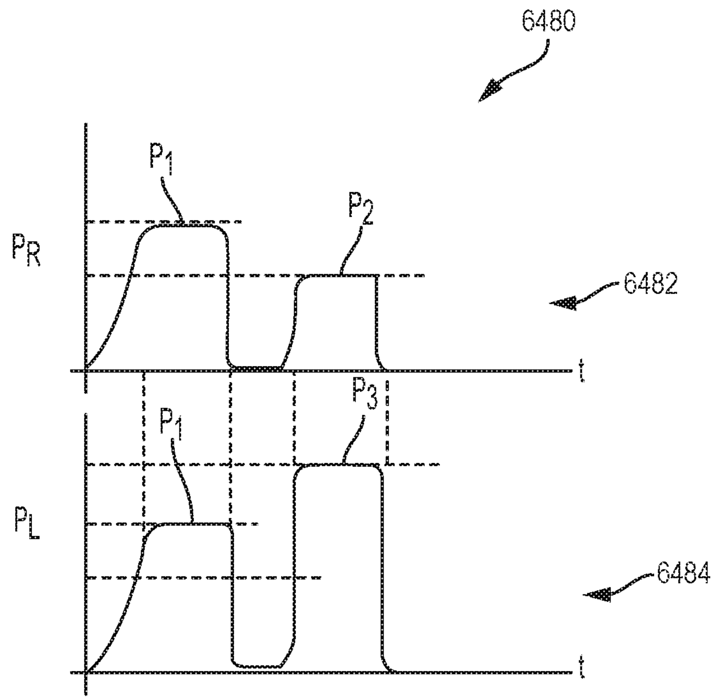

6480
P1
P2
PR
6482
t
P1
P3
PL
6484
t

FIG. 94

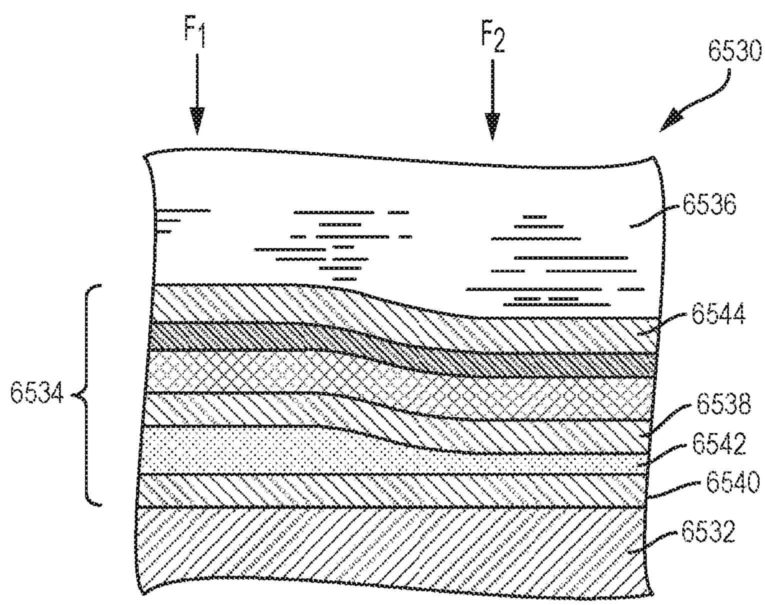

F1
F2
6530
6536
6544
6534
6538
6542
6540
6532

FIG. 95

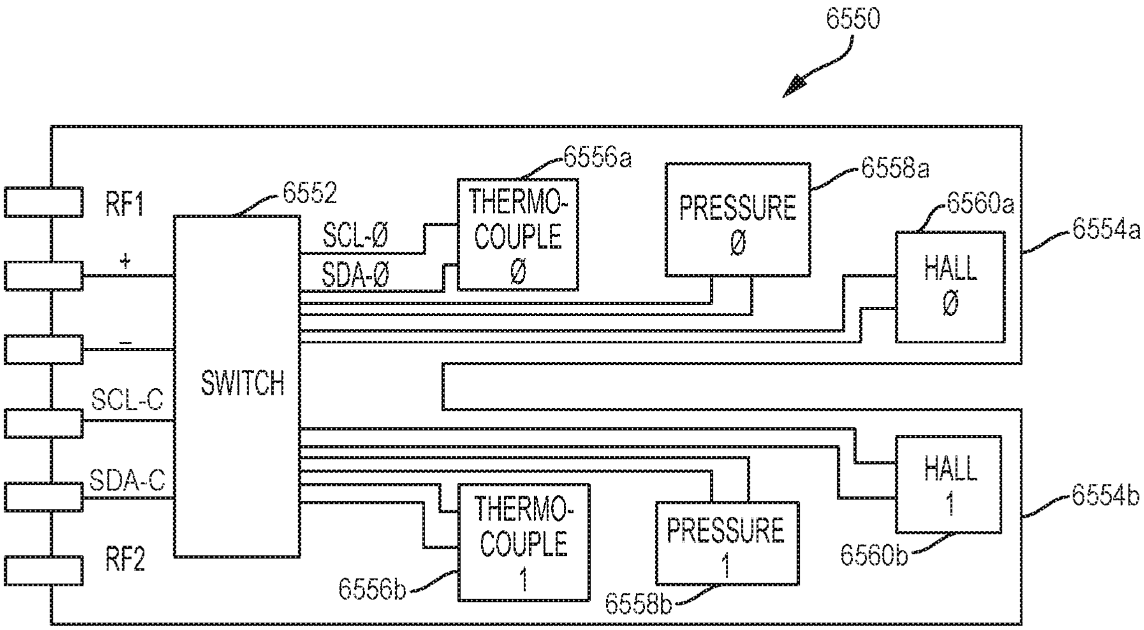

6550
6556a
6558a
6552
6560a
RF1
SCL-Ø
THERMO-
COUPLE
Ø
PRESSURE
Ø
6554a
HALL
Ø
+
SDA-Ø
–
SWITCH
SCL-C
HALL
1
6554b
SDA-C
THERMO-
COUPLE
1
PRESSURE
1
RF2
6560b
6556b
6558b

FIG. 96

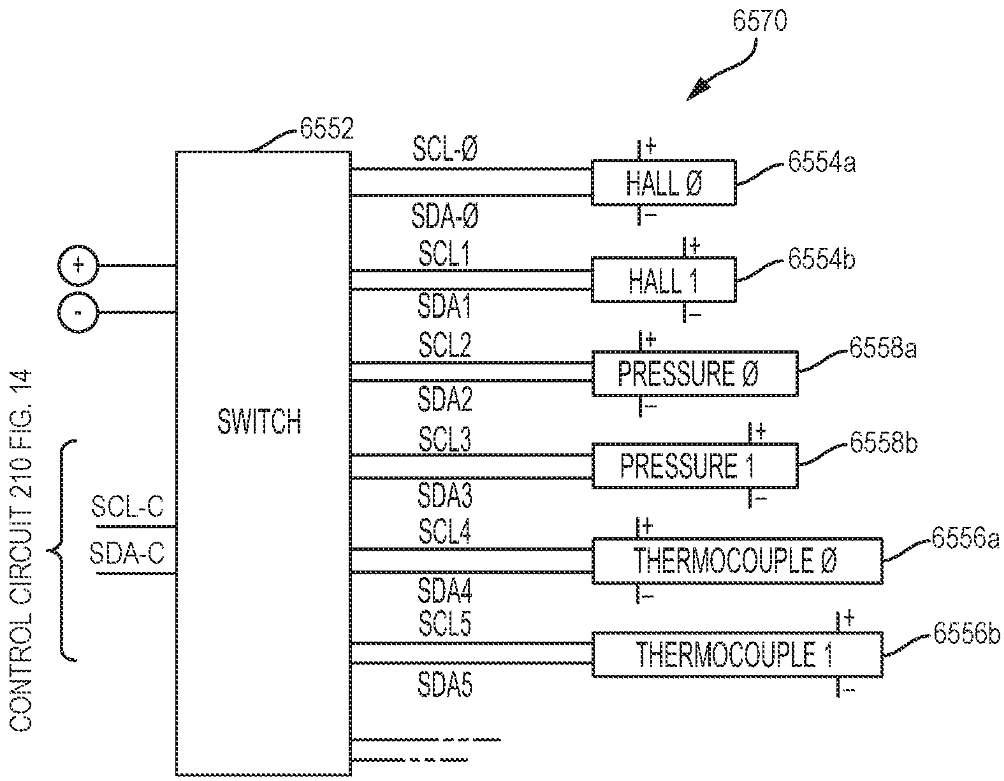

6570
6552
SCL-Ø
HALL Ø
6554a
SDA-Ø
SCL1
HALL 1
6554b
SDA1
SCL2
PRESSURE Ø
6558a
SDA2
SCL3
PRESSURE 1
6558b
SWITCH
SDA3
CONTROL CIRCUIT 210 FIG. 14
SCL-C
SDA-C
SCL4
THERMOCOUPLE Ø
6556a
SDA4
SCL5
THERMOCOUPLE 1
6556b
SDA5

FIG. 97

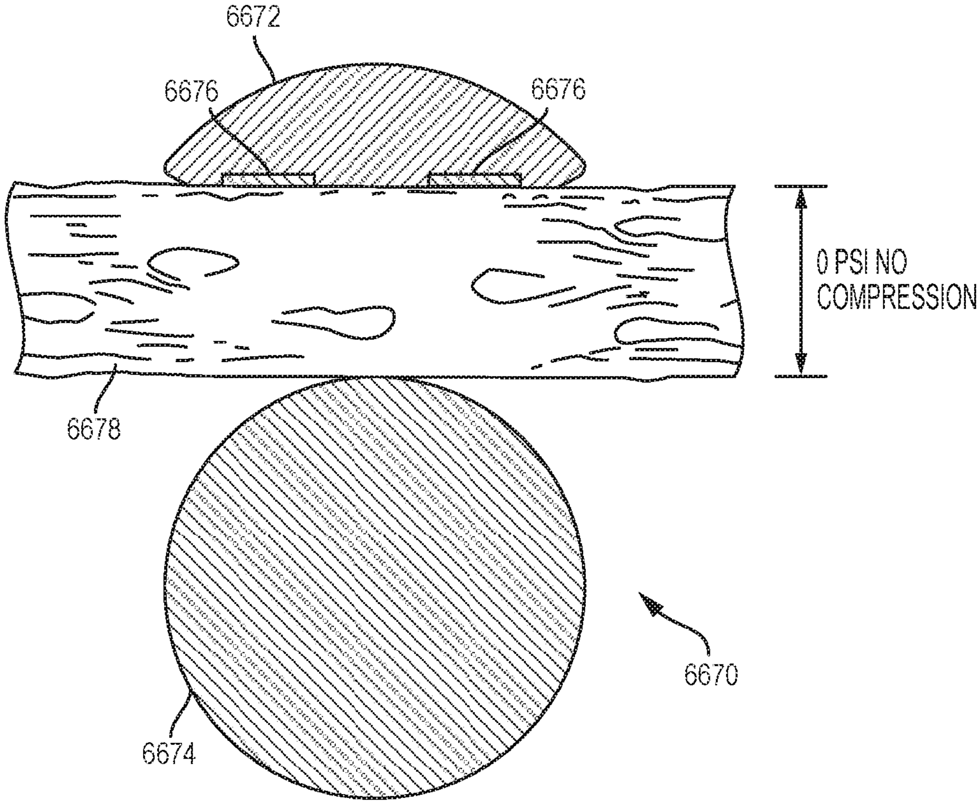


FIG. 98A

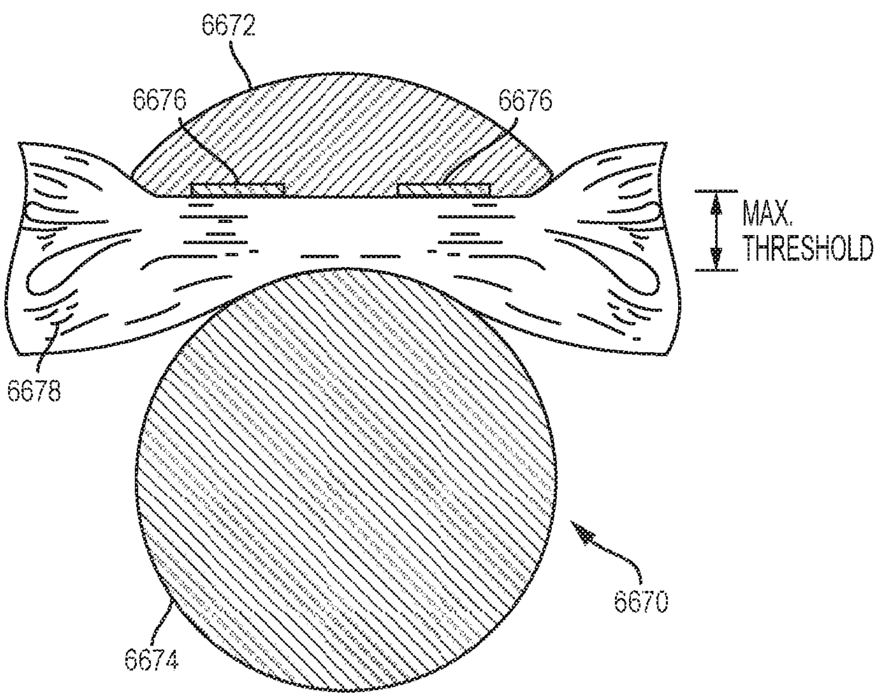


FIG. 98B

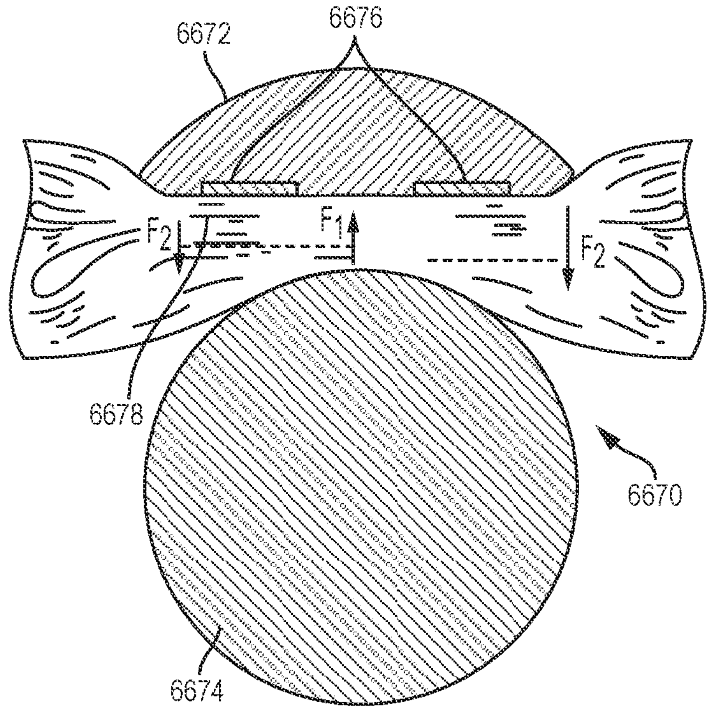


FIG. 99A

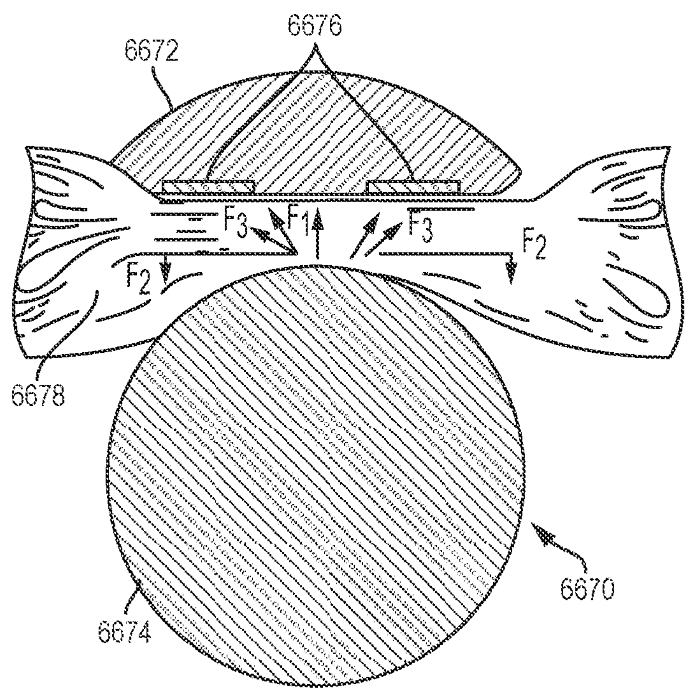


FIG. 99B

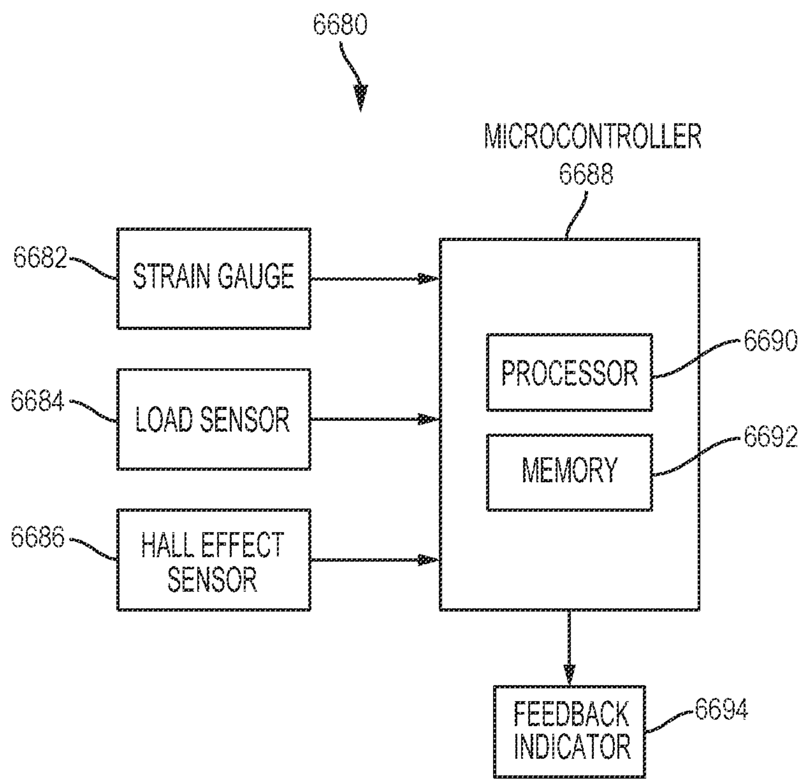

6680
MICROCONTROLLER
6688
6682
STRAIN GAUGE
6684
LOAD SENSOR
6686
HALL EFFECT SENSOR
PROCESSOR
6690
MEMORY
6692
FEEDBACK INDICATOR
6694

FIG. 100

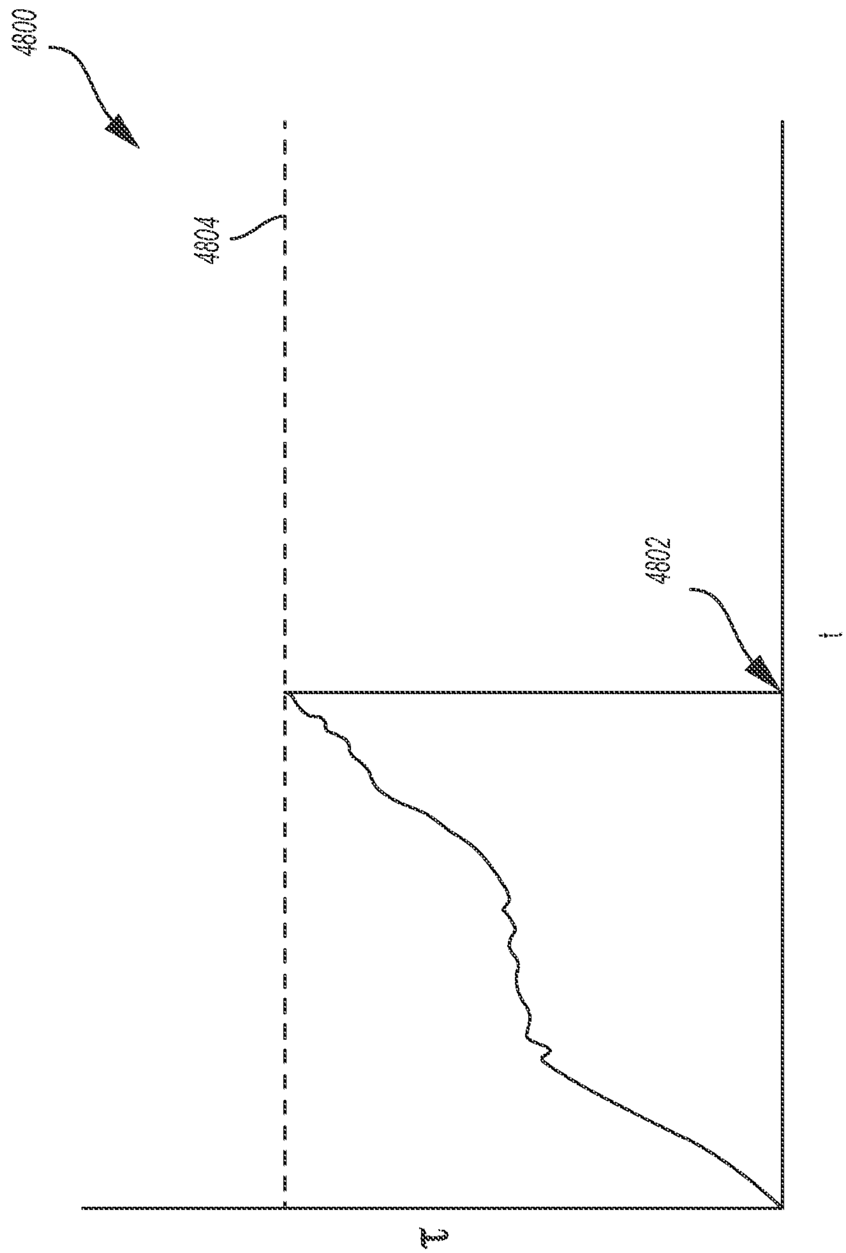


FIG. 101

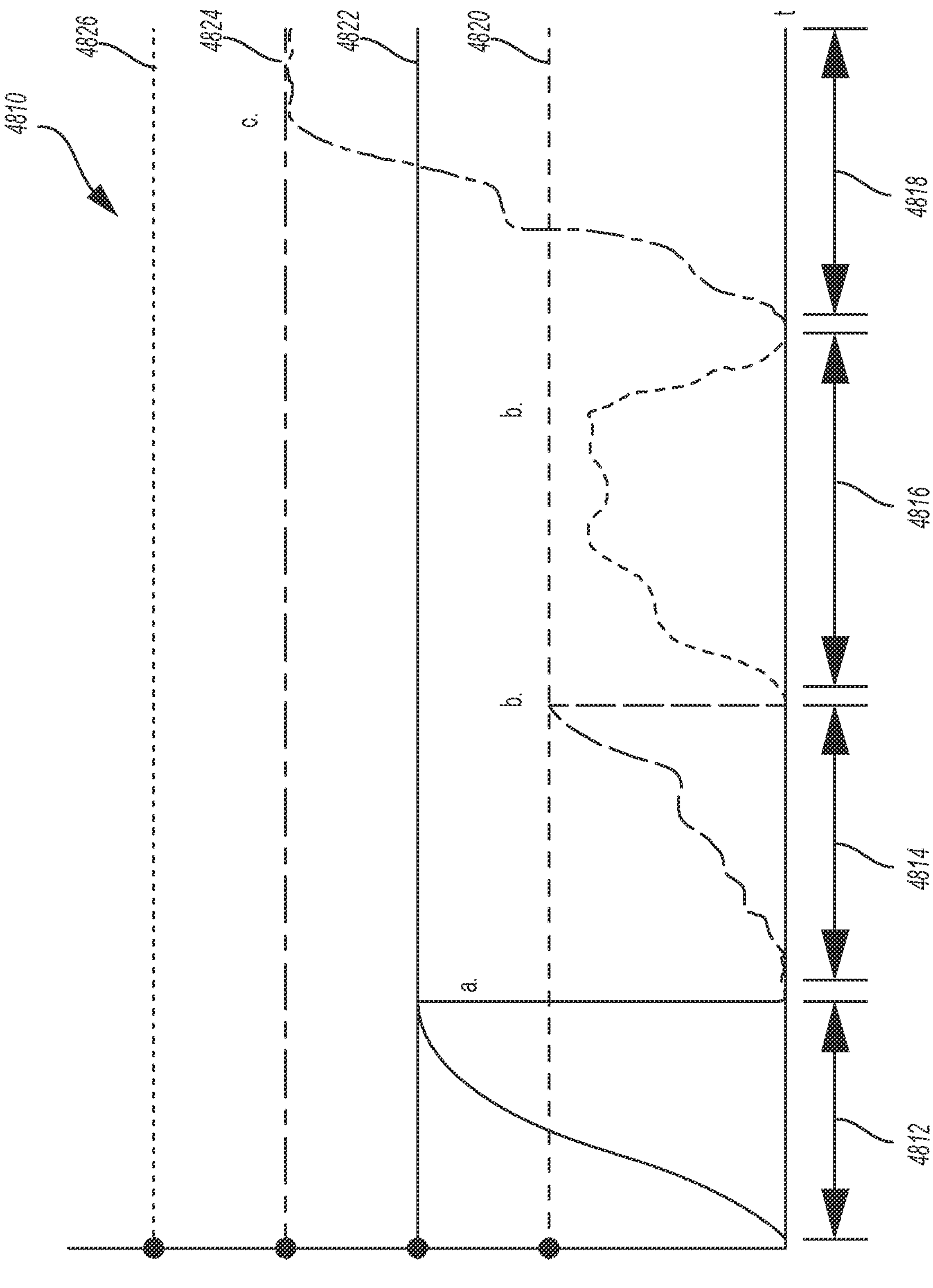


FIG. 102

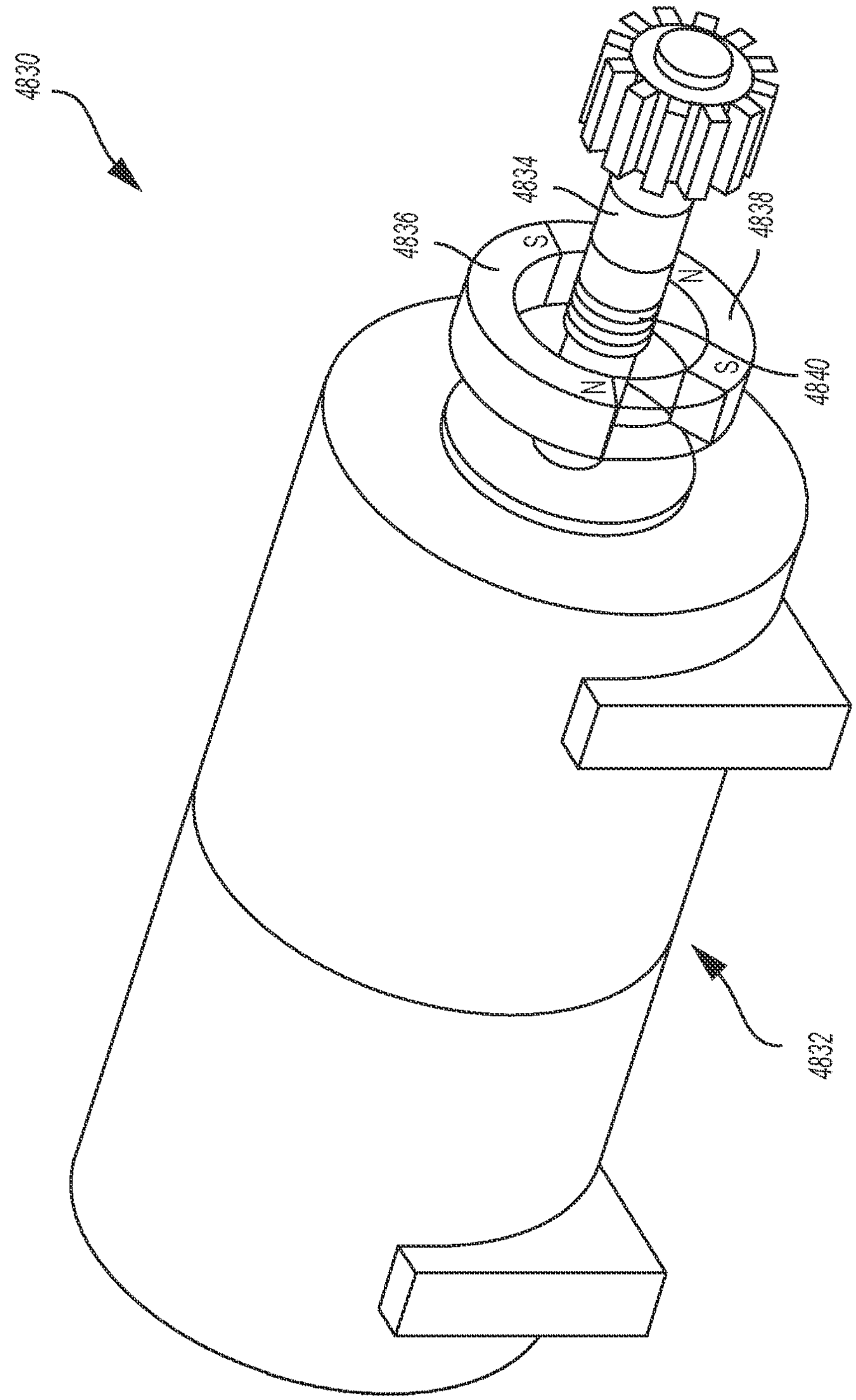


FIG. 103

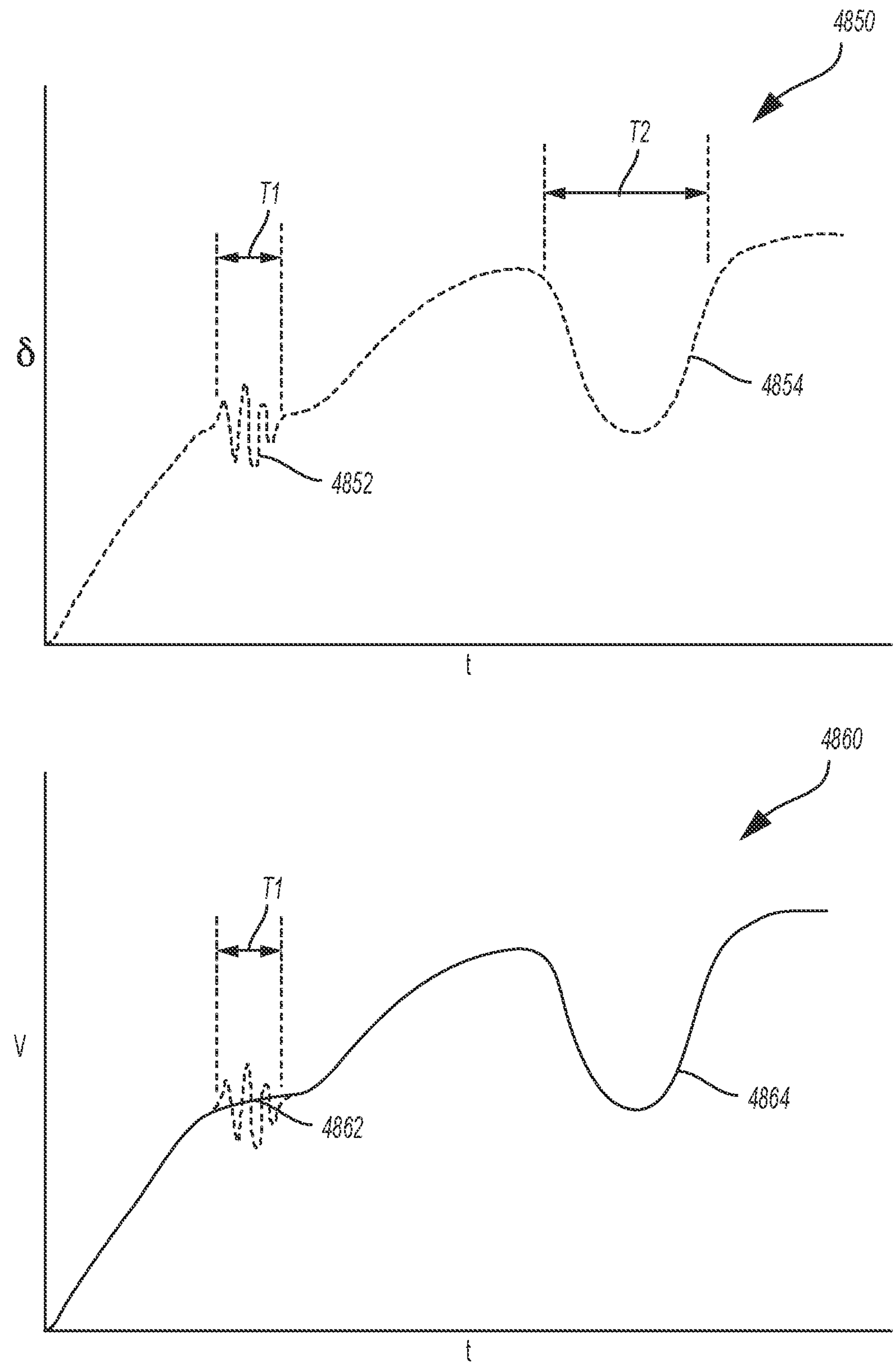


FIG. 104

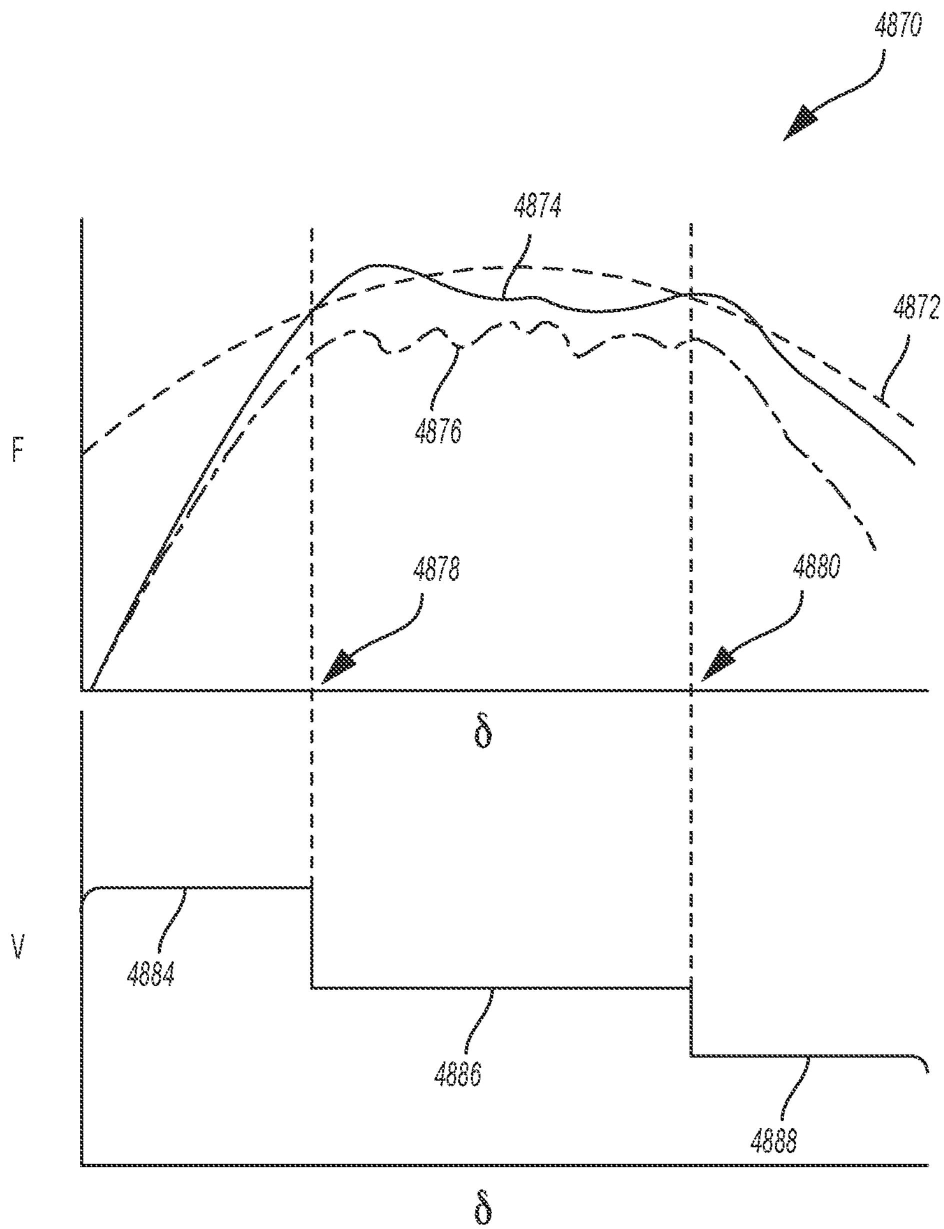


FIG. 105

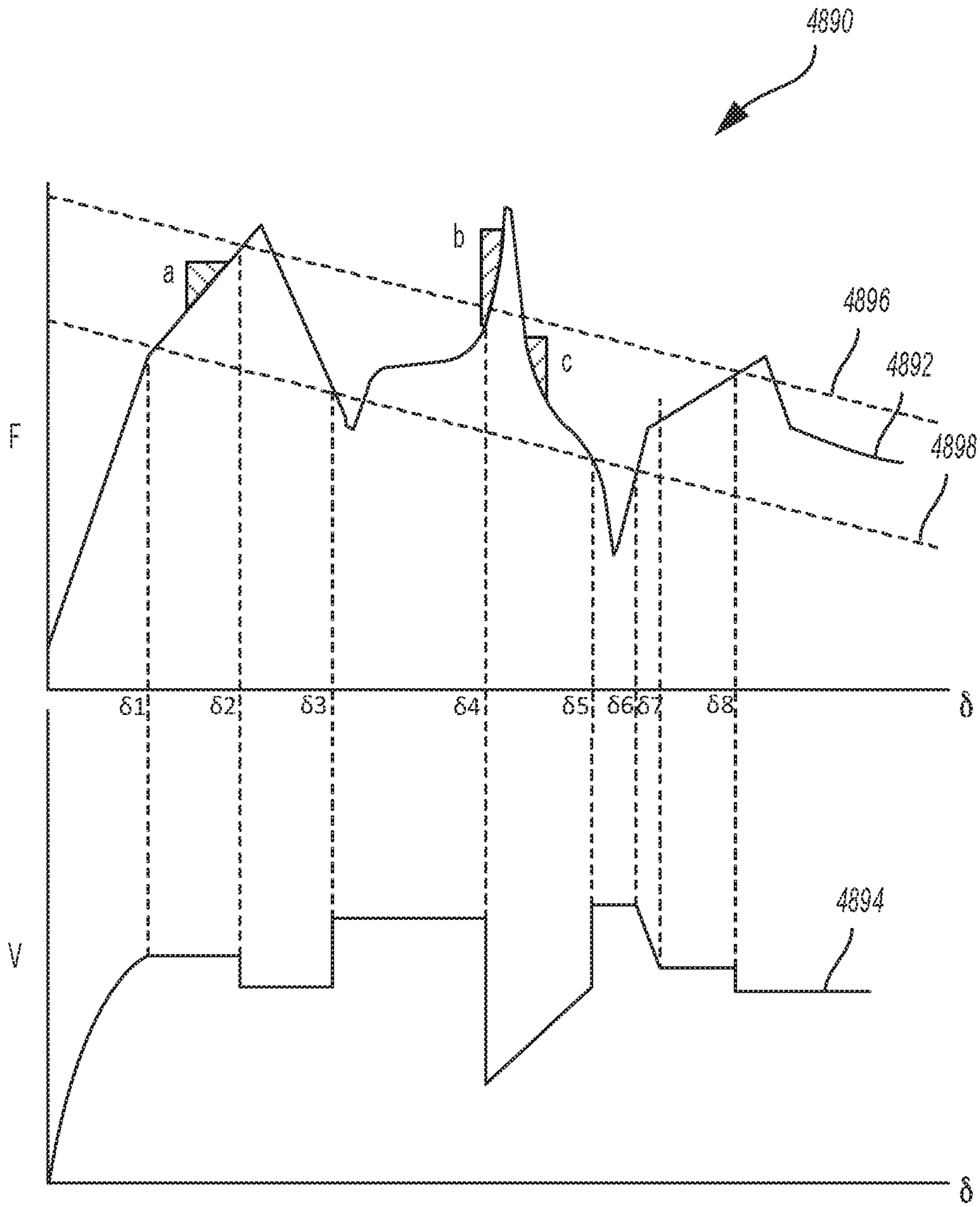


FIG. 106

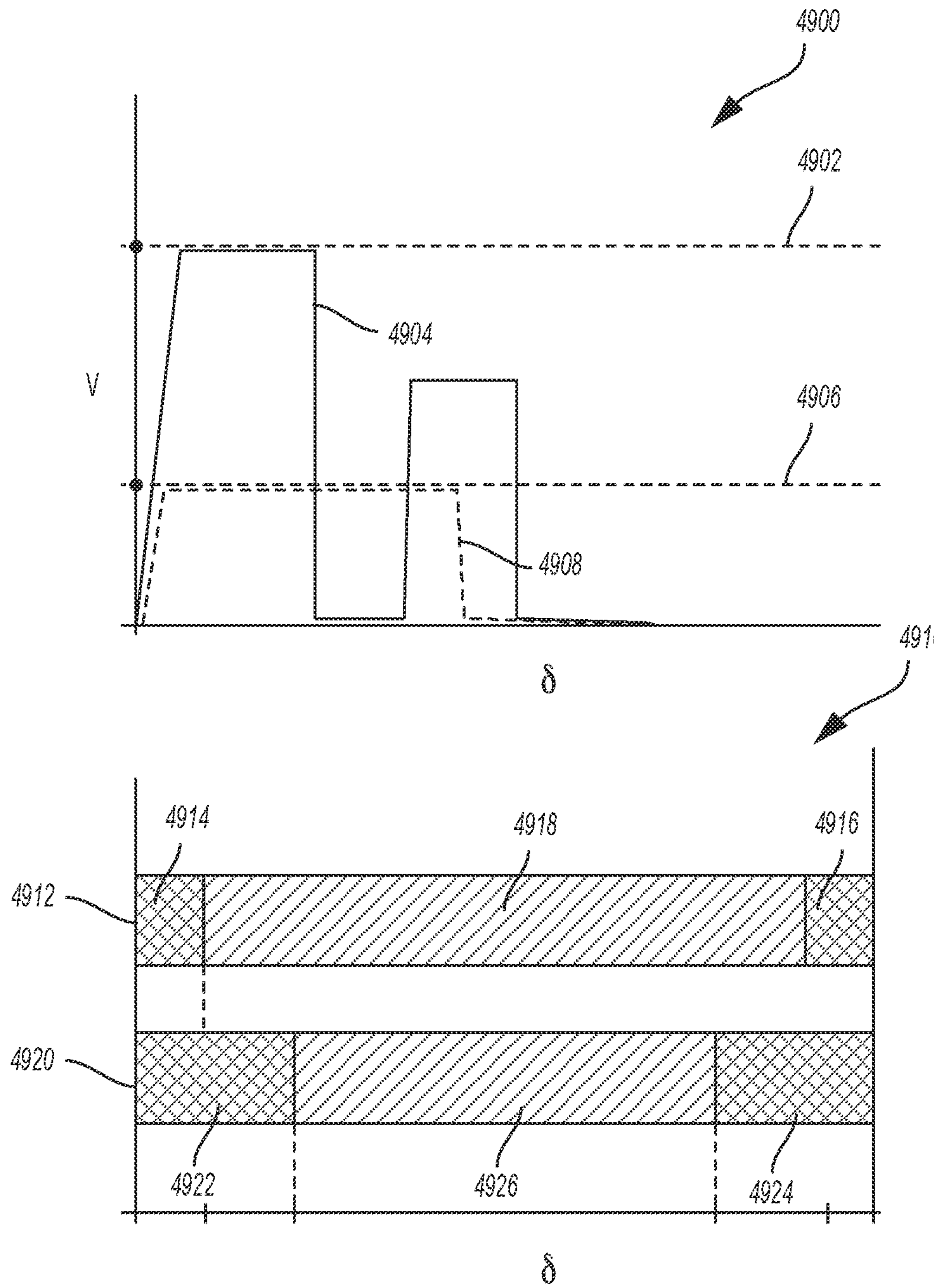

4900
4902
4904
V
4906
4908
δ
4910
4914
4918
4916
4912
4920
4922
4926
4924
δ

FIG. 107

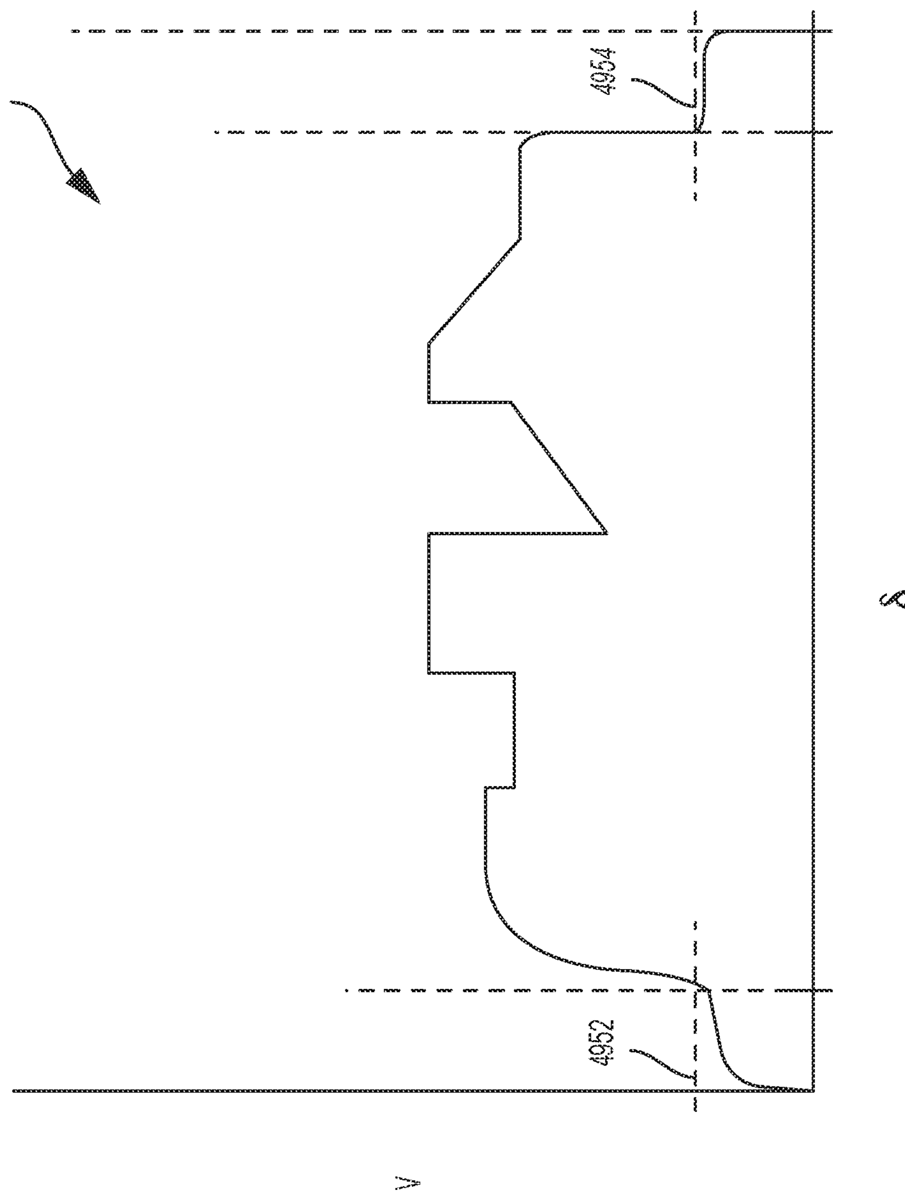


FIG. 108

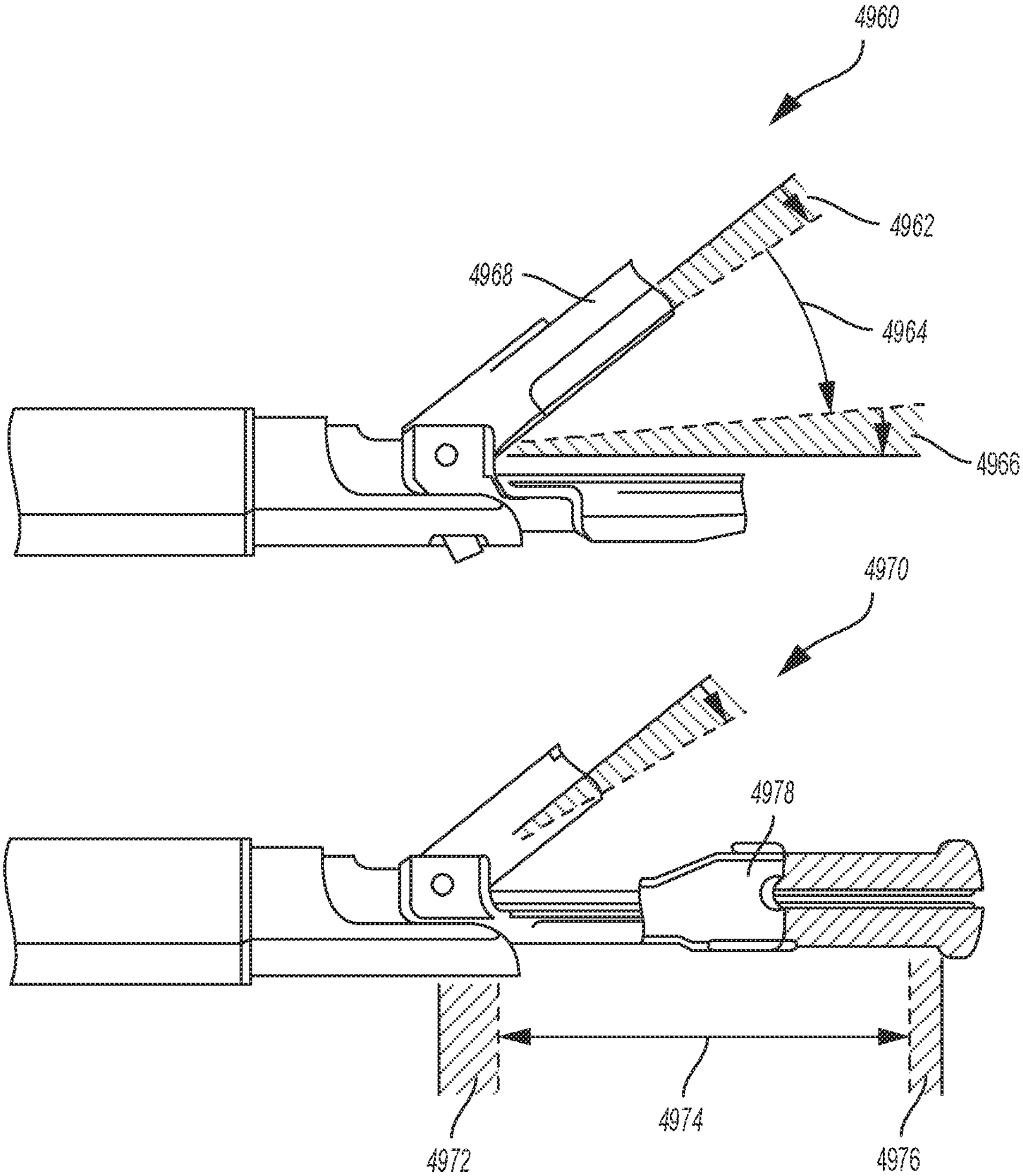


FIG. 109

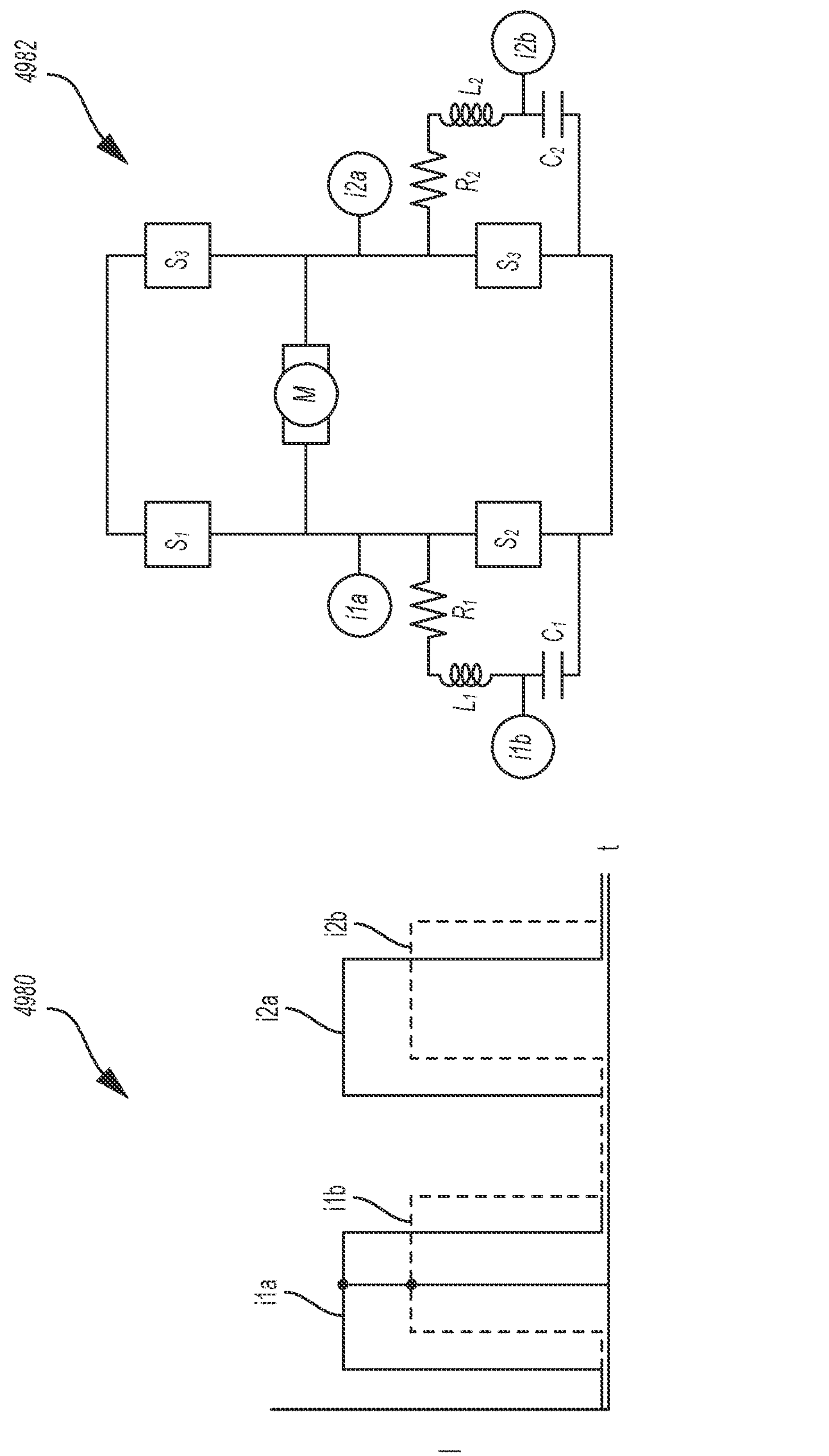

4982
S1
S2
S3
S3
M
i1a
i1b
i2a
i2b
R1
R2
L1
L2
C1
C2
4980
i1a
i1b
i2a
i2b
t
I

FIG. 110

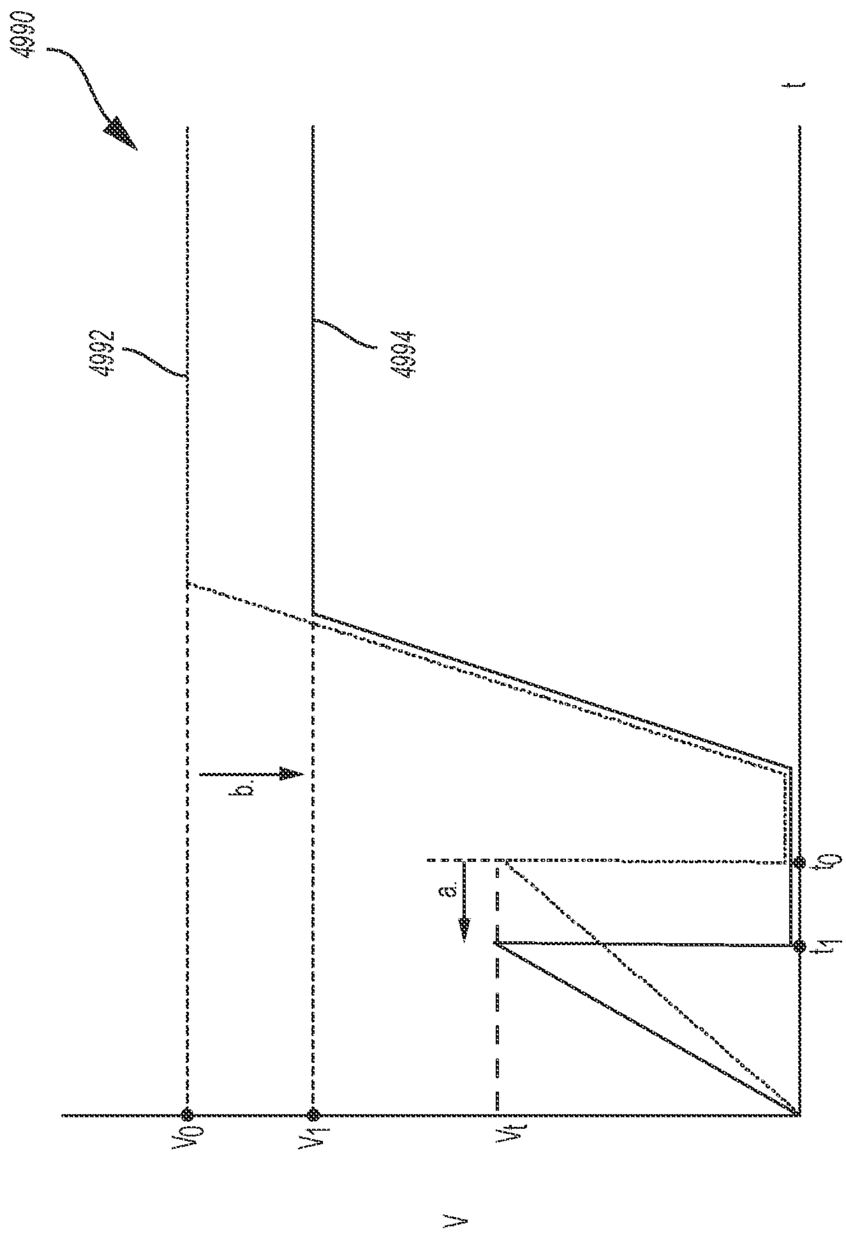


FIG. 111

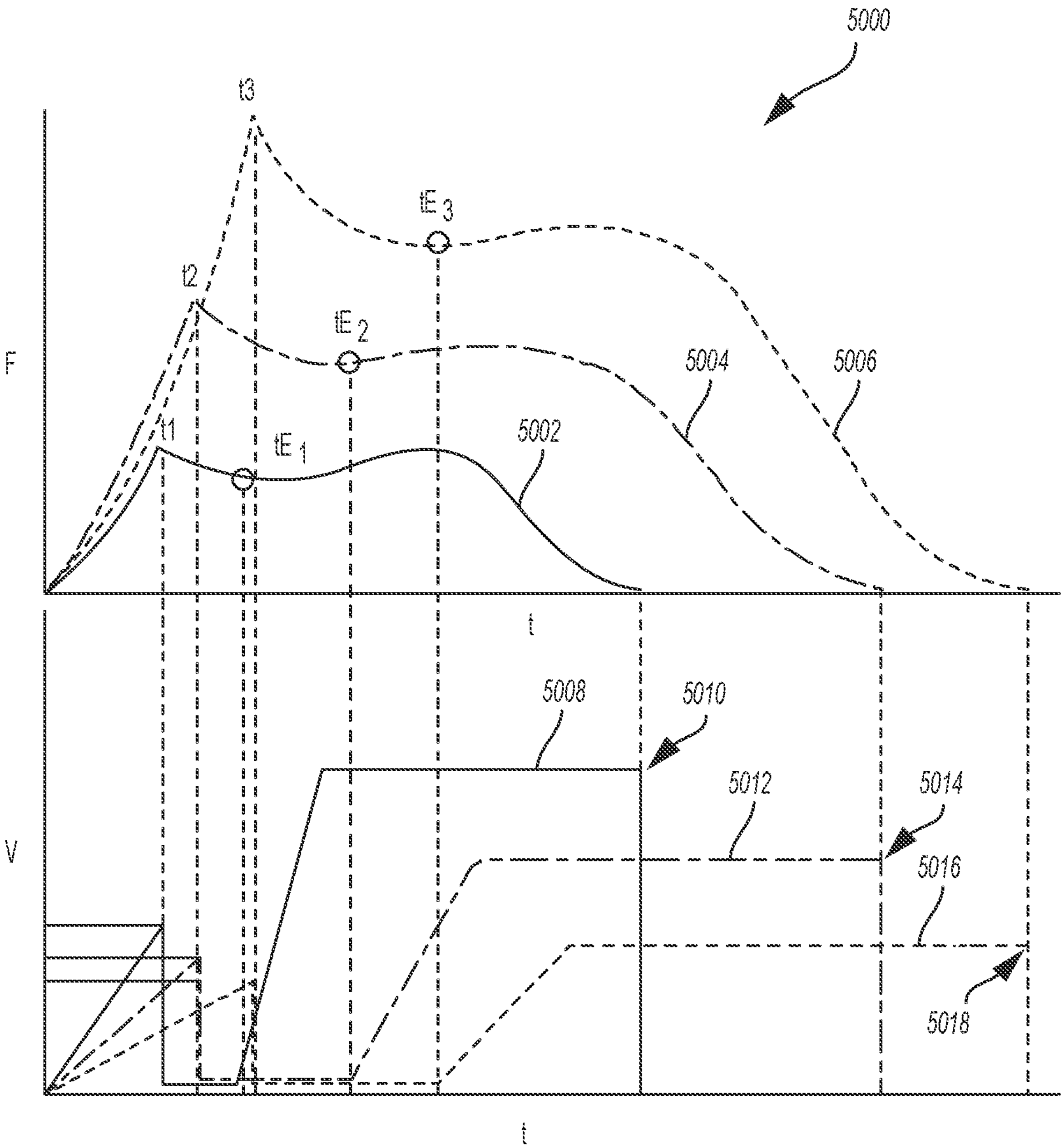

5000
t3
tE 3
t2
tE 2
F
5004
5006
t1
tE 1
5002
t
5008
5010
5012
5014
V
5016
5018
t

FIG. 112

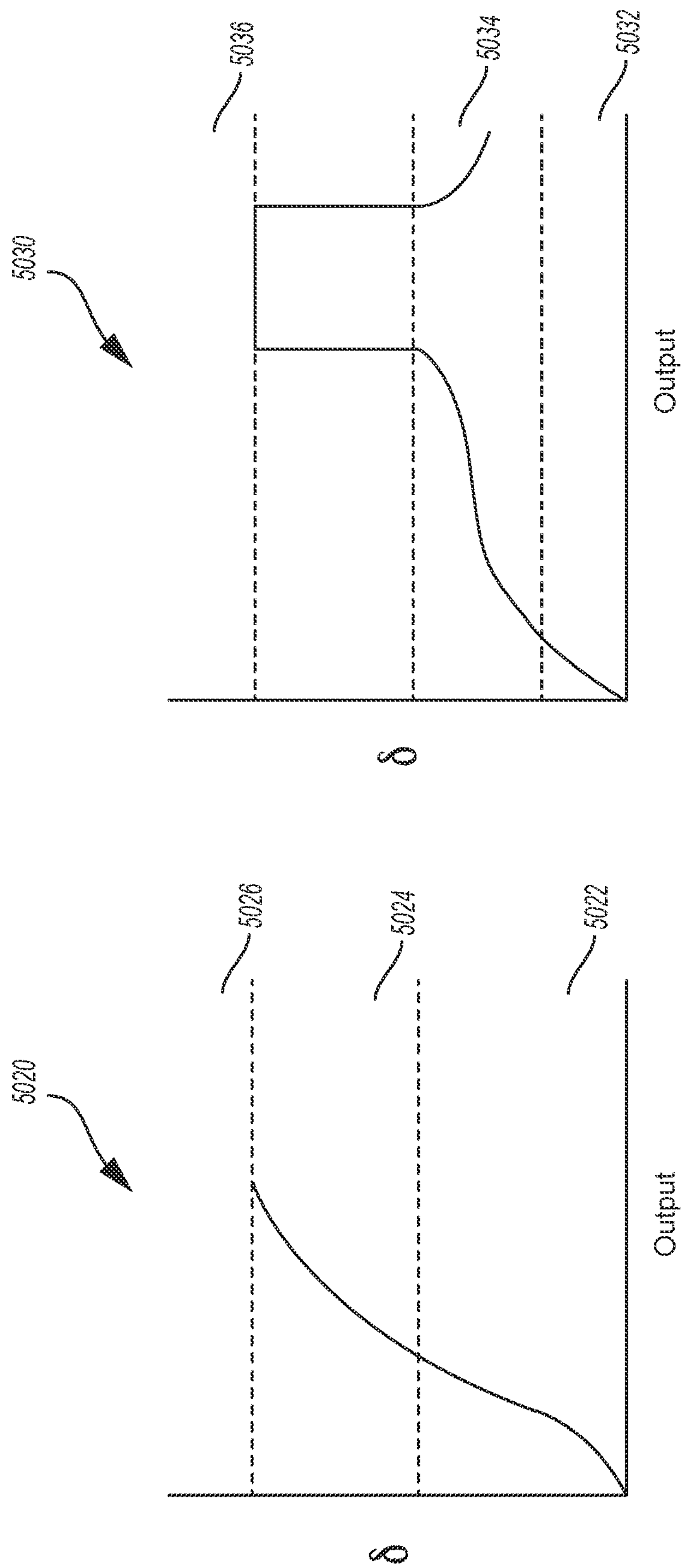


FIG. 113

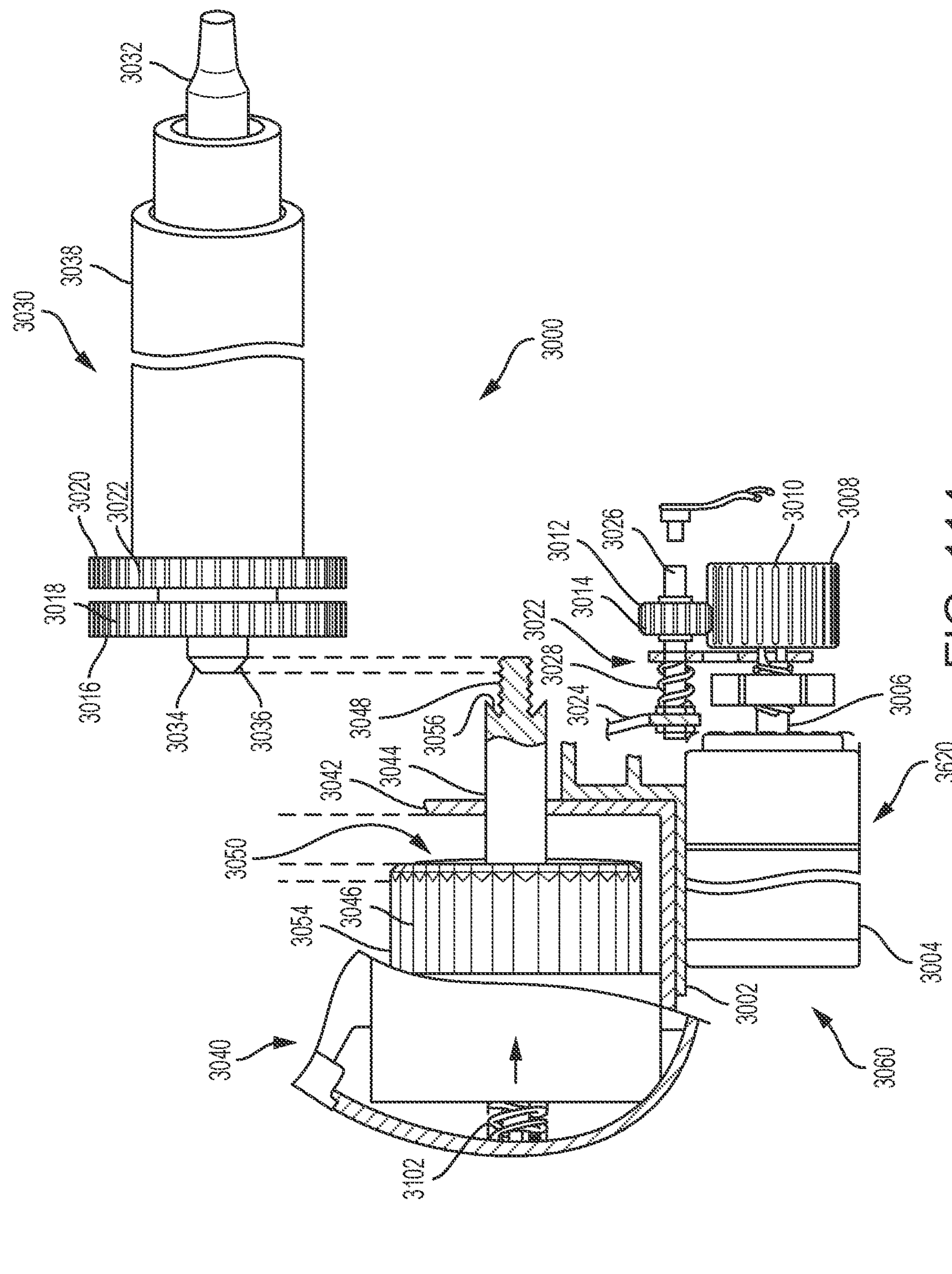

3032
3038
3030
3000
3020
3022
3018
3016
3034
3036
3048
3056
3042
3044
3050
3046
3054
3040
3102
3012
3026
3010
3008
3014
3022
3028
3024
3006
3620
3004
3002
3060

FIG. 114

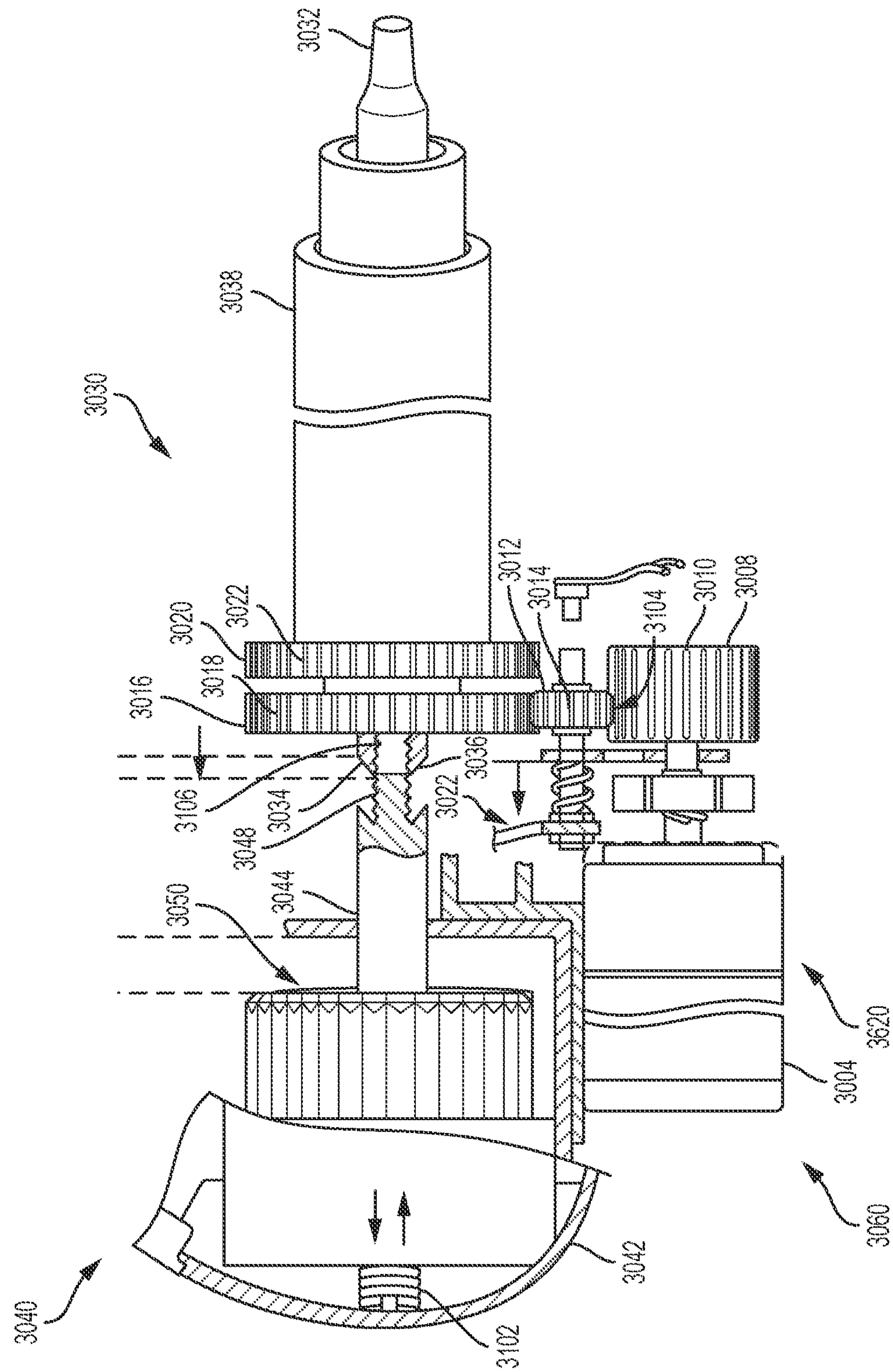

3032
3038
3030
3012
3014
3010
3008
3104
3020
3022
3018
3016
3036
3106
3034
3022
3048
3044
3050
3620
3004
3060
3042
3040
3102

FIG. 115

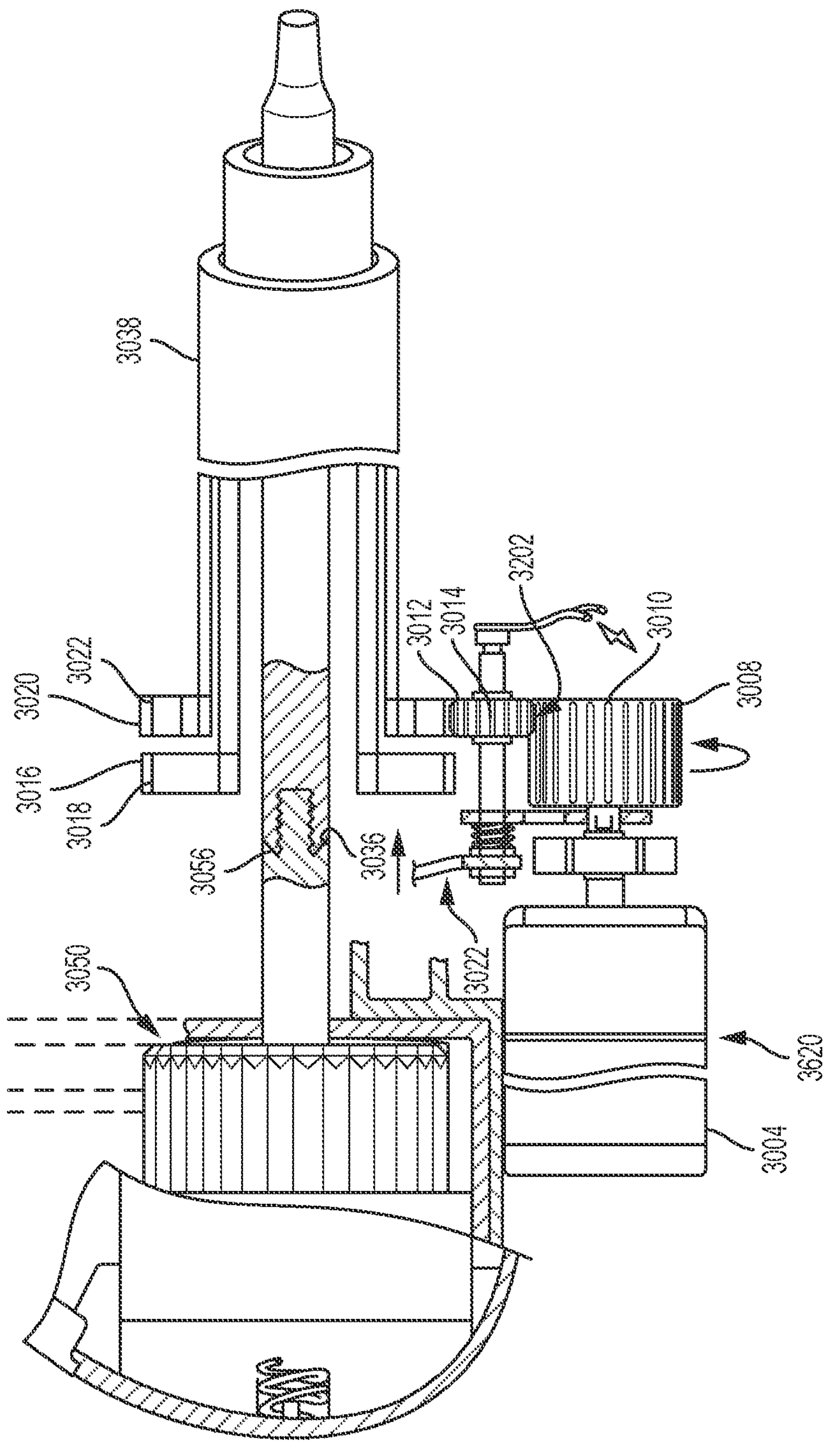

3038
3022
3020
3016
3018
3056
3036
3012
3014
3202
3010
3008
3022
3050
3620
3004

FIG. 116

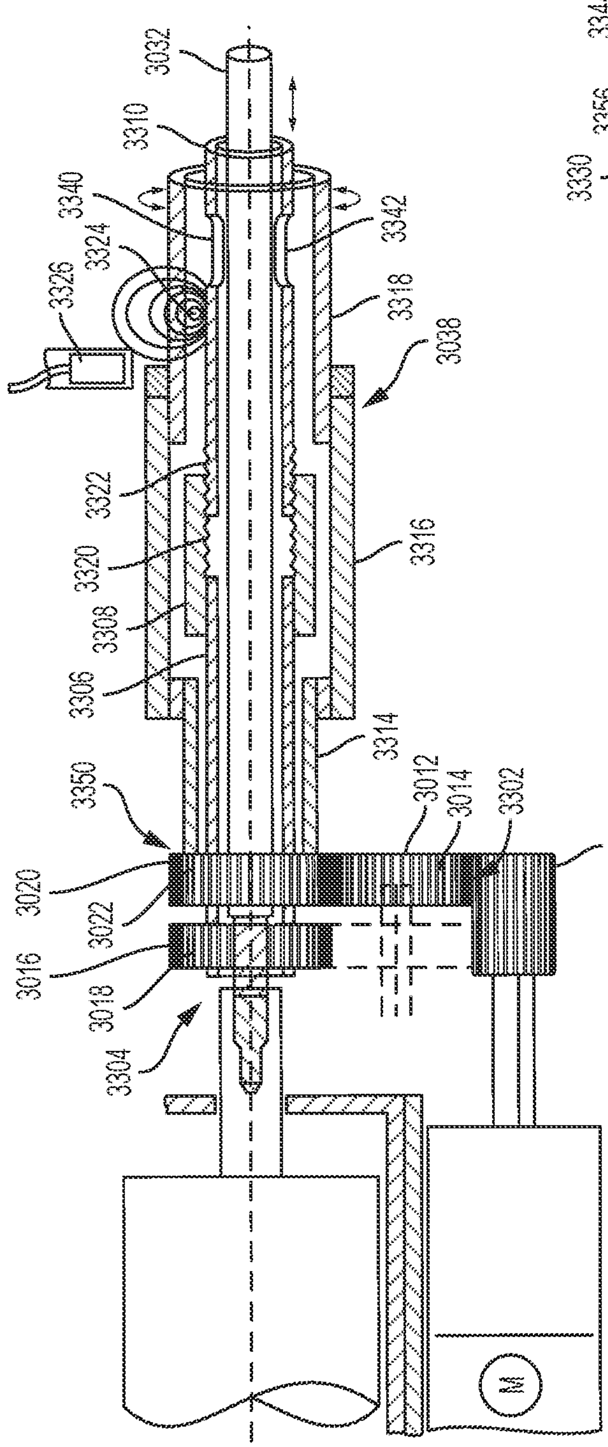


FIG. 117

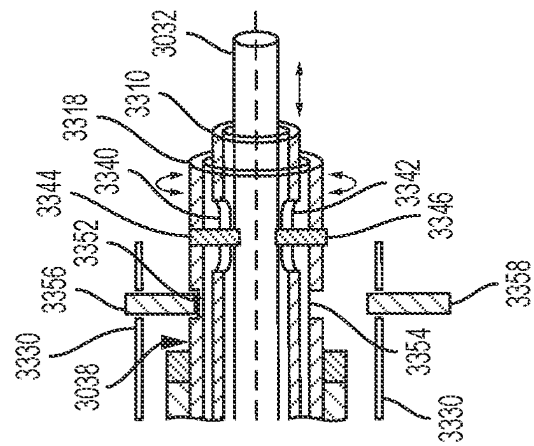


FIG. 119

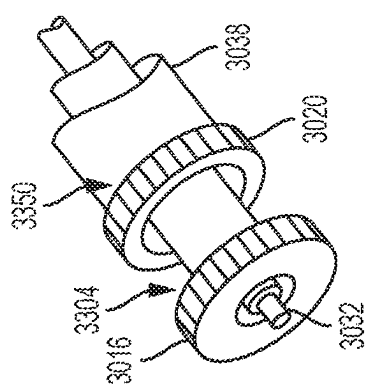


FIG. 118

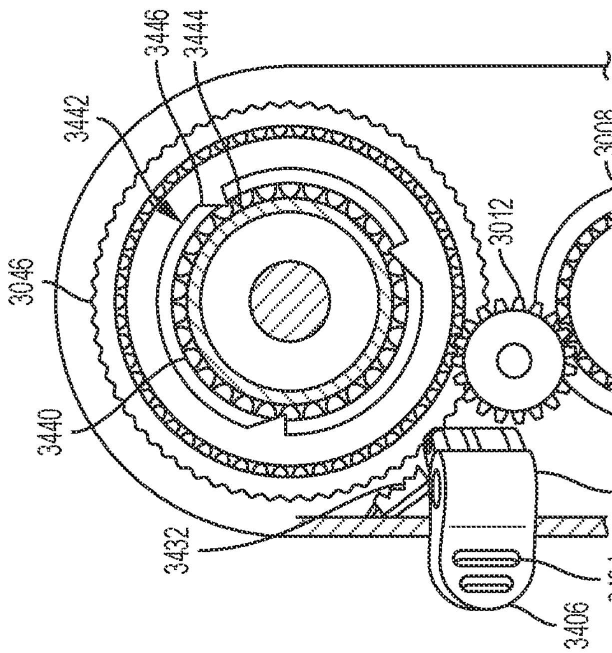


FIG. 121

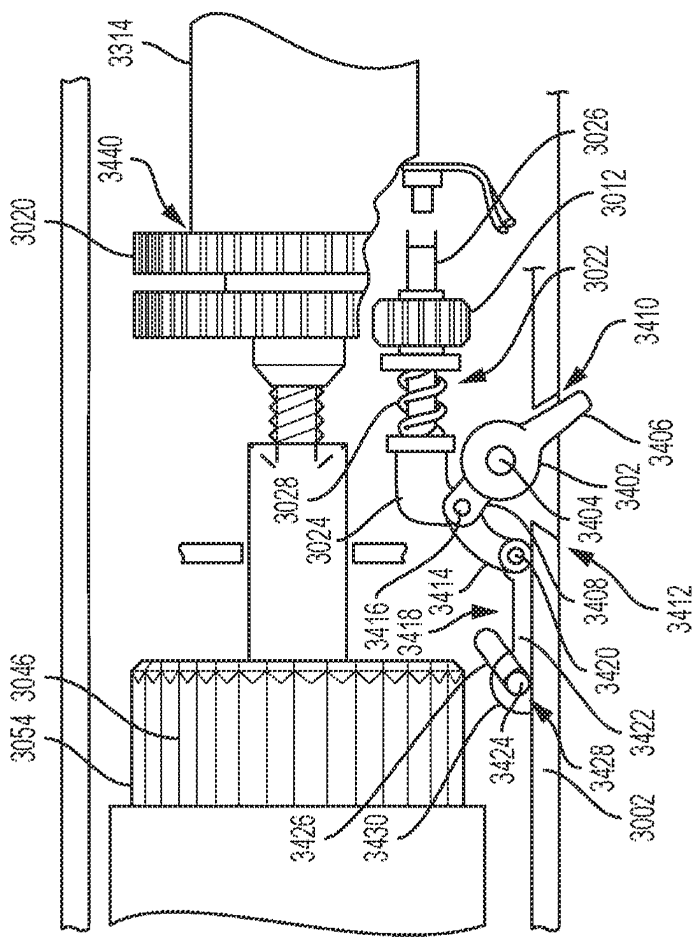
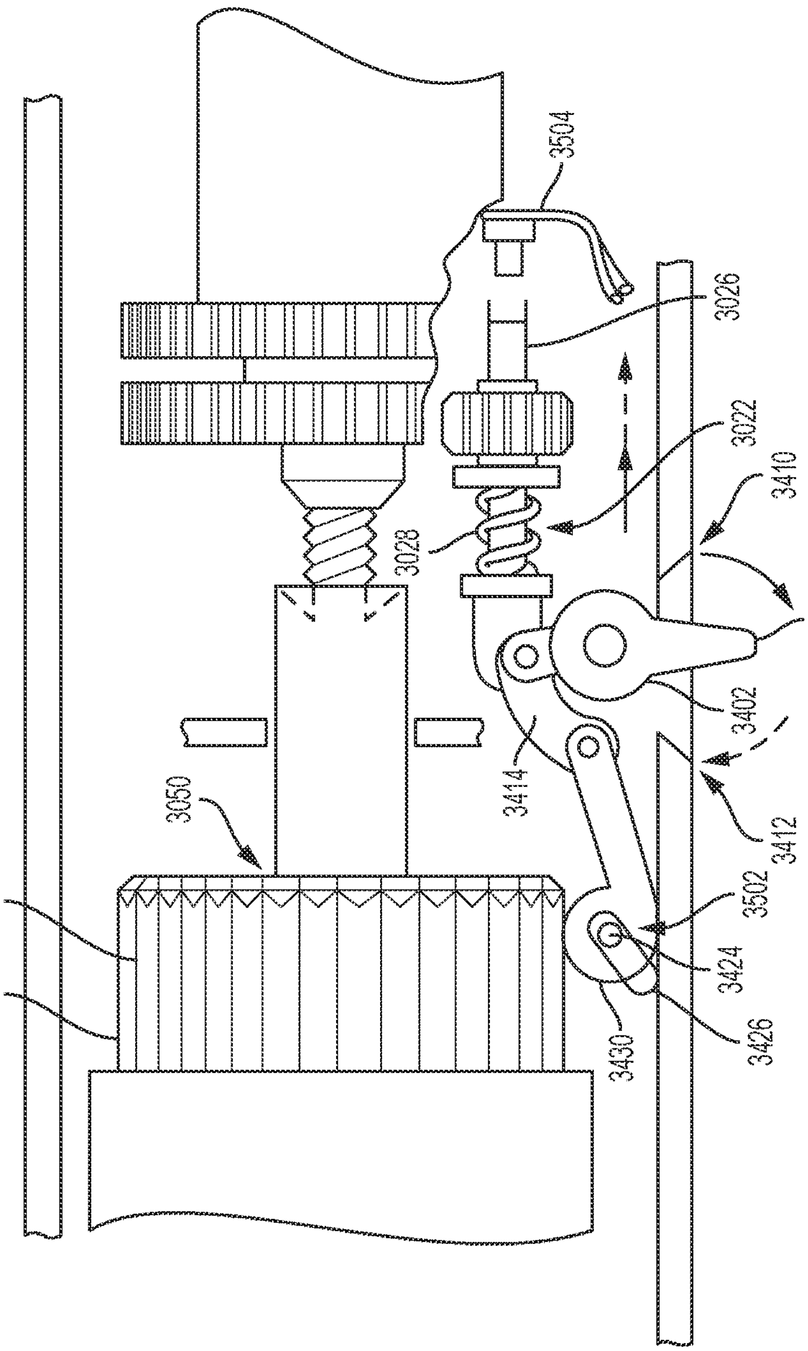


FIG. 120



3504
3026
3022
3410
3028
3406
3402
3414
3050
3412
3502
3046
3054
3424
3426
3430

FIG. 122

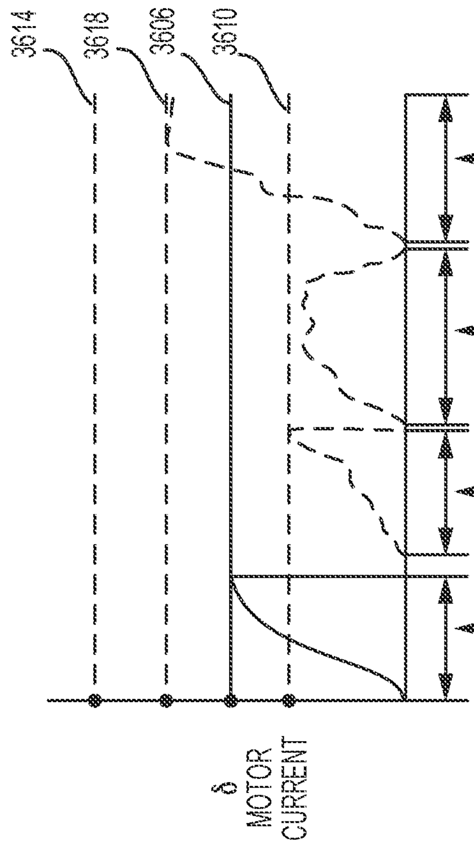


FIG. 124

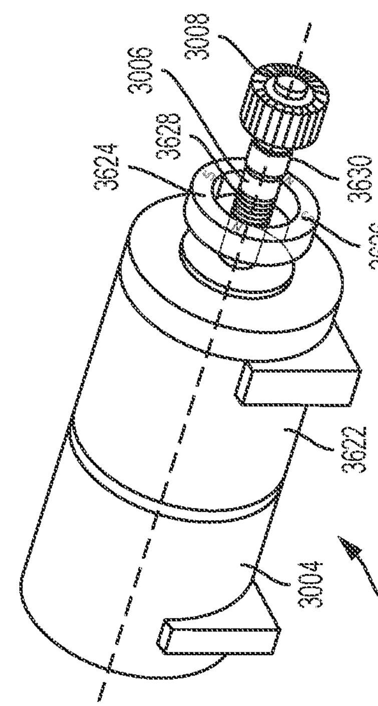


FIG. 125

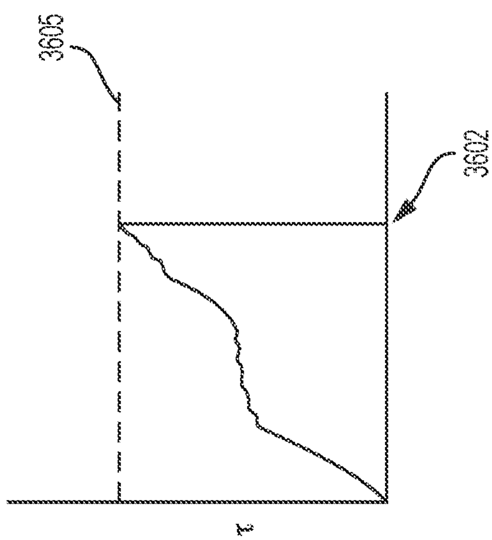


FIG. 123

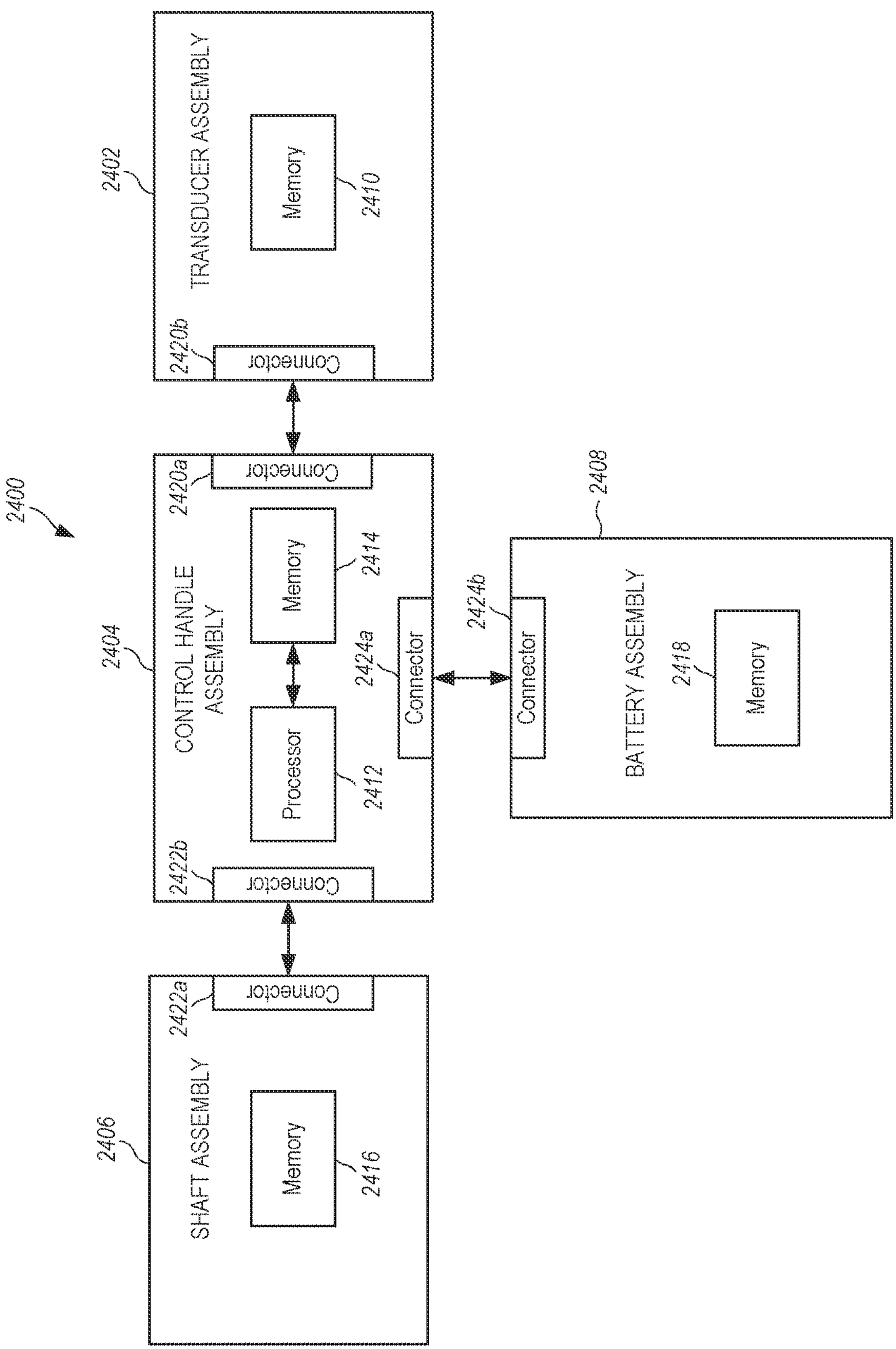

2400
2402
TRANSDUCER ASSEMBLY
Memory
2410
2420b
Connector
2404
CONTROL HANDLE ASSEMBLY
2420a
Connector
Memory
2414
Processor
2412
2424a
Connector
2422b
Connector
2406
SHAFT ASSEMBLY
2422a
Connector
Memory
2416
2408
BATTERY ASSEMBLY
2424b
Connector
2418
Memory

FIG. 126

2500

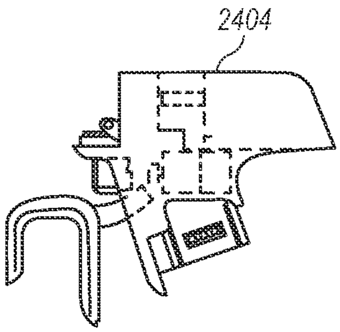
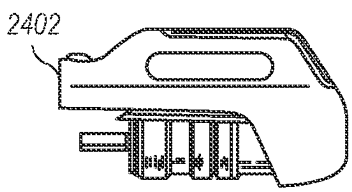
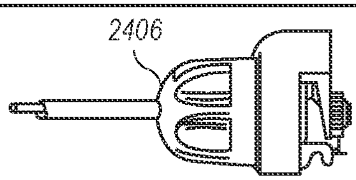
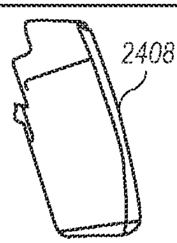
HANDLE CONTROLS COMPONENTS			
HARDWARE	SOFTWARE		
HANDLE			
	RTOS Software		2502
	Motor Control		2504
	Switch		2506
	Safety Control		2508
	Transducer		2509
		55kHz	2510
		31kHz	2512
		RF	2514
	Shaft		2515
		Ultrasonic	2516
		Combo US/RF	2518
		RF I-Blade	2520
		RF Opposable Jaw	2522
TRANSDUCER			
	Component ID		2524
	Usage Counter		2526
	RTOS Update		2528
	Energy Update		2530
SHAFT			
	Component ID		2532
	Usage Counter		2534
	RTOS Update		2536
	Energy Update		2538
BATTERY			
	Usage Counter		2540
	Maximum Number of Uses		2542
	Charge / Drainage		2544
	RTOS Update		2546
	Energy Update		2548

FIG. 127

2600

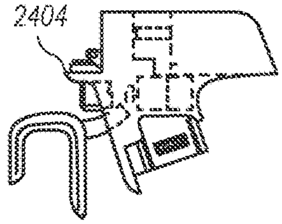
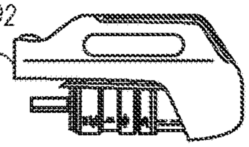
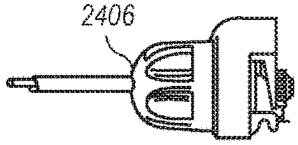
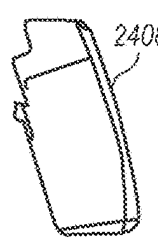
BATTERY CONTROLS COMPONENTS		
HARDWARE	SOFTWARE	
HANDLE		
	NONE	
TRANSDUCER		
	Component ID	2602
	Usage Counter	2604
SHAFT		
	Component ID	2606
	Usage Counter	2608
BATTERY		
	RTOS Software	2502
	Motor Control	2504
	Switch	2506
	Safety Control	2508
	Transducer	2509
	55kHz	2510
	31kHz	2512
	RF	2514
	Shaft	2515
	Ultrasonic	2516
	Combo US/RF	2518
	RF I-Blade	2520
	RF Opposable Jaw	2522
	Usage Counter	2610
	Maximum Number of Uses	2612
	Charge / Drainage	2614
	Energy Update	2616

FIG. 128

2700

COMPONENTS CONTROLS
COMPONENTS

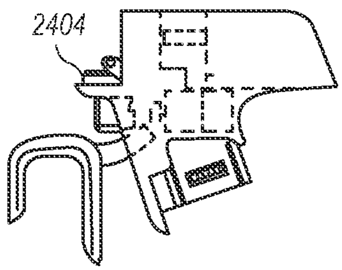
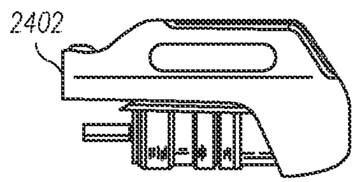
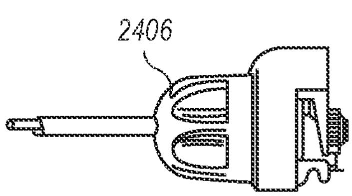
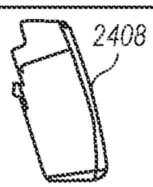
HARDWARE	SOFTWARE	
HANDLE		
	RTOS Software	2502
	Motor Control	2504
	Switch Control	2506
	Safety Control	2508
	Energy Update	2702
TRANSDUCER		
	Component ID	2704
	RTOS Update	2706
	Usage Counter	2708
	Energy Update	2710
	55kHz	2510
	31kHz	2512
	RF	2514
SHAFT		
	Component ID	2712
	Ultrasonic	2516
	Combo US/RF	2518
	RF I-Blade	2520
	RF Opposable Jaw	2522
	RTOS Update	2714
	Usage Counter	2716
	Energy Update	2718
BATTERY		
	Usage Counter	2720
	Maximum Number of Uses	2722
	Charge / Drainage	2724
	RTOS Update	2726
	Energy Update	2728

FIG. 129

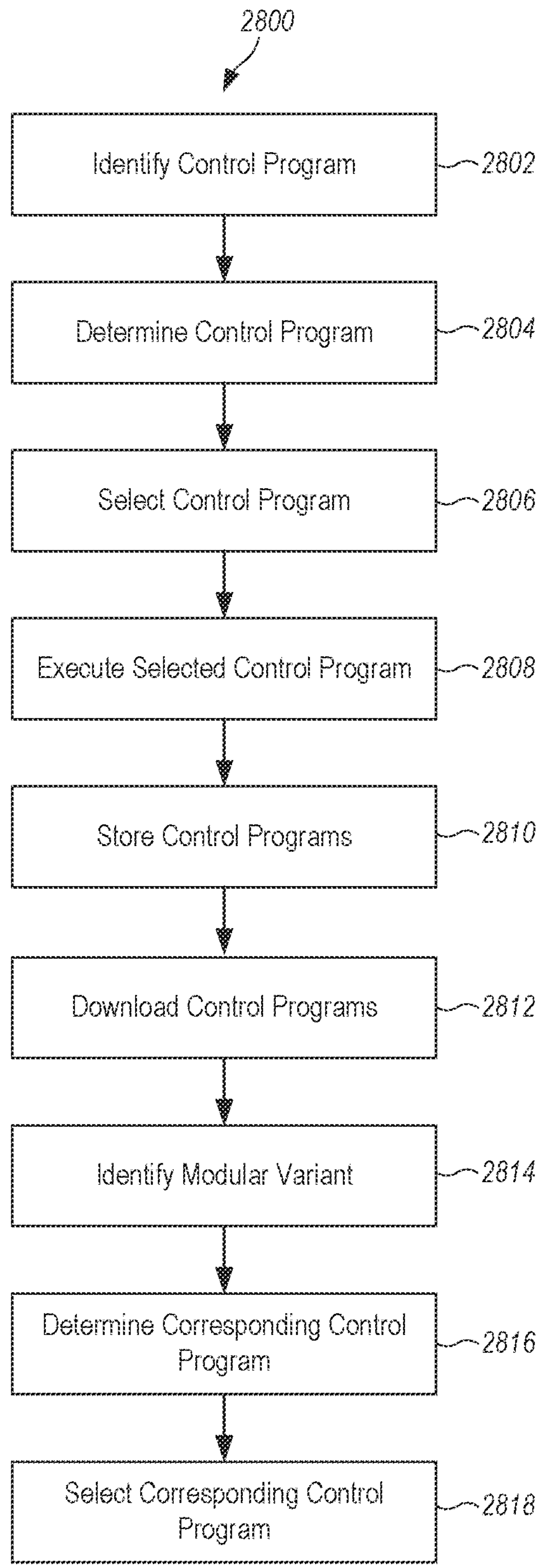

2800
Identify Control Program
2802
Determine Control Program
2804
Select Control Program
2806
Execute Selected Control Program
2808
Store Control Programs
2810
Download Control Programs
2812
Identify Modular Variant
2814
Determine Corresponding Control Program
2816
Select Corresponding Control Program
2818

FIG. 130

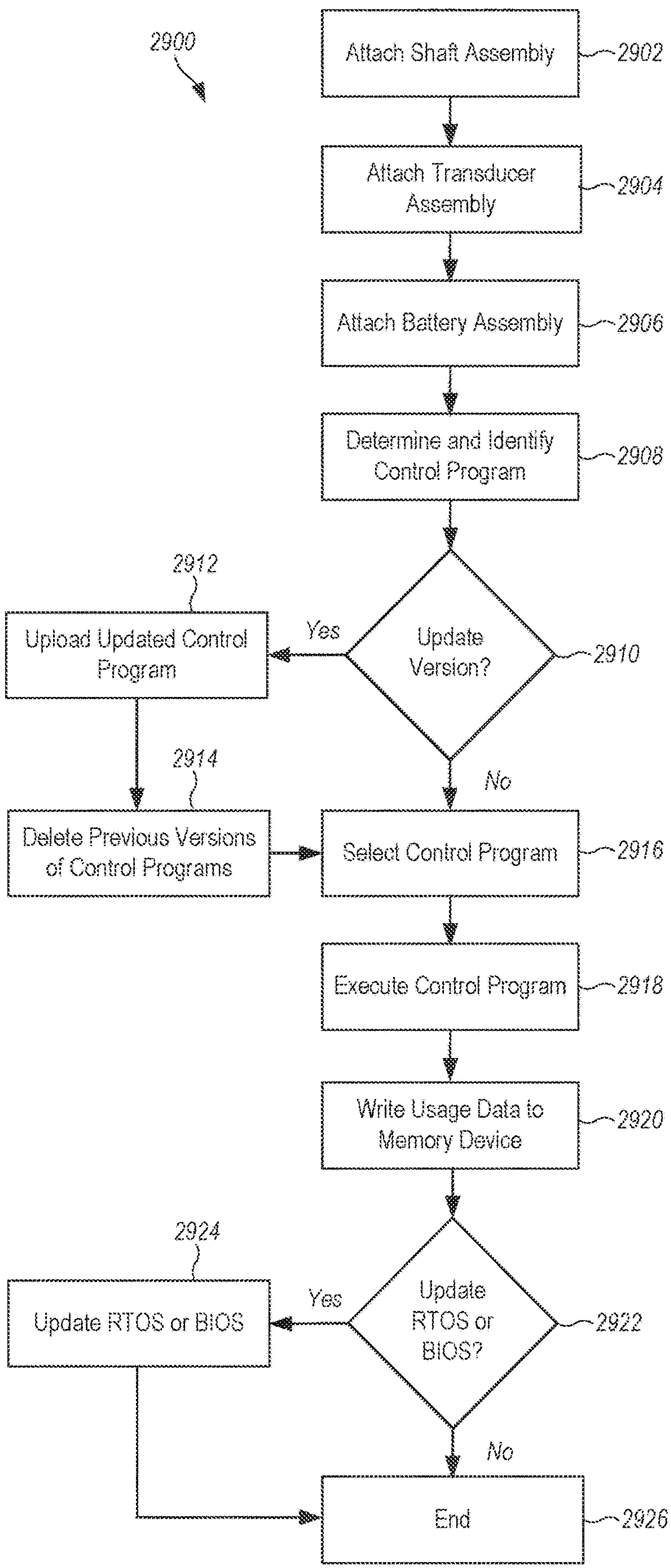

2900
Attach Shaft Assembly
2902
Attach Transducer Assembly
2904
Attach Battery Assembly
2906
Determine and Identify Control Program
2908
Update Version?
2910
Yes
Upload Updated Control Program
2912
Delete Previous Versions of Control Programs
2914
No
Select Control Program
2916
Execute Control Program
2918
Write Usage Data to Memory Device
2920
Update RTOS or BIOS?
2922
Yes
Update RTOS or BIOS
2924
No
End
2926

FIG. 131

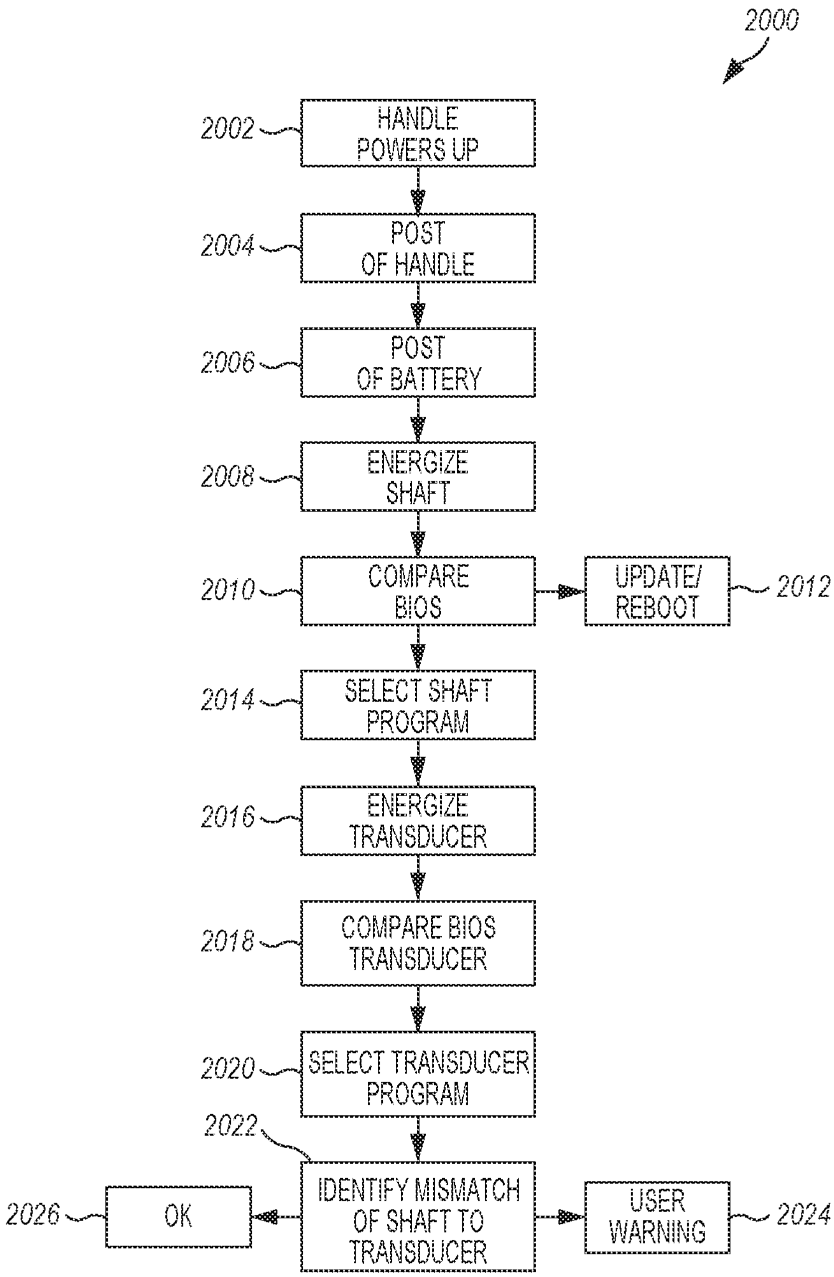

2000
2002
HANDLE POWERS UP
2004
POST OF HANDLE
2006
POST OF BATTERY
2008
ENERGIZE SHAFT
2010
COMPARE BIOS
UPDATE/ REBOOT
2012
2014
SELECT SHAFT PROGRAM
2016
ENERGIZE TRANSDUCER
2018
COMPARE BIOS TRANSDUCER
2020
SELECT TRANSDUCER PROGRAM
2022
IDENTIFY MISMATCH OF SHAFT TO TRANSDUCER
2026
OK
USER WARNING
2024

FIG. 132

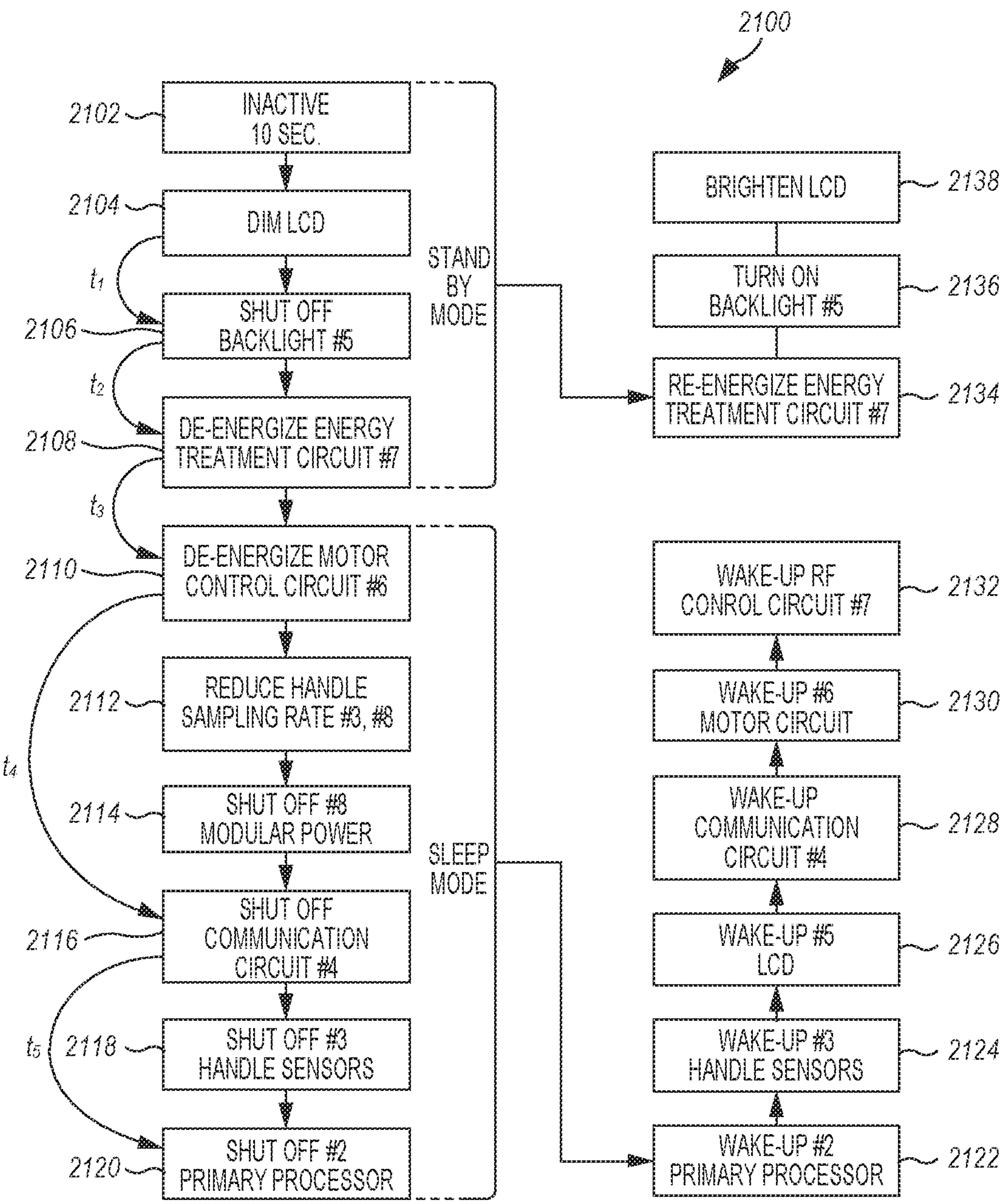

2100
INACTIVE 10 SEC.
2102
DIM LCD
2104
t_1
SHUT OFF BACKLIGHT #5
2106
t_2
DE-ENERGIZE ENERGY TREATMENT CIRCUIT #7
2108
t_3
STAND BY MODE
DE-ENERGIZE MOTOR CONTROL CIRCUIT #6
2110
REDUCE HANDLE SAMPLING RATE #3, #8
2112
t_4
SHUT OFF #8 MODULAR POWER
2114
SHUT OFF COMMUNICATION CIRCUIT #4
2116
t_5
SHUT OFF #3 HANDLE SENSORS
2118
SHUT OFF #2 PRIMARY PROCESSOR
2120
SLEEP MODE
BRIGHTEN LCD
2138
TURN ON BACKLIGHT #5
2136
RE-ENERGIZE ENERGY TREATMENT CIRCUIT #7
2134
WAKE-UP RF CONROL CIRCUIT #7
2132
WAKE-UP #6 MOTOR CIRCUIT
2130
WAKE-UP COMMUNICATION CIRCUIT #4
2128
WAKE-UP #5 LCD
2126
WAKE-UP #3 HANDLE SENSORS
2124
WAKE-UP #2 PRIMARY PROCESSOR
2122

FIG. 133

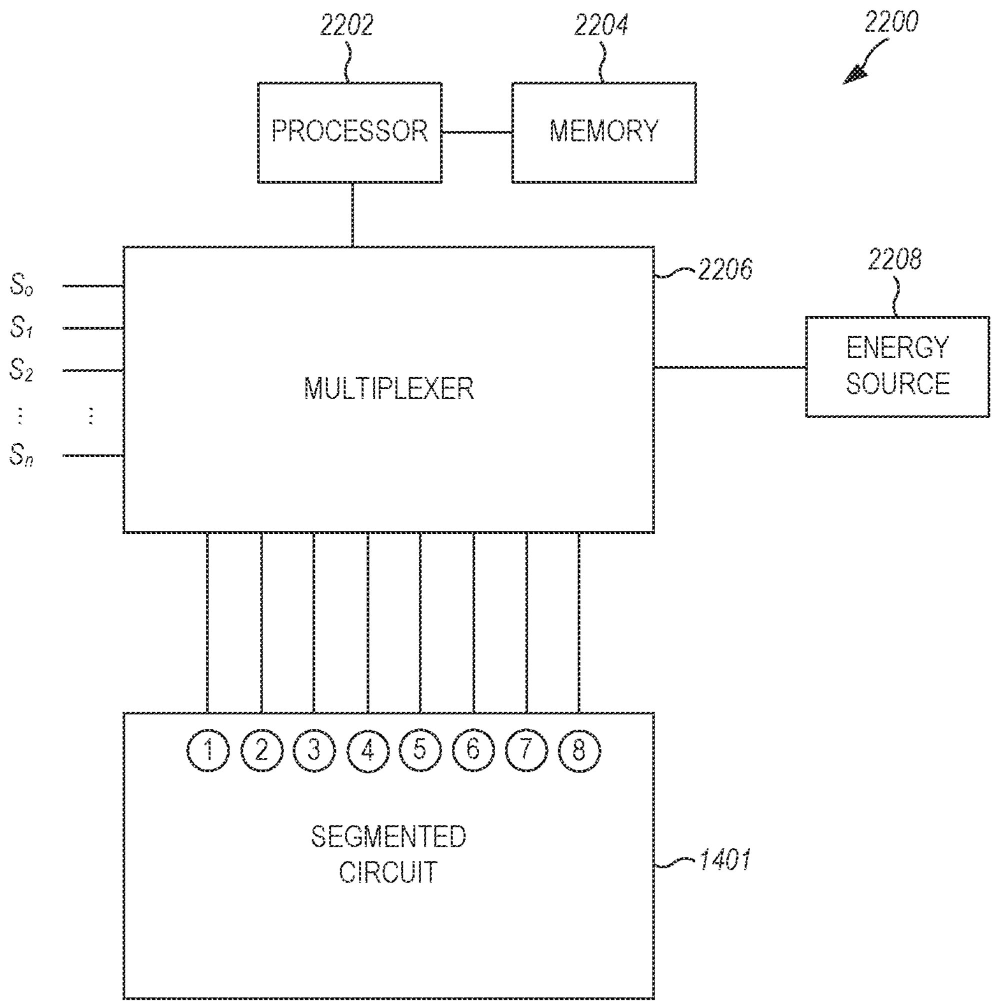
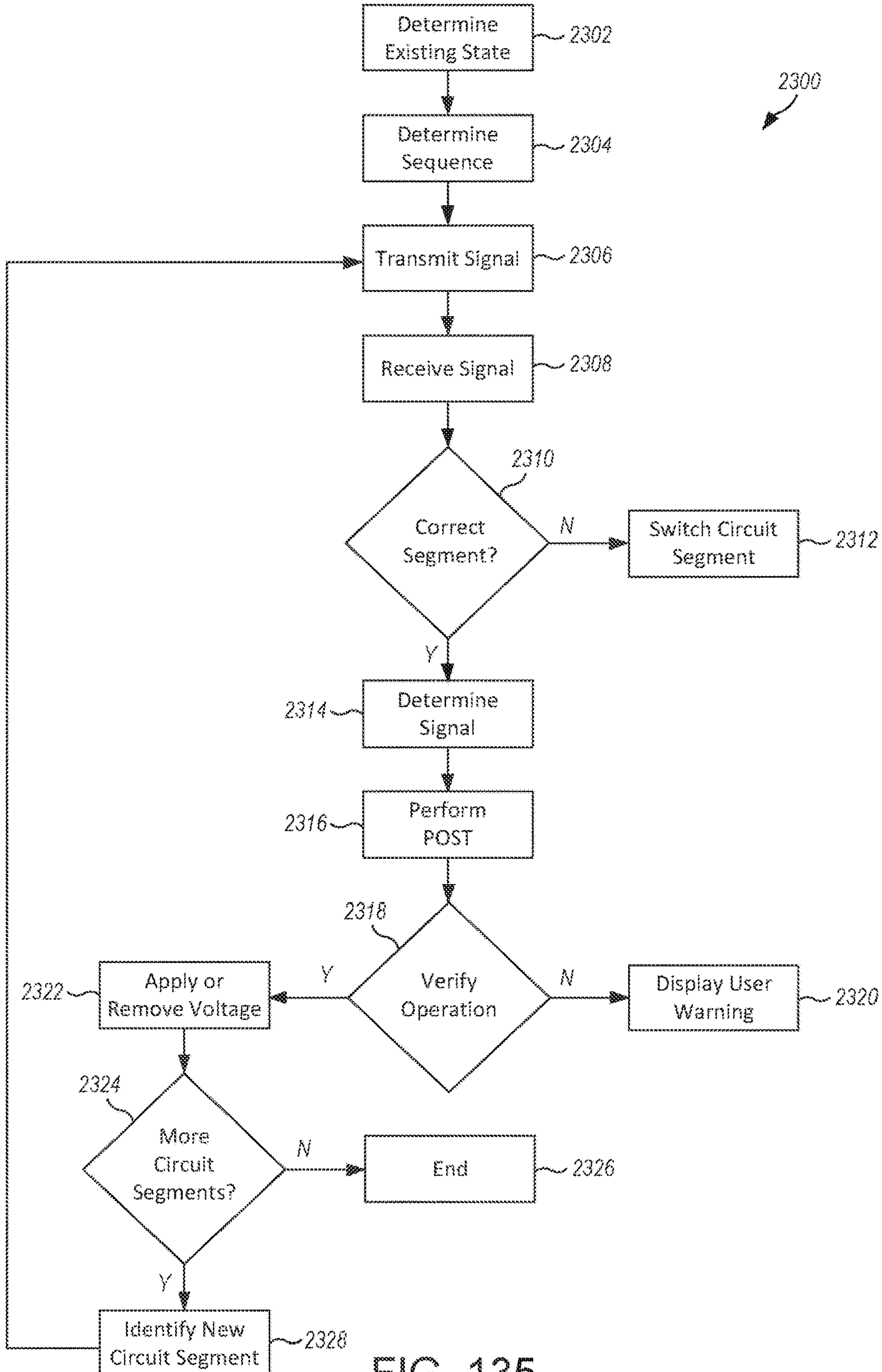

2200
2202
2204
PROCESSOR
MEMORY
2206
2208
S_0
S_1
S_2
S_n
MULTIPLEXER
ENERGY
SOURCE
1
2
3
4
5
6
7
8
SEGMENTED
CIRCUIT
1401

FIG. 134



2300
Determine Existing State
2302
Determine Sequence
2304
Transmit Signal
2306
Receive Signal
2308
Correct Segment?
2310
N
Switch Circuit Segment
2312
Y
Determine Signal
2314
Perform POST
2316
Verify Operation
2318
Y
N
Apply or Remove Voltage
2322
Display User Warning
2320
More Circuit Segments?
2324
N
End
2326
Y
Identify New Circuit Segment
2328

FIG. 135

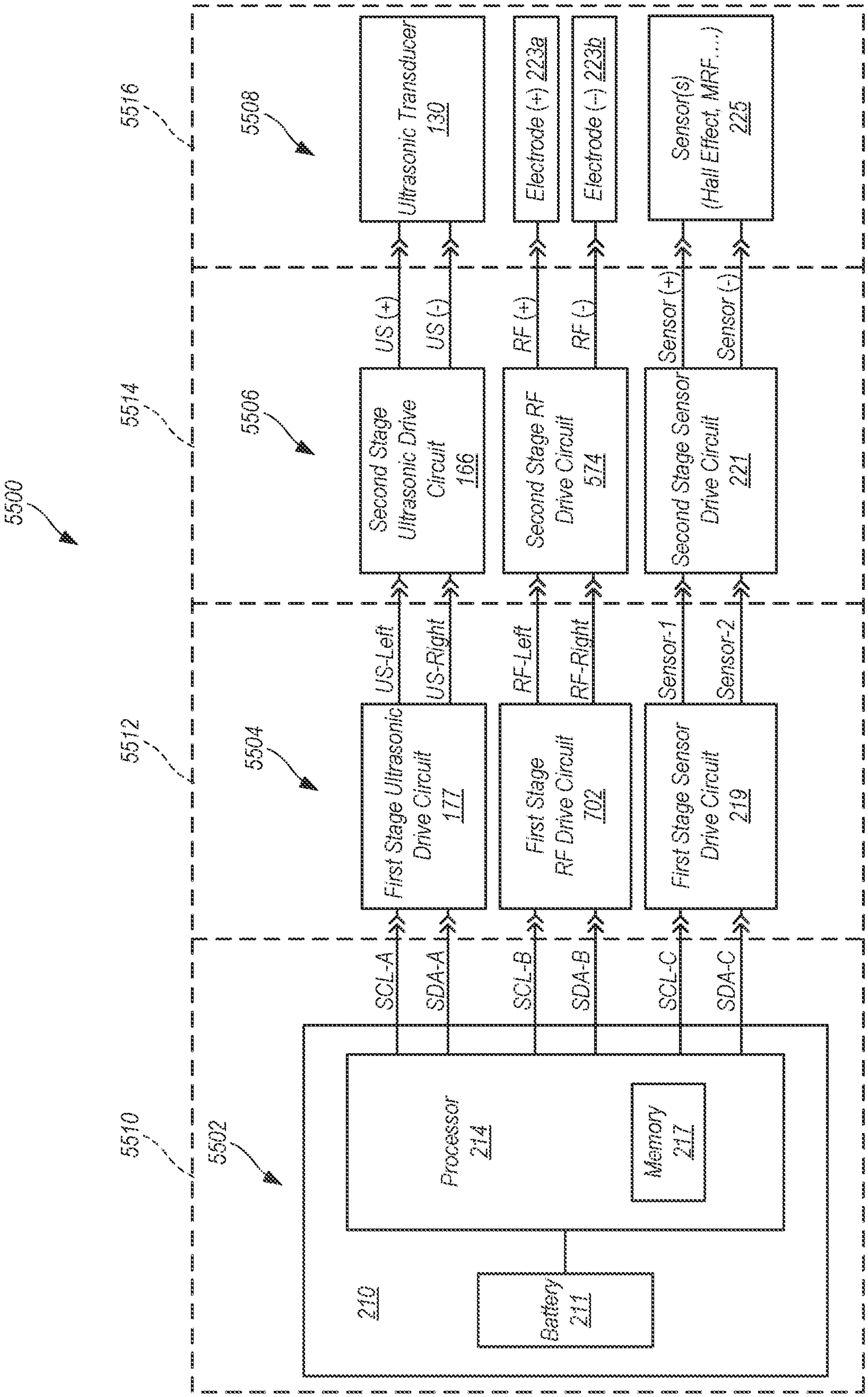

5500
5510
5502
5512
5504
5514
5506
5516
5508
210
Battery 211
Processor 214
Memory 217
SCL-A
SDA-A
SCL-B
SDA-B
SCL-C
SDA-C
First Stage Ultrasonic Drive Circuit 177
First Stage RF Drive Circuit 702
First Stage Sensor Drive Circuit 219
US-Left
US-Right
RF-Left
RF-Right
Sensor-1
Sensor-2
Second Stage Ultrasonic Drive Circuit 166
Second Stage RF Drive Circuit 574
Second Stage Sensor Drive Circuit 221
US (+)
US (-)
RF (+)
RF (-)
Sensor (+)
Sensor (-)
Ultrasonic Transducer 130
Electrode (+) 223a
Electrode (-) 223b
Sensor(s) (Hall Effect, MRF...) 225

FIG. 136

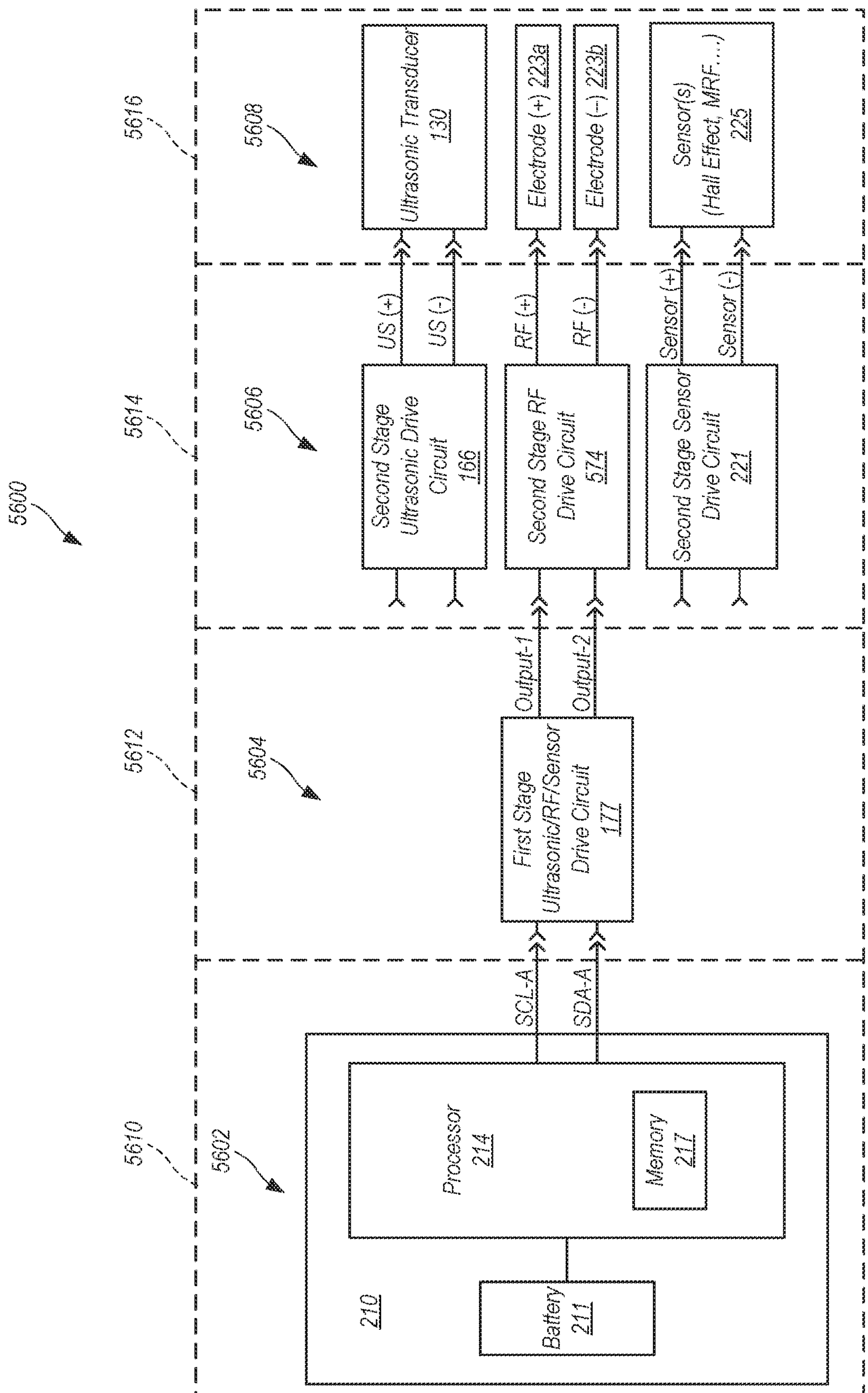

5600
5610
5602
5612
5604
5614
5606
5616
5608
210
Battery
211
Processor
214
Memory
217
SCL-A
SDA-A
First Stage
Ultrasonic/RF/Sensor
Drive Circuit
177
Output-1
Output-2
Second Stage
Ultrasonic Drive
Circuit
166
Second Stage RF
Drive Circuit
574
Second Stage Sensor
Drive Circuit
221
US (+)
US (-)
RF (+)
RF (-)
Sensor (+)
Sensor (-)
Ultrasonic Transducer
130
Electrode (+) 223a
Electrode (−) 223b
Sensor(s)
(Hall Effect, MRF…)
225

FIG. 137

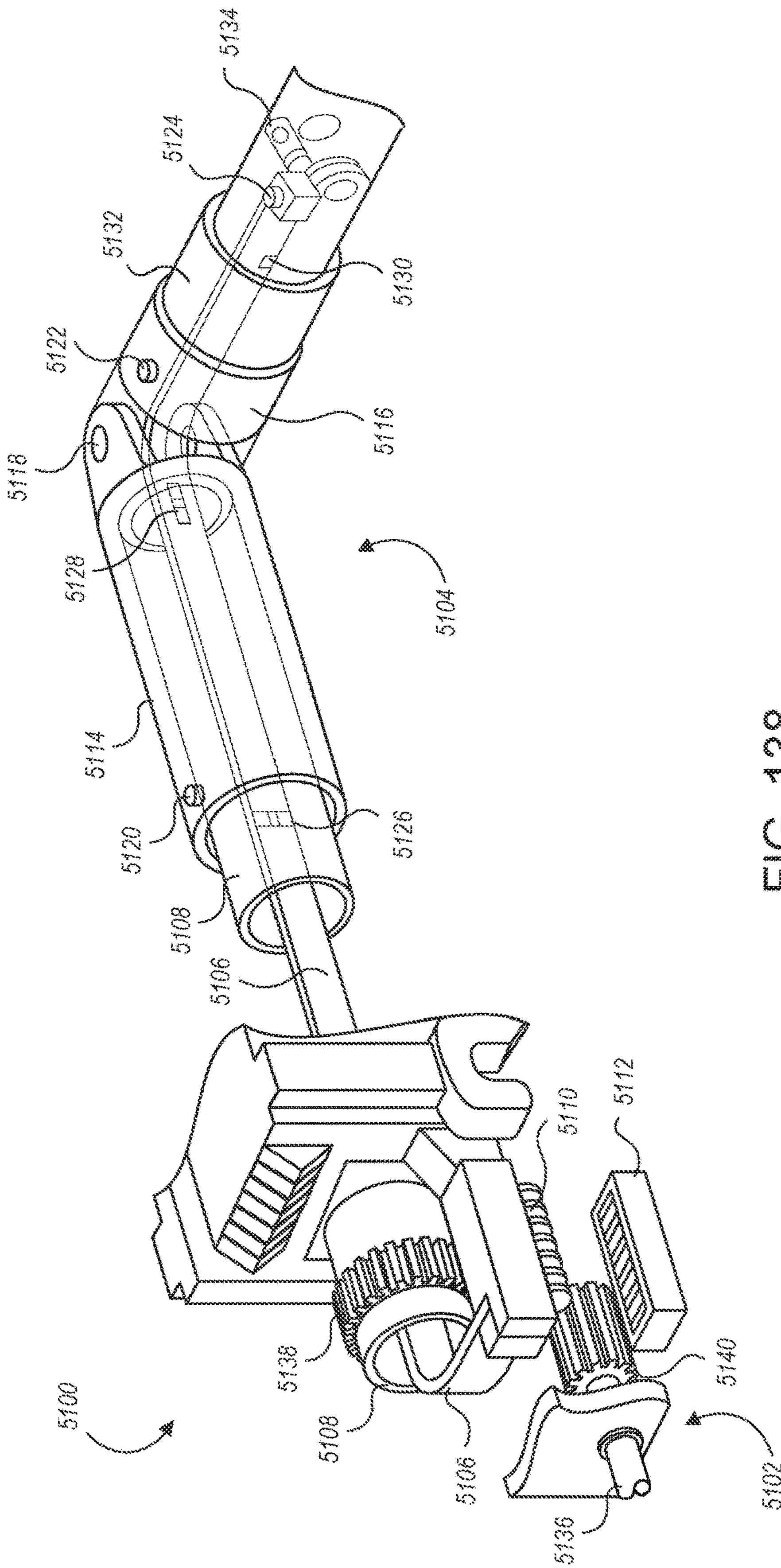

5134
5124
5132
5130
5122
5116
5118
5128
5104
5114
5120
5126
5108
5106
5110
5112
5138
5140
5100
5108
5106
5102
5136

FIG. 138

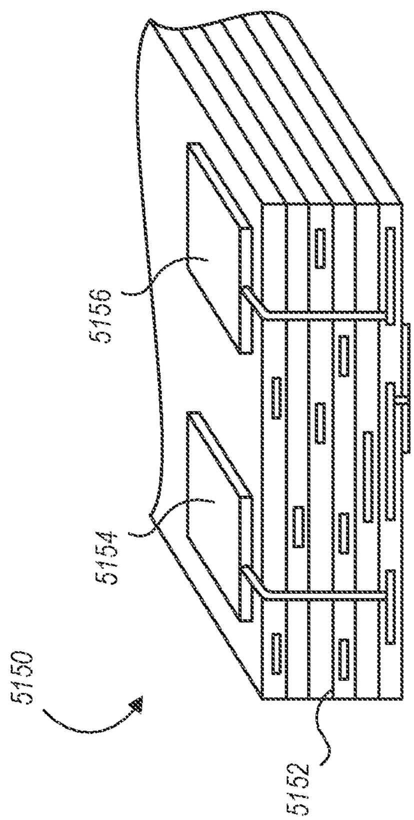


FIG. 139

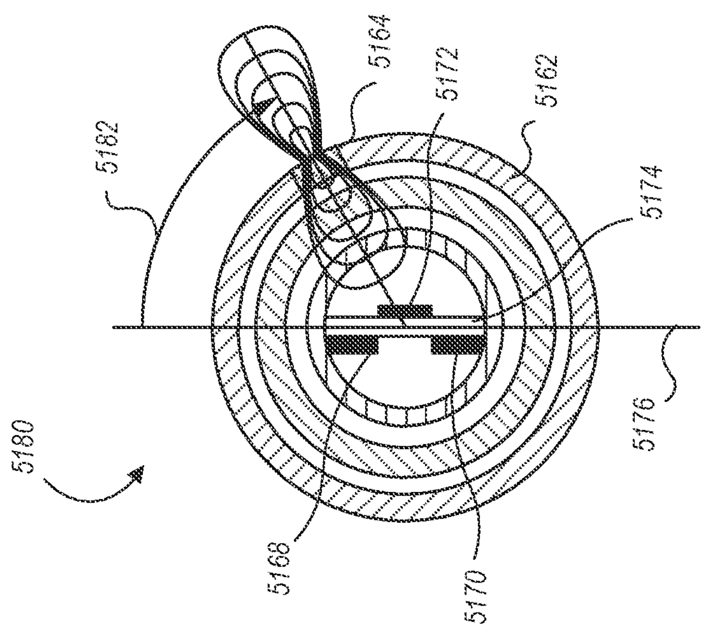


FIG. 140B

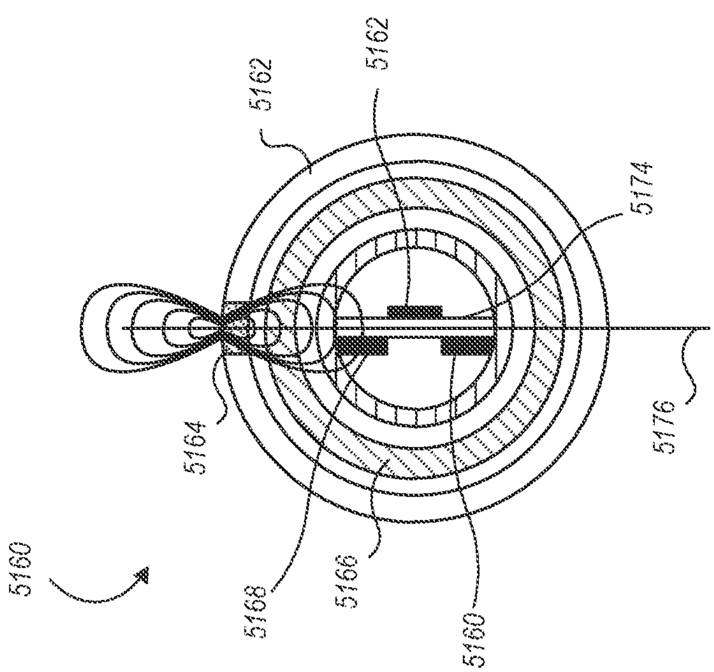


FIG. 140A

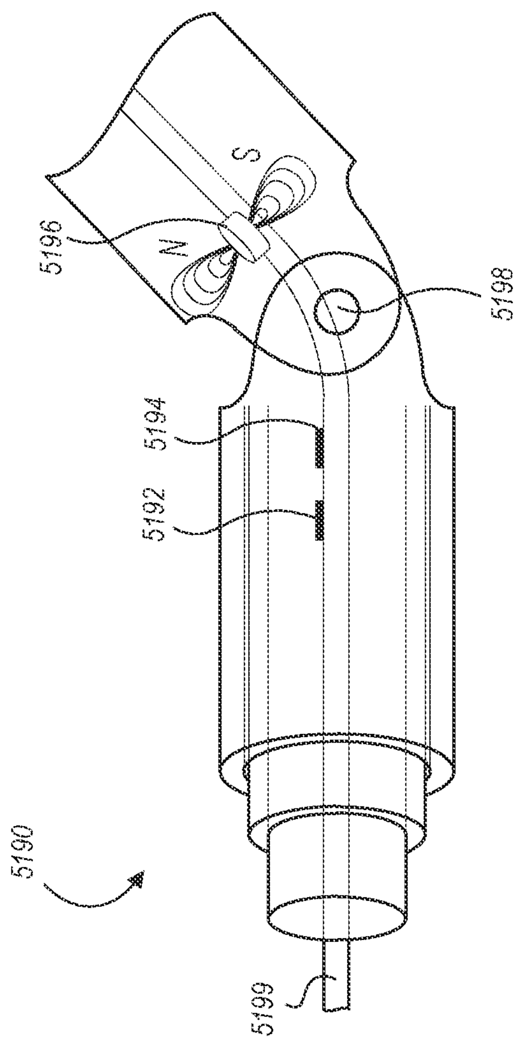


FIG. 141

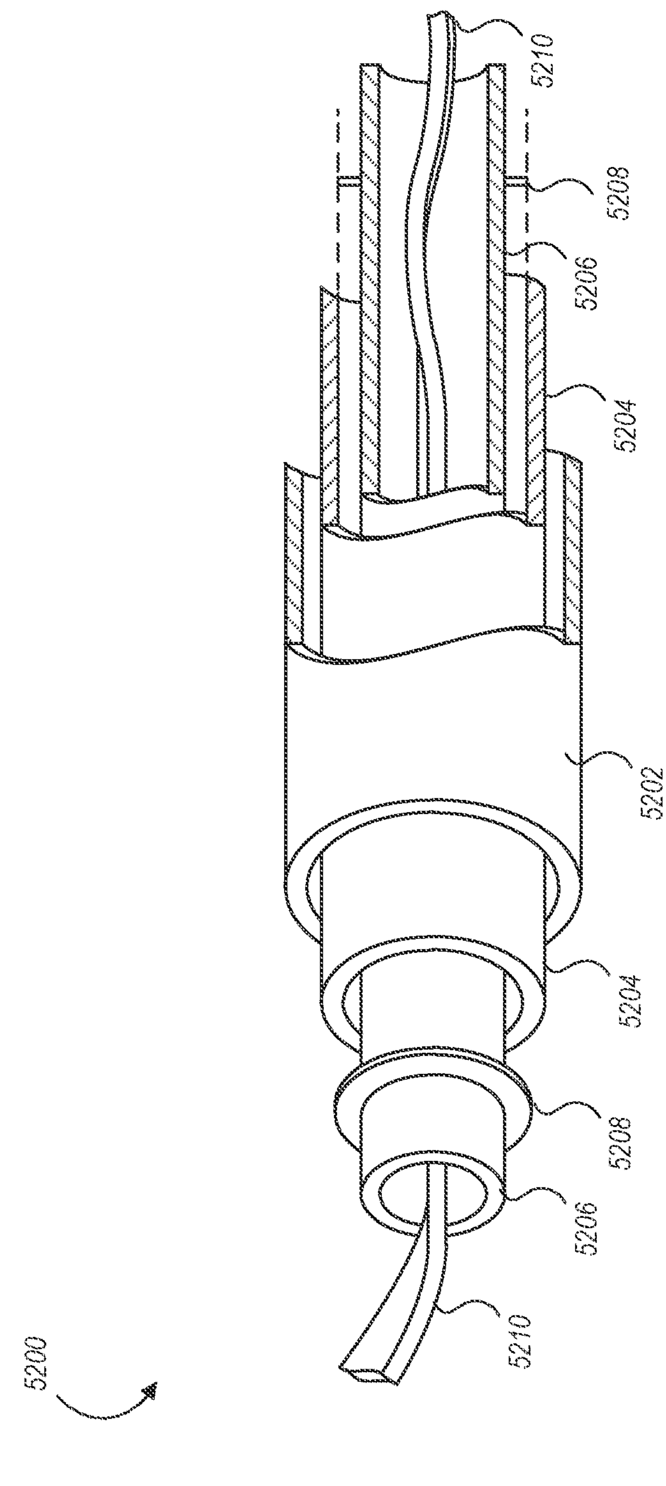


FIG. 142

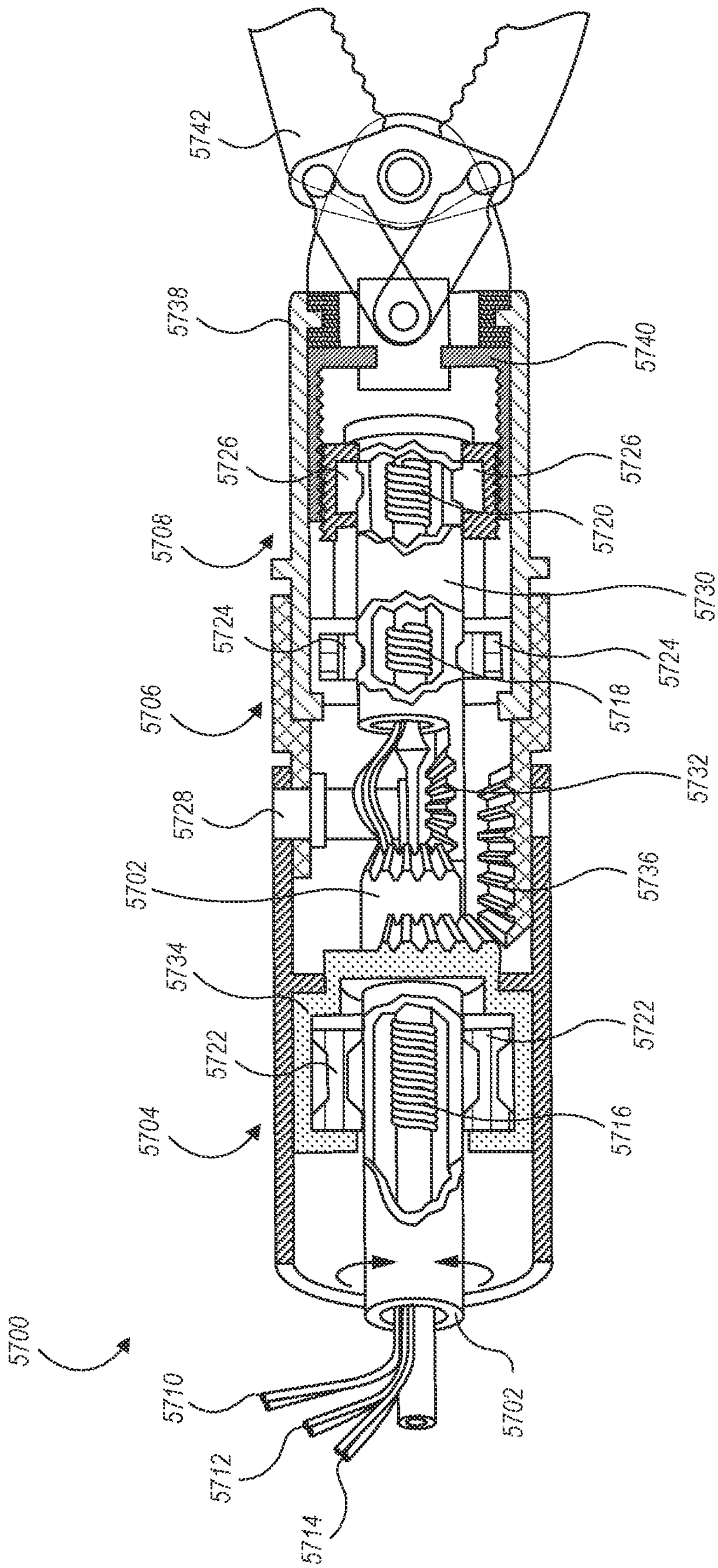

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FIG. 143

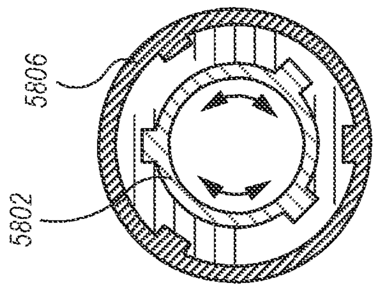

FIG. 144C

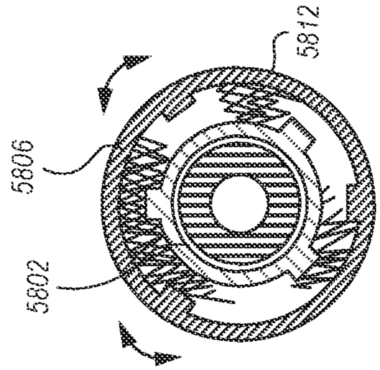

FIG. 144D

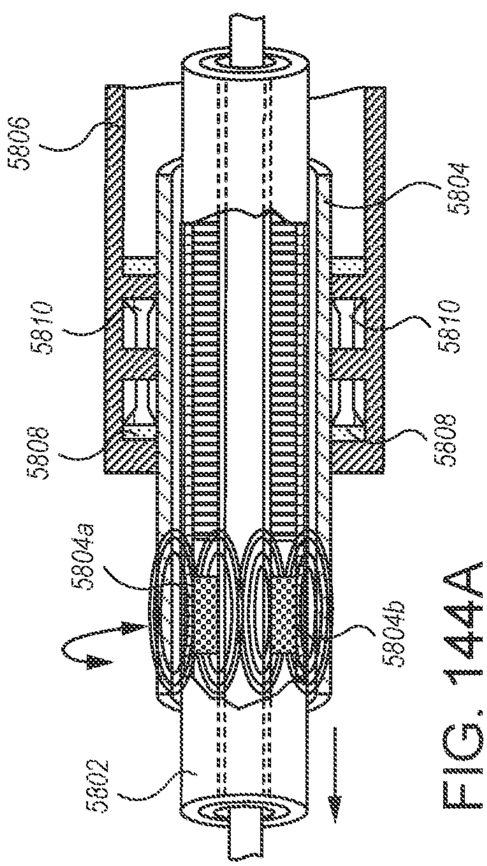

FIG. 144A

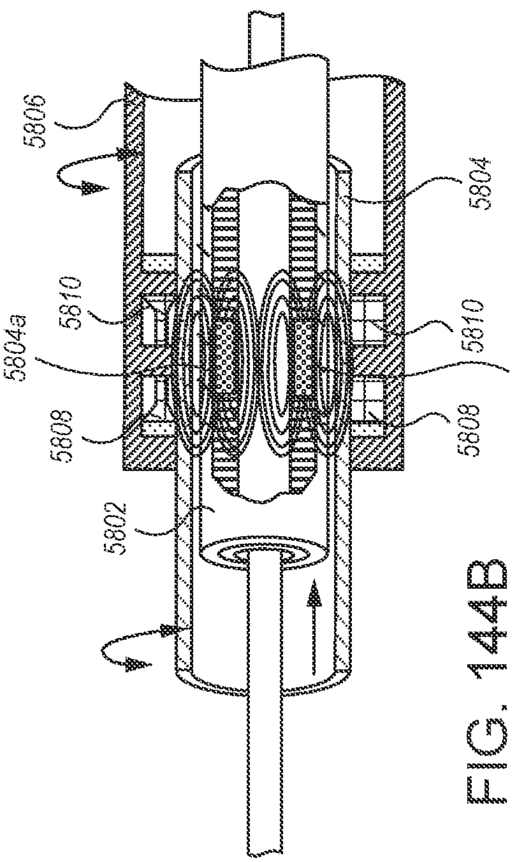

FIG. 144B

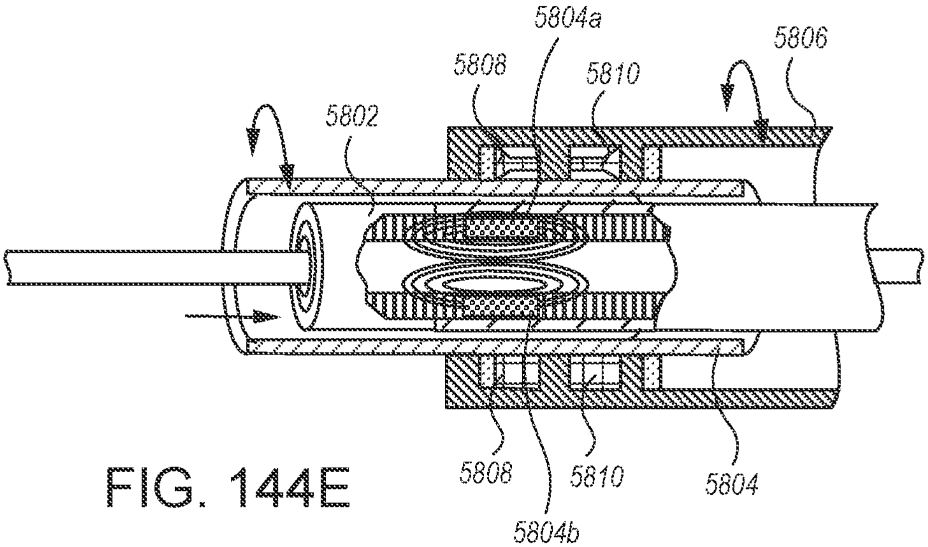

FIG. 144E

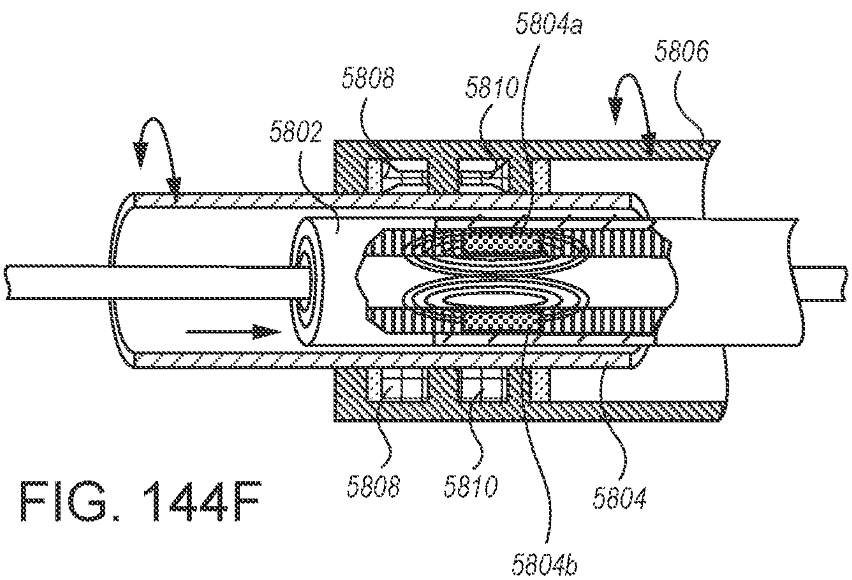

FIG. 144F

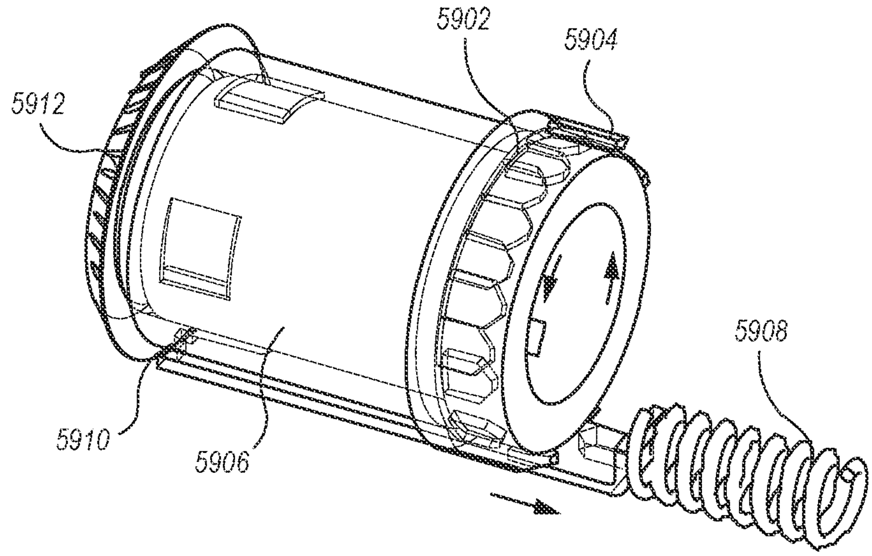

FIG. 145A

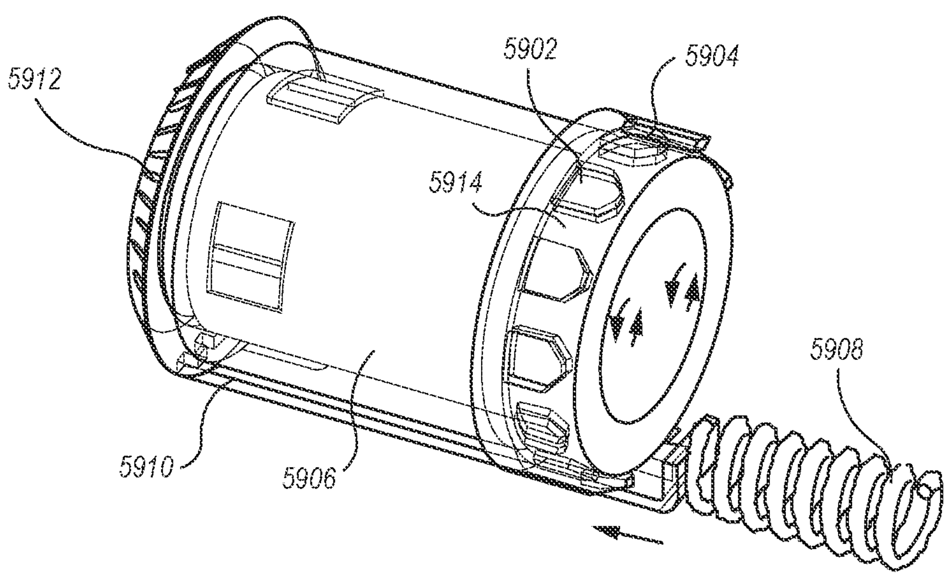

FIG. 145B

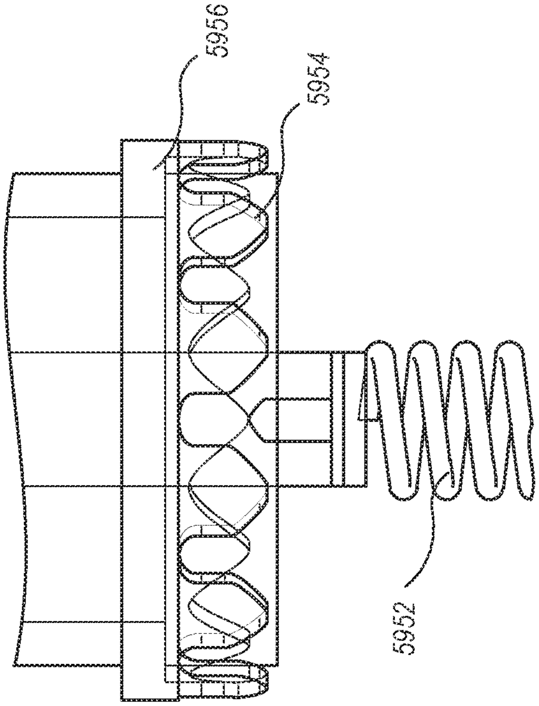

FIG. 146B

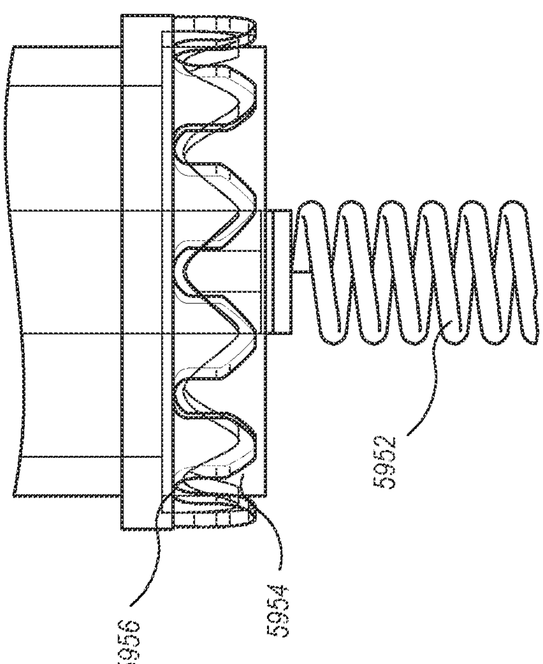

FIG. 146A

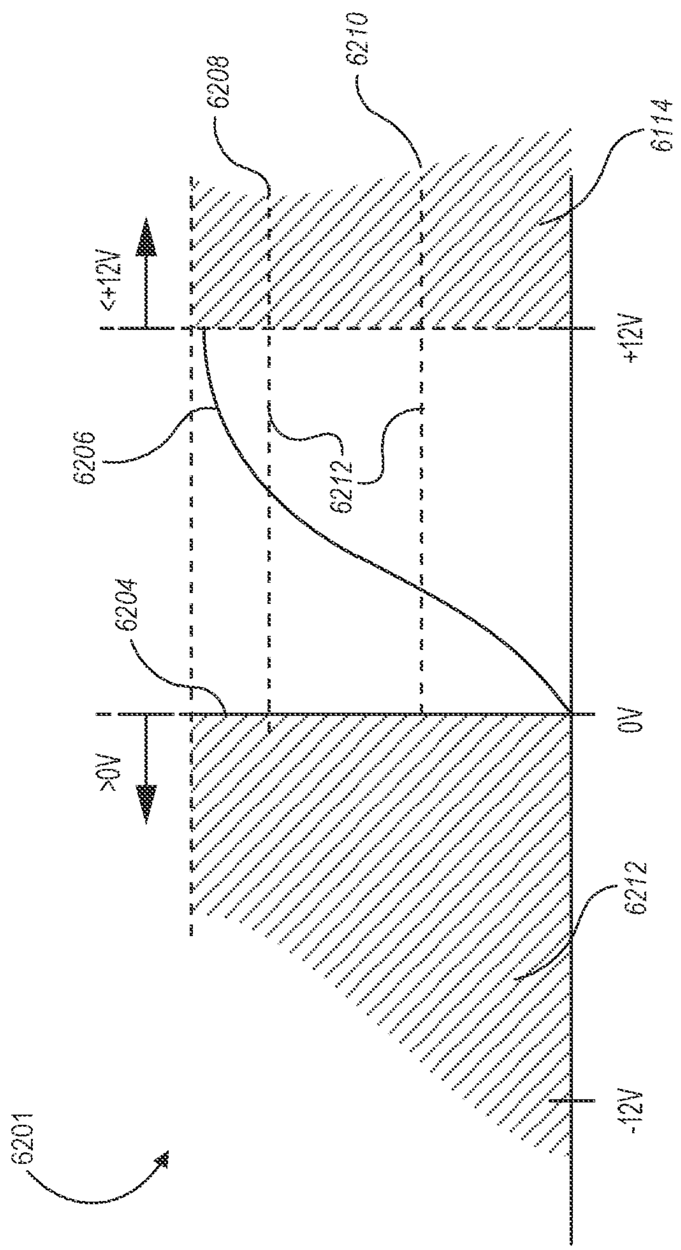

FIG. 147

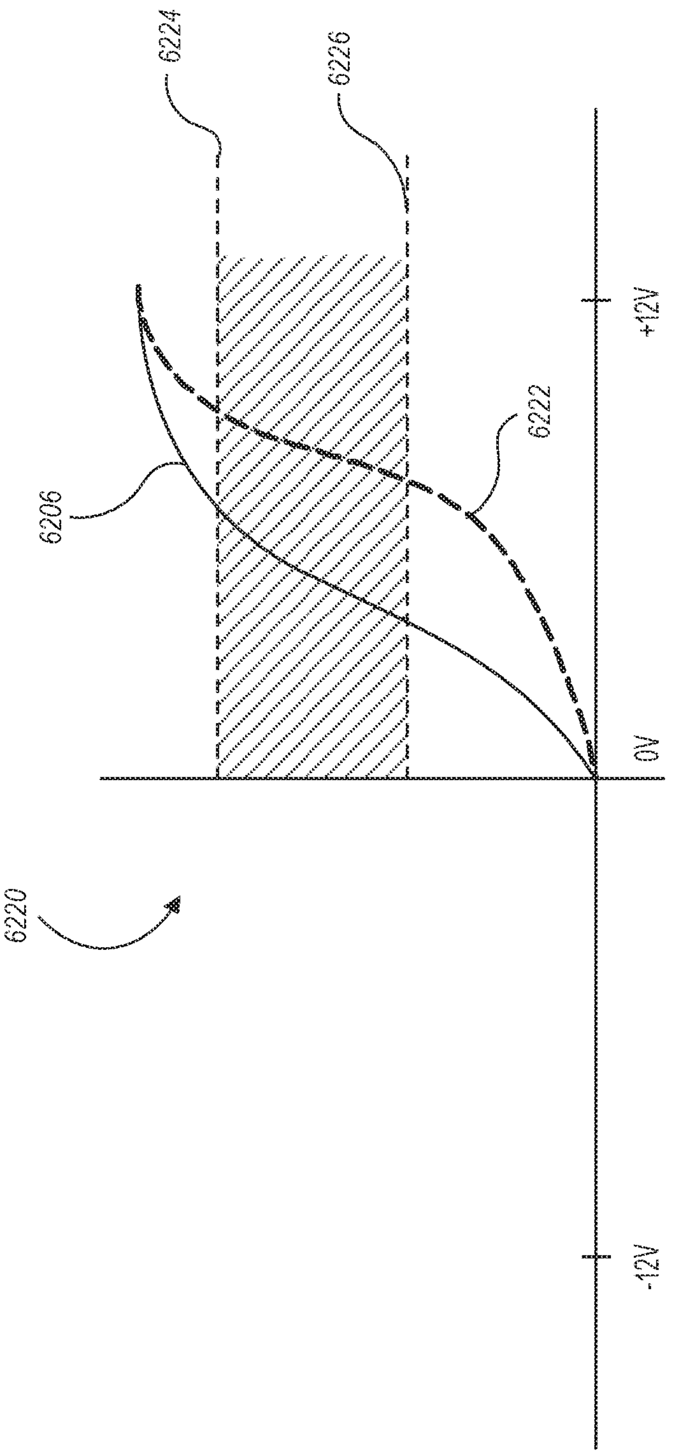

FIG. 148

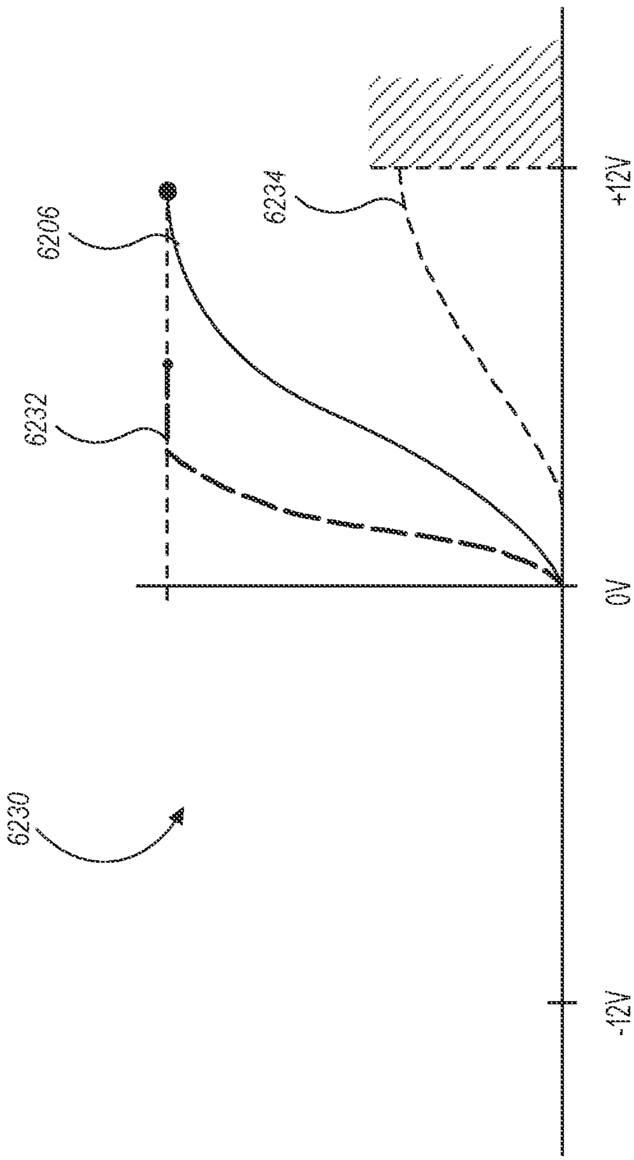

FIG. 149

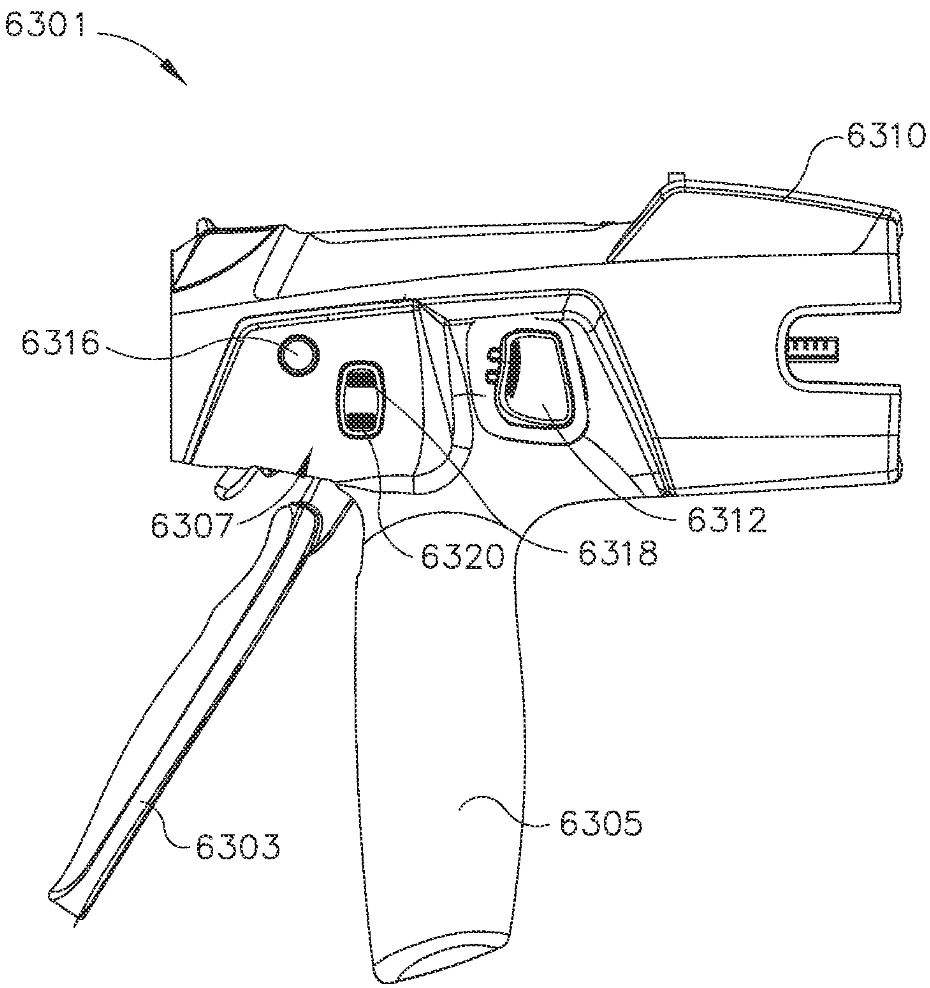

FIG. 150

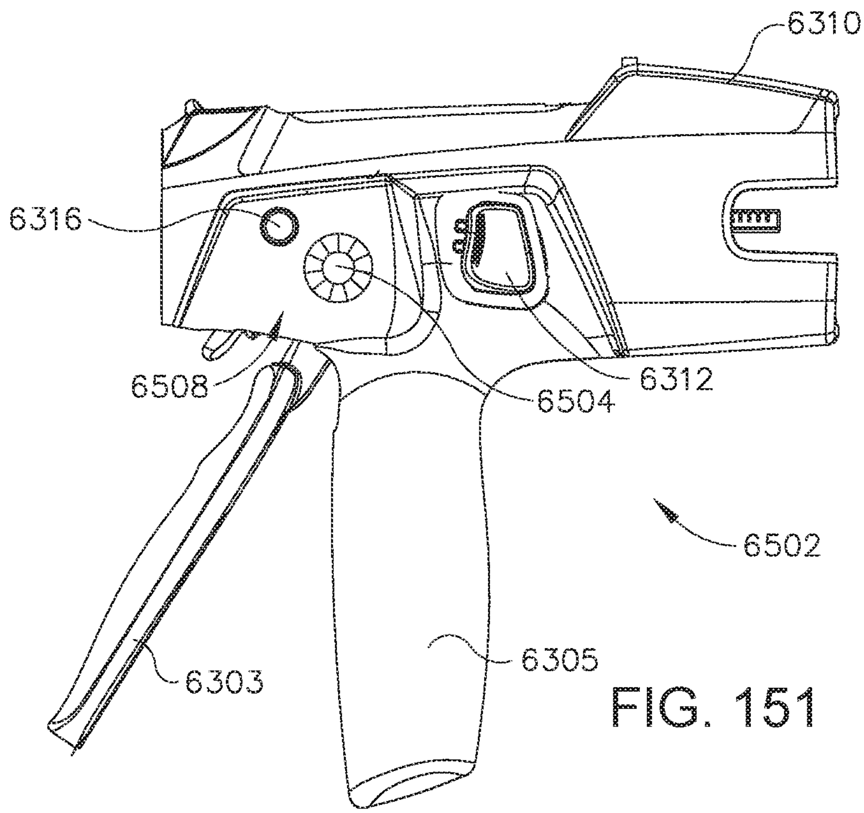

FIG. 151

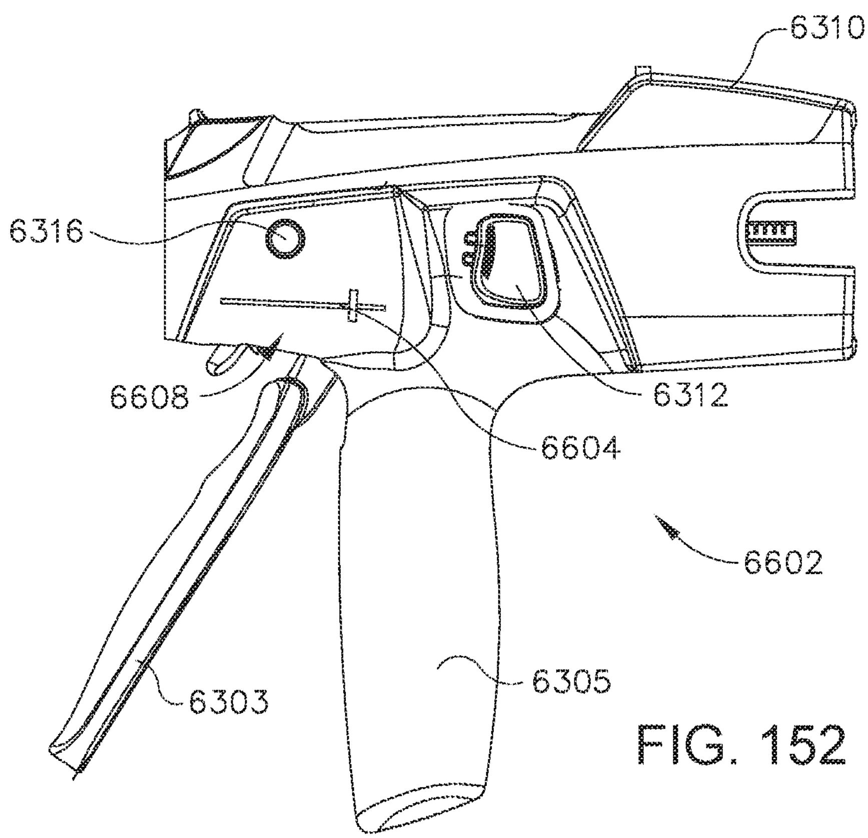

FIG. 152

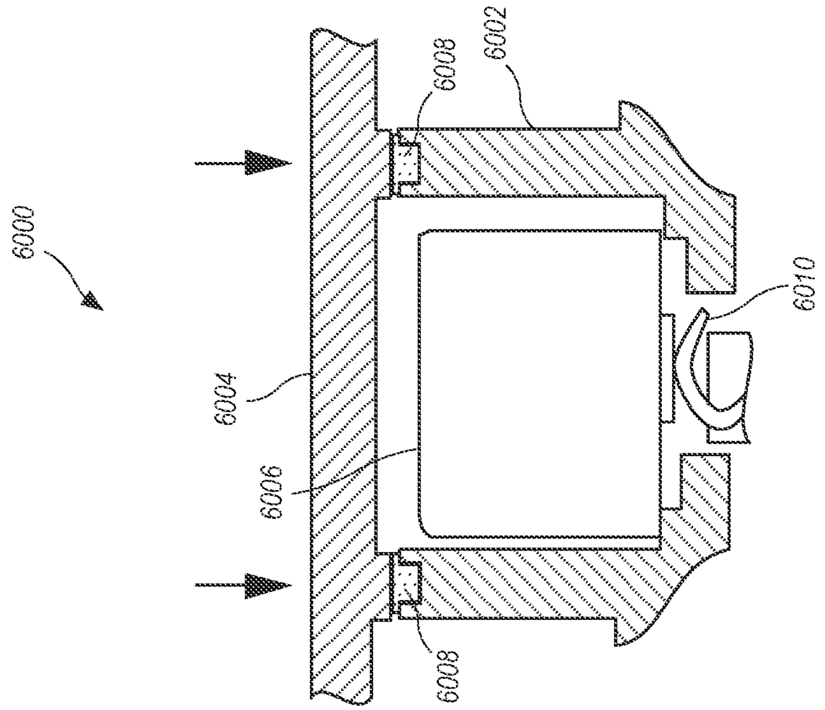

FIG. 153B

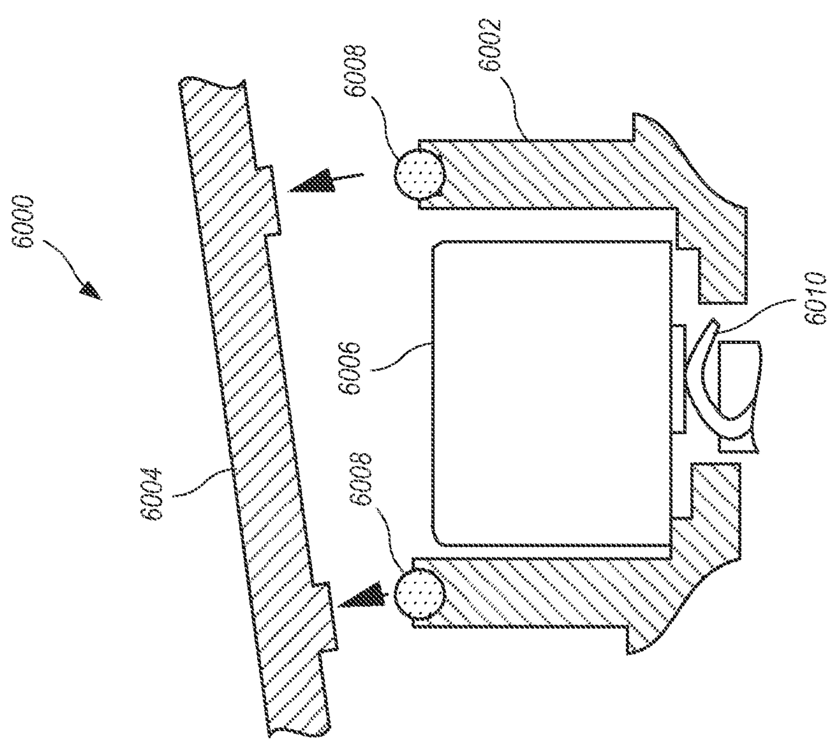

FIG. 153A

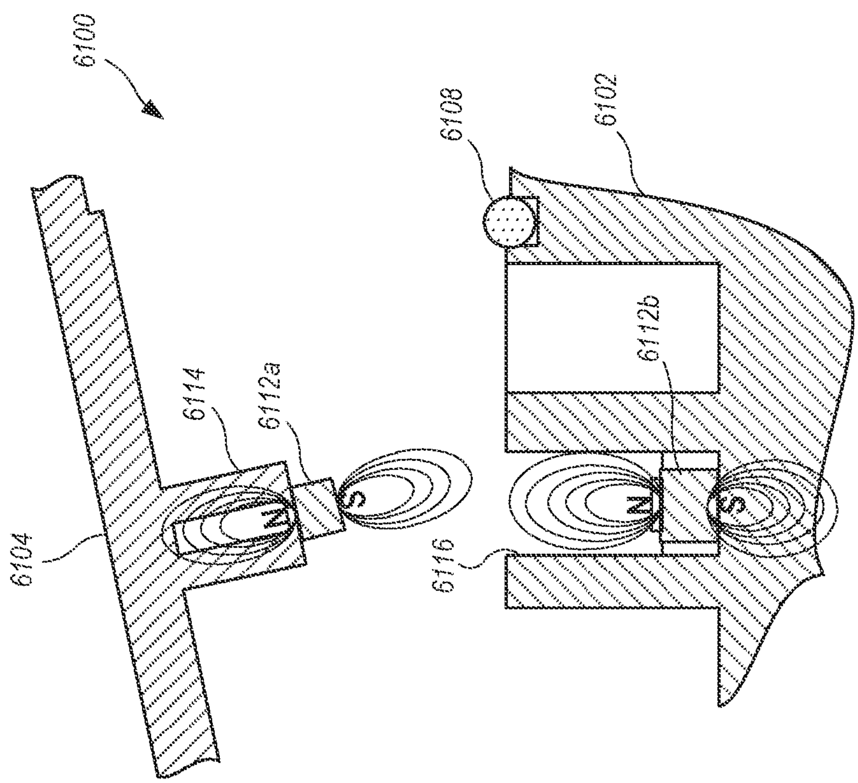

FIG. 154A

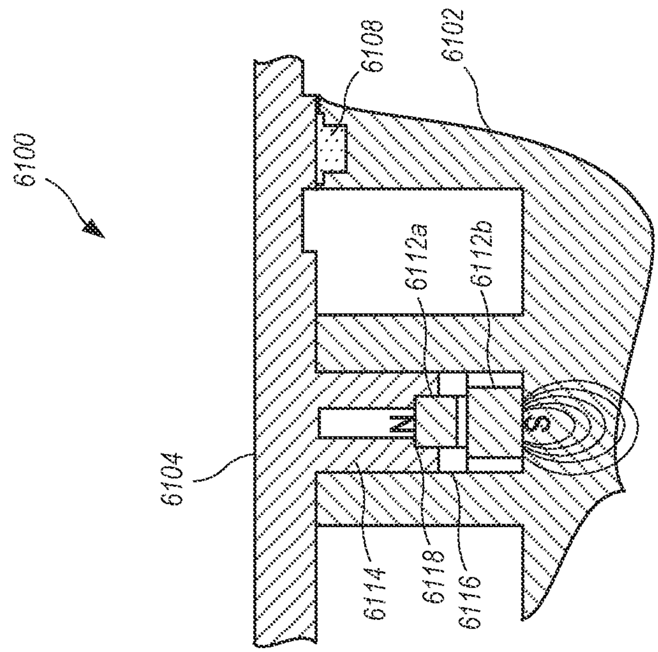

FIG. 154B

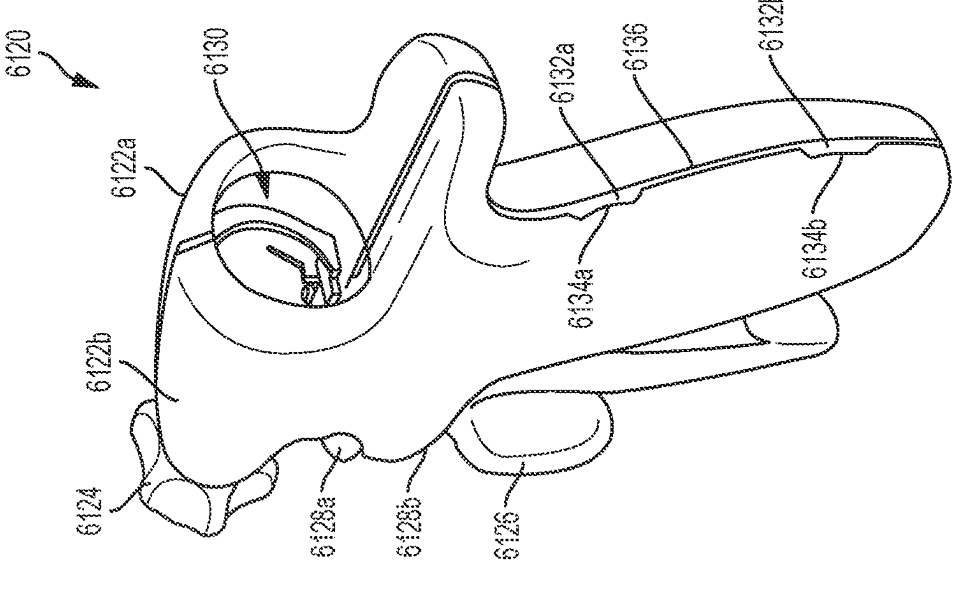

FIG. 155A

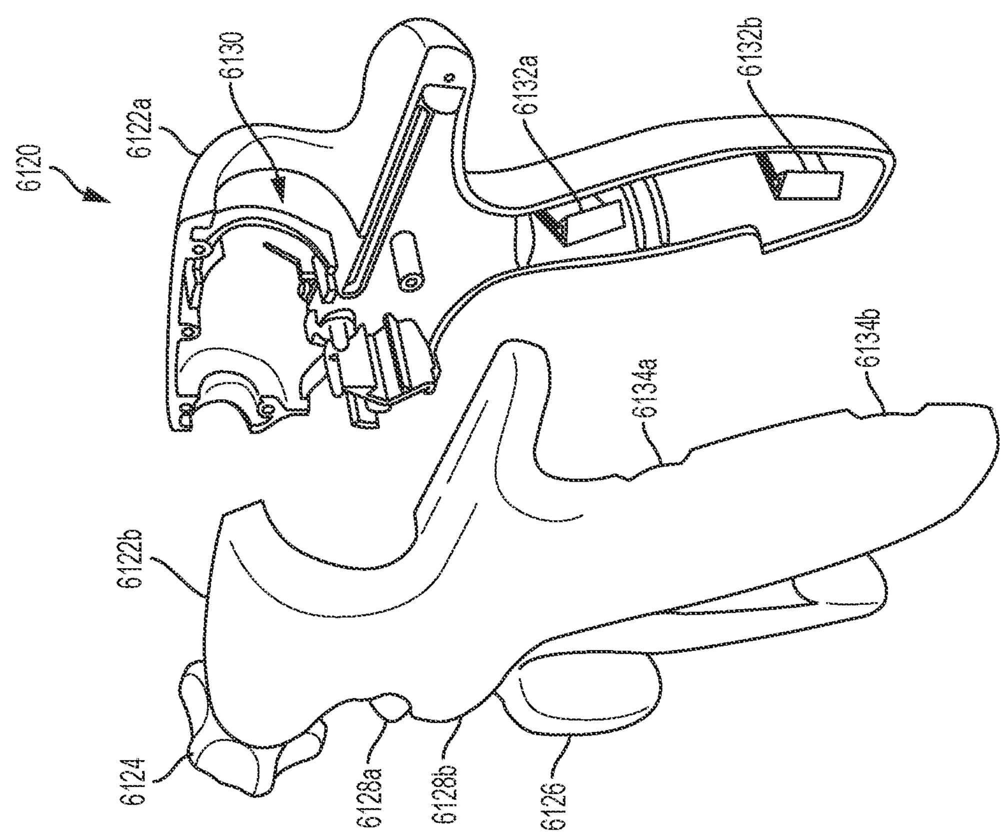

FIG. 155B

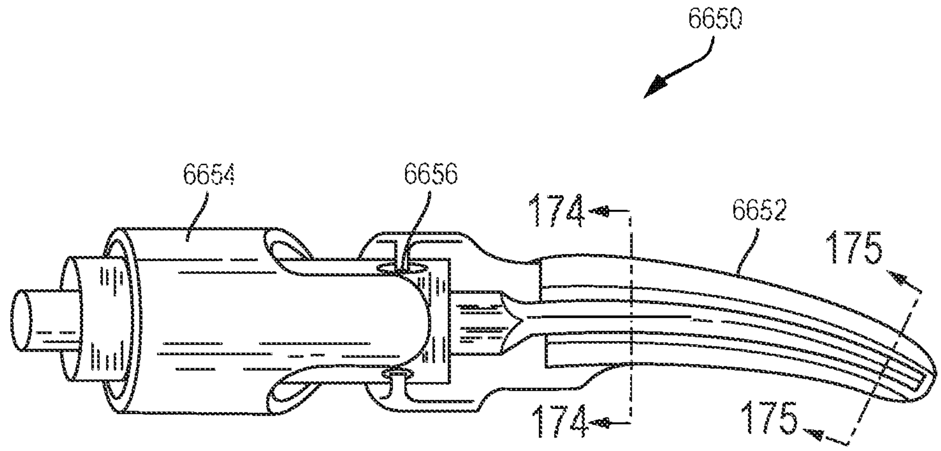

FIG. 156

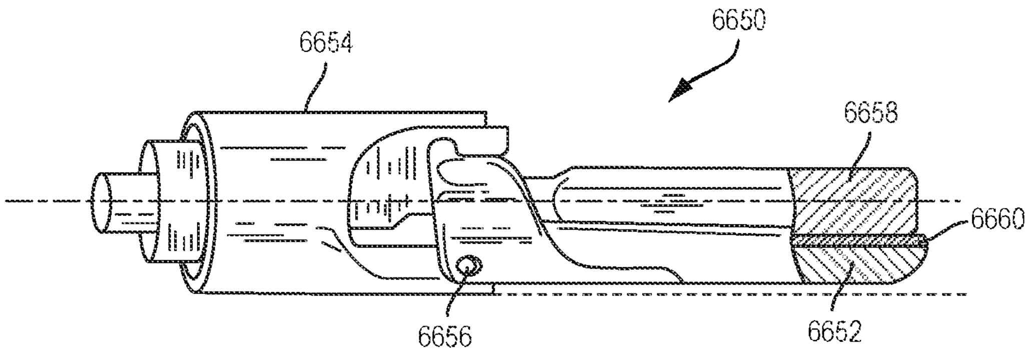

FIG. 157

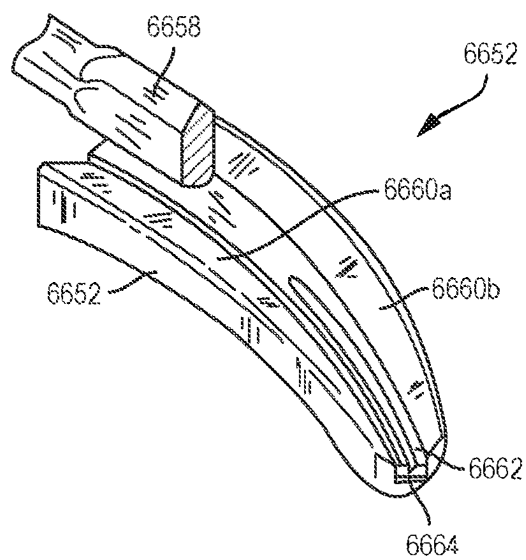

FIG. 158

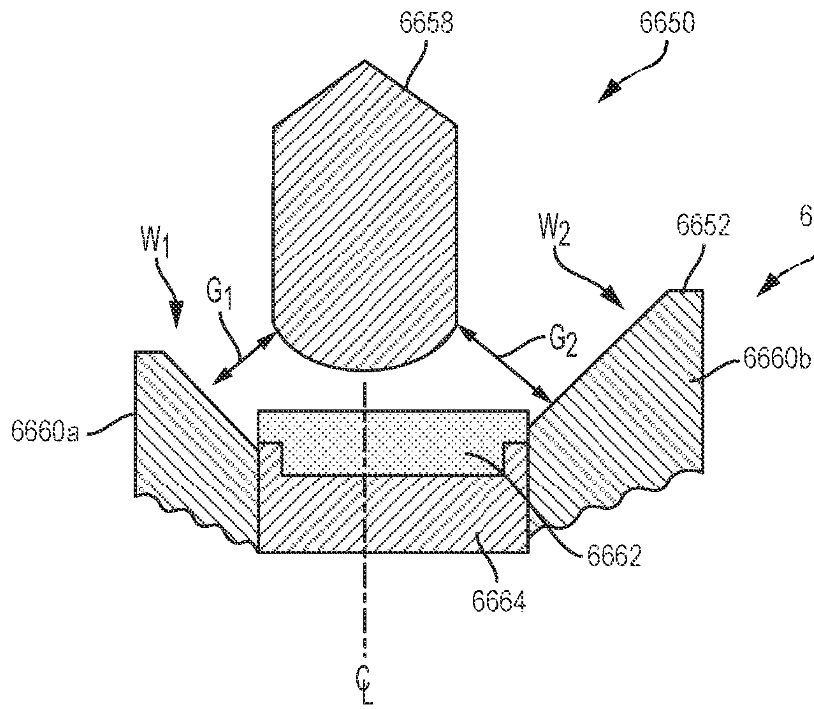

FIG. 159

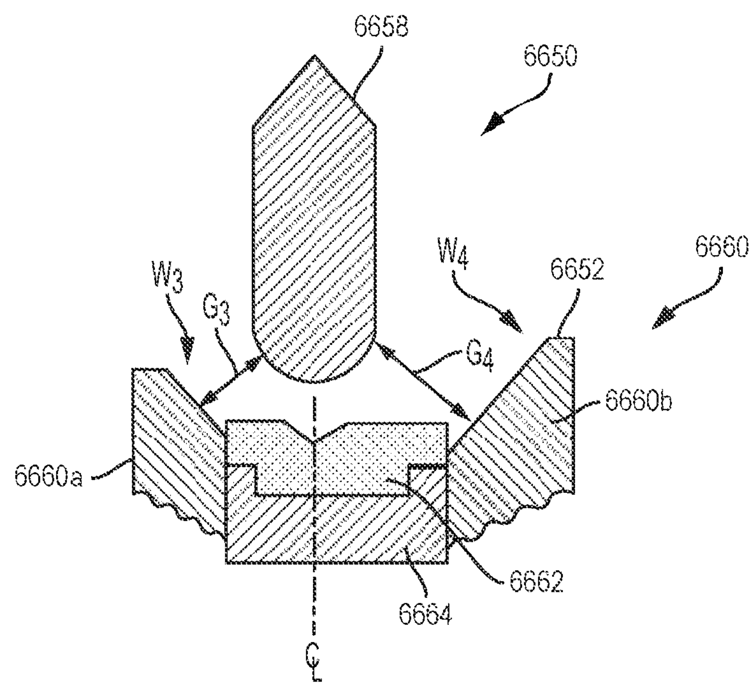

FIG. 160

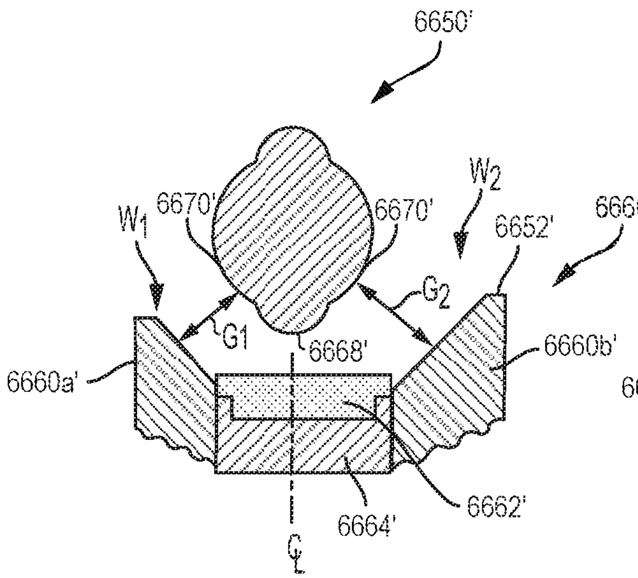

FIG. 161

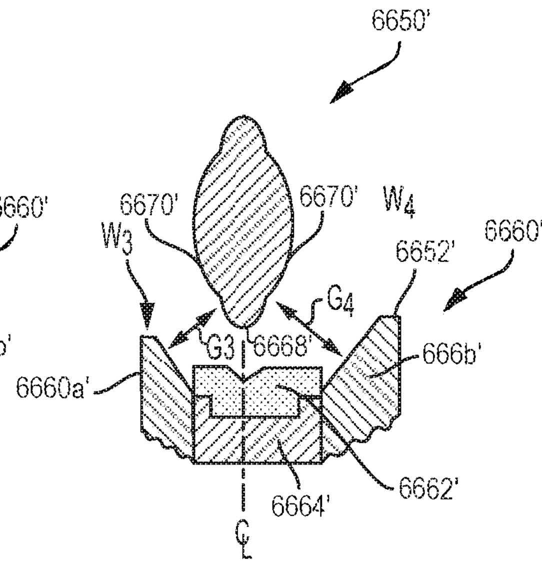

FIG. 162

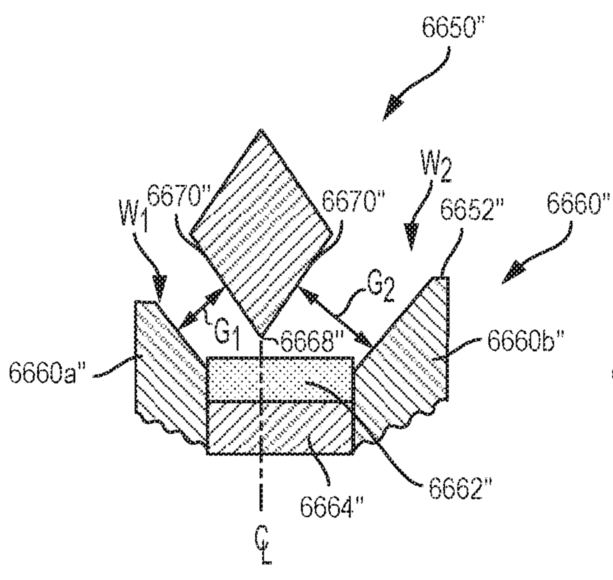

FIG. 163

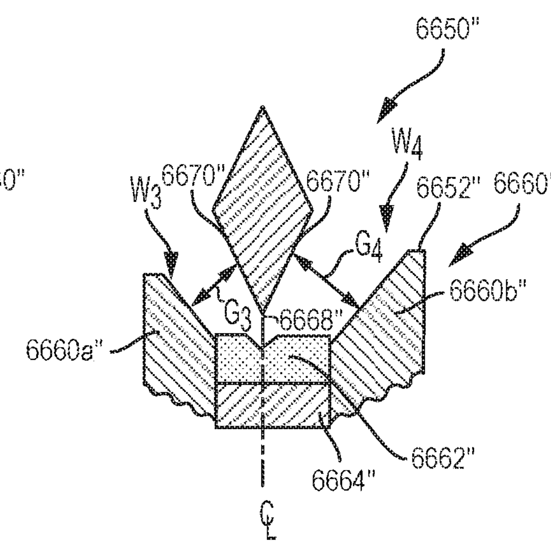

FIG. 164

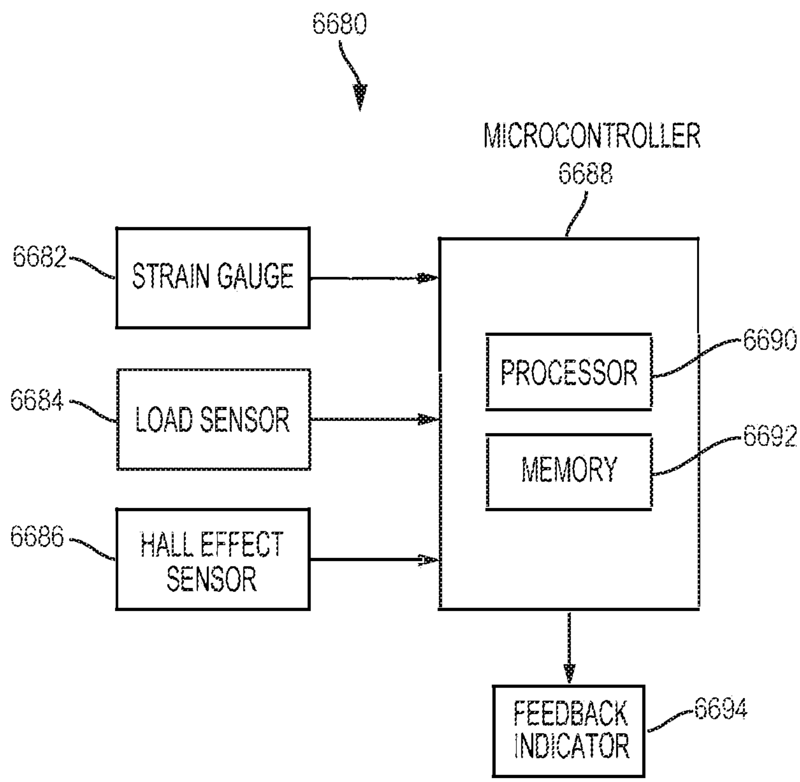
6680
MICROCONTROLLER
6688
6682
STRAIN GAUGE
6684
LOAD SENSOR
6686
HALL EFFECT SENSOR
PROCESSOR
6690
MEMORY
6692
FEEDBACK INDICATOR
6694

FIG. 165

MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT AND METHODS THEREFOR

PRIORITY

This application is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 15/382,515, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT AND METHODS THEREFOR, filed on Dec. 16, 2016, published on Jul. 20, 2017 as U.S. Patent Application Publication No. 2017/0202605, and issued on Nov. 24, 2020 as U.S. Pat. No. 10,842,523, which claims the benefit of U.S. Provisional Patent Application No. 62/279,635, titled MODULAR COMBINATION SURGICAL DEVICE, filed Jan. 15, 2016 and U.S. Provisional Patent Application No. 62/330,669, titled MODULAR COMBINATION SURGICAL DEVICE, filed May 2, 2016, the entire disclosures of which are hereby incorporated by reference herein.

BACKGROUND

The present disclosure is related generally to surgical instruments and associated surgical techniques. More particularly, the present disclosure is related to ultrasonic and electrosurgical systems that allow surgeons to perform cutting and coagulation and to adapt and customize such procedures based on the type of tissue being treated.

Ultrasonic surgical instruments are finding increasingly widespread applications in surgical procedures by virtue of the unique performance characteristics of such instruments. Depending upon specific instrument configurations and operational parameters, ultrasonic surgical instruments can provide simultaneous or near-simultaneous cutting of tissue and hemostasis by coagulation, desirably minimizing patient trauma. The cutting action is typically realized by an-end effector, or blade tip, at the distal end of the instrument, which transmits ultrasonic energy to tissue brought into contact with the end effector. Ultrasonic instruments of this nature can be configured for open surgical use, laparoscopic, or endoscopic surgical procedures including robotic-assisted procedures.

Some surgical instruments utilize ultrasonic energy for both precise cutting and controlled coagulation. Ultrasonic energy cuts and coagulates by vibrating a blade in contact with tissue. Vibrating at high frequencies (e.g., 55,500 times per second), the ultrasonic blade denatures protein in the tissue to form a sticky coagulum. Pressure exerted on tissue with the blade surface collapses blood vessels and allows the coagulum to form a hemostatic seal. The precision of cutting and coagulation is controlled by the surgeon's technique and adjusting the power level, blade edge, tissue traction, and blade pressure.

Electrosurgical instruments for applying electrical energy to tissue in order to treat and/or destroy the tissue are also finding increasingly widespread applications in surgical procedures. An electrosurgical instrument typically includes a hand piece, an instrument having a distally-mounted end effector (e.g., one or more electrodes). The end effector can be positioned against the tissue such that electrical current is introduced into the tissue. Electrosurgical instruments can be configured for bipolar or monopolar operation. During bipolar operation, current is introduced into and returned from the tissue by active and return electrodes, respectively, of the end effector. During monopolar operation, current is introduced into the tissue by an active electrode of the end effector and returned through a return electrode (e.g., a grounding pad) separately located on a patient's body. Heat generated by the current flowing through the tissue may form hemostatic seals within the tissue and/or between tissues and thus may be particularly useful for sealing blood vessels, for example. The end effector of an electrosurgical instrument also may include a cutting member that is movable relative to the tissue and the electrodes to transect the tissue.

Electrical energy applied by an electrosurgical instrument can be transmitted to the instrument by a generator in communication with the hand piece. The electrical energy may be in the form of radio frequency ("RF") energy. RF energy is a form of electrical energy that may be in the frequency range of 200 kilohertz (kHz) to 1 megahertz (MHz). In application, an electrosurgical instrument can transmit low frequency RF energy through tissue, which causes ionic agitation, or friction, in effect resistive heating, thereby increasing the temperature of the tissue. Because a sharp boundary is created between the affected tissue and the surrounding tissue, surgeons can operate with a high level of precision and control, without sacrificing un-targeted adjacent tissue. The low operating temperatures of RF energy is useful for removing, shrinking, or sculpting soft tissue while simultaneously sealing blood vessels. RF energy works particularly well on connective tissue, which is primarily comprised of collagen and shrinks when contacted by heat.

The RF energy may be in a frequency range described in EN 60601-2-2:2009+A11:2011, Definition 201.3.218—HIGH FREQUENCY. For example, the frequency in monopolar RF applications may be typically restricted to less than 5 MHz. However, in bipolar RF applications, the frequency can be almost anything. Frequencies above 200 kHz can be typically used for monopolar applications in order to avoid the unwanted stimulation of nerves and muscles that would result from the use of low frequency current. Lower frequencies may be used for bipolar applications if the risk analysis shows the possibility of neuromuscular stimulation has been mitigated to an acceptable level. Normally, frequencies above 5 MHz are not used in order to minimize the problems associated with high frequency leakage currents. Higher frequencies may, however, be used in the case of bipolar applications. It is generally recognized that 10 mA is the lower threshold of thermal effects on tissue.

A challenge of using these medical devices is the inability to fully control and customize the functions of the surgical instruments. It would be desirable to provide a surgical instrument that overcomes some of the deficiencies of current instruments.

SUMMARY

In one aspect, the present disclosure provides a method of controlling a modular battery powered handheld surgical instrument. The surgical instrument comprising a battery, as user input sensor, a controller, a radio frequency (RF) drive circuit, an ultrasonic transducer, ultrasonic transducer drive circuit, and an end effector. The end effector comprising an electrode electrically coupled to RF drive circuit, an ultrasonic blade acoustically coupled to the ultrasonic transducer, and a sensor to measure tissue parameters. The method comprising applying an RF current drive signal to the electrode by the RF drive circuit; applying an ultrasonic drive signal to the ultrasonic transducer by the ultrasonic transducer drive circuit to acoustically excite the ultrasonic blade; controlling intensity, wave shape, and/or frequency of the RF current drive signal based on a sensed measure of a tissue or user parameter.

In another aspect, the present disclosure provides a method of controlling a modular battery powered handheld surgical instrument. The surgical instrument comprising a battery, a user input sensor, a controller, a radio frequency (RF) drive circuit, an ultrasonic transducer, ultrasonic transducer drive circuit, and an end effector. The end effector comprising an electrode electrically coupled to RF drive circuit, an ultrasonic blade acoustically coupled to the ultrasonic transducer, and a sensor to measure tissue parameters. The method comprising applying an RF current drive signal to the electrode by the RF drive circuit; applying an ultrasonic drive signal to the ultrasonic transducer by the ultrasonic transducer drive circuit to acoustically excite the ultrasonic blade; controlling intensity, wave shape, and/or frequency of the ultrasonic drive signal based on a tissue or user parameter.

In another aspect, the present disclosure provides a method of controlling a modular battery powered handheld surgical instrument. The surgical instrument comprising a battery, a user input sensor, a controller, a radio frequency (RF) drive circuit, an ultrasonic transducer, ultrasonic transducer drive circuit, and an end effector. The end effector comprising an electrode electrically coupled to RF drive circuit, an ultrasonic blade acoustically coupled to the ultrasonic transducer, and a sensor to measure tissue parameters. The method comprising applying an RF current drive signal to the electrode by the RF drive circuit; applying an ultrasonic drive signal to the ultrasonic transducer by the ultrasonic transducer drive circuit to acoustically excite the ultrasonic blade; controlling intensity, wave shape, and/or frequency of the RF current drive signal and the ultrasonic drive signal on a sensed measure of a tissue or user parameter.

In addition to the foregoing, various other method and/or system and/or program product aspects are set forth and described in the teachings such as text (e.g., claims and/or detailed description) and/or drawings of the present disclosure.

The foregoing is a summary and thus may contain simplifications, generalizations, inclusions, and/or omissions of detail; consequently, those skilled in the art will appreciate that the summary is illustrative only and is NOT intended to be in any way limiting. Other aspects, features, and advantages of the devices and/or processes and/or other subject matter described herein will become apparent in the teachings set forth herein.

In one or more various aspects, related systems include but are not limited to circuitry and/or programming for effecting herein-referenced method aspects; the circuitry and/or programming can be virtually any combination of hardware, software, and/or firmware configured to affect the herein-referenced method aspects depending upon the design choices of the system designer. In addition to the foregoing, various other method and/or system aspects are set forth and described in the teachings such as text (e.g., claims and/or detailed description) and/or drawings of the present disclosure.

Further, it is understood that any one or more of the following-described forms, expressions of forms, examples, can be combined with any one or more of the other following-described forms, expressions of forms, and examples.

The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, and features described above, further aspects, and features will become apparent by reference to the drawings and the following detailed description.

FIGURES

The novel features of the various aspects described herein are set forth with particularity in the appended claims. The various aspects, however, both as to organization and methods of operation may be better understood by reference to the following description, taken in conjunction with the accompanying drawings as follows:

FIG. **1** is a diagram of a modular battery powered handheld ultrasonic surgical instrument, according to an aspect of the present disclosure.

FIG. **2** is an exploded view of the surgical instrument shown in FIG. **1**, according to an aspect of the present disclosure.

FIG. **3** is an exploded view of a modular shaft assembly of the surgical instrument shown in FIG. **1**, according to aspect of the present disclosure.

FIG. **4** is a perspective transparent view of the ultrasonic transducer/generator assembly of the surgical instrument shown in FIG. **1**, according to aspect of the present disclosure.

FIG. **5** is an end view of the ultrasonic transducer/generator assembly, according to aspect of the present disclosure.

FIG. **6** is a perspective view of the ultrasonic transducer/generator assembly with the top housing portion removed to expose the ultrasonic generator, according to aspect of the present disclosure.

FIG. **7** is a sectional view of the of the ultrasonic transducer/generator assembly, according to aspect of the present disclosure.

FIG. **8** is an elevation view of an ultrasonic transducer/generator assembly that is configured to operate at 31 kHz resonant frequency, according to one aspect of the present disclosure.

FIG. **9** is an elevation view of an ultrasonic transducer/generator assembly that is configured to operate at 55 kHz resonant frequency, according to one aspect of the present disclosure.

FIGS. **10**A and **10**B illustrate a shifting assembly that selectively rotates the ultrasonic transmission waveguide with respect to the ultrasonic transducer and urges them towards one another, according to one aspect of the present disclosure.

FIG. **11** is a schematic diagram of one aspect of an ultrasonic drive circuit shown in FIG. **4** suitable for driving an ultrasonic transducer, according to one aspect of the present disclosure.

FIG. **12** is a schematic diagram of the transformer coupled to the ultrasonic drive circuit shown in FIG. **11**, according to one aspect of the present disclosure.

FIG. **13** is a schematic diagram of the transformer shown in FIG. **12** coupled to a test circuit, according to one aspect of the present disclosure.

FIG. **14** is a schematic diagram of a control circuit, according to one aspect f the present disclosure.

FIG. **15** shows a simplified block circuit diagram illustrating another electrical circuit contained within a modular ultrasonic surgical instrument, according to one aspect of the present disclosure.

FIG. **16** shows a battery assembly for use with the surgical instrument, according to one aspect of the present disclosure.

FIG. **17** shows a disposable battery assembly for use with the surgical instrument, according to one aspect of the present disclosure.

FIG. **18** shows a reusable battery assembly for use with the surgical instrument, according to one aspect of the present disclosure.

FIG. **19** is an elevated perspective view of a battery assembly with both halves of the housing shell removed exposing battery cells coupled to multiple circuit boards which are coupled to the multi-lead battery terminal in accordance with one aspect of the present disclosure.

FIG. **20** illustrates a battery test circuit, according to one aspect of the present disclosure.

FIG. **21** illustrates a supplemental power source circuit to maintain a minimum output voltage, according to one aspect of the present disclosure.

FIG. **22** illustrates a switch mode power supply circuit for supplying energy to the surgical instrument, according to one aspect of the present disclosure.

FIG. **23** illustrates a discrete version of the switching regulator shown in FIG. **22** for supplying energy to the surgical instrument, according to one aspect of the present disclosure.

FIG. **24** illustrates a linear power supply circuit for supplying energy to the surgical instrument, according to one aspect of the present disclosure.

FIG. **25** is an elevational exploded view of modular handheld ultrasonic surgical instrument showing the left shell half removed from a handle assembly exposing a device identifier communicatively coupled to the multi-lead handle terminal assembly in accordance with one aspect of the present disclosure.

FIG. **26** is a detail view of a trigger portion and switch of the ultrasonic surgical instrument shown in FIG. **25**, according to one aspect of the present disclosure.

FIG. **27** is a fragmentary, enlarged perspective view of an end effector from a distal end with a jaw member in an open position, according to one aspect of the present disclosure.

FIG. **28** illustrates a modular shaft assembly and end effector portions of the surgical instrument, according to one aspect of the present disclosure.

FIG. **29** is a detail view of an inner tube/spring assembly, according to one aspect of present disclosure.

FIG. **30** illustrates a modular battery powered handheld combination ultrasonic/electrosurgical instrument, according to one aspect of the present disclosure.

FIG. **31** is an exploded view of the surgical instrument shown in FIG. **30**, according to one aspect of the present disclosure.

FIG. **32** is a partial perspective view of a modular battery powered handheld combination ultrasonic/RF surgical instrument, according to one aspect of the present disclosure.

FIG. **33** illustrates a nozzle portion of the surgical instruments described in connection with FIGS. **30-32**, according to one aspect of the present disclosure.

FIG. **34** is a schematic diagram of one aspect of a drive circuit configured for driving a high-frequency current (RF), according to one aspect of the present disclosure.

FIG. **35** is a schematic diagram of the transformer coupled to the RF drive circuit shown in FIG. **34**, according to one aspect of the present disclosure.

FIG. **36** is a schematic diagram of a circuit comprising separate power sources for high power energy/drive circuits and low power circuits, according to one aspect of the resent disclosure.

FIG. **37** illustrates a control circuit that allows a dual generator system to switch between the RF generator and the ultrasonic generator energy modalities for the surgical instrument shown in FIGS. **30** and **31**.

FIG. **38** is a sectional view of an end effector, according to one aspect of the present disclosure.

FIG. **39** is a sectional view of an end effector, according to one aspect of the present disclosure.

FIG. **40** is a partial longitudinal sectional side view showing a distal jaw section in a closed state, according to one aspect of the present disclosure.

FIG. **41** is a partial longitudinal sectional side view showing the distal jaw section in an open state, according to one aspect of the present disclosure.

FIG. **42** is a partial longitudinal sectional side view showing a jaw member, according to one aspect of the present disclosure.

FIG. **43** is a cross-sectional view showing the distal jaw section in a normal state, according to one aspect of the present disclosure.

FIG. **44** is a cross-sectional view showing the distal jaw section in a worn state, according to one aspect of the present disclosure.

FIG. **45** illustrates a modular battery powered handheld electrosurgical instrument with distal articulation, according to one aspect of the present disclosure.

FIG. **46** is an exploded view of the surgical instrument shown in FIG. **45**, according to one aspect of the present disclosure.

FIG. **47** is a perspective view of the surgical instrument shown in FIGS. **45** and **46** with a display located on the handle assembly, according to one aspect of the present disclosure.

FIG. **48** is a perspective view of the instrument shown in FIGS. **45** and **46** without a display located on the handle assembly, according to one aspect of the present disclosure.

FIG. **49** is a motor assembly that can be used with the surgical instrument to drive the knife, according to one aspect of the present disclosure.

FIG. **50** is diagram of a motor drive circuit, according to one aspect of the present disclosure.

FIG. **51** illustrates a rotary drive mechanism to drive distal head rotation, articulation, and jaw closure, according to one aspect of the present disclosure.

FIG. **52** is an enlarged, left perspective view of an end effector assembly with the jaw members shown in an open configuration, according to one aspect of the present disclosure.

FIG. **53** is an enlarged, right side view of the end effector assembly of FIG. **52**, according to one aspect of the present disclosure.

FIG. **54** illustrates a modular battery powered handheld electrosurgical instrument with distal articulation, according to one aspect of the present disclosure.

FIG. **55** is an exploded view of the surgical instrument shown in FIG. **54**, according to one aspect of the present disclosure.

FIG. **56** is an enlarged area detail view of an articulation section illustrated in FIG. **54** including electrical connections, according to one aspect of the present disclosure.

FIG. **57** is an enlarged area detail view articulation section illustrated in FIG. **56** including electrical connections, according to one aspect of the present disclosure.

FIG. **58** illustrates a perspective view of components of the shaft assembly, end effector, and cutting member of the surgical instrument of FIG. **54**, according to one aspect of the present disclosure.

FIG. **59** illustrates the articulation section in a second stage of articulation, according to one aspect of the present disclosure.

FIG. **60** illustrates a perspective view of the end effector of the device of FIGS. **54-59** in an open configuration, according to one aspect of the present disclosure.

FIG. **61** illustrates a cross-sectional end view of the end effector of FIG. **60** in a closed configuration and with the blade in a distal position, according to one aspect to the present disclosure.

FIG. **62** illustrates the components of a control circuit of the surgical instrument, according to one aspect of the present disclosure.

FIG. **63** is a system diagram of a segmented circuit comprising a plurality of independently operated circuit segments, according to one aspect of the present disclosure. FIG. **63** is a diagram of one form of a direct digital synthesis circuit.

FIG. **64** illustrates a diagram of one aspect of a surgical instrument comprising a feedback system for use with any one of the surgical instruments described herein in connection with FIGS. **1-61**, which may include or implement many of the features described herein

FIG. **65** illustrates one aspect of a fundamental architecture for a digital synthesis circuit such as a direct digital synthesis (DDS) circuit configured to generate a plurality of wave shapes for the electrical signal waveform for use in any of the surgical instruments described herein in connection with FIGS. **1-61**, according to one aspect of the present disclosure.

FIG. **66** illustrates one aspect of direct digital synthesis (DDS) circuit configured to generate a plurality of wave shapes for the electrical signal waveform for use in any of the surgical instruments described herein in connection with FIGS. **1-61**, according to one aspect of the present disclosure.

FIG. **67** illustrates one cycle of a discrete time digital electrical signal waveform, according to one aspect of the present disclosure of an analog waveform (shown superimposed over a discrete time digital electrical signal waveform for comparison purposes), according to one aspect of the present disclosure.

FIG. **68**A illustrates a circuit comprising a controller comprising one or more processors coupled to at least one memory circuit for use in any of the surgical instruments described herein in connection with FIGS. **1-61**, according to one aspect of the present disclosure.

FIG. **68**B illustrates a circuit comprising a finite state machine comprising a combinational logic circuit configured to implement any of the algorithms, processes, or techniques described herein, according to one aspect of the present disclosure.

FIG. **68**C illustrates a circuit comprising a finite state machine comprising a sequential logic circuit configured to implement any of the algorithms, processes, or techniques described herein, according to one aspect of the present disclosure.

FIG. **69** is a circuit diagram of various components of a surgical instrument with motor control functions, according to one aspect of the present disclosure.

FIG. **70** illustrates a handle assembly with a removable service panel removed to shown internal components of the handle assembly, according to one aspect of the present disclosure.

FIG. **71** is a graphical representation of determining wait time based on tissue thickness, according to aspects of the present disclosure.

FIG. **72** is a force versus time graph for thin, medium, and thick tissue types, according to aspects of the present disclosure.

FIG. **73** is a graph of motor current versus time for different tissue types, according to aspects of the present disclosure.

FIG. **74** is a graphical depiction of impedance bath tub, according to aspects of the present disclosure.

FIG. **75** is a graph depicting one aspect of adjustment of energy switching threshold due to the measurement of a secondary tissue parameter, according to aspects of the present disclosure.

FIG. **76** is a diagram of a process illustrating selective application of radio frequency or ultrasonic treatment energy based on measured tissue characteristics, according to aspects of the present disclosure.

FIG. **77** is a graph depicting a relationship between trigger button displacement and sensor output, according to aspects of the present disclosure.

FIG. **78** is a graph depicting an abnormal relationship between trigger button displacement and sensor output, according to aspects of the present disclosure.

FIG. **79** is a graph depicting an acceptable relationship between trigger button displacement and sensor output, according to aspects of the present disclosure.

FIG. **80** illustrates one aspect of a left-right segmented flexible circuit, according to aspects of the present disclosure.

FIG. **81** is a cross-sectional view of one aspect of a flexible circuit comprising RF electrodes and data sensors embedded therein, according to aspects of the present disclosure.

FIG. **82** is a cross sectional view of an end effector comprising a jaw member, a flexible circuit, and a segmented electrode, according to one aspect of the present disclosure.

FIG. **83** is a detailed view of the end effector shown in FIG. **82**, according to one aspect of the present disclosure.

FIG. **84**A is a cross sectional view of an end effector comprising a rotatable jaw member, a flexible circuit, and an ultrasonic blade positioned in a vertical orientation relative to the jaw member with no tissue located between the jaw member and the ultrasonic blade, according to one aspect of the present disclosure.

FIG. **84**B is a cross sectional view of the end effector shown in FIG. **84**A with tissue located between the jaw member and the ultrasonic blade, according to one aspect of the present disclosure.

FIG. **85**A is a cross sectional view of the effector shown in FIGS. **84**A and **84**B with the ultrasonic blade positioned in a horizontal orientation relative to the jaw member with no tissue located between the jaw member and the ultrasonic blade, according to one aspect of the present disclosure.

FIG. **85**B is a cross sectional view of the end effector shown in FIG. **85**A with tissue located between the jaw member and the ultrasonic blade, according to one aspect of the present disclosure.

FIG. **86** illustrates one aspect of an end effector comprising RF data sensors located on the jaw member, according to one aspect of the present disclosure.

FIG. **87** illustrates one aspect of the flexible circuit shown in FIG. **86** in which the sensors may be mounted to or formed integrally therewith, according to one aspect of the present disclosure.

FIG. **88** is a cross-sectional view of the flexible circuit shown in FIG. **87**, according to one aspect of the present disclosure.

FIG. **89** illustrates one aspect of a segmented flexible circuit configured to fixedly attach to a jaw member of an end effector, according to one aspect of the present disclosure.

FIG. **90** illustrates one aspect of a segmented flexible circuit configured to mount to a jaw member of an end effector, according to one aspect of the present disclosure.

FIG. **91** illustrates one aspect of an end effector configured to measure a tissue gap GT, according to one aspect of the present disclosure.

FIG. **92** illustrates one aspect of an end effector comprising segmented flexible circuit, according to one aspect of the present disclosure.

FIG. **93** illustrates the end effector shown in FIG. **92** with the jaw member clamping tissue between the jaw member and the ultrasonic blade, according to one aspect of the present disclosure.

FIG. **94** illustrates graphs of energy applied by the right and left side of an end effector based on locally sensed tissue parameters, according to one aspect of the present disclosure.

FIG. **95** is a cross-sectional view of one aspect of an end effector configured to sense force or pressure applied to tissue located between a jaw member and an ultrasonic blade, according to one aspect of the present disclosure.

FIG. **96** is a schematic diagram of one aspect of a signal layer of a flexible circuit, according to one aspect of the present disclosure.

FIG. **97** is a schematic diagram of sensor wiring for the flexible circuit shown in FIG. **96**, according to one aspect of the present disclosure.

FIG. **98**A is a graphical representation of one aspect of a medical device surrounding tissue, according to one aspect of the present disclosure.

FIG. **98**B is a graphical representation of one aspect of a medical device compressing tissue, according to one aspect of the present disclosure.

FIG. **99**A is a graphical representation of one aspect of a medical device compressing tissue, according to one aspect of the present disclosure.

FIG. **99**B also depicts example forces exerted by one aspect of an end-effector of a medical device compressing tissue, according to one aspect of the present disclosure.

FIG. **100** illustrates a logic diagram of one aspect of a feedback system, according to one aspect of the present disclosure.

FIG. **101** is a graph showing a limiting operation on a torque applied to a transducer;

FIG. **102** is a graph comparing different limiting conditions and different limiting operations when the motor is driving different moving components, according to one aspect of the present disclosure.

FIG. **103** is a perspective view of a motor comprising a strain gauge used for measuring torque, according to one aspect of the present disclosure.

FIG. **104** are graphs and showing smoothing out of fine trigger button movement by a motor controller, according to one aspect of the present disclosure.

FIG. **105** is a graph showing a non-linear force limit and corresponding velocity adjustment for I-blade RF motor control, according to one aspect of the present disclosure.

FIG. **106** is a graph showing force limits and corresponding velocity adjustments for jaw motor control, according to one aspect of the present disclosure.

FIG. **107** are graphs showing different velocity limits when the motor is driving different moving components, according to one aspect of the present disclosure.

FIG. **108** is a graph showing velocity limits at the beginning and the end of the range of movement, according to one aspect of the present disclosure.

FIG. **109** are diagrams showing velocity limits at the beginning and the end of the range of movement for jaw clamping and I-beam advance, according to one aspect of the present disclosure.

FIG. **110** are diagrams showing measurement of out-of-phase aspect of the current through the motor, according to one aspect of the present disclosure.

FIG. **111** is a graph showing adjustment of velocity limit based on tissue type, according to one aspect of the present disclosure.

FIG. **112** is a graph showing different wait periods and different velocity limits in I-beam motor control for tissues with different thickness, according to one aspect of the present disclosure.

FIG. **113** are graphs depicting a relationship between trigger button displacement and sensor output, according to one aspect of the present disclosure.

FIG. **114** illustrates an elevation view of system comprising a modular ultrasonic transducer assembly attached to a modular handle assembly of a battery powered modular handheld surgical instrument, wherein an ultrasonic waveguide of a modular shaft assembly is positioned for attachment to an ultrasonic transducer shaft of the modular ultrasonic transducer assembly according to an aspect of the present disclosure.

FIG. **115** illustrates an elevation view of the system of FIG. **114**, wherein the ultrasonic waveguide is positioned in axial alignment with the ultrasonic transducer shaft when the modular shaft assembly is attached to the modular ultrasonic transducer assembly and the modular handle assembly of the battery powered modular handheld surgical instrument according to an aspect of the present disclosure.

FIG. **116** illustrates an elevation view of the system of FIG. **114**, wherein the ultrasonic waveguide of the modular shaft assembly is attached to and torqued to the ultrasonic transducer shaft of the modular ultrasonic transducer assembly according to an aspect of the present disclosure.

FIG. **117** illustrates an elevation view of a modular handheld surgical instrument wherein a shifting spur gear has been translated from a first position at a primary drive gear to a second position at a slip clutch gear for a primary gear of a motor to rotate the slip-clutch gear via the shifting spur gear according to an aspect of the present disclosure.

FIG. **118** illustrates an isometric view of a portion of the modular handheld surgical instrument of FIG. **117**, wherein the ultrasonic waveguide is rotatable within a primary rotary drive rotatable via the primary drive gear within an exterior shaft rotatable via a slip clutch subsystem of the slip clutch gear according to an aspect of the present disclosure.

FIG. **119** illustrates an elevation view of a portion of the modular handheld surgical instrument of FIG. **117**, wherein the exterior shaft is coupled to the ultrasonic waveguide and wherein the exterior shaft is coupleable to a fixed external housing of the modular handheld surgical instrument according to an aspect of the present disclosure.

FIG. **120** illustrates a top elevation view of the system of FIG. **114** comprising a transducer locking mechanism to lock the ultrasonic transducer of the modular ultrasonic transducer assembly according to an aspect of the present disclosure.

FIG. **121** illustrates an end elevation view of the system of FIG. **114** that shows the slip clutch subsystem of FIG. **118** and the transducer locking mechanism of FIG. **120** according to an aspect of the present disclosure.

FIG. **122** illustrates a top elevation view of the transducer locking mechanism of FIG. **120** being operated via a torque lever to lock the ultrasonic transducer of the modular ultrasonic transducer assembly according to an aspect of the present disclosure.

FIG. **123** illustrates a graph of torque versus motor current wherein current supplied to the motor is electronically limited when torqueing the ultrasonic waveguide of the modular shaft assembly onto the ultrasonic transducer shaft of the modular ultrasonic transducer assembly according to an aspect of the present disclosure.

FIG. **124** illustrates graphs of torque versus motor current wherein current supplied to the motor is electronically limited based on the function (e.g., torqueing, rotating, articulating, clamping, etc.) performed by the modular hand-held surgical instrument according to an aspect of the present disclosure.

FIG. **125** illustrates an isometric view of a motor of the modular handheld surgical instrument wherein a strain gauge is used to measure torque applied by the motor according to an aspect of the present disclosure.

FIG. **126** is a system schematic diagram illustrating components of a battery powered modular surgical instrument, such as the battery operated modular surgical instruments described herein in connection with FIGS. **1-70**, according to various aspects of the present disclosure.

FIG. **127** describes a distribution of pluralities of control programs according to one aspect of the present disclosure in which the memory device of the control handle assembly stores a plurality of control programs comprising base operating control programs corresponding to the general operation of the modular surgical instrument, according to one aspect of the present disclosure.

FIG. **128** describes a distribution of pluralities of control programs according to one aspect of the present disclosure in which the memory device of the battery assembly stores a plurality of control programs comprising base operating control programs corresponding to the general operation of the modular surgical instrument, according to an aspect of the present disclosure.

FIG. **129** describes a distribution of pluralities of control programs according to one aspect of the present disclosure in which the memory device stores a plurality of control programs comprising base operating control programs corresponding to the general operation of the modular surgical instrument, according to one aspect of the present disclosure.

FIG. **130** is a logic diagram of a process for controlling the operation of a battery assembly operated modular surgical instrument with a plurality of control programs, according to one aspect of the present disclosure.

FIG. **131** is a logic diagram of a process for controlling the operation of a battery assembly operated modular surgical instrument with a plurality of control programs, according to one aspect of the present disclosure.

FIG. **132** is a logic diagram of a process for controlling the operation of a surgical instrument according to the energy conservation method comprising the POST and operation verification, according to one aspect of the present disclosure.

FIG. **133** is a logic diagram of a process for controlling the operation of a surgical instrument according to the energy conservation method comprising a deenergization sequence and an energization sequence, according to one aspect of the present disclosure.

FIG. **134** illustrates a control circuit for selecting a segmented circuit of a surgical instrument, according to an aspect of the present disclosure.

FIG. **135** is a logic diagram of a process for controlling the operation of a surgical instrument according to the energy conservation method comprising a deenergization sequence and an energization sequence that is different from the deenergization sequence, according to one aspect of the present disclosure.

FIG. **136** illustrates a generator circuit partitioned into multiple stages, according to one aspect of the present disclosure.

FIG. **137** illustrates a generator circuit partitioned into multiple stages where a first stage circuit is common to the second stage circuit, according to one aspect of the present disclosure.

FIG. **138** shows an example system of shaft control electronics interconnecting inside a portion of a handle assembly and shaft assembly of a medical device, according to one aspect of the present disclosure.

FIG. **139** provides an example of a zoomed in portion of the flexible circuit, according to some aspects, according to one aspect of the present disclosure.

FIGS. **140**A and **140**B show how a plurality of sensors may be used in combination to measure and monitor movements of the medical device, according to one aspect of the present disclosure.

FIG. **141** provides another example of how multiple sensors can be used to determine a three-dimensional position of one or more magnets, according to one aspect of the present disclosure.

FIG. **142** shows a variation of a shaft assembly of a medical device that includes a seal to protect sensitive electronics, according to one aspect of the present disclosure.

FIG. **143** shows the inner workings of a portion of a shaft assembly that is part of the surgical instrument and includes multiple clutch systems, according to one aspect of the present disclosure.

FIGS. **144**A, **144**B, **144**C, **144**D, **144**E, and **144**F show another variation for including multiple clutch systems, according to one aspect of the present disclosure.

FIGS. **145**A and **145**B show a shaft assembly that may include a locking element to help stabilize engagement of the clutches, according to one aspect of the present disclosure.

FIGS. **146**A and **146**B show side views of the locking element, according to one aspect of the present disclosure.

FIG. **147** shows an example plot of the amount of voltage output by a power switch in relation to its sliding distance within a slot of a self diagnosing control switch, according to one aspect of the present disclosure.

FIG. **148** provides an example of how the displacement vs. voltage profile of a control switch may change over time, due to natural wear and tear of the mechanical and electrical components, according to one aspect of the present disclosure.

FIG. **149** shows how the self diagnosing control switch of the present disclosures may be configured to transmit an alert when it is determined that automatic adjustments to account for wear and tear are no longer possible, according to one aspect of the present disclosure.

FIG. **150** is a side elevational view of one aspect of a handle assembly of a battery powered modular surgical instrument, according to one aspect of the present disclosure.

FIG. **151** is a side elevational view of another aspect of a handle assembly of a battery powered modular surgical instrument, according to one aspect of the present disclosure.

FIG. **152** is a side elevational view of another aspect of a handle assembly of a battery powered modular surgical instrument, according to one aspect of the present disclosure.

FIG. **153**A illustrates a cross sectional view of a reusable and serviceable handle assembly with a service cover in an open position, according to one aspect of the present disclosure.

FIG. **153**B illustrates a cross sectional view of the reusable and serviceable handle assembly shown in FIG. **153**A with the service cover in a closed position, according to one aspect of the present disclosure.

FIG. **154**A illustrates a cross sectional view of a reusable and serviceable handle assembly with a service cover in an open position, according to one aspect of the present disclosure.

FIG. **154**B illustrates a cross sectional view of the reusable and serviceable handle assembly shown in FIG. **154**A with the service cover in a closed position, according to one aspect of the present disclosure.

FIG. **155**A illustrates a handle assembly is a secure fastened configuration, according to one aspect of the present disclosure.

FIG. **155**B illustrates the handle assembly shown in FIG. **155**A in an unlatched configuration, according to one aspect of the present disclosure.

FIG. **156** is a plan view of one aspect of an end effector, according to one aspect of the present disclosure.

FIG. **157** is a side view of the end effector shown in FIG. **156** with a partial cut away view to expose the underlying structure of the jaw member and an ultrasonic blade, according to one aspect of the present disclosure.

FIG. **158** is partial sectional view of the end effector shown in FIGS. **156**, **157** to expose the ultrasonic blade and right and left electrodes, respectively, according to one aspect of the present disclosure.

FIG. **159** is a cross-sectional view taken at section **174-174** of the end effector shown in FIG. **156**, according to one aspect of the present disclosure.

FIG. **160** is cross-sectional view taken at section **175-175** of the end effector shown in FIG. **156**, according to one aspect of the present disclosure.

FIG. **161** is a cross-sectional view taken at section **174-174** of the end effector shown in FIG. **156**, except that the ultrasonic blade has a different geometric configuration, according to one aspect of the present disclosure.

FIG. **162** is cross-sectional view taken at section **175-175** of the end effector shown in FIG. **156**, except that the ultrasonic blade has a different geometric configuration, according to one aspect of the present disclosure.

FIG. **163** is a cross-sectional view taken at section **174-174** of the end effector shown in FIG. **156**, except that the ultrasonic blade has a different geometric configuration, according to one aspect of the present disclosure.

FIG. **164** is cross-sectional view taken at section **175-175** of the end effector shown in FIG. **156**, except that the ultrasonic blade has a different geometric configuration, according to one aspect of the present disclosure.

FIG. **165** illustrates a logic diagram of one aspect of a feedback system, according to one aspect of the present disclosure.

DESCRIPTION

This application is related to following commonly owned patent applications filed on Dec. 16, 2016, the content of each of which is incorporated herein by reference in its entirety:

U.S. patent application Ser. No. 15/382,238, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT WITH SELECTIVE APPLICATION OF ENERGY BASED ON TISSUE CHARACTERIZATION, now U.S. Patent Application Publication No. 2017/0202591.

U.S. patent application Ser. No. 15/382,246, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT WITH SELECTIVE APPLICATION OF ENERGY BASED ON BUTTON DISPLACEMENT, INTENSITY, OR LOCAL TISSUE CHARACTERIZATION, now U.S. Patent Application Publication No. 2017/0202607.

U.S. patent application Ser. No. 15/382,252, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT WITH VARIABLE MOTOR CONTROL LIMITS, now U.S. Pat. No. 10,537,351.

U.S. patent application Ser. No. 15/382,257, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT WITH MOTOR CONTROL LIMIT PROFILE, now U.S. Pat. No. 10,299,821.

U.S. patent application Ser. No. 15/382,265, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT WITH MOTOR CONTROL LIMITS BASED ON TISSUE CHARACTERIZATION, now U.S. Patent Application Publication No. 2017/0202594.

U.S. patent application Ser. No. 15/382,274, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT WITH MULTI-FUNCTION MOTOR VIA SHIFTING GEAR ASSEMBLY, now U.S. Pat. No. 10,251,664.

U.S. patent application Ser. No. 15/382,281, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT WITH A PLURALITY OF CONTROL PROGRAMS, now U.S. Patent Application Publication No. 2017/0202595.

U.S. patent application Ser. No. 15/382,283, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT WITH ENERGY CONSERVATION TECHNIQUES, now U.S. Patent Application Publication No. 2017/0202596.

U.S. patent application Ser. No. 15/382,285, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT WITH VOLTAGE SAG RESISTANT BATTERY PACK, now U.S. Patent Application Publication No. 2017/0207467.

U.S. patent application Ser. No. 15/382,287, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT WITH MULTISTAGE GENERATOR CIRCUITS, now U.S. Patent Application Publication No. 2017/0202597.

U.S. patent application Ser. No. 15/382,288, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT WITH MULTIPLE MAGNETIC POSITION SENSORS, now U.S. Patent Application Publication No. 2017/0202598.

U.S. patent application Ser. No. 15/382,290, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT CONTAINING ELONGATED MULTI-LAYERED SHAFT, now U.S. Patent Application Publication No. 2017/0202608.

U.S. patent application Ser. No. 15/382,292, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT WITH MOTOR DRIVE, now U.S. Patent Application Publication No. 2017/0202572.

U.S. patent application Ser. No. 15/382,297, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT WITH SELF-DIAGNOSING CONTROL SWITCHES FOR REUSABLE HANDLE ASSEMBLY, now U.S. Patent Application Publication No. 2017/0202599.

U.S. patent application Ser. No. 15/382,306, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT WITH REUSABLE ASYMMETRIC HANDLE HOUSING, now U.S. Patent Application Publication No. 2017/0202571.

U.S. patent application Ser. No. 15/382,309, titled MODULAR BATTERY POWERED HANDHELD SURGICAL INSTRUMENT WITH CURVED END EFFECTORS HAVING ASYMMETRIC ENGAGEMENT BETWEEN JAW AND BLADE, now U.S. Patent Application Publication No. 2017/0202609.

In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols and reference characters typically identify similar components throughout the several views, unless context dictates otherwise. The illustrative aspects described in the detailed description, drawings, and claims are not meant to be limiting. Other aspects may be utilized, and other changes may be made, without departing from the scope of the subject matter presented here.

Before explaining the various aspects of the present disclosure in detail, it should be noted that the various aspects disclosed herein are not limited in their application or use to the details of construction and arrangement of parts illustrated in the accompanying drawings and description. Rather, the disclosed aspects may be positioned or incorporated in other aspects, variations and modifications thereof, and may be practiced or carried out in various ways. Accordingly, aspects disclosed herein are illustrative in nature and are not meant to limit the scope or application thereof. Furthermore, unless otherwise indicated, the terms and expressions employed herein have been chosen for the purpose of describing the aspects for the convenience of the reader and are not to limit the scope thereof. In addition, it should be understood that any one or more of the disclosed aspects, expressions of aspects, and/or examples thereof, can be combined with any one or more of the other disclosed aspects, expressions of aspects, and/or examples thereof, without limitation.

Also, in the following description, it is to be understood that terms such as front, back, inside, outside, top, bottom and the like are words of convenience and are not to be construed as limiting terms. Terminology used herein is not meant to be limiting insofar as devices described herein, or portions thereof, may be attached or utilized in other orientations. The various aspects will be described in more detail with reference to the drawings.

In various aspects, the present disclosure is directed to a mixed energy surgical instrument that utilizes both ultrasonic and RF energy modalities. The mixed energy surgical instrument my use modular shafts using that accomplish existing end-effector functions such as ultrasonic functions disclosed in U.S. Pat. No. 9,107,690, which is incorporated herein by reference in its entirety, combination device functions disclosed in U.S. Pat. Nos. 8,696,666 and 8,663,223, which are both incorporated herein by reference in their entireties, RF opposed electrode functions disclosed in U.S. Pat. Nos. 9,028,478 and 9,113,907, which are both incorporated herein by reference in their entireties, and RF I-blade offset electrode functions as disclosed in U.S. Patent Application Publication No. 2013/0023868, now U.S. Pat. No. 9,545,253, which is incorporated herein by reference in its entirety.

In various aspects, the present disclosure is directed to a modular battery powered handheld ultrasonic surgical instrument comprising a first generator, a second generator, and a control circuit for controlling the energy modality applied by the surgical instrument. The surgical instrument is configured to apply at least one energy modality that comprises an ultrasonic energy modality, a radio frequency (RF) energy modality, or a combination ultrasonic and RF energy modalities.

In another aspect, the present disclosure is directed to a modular battery powered handheld surgical instrument that can be configured for ultrasonic energy modality, RF modality, or a combination of ultrasonic and RF energy modalities. A mixed energy surgical instrument utilizes both ultrasonic and RF energy modalities. The mixed energy surgical instrument may use modular shafts that accomplish end effector functions. The energy modality may be selectable based on a measure of specific measured tissue and device parameters, such as, for example, electrical impedance, tissue impedance, electric motor current, jaw gap, tissue thickness, tissue compression, tissue type, temperature, among other parameters, or a combination thereof, to determine a suitable energy modality algorithm to employ ultrasonic vibration and/or electrosurgical high-frequency current to carry out surgical coagulation/cutting treatments on the living tissue based on the measured tissue parameters identified by the surgical instrument. Once the tissue parameters have been identified, the surgical instrument may be configured to control treatment energy applied to the tissue in a single or segmented RF electrode configuration or in an ultrasonic device, through the measurement of specific tissue/device parameters. Tissue treatment algorithms are described in commonly owned U.S. patent application Ser. No. 15/177,430, titled SURGICAL INSTRUMENT WITH USER ADAPTABLE TECHNIQUES, now U.S. Patent Application Publication No. 2017/0000541, which is herein incorporated by reference in its entirety.

In another aspect, the present disclosure is directed to a modular battery powered handheld surgical instrument having a motor and a controller, where a first limiting threshold is used on the motor for the purpose of attaching a modular assembly and a second threshold is used on the motor and is associated with a second assembly step or functionality of the surgical instrument. The surgical instrument may comprise a motor driven actuation mechanism utilizing control of motor speed or torque through measurement of motor current or parameters related to motor current, wherein motor control is adjusted via a non-linear threshold to trigger motor adjustments at different magnitudes based on position, inertia, velocity, acceleration, or a combination thereof. Motor driven actuation of a moving mechanism and a motor controller may be employed to control the motor velocity or torque. A sensor associated with physical properties of the moving mechanism provides feedback to the motor controller. In one aspect, the sensor is employed to adjust a predefined threshold which triggers a change in the operation of the motor controller. A motor may be utilized to drive shaft functions such as shaft rotation and jaw closure and switching that motor to also provide a torque limited waveguide attachment to a transducer. A motor control algorithm may be utilized to generate tactile feedback to a user through a motor drive train for indication of device status and/or limits of the powered actuation. A motor powered modular advanced energy based surgical instrument may comprise a series of control programs or algorithms to operate a series of different shaft modules and transducers. In one aspect, the programs or algorithms reside in a module and are uploaded to a control handle when attached. The motor driven modular battery powered handheld surgical instrument may comprise a primary rotary drive capable of being selectably coupleable to at least two independent actuation functions (first, second, both, neither) and utilize a clutch mechanism located in a distal modular elongated tube.

In another aspect, the present disclosure is directed to modular battery powered handheld surgical instrument comprising energy conservation circuits and techniques using sleep mode de-energizing of a segmented circuit with short cuts to minimize non-use power drain and differing wake-up sequence order than the order of a sleep sequence. A disposable primary cell battery pack may be utilized with a battery powered modular handheld surgical instrument. The disposable primary cell may comprise power management circuits to compensate the battery output voltage with additional voltage to offset voltage sags under load and to prevent the battery pack output voltage from sagging below a predetermined level during operation under load. The circuitry of the surgical instrument comprises radiation tolerant components and amplification of electrical signals may be divided into multiple stages. An ultrasonic transducer housing or RF housing may contain the final amplification stage and may comprise different ratios depending on an energy modality associated with the ultrasonic transducer or RF module.

In another aspect, the present disclosure is directed to a modular battery powered handheld surgical instrument comprising multiple magnetic position sensors along a length of a shaft and paired in different configurations to allow multiple sensors to detect the same magnet in order to determine three dimensional position of actuation components of the shaft from a stationary reference plane and simultaneously diagnosing any error from external sources. Control and sensing electronics may be incorporated in the shaft. A portion of the shaft control electronics may be disposed along the inside of moving shaft components and are separated from other shaft control electronics that are disposed along the outside of the moving shaft components. Control and sensing electronics may be situated and designed such that they act as a shaft seal in the device.

In another aspect, the present disclosure is directed to a modular battery powered handheld surgical instrument comprising self diagnosing control switches within a battery powered, modular, reusable handle. The control switches are capable of adjusting their thresholds for triggering an event as well as being able to indicate external influences on the controls or predict time till replacement needed. The reusable handle housing is configured for use with modular disposable shafts and at least one control and wiring harness. The handle is configured to asymmetrically part when opened so that the switches, wiring harness, and/or control electronics can be supportably housed in one side such that the other side is removably attached to cover the primary housing.

Modular Battery Powered Handheld Surgical Instruments

FIG. **1** is a diagram of a modular battery powered handheld ultrasonic surgical instrument **100**, according to an aspect of the present disclosure. FIG. **2** is an exploded view of the surgical instrument **100** shown in FIG. **1**, according to an aspect of the present disclosure. With reference now to FIGS. **1** and **2**, the surgical instrument **100** comprises a handle assembly **102**, an ultrasonic transducer/generator assembly **104**, a battery assembly **106**, a shaft assembly **110**, and an end effector **112**. The ultrasonic transducer/generator assembly **104**, battery assembly **106**, and shaft assembly **110** are modular components that are removably connectable to the handle assembly **102**. The handle assembly **102** comprises a motor assembly **160**. In addition, some aspects of the surgical instrument **100** include battery assemblies **106** that contain the ultrasonic generator and motor control circuits. The battery assembly **106** includes a first stage generator function with a final stage existing as part of the ultrasonic transducer/generator assembly **104** for driving 55 kHz and 33.1 Khz ultrasonic transducers. A different final stage generator for interchangeable use with the battery assembly **106**, common generator components, and segmented circuits enable battery assembly **106** to power up sections of the drive circuits in a controlled manner and to enable checking of stages of the circuit before powering them up and enabling power management modes. In addition, general purpose controls may be provide in the handle assembly **102** with dedicated shaft assembly **110** controls located on the shafts that have those functions. For instance, an end effector **112** module may comprise distal rotation electronics, the shaft assembly **110** may comprise rotary shaft control along with articulation switches, and the handle assembly **102** may comprise energy activation controls and jaw member **114** trigger **108** controls to clamp and unclamp the end effector **112**.

The ultrasonic transducer/generator assembly **104** comprises a housing **148**, a display **176**, such as a liquid crystal display (LCD), for example, an ultrasonic transducer **130**, and an ultrasonic generator **162** (FIG. **4**). The shaft assembly **110** comprises an outer tube **144** an ultrasonic transmission waveguide **145**, and an inner tube (not shown). The end effector **112** comprises a jaw member **114** and an ultrasonic blade **116**. As described hereinbelow, a motor or other mechanism operated by the trigger **108** may be employed to close the jaw member **114**. The ultrasonic blade **116** is the distal end of the ultrasonic transmission waveguide **145**. The jaw member **114** is pivotally rotatable to grasp tissue between the jaw member and the ultrasonic blade **116**. The jaw member **114** is operably coupled to a trigger **108** such that when the trigger **108** is squeezed the jaw member **114** closes to grasp tissue and when the trigger **108** is released the jaw member **114** opens to release tissue. In a one-stage trigger configuration, the trigger **108** functions to close the jaw member **114** when the trigger **108** is squeezed and to open the jaw member **114** when the trigger **108** is released. Once the jaw member **114** is closed, the switch **120** is activated to energize the ultrasonic generator to seal and cut the tissue. In a two-stage trigger configuration, during the first stage, the trigger **108** is squeezed part of the way to close the jaw member **114** and, during the second stage, the trigger **108** is squeezed the rest of the way to energize the ultrasonic generator to seal and cut the tissue. The jaw member **114***a* opens by releasing the trigger **108** to release the tissue. It will be appreciated that in other aspects, the ultrasonic transducer **103** may be activated without the jaw member **114** being closed.

The battery assembly **106** is electrically connected to the handle assembly **102** by an electrical connector **132**. The handle assembly **102** is provided with a switch **120**. The ultrasonic blade **116** is activated by energizing the ultrasonic transducer/generator circuit by actuating the switch **120**. The battery assembly **106**, according to one aspect, is a rechargeable, reusable battery pack with regulated output. In some cases, as is explained below, the battery assembly **106** facilitates user-interface functions. The handle assembly **102** is a disposable unit that has bays or docks for attachment to the battery assembly **106**, the ultrasonic transducer/generator assembly **104**, and the shaft assembly **110**. The handle assembly **102** also houses various indicators including, for example, a speaker/buzzer and activation switches. In one aspect, the battery assembly is a separate component that is inserted into the housing of the handle assembly through a door or other opening defined by the housing of the handle assembly.

The ultrasonic transducer/generator assembly **104** is a reusable unit that produces high frequency mechanical motion at a distal output. The ultrasonic transducer/generator assembly **104** is mechanically coupled to the shaft assembly **110** and the ultrasonic blade **116** and, during operation of the device, produces movement at the distal output of the ultrasonic blade **116**. In one aspect, the ultravisonic transducer/generator assembly **104** also provides a visual user interface, such as, through a red/green/blue (RGB) light-emitting diode (LED), LCD, or other display. As such, a visual indicator of the battery status is uniquely not located on the battery and is, therefore, remote from the battery.

In accordance with various aspects of the present disclosure, the three components of the surgical instrument **100**, e.g., the ultrasonic transducer/generator assembly **104**, the battery assembly **106**, and the shaft assembly **110**, are advantageously quickly disconnectable from one or more of the others. Each of the three components of the surgical instrument **100** is sterile and can be maintained wholly in a sterile field during use. Because the components of the surgical instrument **100** are separable, the surgical instrument **100** can be composed of one or more portions that are single-use items (e.g., disposable) and others that are multi-use items (e.g., sterilizable for use in multiple surgical procedures). Aspects of the components separate as part of the surgical instrument **100**. In accordance with an additional aspect of the present disclosure, the handle assembly **102**, battery assembly **106**, and shaft assembly **110** components is equivalent in overall weight; each of the handle assembly **102**, battery assembly **106**, and shaft assembly **110** components is balanced so that they weigh the same or substantially the same. The handle assembly **102** overhangs the operator's hand for support, allowing the user's hand to more freely operate the controls of the surgical instrument **100** without bearing the weight. This overhang is set to be very close to the center of gravity. This combined with a triangular assembly configuration, makes the surgical instrument **100** advantageously provided with a center of balance that provides a very natural and comfortable feel to the user operating the device. That is, when held in the hand of the user, the surgical instrument **100** does not have a tendency to tip forward or backward or side-to-side, but remains relatively and dynamically balanced so that the waveguide is held parallel to the ground with very little effort from the user. Of course, the instrument can be placed in non-parallel angles to the ground just as easily.

A rotation knob **118** is operably coupled to the shaft assembly **110**. Rotation of the rotation knob **118** ±360° in the direction indicated by the arrows **126** causes an outer tube **144** to rotate ±360° in the respective direction of the arrows **128**. In one aspect, the rotation knob **118** may be configured to rotate the jaw member **114** while the ultrasonic blade **116** remains stationary and a separate shaft rotation knob may be provided to rotate the outer tube **144** ±360°. In various aspects, the ultrasonic blade **116** does not have to stop at ±360° and can rotate at an angle of rotation that is greater than ±360°. The outer tube **144** may have a diameter D_1 ranging from 5 mm to 10 mm, for example.

The ultrasonic blade **116** is coupled to an ultrasonic transducer **130** (FIG. **2**) portion of the ultrasonic transducer/generator assembly **104** by an ultrasonic transmission waveguide located within the shaft assembly **110**. The ultrasonic blade **116** and the ultrasonic transmission waveguide may be formed as a unit construction from a material suitable for transmission of ultrasonic energy. Examples of such materials include Ti6Al4V (an alloy of Titanium including Aluminum and Vanadium), Aluminum, Stainless Steel, or other suitable materials. Alternately, the ultrasonic blade **116** may be separable (and of differing composition) from the ultrasonic transmission waveguide, and coupled by, for example, a stud, weld, glue, quick connect, or other suitable known methods. The length of the ultrasonic transmission waveguide may be an integral number of one-half wavelengths ($n\lambda/2$), for example. The ultrasonic transmission waveguide may be preferably fabricated from a solid core shaft constructed out of material suitable to propagate ultrasonic energy efficiently, such as the titanium alloy discussed above (i.e., Ti6Al4V) or any suitable aluminum alloy, or other alloys, or other materials such as sapphire, for example.

The ultrasonic transducer/generator assembly **104** also comprises electronic circuitry for driving the ultrasonic transducer **130**. The ultrasonic blade **116** may be operated at a suitable vibrational frequency range may be about 20 Hz to 120 KHz and a well-suited vibrational frequency range may be about 30-100 KHz. A suitable operational vibrational frequency may be approximately 55.5 kHz, for example. The ultrasonic transducer **130** is energized by the actuating the switch **120**.

It will be appreciated that the terms "proximal" and "distal" are used herein with reference to a clinician gripping the handle assembly **102**. Thus, the ultrasonic blade **116** is distal with respect to the handle assembly **102**, which is more proximal. It will be further appreciated that, for convenience and clarity, spatial terms such as "top" and "bottom" also are used herein with respect to the clinician gripping the handle assembly **102**. However, surgical instruments are used in many orientations and positions, and these terms are not intended to be limiting and absolute.

FIG. **3** is an exploded view of a modular shaft assembly **110** of the surgical instrument **100** shown in FIG. **1**, according to aspect of the present disclosure. The surgical instrument **100** uses ultrasonic vibration to carry out a surgical treatment on living tissue. The shaft assembly **110** couples to the handle assembly **102** via slots **142***a*, **142***b* formed on the handle assembly **102** and tabs **134***a*, **134***b* on the shaft assembly **110**. The handle assembly **102** comprises a male coupling member **136** that is received in a corresponding female coupling member in the **138** shaft assembly **110**. The male coupling member **136** is operably coupled to the trigger **108** such that when the trigger **108** is squeezed the male coupling member **136** translates distally to drive a closure tube mechanism **140** that translates an outer tube portion of the shaft assembly **110** to close the jaw member **114**. As previously discussed, when the trigger **108** is released, the jaw member **114** opens. The male coupling member **136** also couples to the ultrasonic transmission waveguide **145** (FIG. **2**) located within the outer tube **144** of the shaft assembly **110** and couples to the ultrasonic transducer **130** (FIG. **2**), which is received within the nozzle **146** of the handle assembly **102**. The shaft assembly **110** is electrically coupled to the handle assembly **102** via electrical contacts **137**.

FIG. **4** is a perspective transparent view of the ultrasonic transducer/generator assembly **104** of the surgical instrument **100** shown in FIG. **1**, according to aspect of the present disclosure. FIG. **5** is an end view of the ultrasonic transducer/generator assembly **104**, FIG. **6** is a perspective view of the ultrasonic transducer/generator assembly **104** with the top housing portion removed to expose the ultrasonic generator **162**, and FIG. **7** is a sectional view of the of the ultrasonic transducer/generator assembly **104**. With reference now to FIGS. **4**-**7**, the ultrasonic transducer/generator assembly **104** comprises an ultrasonic transducer **130**, an ultrasonic generator **162** to drive the ultrasonic transducer **130**, and a housing **148**. A first electrical connector **158** couples the ultrasonic generator **162** to the battery assembly **106** (FIGS. **1** and **2**) and a second electrical connector **161** couples the ultrasonic generator **162** to the nozzle (FIG. **3**). In one aspect, a display **176** may be provided on one side of the ultrasonic transducer/generator assembly **104** housing **148**.

The ultrasonic generator **162** comprises an ultrasonic driver circuit such as the electrical circuit **177** shown in FIG. **11** and, in some aspects, a second stage amplifier circuit **178**. The electrical circuit **177** is configured for driving the ultrasonic transducer **130** and forms a portion of the ultrasonic generator circuit. The electrical circuit **177** comprises a transformer **166** and a blocking capacitor **168**, among other components. The transformer **166** is electrically coupled to the piezoelectric elements **150***a*, **150***b*, **150***c*, **150***d* of the ultrasonic transducer **130**. The electrical circuit **177** is electrically coupled to first electrical connector **158** via a first cable **179**. The first electrical connector **158** is electrically coupled to the battery assembly **106** (FIGS. **1** and **2**). The electrical circuit **177** is electrically coupled to second electrical connector **160** via a second cable **183**. The second electrical connector **160** is electrically coupled to the nozzle **146** (FIG. **3**). In one aspect, the second stage amplifier circuit **178** may be employed in a two stage amplification system.

The ultrasonic transducer **130**, which is known as a "Langevin stack", generally includes a transduction portion comprising piezoelectric elements **150***a*-**150***d*, a first resonator portion or end-bell **164**, and a second resonator portion or fore-bell **152**, and ancillary components. The total construction of these components is a resonator. There are other forms of transducers, such as magnetostrictive transducers, that could also be used. The ultrasonic transducer **130** is preferably an integral number of one-half system wavelengths ($n\lambda/2$; where "n" is any positive integer; e.g., n=1, 2, 3 . . .) in length as will be described in more detail later. An acoustic assembly includes the end-bell **164**, ultrasonic transducer **130**, fore-bell **152**, and a velocity transformer **154**.

The distal end of the end-bell **164** is acoustically coupled to the proximal end of the piezoelectric element **150***a*, and the proximal end of the fore-bell **152** is acoustically coupled to the distal end of the piezoelectric element **150***d*. The fore-bell **152** and the end-bell **164** have a length determined by a number of variables, including the thickness of the transduction portion, the density and modulus of elasticity of the material used to manufacture the end-bell **164** and the fore-bell **152**, and the resonant frequency of the ultrasonic transducer **130**. The fore-bell **152** may be tapered inwardly from its proximal end to its distal end to amplify the ultrasonic vibration amplitude at the velocity transformer **154**, or alternately may have no amplification. A suitable vibrational frequency range may be about 20 Hz to 120 KHz and a well-suited vibrational frequency range may be about 30-100 KHz. A suitable operational vibrational frequency may be approximately 55.5 kHz, for example.

The ultrasonic transducer **130** comprises several piezoelectric elements **150***a*-**150***d* acoustically coupled or stacked to form the transduction portion. The piezoelectric elements **150***a*-**150***d* may be fabricated from any suitable material, such as, for example, lead zirconate-titanate, lead metaniobate, lead titanate, barium titanate, or other piezoelectric ceramic material. Electrically conductive elements **170***a*, **170***b*, **170***c*, **170***d* are inserted between the piezoelectric elements **150***a*-**150***d* to electrically couple the electrical circuit **177** to the piezoelectric elements **150***a*-**150***d*. The electrically conductive element **170***a* located between piezoelectric elements **150***a*, **150***b* and the electrically conductive element **170***d* located between piezoelectric element **150***d* and the fore-bell **152** are electrically coupled to the positive electrode **174***a* of the electrical circuit **177**. The electrically conductive element **170***b* located between piezoelectric elements **150***b*, **150***c* and the electrically conductive element **170***c* located between piezoelectric elements **150***c*, **150***d* are electrically coupled to the negative electrode **174***b* of the electrical circuit **177**. The positive and negative electrodes **174***a*, **174***b* are electrically coupled to the electrical circuit **177** by electrical conductors.

The ultrasonic transducer **130** converts the electrical drive signal from the electrical circuit **177** into mechanical energy that results in primarily a standing acoustic wave of longitudinal vibratory motion of the ultrasonic transducer **130** and the ultrasonic blade **116** (FIGS. **1** and **3**) at ultrasonic frequencies. In another aspect, the vibratory motion of the ultrasonic transducer **130** may act in a different direction. For example, the vibratory motion may comprise a local longitudinal component of a more complicated motion of the ultrasonic blade **116**. When the acoustic assembly is energized, a vibratory motion in the form of a standing wave is generated through the ultrasonic transducer **130** to the ultrasonic blade **116** at a resonance and amplitude determined by various electrical and geometrical parameters. The amplitude of the vibratory motion at any point along the acoustic assembly depends upon the location along the acoustic assembly at which the vibratory motion is measured. A minimum or zero crossing in the vibratory motion standing wave is generally referred to as a node (i.e., where motion is minimal), and a local absolute value maximum or peak in the standing wave is generally referred to as an anti-node (i.e., where local motion is maximal). The distance between an anti-node and its nearest node is one-quarter wavelength ($\lambda/4$).

The wires transmit an electrical drive signal from the electrical circuit **177** to the positive electrode **170***a* and the negative electrode **170***b*. The piezoelectric elements **150***a*-**150***d* are energized by the electrical signal supplied from the electrical circuit **177** in response to an actuator, such as the switch **120**, for example, to produce an acoustic standing wave in the acoustic assembly. The electrical signal causes disturbances in the piezoelectric elements **150***a*-**150***d* in the form of repeated small displacements resulting in large alternating compression and tension forces within the material. The repeated small displacements cause the piezoelectric elements **150***a*-**150***d* to expand and contract in a continuous manner along the axis of the voltage gradient, producing longitudinal waves of ultrasonic energy. The ultrasonic energy is transmitted through the acoustic assembly to the ultrasonic blade **116** (FIGS. **1** and **3**) via a transmission component or an ultrasonic transmission waveguide through the shaft assembly **110** (FIGS. **1-3**).

In order for the acoustic assembly to deliver energy to the ultrasonic blade **116** (FIGS. **1** and **3**), components of the acoustic assembly are acoustically coupled to the ultrasonic blade **116**. A coupling stud **156** of the ultrasonic transducer **130** is acoustically coupled to the ultrasonic transmission waveguide **145** by a threaded connection such as a stud. In one aspect, the ultrasonic transducer **130** may be acoustically coupled to the ultrasonic transmission waveguide **145** as shown in FIGS. **10**A and **10**B.

The components of the acoustic assembly are preferably acoustically tuned such that the length of any assembly is an integral number of one-half wavelengths ($n\lambda/2$), where the wavelength A is the wavelength of a pre-selected or operating longitudinal vibration drive frequency fa of the acoustic assembly. It is also contemplated that the acoustic assembly may incorporate any suitable arrangement of acoustic elements.

The ultrasonic blade **116** (FIGS. **1** and **3**) may have a length that is an integral multiple of one-half system wavelengths ($n\lambda/2$). A distal end of the ultrasonic blade **116** may be disposed near an antinode in order to provide the maximum longitudinal excursion of the distal end. When the ultrasonic transducer **130** is energized, the distal end of the ultrasonic blade **116** may be configured to move in the range of, for example, approximately 10 to 500 microns peak-to-peak, and preferably in the range of about 30 to 150 microns, and in some aspects closer to 100 microns, at a predetermined vibrational frequency of 55 kHz, for example.

FIG. **8** is an elevation view of an ultrasonic transducer/generator assembly **104** that is configured to operate at 31 kHz resonant frequency, according to one aspect of the present disclosure. FIG. **9** is an elevation view of an ultrasonic transducer/generator assembly **104'** that is configured to operate at 55 kHz resonant frequency, according to one aspect of the present disclosure. As can be seen, the ultrasonic transducer/generator assemblies **104**, **104'**, the housings **148** are the same size in order to fit into the nozzle **146** of the surgical instrument **100** shown in FIG. **3**. Nevertheless, the individual ultrasonic transducers **130**, **130'** will vary in size depending on the desired resonant frequency. For example, the ultrasonic transducer **130** shown in FIG. **8** is tuned at a resonant frequency of 31 kHz is physically larger than the ultrasonic transducer **130'** shown in FIG. **9**, which is tuned at a resonant frequency of 55 KHz. The coupling stud **156**, **156'** of the ultrasonic transducer **130**, **130'** may be acoustically coupled to the ultrasonic transmission waveguide **145** by a threaded connection such as a stud.

FIGS. **10**A and **10**B illustrate a shifting assembly **200** that selectively rotates the ultrasonic transmission waveguide **145** with respect to the ultrasonic transducer **130** and urges them towards one another, according to one aspect of the present disclosure. FIG. **10**A illustrates the shifting assembly **200** with the ultrasonic transmission waveguide **145** and the ultrasonic transducer **130** in a disengaged configuration and FIG. **10**B illustrates the shifting assembly **200** with the ultrasonic transmission waveguide **145** and the ultrasonic transducer **130** in an engaged configuration. With reference now to both FIGS. **10**A and **10**B, the shifting assembly **200** is located in the handle assembly **102** of the surgical instrument **100**. One or more sleeves **204** hold the ultrasonic transducer **130** in place within the housing **148**. The distal end of the ultrasonic transducer **130** includes threads **202** that are engaged by a worm gear **206**. As the worm gear **206** rotates the ultrasonic transducer **130** is urged in the direction indicated by the arrow **208** to thread the threaded coupling stud **156** into a threaded end of the ultrasonic transmission waveguide **145**. The worm gear **206** may be driven by a motor located within the handle assembly **102** of the surgical instrument **100**.

In one aspect, the shifting assembly **200** may include a torque limited motor driven attachment of the ultrasonic transmission waveguide **145** via the motor located in the handle assembly **102** that controls shaft actuation of clamping, rotation, and articulation. The shifting assembly **200** in the handle assembly **102** applies the proper torque onto the ultrasonic transmission waveguide **145** into place with a predetermined minimum torque. For instance, the handle assembly **102** may include a transducer torqueing mechanism which shifts the primary motor longitudinally uncoupling the primary drive shaft spur gear and coupling the transducer torqueing gear which rotates the shaft and nozzle therefore screwing the wave guide into the transducer.

FIG. **11** is a schematic diagram of one aspect of a electrical circuit **177** shown in FIG. **4**, suitable for driving an ultrasonic transducer **130**, according to one aspect of the present disclosure. The electrical circuit **177** comprises an analog multiplexer **180**. The analog multiplexer **180** multiplexes various signals from the upstream channels SCL-A/SDA-A such as ultrasonic, battery, and power control circuit. A current sensor **182** is coupled in series with the return or ground leg of the power supply circuit to measure the current supplied by the power supply. A field effect transistor (FET) temperature sensor **184** provides the ambient temperature. A pulse width modulation (PWM) watchdog timer **188** automatically generates a system reset if the main program neglects to periodically service it. It is provided to automatically reset the electrical circuit **177** when it hangs or freezes because of a software or hardware fault. It will be appreciated that the electrical circuit **177** may be configured as an RF driver circuit for driving the ultrasonic transducer **130** or for driving RF electrodes such as the electrical circuit **702** shown in FIG. **34**, for example. Accordingly, with reference now back to FIG. **11**, the electrical circuit **177** can be used to drive both ultrasonic transducers and RF electrodes interchangeably. If driven simultaneously, filter circuits may be provided in the corresponding first stage circuits **5504** to select either the ultrasonic waveform or the RF waveform. Such filtering techniques are described in commonly owned U.S. patent application Ser. No. 15/265,293, titled TECHNIQUES FOR CIRCUIT TOPOLOGIES FOR COMBINED GENERATOR, now U.S. Pat. No. 10,610,286, which is herein incorporated by reference in its entirety.

A drive circuit **186** provides left and right ultrasonic energy outputs. A digital signal the represents the signal waveform is provided to the SCL-A/SDA-A inputs of the analog multiplexer **180** from a control circuit, such as the control circuit **210** (FIG. **14**). A digital-to-analog converter **190** (DAC) converts the digital input to an analog output to drive a PWM circuit **192** coupled to an oscillator **194**. The PWM circuit **192** provides a first signal to a first gate drive circuit **196***a* coupled to a first transistor output stage **198***a* to drive a first ultrasonic (Left) energy output. The PWM circuit **192** also provides a second signal to a second gate drive circuit **196***b* coupled to a second transistor output stage **198***b* to drive a second ultrasonic (Right) energy output. A voltage sensor **199** is coupled between the Ultrasonic Left/Right output terminals to measure the output voltage. The drive circuit **186**, the first and second drive circuits **196***a*, **196***b*, and the first and second transistor output stages **198***a*, **198***b* define a first stage amplifier circuit. In operation, the control circuit **210** (FIG. **14**) generates a digital waveform

1800 (FIG. **67**) employing circuits such as direct digital synthesis (DDS) circuits **1500**, **1600** (FIGS. **65** and **66**). The DAC **190** receives the digital waveform **1800** and converts it into an analog waveform, which is received and amplified by the first stage amplifier circuit.

FIG. **12** is a schematic diagram of the transformer **166** coupled to the electrical circuit **177** shown in FIG. **11**, according to one aspect of the present disclosure. The Ultrasonic Left/Right input terminals (primary winding) of the transformer **166** are electrically coupled to the Ultrasonic Left/Right output terminals of the electrical circuit **177**. The secondary winding of the transformer **166** are coupled to the positive and negative electrodes **174***a*, **174***b*. The positive and negative electrodes **174***a*, **174***b* of the transformer **166** are coupled to the positive terminal **170***a* (Stack 1) and the negative terminal **170***b* (Stack 2) of the ultrasonic transducer **130** (FIG. **4**). In one aspect, the transformer **166** has a turns-ratio of n1:n2 of 1:50.

FIG. **13** is a schematic diagram of the transformer **166** shown in FIG. **12** coupled to a test circuit **165**, according to one aspect of the present disclosure. The test circuit **165** is coupled to the positive and negative electrodes **174***a*, **174***b*. A switch **167** is placed in series with an inductor/capacitor/resistor (LCR) load that simulates the load of an ultrasonic transducer.

FIG. **14** is a schematic diagram of a control circuit **210**, according to one aspect f the present disclosure. The control circuit **210** is located within a housing of the battery assembly **106**. The battery assembly **106** is the energy source for a variety of local power supplies **215**. The control circuit comprises a main processor **214** coupled via an interface master **218** to various downstream circuits by way of outputs SCL-A/SDA-A, SCL-B/SDA-B, SCL-C/SDA-C, for example. In one aspect, the interface master **218** is a general purpose serial interface such as an I^2C serial interface. The main processor **214** also is configured to drive switches **224** through general purposes input output **220** (GPIO), a display **226** (e.g., and LCD display), and various indicators **228** trough GPIO **222**. A watchdog processor **216** is provided to control the main processor **214**. A switch **230** is provided in series with the battery **211** to activate the control circuit **212** upon insertion of the battery assembly **106** into the handle assembly **102** (FIGS. **1-3**).

In one aspect, the main processor **214** is coupled to the electrical circuit **177** (FIGS. **4** and **11**) by way of output terminals SCL-A/SDA-A. The main processor **214** comprises a memory for storing tables of digitized drive signals or waveforms that are transmitted to the electrical circuit **177** for driving the ultrasonic transducer **130** (FIGS. **4-8**), for example. In other aspects, the main processor **214** may generate a digital waveform and transmit it to the electrical circuit **177** or may store the digital waveform for later transmission to the electrical circuit **177**. The main processor **214** also may provide RF drive by way of output terminals SCL-B/SDA-B and various sensors (e.g., Hall-effect sensors, magnetorheological fluid (MRF) sensors, etc.) by way of output terminals SCL-C/SDA-C. In one aspect, the main processor **214** is configured to sense the presence of ultrasonic drive circuitry and/or RF drive circuitry to enable appropriate software and user interface functionality.

In one aspect, the main processor **214** may be an LM 4F230H5QR, available from Texas Instruments, for example. In at least one example, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHZ, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle serial random access memory (SRAM), internal read-only memory (ROM) loaded with StellarisWare® software, 2 KB electrically erasable programmable read-only memory (EEPROM), one or more pulse width modulation (PWM) modules, one or more quadrature encoder inputs (QED analog, one or more 12-bit Analog-to-Digital Converters (ADC) with 12 analog input channels, among other features that are readily available for the product datasheet. Other processors may be readily substituted and, accordingly, the present disclosure should not be limited in this context.

FIG. **15** shows a simplified block circuit diagram illustrating another electrical circuit **300** contained within a modular ultrasonic surgical instrument **334**, according to one aspect of the present disclosure. The electrical circuit **300** includes a processor **302**, a clock **330**, a memory **326**, a power supply **304** (e.g., a battery), a switch **306**, such as a metal-oxide semiconductor field effect transistor (MOSFET) power switch, a drive circuit **308** (PLL), a transformer **310**, a signal smoothing circuit **312** (also referred to as a matching circuit and can be, e.g., a tank circuit), a sensing circuit **314**, a transducer **130**, and a shaft assembly **110** comprising an ultrasonic transmission waveguide that terminates at an ultrasonic blade **116**, which may be referred to herein simply as the waveguide.

One feature of the present disclosure that severs dependency on high voltage (120 VAC) input power (a characteristic of general ultrasonic cutting devices) is the utilization of low-voltage switching throughout the wave-forming process and the amplification of the driving signal only directly before the transformer stage. For this reason, in one aspect of the present disclosure, power is derived from only a battery, or a group of batteries, small enough to fit either within the handle assembly **102** (FIGS. **1-3**). State-of-the-art battery technology provides powerful batteries of a few centimeters in height and width and a few millimeters in depth. By combining the features of the present disclosure to provide a self-contained and self-powered ultrasonic device, a reduction in manufacturing cost may be achieved.

The output of the power supply **304** is fed to and powers the processor **302**. The processor **302** receives and outputs signals and, as will be described below, functions according to custom logic or in accordance with computer programs that are executed by the processor **302**. The electrical circuit **300** can also include a memory **326**, preferably, random access memory (RAM), that stores computer-readable instructions and data.

The output of the power supply **304** also is directed to a switch **306** having a duty cycle controlled by the processor **302**. By controlling the on-time for the switch **306**, the processor **302** is able to dictate the total amount of power that is ultimately delivered to the transducer **316**. In one aspect, the switch **306** is a MOSFET, although other switches and switching configurations are adaptable as well. The output of the switch **306** is fed to a drive circuit **308** that contains, for example, a phase detecting phase-locked loop (PLL) and/or a low-pass filter and/or a voltage-controlled oscillator. The output of the switch **306** is sampled by the processor **302** to determine the voltage and current of the output signal (V IN and I IN, respectively). These values are used in a feedback architecture to adjust the pulse width modulation of the switch **306**. For instance, the duty cycle of the switch **306** can vary from about 20% to about 80%, depending on the desired and actual output from the switch **306**.

The drive circuit **308**, which receives the signal from the switch **306**, includes an oscillatory circuit that turns the output of the switch **306** into an electrical signal having an ultrasonic frequency, e.g., 55 kHz (VCO). As explained above, a smoothed-out version of this ultrasonic waveform is ultimately fed to the ultrasonic transducer **130** to produce a resonant sine wave along the ultrasonic transmission waveguide **145** (FIG. **2**).

At the output of the drive circuit **308** is a transformer **310** that is able to step up the low voltage signal(s) to a higher voltage. It is noted that upstream switching, prior to the transformer **310**, is performed at low (e.g., battery driven) voltages, something that, to date, has not been possible for ultrasonic cutting and cautery devices. This is at least partially due to the fact that the device advantageously uses low on-resistance MOSFET switching devices. Low on-resistance MOSFET switches are advantageous, as they produce lower switching losses and less heat than a traditional MOSFET device and allow higher current to pass through. Therefore, the switching stage (pre-transformer) can be characterized as low voltage/high current. To ensure the lower on-resistance of the amplifier MOSFET(s), the MOSFET(s) are run, for example, at 10 V. In such a case, a separate 10 VDC power supply can be used to feed the MOSFET gate, which ensures that the MOSFET is fully on and a reasonably low on resistance is achieved. In one aspect of the present disclosure, the transformer **310** steps up the battery voltage to 120V root-mean-square (RMS). Transformers are known in the art and are, therefore, not explained here in detail.

In the circuit configurations described, circuit component degradation can negatively impact the circuit performance of the circuit. One factor that directly affects component performance is heat. Known circuits generally monitor switching temperatures (e.g., MOSFET temperatures). However, because of the technological advancements in MOSFET designs, and the corresponding reduction in size, MOSFET temperatures are no longer a valid indicator of circuit loads and heat. For this reason, according to one aspect of the present disclosure, a sensing circuit **314** senses the temperature of the transformer **310**. This temperature sensing is advantageous as the transformer **310** is run at or very close to its maximum temperature during use of the device. Additional temperature will cause the core material, e.g., the ferrite, to break down and permanent damage can occur. The present disclosure can respond to a maximum temperature of the transformer **310** by, for example, reducing the driving power in the transformer **310**, signaling the user, turning the power off, pulsing the power, or other appropriate responses.

In one aspect of the present disclosure, the processor **302** is communicatively coupled to the end effector **112**, which is used to place material in physical contact with the ultrasonic blade **116**, e.g., the clamping mechanism shown in FIG. **1**. Sensors are provided that measure, at the end effector **112**, a clamping force value (existing within a known range) and, based upon the received clamping force value, the processor **302** varies the motional voltage VM. Because high force values combined with a set motional rate can result in high blade temperatures, a temperature sensor **336** can be communicatively coupled to the processor **302**, where the processor **302** is operable to receive and interpret a signal indicating a current temperature of the blade from the temperature sensor **336** and to determine a target frequency of blade movement based upon the received temperature. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to the trigger **108** to measure the force applied to the trigger **108** by the user. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to the switch **120** button such that displacement intensity corresponds to the force applied by the user to the switch **120** button.

According to one aspect of the present disclosure, the PLL portion of the drive circuit **308**, which is coupled to the processor **302**, is able to determine a frequency of waveguide movement and communicate that frequency to the processor **302**. The processor **302** stores this frequency value in the memory **326** when the device is turned off. By reading the clock **330**, the processor **302** is able to determine an elapsed time after the device is shut off and retrieve the last frequency of waveguide movement if the elapsed time is less than a predetermined value. The device can then start up at the last frequency, which, presumably, is the optimum frequency for the current load.

FIG. **16** shows a battery assembly **400** for use with the surgical instrument **100**, according to one aspect of the present disclosure. The battery assembly **400** comprises a housing **402** sized and configured to contain various energy cells. The energy cells may include rechargeable and non-rechargeable batteries. In one aspect, the battery assembly **400** includes four Li-ion non-rechargeable batteries **404***a*, **404***b*, **404***c*, **404***d* and two nickel metal hydride (NiMH) rechargeable batteries **406***a* (the second battery is not shown). The housing **402** comprises tabs **408***a*, **408***b* to removably connect the battery assembly **400** to the handle assembly **102** of the surgical instrument **100** (FIGS. **1** and **2**).

FIG. **17** shows a disposable battery assembly **410** for use with the surgical instrument **100**, according to one aspect of the present disclosure. In one aspect, the disposable battery assembly **410** comprises a primary cell battery pack for use with a battery powered advanced energy instrument such as the surgical instrument **100** (FIGS. **1** and **2**), comprising compensating electronics with additional voltage to offset a voltage sag from the disposable battery assembly **410** to prevent the output voltage from sagging below a predetermined level during operation under load. The disposable battery assembly **410** comprises a housing **412** sized and configured to contain various energy cells. The energy cells may include rechargeable and non-rechargeable batteries. In one aspect, the disposable battery assembly **410** includes four primary Lithium-ion (Li-ion) non-rechargeable batteries **414***a*, **414***b*, **414***c*, **414***d* and two secondary NiMH or Nickel Cadmium (NiCd) rechargeable batteries **416***a*, **416***b*. The housing **412** comprises electrical contact **418** to electrically couple the disposable battery assembly **410** to the handle assembly **102** of the surgical instrument **100**. In the illustrated example the electrical contact **418** comprises four metal contacts. The disposable battery assembly **410** also includes electrical circuits **419** such as the control circuit **210** (FIG. **14**) and/or the electrical circuit **300** (FIG. **15**). The electrical circuits **419** are radiated hardened.

In one aspect, the disposable battery assembly **410** includes batteries **414***a-d*, electrical circuits **419**, and other componentry that is resistant to gamma or other radiation sterilization. For instance, a switching mode power supply **460** (FIG. **22**) or a linear power supply **470** (FIG. **24**) and an optional charge circuit may be incorporated within the housing **412** of the disposable battery assembly **410** to reduce voltage sag of the primary Li-ion batteries **414***a-d* and to allow the secondary NiMH batteries **416***a*, **416***b* to be used to reduce voltage sag. This guarantees full charged cells at the beginning of each surgery that are easy to introduce into the sterile field. A dual type battery assembly including primary Li-ion batteries **414***a-d* and secondary NiMH batteries **416***a-b* can be used with dedicated energy cells **416***a-b* to control the electronics from dedicated energy cells **414***a-d* that run the generator and motor control circuits. In one aspect, the system pulls from the batteries involved in driving the electronics circuits in the case that batteries involved are dropping low. In one aspect, the system would include a one way diode system that would not allow for current to flow in the opposite direction, for example, from the batteries involved in driving the energy and/or motor control circuits to the batteries involved in driving the electronic circuits. In one additional aspect, the system may comprise a gamma friendly charge circuit and switch mode power supply using diodes and vacuum tube components that would minimize voltage sag at a predetermined level. The switch mode power supply may be eliminated by including a minimum sag voltage that is a division of the NiMH voltages (e.g., three NiMH cells). In another aspect, a modular system can be made wherein the radiation hardened components are located in a module, making this module sterilizable by radiation sterilization. Other non-radiation hardened components are included in other modular components and connections are made between the modular components such that the componentry operate together as if the components were located together on the same circuit board. If only two cells of the secondary NiMH batteries **416***a-b* are desired the switch mode power supply based on diodes and vacuum tubes allows for sterilizable electronics within the disposable primary Li-ion batteries **414***a-d*.

FIG. **18** shows a reusable battery assembly **420** for use with the surgical instrument **100**, according to one aspect of the present disclosure. The reusable battery assembly **420** comprises a housing **422** sized and configured to contain various rechargeable energy cells. The energy cells may include rechargeable batteries. In one aspect, the reusable battery assembly **420** includes five laminated NiMH rechargeable batteries **424***a*, **424***b*, **424***c*, **424***d*, **424***e*. The housing **422** comprises electrical contact **428** to electrically couple the reusable battery assembly **420** to the handle assembly **102** of the surgical instrument **100** (FIGS. **1** and **2**). In the illustrated example, the electrical contact **428** comprises six metal contacts. The reusable battery assembly **420** also includes up to six circuit boards **429***a*, **429***b*, **429***c*, **429***d*, **429***e*, **429***f* that may include electrical circuits such as the control circuit **210** (FIG. **14**) and/or the electrical circuit **300** (FIG. **15**). In one aspect, the reusable battery assembly **420** comprises drive FET transistors and associated circuitry **429***a-f* in the housing **422** for easy swap and no need to shut down the surgical instrument **100** (FIGS. **1** and **2**) to replace the reusable battery assembly **420** with energy delivery.

The reusable battery assembly **420** comprises a battery test switch **426** and up to three LED indicators **427***a*, **427***b*, **427***c* to determine the health of the batteries **424***a-e* in the reusable battery assembly **420**. The first LED indicator **427***a* may indicate fully charged batteries **424***a-e* that is ready to use. The second LED indicator **427***b* may indicate that the battery needs to be recharged. The third LED indicator **427***c* may indicate that battery is not good and to dispose. The reusable battery assembly **420** health indication to allow the user to determine the specific health and capabilities of the batteries **424***a-e* before it is inserted and used. For instance, charge status of the rechargeable secondary cells, sag voltage, primary cell voltage are checked by the activation of the battery test switch **426** which could measure these in an unload state or with a predefined resistive load placed on the system. The voltages could have at least one but more preferably three thresholds to compare the resulting voltages checks to. In the case of the first indicator **427***a*, the batteries **424***a-e* indicating whether or not they are suitable to use. With three levels the reusable battery assembly **420** could display full charge, minimum charge, and some marginal but limited charge status. This battery **424***a-e* health monitor would be useful for either the disposable battery assembly **410** (FIG. **17**) or the reusable battery assembly **420**. In the case of the disposable battery assembly **410** it is a ready/damaged indicator. In the case of the reusable battery assembly **420** it could indicate life remaining, recharge capacity, even age before failure in addition to ready/not ready.

FIG. **19** is an elevated perspective view of a removable battery assembly **430** with both halves of the housing shell removed exposing battery cells coupled to multiple circuit boards which are coupled to the multi-lead battery terminal in accordance with an aspect of the present disclosure. Further, more than or less than three circuit boards is possible to provide expanded or limited functionality. As shown in FIG. **19**, the multiple circuit boards **432**, **434**, **436** may be positioned in a stacked architecture, which provides a number of advantages. For example, due to the smaller layout size, the circuit boards have a reduced footprint within the removable battery assembly **430**, thereby allowing for a smaller battery. In addition, in this configuration, is possible to easily isolate power boards from digital boards to prevent any noise originating from the power boards to cause harm to the digital boards. Also, the stacked configuration allows for direct connect features between the boards, thereby reducing the presence of wires. Furthermore, the circuit boards can be configured as part of a rigid-flex-rigid circuit to allow the rigid parts to be "fanned" into a smaller volumetric area.

According to aspects of the present disclosure, the circuit board **432**, **434**, **436** provides a specific function. For instance, one circuit board **432** can provide the components for carrying out the battery protection circuitry. Similarly, another circuit board **434** can provide the components for carrying out the battery controller. Another circuit board **436** can, for example, provide high power buck controller components. Finally, the battery protection circuitry can provide connection paths for coupling the battery cells **438***a-n*. By placing the circuit boards in a stacked configuration and separating the boards by their respective functions, the boards may be strategically placed in a specific order that best handles their individual noise and heat generation. For example, the circuit board having the high-power buck controller components produces the most heat and, therefore, it can be isolated from the other boards and placed in the center of the stack. In this way, the heat can be kept away from the outer surface of the device in an effort to prevent the heat from being felt by the physician or operator of the device. In addition, the battery board grounds may be configured in a star topology with the center located at the buck controller board to reduce the noise created by ground loops.

The strategically stacked circuit boards, the low thermal conductivity path from the circuit boards to the multi-lead battery terminal assembly, and a flex circuit **3516** are features that assist in preventing heat from reaching the exterior surface of the device. The battery cells and buck components are thermally connected to a flex circuit within the handle assembly **102** (FIGS. **1** and **2**) so that the heat generated by the cells and buck components enter a portion away from the physician's hand. The flex circuit presents a relatively high thermal mass, due to its broad area of exposure and the advantageous conduction characteristics of the copper, which redirects, absorbs, and/or dissipates heat across a broader area thereby slowing the concentration of heat and limiting high spot temperatures on the exterior surface of the device. Other techniques may be implemented as well, including, but not limited to, larger heat wells, sinks or insulators, a metal connector cap and heavier copper content in the flex circuit or the handle assembly **102** of the device.

Another advantage of the removable battery assembly **430** is realized when Li-ion batteries are used. As previously stated, Li-ion batteries should not be charged in a parallel configuration of multiple cells. This is because, as the voltage increases in a particular cell, it begins to accept additional charge faster than the other lower-voltage cells. Therefore, the cells are monitored so that a charge to that cell can be controlled individually. When a Li-ion battery is formed from a group of cells **438***a-n*, a multitude of wires extending from the exterior of the device to the batteries **438***a-n* is needed (at least one additional wire for each battery cell beyond the first). By having a removable battery assembly **430**, a battery cell **438***a-n* can, in one aspect, have its own exposed set of contacts and, when the removable battery assembly **430** is not present inside the handle assembly **102** (FIGS. **1** and **2**), a set of contacts can be coupled to a corresponding set of contacts in an external, non-sterile, battery-charging device. In another aspect, a battery cell **438***a-n* can be electrically connected to the battery protection circuitry to allow the battery protection circuitry to control and regulate recharging of a cell **438***a-n*. The removable battery assembly **430** is provided with circuitry to prevent use of the removable battery assembly **430** past an expected term-of-life. This term is not only dictated by the cells but is also dictated by the outer surfaces, including the battery casing or shell and the upper contact assembly. Such circuitry will be explained in further detail below and includes, for example, a use count, a recharge count, and an absolute time from manufacture count.

FIG. **19** also shows a multi-lead battery terminal assembly **433**, which is an interface that electrically couples the components within the removable battery assembly **430** to an electrical interface of the handle assembly **102** (FIGS. **1** and **2**). It is through the handle assembly **102** that the removable battery assembly **430** is able to electrically (and mechanically) couple with the ultrasonic transducer/generator assembly **104** (FIG. **4**). As is explained above, the removable battery assembly **430**, through the multi-lead battery terminal assembly **433**, provides power to the surgical instrument **100** (FIGS. **1** and **2**), as well as other functionality described herein. The multi-lead battery terminal assembly **433** includes a plurality of contacts pads **435***a-n* capable of separately electrically connecting a terminal within the removable battery assembly **430** to another terminal provided by a docking bay of the handle assembly **102**. One example of such electrical connections coupled to the plurality of contact pads **435***a-n* as power and communication signal paths. In the aspect of the multi-lead battery terminal assembly **433**, sixteen different contact pads **435***a-n* are shown. This number is merely illustrative. In an aspect, an interior side of the battery terminal assembly **433** has a well formed on the molded terminal holder that can be filled with potting materials to create a gas tight seal. The contact pads **435***a-n* are overmolded in the lid and extend through the potting well into the interior of the battery **430**. Here a flex circuit can be used to rearrange the array of pins and provide an electrical connection to the circuit boards. In one example, a 4×4 array is converted to a 2×8 array. In one example the multi-lead battery terminal assembly **433**, a plurality of contact pads **435***a-n* of the multi-lead battery terminal assembly **2804** include a corresponding plurality of interior contact pins **437***a-n*. A contact pin **437***a* provides a direct electrical coupling to a corresponding one of the contact pads **435***a*.

FIG. **20** illustrates a battery test circuit **440**, according to one aspect of the present disclosure. The battery test circuit **440** includes the battery test switch **426** as described in FIG. **18**. The battery test switch **426** is a switch that engages an LCR dummy load that simulates a transducer or shaft assembly electronics. As described in FIG. **18**, additional indicator circuits may be coupled to the battery test circuit **440** to provide a suitable indication of the capacity of the batteries in the reusable battery assembly **420**. The illustrated battery test circuit **440** may be employed in any of the battery assemblies **400**, **410**, **420**, **430** described in connection with FIGS. **16-19**, respectively.

FIG. **21** illustrates a supplemental power source circuit **450** to maintain a minimum output voltage, according to one aspect of the present disclosure. The supplemental power source circuit **450** may be included in any of the battery assemblies **400**, **410**, **420**, **430** described in connection with FIGS. **16-19**. The supplemental power source circuit **450** prevents the output voltage V_o from sagging under high load conditions. The supplemental power source circuit **450** includes a set of four primary batteries **452***a-b*, **452***c-d* (up to n batteries may be used) that are activated when the switch **453** closes upon insertion of the battery assembly **400**, **410**, **420**, **430** into the handle assembly **102** of the surgical instrument **100** (FIGS. **1** and **2**). The primary batteries **452***a-d* may be Li-ion batteries such as CR123A Li-ion batteries. Under load, the primary batteries **452***a-d* provide the output voltage V_o while the secondary rechargeable battery **454** is charged by the battery charger **455**. In one aspect, the secondary rechargeable battery **454** in a NiMH battery and the battery charger **455** is a suitable NiMH charger. When the output voltage V_o sags or droops due to high load conditions the voltage V_x operates the switch mode power supply **456** to restore the output voltage V_o by supplying the additional current into the load. The diode **458** is provided to prevent current from flowing into the output of the switch mode power supply **456**. Accordingly, the output voltage V_o of the switch mode power supply **456** must exceed the voltage drop across the diode **458** (~0.7V) before the supplemental current can flow into the load. Optionally, a battery test switch **459** and test resistor R_{Test} may be provided to test the supplemental power source circuit **450** under load conditions. In particular, in view of FIG. **21**, the battery assemblies **400**, **410**, **420**, **430** may comprise a test circuit **457***a* comprising a switch **457***b* and a resistor **457***c* such that when the switch **457***b* is closed (e.g., via the test button **426**), the resistor **457***c* tests whether the primary batteries **452***a-d* are capable of delivering the output voltage V_o. Otherwise, the resistor **457** tests whether the secondary battery **454**, via operation of the switch mode power supply **456**, is capable of delivering a V_b such that supplemental current passing through the diode **458** restores the output voltage V_o.

FIG. **22** illustrates a switch mode power supply circuit **460** for supplying energy to the surgical instrument **100**, according to one aspect of the present disclosure. The switch mode power supply circuit **460** may be disposed within any one of the battery assemblies **400**, **410**, **430** described in connection with FIGS. **16**, **17**, and **19**, respectively. In the illustrated example, the switch mode power supply circuit **460** comprises primary Li cell batteries **429***a-d* where the positive (+) output voltage is coupled to an input terminal V_{IN} of a switching regulator **464**. It will be appreciated that any suitable number of primary cells may be employed. The switch mode power supply circuit **460** includes a remote ON/OFF switch. The input V_{IN} of the switching regulator **464** also includes an input filter represented by capacitor C_i. The output V_{OUT} of the switching regulator **464** is coupled to an inductor L and an output filter represented by capacitor C_o. A catch diode D is disposed between V_{OUT} and ground. A feedback signal is provided from the output filter C_o to the FB input of the switching regulator **464**. A load resistor R_L represents a load. In one aspect, the minimum load is about 200 mA. In one aspect, the output voltage V_{OUT} is 3.3 VDC at 800 mA.

FIG. **23** illustrates a discrete version of the switching regulator **464** shown in FIG. **22** for supplying energy to the surgical instrument **100**, according to one aspect of the present disclosure. The switching regulator **464** receives the input voltage from a battery assembly **400**, **410**, **420**, **430** at the V_{IN} terminal. The signal at the ON/OFF input enables or disables the operation of the switching regulator **464** by controlling the state of the switch **471**. A feedback signal is received from the load at the FB input where is divided by a voltage divider circuit **463**. The voltage from the voltage divider **463** is applied to the positive input of a fixed gain amplifier **465**. The negative input of the fixed gain amplifier **465** is coupled to a bandgap reference diode **469** (e.g., 1.23V). The amplified output of the fixed gain amplifier **465** is applied to the positive input of a comparator **466**. The negative input of the comparator **466** receives a 50 KHz oscillator **467** input. The output of the comparator **466** is applied to a driver **468** which drives and output transistor **461**. The output transistor **461** supplies voltage and current to the load via the V_{OUT} terminal.

FIG. **24** illustrates a linear power supply circuit **470** for supplying energy to the surgical instrument **100**, according to one aspect of the present disclosure. The linear power supply circuit **470** may be disposed within any one of the battery assemblies **400**, **410**, **420**, **430** described in connection with FIGS. **16**, **17**, **18**, and **19**, respectively. In the illustrated example, the linear power supply circuit **470** comprises primary Li-ion cell batteries **462***a-d* where the positive (+) output voltage is coupled to the V_{IN} terminal of transistor **472**. The output of the transistor **472** supplies the current and voltage to the load via the V_{OUT} terminal of the linear power supply circuit **470**. An input filter C_i is provided at the input side and an output filter C_o is provided at an output side. A Zener diode D_Z applies a regulated voltage to the base of the transistor **472**. A bias resistor biases the Zener diode D_Z and the transistor **472**.

FIG. **25** is an elevational exploded view of modular handheld ultrasonic surgical instrument **480** showing the left shell half removed from a handle assembly **482** exposing a device identifier communicatively coupled to the multi-lead handle terminal assembly in accordance with one aspect of the present disclosure. In additional aspects of the present disclosure, an intelligent or smart battery is used to power the modular handheld ultrasonic surgical instrument **480**. However, the smart battery is not limited to the modular handheld ultrasonic surgical instrument **480** and, as will be explained, can be used in a variety of devices, which may or may not have power requirements (e.g., current and voltage) that vary from one another. The smart battery assembly **486**, in accordance with one aspect of the present disclosure, is advantageously able to identify the particular device to which it is electrically coupled. It does this through encrypted or unencrypted identification methods. For instance, a smart battery assembly **486** can have a connection portion, such as connection portion **488**. The handle assembly **482** can also be provided with a device identifier communicatively coupled to the multi-lead handle terminal assembly **491** and operable to communicate at least one piece of information about the handle assembly **482**. This information can pertain to the number of times the handle assembly **482** has been used, the number of times an ultrasonic transducer/generator assembly **484** (presently disconnected from the handle assembly **482**) has been used, the number of times a waveguide shaft assembly **490** (presently connected to the handle assembly **482**) has been used, the type of the waveguide shaft assembly **490** that is presently connected to the handle assembly **482**, the type or identity of the ultrasonic transducer/generator assembly **484** that is presently connected to the handle assembly **482**, and/or many other characteristics. When the smart battery assembly **486** is inserted in the handle assembly **482**, the connection portion **488** within the smart battery assembly **486** makes communicating contact with the device identifier of the handle assembly **482**. The handle assembly **482**, through hardware, software, or a combination thereof, is able to transmit information to the smart battery assembly **486** (whether by self-initiation or in response to a request from the smart battery assembly **486**). This communicated identifier is received by the connection portion **488** of the smart battery assembly **486**. In one aspect, once the smart battery assembly **486** receives the information, the communication portion is operable to control the output of the smart battery assembly **486** to comply with the device's specific power requirements.

In one aspect, the communication portion includes a processor **493** and a memory **497**, which may be separate or a single component. The processor **493**, in combination with the memory, is able to provide intelligent power management for the modular handheld ultrasonic surgical instrument **480**. This aspect is particularly advantageous because an ultrasonic device, such as the modular handheld ultrasonic surgical instrument **480**, has a power requirement (frequency, current, and voltage) that may be unique to the modular handheld ultrasonic surgical instrument **480**. In fact, the modular handheld ultrasonic surgical instrument **480** may have a particular power requirement or limitation for one dimension or type of outer tube **494** and a second different power requirement for a second type of waveguide having a different dimension, shape, and/or configuration.

A smart battery assembly **486**, according to one aspect of the present disclosure, therefore, allows a battery assembly to be used amongst several surgical instruments. Because the smart battery assembly **486** is able to identify to which device it is attached and is able to alter its output accordingly, the operators of various different surgical instruments utilizing the smart battery assembly **486** no longer need be concerned about which power source they are attempting to install within the electronic device being used. This is particularly advantageous in an operating environment where a battery assembly needs to be replaced or interchanged with another surgical instrument in the middle of a complex surgical procedure.

In a further aspect of the present disclosure, the smart battery assembly **486** stores in a memory **497** a record of each time a particular device is used. This record can be useful for assessing the end of a device's useful or permitted life. For instance, once a device is used 20 times, such batteries in the smart battery assembly **486** connected to the device will refuse to supply power thereto—because the device is defined as a "no longer reliable" surgical instrument. Reliability is determined based on a number of factors. One factor can be wear, which can be estimated in a number of ways including the number of times the device has been used or activated. After a certain number of uses, the parts of the device can become worn and tolerances between parts exceeded. For instance, the smart battery assembly **486** can sense the number of button pushes received by the handle assembly **482** and can determine when a maximum number of button pushes has been met or exceeded. The smart battery assembly **486** can also monitor an impedance of the button mechanism which can change, for instance, if the handle gets contaminated, for example, with saline.

This wear can lead to an unacceptable failure during a procedure. In some aspects, the smart battery assembly **486** can recognize which parts are combined together in a device and even how many uses a part has experienced. For instance, if the smart battery assembly **486** is a smart battery according to the present disclosure, it can identify the handle assembly **482**, the waveguide shaft assembly **490**, as well as the ultrasonic transducer/generator assembly **484**, well before the user attempts use of the composite device. The memory **497** within the smart battery assembly **486** can, for example, record a time when the ultrasonic transducer/generator assembly **484** is operated, and how, when, and for how long it is operated. If the ultrasonic transducer/generator assembly **484** has an individual identifier, the smart battery assembly **486** can keep track of uses of the ultrasonic transducer/generator assembly **484** and refuse to supply power to that the ultrasonic transducer/generator assembly **484** once the handle assembly **482** or the ultrasonic transducer/generator assembly **484** exceeds its maximum number of uses. The ultrasonic transducer/generator assembly **484**, the handle assembly **482**, the waveguide shaft assembly **490**, or other components can include a memory chip that records this information as well. In this way, any number of smart batteries in the smart battery assembly **486** can be used with any number of ultrasonic transducer/generator assemblies **484**, staplers, vessel sealers, etc. and still be able to determine the total number of uses, or the total time of use (through use of the clock), or the total number of actuations, etc. of the ultrasonic transducer/generator assembly **484**, the stapler, the vessel sealer, etc. or charge or discharge cycles. Smart functionality may reside outside the battery assembly **486** and may reside in the handle assembly **482**, the ultrasonic transducer/generator assembly **484**, and/or the shaft assembly **490**, for example.

When counting uses of the ultrasonic transducer/generator assembly **484**, to intelligently terminate the life of the ultrasonic transducer/generator assembly **484**, the surgical instrument accurately distinguishes between completion of an actual use of the ultrasonic transducer/generator assembly **484** in a surgical procedure and a momentary lapse in actuation of the ultrasonic transducer/generator assembly **484** due to, for example, a battery change or a temporary delay in the surgical procedure. Therefore, as an alternative to simply counting the number of activations of the ultrasonic transducer/generator assembly **484**, a real-time clock (RTC) circuit can be implemented to keep track of the amount of time the ultrasonic transducer/generator assembly **484** actually is shut down. From the length of time measured, it can be determined through appropriate logic if the shutdown was significant enough to be considered the end of one actual use or if the shutdown was too short in time to be considered the end of one use. Thus, in some applications, this method may be a more accurate determination of the useful life of the ultrasonic transducer/generator assembly **484** than a simple "activations-based" algorithm, which for example, may provide that ten "activations" occur in a surgical procedure and, therefore, ten activations should indicate that the counter is incremented by one. Generally, this type and system of internal clocking will prevent misuse of the device that is designed to deceive a simple "activations-based" algorithm and will prevent incorrect logging of a complete use in instances when there was only a simple de-mating of the ultrasonic transducer/generator assembly **484** or the smart battery assembly **486** that was required for legitimate reasons.

Although the ultrasonic transducer/generator assemblies **484** of the surgical instrument **480** are reusable, in one aspect a finite number of uses may be set because the surgical instrument **480** is subjected to harsh conditions during cleaning and sterilization. More specifically, the battery pack is configured to be sterilized. Regardless of the material employed for the outer surfaces, there is a limited expected life for the actual materials used. This life is determined by various characteristics which could include, for example, the amount of times the pack has actually been sterilized, the time from which the pack was manufactured, and the number of times the pack has been recharged, to name a few. Also, the life of the battery cells themselves is limited. Software of the present disclosure incorporates inventive algorithms that verify the number of uses of the ultrasonic transducer/generator assembly **484** and smart battery assembly **486** and disables the device when this number of uses has been reached or exceeded. Analysis of the battery pack exterior in each of the possible sterilizing methods can be performed. Based on the harshest sterilization procedure, a maximum number of permitted sterilizations can be defined and that number can be stored in a memory of the smart battery assembly **486**. If it is assumed that a charger is non-sterile and that the smart battery assembly **486** is to be used after it is charged, then the charge count can be defined as being equal to the number of sterilizations encountered by that particular pack.

In one aspect, the hardware in the battery pack may be to disabled to minimize or eliminate safety concerns due to continuous drain in from the battery cells after the pack has been disabled by software. A situation can exist where the battery's internal hardware is incapable of disabling the battery under certain low voltage conditions. In such a situation, in an aspect, the charger can be used to "kill" the battery. Due to the fact that the battery microcontroller is OFF while the battery is in its charger, a non-volatile, System Management Bus (SMB) based electrically erasable programmable read only memory (EEPROM) can be used to exchange information between the battery microcontroller and the charger. Thus, a serial EEPROM can be used to store information that can be written and read even when the battery microcontroller is OFF, which is very beneficial when trying to exchange information with the charger or other peripheral devices. This example EEPROM can be configured to contain enough memory registers to store at least (a) a use-count limit at which point the battery should be disabled (Battery Use Count), (b) the number of procedures the battery has undergone (Battery Procedure Count), and/or (c) a number of charges the battery has undergone (Charge Count), to name a few. Some of the information stored in the EEPROM, such as the Use Count Register and Charge Count Register are stored in write-protected sections of the EEPROM to prevent users from altering the information. In an aspect, the use and counters are stored with corresponding bit-inverted minor registers to detect data corruption.

Any residual voltage in the SMBus lines could damage the microcontroller and corrupt the SMBus signal. Therefore, to ensure that the SMBus lines of the battery controller

703 do not carry a voltage while the microcontroller is OFF, relays are provided between the external SMBus lines and the battery microcontroller board.

During charging of the smart battery assembly **486**, an "end-of-charge" condition of the batteries within the smart battery assembly **486** is determined when, for example, the current flowing into the battery falls below a given threshold in a tapering manner when employing a constant-current/constant-voltage charging scheme. To accurately detect this "end-of-charge" condition, the battery microcontroller and buck boards are powered down and turned OFF during charging of the battery to reduce any current drain that may be caused by the boards and that may interfere with the tapering current detection. Additionally, the microcontroller and buck boards are powered down during charging to prevent any resulting corruption of the SMBus signal.

With regard to the charger, in one aspect the smart battery assembly **486** is prevented from being inserted into the charger in any way other than the correct insertion position. Accordingly, the exterior of the smart battery assembly **486** is provided with charger-holding features. A cup for holding the smart battery assembly **486** securely in the charger is configured with a contour-matching taper geometry to prevent the accidental insertion of the smart battery assembly **486** in any way other than the correct (intended) way. It is further contemplated that the presence of the smart battery assembly **486** may be detectable by the charger itself. For example, the charger may be configured to detect the presence of the SMBus transmission from the battery protection circuit, as well as resistors that are located in the protection board. In such case, the charger would be enabled to control the voltage that is exposed at the charger's pins until the smart battery assembly **486** is correctly seated or in place at the charger. This is because an exposed voltage at the charger's pins would present a hazard and a risk that an electrical short could occur across the pins and cause the charger to inadvertently begin charging.

In some aspects, the smart battery assembly **486** can communicate to the user through audio and/or visual feedback. For example, the smart battery assembly **486** can cause the LEDs to light in a pre-set way. In such a case, even though the microcontroller in the ultrasonic transducer/generator assembly **484** controls the LEDs, the microcontroller receives instructions to be carried out directly from the smart battery assembly **486**.

In yet a further aspect of the present disclosure, the microcontroller in the ultrasonic transducer/generator assembly **484**, when not in use for a predetermined period of time, goes into a sleep mode. Advantageously, when in the sleep mode, the clock speed of the microcontroller is reduced, cutting the current drain significantly. Some current continues to be consumed because the processor continues pinging waiting to sense an input. Advantageously, when the microcontroller is in this power-saving sleep mode, the microcontroller and the battery controller can directly control the LEDs. For example, a decoder circuit could be built into the ultrasonic transducer/generator assembly **484** and connected to the communication lines such that the LEDs can be controlled independently by the processor **493** while the ultrasonic transducer/generator assembly **484** microcontroller is "OFF" or in a "sleep mode." This is a power-saving feature that eliminates the need for waking up the microcontroller in the ultrasonic transducer/generator assembly **484**. Power is conserved by allowing the generator to be turned off while still being able to actively control the user-interface indicators.

Another aspect slows down one or more of the microcontrollers to conserve power when not in use. For example, the clock frequencies of both microcontrollers can be reduced to save power. To maintain synchronized operation, the microcontrollers coordinate the changing of their respective clock frequencies to occur at about the same time, both the reduction and, then, the subsequent increase in frequency when full speed operation is required. For example, when entering the idle mode, the clock frequencies are decreased and, when exiting the idle mode, the frequencies are increased.

In an additional aspect, the smart battery assembly **486** is able to determine the amount of usable power left within its cells and is programmed to only operate the surgical instrument to which it is attached if it determines there is enough battery power remaining to predictably operate the device throughout the anticipated procedure. For example, the smart battery assembly **486** is able to remain in a non-operational state if there is not enough power within the cells to operate the surgical instrument for 20 seconds. According to one aspect, the smart battery assembly **486** determines the amount of power remaining within the cells at the end of its most recent preceding function, e.g., a surgical cutting. In this aspect, therefore, the smart battery assembly **486** would not allow a subsequent function to be carried out if, for example, during that procedure, it determines that the cells have insufficient power. Alternatively, if the smart battery assembly **486** determines that there is sufficient power for a subsequent procedure and goes below that threshold during the procedure, it would not interrupt the ongoing procedure and, instead, will allow it to finish and thereafter prevent additional procedures from occurring.

The following explains an advantage to maximizing use of the device with the smart battery assembly **486** of the present disclosure. In this example, a set of different devices have different ultrasonic transmission waveguides. By definition, the waveguides could have a respective maximum allowable power limit where exceeding that power limit overstresses the waveguide and eventually causes it to fracture. One waveguide from the set of waveguides will naturally have the smallest maximum power tolerance. Because prior-art batteries lack intelligent battery power management, the output of prior-art batteries must be limited by a value of the smallest maximum allowable power input for the smallest/thinnest/most-frail waveguide in the set that is envisioned to be used with the device/battery. This would be true even though larger, thicker waveguides could later be attached to that handle and, by definition, allow a greater force to be applied. This limitation is also true for maximum battery power. For example, if one battery is designed to be used in multiple devices, its maximum output power will be limited to the lowest maximum power rating of any of the devices in which it is to be used. With such a configuration, one or more devices or device configurations would not be able to maximize use of the battery because the battery does not know the particular device's specific limits.

In one aspect, the smart battery assembly **486** may be employed to intelligently circumvent the above-mentioned ultrasonic device limitations. The smart battery assembly **486** can produce one output for one device or a particular device configuration and the same smart battery assembly **486** can later produce a different output for a second device or device configuration. This universal smart battery surgical system lends itself well to the modern operating room where space and time are at a premium. By having a smart battery pack operate many different devices, the nurses can easily manage the storage, retrieval, and inventory of these packs.

Advantageously, in one aspect the smart battery system according to the present disclosure may employ one type of charging station, thus increasing ease and efficiency of use and decreasing cost of surgical room charging equipment.

In addition, other surgical instruments, such as an electric stapler, may have a different power requirement than that of the modular handheld ultrasonic surgical instrument **480**. In accordance with various aspects of the present disclosure, a smart battery assembly **486** can be used with any one of a series of surgical instruments and can be made to tailor its own power output to the particular device in which it is installed. In one aspect, this power tailoring is performed by controlling the duty cycle of a switched mode power supply, such as buck, buck-boost, boost, or other configuration, integral with or otherwise coupled to and controlled by the smart battery assembly **486**. In other aspects, the smart battery assembly **486** can dynamically change its power output during device operation. For instance, in vessel sealing devices, power management provides improved tissue sealing. In these devices, large constant current values are needed. The total power output needs to be adjusted dynamically because, as the tissue is sealed, its impedance changes. Aspects of the present disclosure provide the smart battery assembly **486** with a variable maximum current limit. The current limit can vary from one application (or device) to another, based on the requirements of the application or device.

FIG. **26** is a detail view of a trigger **483** portion and switch of the ultrasonic surgical instrument **480** shown in FIG. **25**, according to one aspect of the present disclosure. The trigger **483** is operably coupled to the jaw member **495** of the end effector **492**. The ultrasonic blade **496** is energized by the ultrasonic transducer/generator assembly **484** upon activating the activation switch **485**. Continuing now with FIG. **25** and also looking to FIG. **26**, the trigger **483** and the activation switch **485** are shown as components of the handle assembly **482**. The trigger **483** activates the end effector **492**, which has a cooperative association with the ultrasonic blade **496** of the waveguide shaft assembly **490** to enable various kinds of contact between the end effector jaw member **495** and the ultrasonic blade **496** with tissue and/or other substances. The jaw member **495** of the end effector **492** is usually a pivoting jaw that acts to grasp or clamp onto tissue disposed between the jaw and the ultrasonic blade **496**. In one aspect, an audible feedback is provided in the trigger that clicks when the trigger is fully depressed. The noise can be generated by a thin metal part that the trigger snaps over while closing. This feature adds an audible component to user feedback that informs the user that the jaw is fully compressed against the waveguide and that sufficient clamping pressure is being applied to accomplish vessel sealing. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to the trigger **483** to measure the force applied to the trigger **483** by the user. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to the switch **485** button such that displacement intensity corresponds to the force applied by the user to the switch **485** button.

The activation switch **485**, when depressed, places the modular handheld ultrasonic surgical instrument **480** into an ultrasonic operating mode, which causes ultrasonic motion at the waveguide shaft assembly **490**. In one aspect, depression of the activation switch **485** causes electrical contacts within a switch to close, thereby completing a circuit between the smart battery assembly **486** and the ultrasonic transducer/generator assembly **484** so that electrical power is applied to the ultrasonic transducer, as previously described. In another aspect, depression of the activation switch **485** closes electrical contacts to the smart battery assembly **486**. Of course, the description of closing electrical contacts in a circuit is, here, merely an example general description of switch operation. There are many alternative aspects that can include opening contacts or processor-controlled power delivery that receives information from the switch and directs a corresponding circuit reaction based on the information.

FIG. **27** is a fragmentary, enlarged perspective view of an end effector **492**, according to one aspect of the present disclosure, from a distal end with a jaw member **495** in an open position. Referring to FIG. **27**, a perspective partial view of the distal end **498** of the waveguide shaft assembly **490** is shown. The waveguide shaft assembly **490** includes an outer tube **494** surrounding a portion of the waveguide. The ultrasonic blade **496** portion of the waveguide **499** protrudes from the distal end **498** of the outer tube **494**. It is the ultrasonic blade **496** portion that contacts the tissue during a medical procedure and transfers its ultrasonic energy to the tissue. The waveguide shaft assembly **490** also includes a jaw member **495** that is coupled to the outer tube **494** and an inner tube (not visible in this view). The jaw member **495**, together with the inner and outer tubes and the ultrasonic blade **496** portion of the waveguide **499**, can be referred to as an end effector **492**. As will be explained below, the outer tube **494** and the non-illustrated inner tube slide longitudinally with respect to each other. As the relative movement between the outer tube **494** and the non-illustrated inner tube occurs, the jaw member **495** pivots upon a pivot point, thereby causing the jaw member **495** to open and close. When closed, the jaw member **495** imparts a pinching force on tissue located between the jaw member **495** and the ultrasonic blade **496**, insuring positive and efficient blade-to-tissue contact.

FIG. **28** illustrates a modular shaft assembly **110** and end effector **112** portions of the surgical instrument **100**, according to one aspect of the present disclosure. The shaft assembly **110** comprises an outer tube **144**, an inner tube **147**, and an ultrasonic transmission waveguide **145**. The shaft assembly **110** is removably mounted to the handle assembly **102**. The inner tube **147** is slidably received within the outer tube **144**. The ultrasonic transmission waveguide **145** is positioned within the inner tube **147**. The jaw member **114** of the end effector **112** is pivotally coupled to the outer tube **144** at a pivot point **151**. The jaw member **114** also is coupled to inner tube **147** by a pin **153** such that as the inner tube **147** slides within the slot **155**, the jaw member opens and closes. In the illustrated configuration, the inner tube **147** is in its distal position and the jaw member **114** is open. To close the jaw member **114**, the inner tube **147** is retracted in the proximal direction **157** and to open the jaw member is advanced in the distal direction **159**. The proximal end of the shaft assembly **110** comprises a jaw member tube (e.g., inner tube)/spring assembly **141**. A spring **139** is provided to apply a constant force control mechanism for use with different shaft assemblies, motor closures to control constant force closures, two bar mechanism to drive closure systems, cam lobes to push and pull closure system, drive screw designs to drive closure or wave spring designs to control constant force.

FIG. **29** is a detail view of the inner tube/spring assembly **141**. A closure mechanism **149** is operably coupled to the trigger **108** (FIGS. **1-3**). Accordingly, as the trigger **108** is squeezed, the inner tube **143** is retracted in the proximal direction **157** to close the jaw member **114**. Accordingly, as the trigger **108** is released, the inner tube **143** is advanced in the distal direction **159** to open the jaw member **114**.

For a more detailed description of a combination ultrasonic/electrosurgical instrument, reference is made to U.S. Pat. No. 9,107,690, which is herein incorporated by reference.

FIG. **30** illustrates a modular battery powered handheld combination ultrasonic/electrosurgical instrument **500**, according to one aspect of the present disclosure. FIG. **31** is an exploded view of the surgical instrument **500** shown in FIG. **30**, according to one aspect of the present disclosure. With reference now to FIGS. **30** and **31**, the surgical instrument **500** comprises a handle assembly **502**, an ultrasonic transducer/RF generator assembly **504**, a battery assembly **506**, a shaft assembly **510**, and an end effector **512**. The ultrasonic transducer/RF generator assembly **504**, battery assembly **506**, and shaft assembly **510** are modular components that are removably connectable to the handle assembly **502**. The handle assembly **502** also comprises a motor assembly **560**. The surgical instrument **500** is configured to use both ultrasonic vibration and electrosurgical high-frequency current to carry out surgical coagulation/cutting treatments on living tissue, and uses high-frequency current to carry out a surgical coagulation treatment on living tissue. The ultrasonic vibrations and the high-frequency (e.g., RF) current can be applied independently or in combination according to algorithms or user input control.

The ultrasonic transducer/RF generator assembly **504** comprises a housing **548**, a display **576**, such as an LCD display, for example, an ultrasonic transducer **530**, an electrical circuit **177** (FIGS. **4**, **10** and/or electrical circuit **300** in FIG. **14**), and a electrical circuit **702** (FIG. **34**) configured to drive an RF electrode and forms a portion of an RF generator circuit. The shaft assembly **510** comprises an outer tube **544** an ultrasonic transmission waveguide **545**, and an inner tube (not shown). The end effector **512** comprises a jaw member **514** and an ultrasonic blade **516**. The jaw member **514** comprises an electrode **515** that is coupled to an RF generator circuit. The ultrasonic blade **516** is the distal end of the ultrasonic transmission waveguide **545**. The jaw member **514** is pivotally rotatable to grasp tissue between the jaw member **514** and the ultrasonic blade **516**. The jaw member **514** is operably coupled to a trigger **508**. The trigger **508** functions to close the jaw member **514** when the trigger **508** is squeezed and to open the jaw member **514** when the trigger **508** is released to release the tissue. In a one-stage trigger configuration, the trigger **508** is squeezed to close the jaw member **514** and, once the jaw member **514** is closed, a first switch **521***a* of a switch section is activated to energize the RF generator to seal the tissue. After the tissue is sealed, a second switch **521***b* of the switch section **520** is activated to energize the ultrasonic generator to cut the tissue. In various aspects, the trigger **508** may be a two-stage, or a multi-stage, trigger. In a two-stage trigger configuration, during the first stage, the trigger **508** is squeezed part of the way to close the jaw member **514** and, during the second stage, the trigger **508** is squeezed the rest of the way to energize the RF generator circuit to seal the tissue. After the tissue is sealed, one of the switches **521***a*, **521***b* can be activated to energize the ultrasonic generator to cut the tissue. After the tissue is cut, the jaw member **514** is opened by releasing the trigger **508** to release the tissue. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to the trigger **508** to measure the force applied to the trigger **508** by the user. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to the switch **520** button such that displacement intensity corresponds to the force applied by the user to the switch **520** button.

The battery assembly **506** is electrically connected to the handle assembly **502** by an electrical connector **532**. The handle assembly **502** is provided with a switch section **520**. A first switch **520***a* and a second switch **520***b* are provided in the switch section **520**. The RF generator is activated by actuating the first switch **520***a* and the ultrasonic blade **516** is activated by actuating the second switch **520***b*. Accordingly, the first switch **520***a* energizes the RF circuit to drive high-frequency current through the tissue to form a seal and the second switch **520***b* energizes the ultrasonic transducer **530** to vibrate the ultrasonic blade **516** and cut the tissue.

A rotation knob **518** is operably coupled to the shaft assembly **510**. Rotation of the rotation knob **518** ±360° in the direction indicated by the arrows **526** causes an outer tube **544** to rotate ±360° in the respective direction of the arrows **528**. In one aspect, another rotation knob **522** may be configured to rotate the jaw member **514** while the ultrasonic blade **516** remains stationary and the rotation knob **518** rotates the outer tube **144** ±360°. The outer tube **144** may have a diameter D_1 ranging from 5 mm to 10 mm, for example.

FIG. **32** is a partial perspective view of a modular battery powered handheld combination ultrasonic/RF surgical instrument **600**, according to one aspect of the present disclosure. The surgical instrument **600** is configured to use both ultrasonic vibration and high-frequency current to carry out surgical coagulation/cutting treatments on living tissue, and uses high-frequency current to carry out a surgical coagulation treatment on living tissue. The ultrasonic vibrations and the high-frequency (e.g., RF) current can be applied independently or in combination according to algorithms or user input control. The surgical instrument **600** comprises a handle assembly **602**, an ultrasonic transducer/RF generator assembly **604**, a battery assembly **606**, a shaft assembly (not shown), and an end effector (not shown). The ultrasonic transducer/RF generator assembly **604**, battery assembly **606**, and shaft assembly are modular components that are removably connectable to the handle assembly **602**. A trigger **608** is operatively coupled to the handle assembly **602**. As previously described, the trigger operates the end effector.

The ultrasonic transducer/RF generator assembly **604** comprises a housing **648**, a display **676**, such as an LCD display, for example. The display **676** provides a visual display of surgical procedure parameters such as tissue thickness, status of seal, status of cut, tissue thickness, tissue impedance, algorithm being executed, battery capacity, energy being applied (either ultrasonic vibration or RF current), among other parameters. The ultrasonic transducer/RF generator assembly **604** also comprises two visual feedback indicators **678**, **679** to indicate the energy modality currently being applied in the surgical procedure. For example, one indicator **678** shows when RF energy is being used and another indicator **679** shows when ultrasonic energy is being used. It will be appreciated that when both energy modalities RF and ultrasonic are being applied, both indicators will show this condition. The surgical instrument **600** also comprises an ultrasonic transducer, an ultrasonic generator circuit and/or electrical circuit, a shaft assembly, and an end effector comprising a jaw member and an ultrasonic blade, the modular components being similar to those described in connection with FIGS. **30** and **31** and the description will not be repeated here for conciseness and clarity of disclosure.

The battery assembly **606** is electrically connected to the handle assembly **602** by an electrical connector. The handle assembly **602** is provided with a switch section **620**. A first switch **620***a* and a second switch **620***b* are provided in the switch section **620**. The ultrasonic blade is activated by actuating the first switch **620***a* and the RF generator is activated by actuating the second switch **620***b*. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to the trigger **608** to measure the force applied to the trigger **608** by the user. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to the switch **620** button such that displacement intensity corresponds to the force applied by the user to the switch **620** button.

A rotation knob **618** is operably coupled to the shaft assembly. Rotation of the rotation knob **618** ±360° causes an outer tube to rotate ±360° in the respective direction, as described herein in connection with FIGS. **30** and **31**. In one aspect, another rotation knob may be configured to rotate the jaw member while the ultrasonic blade remains stationary and the rotation knob **618** rotates the outer tube ±360°. A button **673** is used to connect and retain the shaft assembly to the handle assembly **602**. Another slide switch **675** is used to lock in and release the ultrasonic transducer/RF generator assembly **604**.

In one aspect, the surgical instrument **500**, **600** includes a battery powered advanced energy (ultrasonic vibration plus high-frequency current) with driver amplification broken into multiple stages. The different stages of amplification may reside in different modular components of the surgical instrument **500**, **600** such as the handle assembly **502**, **602** ultrasonic transducer/RF generator assembly **504**, **604**, battery assembly **506**, **606**, shaft assembly **510**, and/or the end effector **112**. In one aspect, the ultrasonic transducer/RF generator assembly **504**, **604** may include an amplification stage in the ultrasonic transducer and/or RF electronic circuits within the housing **548**, **648** and different ratios of amplification based on the energy modality associated with the particular energy mode. The final stage may be controlled via signals from the electronic system of the surgical instrument **100** located in the handle assembly **502**, **602** and/or the battery assembly **506**, **606** through a bus structure, such as I^2C, as previously described. Final stage switches system may be employed to apply power to the transformer and blocking capacitors to form the RF waveform. Measurements of the RF output, such as voltage and current, are fed back to the electronic system over the bus. The handle assembly **502**, **602** and/or battery assembly **506**, **606** may contain the majority of the primary amplification circuits including any electrical isolation components, motor control, and waveform generator. The two differing ultrasonic transducers (e.g., ultrasonic transducer **130**, **130'** shown in FIGS. **8** and **9**) and the RF transducer contain the electronics to utilize the preconditions generator signals and perform the final conditioning to power different frequency transducers of RF signals in the desired frequency ranges and amplitudes. This minimizes the weight size and cost of the electronics residing only in the transducers themselves. It also allows the primary processor boards to occupy the areas of the handle that have the most useful space which is rarely where the transducer is, due to its size. It also allows the electronics to be divided in such a way as the high wear high duty cycle elements could be only connectively attached to the primary electronics enabling it to be more serviceable and repairable since the system is designed for high repeated use before disposal.

The surgical instruments **500**, **600** described in connection with FIGS. **30-32** are configured to use high-frequency current to carry out surgical coagulation/cutting treatments on living tissue, and uses high-frequency current to carry out a surgical coagulation treatment on living tissue. Accordingly, additional structural and functional components to carry out this additional functionality will be described hereinbelow in connection with FIGS. **33-44**.

The structural and functional aspects of the battery assembly **506**, **606** are similar to those of the battery assembly **106** for the surgical instrument **100** described in connection with FIGS. **1**, **2**, and **16-24**, including the battery circuits described in connection with FIGS. **20-24**. Accordingly, for conciseness and clarity of disclosure, such the structural and functional aspects of the battery assembly **106** are incorporated herein by reference and will not be repeated here. Similarly, unless otherwise noted, the structural and functional aspects of the shaft assembly **510** are similar to those of the shaft assembly **110** for the surgical instrument **100** described in connection with FIGS. **1-3**. Accordingly, for conciseness and clarity of disclosure, such the structural and functional aspects of the shaft assembly **110** are incorporated herein by reference and will not be repeated here. Furthermore, the structural and functional aspects of the ultrasonic transducer **530** generator circuits are similar to those of the ultrasonic transducer **130** generator circuits for the surgical instrument **100** described in connection with FIGS. **1**, **2**, and **4-15**. Accordingly, for conciseness and clarity of disclosure, such the structural and functional aspects of the ultrasonic transducer **130** and generator circuits are incorporated herein by reference and will not be repeated here. Furthermore, the surgical instruments **500**, **600** include the circuits described in connection with FIGS. **12-15**, including, for example, the control circuit **210** described in connection with FIG. **14** and the electrical circuit **300** described in connection withe FIG. **15**. Accordingly, for conciseness and clarity of disclosure, the description of the circuits described in connection with FIGS. **12-15** is incorporated herein by reference and will not be repeated here.

Turning now to FIG. **33**, there is shown a nozzle **700** portion of the surgical instruments **500**, **600** described in connection with FIGS. **30-32**, according to one aspect of the present disclosure. The nozzle **700** contains an electrical circuit **702** configured to drive the high-frequency RF current to an electrode located in the end effector as described hereinbelow in connection with FIGS. **38-44**. The electrical circuit **702** is coupled to the primary winding of a transformer **704**. The positive side of the secondary winding of the transformer **704** is coupled to series connected first and second blocking capacitors **706**, **708**. The load side of the second blocking capacitor **708** is coupled to the positive RF(+) terminal which is coupled to the positive side of the end effector electrode. The negative side of the secondary winding of the transformer **704** is coupled to the negative RF(−) terminal, otherwise referred to as ground. It will be appreciated that the RF(−) or ground terminal of the RF energy circuit is coupled to an outer tube **744**, which is formed of an electrically conductive metal. Accordingly, in use, high-frequency current is conducted from the end effector electrode RF(+), through the tissue, and returns through the negative electrode RF(−).

With reference now also to FIGS. **30**, **31**, in one aspect, the outer tube **744** is operably coupled to the jaw member **514** portion of the end effector **512** such that the jaw member **514** opens when the outer tube **744** is advanced in the distal direction **722** and the jaw member **514** closes when the outer tube **744** is retracted in the proximal direction **724**. Although not shown in FIG. **33**, the outer tube **744** is operably coupled to the trigger **508**, which is used to open and close the jaw member **514** portion of the end effector **512**. Examples of actuation mechanisms for use with ultrasonic surgical instruments as described herein are disclosed in U.S. Patent Application Publication No. 2006/0079879 and U.S. Patent Application Publication No. 2015/0164532, now U.S. Pat. No. 9,724,120, each of which is herein incorporated by reference.

Still with reference to FIGS. **30**, **31**, and **33**, in one aspect, an inner tube **714** is slidably disposed within the outer tube **744**. The inner tube **714** is operably coupled to the jaw member **514** to rotate the jaw member **514** while maintaining the ultrasonic blade **516** stationary. In the aspect shown in FIGS. **30** and **31** the inner tube **714** is rotated by the rotation knob **522**. In the aspect shown in FIG. **33**, a motor **719** may be provided within the handle assembly **502** to engage a gear **721** on the proximal end of the outer tube **744**, optionally through an idler gear **725**.

Still with reference to FIGS. **30**, **31**, and **33**, in one aspect, an inner electrically insulative (e.g., rubber, plastic) tube **716** is slidably disposed within the inner tube **714**. A flex circuit **728** may be disposed within the inner electrically insulative tube **716** to electrically couple energy and sensor circuits to the end effector **512**. For example, the jaw member **514** may comprise an electrode coupled to conductors in the flex circuit **728**. In other aspects, the end effector **512**, jaw member **514**, or the ultrasonic blade **516** may comprise various sensors or other electrical elements that can be interconnected to electrical circuits and components in the shaft assembly **510**, the handle assembly **502**, the ultrasonic transducer/RF generator assembly **504**, and/or the battery assembly **506**, for example.

Still with reference to FIGS. **30**, **31**, and **33**, in one aspect, the ultrasonic transmission waveguide **545** (shown in FIG. **32** only; not shown in FIG. **33** for clarity) is disposed within the inner electrically insulative tube **716**. In one aspect, the positive electrode RF(+) of the electrical circuit **702** is electrically coupled to the ultrasonic transmission waveguide **545** and the negative electrode RF(−) of the electrical circuit **702** is electrically coupled to an electrode disposed in the jaw member **514**, which is electrically coupled to the outer tube **744**. In operation, after tissue is grasped between the ultrasonic blade **516** and the jaw member **514**, control circuits of the surgical instrument **500** can execute various algorithms to seal and the cut the tissue. The ultrasonic vibrations and high-frequency energy may be applied to the tissue in accordance with monitored tissue conditions such as tissue impedance, friction, and the like. In some situations, high-frequency current is applied to the tissue through the ultrasonic blade **516** and back to the outer tube **744** return path. The tissue impedance is monitored and when a tissue seal is formed, as may be determined by the tissue impedance, the ultrasonic blade **516** is mechanically energized to induce vibrational energy into the tissue to cut the tissue. In other aspects ultrasonic vibrations and high-frequency may be applied by pulsing these energy modalities, applying the energy modalities alternatively or simultaneously. In somewhat unique situations, an algorithm can detect when the tissue impedance is extremely low to deliver energy to the tissue. In response, the algorithm energizes the ultrasonic blade **516** mechanically to apply vibratory energy to the tissue until such time that the impedance rises above a threshold suitable for the application of the high-frequency current. Upon reaching this threshold, the algorithm switches energy delivery mode to high-frequency current to seal the tissue.

FIG. **34** is a schematic diagram of one aspect of an electrical circuit **702** configured to drive a high-frequency current (RF), according to one aspect of the present disclosure. The electrical circuit **702** comprises an analog multiplexer **580**. The analog multiplexer **580** multiplexes various signals from the upstream channels SCL-A/SDA-A such as RF, battery, and power control circuit. A current sensor **582** is coupled in series with the return or ground leg of the power supply circuit to measure the current supplied by the power supply. A field effect transistor (FET) temperature sensor **584** provides the ambient temperature. A pulse width modulation (PWM) watchdog timer **588** automatically generates a system reset if the main program neglects to periodically service it. It is provided to automatically reset the electrical circuit **702** when it hangs or freezes because of a software or hardware fault. It will be appreciated that the electrical circuit **702** may be configured for driving RF electrodes or for driving the ultrasonic transducer **130** as described in connection with FIG. **11**, for example. Accordingly, with reference now back to FIG. **34**, the electrical circuit **702** can be used to drive both ultrasonic and RF electrodes interchangeably.

A drive circuit **586** provides left and right RF energy outputs. A digital signal that represents the signal waveform is provided to the SCL-A/SDA-A inputs of the analog multiplexer **580** from a control circuit, such as the control circuit **210** (FIG. **14**). A digital-to-analog converter **590** (DAC) converts the digital input to an analog output to drive a PWM circuit **592** coupled to an oscillator **594**. The PWM circuit **592** provides a first signal to a first gate drive circuit **596***a* coupled to a first transistor output stage **598***a* to drive a first RF+ (Left) energy output. The PWM circuit **592** also provides a second signal to a second gate drive circuit **596***b* coupled to a second transistor output stage **598***b* to drive a second RF− (Right) energy output. A voltage sensor **599** is coupled between the RF Left/RF output terminals to measure the output voltage. The drive circuit **586**, the first and second drive circuits **596***a*, **596***b*, and the first and second transistor output stages **598***a*, **598***b* define a first stage amplifier circuit. In operation, the control circuit **210** (FIG. **14**) generates a digital waveform **1800** (FIG. **67**) employing circuits such as direct digital synthesis (DDS) circuits **1500**, **1600** (FIGS. **65** and **66**). The DAC **590** receives the digital waveform **1800** and converts it into an analog waveform, which is received and amplified by the first stage amplifier circuit.

FIG. **35** is a schematic diagram of the transformer **704** coupled to the electrical circuit **702** shown in FIG. **34**, according to one aspect of the present disclosure. The RF Left/RF input terminals (primary winding) of the transformer **704** are electrically coupled to the RF Left/RF output terminals of the electrical circuit **702**. One side of the secondary winding is coupled in series with first and second blocking capacitors **706**, **708**. The second blocking capacitor is coupled to the RF+ **574***a* terminal. The other side of the secondary winding is coupled to the RF− **574***b* terminal. As previously discussed, the RF+ **574***a* output is coupled to the ultrasonic blade **516** (FIG. **30**) and the RF− **574***b* ground terminal is coupled to the outer tube **544** (FIG. **30**). In one aspect, the transformer **166** has a turns-ratio of n1:n2 of 1:50.

FIG. **36** is a schematic diagram of a circuit **710** comprising separate power sources for high power energy/drive circuits and low power circuits, according to one aspect of the present disclosure. A power supply **712** includes a primary battery pack comprising first and second primary batteries **715**, **717** (e.g., Li-ion batteries) that are connected into the circuit **710** by a switch **718** and a secondary battery pack comprising a secondary battery **720** that is connected into the circuit by a switch **723** when the power supply **712** is inserted into the battery assembly. The secondary battery **720** is a sag preventing battery that has componentry resistant to gamma or other radiation sterilization. For instance, a switch mode power supply **727** and optional charge circuit within the battery assembly can be incorporated to allow the secondary battery **720** to reduce the voltage sag of the primary batteries **715**, **717**. This guarantees full charged cells at the beginning of a surgery that are easy to introduce into the sterile field. The primary batteries **715**, **717** can be used to power the motor control circuits **726** and the energy circuits **732** directly. The power supply/battery pack **712** may comprise a dual type battery assembly including primary Li-ion batteries **715**, **717** and secondary NiMH batteries **720** with dedicated energy cells **720** to control the handle electronics circuits **730** from dedicated energy cells **715**, **717** to run the motor control circuits **726** and the energy circuits **732**. In this case the circuit **710** pulls from the secondary batteries **720** involved in driving the handle electronics circuits **730** when the primary batteries **715**, **717** involved in driving the energy circuits **732** and/or motor control circuits **726** are dropping low. In one various aspect, the circuit **710** may include a one way diode that would not allow for current to flow in the opposite direction (e.g., from the batteries involved in driving the energy and/or motor control circuits to the batteries involved in driving the electronics circuits).

Additionally, a gamma friendly charge circuit may be provided that includes a switch mode power supply **727** using diodes and vacuum tube components to minimize voltage sag at a predetermined level. With the inclusion of a minimum sag voltage that is a division of the NiMH voltages (3 NiMH cells) the switch mode power supply **727** could be eliminated. Additionally a modular system may be provided wherein the radiation hardened components are located in a module, making the module sterilizable by radiation sterilization. Other non-radiation hardened components may be included in other modular components and connections made between the modular components such that the componentry operates together as if the components were located together on the same circuit board. If only two NiMH cells are desired the switch mode power supply **727** based on diodes and vacuum tubes allows for sterilizable electronics within the disposable primary battery pack.

Turning now to FIG. **37**, there is shown a control circuit **800** for operating a battery **801** powered RF generator circuit **802** for use with the surgical instrument **500** shown in FIGS. **30** and **31**, according to one aspect of the present disclosure. The surgical instrument **500** is configured to use both ultrasonic vibration and high-frequency current to carry out surgical coagulation/cutting treatments on living tissue, and uses high-frequency current to carry out a surgical coagulation treatment on living tissue.

FIG. **37** illustrates a control circuit **800** that allows a dual generator system to switch between the RF generator circuit **802** and the ultrasonic generator circuit **820** (similar to the electrical circuit **177** shown in FIGS. **11** and **12**) energy modalities for the surgical instrument **500** shown in FIGS. **30** and **31**. In one aspect, a current threshold in an RF signal is detected. When the impedance of the tissue is low the high-frequency current through tissue is high when RF energy is used as the treatment source for the tissue. According to one aspect, a visual indicator **812** or light located on the surgical instrument **500** may be configured to be in an on-state during this high current period. When the current falls below a threshold, the visual indicator **812** is in an off-state. Accordingly, a photo-transistor **814** may be configured to detect the transition from an on-state to an off-state and disengages the RF energy as shown in the control circuit **800** shown in FIG. **37**. Therefore, when the energy button is released and the energy switch **826** is opened, the control circuit **800** is reset and both the RF and ultrasonic generator circuits **802**, **820** are held off.

With reference to FIGS. **30-33** and **37**, in one aspect, a method of managing an RF generator circuit **802** and ultrasound generator circuit **820** is provided. As previously described the RF generator circuit **802** and/or the ultrasound generator circuit **820** may be located in the handle assembly **502**, the ultrasonic transducer/RF generator assembly **504**, the battery assembly **506**, the shaft assembly **510**, and/or the nozzle **700**. The control circuit **800** is held in a reset state if the energy switch **826** is off (e.g., open). Thus, when the energy switch **826** is opened, the control circuit **800** is reset and both the RF and ultrasonic generator circuits **802**, **820** are turned off. When the energy switch **826** is squeezed and the energy switch **826** is engaged (e.g., closed), RF energy is delivered to the tissue and a visual indicator **812** operated by a current sensing step-up transformer **804** will be lit while the tissue impedance is low. The light from the visual indicator **812** provides a logic signal to keep the ultrasonic generator circuit **820** in the off state. Once the tissue impedance increases above a threshold and the high-frequency current through the tissue decreases below a threshold, the visual indicator **812** turns off and the light transitions to an off-state. A logic signal generated by this transition turns off the relay **808**, whereby the RF generator circuit **802** is turned off and the ultrasonic generator circuit **820** is turned on, to complete the coagulation and cut cycle.

Still with reference to FIGS. **30-33** and **37**, in one aspect, the dual generator circuit **802**, **820** configuration employs an on-board RF generator circuit **802**, which is battery **801** powered, for one modality and a second, on-board ultrasound generator circuit **820**, which may be on-board in the handle assembly **502**, battery assembly **506**, shaft assembly **510**, nozzle **700**, and/or the ultrasonic transducer/RF generator assembly **504**. The ultrasonic generator circuit **820** also is battery **801** operated. In various aspects, the RF generator circuit **802** and the ultrasonic generator circuit **820** may be an integrated or separable component of the handle assembly **502**. According to various aspects, having the dual RF/ultrasonic generator circuits **802**, **820** as part of the handle assembly **502** may eliminate the need for complicated wiring in an environment where the surgical instrument **500**. The RF/ultrasonic generator circuits **802**, **820** may be configured to provide the full capabilities of an existing generator while utilizing the capabilities of a cordless generator system simultaneously.

Either type of system can have separate controls for the modalities that are not communicating with each other. The surgeon activates the RF and Ultrasonic separately and at their discretion. Another approach would be to provide fully integrated communication schemes that share buttons, tissue status, instrument operating parameters (such as jaw closure, forces, etc.) and algorithms to manage tissue treatment. Various combinations of this integration can be implemented to provide the appropriate level of function and performance.

In one aspect, the control circuit **800** includes a battery **801** powered RF generator circuit **802** comprising a battery as an energy source. As shown, RF generator circuit **802** is coupled to two electrically conductive surfaces referred to herein as electrodes **806***a*, **806***b* and is configured to drive the electrodes **806***a*, **806***b* with RF energy (e.g., high-frequency current). A first winding **810***a* of a step-up transformer **804** is connected in series with one pole of the bipolar RF generator circuit **802** and the return electrode **806***b*. In one aspect, the first winding **810***a* and the return electrode **806***b* are connected to the negative pole of the bipolar RF generator circuit **802**. The other pole of the bipolar RF generator circuit **802** is connected to the active electrode **806***a* through a switch contact **809** of a relay **808**, or any suitable electromagnetic switching device comprising an armature which is moved by an electromagnet **836** to operate the switch contact **809**. The switch contact **809** is closed when the electromagnet **836** is energized and the switch contact **809** is open when the electromagnet **836** is de-energized. When the switch contact is closed, RF current flows through conductive tissue (not shown) located between the electrodes **806***a*, **806***b*. It will be appreciated, that in one aspect, the active electrode **806***a* is connected to the positive pole of the bipolar RF generator circuit **802**.

A visual indicator circuit **805** comprises a step-up transformer **804**, a series resistor R**2**, and a visual indicator **812**. The visual indicator **812** can be adapted for use with the surgical instrument **500** and other electrosurgical systems and tools, such as those described herein. The first winding **810***a* of the step-up transformer **804** is connected in series with the return electrode **806***b* and a second winding **810***b* of the step-up transformer **804** is connected in series with a resistor R**2** and a visual indicator **812** comprising a type NE-2 neon bulb, for example.

In operation, when the switch contact **809** of the relay **808** is open, the active electrode **806***a* is disconnected from the positive pole of the bipolar RF generator circuit **802** and no current flows through the tissue, the return electrode **806***b*, and the first winding **810***a* of the step-up transformer **804**. Accordingly, the visual indicator **812** is not energized and does not emit light. When the switch contact **809** of the relay **808** is closed, the active electrode **806***a* is connected to the positive pole of the bipolar RF generator circuit **802** enabling current to flow through tissue, the return electrode **806***b*, and the first winding **810***a* of the step-up transformer **804** to operate on tissue, for example cut and cauterize the tissue.

A first current flows through the first winding **810***a* as a function of the impedance of the tissue located between the active and return electrodes **806***a*, **806***b* providing a first voltage across the first winding **810***a* of the step-up transformer **804**. A stepped up second voltage is induced across the second winding **810***b* of the step-up transformer **804**. The secondary voltage appears across the resistor R**2** and energizes the visual indicator **812** causing the neon bulb to light when the current through the tissue is greater than a predetermined threshold. It will be appreciated that the circuit and component values are illustrative and not limited thereto. When the switch contact **809** of the relay **808** is closed, current flows through the tissue and the visual indicator **812** is turned on.

Turning now to the energy switch **826** portion of the control circuit **800**, when the energy switch **826** is open position, a logic high is applied to the input of a first inverter **828** and a logic low is applied of one of the two inputs of the AND gate **832**. Thus, the output of the AND gate **832** is low and the transistor **834** is off to prevent current from flowing through the winding of the electromagnet **836**. With the electromagnet **836** in the de-energized state, the switch contact **809** of the relay **808** remains open and prevents current from flowing through the electrodes **806***a*, **806***b*. The logic low output of the first inverter **828** also is applied to a second inverter **830** causing the output to go high and resetting a flip-flop **818** (e.g., a D-Type flip-flop). At which time, the Q output goes low to turn off the ultrasound generator circuit **820** circuit and the $\overline{Q}$ output goes high and is applied to the other input of the AND gate **832**.

When the user presses the energy switch **826** on the instrument handle to apply energy to the tissue between the electrodes **806***a*, **806***b*, the energy switch **826** closes and applies a logic low at the input of the first inverter **828**, which applies a logic high to other input of the AND gate **832** causing the output of the AND gate **832** to go high and turns on the transistor **834**. In the on state, the transistor **834** conducts and sinks current through the winding of the electromagnet **836** to energize the electromagnet **836** and close the switch contact **809** of the relay **808**. As discussed above, when the switch contact **809** is closed, current can flow through the electrodes **806***a*, **806***b* and the first winding **810***a* of the step-up transformer **804** when tissue is located between the electrodes **806***a*, **806***b*.

As discussed above, the magnitude of the current flowing through the electrodes **806***a*, **806***b* depends on the impedance of the tissue located between the electrodes **806***a*, **806***b*. Initially, the tissue impedance is low and the magnitude of the current high through the tissue and the first winding **810***a*. Consequently, the voltage impressed on the second winding **810***b* is high enough to turn on the visual indicator **812**. The light emitted by the visual indicator **812** turns on the phototransistor **814**, which pulls the input of the inverter **816** low and causes the output of the inverter **816** to go high. A high input applied to the CLK of the flip-flop **818** has no effect on the Q or the $\overline{Q}$ outputs of the flip-flop **818** and Q output remains low and the $\overline{Q}$ output remains high. Accordingly, while the visual indicator **812** remains energized, the ultrasound generator circuit **820** is turned OFF and the ultrasonic transducer **822** and ultrasonic blade **824** are not activated.

As the tissue between the electrodes **806***a*, **806***b* dries up, due to the heat generated by the current flowing through the tissue, the impedance of the tissue increases and the current therethrough decreases. When the current through the first winding **810***a* decreases, the voltage across the second winding **810***b* also decreases and when the voltage drops below a minimum threshold required to operate the visual indicator **812**, the visual indicator **812** and the phototransistor **814** turn off. When the phototransistor **814** turns off, a logic high is applied to the input of the inverter **816** and a logic low is applied to the CLK input of the flip-flop **818** to clock a logic high to the Q output and a logic low to the $\overline{Q}$ output. The logic high at the Q output turns on the ultrasound generator circuit **820** to activate the ultrasonic transducer **822** and the ultrasonic blade **824** to initiate cutting the tissue located between the electrodes **806***a*, **806***a*. Simultaneously or near simultaneously with the ultrasound generator circuit **820** turning on, the $\overline{Q}$ output of the flip-flop **818** goes low and causes the output of the AND gate **832** to go low and turn off the transistor **834**, thereby de-energizing the electromagnet **836** and opening the switch contact **809** of the relay **808** to cut off the flow of current through the electrodes **806***a*, **806***b*.

While the switch contact **809** of the relay **808** is open, no current flows through the electrodes **806***a*, **806***b*, tissue, and the first winding **810***a* of the step-up transformer **804**. Therefore, no voltage is developed across the second winding **810***b* and no current flows through the visual indicator **812**.

The state of the Q and the $\overline{Q}$ outputs of the flip-flop **818** remain the same while the user squeezes the energy switch

826 on the instrument handle to maintain the energy switch **826** closed. Thus, the ultrasonic blade **824** remains activated and continues cutting the tissue between the jaws of the end effector while no current flows through the electrodes **806***a*, **806***b* from the bipolar RF generator circuit **802**. When the user releases the energy switch **826** on the instrument handle, the energy switch **826** opens and the output of the first inverter **828** goes low and the output of the second inverter **830** goes high to reset the flip-flop **818** causing the Q output to go low and turn off the ultrasound generator circuit **820**. At the same time, the $\overline{Q}$ output goes high and the circuit is now in an off state and ready for the user to actuate the energy switch **826** on the instrument handle to close the energy switch **826**, apply current to the tissue located between the electrodes **806***a*, **806***b*, and repeat the cycle of applying RF energy to the tissue and ultrasonic energy to the tissue as described above.

FIG. **38** is a sectional view of an end effector **900**, according to one aspect of the present disclosure. The end effector **900** comprises an ultrasonic blade **902** and a jaw member **904**. The jaw member **904** has a channel-shaped groove **906** in which part of the end effector **900** is engaged, along an axial direction. The channel-shaped groove **906** has a wide channel shape with a wide opening in a section orthogonal to an axis of the jaw member **904**. The jaw member **904** is made of a conductive material, and an insulating member **910** is provided in a range where the ultrasonic blade **902** is in contact along the axial direction on a bottom surface portion **912** of the channel shape.

The ultrasonic blade **902** has a rhombic shape partially cut out in the section orthogonal to the axial direction. The sectional shape of the ultrasonic blade **902** is a shape which is cut out in the direction orthogonal to a longer diagonal line of the rhombic shape as shown in FIG. **38**. The ultrasonic blade **902** with part of the rhombic shape cut out in the sectional shape has a trapezoidal portion **914** which is engaged in the channel-shaped groove **906** of the jaw member **904**. A portion in which part of the rhombic shape is not cut out in the sectional shape is an isosceles triangle portion **916** of the ultrasonic blade **902**.

When the trigger of the handle assembly is closed, the ultrasonic blade **902** and the jaw member **904** are fitted to each other. When they are fitted, the bottom surface portion **912** of the channel-shaped groove **906** abuts on a top surface portion **918** of the trapezoidal portion **914** of the ultrasonic blade **902**, and two inner wall portions **920** of the channel-shaped groove **906** abut on inclined surface portions **922** of the trapezoidal portion **914**.

Further, an apex portion **924** of the isosceles triangle portion **916** of the ultrasonic blade **902** is formed to be rounded, but the apex portion **924** has a slightly sharp angle.

When the surgical instrument is used as a spatulate ultrasound treatment instrument, the ultrasonic blade **902** acts as an ultrasound vibration treatment portion, and the apex portion **924** and its peripheral portion (shown by the dotted line) particularly act as a scalpel knife to the tissue of the treatment object.

Further, when the surgical instrument is used as a spatulate high-frequency treatment instrument, the apex portion **924** and its peripheral portion (shown by the dotted line) act as an electric scalpel knife to the tissue of the treatment object.

In one aspect, the bottom surface portion **912** and the inner wall portions **920**, and the top surface portion **918** and the inclined surface portions **922** act as the working surfaces of an ultrasound vibration.

Further, in one aspect, the inner wall portions **920** and the inclined surface portions **922** act as the working surfaces of a bipolar high-frequency current.

In one aspect, the surgical instrument may be used as a spatulate treatment instrument of simultaneous output of ultrasound and high-frequency current, the ultrasonic blade **902** acts as the ultrasound vibration treatment portion, and the apex portion **924** and its peripheral portion (shown by the dotted line) particularly act as an electrical scalpel knife to the tissue of the treatment object.

Further, when the surgical instrument provides simultaneous output of ultrasound and high-frequency current, the bottom surface portion **912** and the top surface portion **918** act as the working surfaces of an ultrasound vibration, and the inner wall portions **920** and the inclined surface portions **922** act as the working surfaces of a bipolar high-frequency current.

Consequently, according to the configuration of the treatment portion shown in FIG. **37**, excellent operability is provided not only in the case of use of the surgical instrument as an ultrasound treatment instrument or a high-frequency treatment instrument, but also in the case of use of the surgical instrument as an ultrasound treatment instrument or high-frequency current treatment instrument, and further in the case of use of the surgical instrument for the time of simultaneous output of ultrasound and high frequency.

When the surgical instrument performs high-frequency current output or simultaneous output of high-frequency current and ultrasound, monopolar output may be enabled instead of a bipolar output as the high-frequency output.

FIG. **39** is a sectional view of an end effector **930**, according to one aspect of the present disclosure. The jaw member **932** is made of a conductive material, and an insulating member **934** is provided along the axial direction on a bottom surface portion **936** of the channel shape.

The ultrasonic blade **938** has a rhombic shape partially cut out in the section orthogonal to the axial direction. The sectional shape of the ultrasonic blade **938** is a shape in which part of the rhombic shape is cut out in the direction orthogonal to one diagonal line as shown in FIG. **39**. The ultrasonic blade **938** with part of the rhombic shape cut out in the sectional shape has a trapezoidal portion **940** which is engaged in a channel-shaped groove **942** of the jaw member **932**. A portion in which part of the rhombic shape is not cut out in the sectional shape is an isosceles triangle portion **944** of the end effector **900**.

When the trigger of the handle assembly is closed, the ultrasonic blade **938** and the jaw member **906** are fitted to each other. When they are fitted, the bottom surface portion **936** of the channel-shaped groove **942** abuts on a top surface portion **946** of the trapezoidal portion **940** of the ultrasonic blade **938**, and two inner wall portions **954** of the channel-shaped groove **932** abut on inclined surface portions **948** of the trapezoidal portion **940**.

Further, an apex portion **950** of the isosceles triangle portion **944** of the ultrasonic blade **938** is formed to be rounded, but an apex portion **952** of the inner side of the hook shape has a slightly sharp angle. An angle θ of the apex portion **952** is preferably 45° to 100°. 45° is a strength limit of the ultrasonic blade **938**. As above, the apex portion **952** of the ultrasonic blade **938** configures a protruding portion having a predetermined angle at the inner side of the hook-shaped portion, that is, an edge portion.

The treatment portion in the hook shape is often used for dissection. The apex portion **952** of the end effector **930** becomes a working portion at the time of dissection. Since the apex portion **952** has the slightly sharp angle θ, the apex portion **952** is effective for dissection treatment.

The ultrasonic blade **938** and the jaw member **932** shown in FIG. **39** perform the same operation as the ultrasonic blade **938** and the jaw member **932** shown in FIG. **38** at the time of ultrasound output, at the time of high-frequency output, and at the time of simultaneous output of ultrasound and high frequency respectively, except for the aforementioned operation at the time of dissection.

Referring now to FIGS. **40-43**, there is shown and end effector **1000** operably coupled to an insertion sheath **1001**, which is formed by an outer sheath **1002** and an inner sheath **1004**. The end effector **1000** comprises an ultrasonic blade **1006** and a jaw member **1014**. In the outer sheath **1002**, the outside of a conductive metal pipe is covered with an insulating resin tube. The inner sheath **1004** is a conductive metal pipe. The inner sheath **1004** can be axially moved back and forth relative to the outer sheath **1002**.

The ultrasonic blade **1006** is made of a conductive material having high acoustic effects and biocompatibility, for example, a titanium alloy such as a Ti-6Al-4V alloy. In the ultrasonic blade **1006**, an insulating and elastic rubber lining **1008** is externally equipped in the position of nodes of the ultrasonic vibration. The rubber lining **1008** is disposed between the inner sheath **1004** and the ultrasonic blade **1006** in a compressed state. The ultrasonic blade **1006** is held to the inner sheath **1004** by the rubber lining **1008**. A clearance is maintained between the inner sheath **1004** and the ultrasonic blade **1006**.

An abutting portion **1010** is formed by the part of the ultrasonic blade **1012** facing the jaw member **1014** at the distal end portion of the ultrasonic blade **1006**. Here, the ultrasonic blade **1012** is octagonal in its cross section perpendicular to the axial directions of the ultrasonic blade **1006**. An abutting surface **1016** is formed by one surface of the abutting portion **1010** facing the jaw member **1014**. A pair of electrode surfaces **1018** is formed by surfaces provided to the sides of the abutting surface **1016**.

The jaw member **1014** is formed by a body member **1020**, an electrode member **1022**, a pad member **1024**, and a regulating member **1026** as a regulating section.

The body member **1020** is made of a hard and conductive material. A proximal end portion of the body member **1020** constitutes a pivot connection portion **1028**. The pivot connection portion **1028** is pivotally connected to a distal end portion of the outer sheath **1002** via a pivot connection shaft **1030**. The pivot connection shaft **1030** extends in width directions perpendicular to the axial directions and the opening/closing directions. The body member **1020** can turn about the pivot connection shaft **1030** in the opening/closing directions relative to the outer sheath **1002**. A distal end portion of the inner sheath **1004** is pivotally connected to the pivot connection portion **1028** of the body member **1020** at a position provided to the distal side and the opening-direction side of the pivot connection shaft **1030**. If the movable handle is turned relative to the fixed handle in the handle unit, the inner sheath **1004** is moved back and forth relative to the outer sheath **1002**, and the body member **1020** is driven by the inner sheath **1004** to turn about the pivot connection shaft **1030** in the opening/closing directions relative to the outer sheath **1002**. In one aspect, a distal part of the body member **1020** constitutes a pair of pivot bearings **1032**. The pair of pivot bearings **1032** are in the form of plates which extend in the axial directions and which are perpendicular to the width directions, and are disposed apart from each other in the width directions.

The electrode member **1022** is made of a hard and conductive material. The part of the electrode member **1022** provided on the opening-direction side constitutes a pivot support **1034**. An insertion hole **1036** is formed through the pivot support **1034** in the width directions. A pivot support shaft **1038** is inserted through the insertion hole **1036** and extends in the width directions. The pivot support **1034** is disposed between the pair of pivot bearings **1032** of the body member **1020**, and is pivotally supported on the pair of pivot bearings **1032** via the pivot support shaft **1038**. The electrode member **1022** can oscillate about the pivot support shaft **1038** relative to the body member **1020**. Further, the part of the electrode member **1022** provided on the closing-direction side constitutes an electrode section **1040**. The electrode section **1040** extends in the axial directions and projects to the sides in the width directions. A recessed groove **1042** which is open toward the closing direction extends in the axial directions in the part of the electrode section **1040** provided on the closing-direction side. Teeth are axially provided in the parts of the groove **1042** provided in the closing direction side, thus forming a tooth portion **1044**. The side surfaces that define the groove **1042** constitute a pair of electrode receiving surfaces **1046** that are inclined from the closing direction toward the sides in the width directions. A recessed mating receptacle **1048** which is open toward the closing direction axially extends in a bottom portion that defines the groove **1042**. An embedding hole **1050** is formed through the pivot support **1034** of the electrode member **1022** in the opening/closing directions perpendicularly to the insertion hole **1036**. The embedding hole **1050** is open to the mating receptacle **1048**.

The pad member **1024** is softer than the ultrasonic blade **1006**, and is made of an insulating material having biocompatibility such as polytetrafluorethylene. The pad member **1024** is mated with the mating receptacle **1048** of the electrode member **1022**. The part of the pad member **1024** provided on the closing-direction side protrudes from the electrode member **1022** to the closing direction, thus forming an abutting receptacle **1052**. In the cross section perpendicular to the axial directions, the abutting receptacle **1052** is in a recessed shape corresponding to the projecting shape of the abutting portion **1010** of the ultrasonic blade **1012**. When the jaw member **1014** is closed relative to the ultrasonic blade **1012**, the abutting portion **1010** of the ultrasonic blade **1012** abuts onto and engages with the abutting receptacle **1052** of the pad member **1024**. The pair of electrode surfaces **1018** of the ultrasonic blade **1012** are arranged parallel to the pair of electrode receiving surfaces **1046** of the electrode section **1040**, and a clearance is maintained between the electrode section **1040** and the ultrasonic blade **1012**.

The regulating member **1026** is harder than the ultrasonic blade **1006**, and is made of an insulating high-strength material such as ceramics. The regulating pad member **1024** is pin-shaped. The regulating pad member **1024** is inserted into the embedding hole **1050** of the pivot support **1034** of the electrode member **1022**, protrudes toward the mating receptacle **1048** of the electrode section **1040**, and is embedded in the abutting receptacle **1052** of the pad member **1024** in the mating receptacle **1048**. A closing-direction end of the regulating member **1026** constitutes a regulating end **1054**. The regulating end **1054** does not protrude from the abutting receptacle **1052** to the closing direction, and is accommodated in the abutting receptacle **1052**. The insertion hole **1036** is also formed through the regulating member **1026**, and the pivot support shaft **1038** is inserted through the insertion hole **1036** of the regulating member **1026**.

Here, the inner sheath **1004**, the body member **1020**, and the electrode member **1022** are electrically connected to one another, and constitute the first electrical path **1056** used in a high-frequency surgical treatment. The electrode section **1040** of the electrode member **1022** functions as one of bipolar electrodes used in a high-frequency surgical treatment. In one aspect, the ultrasonic blade **1006** constitutes the second electrical path **1058** used in the high-frequency treatment. The ultrasonic blade **1012** provided to the distal end portion of the ultrasonic blade **1006** functions as the other of the bipolar electrodes used in a high-frequency treatment. As described above, the ultrasonic blade **1006** is held to the inner sheath **1004** by the insulating rubber lining **1008**, and the clearance is maintained between the inner sheath **1004** and the ultrasonic blade **1006**. This prevents a short circuit between the inner sheath **1004** and the ultrasonic blade **1006**. When the jaw member **1014** is closed relative to the ultrasonic blade **1012**, the abutting portion **1010** of the ultrasonic blade **1012** abuts onto and engages with the abutting receptacle **1052** of the pad member **1024**. Thus, the pair of electrode surfaces **1018** of the ultrasonic blade **1012** are arranged parallel to the pair of electrode receiving surfaces **1046** of the electrode section **1040**, and the clearance is maintained between the electrode section **1040** and the ultrasonic blade **1012**. This prevents a short circuit between the electrode section **1040** and the ultrasonic blade **1012**.

Referring to FIG. **44**, the pad member **1024** is softer than the ultrasonic blade **1006**. Therefore, the abutting receptacle **1052** is worn by the ultrasonic blade **1012** in the case where the ultrasonic blade **1012** is ultrasonically vibrated when the jaw member **1014** is closed relative to the ultrasonic blade **1012** and the abutting portion **1010** of the ultrasonic blade **1012** abuts onto and engages with the abutting receptacle **1052** of the pad member **1024**. As the abutting receptacle **1052** is worn, the clearance between the electrode section **1040** and the ultrasonic blade **1012** is gradually reduced when the abutting portion **1010** is in a frictional engagement with the abutting receptacle **1052**. When the abutting receptacle **1052** is worn more than a predetermined amount, the regulating end **1054** of the regulating member **1026** is exposed from the abutting receptacle **1052** in the closing direction. When the regulating end **1054** is exposed from the abutting receptacle **1052** in the closing direction, the regulating end **1054** contacts the ultrasonic blade **1012** before the electrode section **1040** contacts the ultrasonic blade **1012** if the jaw member **1014** is closed relative to the ultrasonic blade **1012**. As a result, the contact between the ultrasonic blade **1012** and the electrode section **1040** is regulated. Here, the electrode section **1040** and the ultrasonic blade **1012** are hard. Therefore, when the ultrasonically vibrated ultrasonic blade **1012** contacts the electrode section **1040**, the ultrasonic blade **1012** rapidly and repetitively comes in and out of contact with the electrode section **58**. When a high-frequency voltage is applied between the electrode section **1040** and the ultrasonic blade **1012**, sparking occurs between the ultrasonic blade **1012** and the electrode section **1040**. In one aspect, the contact between the ultrasonic blade **1012** and the electrode section **1040** is regulated by the regulating end **1054** of the regulating member **1026**, so that sparking is prevented. The regulating member **1026** is made of an insulating material, and is electrically insulated relative to the electrode member **1022**. Thus, if the ultrasonically vibrated ultrasonic blade **1012** contacts the regulating end **1054** of the regulating member **1026**, no sparking occurs between the regulating end **1054** and the ultrasonic blade **1012** even when the ultrasonic blade **1012** rapidly and repetitively comes in and out of contact with the regulating end **1054**. This prevents sparking between the ultrasonic blade **1012** and the jaw member **1014**.

The regulating member **1026** is made of a high-strength material harder than the ultrasonic blade **1006**. Therefore, when the regulating end **1054** contacts the ultrasonically vibrated ultrasonic blade **1012**, the regulating member **1026** is not worn, and the ultrasonic blade **1006** cracks. In the surgical treatment system according to one aspect, when the abutting receptacle **1052** is worn more than a predetermined amount, the regulating end **1054** contacts the ultrasonic blade **1012** to intentionally crack the ultrasonic blade **1006**. By detecting this crack, the end of the life of the surgical treatment instrument is detected. Therefore, the position of the contact between the ultrasonic blade **1012** and the regulating end **1054** is set at the stress concentration region in the ultrasonic blade **1012** to ensure that the ultrasonic blade **1006** cracks when the regulating end **1054** contacts the ultrasonic blade **1012**. In a linear ultrasonic blade **1006**, stress concentrates in the positions of the nodes of the ultrasonic vibration, and a stress concentration region is located at the proximal end portion of the ultrasonic blade **1012**.

For a more detailed description of a combination ultrasonic/electrosurgical instrument, reference is made to U.S. Pat. Nos. 8,696,666 and 8,663,223, each of which is herein incorporated by reference.

FIG. **45** illustrates a modular battery powered handheld electrosurgical instrument **1100** with distal articulation, according to one aspect of the present disclosure. The surgical instrument **1100** comprises having a handle assembly **1102**, a knife drive assembly **1104**, a battery assembly **1106**, a shaft assembly **1110**, and an end effector **1112**. The end effector **1112** comprises a pair of jaw members **1114***a*, **1114***b* in opposing relationship affixed to a distal end thereof. The end effector **1112** is configured to articulate and rotate. FIG. **46** is an exploded view of the surgical instrument **1100** shown in FIG. **45**, according to one aspect of the present disclosure. The end effector **1112** for use with the surgical instrument **1100** for sealing and cutting tissue includes a pair of jaw members **1114***a*, **1114***b* that in opposing relationship and movable relative to each other to grasp tissue therebetween. A jaw member **1114***a*, **1114***b* includes a jaw housing and an electrically conductive surface **1116***a*, **1116***b*, e.g., electrodes, adapted to connect to a source of electrosurgical energy (RF source) such that the electrically conductive surfaces are capable of conducting electrosurgical energy through tissue held therebetween to effect a tissue seal. One of the electrically conductive surfaces **1116***b* includes a channel defined therein and extending along a length thereof that communicates with a drive rod **1145** connected to a motor disposed in the knife drive assembly **1104**. The knife is configured to translate and reciprocate along the channel to cut tissue grasped between the jaw members **1114***a*, **1114***b*.

FIG. **47** is a perspective view of the surgical instrument **1100** shown in FIGS. **45** and **46** with a display located on the handle assembly **1102**, according to one aspect of the present disclosure. The handle assembly **1102** of the surgical instrument shown in FIGS. **45-47** comprises a motor assembly **1160** and a display assembly. The display assembly comprises a display **1176**, such as an LCD display, for example, which is removably connectable to a housing **1148** portion of the handle assembly **1102**. The display **1176** provides a visual display of surgical procedure parameters such as tissue thickness, status of seal, status of cut, tissue thickness, tissue impedance, algorithm being executed, battery capacity, among other parameters.

FIG. **48** is a perspective view of the instrument shown in FIGS. **45** and **46** without a display located on the handle assembly **1102**, according to one aspect of the present disclosure. The handle assembly **1102** of the surgical instrument **1150** shown in FIG. **48** includes a different display assembly **1154** on a separate housing **1156**. With reference now to FIGS. **45-48**, the surgical instrument **1100**, **1150** is configured to use high-frequency (RF) current and a knife to carry out surgical coagulation/cutting treatments on living tissue, and uses high-frequency current to carry out a surgical coagulation treatment on living tissue. The high-frequency (RF) current can be applied independently or in combination with algorithms or user input control. The display assembly, battery assembly **1106**, and shaft assembly **1110** are modular components that are removably connectable to the handle assembly **1102**. A motor **1140** is located within the handle assembly **1102**. RF generator circuits and motor drive circuits are described herein in connection with FIGS. **34-37** and **50**, for example, is located within the housing **1148**.

The shaft assembly **1110** comprises an outer tube **1144**, a knife drive rod **1145**, and an inner tube (not shown). The shaft assembly **1110** comprises an articulation section **1130** and a distal rotation section **1134**. The end effector **1112** comprises jaw members **1114***a*, **1114***b* in opposing relationship and a motor driven knife. The jaw member **1114***a*, **1114***b* comprises an electrically conductive surface **1116***a*, **1116***b* coupled to the RF generator circuit for delivering high-frequency current to tissue grasped between the opposed jaw members **1114***a*, **1114***b*. The jaw members **1114***a*, **1114***b* are pivotally rotatable about a pivot pin **1136** to grasp tissue between the jaw members **1114***a*, **1114***b*. The jaw members **1114***a*, **1114***b* are operably coupled to a trigger **1108** such that when the trigger **1108** is squeezed the jaw members **1114***a*, **1114***b* close to grasp tissue and when the trigger **1108** is released the jaw members **1114***a*, **1114***b* open to release tissue.

The jaw members **1114***a*, **1114***b* are operably coupled to a trigger **1108** such that when the trigger **1108** is squeezed the jaw members **1114***a*, **1114***b* close to grasp tissue and when the trigger **1108** is released the jaw members **1114***a*, **1114***b* open to release tissue. In a one-stage trigger configuration, the trigger **1108** is squeezed to close the jaw members **1114***a*, **1114***b* and, once the jaw members **1114***a*, **1114***b* are closed, a first switch **1121***a* of a switch section **1121** is activated to energize the RF generator to seal the tissue. After the tissue is sealed, a second switch **1121***b* of the switch section **1120** is activated to advance a knife to cut the tissue. In various aspects, the trigger **1108** may be a two-stage, or a multi-stage, trigger. In a two-stage trigger configuration, during the first stage, the trigger **1108** is squeezed part of the way to close the jaw members **1114***a*, **1114***b* and, during the second stage, the trigger **1108** is squeezed the rest of the way to energize the RF generator circuit to seal the tissue. After the tissue is sealed, one of the first and second switches **1121***a*, **1121***b* can be activated to advance the knife to cut the tissue. After the tissue is cut, the jaw members **1114***a*, **1114***b* are opened by releasing the trigger **1108** to release the tissue. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to the trigger **1108** to measure the force applied to the trigger **1108** by the user. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to the switch section **1120** first and second switch **1121***a*, **1121***b* buttons such that displacement intensity corresponds to the force applied by the user to the switch section **1120** first and second switch **1121***a*, **1121***b* buttons.

The battery assembly **1106** is electrically connected to the handle assembly **1102** by an electrical connector **1132**. The handle assembly **1102** is provided with a switch section **1120**. A first switch **1121***a* and a second switch **1121***b* are provided in the switch section **1120**. The RF generator is energized by actuating the first switch **1121***a* and the knife is activated by energizing the motor **1140** by actuating the second switch **1121***b*. Accordingly, the first switch **1121***a* energizes the RF circuit to drive the high-frequency current through the tissue to form a seal and the second switch **1121***b* energizes the motor to drive the knife to cut the tissue. The structural and functional aspects of the battery assembly **1106** are similar to those of the battery assembly **106** for the surgical instrument **100** described in connection with FIGS. **1**, **2**, and **16-24**. Accordingly, for conciseness and clarity of disclosure, such the structural and functional aspects of the battery assembly **106** are incorporated herein by reference and will not be repeated here.

A rotation knob **1118** is operably coupled to the shaft assembly **1110**. Rotation of the rotation knob **1118** ±360° in the direction indicated by the arrows **1126** causes the outer tube **1144** to rotate ±360° in the respective direction of the arrows **1119**. In one aspect, another rotation knob **1122** may be configured to rotate the end effector **1112** ±360° in the direction indicated by the arrows **1128** independently of the rotation of the outer tube **1144**. The end effector **1112** may be articulated by way of first and second control switches **1124***a*, **1124***b* such that actuation of the first control switch **1124***a* articulates the end effector **1112** about a pivot **1138** in the direction indicated by the arrow **1132***a* and actuation of the second control switch **1124***b* articulates the end effector **1112** about the pivot **1138** in the direction indicated by the arrow **1132***b*. Further, the outer tube **1144** may have a diameter D_3 ranging from 5 mm to 10 mm, for example.

FIG. **49** is a motor assembly **1160** that can be used with the surgical instrument **1100**, **1150** to drive the knife, according to one aspect of the present disclosure. The motor assembly **1160** comprises a motor **1162**, a planetary gear **1164**, a shaft **1166**, and a drive gear **1168**. The gear may be operably coupled to drive the knife bar **1145** (FIG. **46**). In one aspect, the drive gear **1168** or the shaft **1166** is operably coupled to a rotary drive mechanism **1170** described in connection with FIG. **50** to drive distal head rotation, articulation, and jaw closure.

FIG. **50** is diagram of a motor drive circuit **1165**, according to one aspect of the present disclosure. The motor drive circuit **1165** is suitable for driving the motor M, which may be employed in the surgical instruments **1100**, **1150** described herein. The motor M is driven by an H-bridge comprising four switches S_1-S_4. The switches S_1-S_4 are generally solid state switches such as MOSFET switches. To turn the motor M in one direction, two switches S_1, S_4 are turned on and the other two switches S_3, S_1 are turned off. To reverse the direction of the motor M, the state of the switches S_1-S_4 is reversed such that the switches S_1, S_4 are turned off and the other two switches S_3, S_1 are turned on. Current sensing circuits can be placed in the motor drive circuit **1165** to sense motor currents i_{1a}, i_{2a}, i_{1b}, i_{2b}.

FIG. **51** illustrates a rotary drive mechanism **1170** to drive distal head rotation, articulation, and jaw closure, according to one aspect of the present disclosure. The rotary drive mechanism **1170** has a primary rotary drive shaft **1172** that is operably coupled to the motor assembly **1160**. The primary rotary drive shaft **1172** is capable of being selectively coupled to at least two independent actuation mechanisms (first, second, both, neither) with a clutch mechanism located within the outer tube **1144** of the shaft assembly **1110**. The primary rotary drive shaft **1172** is coupled to independent clutches that allow the shaft functions to be independently coupled to the rotary drive shaft **1172**. For example, the articulation clutch **1174** is engaged to articulate the shaft assembly **1110** about the articulation axis **1175** of the articulation section **1130**. The distal head rotation clutch **1178** is engaged to rotate the distal rotation section **1134** and the jaw closure clutch **1179** is engaged to close the jaw members **1114***a*, **1114***b* of the end effector **1112**. The knife is advanced and retracted by the knife drive rod **1145**. All, none, or any combination of rotary mechanisms can be couple at any one time.

In one aspect, a micro-electrical clutching configuration enables rotation of the distal rotation section **1134** and articulation of the articulation section **1130** about pivot **1138** and articulation axis **1175**. In one aspect, a ferro-fluid clutch couples the clutch to the primary rotary drive shaft **1172** via a fluid pump. The clutch ferro-fluid is activated by electrical coils **1181**, **1183**, **1185** which are wrapped around the knife drive rod **1145**. The other ends of the coils **1181**, **1183**, **1185** are connected to three separate control circuits to independently actuate the clutches **1174**, **1178**, **1179**. In operation, when the coils **1181**, **1183**, **1185** are not energized, the clutches **1174**, **1178**, **1179** are disengaged and there is no articulation, rotation, or jaw movements.

When the articulation clutch **1174** is engaged by energizing the coil **1181** and the distal head rotation clutch **1178** and the jaw closure clutch **1179** are disengaged by de-energizing the coils **1183**, **1185**, a gear **1180** is mechanically coupled to the primary rotary drive shaft **1172** to articulate the articulation section **1130**. In the illustrated orientation, when the primary rotary drive shaft **1172** rotates clockwise, the gear **1180** rotates clockwise and the shaft articulates in the right direction about the articulation axis **1175** and when the primary rotary drive shaft **1172** rotates counter clockwise, the gear **1180** rotates counter clockwise and the shaft articulates in the left direction about the articulation axis **1175**. It will be appreciated that left/right articulation depends on the orientation of the surgical instrument **1100**, **1150**.

When the articulation clutch **1174** and the jaw closure clutch **1179** are disengaged by de-energizing the coils **1181**, **1185**, and the distal head rotation clutch **1178** is engaged by energizing the coil **1183**, the primary rotary drive shaft **1172** rotates the distal rotation section **1134** in the same direction of rotation. When the coil **1183** is energized, the distal head rotation clutch **1178** engages the primary rotary drive shaft **1172** with the distal rotation section **1134**. Accordingly, the distal rotation section **1134** rotates with the primary rotary drive shaft **1172**.

When the articulation clutch **1174** and the distal head rotation clutch **1178** are disengaged by de-energizing the coils **1181**, **1183**, and the jaw closure clutch **1179** is engaged by energizing the coil **1185**, the jaw members **1114***a*, **114***b* can be opened or closed depending on the rotation of the primary rotary drive shaft **1172**. When the coil **1185** is energized, the jaw closure clutch **1179** engages a captive inner threaded drive member **1186**, which rotates in place in the direction of the primary rotary drive shaft **1172**. The captive inner threaded drive member **1186** includes outer threads that are in threaded engagement with an outer threaded drive member **1188**, which includes an inner threaded surface. As the primary rotary drive shaft **1172** rotates clockwise, the outer threaded drive member **1188** that is in threaded engagement with the captive inner threaded drive member **1186** will be driven in a proximal direction **1187** to close the jaw members **1114***a*, **1114***b*. As the primary rotary drive shaft **1172** rotates counterclockwise, the outer threaded drive member **1188** that is in threaded engagement with the captive inner threaded drive member **1186** will be driven in a distal direction **1189** to open the jaw members **1114***a*, **1114***b*.

FIG. **52** is an enlarged, left perspective view of an end effector assembly with the jaw members shown in an open configuration, according to one aspect of the present disclosure. FIG. **53** is an enlarged, right side view of the end effector assembly of FIG. **52**, according to one aspect of the present disclosure. Referring now to FIGS. **52** and **53**, enlarged views of an end effector **1112** shown in an open position for approximating tissue. Jaw members **1114**, **1114***b* are generally symmetrical and include similar component features which cooperate to permit facile rotation about pivot pin **1136** to effect the sealing and dividing of tissue. As a result and unless otherwise noted, only the jaw member **1114***a* and the operative features associated therewith are describe in detail herein but as can be appreciated, many of these features apply to the other jaw member **1114***b* as well.

The jaw member **1114***a* also includes a jaw housing **1115***a*, an insulative substrate or insulator **1117***a* and an electrically conductive surface **1116***a*. The insulator **1117***a* is configured to securely engage the electrically conductive sealing surface **1116***a*. This may be accomplished by stamping, by overmolding, by overmolding a stamped electrically conductive sealing plate and/or by overmolding a metal injection molded seal plate. These manufacturing techniques produce an electrode having an electrically conductive surface **1116***a* that is surrounded by an insulator **1117***a*.

As mentioned above, the jaw member **1114***a* includes similar elements which include: a jaw housing **1115***b*; insulator **1117***b*; and an electrically conductive surface **1116***b* that is dimensioned to securely engage the insulator **1117***b*. Electrically conductive surface **1116***b* and the insulator **1117***b*, when assembled, form a longitudinally-oriented knife channel **1113** defined therethrough for reciprocation of the knife blade **1123**. The knife channel **1113** facilitates longitudinal reciprocation of the knife blade **1123** along a predetermined cutting plane to effectively and accurately separate the tissue along the formed tissue seal. Although not shown, the jaw member **1114***a* may also include a knife channel that cooperates with the knife channel **1113** to facilitate translation of the knife through tissue.

The jaw members **1114***a*, **1114***b* are electrically isolated from one another such that electrosurgical energy can be effectively transferred through the tissue to form a tissue seal. The electrically conductive surfaces **1116***a*, **1116***b* are also insolated from the remaining operative components of the end effector **1112** and the outer tube **1144**. A plurality of stop members may be employed to regulate the gap distance between the electrically conductive surfaces **1116***a*, **1116***b* to insure accurate, consistent, and reliable tissue seals.

The structural and functional aspects of the battery assembly **1106** are similar to those of the battery assembly **106** for the surgical instrument **100** described in connection with FIGS. **1**, **2**, and **16-24**, including the battery circuits described in connection with FIGS. **20-24**. Accordingly, for conciseness and clarity of disclosure, such the structural and functional aspects of the battery assembly **106** are incorporated herein by reference and will not be repeated here. Furthermore, the structural and functional aspects of the RF generator circuits are similar to those of the RF generator circuits described in for the surgical instruments **500**, **600** described in connection with FIGS. **34-37**. Accordingly, for conciseness and clarity of disclosure, such the structural and functional aspects of the RF generator circuits are incorporated herein by reference and will not be repeated here. Furthermore, the surgical instrument **1100** includes the battery and control circuits described in connection with FIGS. **12-15**, including, for example, the control circuit **210** described in connection with FIG. **14** and the electrical circuit **300** described in connection withe FIG. **15**. Accordingly, for conciseness and clarity of disclosure, the description of the circuits described in connection with FIGS. **12-15** is incorporated herein by reference and will not be repeated here.

For a more detailed description of an electrosurgical instrument comprising a cutting mechanism and an articulation section that is operable to deflect the end effector away from the longitudinal axis of the shaft, reference is made to U.S. Pat. Nos. 9,028,478 and 9,113,907, each of which is herein incorporated by reference.

FIG. **54** illustrates a modular battery powered handheld electrosurgical instrument **1200** with distal articulation, according to one aspect of the present disclosure. The surgical instrument **1200** comprises a handle assembly **1202**, a knife drive assembly **1204**, a battery assembly **1206**, a shaft assembly **1210**, and an end effector **1212**. The end effector **1212** comprises a pair of jaw members **1214***a*, **1214***b* in opposing relationship affixed to a distal end thereof. The end effector **1212** is configured to articulate and rotate. FIG. **55** is an exploded view of the surgical instrument **1200** shown in FIG. **54**, according to one aspect of the present disclosure. The end effector **1212** for use with the surgical instrument **1200** for sealing and cutting tissue includes a pair of jaw members **1214***a*, **1214***b* in opposing relationship movable relative to each other to grasp tissue therebetween. Either jaw member **1214***a*, **1214***b* may include a jaw housing and an electrically conductive surface **1216***a*, **1216***b*, e.g., electrodes, adapted to connect to a source of electrosurgical energy (RF source) such that the electrically conductive surfaces are capable of conducting electrosurgical energy through tissue held therebetween to effect a tissue seal. The jaw members **1214***a*, **1214***b* and the electrically conductive surfaces **1216***a*, **1216***b* include a channel defined therein and extending along a length thereof that communicates with a knife drive rod **1245** connected to a knife drive assembly **1204**. The knife **1274** (FIGS. **60-61**) is configured to translate and reciprocate along the channels to cut tissue grasped between the jaw members **1214***a*, **1214***b*. The knife has an I-beam configuration such that the jaw members **1214***a*, **1214***b* are brought closer together as the knife **1274** advances through the channels. In one aspect, the electrically conductive surfaces **1216***a*, **1216***b* are offset relative to each other. The knife **1274** includes a sharp distal end.

The handle assembly **1202** of the surgical instrument shown in FIGS. **54-55** comprises a motor assembly **1260** and a knife drive assembly **1204**. In one aspect, a display assembly may be provided on the housing **1248**. The display assembly may comprise a display, such as an LCD display, for example, which is removably connectable to a housing **1248** portion of the handle assembly **1202**. The LCD display provides a visual display of surgical procedure parameters such as tissue thickness, status of seal, status of cut, tissue thickness, tissue impedance, algorithm being executed, battery capacity, among other parameters. With reference now to FIGS. **54-55**, the surgical instrument **1200** is configured to use high-frequency (RF) current and a knife **1274** (FIGS. **60-61**) to carry out surgical coagulation/cutting treatments on living tissue, and uses high-frequency current to carry out a surgical coagulation treatment on living tissue. The high-frequency (RF) current can be applied independently or in combination with algorithms or user input control. The knife drive assembly **1204**, battery assembly **1206**, and shaft assembly **1210** are modular components that are removably connectable to the handle assembly **1202**. A motor assembly **1240** may be located within the handle assembly **1202**. The RF generator and motor drive circuits are described in connection with FIGS. **34-37** and **50**, for example, are located within the housing **1248**. The housing **1248** includes a removable cover plate **1276** to provide access to the circuits and mechanisms located within the housing **1248**. The knife drive assembly **1204** includes gears and linkages operably coupled to the handle assembly **1202** and the switch section **1220** to activate and drive the knife **1274**. As discussed in more detail hereinbelow, the knife **1274** has an I-beam configuration.

The shaft assembly **1210** comprises an outer tube **1244**, a knife drive rod **1245**, and an inner tube (not shown). The shaft assembly **1210** comprises an articulation section **1230**. The end effector **1212** comprises a pair of jaw members **1214***a*, **1214***b* and a knife **1274** configured to reciprocate with channels formed in the jaw members **1214***a*, **1214***b*. In one aspect, the knife **1274** may be driven by a motor. The jaw member **1214***a*, **1214***b* comprises an electrically conductive surface **1216***a*, **1216***b* coupled to the RF generator circuit for delivering high-frequency current to tissue grasped between the jaw members **1214***a*, **1214***b*. The jaw members **1214***a*, **1214***b* are pivotally rotatable about a pivot pin **1235** to grasp tissue between the jaw members **1214***a*, **1214***b*. The jaw members **1214***a*, **1214***b* are operably coupled to a trigger **1208** such that when the trigger **1208** is squeezed one or both of the jaw members **1214***a*, **1214***b* close to grasp tissue and when the trigger **1208** is released the jaw members **1214***a*, **1214***b* open to release tissue. In the illustrated example, one jaw member **1214***a* is movable relative to the other jaw member **1214***b*. In other aspects, both jaw members **1214***a*, **1214***b* may be movable relative to each other. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to the trigger **1208** to measure the force applied to the trigger **1208** by the user. In another aspect, force sensors such as strain gages or pressure sensors may be coupled to the switch section **1220** first and second switch **1221***a*, **1221***b* buttons such that displacement intensity corresponds to the force applied by the user to the switch section **1220** first and second switch **1221***a*, **1221***b* buttons.

The jaw member **1214***a* is operably coupled to a trigger **1208** such that when the trigger **1208** is squeezed the jaw member **1214***a* closes to grasp tissue and when the trigger **1208** is released the jaw member **1214***a* opens to release tissue. In a one-stage trigger configuration, the trigger **1208** is squeezed to close the jaw member **1214***a* and, once the jaw member **1214***a* is closed, a first switch **1221***a* of a switch section **1220** is activated to energize the RF generator to seal the tissue. After the tissue is sealed, a second switch **1221***b* of the switch section **1220** is activated to advance a knife to cut the tissue. In various aspects, the trigger **1208** may be a two-stage, or a multi-stage, trigger. In a two-stage trigger configuration, during the first stage, the trigger **1208** is squeezed part of the way to close the jaw member **1214***a* and during the second stage, the trigger **1208** is squeezed the rest of the way to energize the RF generator circuit to seal the tissue. After the tissue is sealed, one of the switches **1221***a*, **1221***b* can be activated to advance the knife to cut the tissue. After the tissue is cut, the jaw member **1214***a* is opened by releasing the trigger **1208** to release the tissue.

The shaft assembly **1210** includes an articulation section **1230** that is operable to deflect the end effector **1212** away from the longitudinal axis "A" of the shaft assembly **1210**. The dials **1232***a*, **1232***b* are operable to pivot the articulation section **1230** at the distal end of the elongated shaft assembly **1210** to various articulated orientations with respect to the longitudinal axis A-A. More particularly, the articulation dials **1232***a*, **1232***b* operably couple to a plurality of cables or tendons that are in operative communication with the articulation section **1230** of the shaft assembly **1210**, as described in greater detail below. One articulation dial **1232***a* may be rotated in the direction of arrows "C0" to induce pivotal movement in a first plane, e.g., a vertical plane, as indicated by arrows "C1". Similarly, another articulation dial **1232***b* may be rotated in the direction of arrows "D0" to induce pivotal movement in a second plane, e.g., a horizontal plane, as indicated by arrows "D1". Rotation of the articulation dials **1232***a*, **1232***b* in either direction of arrows "C0" or "D0" results in the tendons pivoting or articulating the shaft assembly **1210** about the articulation section **1230**.

The battery assembly **1206** is electrically connected to the handle assembly **1202** by an electrical connector **1231**. The handle assembly **1202** is provided with a switch section **1220**. A first switch **1221***a* and a second switch **1221***b* are provided in the switch section **1220**. The RF generator is energized by actuating the first switch **1221***a* and the knife **1274** may be activated by energizing the motor assembly **1240** by actuating the second switch **1221***b*. Accordingly, the first switch **1221***a* energizes the RF circuit to drive the high-frequency current through the tissue to form a seal and the second switch **1221***b* energizes the motor to drive the knife **1274** to cut the tissue. In other aspects, the knife **1274** may be fired manually using a two-stage trigger **1208** configuration. The structural and functional aspects of the battery assembly **1206** are similar to those of the battery assembly **106** for the surgical instrument **100** described in connection with FIGS. **1**, **2**, and **16**-**24**. Accordingly, for conciseness and clarity of disclosure, such the structural and functional aspects of the battery assembly **106** are incorporated herein by reference and will not be repeated here.

A rotation knob **1218** is operably coupled to the shaft assembly **1210**. Rotation of the rotation knob **1218** ±360° in the direction indicated by the arrows **1226** causes the outer tube **1244** to rotate ±360° in the respective direction of the arrows **1228**. The end effector **1212** may be articulated by way of control buttons such that actuation of control buttons articulates the end effector **1212** in one direction indicated by arrows C1 and D1. Further, the outer tube **1244** may have a diameter D_3 ranging from 5 mm to 10 mm, for example.

FIG. **56** is an enlarged area detail view of an articulation section illustrated in FIG. **54** including electrical connections, according to one aspect of the present disclosure. FIG. **57** is an enlarged area detail view articulation section illustrated in FIG. **56** including electrical connections, according to one aspect of the present disclosure. With reference now to FIGS. **56**-**57**, there is shown the articulation section **1230** is operably disposed on or coupled to the shaft assembly **1210** between the proximal end and the distal end **1222**, respectively. In the aspect illustrated in FIGS. **56**-**57**, the articulation section **1230** is defined by a plurality of articulating links **1233** (links **1233**). The links **1233** are configured to articulate the shaft assembly **1210** transversely across the longitudinal axis "A-A" in either a horizontal or vertical plane, see FIG. **54**. For illustrative purposes, the shaft assembly **1210** is shown articulated across the horizontal plane.

The links **1233** collectively define a central annulus **1238** therethrough that is configured to receive a drive mechanism, e.g., a drive rod, therethrough. As can be appreciated, the configuration of the central annulus **1238** provides adequate clearance for the drive rod therethrough. The central annulus **1238** defines an axis "B-B" therethrough that is parallel to the longitudinal axis "A-A" when the shaft assembly **1210** is in a non-articulated configuration, see FIG. **54**.

Continuing with reference to FIGS. **56**-**57**, the links **1233** are operably coupled to the articulation dials **1232***a*, **1232***b* via tendons **1234**. For illustrative purposes, four (4) tendons **1234** are shown. The tendons **1234** may be constructed of stainless steel wire or other material suitable for transmitting tensile forces to a distal-most link of links **1233**. Regardless of the construction materials, the tendons **1234** exhibit a spring rate that is amplified over the length of the tendons **1234** and thus, the tendons **1234** may tend to stretch when external loads are applied to the elongated shaft assembly **1210**. This tendency to stretch may be associated with an unintended change in orientation of the distal end **1222** of the elongated shaft assembly **1210**, e.g., without a corresponding movement of the articulation dials **1232***a*, **1232***b* initiated by the surgeon.

The tendons **1234** operably couple to the articulating dials **1232***a*, **1232***b* that are configured to actuate the tendons **1234**, e.g., "pull" the tendons **1234**, when the articulating dials **1232***a*, **1232***b* are rotated. The plurality of tendons **1234** operably couple to the links **1233** via one or more suitable coupling methods. More particularly, the link **1233** includes a corresponding plurality of first apertures or bores **1236***a* defined therein (four (4) bores **1236***a* are shown in the representative figures) that are radially disposed along the links **1233** and centrally aligned along a common axis, see FIG. **56**. A bore of the plurality of bores **1236***a* is configured to receive a tendon **1234**. A distal end of a tendon **1234** is operably coupled to a distal most link of the links **1233** by suitable methods, e.g., one or more of the coupling methods described above.

Continuing with reference to FIGS. **56**-**57** a link **1233** includes a second plurality of bores **1236***b* (four (4) bores **1236***b* are shown in the representative drawings, as best seen in FIG. **56**). A bore **1236***b* is configured to receive a corresponding conductive lead of a plurality of conductive leads **1237** (four (4) conductive leads **1237** are shown in the representative drawings). The conductive leads **1237** are configured to transition between first and second states within the second plurality of bores **1236***b*. To facilitate transitioning of the conductive leads **1237**, a bore **1236***b* includes a diameter that is greater than a diameter of the conductive leads **1237** when the conductive leads **1237** are in the first state.

The surgical instrument **1220** includes electrical circuitry that is configured to selectively induce a voltage and current flow to the plurality of conductive leads **1237** such that a conductive lead **1237** transitions from the first state to the second state. To this end, the generator G provides a voltage potential Eo of suitable proportion. A voltage is induced in a conductive lead **1237** and current flow therethrough. The current flowing through a conductive lead **1237** causes the conductive lead **1237** to transition from the first state (FIG. **56**) to the second state (FIG. **57**). In the second state, the conductive lead **1237** provides an interference fit between the conductive lead **1237** and the corresponding bores **1236***b*, as best seen in FIG. **57**.

FIG. **58** illustrates a perspective view of components of the shaft assembly **1210**, end effector **1212**, and cutting member **1254** of the surgical instrument **1200** of FIG. **54**, according to one aspect of the present disclosure. FIG. **59** illustrates the articulation section in a second stage of articulation, according to one aspect of the present disclosure. With reference now to FIGS. **58-59**, one articulation band **1256***a* is slidably disposed in one side recess of a separator **1261** while a second articulation band **1256***b* (FIG. **59**) is slidably disposed in the other side recess of the separator **1261**. A cutting member driver tube is movable longitudinally to drive a driver block **1258** longitudinally, to thereby move cutting member **1254** longitudinally. The side recesses include longitudinally extending grooves that are configured to reduce the contact surface area with articulation bands **1256***a*, **1256***b*, thereby reducing friction between separator **1261** and articulation bands **1256***a*, **1256***b*. The separator **1261** also may be formed of a low friction material and/or include a surface treatment to reduce friction. Articulation bands **1256***a*, **1256***b* extend longitudinally along the length of the shaft assembly **1210**, including through the articulation section **1230**. The distal end **1252** of one articulation band **1256***a* is secured to one side of the proximal portion **1250** of end effector **1212** at an anchor point. The distal end **1262** of the second articulation band **1256***b* is secured to the other side of proximal portion **1250** of end effector **1212** at an anchor point. A rotary articulation knob is operable to selectively advance the articulation band **1256***a* distally while simultaneously retracting the second articulation band **1256***b* proximally, and vice-versa. It should be understood that this opposing translation will cause articulation section **1230** to bend, thereby articulating end effector **1212**. In particular, the end effector **1212** will deflect toward whichever articulation band **1256***a*, **1256***b* is being retracted proximally; and away from whichever articulation band **1256***a*, **1256***b* is being advanced distally.

With continued referenced to FIGS. **58-59**, several of the above described components are shown interacting to bend the articulation section **1230** to articulate end effector **1212**. In FIG. **58**, articulation **1230** is in a straight configuration. Then, one of the articulation dials **1232***a*, **1232***b* (FIGS. **54-55**) is rotated, which causes a lead screw to translate proximally and another lead screw to advance distally. This proximal translation of one lead screw pulls the articulation band **1256***b* proximally, which causes articulation section **1230** to start bending as shown in FIG. **59**. This bending of articulation section **1230** pulls the other articulation band **1256***a* distally. The distal advancement of lead screw in response to rotation of the articulation dials **1232***a*, **1232***b* enables the articulation band **1256***a* and the drive member to advance distally. In some other versions, the distal advancement of the lead screw actively drives drive member and articulation band **1256***a* distally. As the user continues rotating one of the articulation dials **1232***a*, **1232***b*, the above described interactions continue in the same fashion, resulting in further bending of articulation section **1230** as shown in FIG. **59**. It should be understand that rotating the articulation dials **1232***a*, **1232***b* in the opposite direction will cause articulation section **1230** to straighten, and further rotation in the opposite direction will cause articulation section **1230** to bend in the opposite direction.

FIG. **60** illustrates a perspective view of the end effector **1212** of the device of FIGS. **54-59** in an open configuration, according to one aspect of the present disclosure. The end effector **1212** of the present example comprises a pair of jaw members **1214***a*, **1214***b*. In the present example, one jaw member **1214***b* is fixed relative to shaft assembly; while the other jaw member **1214***a* pivots relative to shaft assembly, toward and away from the other jaw member **1214***b*. In some versions, actuators such as rods or cables, etc., may extend through a sheath and be joined with one jaw member **1214***a* at a pivotal coupling, such that longitudinal movement of the actuator rods/cables/etc. through the shaft assembly provides pivoting of the jaw member **1214***a* relative to shaft assembly and relative to the second jaw member **1214***b*. Of course, the jaw members **1214***a*, **1214***b* instead may have any other suitable kind of movement and may be actuated in any other suitable fashion. By way of example only, the jaw members **1214***a*, **1214***b* may be actuated and thus closed by longitudinal translation of a firing beam **1266**, such that actuator rods/cables/etc. may simply be eliminated in some versions. The upper side of one jaw member **1214***a* including a plurality of teeth serrations **1272**. It should be understood that the lower side of the other jaw member **1214***b* may include complementary serrations **1277** that nest with the serrations **1272**, to enhance gripping of tissue captured between the jaw members **1214***a*, **1214***b* of the end effector **1212** without necessarily tearing the tissue.

FIG. **61** illustrates a cross-sectional end view of the end effector **1212** of FIG. **60** in a closed configuration and with the blade **1274** in a distal position, according to one aspect to the present disclosure. With reference now to FIGS. **60-61**, one jaw member **1214***a* defines a longitudinally extending elongate slot **1268**; while the other jaw member **1214***b* also defines a longitudinally extending elongate slot **1270**. In addition, the underside of one jaw member **1214***a* presents an electrically conductive surface **1216***a*; while the top side of the other jaw member **1214***b* presents another electrically conductive surface **1216***b*. The electrically conductive surfaces **1216***a*, **1216***b* are in communication with an electrical source **1278** and a controller **1280** via one or more conductors (not shown) that extend along the length of shaft assembly. The electrical source **1278** is operable to deliver RF energy to first electrically conductive surface **1216***b* at a first polarity and to second electrically conductive surface **1216***a* at a second (opposite) polarity, such that RF current flows between electrically conductive surfaces **1216***a*, **1216***b* and thereby through tissue captured between the jaw members **1214***a*, **1214***b*. In some versions, firing beam **1266** serves as an electrical conductor that cooperates with the electrically conductive surfaces **1216***a*, **1216***b* (e.g., as a ground return) for delivery of bipolar RF energy captured between the jaw members **1214***a*, **1214***b*. The electrical source **1278** may be external to surgical instrument **1200** or may be integral with surgical instrument **1200** (e.g., in the handle assembly **1202**, etc.), as described in one or more references cited herein or otherwise. A controller **1280** regulates delivery of power from electrical source **1278** to the electrically conductive surfaces **1216***a*, **1216***b*. The controller **1280** may also be external to surgical instrument **1200** or may be integral with surgical instrument **1200** (e.g., in handle assembly **1202**, etc.), as described in one or more references cited herein or otherwise. It should also be understood that the electrically conductive surfaces **1216***a*, **1216***b* may be provided in a variety of alternative locations, configurations, and relationships.

Still with reference to FIGS. **60-61**, the surgical instrument **1200** of the present example includes a firing beam **1266** that is longitudinally movable along part of the length of end effector **1212**. The firing beam **1266** is coaxially positioned within the shaft assembly **1210**, extends along the length of the shaft assembly **1210**, and translates longitudinally within the shaft assembly **1210** (including the articulation section **1230** in the present example), though it should be understood that firing beam **12660** and the shaft assembly **1210** may have any other suitable relationship. The firing beam **1266** includes a knife **1274** with a sharp distal end, an upper flange **1281**, and a lower flange **1282**. As best seen in FIG. **61**, the knife **1274** extends through slots **1268**, **1270** of the jaw members **1214***a*, **1214***b*, with the upper flange **1281** being located above the jaw member **1214***a* in a recess **1284** and the lower flange **1282** being located below the jaw member **1214***b* in a recess **1286**. The configuration of the knife **1274** and the flanges **1281**, **1282** provides an "I-beam" type of cross section at the distal end of firing beam **1266**. While the flanges **1281**, **1282** extend longitudinally only along a small portion of the length of firing beam **1266** in the present example, it should be understood that the flanges **1281**, **1282** may extend longitudinally along any suitable length of firing beam **1266**. In addition, while the flanges **1281**, **1282** are positioned along the exterior of the jaw members **1214***a*, **1214***b*, the flanges **1281**, **1282** may alternatively be disposed in corresponding slots formed within jaw members **1214***a*, **1214***b*. For instance, the jaw members **1214***a*, **1214***b* may define a "T"-shaped slot, with parts of the knife **1274** being disposed in one vertical portion of a "T"-shaped slot and with the flanges **1281**, **1282** being disposed in the horizontal portions of the "T"-shaped slots. Various other suitable configurations and relationships will be apparent to those of ordinary skill in the art in view of the teachings herein. By way of example only, the end effector **1212** may include one or more positive temperature coefficient (PTC) thermistor bodies **1288**, **1290** (e.g., PTC polymer, etc.), located adjacent to the electrically conductive surfaces **1216***a*, **1216***b* and/or elsewhere.

The structural and functional aspects of the battery assembly **1206** are similar to those of the battery assembly **106** for the surgical instrument **100** described in connection with FIGS. **1**, **2**, and **16-24**, including the battery circuits described in connection with FIGS. **20-24**. Accordingly, for conciseness and clarity of disclosure, such the structural and functional aspects of the battery assembly **106** are incorporated herein by reference and will not be repeated here. Furthermore, the structural and functional aspects of the RF generator circuits are similar to those of the RF generator circuits described in for the surgical instruments **500**, **600** described in connection with FIGS. **34-37**. Accordingly, for conciseness and clarity of disclosure, such the structural and functional aspects of the RF generator circuits are incorporated herein by reference and will not be repeated here. Furthermore, the surgical instrument **1200** includes the battery and control circuits described in connection with FIGS. **12-15**, including, for example, the control circuit **210** described in connection with FIG. **14** and the electrical circuit **300** described in connection withe FIG. **15**. Accordingly, for conciseness and clarity of disclosure, the description of the circuits described in connection with FIGS. **12-15** is incorporated herein by reference and will not be repeated here.

For a more detailed description of an electrosurgical instrument comprising a cutting mechanism and an articulation section that is operable to deflect the end effector away from the longitudinal axis of the shaft, reference is made to U.S. Patent Application Publication No. 2013/0023868, now U.S. Pat. No. 9,545,253, which is herein incorporated by reference.

It should also be understood that any of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein may be modified to include a motor or other electrically powered device to drive an otherwise manually moved component. Various examples of such modifications are described in U.S. Patent Application Publication No. 2012/0116379, now U.S. Pat. No. 9,161,803, and U.S. Patent Application Publication No. 2016/0256184, now U.S. Pat. No. 9,808,246, each of which is incorporated herein by reference. Various other suitable ways in which a motor or other electrically powered device may be incorporated into any of the devices herein will be apparent to those of ordinary skill in the art in view of the teachings herein.

It should also be understood that the circuits described in connection with FIGS. **11-15**, **20-24**, **34-37**, and **50** may be configured to operate either alone or in combination with any of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein.

FIGS. **62-70** describe various circuits that are configured to operate with any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described in connections with FIGS. **1-61**. Turning now to FIG. **62**, there is shown the components of a control circuit **1300** of the surgical instrument, according to one aspect of the present disclosure. The control circuit **1300** comprises a processor **1302** coupled to a volatile memory **1304**, one or more sensors **1306**, a nonvolatile memory **1308** and a battery **1310**. In one aspect, the surgical instrument may comprise a handle housing to house the control circuit **1300** and to contain general purpose controls to implement the power conservation mode. In some aspects, the processor **1302** may be a primary processor of the surgical instrument that includes one or more secondary processors. In some aspects, the processor **1302** may be stored within the battery **1310**. The processor **1302** is configured to control various operations and functions of the surgical instrument by executing machine executable instructions, such as control programs or other software modules. For example, execution of an energy modality control program by the processor **1302** enables selection of a particular type of energy to be applied to patient tissue by a surgeon using the surgical instrument. The surgical instrument may comprise an energy modality actuator located on the handle of the surgical instrument. The actuator may be a slider, a toggle switch, a segmented momentary contact switch, or some other type of actuator. Actuation of the energy modality actuator causes the processor **1302** to activate an energy modality corresponding to a selected type of energy. The type of energy can be ultrasonic, RF, or a combination of ultrasonic and RF energy. In various aspects general, the processor **1302** is electrically coupled to the plurality of circuit segments of the surgical instrument as illustrated in FIG. **63** to activate or deactivate the circuit segments in accordance with energization and deenergization sequences.

The volatile memory **1304**, such as a random-access memory (RAM), temporarily stores selected control programs or other software modules while the processor **1302** is in operation, such as when the processor **1302** executes a control program or software module. The one or more sensors **1306** may include force sensors, temperature sensors, current sensors or motion sensors. In some aspects, the one or more sensors **1306** may be located at the shaft, end effector, battery, or handle, or any combination or subcombination thereof. The one or more sensors **1306** transmit data associated with the operation of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described in connection with FIGS. **1-61**, such as the presence of tissue grasped by the jaws of the end effector or the force applied by the motor. In one aspect, the one or more sensors **1306** may include an accelerometer to verify the function or operation of the circuit segments, based on a safety check and a Power On Self Test (POST). Machine executable instructions such as control programs or other software modules are stored in the nonvolatile memory **1308**. For example, the nonvolatile memory **1308** stores the Basic Input/Output System (BIOS) program. The nonvolatile memory **1308** may be a read-only memory, erasable programmable ROM (EPROM), an EEPROM, flash memory or some other type of nonvolatile memory device. Various examples of control programs are described in U.S. Patent Application Publication No. 2015/0272578, now U.S. Pat. No. 9,690,362, which is incorporated herein by reference in its entirety. The battery **1310** powers the surgical instrument by providing a source voltage that causes a current. The battery **1310** may comprise the motor control circuit segment **1428** illustrated in FIG. **63**.

In one aspect, the processor **1302** may be any single core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In one aspect, the processor **1302** may be implemented as a safety processor comprising two microcontroller-based families such as TMS570 and RM4x known under the trade name Hercules ARM Cortex R4, also by Texas Instruments. Nevertheless, other suitable substitutes for microcontrollers and safety processor may be employed, without limitation. In one aspect, the safety processor may be configured specifically for IEC 61508 and ISO 26262 safety critical applications, among others, to provide advanced integrated safety features while delivering scalable performance, connectivity, and memory options.

In certain aspects, the processor **1302** may be an LM 4F230H5QR, available from Texas Instruments, for example. In at least one example, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHZ, a prefetch buffer to improve performance above 40 MHZ, a 32 KB single-cycle serial random access memory (SRAM), internal read-only memory (ROM) loaded with StellarisWare® software, 2 KB electrically erasable programmable read-only memory (EEPROM), one or more pulse width modulation (PWM) modules, one or more quadrature encoder inputs (QED analog, one or more 12-bit Analog-to-Digital Converters (ADC) with 12 analog input channels, among other features that are readily available for the product datasheet. Other processors may be readily substituted and, accordingly, the present disclosure should not be limited in this context.

FIG. **63** is a system diagram **1400** of a segmented circuit **1401** comprising a plurality of independently operated circuit segments **1402**, **1414**, **1416**, **1420**, **1424**, **1428**, **1434**, **1440**, according to one aspect of the present disclosure. A circuit segment of the plurality of circuit segments of the segmented circuit **1401** comprises one or more circuits and one or more sets of machine executable instructions stored in one or more memory devices. The one or more circuits of a circuit segment are coupled to for electrical communication through one or more wired or wireless connection media. The plurality of circuit segments are configured to transition between three modes comprising a sleep mode, a standby mode and an operational mode.

In one aspect shown, the plurality of circuit segments **1402**, **1414**, **1416**, **1420**, **1424**, **1428**, **1434**, **1440** start first in the standby mode, transition second to the sleep mode, and transition third to the operational mode. However, in other aspects, the plurality of circuit segments may transition from any one of the three modes to any other one of the three modes. For example, the plurality of circuit segments may transition directly from the standby mode to the operational mode. Individual circuit segments may be placed in a particular state by the voltage control circuit **1408** based on the execution by the processor **1302** of machine executable instructions. The states comprise a deenergized state, a low energy state, and an energized state. The deenergized state corresponds to the sleep mode, the low energy state corresponds to the standby mode, and the energized state corresponds to the operational mode. Transition to the low energy state may be achieved by, for example, the use of a potentiometer.

In one aspect, the plurality of circuit segments **1402**, **1414**, **1416**, **1420**, **1424**, **1428**, **1434**, **1440** may transition from the sleep mode or the standby mode to the operational mode in accordance with an energization sequence. The plurality of circuit segments also may transition from the operational mode to the standby mode or the sleep mode in accordance with a deenergization sequence. The energization sequence and the deenergization sequence may be different. In some aspects, the energization sequence comprises energizing only a subset of circuit segments of the plurality of circuit segments. In some aspects, the deenergization sequence comprises deenergizing only a subset of circuit segments of the plurality of circuit segments.

Referring back to the system diagram **1400** in FIG. **63**, the segmented circuit **1401** comprise a plurality of circuit segments comprising a transition circuit segment **1402**, a processor circuit segment **1414**, a handle circuit segment **1416**, a communication circuit segment **1420**, a display circuit segment **1424**, a motor control circuit segment **1428**, an energy treatment circuit segment **1434**, and a shaft circuit segment **1440**. The transition circuit segment comprises a wake up circuit **1404**, a boost current circuit **1406**, a voltage control circuit **1408**, a safety controller **1410** and a POST controller **1412**. The transition circuit segment **1402** is configured to implement a deenergization and an energization sequence, a safety detection protocol, and a POST.

In some aspects, the wake up circuit **1404** comprises an accelerometer button sensor **1405**. In aspects, the transition circuit segment **1402** is configured to be in an energized state while other circuit segments of the plurality of circuit segments of the segmented circuit **1401** are configured to be in a low energy state, a deenergized state or an energized state. The accelerometer button sensor **1405** may monitor movement or acceleration of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61**. For example, the movement may be a change in orientation or rotation of the surgical instrument. The surgical instrument may be moved in any direction relative to a three dimensional Euclidean space by for example, a user of the surgical instrument. When the accelerometer button sensor **1405** senses movement or acceleration, the accelerometer button sensor **1405** sends a signal to the voltage control circuit **1408** to cause the voltage control circuit **1408** to apply voltage to the processor circuit segment **1414** to transition the processor **1302** and the volatile memory **1304** to an energized state. In aspects, the processor **1302** and the volatile memory **1304** are in an energized state before the voltage control circuit **1409** applies voltage to the processor **1302** and the volatile memory **1304**. In the operational mode, the processor **1302** may initiate an energization sequence or a deenergization sequence. In various aspects, the accelerometer button sensor **1405** may also send a signal to the processor **1302** to cause the processor **1302** to initiate an energization sequence or a deenergization sequence. In some aspects, the processor **1302** initiates an energization sequence when the majority of individual circuit segments are in a low energy state or a deenergized state. In other aspects, the processor **1302** initiates a deenergization sequence when the majority of individual circuit segments are in an energized state.

Additionally or alternatively, the accelerometer button sensor **1405** may sense external movement within a predetermined vicinity of the surgical instrument. For example, the accelerometer button sensor **1405** may sense a user of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61** moving a hand of the user within the predetermined vicinity. When the accelerometer button sensor **1405** senses this external movement, the accelerometer button sensor **1405** may send a signal to the voltage control circuit **1408** and a signal to the processor **1302**, as previously described. After receiving the sent signal, the processor **1302** may initiate an energization sequence or a deenergization sequence to transition one or more circuit segments between the three modes. In aspects, the signal sent to the voltage control circuit **1408** is sent to verify that the processor **1302** is in operational mode. In some aspects, the accelerometer button sensor **1405** may sense when the surgical instrument has been dropped and send a signal to the processor **1302** based on the sensed drop. For example, the signal can indicate an error in the operation of an individual circuit segment. The one or more sensors **1306** may sense damage or malfunctioning of the affected individual circuit segments. Based on the sensed damage or malfunctioning, the POST controller **1412** may perform a POST of the corresponding individual circuit segments.

An energization sequence or a deenergization sequence may be defined based on the accelerometer button sensor **1405**. For example, the accelerometer button sensor **1405** may sense a particular motion or a sequence of motions that indicates the selection of a particular circuit segment of the plurality of circuit segments. Based on the sensed motion or series of sensed motions, the accelerometer button sensor **1405** may transmit a signal comprising an indication of one or more circuit segments of the plurality of circuit segments to the processor **1302** when the processor **1302** is in an energized state. Based on the signal, the processor **1302** determines an energization sequence comprising the selected one or more circuit segments. Additionally or alternatively, a user of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61** may select a number and order of circuit segments to define an energization sequence or a deenergization sequence based on interaction with a graphical user interface (GUI) of the surgical instrument.

In various aspects, the accelerometer button sensor **1405** may send a signal to the voltage control circuit **1408** and a signal to the processor **1302** only when the accelerometer button sensor **1405** detects movement of any one the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61** or external movement within a predetermined vicinity above a predetermined threshold. For example, a signal may only be sent if movement is sensed for 5 or more seconds or if the surgical instrument is moved 5 or more inches. In other aspects, the accelerometer button sensor **1405** may send a signal to the voltage control circuit **1408** and a signal to the processor **1302** only when the accelerometer button sensor **1405** detects oscillating movement of the surgical instrument. A predetermined threshold reduces inadvertent transition of circuit segments of the surgical instrument. As previously described, the transition may comprise a transition to operational mode according to an energization sequence, a transition to low energy mode according to a deenergization sequence, or a transition to sleep mode according to a deenergization sequence. In some aspects, the surgical instrument comprises an actuator that may be actuated by a user of the surgical instrument. The actuation is sensed by the accelerometer button sensor **1405**. The actuator may be a slider, a toggle switch, or a momentary contact switch. Based on the sensed actuation, the accelerometer button sensor **1405** may send a signal to the voltage control circuit **1408** and a signal to the processor **1302**.

The boost current circuit **1406** is coupled to the battery **1310**. The boost current circuit **1406** is a current amplifier, such as a relay or transistor, and is configured to amplify the magnitude of a current of an individual circuit segment. The initial magnitude of the current corresponds to the source voltage provided by the battery **1310** to the segmented circuit **1401**. Suitable relays include solenoids. Suitable transistors include field-effect transistors (FET), MOSFET, and bipolar junction transistors (BJT). The boost current circuit **1406** may amplify the magnitude of the current corresponding to an individual circuit segment or circuit which requires more current draw during operation of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described in connection with FIGS. **1-61**. For example, an increase in current to the motor control circuit segment **1428** may be provided when a motor of the surgical instrument requires more input power. The increase in current provided to an individual circuit segment may cause a corresponding decrease in current of another circuit segment or circuit segments. Additionally or alternatively, the increase in current may correspond to voltage provided by an additional voltage source operating in conjunction with the battery **1310**.

The voltage control circuit **1408** is coupled to the battery **1310**. The voltage control circuit **1408** is configured to provide voltage to or remove voltage from the plurality of circuit segments. The voltage control circuit **1408** is also configured to increase or reduce voltage provided to the plurality of circuit segments of the segmented circuit **1401**. In various aspects, the voltage control circuit **1408** comprises a combinational logic circuit such as a multiplexer (MUX) to select inputs, a plurality of electronic switches, and a plurality of voltage converters. An electronic switch of the plurality of electronic switches may be configured to switch between an open and closed configuration to disconnect or connect an individual circuit segment to or from the battery **1310**. The plurality of electronic switches may be solid state devices such as transistors or other types of switches such as wireless switches, ultrasonic switches, accelerometers, inertial sensors, among others. The combinational logic circuit is configured to select an individual electronic switch for switching to an open configuration to enable application of voltage to the corresponding circuit segment. The combination logic circuit also is configured to select an individual electronic switch for switching to a closed configuration to enable removal of voltage from the corresponding circuit segment. By selecting a plurality of individual electronic switches, the combination logic circuit may implement a deenergization sequence or an energization sequence. The plurality of voltage converters may provide a stepped-up voltage or a stepped-down voltage to the plurality of circuit segments. The voltage control circuit **1408** may also comprise a microprocessor and memory device, as illustrated in FIG. **62**.

The safety controller **1410** is configured to perform safety checks for the circuit segments. In some aspects, the safety controller **1410** performs the safety checks when one or more individual circuit segments are in the operational mode. The safety checks may be performed to determine whether there are any errors or defects in the functioning or operation of the circuit segments. The safety controller **1410** may monitor one or more parameters of the plurality of circuit segments. The safety controller **1410** may verify the identity and operation of the plurality of circuit segments by comparing the one or more parameters with predefined parameters. For example, if an RF energy modality is selected, the safety controller **1410** may verify that an articulation parameter of the shaft matches a predefined articulation parameter to verify the operation of the RF energy modality of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described in connection with FIGS. **1-61**. In some aspects, the safety controller **1410** may monitor, by the sensors **1306**, a predetermined relationship between one or more properties of the surgical instrument to detect a fault. A fault may arise when the one or more properties are inconsistent with the predetermined relationship. When the safety controller **1410** determines that a fault exists, an error exists, or that some operation of the plurality of circuit segments was not verified, the safety controller **1410** prevents or disables operation of the particular circuit segment where the fault, error or verification failure originated.

The POST controller **1412** performs a POST to verify proper operation of the plurality of circuit segments. In some aspects, the POST is performed for an individual circuit segment of the plurality of circuit segments prior to the voltage control circuit **1408** applying a voltage to the individual circuit segment to transition the individual circuit segment from standby mode or sleep mode to operational mode. If the individual circuit segment does not pass the POST, the particular circuit segment does not transition from standby mode or sleep mode to operational mode. POST of the handle circuit segment **1416** may comprise, for example, testing whether the handle control sensors **1418** sense an actuation of a handle control of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described in connection with FIGS. **1-61**. In some aspects, the POST controller **1412** may transmit a signal to the accelerometer button sensor **1405** to verify the operation of the individual circuit segment as part of the POST. For example, after receiving the signal, the accelerometer button sensor **1405** may prompt a user of the surgical instrument to move the surgical instrument to a plurality of varying locations to confirm operation of the surgical instrument. The accelerometer button sensor **1405** may also monitor an output of a circuit segment or a circuit of a circuit segment as part of the POST. For example, the accelerometer button sensor **1405** can sense an incremental motor pulse generated by the motor **1432** to verify operation. A motor controller of the motor control circuit **1430** may be used to control the motor **1432** to generate the incremental motor pulse.

In various aspects, any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described in connection with FIGS. **1-61** may comprise additional accelerometer button sensors may be used. The POST controller **1412** may also execute a control program stored in the memory device of the voltage control circuit **1408**. The control program may cause the POST controller **1412** to transmit a signal requesting a matching encrypted parameter from a plurality of circuit segments. Failure to receive a matching encrypted parameter from an individual circuit segment indicates to the POST controller **1412** that the corresponding circuit segment is damaged or malfunctioning. In some aspects, if the POST controller **1412** determines based on the POST that the processor **1302** is damaged or malfunctioning, the POST controller **1412** may send a signal to one or more secondary processors to cause one or more secondary processors to perform critical functions that the processor **1302** is unable to perform. In some aspects, if the POST controller **1412** determines based on the POST that one or more circuit segments do not operate properly, the POST controller **1412** may initiate a reduced performance mode of those circuit segments operating properly while locking out those circuit segments that fail POST or do not operate properly. A locked out circuit segment may function similarly to a circuit segment in standby mode or sleep mode.

The processor circuit segment **1414** comprises the processor **1302** and the volatile memory **1304** described with reference to FIG. **62**. The processor **1302** is configured to initiate an energization or a deenergization sequence. To initiate the energization sequence, the processor **1302** transmits an energizing signal to the voltage control circuit **1408** to cause the voltage control circuit **1408** to apply voltage to the plurality or a subset of the plurality of circuit segments in accordance with the energization sequence. To initiate the deenergization sequence, the processor **1302** transmits a deenergizing signal to the voltage control circuit **1408** to cause the voltage control circuit **1408** to remove voltage from the plurality or a subset of the plurality of circuit segments in accordance with the deenergization sequence.

The handle circuit segment **1416** comprises handle control sensors **1418**. The handle control sensors **1418** may sense an actuation of one or more handle controls of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61**. In various aspects, the one or more handle controls comprise a clamp control, a release button, an articulation switch, an energy activation button, and/or any other suitable handle control. The user may activate the energy activation button to select between an RF energy mode, an ultrasonic energy mode or a combination RF and ultrasonic energy mode. The handle control sensors **1418** may also facilitate attaching a modular handle to the surgical instrument. For example, the handle control sensors **1418** may sense proper attachment of the modular handle to the surgical instrument and indicate the sensed attachment to a user of the surgical instrument. The LCD display **1426** may provide a graphical indication of the sensed attachment. In some aspects, the handle control sensors **1418** senses actuation of the one or more handle controls. Based on the sensed actuation, the processor **1302** may initiate either an energization sequence or a deenergization sequence.

The communication circuit segment **1420** comprises a communication circuit **1422**. The communication circuit **1422** comprises a communication interface to facilitate signal communication between the individual circuit segments of the plurality of circuit segments. In some aspects, the communication circuit **1422** provides a path for the modular components of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61** to communicate electrically. For example, a modular shaft and a modular transducer, when attached together to the handle of the surgical instrument, can upload control programs to the handle through the communication circuit **1422**.

The display circuit segment **1424** comprises a LCD display **1426**. The LCD display **1426** may comprise a liquid crystal display screen, LED indicators, etc. In some aspects, the LCD display **1426** is an organic light-emitting diode (OLED) screen. The Display **226** may be placed on, embedded in, or located remotely from any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61**. For example, the

Display **226** can be placed on the handle of the surgical instrument. The Display **226** is configured to provide sensory feedback to a user. In various aspects, the LCD display **1426** further comprises a backlight. In some aspects, the surgical instrument may also comprise audio feedback devices such as a speaker or a buzzer and tactile feedback devices such as a haptic actuator.

The motor control circuit segment **1428** comprises a motor control circuit **1430** coupled to a motor **1432**. The motor **1432** is coupled to the processor **1302** by a driver and a transistor, such as a FET. In various aspects, the motor control circuit **1430** comprises a motor current sensor in signal communication with the processor **1302** to provide a signal indicative of a measurement of the current draw of the motor to the processor **1302**. The processor transmits the signal to the Display **226**. The Display **226** receives the signal and displays the measurement of the current draw of the motor **1432**. The processor **1302** may use the signal, for example, to monitor that the current draw of the motor **1432** exists within an acceptable range, to compare the current draw to one or more parameters of the plurality of circuit segments, and to determine one or more parameters of a patient treatment site. In various aspects, the motor control circuit **1430** comprises a motor controller to control the operation of the motor. For example, the motor control circuit **1430** controls various motor parameters, such as by adjusting the velocity, torque and acceleration of the motor **1432**. The adjusting is done based on the current through the motor **1432** measured by the motor current sensor.

In various aspects, the motor control circuit **1430** comprises a force sensor to measure the force and torque generated by the motor **1432**. The motor **1432** is configured to actuate a mechanism of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61**. For example, the motor **1432** is configured to control actuation of the shaft of the surgical instrument to realize clamping, rotation and articulation functionality. For example, the motor **1432** may actuate the shaft to realize a clamping motion with jaws of the surgical instrument. The motor controller may determine whether the material clamped by the jaws is tissue or metal. The motor controller may also determine the extent to which the jaws clamp the material. For example, the motor controller may determine how open or closed the jaws are based on the derivative of sensed motor current or motor voltage. In some aspects, the motor **1432** is configured to actuate the transducer to cause the transducer to apply torque to the handle or to control articulation of the surgical instrument. The motor current sensor may interact with the motor controller to set a motor current limit. When the current meets the predefined threshold limit, the motor controller initiates a corresponding change in a motor control operation. For example, exceeding the motor current limit causes the motor controller to reduce the current draw of the motor.

The energy treatment circuit segment **1434** comprises a RF amplifier and safety circuit **1436** and an ultrasonic signal generator circuit **1438** to implement the energy modular functionality of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described in connection with FIGS. **1-61**. In various aspects, the RF amplifier and safety circuit **1436** is configured to control the RF modality of the surgical instrument by generating an RF signal. The ultrasonic signal generator circuit **1438** is configured to control the ultrasonic energy modality by generating an ultrasonic signal. The RF amplifier and safety circuit **1436** and an ultrasonic signal generator circuit **1438** may operate in conjunction to control the combination RF and ultrasonic energy modality.

The shaft circuit segment **1440** comprises a shaft module controller **1442**, a modular control actuator **1444**, one or more end effector sensors **1446**, and a non volatile memory **1448**. The shaft module controller **1442** is configured to control a plurality of shaft modules comprising the control programs to be executed by the processor **1302**. The plurality of shaft modules implements a shaft modality, such as ultrasonic, combination ultrasonic and RF, RF I-blade, and RF-opposable jaw. The shaft module controller **1442** can select shaft modality by selecting the corresponding shaft module for the processor **1302** to execute. The modular control actuator **1444** is configured to actuate the shaft according to the selected shaft modality. After actuation is initiated, the shaft articulates the end effector according to the one or more parameters, routines or programs specific to the selected shaft modality and the selected end effector modality. The one or more end effector sensors **1446** located at the end effector may include force sensors, temperature sensors, current sensors or motion sensors. The one or more end effector sensors **1446** transmit data about one or more operations of the end effector, based on the energy modality implemented by the end effector. In various aspects, the energy modalities include an ultrasonic energy modality, a RF energy modality, or a combination of the ultrasonic energy modality and the RF energy modality. The non volatile memory **1448** stores the shaft control programs. A control program comprises one or more parameters, routines or programs specific to the shaft. In various aspects, the non volatile memory **1448** may be an ROM, EPROM, EEPROM or flash memory. The non volatile memory **1448** stores the shaft modules corresponding to the selected shaft of nay one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61**. The shaft modules may be changed or upgraded in the non volatile memory **1448** by the shaft module controller **1442**, depending on the surgical instrument shaft to be used in operation.

FIG. **64** illustrates a diagram of one aspect of a surgical instrument **1500** comprising a feedback system for use with any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61**, which may include or implement many of the features described herein. For example, in one aspect, the surgical instrument **1500** may be similar to or representative of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200**. The surgical instrument **1500** may include a generator **1502**. The surgical instrument **1500** also may include an end effector **1506**, which may be activated when a clinician operates a trigger **1510**. In various aspects, the end effector **1506** may include an ultrasonic blade to deliver ultrasonic vibration to carry out surgical coagulation/cutting treatments on living tissue. In other aspects the end effector **1506** may include electrically conductive elements coupled to an electrosurgical high-frequency current energy source to carry out surgical coagulation or cauterization treatments on living tissue and either a mechanical knife with a sharp edge or an ultrasonic blade to carry out cutting treatments on living tissue. When the trigger **1510** is actuated, a force sensor **1512** may generate a signal indicating the amount of force being applied to the trigger **1510**. In addition to, or instead of a force sensor **1512**, the surgical instrument **1500** may include a position sensor **1513**, which may generate a signal indicating the position of the trigger **1510** (e.g., how far the trigger has been depressed or otherwise actuated). In one aspect, the position sensor **1513** may be a sensor positioned with the outer tubular sheath described above or reciprocating tubular actuating member located within the outer tubular sheath described above. In one aspect, the sensor may be a Hall-effect sensor or any suitable transducer that varies its output voltage in response to a magnetic field. The Hall-effect sensor may be used for proximity switching, positioning, speed detection, and current sensing applications. In one aspect, the Hall-effect sensor operates as an analog transducer, directly returning a voltage. With a known magnetic field, its distance from the Hall plate can be determined.

A control circuit **1508** may receive the signals from the sensors **1512** and/or **1513**. The control circuit **1508** may include any suitable analog or digital circuit components. The control circuit **1508** also may communicate with the generator **1502** and/or the transducer **1504** to modulate the power delivered to the end effector **1506** and/or the generator level or ultrasonic blade amplitude of the end effector **1506** based on the force applied to the trigger **1510** and/or the position of the trigger **1510** and/or the position of the outer tubular sheath described above relative to the reciprocating tubular actuating member **58** located within the outer tubular sheath **56** described above (e.g., as measured by a Hall-effect sensor and magnet combination). For example, as more force is applied to the trigger **1510**, more power and/or a higher ultrasonic blade amplitude may be delivered to the end effector **1506**. According to various aspects, the force sensor **1512** may be replaced by a multi-position switch.

According to various aspects, the end effector **1506** may include a clamp or clamping mechanism, for example, such as that described above with respect to FIGS. **1-5**. When the trigger **1510** is initially actuated, the clamping mechanism may close, clamping tissue between a clamp arm and the end effector **1506**. As the force applied to the trigger increases (e.g., as sensed by force sensor **1512**) the control circuit **1508** may increase the power delivered to the end effector **1506** by the transducer **1504** and/or the generator level or ultrasonic blade amplitude brought about in the end effector **1506**. In one aspect, trigger position, as sensed by position sensor **1513** or clamp or clamp arm position, as sensed by position sensor **1513** (e.g., with a Hall-effect sensor), may be used by the control circuit **1508** to set the power and/or amplitude of the end effector **1506**. For example, as the trigger is moved further towards a fully actuated position, or the clamp or clamp arm moves further towards the ultrasonic blade (or end effector **1506**), the power and/or amplitude of the end effector **1506** may be increased.

According to various aspects, the surgical instrument **1500** also may include one or more feedback devices for indicating the amount of power delivered to the end effector **1506**. For example, a speaker **1514** may emit a signal indicative of the end effector power. According to various aspects, the speaker **1514** may emit a series of pulse sounds, where the frequency of the sounds indicates power. In addition to, or instead of the speaker **1514**, the surgical instrument **1500** may include a visual display **1516**. The visual display **1516** may indicate end effector power according to any suitable method. For example, the visual display **1516** may include a series of LEDs, where end effector power is indicated by the number of illuminated LEDs. The speaker **1514** and/or visual display **1516** may be driven by the control circuit **1508**. According to various aspects, the surgical instrument **1500** may include a ratcheting device (not shown) connected to the trigger **1510**. The ratcheting device may generate an audible sound as more force is applied to the trigger **1510**, providing an indirect indication of end effector power. The surgical instrument **1500** may include other features that may enhance safety. For example, the control circuit **1508** may be configured to prevent power from being delivered to the end effector **1506** in excess of a predetermined threshold. Also, the control circuit **1508** may implement a delay between the time when a change in end effector power is indicated (e.g., by speaker **1514** or visual display **1516**), and the time when the change in end effector power is delivered. In this way, a clinician may have ample warning that the level of ultrasonic power that is to be delivered to the end effector **1506** is about to change.

In one aspect, the ultrasonic or high-frequency current generators of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61** may be configured to generate the electrical signal waveform digitally such that the desired using a predetermined number of phase points stored in a lookup table to digitize the wave shape. The phase points may be stored in a table defined in a memory, a field programmable gate array (FPGA), or any suitable non-volatile memory. FIG. **65** illustrates one aspect of a fundamental architecture for a digital synthesis circuit such as a direct digital synthesis (DDS) circuit **1600** configured to generate a plurality of wave shapes for the electrical signal waveform. The generator software and digital controls may command the FPGA to scan the addresses in the lookup table **1604** which in turn provides varying digital input values to a DAC circuit **1608** that feeds a power amplifier. The addresses may be scanned according to a frequency of interest. Using such a lookup table **1604** enables generating various types of wave shapes that can be fed into tissue or into a transducer, an RF electrode, multiple transducers simultaneously, multiple RF electrodes simultaneously, or a combination of RF and ultrasonic instruments. Furthermore, multiple lookup tables **1604** that represent multiple wave shapes can be created, stored, and applied to tissue from a generator.

The waveform signal may be configured to control at least one of an output current, an output voltage, or an output power of an ultrasonic transducer and/or an RF electrode, or multiples thereof (e.g. two or more ultrasonic transducers and/or two or more RF electrodes). Further, where the surgical instrument comprises an ultrasonic components, the waveform signal may be configured to drive at least two vibration modes of an ultrasonic transducer of the at least one surgical instrument. Accordingly, a generator may be configured to provide a waveform signal to at least one surgical instrument wherein the waveform signal corresponds to at least one wave shape of a plurality of wave shapes in a table. Further, the waveform signal provided to the two surgical instruments may comprise two or more wave shapes. The table may comprise information associated with a plurality of wave shapes and the table may be stored within the generator. In one aspect or example, the table may be a direct digital synthesis table, which may be stored in an FPGA of the generator. The table may be addressed by anyway that is convenient for categorizing wave shapes. According to one aspect, the table, which may be a direct digital synthesis table, is addressed according to a frequency of the waveform signal. Additionally, the information associated with the plurality of wave shapes may be stored as digital information in the table.

The analog electrical signal waveform may be configured to control at least one of an output current, an output voltage, or an output power of an ultrasonic transducer and/or an RF electrode, or multiples thereof (e.g., two or more ultrasonic transducers and/or two or more RF electrodes). Further, where the surgical instrument comprises ultrasonic components, the analog electrical signal waveform may be configured to drive at least two vibration modes of an ultrasonic transducer of the at least one surgical instrument. Accordingly, the generator circuit may be configured to provide an analog electrical signal waveform to at least one surgical instrument wherein the analog electrical signal waveform corresponds to at least one wave shape of a plurality of wave shapes stored in a lookup table **1604**. Further, the analog electrical signal waveform provided to the two surgical instruments may comprise two or more wave shapes. The lookup table **1604** may comprise information associated with a plurality of wave shapes and the lookup table **1604** may be stored either within the generator circuit or the surgical instrument. In one aspect or example, the lookup table **1604** may be a direct digital synthesis table, which may be stored in an FPGA of the generator circuit or the surgical instrument. The lookup table **1604** may be addressed by anyway that is convenient for categorizing wave shapes. According to one aspect, the lookup table **1604**, which may be a direct digital synthesis table, is addressed according to a frequency of the desired analog electrical signal waveform. Additionally, the information associated with the plurality of wave shapes may be stored as digital information in the lookup table **1604**.

With the widespread use of digital techniques in instrumentation and communications systems, a digitally-controlled method of generating multiple frequencies from a reference frequency source has evolved and is referred to as direct digital synthesis. The basic architecture is shown in FIG. **65**. In this simplified block diagram, a DDS circuit is coupled to a processor, controller, or a logic device of the generator circuit and to a memory circuit located either in the generator circuit of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61**. The DDS circuit **1600** comprises an address counter **1602**, lookup table **1604**, a register **1606**, a DAC circuit **1608**, and a filter **1612**. A stable clock f_c is received by the address counter **1602** and the register **1606** drives a programmable-read-only-memory (PROM) which stores one or more integral number of cycles of a sinewave (or other arbitrary waveform) in a lookup table **1604**. As the address counter **1602** steps through memory locations, values stored in the lookup table **1604** are written to a register **1606**, which is coupled to a DAC circuit **1608**. The corresponding digital amplitude of the signal at the memory location of the lookup table **1604** drives the DAC circuit **1608**, which in turn generates an analog output signal **1610**. The spectral purity of the analog output signal **1610** is determined primarily by the DAC circuit **1608**. The phase noise is basically that of the reference clock f_c. The first analog output signal **1610** output from the DAC circuit **1608** is filtered by the filter **1612** and a second analog output signal **1614** output by the filter **1612** is provided to an amplifier having an output coupled to the output of the generator circuit. The second analog output signal has a frequency f_{out}.

Because the DDS circuit **1600** is a sampled data system, issues involved in sampling must be considered: quantization noise, aliasing, filtering, etc. For instance, the higher order harmonics of the DAC circuit **1608** output frequencies fold back into the Nyquist bandwidth, making them unfilterable, whereas, the higher order harmonics of the output of phase-locked-loop (PLL) based synthesizers can be filtered. The lookup table **1604** contains signal data for an integral number of cycles. The final output frequency f_{out} can be changed changing the reference clock frequency f_c or by reprogramming the PROM.

The DDS circuit **1600** may comprise multiple lookup tables **1604** where the lookup table **1604** stores a waveform represented by a predetermined number of samples, wherein the samples define a predetermined shape of the waveform. Thus multiple waveforms having a unique shape can be stored in multiple lookup tables **1604** to provide different tissue treatments based on instrument settings or tissue feedback. Examples of waveforms include high crest factor RF electrical signal waveforms for surface tissue coagulation, low crest factor RF electrical signal waveform for deeper tissue penetration, and electrical signal waveforms that promote efficient touch-up coagulation. In one aspect, the DDS circuit **1600** can create multiple wave shape lookup tables **1604** and during a tissue treatment procedure (e.g., "on-the-fly" or in virtual real time based on user or sensor inputs) switch between different wave shapes stored in separate lookup tables **1604** based on the tissue effect desired and/or tissue feedback. Accordingly, switching between wave shapes can be based on tissue impedance and other factors, for example. In other aspects, the lookup tables **1604** can store electrical signal waveforms shaped to maximize the power delivered into the tissue per cycle (i.e., trapezoidal or square wave). In other aspects, the lookup tables **1604** can store wave shapes synchronized in such way that they make maximizing power delivery by the multifunction surgical instrument any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61** while delivering RF and ultrasonic drive signals. In yet other aspects, the lookup tables **1604** can store electrical signal waveforms to drive ultrasonic and RF therapeutic, and/or sub-therapeutic, energy simultaneously while maintaining ultrasonic frequency lock. Custom wave shapes specific to different instruments and their tissue effects can be stored in the non-volatile memory of the generator circuit or in the non-volatile memory (e.g., EEPROM) of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61** and be fetched upon connecting the multifunction surgical instrument to the generator circuit. An example of an exponentially damped sinusoid, as used in many high crest factor "coagulation" waveforms is shown in FIG. **67**.

A more flexible and efficient implementation of the DDS circuit **1600** employs a digital circuit called a Numerically Controlled Oscillator (NCO). A block diagram of a more flexible and efficient digital synthesis circuit such as a DDS circuit **1700** is shown in FIG. **66**. In this simplified block diagram, a DDS circuit **1700** is coupled to a processor, controller, or a logic device of the generator and to a memory circuit located either in the generator or in any of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61**. The DDS circuit **1700** comprises a load register **1702**, a parallel delta phase register **1704**, an adder circuit **1716**, a phase register **1708**, a lookup table **1710** (phase-to-amplitude converter), a DAC circuit **1712**, and a filter **1714**. The adder circuit **1716** and the phase register **1708** a form part of a phase accumulator **1706**. A clock signal f_c is applied to the phase register **1708** and the DAC circuit **1712**. The load register **1702** receives a tuning word that specifies output frequency as a fraction of the reference clock frequency f_c. The output of the load register **1702** is provided to a parallel delta phase register **1704** with a tuning word M.

The DDS circuit **1700** includes a sample clock that generates a clock frequency f_c, a phase accumulator **1706**, and a lookup table **1710** (e.g., phase to amplitude converter). The content of the phase accumulator **1706** is updated once per clock cycle f_c. When time the phase accumulator **1706** is updated, the digital number, M, stored in the parallel delta phase register **1704** is added to the number in the phase register **1708** by an adder circuit **1716**. Assuming that the number in the parallel delta phase register **1704** is 00 . . . 01 and that the initial contents of the phase accumulator **1706** is 00 . . . 00. The phase accumulator **1706** is updated by 00 . . . 01 per clock cycle. If the phase accumulator **1706** is 32-bits wide, 232 clock cycles (over 4 billion) are required before the phase accumulator **1706** returns to 00 . . . 00, and the cycle repeats.

The truncated output **1718** of the phase accumulator **1706** is provided to a phase-to amplitude converter lookup table **1710** and the output of the lookup table **1710** is coupled to a DAC circuit **1712**. The truncated output **1718** of the phase accumulator **1706** serves as the address to a sine (or cosine) lookup table. An address in the lookup table corresponds to a phase point on the sinewave from 0° to 360°. The lookup table **1710** contains the corresponding digital amplitude information for one complete cycle of a sinewave. The lookup table **1710** therefore maps the phase information from the phase accumulator **1706** into a digital amplitude word, which in turn drives the DAC circuit **1712**. The output of the DAC circuit is a first analog signal **1720** and is filtered by a filter **1714**. The output of the filter **1714** is a second analog signal **1722**, which is provided to a power amplifier coupled to the output of the generator circuit.

In one aspect, the electrical signal waveform may be digitized into 1024 (210) phase points, although the wave shape may be digitized is any suitable number of 2n phase points ranging from 256 (28) to 281,474,976,710,656 (248), where n is a positive integer, as shown in TABLE 1. The electrical signal waveform may be expressed as An(θn), where a normalized amplitude An at a point n is represented by a phase angle θn is referred to as a phase point at point n. The number of discrete phase points n determines the tuning resolution of the DDS circuit **1700** (as well as the DDS circuit **1600** shown in FIG. **65**).

TABLE 1

N	Number of Phase Points 2^n
8	256
10	1,024
12	4,096
14	16,384
16	65,536
18	262,144
20	1,048,576
22	4,194,304
24	16,777,216
26	67,108,864
28	268,435,456
. . .	. . .
32	4,294,967,296
. . .	. . .
48	281,474,976,710,656
. . .	. . .

The generator circuit algorithms and digital control circuits scan the addresses in the lookup table **1710**, which in turn provides varying digital input values to the DAC circuit **1712** that feeds the filter **1714** and the power amplifier. The addresses may be scanned according to a frequency of interest. Using the lookup table enables generating various types of shapes that can be converted into an analog output signal by the DAC circuit **1712**, filtered by the filter **1714**, amplified by the power amplifier coupled to the output of the generator circuit, and fed to the tissue in the form of RF energy or fed to an ultrasonic transducer and applied to the tissue in the form of ultrasonic vibrations which deliver energy to the tissue in the form of heat. The output of the amplifier can be applied to an RF electrode, multiple RF electrodes simultaneously, an ultrasonic transducer, multiple ultrasonic transducers simultaneously, or a combination of RF and ultrasonic transducers, for example. Furthermore, multiple wave shape tables can be created, stored, and applied to tissue from a generator circuit.

With reference back to FIG. **65**, for n=32, and M=1, the phase accumulator **1706** steps through **232** possible outputs before it overflows and restarts. The corresponding output wave frequency is equal to the input clock frequency divided by 232. If M=2, then the phase register **1708** "rolls over" twice as fast, and the output frequency is doubled. This can be generalized as follows.

For a phase accumulator **1706** configured to accumulate n-bits (n generally ranges from 24 to 32 in most DDS systems, but as previously discussed n may be selected from a wide range of options), there are 2^n possible phase points. The digital word in the delta phase register, M, represents the amount the phase accumulator is incremented per clock cycle. If fc is the clock frequency, then the frequency of the output sinewave is equal to:

$$f_{o}=\frac{M\cdot f_c}{2^n} \quad \text{Eq. 1}$$

Equation 1 is known as the DDS "tuning equation" Note that the frequency resolution of the system is equal to

$$\frac{f_0}{2^n}.$$

For n=32, the resolution is greater than one part in four billion. In one aspect of the DDS circuit **1700**, not all of the bits out of the phase accumulator **1706** are passed on to the lookup table **1710**, but are truncated, leaving only the first **13** to **15** most significant bits (MSBs), for example. This reduces the size of the lookup table **1710** and does not affect the frequency resolution. The phase truncation only adds a small but acceptable amount of phase noise to the final output.

The electrical signal waveform may be characterized by a current, voltage, or power at a predetermined frequency. Further, where any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61** comprises ultrasonic components, the electrical signal waveform may be configured to drive at least two vibration modes of an ultrasonic transducer of the at least one surgical instrument. Accordingly, the generator circuit may be configured to provide an electrical signal waveform to at least one surgical instrument wherein the electrical signal waveform is characterized by a predetermined wave shape stored in the lookup table **1710** (or lookup table **1604** FIG. **65**). Further, the electrical signal waveform may be a combination of two or more wave shapes. The lookup table **1710** may comprise information associated with a plurality of wave shapes. In one aspect or example, the lookup table **1710** may be generated by the DDS circuit

1700 and may be referred to as a direct digital synthesis table. DDS works by first storing a large repetitive waveform in onboard memory. A cycle of a waveform (sine, triangle, square, arbitrary) can be represented by a predetermined number of phase points as shown in TABLE 1 and stored into memory. Once the waveform is stored into memory, it can be generated at very precise frequencies. The direct digital synthesis table may be stored in a non-volatile memory of the generator circuit and/or may be implemented with a FPGA circuit in the generator circuit. The lookup table **1710** may be addressed by any suitable technique that is convenient for categorizing wave shapes. According to one aspect, the lookup table **1710** is addressed according to a frequency of the electrical signal waveform. Additionally, the information associated with the plurality of wave shapes may be stored as digital information in a memory or as part of the lookup table **1710**.

In one aspect, the generator circuit may be configured to provide electrical signal waveforms to at least two surgical instruments simultaneously. The generator circuit also may be configured to provide the electrical signal waveform, which may be characterized two or more wave shapes, via an output channel of the generator circuit to the two surgical instruments simultaneously. For example, in one aspect the electrical signal waveform comprises a first electrical signal to drive an ultrasonic transducer (e.g., ultrasonic drive signal), a second RF drive signal, and/or a combination thereof. In addition, an electrical signal waveform may comprise a plurality of ultrasonic drive signals, a plurality of RF drive signals, and/or a combination of a plurality of ultrasonic and RF drive signals.

In addition, a method of operating the generator circuit according to the present disclosure comprises generating an electrical signal waveform and providing the generated electrical signal waveform to any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61**, where generating the electrical signal waveform comprises receiving information associated with the electrical signal waveform from a memory. The generated electrical signal waveform comprises at least one wave shape. Furthermore, providing the generated electrical signal waveform to the at least one surgical instrument comprises providing the electrical signal waveform to at least two surgical instruments simultaneously.

The generator circuit as described herein may allow for the generation of various types of direct digital synthesis tables. Examples of wave shapes for RF/Electrosurgery signals suitable for treating a variety of tissue generated by the generator circuit include RF signals with a high crest factor (which may be used for surface coagulation in RF mode), a low crest factor RF signals (which may be used for deeper tissue penetration), and waveforms that promote efficient touch-up coagulation. The generator circuit also may generate multiple wave shapes employing a direct digital synthesis lookup table **1710** and, on the fly, can switch between particular wave shapes based on the desired tissue effect. Switching may be based on tissue impedance and/or other factors.

In addition to traditional sine/cosine wave shapes, the generator circuit may be configured to generate wave shape(s) that maximize the power into tissue per cycle (i.e., trapezoidal or square wave). The generator circuit may provide wave shape(s) that are synchronized to maximize the power delivered to the load when driving RF and ultrasonic signals simultaneously and to maintain ultrasonic frequency lock, provided that the generator circuit includes a circuit topology that enables simultaneously driving RF and ultrasonic signals. Further, custom wave shapes specific to instruments and their tissue effects can be stored in a non-volatile memory (NVM) or an instrument EEPROM and can be fetched upon connecting any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61** to the generator circuit.

The DDS circuit **1700** may comprise multiple lookup tables **1604** where the lookup table **1710** stores a waveform represented by a predetermined number of phase points (also may be referred to as samples), wherein the phase points define a predetermined shape of the waveform. Thus multiple waveforms having a unique shape can be stored in multiple lookup tables **1710** to provide different tissue treatments based on instrument settings or tissue feedback. Examples of waveforms include high crest factor RF electrical signal waveforms for surface tissue coagulation, low crest factor RF electrical signal waveform for deeper tissue penetration, and electrical signal waveforms that promote efficient touch-up coagulation. In one aspect, the DDS circuit **1700** can create multiple wave shape lookup tables **1710** and during a tissue treatment procedure (e.g., "on-the-fly" or in virtual real time based on user or sensor inputs) switch between different wave shapes stored in different lookup tables **1710** based on the tissue effect desired and/or tissue feedback. Accordingly, switching between wave shapes can be based on tissue impedance and other factors, for example. In other aspects, the lookup tables **1710** can store electrical signal waveforms shaped to maximize the power delivered into the tissue per cycle (i.e., trapezoidal or square wave). In other aspects, the lookup tables **1710** can store wave shapes synchronized in such way that they make maximizing power delivery by any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-61** when delivering RF and ultrasonic drive signals. In yet other aspects, the lookup tables **1710** can store electrical signal waveforms to drive ultrasonic and RF therapeutic, and/or sub-therapeutic, energy simultaneously while maintaining ultrasonic frequency lock. Generally, the output wave shape may be in the form of a sine wave, cosine wave, pulse wave, square wave, and the like. Nevertheless, the more complex and custom wave shapes specific to different instruments and their tissue effects can be stored in the non-volatile memory of the generator circuit or in the non-volatile memory (e.g., EEPROM) of the surgical instrument and be fetched upon connecting the surgical instrument to the generator circuit. One example of a custom wave shape is an exponentially damped sinusoid as used in many high crest factor "coagulation" waveforms, as shown in FIG. **67**.

FIG. **67** illustrates one cycle of a discrete time digital electrical signal waveform **1800**, according to one aspect of the present disclosure of an analog waveform **1804** (shown superimposed over the discrete time digital electrical signal waveform **1800** for comparison purposes). The horizontal axis represents Time (t) and the vertical axis represents digital phase points. The digital electrical signal waveform **1800** is a digital discrete time version of the desired analog waveform **1804**, for example. The digital electrical signal waveform **1800** is generated by storing an amplitude phase point **1802** that represents the amplitude per clock cycle Talk over one cycle or period T_o. The digital electrical signal waveform **1800** is generated over one period T_o by any suitable digital processing circuit. The amplitude phase points are digital words stored in a memory circuit. In the example illustrated in FIGS. **65**, **66**, the digital word is a six-bit word that is capable of storing the amplitude phase points with a resolution of 26 or 64 bits. It will be appreciated that the examples shown in FIGS. **65**, **66** is for illustrative purposes and in actual implementations the resolution can be much higher. The digital amplitude phase points **1802** over one cycle T_o are stored in the memory as a string of string words in a lookup table **1604**, **1710** as described in connection with FIGS. **65**, **66**, for example. To generate the analog version of the analog waveform **1804**, the amplitude phase points **1802** are read sequentially from the memory from 0 to T_o per clock cycle T_{clk} and are converted by a DAC circuit **1608**, **1712**, also described in connection with FIGS. **65**, **66**. Additional cycles can be generated by repeatedly reading the amplitude phase points **1802** of the digital electrical signal waveform **1800** the from 0 to T_o for as many cycles or periods as may be desired. The smooth analog version of the analog waveform **1804** is achieved by filtering the output of the DAC circuit **1608**, **1712** by a filter **1612**, **1714** (FIGS. **65** and **66**). The filtered analog output signal **1614**, **1722** (FIGS. **65** and **66**) is applied to the input of a power amplifier.

In one aspect, as illustrated in FIG. **68**A, a circuit **1900** may comprise a controller comprising one or more processors **1902** (e.g., microprocessor, microcontroller) coupled to at least one memory circuit **1904**. The at least one memory circuit **1904** stores machine executable instructions that when executed by the processor **1902**, cause the processor **1902** to execute machine instructions to implement any of the algorithms, processes, or techniques described herein.

The processor **1902** may be any one of a number of single or multi-core processors known in the art. The memory circuit **1904** may comprise volatile and non-volatile storage media. In one aspect, as illustrated in FIG. **68**A, the processor **1902** may include an instruction processing unit **1906** and an arithmetic unit **1908**. The instruction processing unit may be configured to receive instructions from the one memory circuit **1904**.

In one aspect, a circuit **1910** may comprise a finite state machine comprising a combinational logic circuit **1912**, as illustrated in FIG. **68**B, configured to implement any of the algorithms, processes, or techniques described herein. In one aspect, a circuit **1920** may comprise a finite state machine comprising a sequential logic circuit, as illustrated in FIG. **68**C. The sequential logic circuit **1920** may comprise the combinational logic circuit **1912** and at least one memory circuit **1914**, for example. The at least one memory circuit **1914** can store a current state of the finite state machine, as illustrated in FIG. **68**C. The sequential logic circuit **1920** or the combinational logic circuit **1912** can be configured to implement any of the algorithms, processes, or techniques described herein. In certain instances, the sequential logic circuit **1920** may be synchronous or asynchronous.

In other aspects, the circuit may comprise a combination of the processor **1902** and the finite state machine to implement any of the algorithms, processes, or techniques described herein. In other aspects, the finite state machine may comprise a combination of the combinational logic circuit **1910** and the sequential logic circuit **1920**.

FIG. **69** is a schematic diagram of a circuit **1925** of various components of a surgical instrument with motor control functions, according to one aspect of the present disclosure. In various aspects, the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1**-**68**C may include a drive mechanism **1930** which is configured to drive shafts and/or gear components in order to perform the various operations associated with the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200**. In one aspect, the drive mechanism **1930 160** includes a rotation drivetrain **1932** configured to rotate end effector **112**, **512**, **1000**, **1112**, **1212** as described in connection with FIGS. **1**, **20**, **40**, **41**, **45**, **54**, for example, about a longitudinal axis relative to handle housing. The drive mechanism **1930** further includes a closure drivetrain **1934** configured to close a jaw member to grasp tissue with the end effector. In addition, the drive mechanism **1930** includes a firing drive train **1936** configured to fire an I-beam knife of the end effector to cut tissue grasped by the end effector.

The drive mechanism **1930** includes a selector gearbox assembly **1938** that can be located in the handle assembly of the surgical instrument. Proximal to the selector gearbox assembly **1938** is a function selection module which includes a first motor **1942** that functions to selectively move gear elements within the selector gearbox assembly **1938** to selectively position one of the drivetrains **1932**, **1934**, **1936** into engagement with an input drive component of an optional second motor **1944** and motor drive circuit **1946** (shown in dashed line to indicate that the second motor **1944** and motor drive circuit **1946** are optional components).

Still referring to FIG. **69**, the motors **1942**, **1944** are coupled to motor control circuits **1946**, **1948**, respectively, which are configured to control the operation of the motors **1942**, **1944** including the flow of electrical energy from a power source **1950** to the motors **1942**, **1944**. The power source **1950** may be a DC battery (e.g., rechargeable lead-based, nickel-based, lithium-ion based, battery etc.) or any other power source suitable for providing electrical energy to the surgical instrument.

The surgical instrument further includes a microcontroller **1952** ("controller"). In certain instances, the controller **1952** may include a microprocessor **1954** ("processor") and one or more computer readable mediums or memory units **1956** ("memory"). In certain instances, the memory **1956** may store various program instructions, which when executed may cause the processor **1954** to perform a plurality of functions and/or calculations described herein. The power source **1950** can be configured to supply power to the controller **1952**, for example.

The processor **1954** be in communication with the motor control circuit **1946**. In addition, the memory **1956** may store program instructions, which when executed by the processor **1954** in response to a user input **1958** or feedback elements **1960**, may cause the motor control circuit **1946** to motivate the motor **1942** to generate at least one rotational motion to selectively move gear elements within the selector gearbox assembly **1938** to selectively position one of the drivetrains **1932**, **1934**, **1936** into engagement with the input drive component of the second motor **1944**. Furthermore, the processor **1954** can be in communication with the motor control circuit **1948**. The memory **1956** also may store program instructions, which when executed by the processor **1954** in response to a user input **1958**, may cause the motor control circuit **1948** to motivate the motor **1944** to generate at least one rotational motion to drive the drivetrain engaged with the input drive component of the second motor **1948**, for example.

The controller **1952** and/or other controllers of the present disclosure may be implemented using integrated and/or discrete hardware elements, software elements, and/or a combination of both. Examples of integrated hardware elements may include processors, microprocessors, microcontrollers, integrated circuits, ASICs, PLDs, DSPs, FPGAs, logic gates, registers, semiconductor devices, chips, microchips, chip sets, microcontrollers, system on a chip (SoC), and/or single in-line package (SIP). Examples of discrete hardware elements may include circuits and/or circuit elements such as logic gates, field effect transistors, bipolar transistors, resistors, capacitors, inductors, and/or relays. In certain instances, the controller **1952** may include a hybrid circuit comprising discrete and integrated circuit elements or components on one or more substrates, for example.

In certain instances, the controller **1952** and/or other controllers of the present disclosure may be an LM 4F230H5QR, available from Texas Instruments, for example. In certain instances, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle SRAM, internal ROM loaded with StellarisWare® software, 2 KB EEPROM, one or more PWM modules, one or more QEI analog, one or more 12-bit ADC with 12 analog input channels, among other features that are readily available. Other microcontrollers may be readily substituted for use with the present disclosure. Accordingly, the present disclosure should not be limited in this context.

In various instances, one or more of the various steps described herein can be performed by a finite state machine comprising either a combinational logic circuit or a sequential logic circuit, where either the combinational logic circuit or the sequential logic circuit is coupled to at least one memory circuit. The at least one memory circuit stores a current state of the finite state machine. The combinational or sequential logic circuit is configured to cause the finite state machine to the steps. The sequential logic circuit may be synchronous or asynchronous. In other instances, one or more of the various steps described herein can be performed by a circuit that includes a combination of the processor **1958** and the finite state machine, for example.

In various instances, it can be advantageous to be able to assess the state of the functionality of a surgical instrument to ensure its proper function. It is possible, for example, for the drive mechanism, as explained above, which is configured to include various motors, drivetrains, and/or gear components in order to perform the various operations of the surgical instrument, to wear out over time. This can occur through normal use, and in some instances the drive mechanism can wear out faster due to abuse conditions. In certain instances, a surgical instrument can be configured to perform self-assessments to determine the state, e.g. health, of the drive mechanism and it various components.

For example, the self-assessment can be used to determine when the surgical instrument is capable of performing its function before a re-sterilization or when some of the components should be replaced and/or repaired. Assessment of the drive mechanism and its components, including but not limited to the rotation drivetrain **1932**, the closure drivetrain **1934**, and/or the firing drivetrain **1936**, can be accomplished in a variety of ways. The magnitude of deviation from a predicted performance can be used to determine the likelihood of a sensed failure and the severity of such failure. Several metrics can be used including: Periodic analysis of repeatably predictable events, Peaks or drops that exceed an expected threshold, and width of the failure.

In various instances, a signature waveform of a properly functioning drive mechanism or one or more of its components can be employed to assess the state of the drive mechanism or the one or more of its components. One or more vibration sensors can be arranged with respect to a properly functioning drive mechanism or one or more of its components to record various vibrations that occur during operation of the properly functioning drive mechanism or the one or more of its components. The recorded vibrations can be employed to create the signature waveform. Future waveforms can be compared against the signature waveform to assess the state of the drive mechanism and its components.

Still referring to FIG. **69**, the surgical instrument **1930** includes a drivetrain failure detection module **1962** configured to record and analyze one or more acoustic outputs of one or more of the drivetrains **1932**, **1934**, **1936**. The processor **1954** can be in communication with or otherwise control the module **1962**. As described below in greater detail, the module **1962** can be embodied as various means, such as circuitry, hardware, a computer program product comprising a computer readable medium (for example, the memory **1956**) storing computer readable program instructions that are executable by a processing device (for example, the processor **1954**), or some combination thereof. In some aspects, the processor **36** can include, or otherwise control the module **1962**.

FIG. **70** illustrates a handle assembly **1970** with a removable service panel **1972** removed to shown internal components of the handle assembly, according to one aspect of the present disclosure. The removable service panel **1972**, or removable service cover, also includes reinforcing ribs **1990** for strength. The removable service panel **1972** comprises a plurality of fasteners **1988** that mate with a plurality of fasteners **1986** on the handle housing **1974** to removably attach the removable service panel **1972** to the handle housing **1974**. In one aspect, the fasteners **1988** in the removable service panel **1972** comprise a first set of magnets and the handle housing **1974** comprises a second set of magnets that magnetically latch the service panel **1972** to the handle housing **1974**. In one aspect, the first and second set of magnets **6112***a*, **6112***b* are rare-earth permanent magnets.

In FIG. **70**, the removable service panel **1972** is shown removed from the handle housing **1974** to show the location of electrical and mechanical components of the surgical instrument such as the motor **1976** and electrical contacts **1984** to electrically couple the battery assembly or flexible circuits to the handle housing **1974**. The motor **1976** and the electrical contacts **1984** are also removable from the handle housing **1974**. The handle assembly **1970** also comprises a trigger **1982** and an actuation switch **1980**, each of which is removable from the handle housing **1974**. As previously described, the removable trigger **1982** may have multiple stages of operation to close the jaw member, fire the knife, activate the ultrasonic transducer, activate the high-frequency current, and/or open the jaw member. The actuation switch **1980** may be replaced with multiple switches to activate different functions such as, for example, close the jaw member, fire the knife, activate the ultrasonic transducer, activate the high-frequency current, and/or open the jaw member. As shown in FIG. **70**, the handle assembly **1970** includes electrical contacts **1978** to electrically couple the handle assembly **1970** to the shaft assembly, where the electrical contacts **1978** are removable from the handle housing **1974**. The handle housing **1974** also defines a space to receive a removable ultrasonic transducer assembly, ultrasonic transducer, ultrasonic transducer drive circuits, high-frequency current drive circuits, and/or display assembly, as previously discussed herein.

Modular Battery Powered Handheld Surgical Instrument with Selective Application of Energy Based on Tissue Characterization and with Selective Application of Energy Based on Button Displacement, Intensity, or Local Tissue Characterization

In one aspect, the present disclosure provides a modular battery powered handheld surgical instrument with selective application of energy based on tissue characterization. Disclosed is a surgical instrument that includes a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft; a handle assembly coupled to a proximal end of the shaft; a battery assembly coupled to the handle assembly; a radio frequency (RF) energy output powered by the battery assembly and configured to apply RF energy to a tissue; an ultrasonic energy output powered by the battery assembly and configured to apply ultrasonic energy to the tissue; and a controller configured to, based at least in part on a measured tissue characteristic, start application of RF energy by the RF energy output or application of ultrasonic energy by the ultrasonic energy output at a first time.

In another aspect, the present disclosure provides a modular battery powered handheld surgical instrument with selective application of energy based on button displacement, intensity, or local tissue characterization. Disclosed is a surgical instrument that includes a controller configured to control application of RF or ultrasonic energy at a low level when displacement or intensity of a button is above a first threshold but below a second threshold higher than the first threshold, and control application of RF or ultrasonic energy at a high level when the displacement or intensity exceeds the second threshold. In another aspect, a surgical instrument comprises a first sensor configured to measure a tissue characteristic at a first location, a second sensor configured to measure the tissue characteristic at a second location, and a controller configured to, based at least in part on the measured tissue characteristic at the first location and the second location, control application of RF or ultrasonic energy.

FIGS. **71**-**81** illustrate one aspect of the present disclosure that is directed to switching between energy modalities such as high-frequency (e.g., RF), ultrasonic, or a combination of high-frequency current and ultrasonic energy modalities automatically based on a sensed/calculated measure of a parameter of the surgical instrument **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1**-**70** implying tissue thickness and/or type based on (impendence, current from the motor, jaw gap sensing, tissue compression, temperature, and the like. The first portion describes an example system wherein a change of energy modality is done based on the measure of tissue thickness by the combination of at least two measures of tissue parameters (impendence, current from the motor, jaw gap sensing, tissue compression, temperature). In one aspect, impedance, force, and velocity/displacement are measured to control energy modality of a combo ultrasonic/RF device. Measurement of force is used to determine the type of energy modality that can be used and indicate the timing at which the user can selectively switch if desired in an ultrasonic/RF combo device. One technique to accomplish this is by utilizing the slope of the motor current to dictate the ultrasonic, RF, or both energy modes. Another option is to use the rate of change of a measurable tissue characteristic to determine the energy modality which can be used (RF or Ultrasonic) or where in the cycle to start or stop using a specific energy modality. Again, the slope of the impedance may be utilized to dictate the ultrasonic, RF, or both energy modes.

Another technique to accomplish control of energy modality is by sensing of tissue gap by a rotary encoder, attached to the trigger or the clamp arm of the device or by measuring tissue thickness to set modality decision of the energy mode. In this scenario, wider gap indicates touch-up or debulking function is required while narrow gap indicates vessel sealing mode. Additionally, the maximum applicable power may be changed based on the slope and intensity of the impedance measured in order to only effect the raising portion of the impedance bath tub.

Yet another technique to accomplish control of energy modality is through motor current thresholds which indicate thickness of tissue and define energy modality options available and/or initiate switching of energy application levels or modes based on predefined levels. For instance, specific motor controls can be based on tissue parameters. As an example, wait and energy profile changes can be made due to sensing different tissue characteristics and types. Or a motor control circuit can be employed which increases the motor current for a motorized device closure and therefore increases forces at the end of the impedance curve for an ultrasonic and increases closure force in order to finish the cut cleanly. Although many of these embodiments may be done using tissue measurements, an alternative embodiment would be to measure the forces on the clamp arm directly through some form of force transducer. Some methods to measure tissue type, thickness and other parameters include tissue thickness sensing as part of closure. This may be done by pre-defining a time and measuring the displacement that the knife or closure system can reach within the pre-defined starting time interval to determine the thickness and compressibility of the tissue encountered, or by pre-defining a constant force level and determining the time that is required to reach that force at a pre-defined speed or acceleration.

Another method to accomplish control of energy modality is by using impedance measurements and force or velocity measures to correlate the density, conductivity and force resistance of the tissue to determine the type and thickness of the tissue as well as any irregularities that should impact rate of advance, wait, or energy density. Using combination of motor force closure measurements to determine if the jaw members of the end effector are closed on something that is likely to cause a short circuit (e.g., staple, clip, etc.) is also contemplated as is combining motor closure force with segmented flex force sensors that sense how much of the jaw member is filled in order to discriminate between large bites of softer tissue and smaller bites of harder tissue. For instance, the motor allows us new ways to determine if there is tissue or metal in the jaw members.

FIG. **71** is a graphical representation **3700** of determining wait time based on tissue thickness. A first graph **3702** represents tissue impedance Z versus time (t) where the horizontal axis represents time (t) and the vertical axis represents tissue impedance Z. A second graph **3704** represents change in gap distance Δgap versus time (t) where the horizontal axis represents time (t) and the vertical axis represents change in gap distance Δgap. A third graph **3706** represents force F versus time (t) where the horizontal axis represents time (t) and the vertical axis represents force F. A constant force F applied to tissue and impedance Z interrogation define a wait period, energy modality (e.g., RF and ultrasonic) and motor control parameters. Displacement at a time provides velocity. With reference to the three graphs **3702**, **3704**, **3706**, impedance sensing energy is applied during a first period to determine the tissue type such as thin mesentery tissue (solid line), intermediate thickness vessel tissue (dashed line), or thick uterus/bowel tissue (dash-dot line).

Using the thin mesentery tissue (solid line) as an example, as shown in the third graph **3706**, the clamp arm initially applies a force which ramps up from zero until it reaches a constant force **3724** at or about a first time t1. As shown in the first and second graphs **3702**, **3704**, from the time the clamp force is applied to the mesentery tissue until the first time t1, the gap distance Δgap curve **3712** decreases and the tissue impedance **3718** also decreases until the first time t1 is reached. From the first time t1, a short wait period **3728** is applied before treatment energy, e.g., RF, is applied to the mesentery tissue at tE1. Treatment energy is applied for a second period **3710**, after which the tissue may be ready for a cut operation.

As shown in the first and second graphs **3702**, **3704**, for intermediate thickness vessel tissue (dashed line), similar operations are performed. However, a medium wait period **3730** is applied before treatment energy is applied to the tissue at tE2.

As shown in the first and second graphs **3702**, **3704**, for thick uterus/bowel tissue (dash-dot line), similar operations are performed. However, a long wait period **3726** is applied before treatment energy is applied to the tissue at tE3.

Therefore, different wait periods may be applied based on the thickness of the tissue. The thickness of the tissue may be determined based on different gap distance behavior or impedance behavior before the time the constant force is reached. For example, as shown in the second graph **3704**, depending on the minimum gap distance reached when the constant force is reached, i.e., small gap, medium gap, or large gap, the tissue is determined as a thin tissue, an intermediate thickness tissue, or a thick tissue, respectively. As shown in the first graph **3702**, depending on the minimum impedance reached when the constant force is reached, e.g., small impedance, medium impedance, or large impedance, the tissue is determined as a thick tissue, an intermediate thickness tissue, or a thin tissue, respectively.

Alternatively, as shown in the second graph **3704**, the thin tissue has a relatively steep gap distance slope, the intermediate thickness tissue has a medium gap distance slope, and the thick tissue has a relatively flat gap distance slope. As shown in the first graph **3702**, the thin tissue has a relatively flat impedance slope, and the intermediate thickness and thick tissues have relatively steep impedance slopes. Tissue thickness may be determined accordingly.

The thickness of the tissue may also be determined as follows with reference to FIG. **72**. FIG. **72** is a force versus time graph **3800** for thin, medium, and thick tissue types. The horizontal axis represents time (t) and the vertical axis represents force (F) applied by the clamp arm to the tissue. The graph **3800** depicts three curves, one for thin tissue **3802** shown in solid line, one for medium thickness tissue **3804** shown in dash-dot line, and one for thick tissue **3806** in dashed line. The graph **3800** depicts measuring time required to reach the preset force as an alternative to tissue gap to control delayed energy mode and other control parameters. Accordingly, the time to preset force **3808** for thick tissue **3806** is t1a, the time to preset force **3808** for medium thickness tissue **3804** is t1b, and the time to preset force **3808** for thin tissue **3802** is t1c.

Once the force reaches the preset force **3808**, energy is applied to the tissue. For thin tissue **3802** the time to preset force t1c>0.5 seconds, and then RF energy is applied for an energizing period of about 1-3 seconds. For thick tissue **3806** the time to preset force t1a<0.5 seconds, and then RF energy is applied for an energizing period of about 5-9 seconds. For medium thickness tissue **3804** the time to preset force t1b is about 0.5 seconds and then RF energy is applied for an energizing period of about 3 to 5 seconds. These specific time periods may be adjusted without departing from the scope of the present disclosure.

Alternatively, instead of predefining a constant force **3808**, a time period may be predefined. The force, gap distance, or impedance reached after the predefined time period may be measured, and may be used to determine the thickness of the tissue.

The gap distance referred to in the above examples may be a gap distance between two jaws of an end effector of a surgical device. As discussed above, the gap distance may be measured with a rotary encoder attached to one or both of the jaws, or attached to a trigger used to operate the jaws.

The force referred to in the above examples may be a force applied by one or both of the jaws on the tissue. As discussed above, the force may be measured using a current of a motor driving the jaws. Alternatively, the force may be measured directly using a force transducer.

The impedance referred to in the above examples may be an impedance between the jaws across the tissue. The impedance may be measured using any conventional electrical techniques.

FIG. **73** is a graph **3900** of motor current I_{motor} versus time t for different tissue types. Here, motor current I_{motor} may be a measure of force applied by one or both of the jaws on the tissue. A first curve **3910** shown in solid line is a motor current versus time curve for a thick tissue. A second curve **3920** shown in dashed line is a motor current versus time curve for a thin tissue. As shown by a first portion **3912** of the first curve **3910**, the motor current I_{motor} increases initially. Thereafter, as shown by a second portion **3914** (shadowed region) of the first curve **3910**, ultrasonic energy is applied, and the motor current I_{motor} decreases sharply. When the motor current decreases below a threshold **3930**, or when it reaches certain amount or certain percentage **3932** below the threshold **3930**, energy is switched from ultrasonic to RF. The switching may also occur when the slope of the motor current becomes relatively flat. As shown by a third portion **3916** of the first curve **3910**, RF energy is applied, and the motor current I_{motor} decreases slowly. In contrast, as shown in the second curve **3920** for a thin tissue, the motor current I_{motor} never increases beyond the threshold **3930**, and thus ultrasonic energy is not applied.

FIG. **74** is a graphical depiction of impedance bath tub (e.g., the tissue impedance versus time initially decreases, stabilizes, and finally increases and the curve resembles a bath tub shape). A graph **4000** comprises three graphs **4002**, **4004**, **4006**, where the first graph **4002** represents RF power (P), RF voltage (V_{RF}), and RF current (I_{RF}) versus tissue impedance (Z), the second graph **4004** and third graph **4006** represent tissue impedance (Z) versus time (t). The first graph **4002** illustrates the application of power (P) for thick tissue impedance range **4010** and thin tissue impedance range **4012**. As the tissue impedance Z increases, the current I_{RF} decreases and the voltage V_{RF} increases. The power P increases until it reaches a maximum power output **4008**. When the RF power P is not high enough, for example as shown in the impedance range **4010**, RF energy may not be enough to treat tissues, therefore ultrasonic energy is applied instead.

The second graph **4004** represents the measured tissue impedance Z versus time (t). The tissue impedance threshold limit **4020** is the cross over limit for switching between the

RF and ultrasonic energy modalities. For example, as shown in the third graph **4006**, RF energy is applied while the tissue impedance is above the tissue impedance threshold limit **4020** and ultrasonic energy **4024** is applied while the tissue impedance is below the tissue impedance threshold limit **4020**. Accordingly, with reference back to the second graph **4004**, the tissue impedance of the thin tissue curve **4016** remains above the tissue impedance threshold limit **4020**, thus only RF energy modality is applied to the tissue. On the other hand, for the thick tissue curve **418**, RF energy modality is applied to the tissue while the impedance is above the tissue impedance threshold limit **4020** and ultrasonic energy is applied to the tissue when the impedance is below the tissue impedance threshold limit **4020**.

Accordingly, the energy modality switches from RF to ultrasonic when the tissue impedance falls below the tissue impedance threshold limit **4020** and thus RF power P is low, and the energy modality switches from ultrasonic to RF when the tissue impedance rises above the tissue impedance threshold limit **4020** and thus RF power P is high enough. As shown in the third graph **4006**, the switching from ultrasonic to RF may be set to occur when the impedance reaches a certain amount or certain percentage above the threshold limit **4020**.

Measurement of current, velocity, or torque of the motor related to the compression applied to the tissue can be used to change the impedance threshold that triggers the control of the treatment energy applied to the tissue. FIG. **75** illustrates a graph **4100** depicting one aspect of adjustment of energy switching threshold due to the measurement of a secondary tissue parameter such as continuity, temperature, pressure, and the like. The horizontal axis of the graph **4100** is time (t) and the vertical axis is tissue impedance (Z). The curve **4112** represents the change of tissue impedance (Z) over time (t) as different energy modalities are applied to the tissue. For example, the threshold may be adjusted depending on whether tissue is present at all parts of the jaws or present at only a portion of the jaws. Accordingly, once the tissue is located in particular segments (zones) the control circuit in the generator adjusts the threshold accordingly. Reference is made to discussion below in connection with FIG. **80** for segmented measurement of tissue presence.

As shown in FIG. **75**, similar to the example described with reference to FIG. **74**, the curve **4112** includes three separate sections **4106**, **4108**, **4110**. The first section **4106** of the curve **4112** represents the time when RF energy is applied to the tissue until the tissue impedance drops below the adjusted threshold **4104**. At that point **4114**, the energy modality applied to tissue is changed from RF energy to ultrasonic energy. The ultrasonic energy is then applied in the second section **4108**.

Yet another embodiment of this concept may cause the wave shape to change in the RF signal based on the thickness measured by the force or force/position slope to determine whether to apply debulking or coagulation. For instance, sine wave or square waves are used to pre-heat the tissue and high voltage peak waves are used to coagulate the tissue.

According to aspects of the present disclosure, a tissue short circuit condition may be detected. Detecting metal in the end effector (such as a staple or clip) avoid short circuits in the end effector that can divert current through the short circuit and render the RF therapy or sensing signal ineffective. A small piece of metal, such as a staple, can become quite hot with therapeutic RF current flowing through it. This could result in undesired effects in the tissue. Metal in contact with a vibrating ultrasonic blade can cause complications with the blade staying in resonance or possibly damage the blade or metal piece. Metal in the jaws can damage the pad that opposes a vibrating blade in the case of a clamped device. Metal can damage the closure mechanism due to over-stress of components while trying to close. Metal can damage a knife blade that may be forced to come in contact with the metal or attempt to cut through it.

By using a motor, it is known (approximately) how open or closed the jaws are. If the jaws are open, the condition of how open or closed the jaws are can be identified in a variety of different methods—it could be the encoder count, the current going to the motor, a drop in motor voltage, etc. This can further be refined by looking at the derivative of either motor current or motor voltage. A short circuit is detected when the calculated impedance from the RF energy, is determined to be below a certain threshold. Due to cables and instrument design, this exact value varies. If the impedance is below or near this threshold, any of the following could aid in detecting a short circuit:

Encoder count—if the jaws are still open, this implies there is tissue. If the impedance is at or below threshold, this is indicative that all energy is going through metal.

Motor Current—if the motor has yet to detect its end of travel and it is experiencing high loads, the current increases (this is a method of force determination/calculation). As the current increases to a maximum, this coupled with the impedance measurement, could indicate a piece of metal is in the jaws. It takes more force to cut through a metal staple than it does of any tissue type. High current with low impedance (at or below threshold) implies possible short circuit.

Motor Voltage—similar to the motor current example. If the motor current goes high and the encoder count slows down, the voltage decreases. Thus, it's possible that the motor voltage, coupled with impedance, could imply a short circuit.

Derivative of Motor Current—this indicates the trend of the current, and is faster at predicting if the current is going to increase or decrease, based on previous performance. If the derivative of the current indicates more current will be going to the motor and the impedance is low, it is likely a short circuit.

Derivative of Motor Voltage—this indicates the trend of the voltage, and is faster at predicting if the voltage is going to increase or decrease, based on previous performance. If the derivative of the voltage indicates less voltage will be going to the motor and the impedance is low, it is likely a short circuit.

Combinations of the above are contemplated. In summary, a short circuit equation could be enhanced by monitoring any of the following conditions:

Encoder Count+Impedance

Encoder Count+Motor Current+Impedance

Encoder Count+Motor Current+Motor Voltage+Impedance

Encoder Count+Derivative of Motor Current+Motor Current+Impedance

Encoder Count+Derivative of Motor Voltage+Motor Current+Impedance

Motor Current+Impedance

Motor Voltage+Impedance

Derivative of Motor Current+Impedance

Derivative of Motor Voltage+Impedance

Encoder Count+Motor Voltage+Impedance

Encoder Count+Derivative of Motor Current+Motor Voltage+Impedance

Encoder Count+Derivative of Motor Voltage+Motor Voltage+Impedance

Encoder Count+Derivative of Motor Current+Impedance

Encoder Count+Derivative of Motor Current+Motor Voltage+Motor Current+Impedance

Encoder Count+Derivative of Motor Voltage+Impedance

Encoder Count+Derivative of Motor Voltage+Motor Voltage+Motor Current+Impedance

Encoder Count+Derivative of Motor Voltage+Derivative of Motor Current+Motor Voltage+Motor Current+Impedance

It is worthwhile noting that there are 5 separate conditions. This implies $2^5=32$ different combinations of short circuit detection based on coupling impedance measurement to the 5 different conditions.

FIG. **76** is a diagram of a process **4200** illustrating selective application of radio frequency or ultrasonic treatment energy based on measured tissue characteristics according to aspects of the present disclosure. One or more parameters, e.g., impedance, gap distance, force, temperature or their derivatives, may be measured **4210**. Based on the measured one or more parameters, one or more tissue characteristics, e.g., thickness, compressibility or short circuit condition may be determined **4220**. A controller of energy application may start application of RF or ultrasonic energy at a first time based at least in part on the one or more tissue characteristics **4230**. Optionally, the controller may switch between RF and ultrasonic energy at a second time based at least in part on the one or more tissue characteristics **4240**. It should be noted that the measuring of the parameters **4210** and the determination of the tissue characteristics **4220** may occur during the application of energy **4230**, **4240**, and not necessarily prior to the application of energy **4230**, **4240**.

According to aspects of the present disclosure, specific energy control algorithms can be employed. For instance, measuring the force of the user input on the energy activation control button can be utilized. The user control button may comprise a continuous measure button sensing that allows the device to set the on/off threshold as well as sense button degradation and user intensity. A force capacitive or resistive contact may be used that gives a continuous signal (not a interrupt/contact signal) which has a predefined force threshold which technique activate, a separate threshold meaning deactivate, and another intensity threshold above activate which indicates the need for a higher desired energy level. In one embodiment of this, the higher energy level could indicate the desire to activate both energy modalities simultaneously.

Additionally or alternatively, a Hall sensor or other displacement based sensor on the energy activation button may also be utilized to get a continuous displacement of the button having a predefined activation position and a different energy deactivation threshold. In yet another embodiment, a control processor monitors the buttons use with a procedure and in-between procedures recording certain parameters of the button outputs, thereby allowing it to adjust the threshold to compensate for sensor wear and degradation, thus prolonging its useful life.

In one aspect, the RF or ultrasonic energy may be terminated by the controller at a specific time. In some instances, the RF energy may desiccate the tissue to the point that application of ultrasonic energy comes too late to make a cut because the tissue is too dried out. For this type of event, the controller may be configured to terminate RF energy once a specific tissue impedance is met and going forward to apply only ultrasonic energy until the seal and cut is complete. For completeness, the present disclosure also contemplates terminating ultrasonic energy prior to terminating the RF energy to seal the tissue. Accordingly, in addition to switching between RF and ultrasonic energy, the present disclosure contemplates applying both RF and ultrasonic energy to the tissue simultaneously to achieve a seal and cut. In other aspects, the present disclosure contemplates applying both RF and ultrasonic energy to the tissue simultaneously and then terminating the RF energy at a predetermined time. This may be advantageous, for example, to prevent the desiccating the tissue to a point that would render the application of ultrasonic energy ineffective for cutting tissue. In yet another aspect, the intensity of the RF energy may be reduced from a therapeutic level to a non-therapeutic level suitable for sensing during the sealing process to measure the tissue impedance, for example, using RF sensing without having an RF therapeutic effect on the tissue when this is desired.

FIG. **77** is a graph **4300** depicting a relationship between trigger button displacement and sensor output. The vertical axis **4370** represents displacement of a trigger button. The trigger button, for example, may be located at a handle assembly or module and is used by a user to control application of RF and/or ultrasonic energy. The horizontal axis **4380** represents output of a displacement sensor, for example a Hall sensor. Shadowed zones **4350**, **4360** represent out-of-bounds zones. As shown in the curve **4310** in FIG. **77**, the sensor output is roughly proportional to the button displacement. A first zone **4320** may be an "OFF" zone, where button displacement is small and no energy is applied. A second zone **4330** may be an "ON" zone, where button displacement is medium and energy is applied. A third zone **4340** may be a "HIGH" zone, where button displacement is large and energy with high intensity is applied. Alternatively, the third zone **4340** may be a "HYBRID" zone, where both RF energy and ultrasonic energy are applied. Although the range of the sensor output is shown as 12V, any appropriate voltage range may be used.

FIG. **78** is a graph **4400** depicting an abnormal relationship between trigger button displacement and sensor output. The vertical axis **4470** represents displacement of a trigger button. The horizontal axis **4480** represents output of a displacement sensor. A first curve **4410** represents a normal relationship between trigger button displacement and sensor output. A second curve **4420** represents an abnormal relationship between trigger button displacement and sensor output, where the sensor output does not reach its maximum value when the button is pressed all the way down. A third curve **4430** represents another abnormal relationship between trigger button displacement and sensor output, where the sensor output reaches its maximum value when the button is only pressed about half way. These abnormal situations may be detected during servicing or sterilization, and may indicate button wear or damage. Upon detection of these abnormal situations, the sensor may be recalibrated to compensate for the wear or damage.

FIG. **79** is a graph **A900** depicting an acceptable relationship between trigger button displacement and sensor output. The vertical axis **4570** represents displacement of a trigger button. The horizontal axis **4580** represents output of a displacement sensor. A first curve **4510** represents an as-manufactured relationship between trigger button displacement and sensor output. A second curve **4515** represents a changed relationship between trigger button displacement and sensor output due to aging. This relationship is acceptable because the user can still activate the three zones **4520**, **4530**, **4540**.

Another embodiment allows for local influencing of the RF power by using other local sensors within the flex circuit to either dampen power output or redirect power to another electrode. For instance, local measurement of temperature within a specific segmented electrode pair is used to influence the balance of power available to each side of the electrode pair. Local measurement of force is used to direct more power to the heavier loaded pairs of electrodes.

FIG. **80** illustrates one aspect of a left-right segmented flexible circuit **4600**. The left-right segmented flexible circuit **4600** comprises a plurality of segments L1-L5 on the left side of the left-right segmented flexible circuit **4600** and a plurality of segments R1-R5 on the right side of the left-right segmented flexible circuit **4600**. Each of the segments L1-L5 and R1-R5 comprise temperature sensors and/or force sensors to sense tissue parameters locally within each segment L1-L5 and R1-R5. The left-right segmented flexible circuit **4600** is configured to influence the RF treatment energy based on tissue parameters sensed locally within each of the segments L1-L5 and R1-R5.

FIG. **81** is a cross-sectional view of one aspect of a flexible circuit A**1100** comprising RF electrodes and data sensors embedded therein. The flexible circuit **4700** can be mounted to the right or left portion of an RF clamp arm A**1102**, which is made of electrically conductive material such as metal. Below the RF clamp arm **4702**, down (vertical) force/pressure sensors **4706***a*, **4706***b* are embedded below a laminate layer **4704**. A transverse force/pressure sensor **4708** is located below the down (vertical) force/pressure sensor **4706***a*, **4706***b* layer and a temperature sensor **4710** is located below the transverse force/pressure sensor **4708**. An electrode **4712** is electrically coupled to the generator and configured to apply RF energy to the tissue **4714** located below the temperature sensor **4710**.

FIG. **82** is a cross sectional view of an end effector **6200** comprising a jaw member **6202**, flexible circuits **6204***a*, **6204***b*, and segmented electrodes **6206***a*, **6206***b* provided on each flexible circuit **6204***a*, **6204***b*, according to one aspect of the present disclosure. FIG. **83** is a detailed view of the end effector **6200** shown in FIG. **82**, according to one aspect of the present disclosure. As previously discussed, it may be advantageous to provide general purpose controls on the primary handle assembly housing of the surgical instrument with dedicated shaft assembly controls located only on the shafts. For instance, an RF instrument may include a distal head rotation electronic rotary shaft control along with articulation buttons while the primary handle includes energy activation controls and jaw member clamp/unclamp trigger controls. In addition, sensors and end effector measurement elements can be employed. Segmented electrodes can be employed that allow for the instrument to sense where in the jaw members tissue is present. Such systems also may employ asymmetric flexible circuit electrodes that sense multiple tissue parameters and have built in electrodes as well as pressure elements for the measurement of pressure against the ultrasonic blade. These systems may also employ flex electrodes that allow a combo device to have sensors built into each of the two electrodes layered within the flex electrode stack.

Turning now to FIGS. **82** and **83**, the end effector **6200** comprises a jaw member **6202**, flexible circuits **6204***a*, **6204***b*, and segmented electrodes **6206***a*, **6206***b* provided on each flexible circuit **6204***a*, **6204***b*. Each segmented electrode **6206***a*, **6206***b* comprises several segments. As shown, a first segmented electrode **6206***a* comprises first and second segment electrode segments **6208***a*, **6208***b* and a second segmented electrode **6206***b* comprises first and second segment electrode segments **6210***a*, **6210***b*. As shown particularly in FIG. **83** the jaw member **6202** is made of metal and conducts heat to maintain the jaw member **6202** cool. Each of the flexible circuits **6204***a*, **6204***b* comprises electrically conductive elements **6214***a*, **6214***b* made of metal or other electrical conductor materials and are electrically insulated from the metal jaw member **6202** by an electrically insulative laminate **6216**. The conductive elements **6214***a*, **6214***b* are coupled to electrical circuits located either in the shaft assembly, handle assembly, transducer assembly, or battery assembly of any one of the combination ultrasonic/electrosurgical instruments **500**, **600**, **700** described herein in connection with FIGS. **30-44**.

FIG. **84**A is a cross sectional view of an end effector **6300** comprising a rotatable jaw member **6302**, a flexible circuit **6304**, and an ultrasonic blade **6306** positioned in a vertical orientation relative to the jaw member with no tissue located between the jaw member **6302** and the ultrasonic blade **6306**, according to one aspect of the present disclosure. FIG. **84**B is a cross sectional view of the end effector **6300** shown in FIG. **84**A with tissue **6308** located between the jaw member **6302** and the ultrasonic blade **6306**, according to one aspect of the present disclosure. The ultrasonic blade **6306** comprises side lobe sections **6310***a*, **6310***b* to enhance tissue dissection and uniform sections **6312***a*, **6312***b* to enhance tissue sealing. In the vertical orientation shown in FIGS. **84**A and **84**B, the ultrasonic blade **6308** is configured for tissue dissection.

FIG. **85**A is a cross sectional view of the end effector **6300** shown in FIGS. **84**A and **84**B comprising a rotatable jaw member **6302**, a flexible circuit **6304**, and an ultrasonic blade **6306** positioned in a horizontal orientation relative to the jaw member **6302** with no tissue located between the jaw member **6302** and the ultrasonic blade **6306**, according to one aspect of the present disclosure. FIG. **84**B is a cross sectional view of the end effector **6300** shown in FIG. **84**A with tissue **6308** located between the jaw member **6302** and the ultrasonic blade **6306**, according to one aspect of the present disclosure. In the horizontal orientation shown in FIGS. **85**A and **85**B, the ultrasonic blade **6308** is configured for tissue sealing (e.g., cauterization).

With reference to FIGS. **84**A-**85**B, the flexible circuit **6304** includes electrodes configured to deliver high-frequency (e.g., RF) current to the tissue **6308** grasped between the jaw member **6302** and the ultrasonic blade **6306**. In one aspect, the electrodes may be segmented electrodes as described herein in connection with FIGS. **82-83** and **86-93**. The flexible circuit **6304** is coupled to a high-frequency (e.g., RF) current drive circuit **702** shown in connection with FIGS. **33-37**. In the illustrated example, the flexible circuit electrodes **6304** are coupled to the positive pole of the high-frequency (e.g., RF) current energy source and the ultrasonic blade **6306** is coupled to the negative (e.g., return) pole of the high-frequency (e.g., RF) current energy source. It will be appreciated that in some configurations, the positive and negative poles may be reversed such that the flexible circuit **6304** electrodes are coupled to the negative pole and the ultrasonic blade **6306** is coupled to the positive pole. The ultrasonic blade **6306** is acoustically coupled to an ultrasonic transducer **130**, **130'** as shown in connection with FIGS. **4-9**. In operation, the high-frequency (e.g., RF) current is employed to seal the tissue **6308** and the ultrasonic blade **6306** is used to dissect tissue using ultrasonic vibrations.

In the example illustrated in FIGS. **784**, **84**B, **85**A, and **85**B the jaw member **6302** is rotatable about a stationary ultrasonic blade **6306**. The jaw member **6302** may rotate 90°

relative to the ultrasonic blade **6306**. In another aspect, the jaw member **6302** may rotate greater than or equal to 360° relative to the ultrasonic blade **6306**. In various other aspects, the ultrasonic blade **6306** is rotatable about a stationary jaw member **6302**. The ultrasonic blade **6306** may rotate 90° relative to the jaw member **6302**. In another aspect, the ultrasonic blade **6306** may rotate greater than or equal to 360° relative to the jaw member **6302**.

Turning now to FIG. **86**, the end effector **6400** comprises RF data sensors **6406**, **6408***a*, **6408***b* located on the jaw member **6402**. The end effector **6400** comprises a jaw member **6402** and an ultrasonic blade **6404**. The jaw member **6402** is shown clamping tissue **6410** located between the jaw member **6402** and the ultrasonic blade **6404**. A first sensor **6406** is located in a center portion of the jaw member **6402**. Second and third sensors **6408***a*, **6408***b* are located on lateral portions of the jaw member **6402**. The sensors **6406**, **6408***a*, **6408***b* are mounted or formed integrally with a flexible circuit **6412** (shown more particularly in FIG. **87**) configured to be fixedly mounted to the jaw member **6402**.

The end effector **6400** is an example end effector for the surgical instruments **500**, **600**, **700** described herein in connection in FIGS. **30-44**. The sensors **6406**, **6408***a*, **6408***b* are electrically connected to a control circuit such as the control circuit **210** (FIG. **14**), **1300** (FIG. **62**), **1400** (FIG. **63**), **1500** (FIG. **64**) via interface circuits such as circuits **6550**, **6570** (FIGS. **96-97**), for example. The sensors **6406**, **6408***a*, **6408***b* are battery powered and the signals generated by the sensors **6406**, **6408***a*, **6408***b* are provided to analog and/or digital processing circuits of the control circuit.

In one aspect, the first sensor **6406** is a force sensor to measure a normal force F_3 applied to the tissue **6410** by the jaw member **6402**. The second and third sensors **6408***a*, **6408***b* include one or more elements to apply RF energy to the tissue **6410**, measure tissue impedance, down force F_1, transverse forces F_2, and temperature, among other parameters. Electrodes **6409***a*, **6409***b* are electrically coupled to an energy source such as the electrical circuit **702** (FIG. **34**) and apply RF energy to the tissue **6410**. In one aspect, the first sensor **6406** and the second and third sensors **6408***a*, **6408***b* are strain gauges to measure force or force per unit area. It will be appreciated that the measurements of the down force F_1, the lateral forces F_2, and the normal force F_3 may be readily converted to pressure by determining the surface area upon which the force sensors **6406**, **6408***a*, **6408***b* are acting upon. Additionally, as described with particularity herein, the flexible circuit **6412** may comprise temperature sensors embedded in one or more layers of the flexible circuit **6412**. The one or more temperature sensors may be arranged symmetrically or asymmetrically and provide tissue **6410** temperature feedback to control circuits of the ultrasonic drive circuit **177** and the RF drive circuit **702**.

FIG. **87** illustrates one aspect of the flexible circuit **6412** shown in FIG. **86** in which the sensors **6406**, **6408***a*, **6408***b* may be mounted to or formed integrally therewith. The flexible circuit **6412** is configured to fixedly attach to the jaw member **6402**. As shown particularly in FIG. **87**, asymmetric temperature sensors **6414***a*, **6414***b* are mounted to the flexible circuit **6412** to enable measuring the temperature of the tissue **6410** (FIG. **86**).

FIG. **88** is a cross-sectional view of the flexible circuit **6412** shown in FIG. **87**. The flexible circuit **6412** comprises multiple layers and is fixedly attached to the jaw member **6402**. A top layer of the flexible circuit **6412** is an electrode **6409***a*, which is electrically coupled to an energy source such as the electrical circuit **702** (FIG. **34**) to apply RF energy to the tissue **6410** (FIG. **86**). A layer of electrical insulation **6418** is provided below the electrode **6409***a* layer to electrically isolate the sensors **6414***a*, **6406**, **6408***a* from the electrode **6409***a*. The temperature sensors **6414***a* are disposed below the layer of electrical insulation **6418**. The first force (pressure) sensor **6406** is located below the layer containing the temperature sensors **6414***a* and above a compressive layer **6420**. The second force (pressure) sensor **6408***a* is located below the compressive layer **6420** and above the jaw member **6402** frame.

FIG. **89** illustrates one aspect of a segmented flexible circuit **6430** configured to fixedly attach to a jaw member **6434** of an end effector. The segmented flexible circuit **6430** comprises a distal segment **6432***a* and lateral segments **6432***b*, **6432***c* that include individually addressable sensors to provide local tissue control. The segments **6432***a*, **6432***b*, **6432***c* are individually addressable to treat tissue and to measure tissue parameters based on individual sensors located within each of the segments **6432***a*, **6432***b*, **6432***c*. The segments **6432***a*, **6432***b*, **6432***c* of the segmented flexible circuit **6430** are mounted to the jaw member **6434** and are electrically coupled to an energy source such as the electrical circuit **702** (FIG. **34**) via electrical conductive elements **6436**. A Hall effect sensor **6438**, or any suitable magnetic sensor, is located on a distal end of the jaw member **6434**. The Hall effect sensor **6438** operates in conjunction with a magnet to provide a measurement of an aperture defined by the jaw member **6434**, which otherwise may be referred to as a tissue gap, as shown with particularity in FIG. **91**.

FIG. **90** illustrates one aspect of a segmented flexible circuit **6440** configured to mount to a jaw member **6444** of an end effector. The segmented flexible circuit **6580** comprises a distal segment **6442***a* and lateral segments **6442***b*, **6442***c* that include individually addressable sensors for tissue control. The segments **6442***a*, **6442***b*, **6442***c* are individually addressable to treat tissue and to read individual sensors located within each of the segments **6442***a*, **6442***b*, **6442***c*. The segments **6442***a*, **6442***b*, **6442***c* of the segmented flexible circuit **6440** are mounted to the jaw member **6444** and are electrically coupled to an energy source such as the electrical circuit **702** (FIG. **34**), via electrical conductive elements **6446**. A Hall effect sensor **6448**, or other suitable magnetic sensor, is provided on a distal end of the jaw member **6444**. The Hall effect sensor **6448** operates in conjunction with a magnet to provide a measurement of an aperture defined by the jaw member **6444** of the end effector or tissue gap as shown with particularity in FIG. **91**. In addition, a plurality of lateral asymmetric temperature sensors **6450***a*, **6450***b* are mounted on or formally integrally with the segmented flexible circuit **6440** to provide tissue temperature feedback to control circuits in the ultrasonic drive circuit **177** and the RF drive circuit **702**.

FIG. **91** illustrates one aspect of an end effector **6460** configured to measure a tissue gap GT. The end effector **6460** comprises a jaw member **6462** and a jaw member **6444**. The flexible circuit **6440** as described in FIG. **90**, is mounted to the jaw member **6444**. The flexible circuit **6440** comprises a Hall effect sensor **6448** that operates with a magnet **6464** mounted to the jaw member **6462** to measure the tissue gap GT. This technique can be employed to measure the aperture defined between the jaw member **6444** and the jaw member **6462**. The jaw member **6462** may be an ultrasonic blade.

FIG. **92** illustrates one aspect of an end effector **6470** comprising segmented flexible circuit **6468** as shown in FIG. **80**. The end effector **6470** comprises a jaw member **6472** and an ultrasonic blade **6474**. The segmented flexible circuit **6468** is mounted to the jaw member **6472**. Each of the sensors disposed within the segments **1-5** are configured to detect the presence of tissue positioned between the jaw member **6472** and the ultrasonic blade **6474** and represent tissue zones **1-5**. In the configuration shown in FIG. **92**, the end effector **6470** is shown in an open position ready to receive or grasp tissue between the jaw member **6472** and the ultrasonic blade **6474**.

FIG. **93** illustrates the end effector **6470** shown in FIG. **92** with the jaw member **6472** clamping tissue **6476** between the jaw member **6472** and the ultrasonic blade **6474**. As shown in FIG. **93**, the tissue **6476** is positioned between segments **1-3** and represents tissue zones **1-3**. Accordingly, tissue **6476** is detected by the sensors in segments **1-3** and the absence of tissue (empty) is detected in section **6478** by segments **4-5**. The information regarding the presence and absence of tissue **6476** positioned within certain segments **1-3** and **4-5**, respectively, is communicated to a control circuit such as such as the control circuits **210** (FIG. **14**), **1300** (FIG. **62**), **1400** (FIG. **63**), **1500** (FIG. **64**) via interface circuits such as circuits **6550**, **6570** (FIGS. **96-97**), for example. The control circuit is configured to energize only the segments **1-3** where tissue **6476** is detected and does not energize the segments **4-5** where tissue is not detected. It will be appreciated that the segments **1-5** may contain any suitable temperature, force/pressure, and/or Hall effect magnetic sensors to measure tissue parameters of tissue located within certain segments **1-5** and electrodes to deliver RF energy to tissue located in certain segments **1-5**.

FIG. **94** illustrates graphs **6480** of energy applied by the right and left side of an end effector based on locally sensed tissue parameters. As discussed herein, the jaw member of an end effector may comprise temperature sensors, force/pressure sensors, Hall effector sensors, among others, along the right and left sides of the jaw member. Thus, RF energy can be selectively applied to tissue positioned between the clam jaw and the ultrasonic blade. The top graph **6482** depicts power PR applied to a right side segment of the jaw member versus time (t) based on locally sensed tissue parameters. Thus, the control circuit such as such as the control circuits **210** (FIG. **14**), **1300** (FIG. **62**), **1400** (FIG. **63**), **1500** (FIG. **64**) via interface circuits such as circuits **6550**, **6570** (FIGS. **96-97**), for example, is configured to measure the sensed tissue parameters and to apply power PR to a right side segment of the jaw member. The RF drive circuit **702** (FIG. **34**) delivers an initial power level P_1 to the tissue via the right side segment and then decreases the power level to P_2 based on local sensing of tissue parameters (e.g., temperature, force/pressure, thickness) in one or more segments. The bottom graph **6484** depicts power P_L applied to a left side segment of the jaw member versus time (t) based on locally sensed tissue parameters. The RF drive circuit **702** delivers an initial power level of P_1 to the tissue via the left side segment and then increases the power level to P_3 based local sensing of tissue parameters (e.g., temperature, force/pressure, thickness). As depicted in the bottom graph **6484**, the RF drive circuit **702** is configured to re-adjust the energy delivered P_3 based on sensing of tissue parameters (e.g., temperature, force/pressure, thickness).

FIG. **95** is a cross-sectional view of one aspect of an end effector **6530** configured to sense force or pressure applied to tissue located between a jaw member and an ultrasonic blade. The end effector **6530** comprises a clamp jaw **6532** and a flexible circuit **6534** fixedly mounted to the jaw member **6532**. The jaw member **6532** applies forces F_1 and F_2 to the tissue **6536** of variable density and thickness, which can be measure by first and second force/pressure sensors **6538**, **6540** located in different layers of the flexible circuit **6534**. A compressive layer **6542** is sandwiched between the first and second force/pressure sensors **6538**, **6540**. An electrode **6544** is located on outer portion of the flexible circuit **6534** which contacts the tissue. As described herein, other layers of the flexible circuit **6534** may comprise additional sensors such temperature sensors, thickness sensors, and the like.

FIGS. **96-97** illustrate various schematic diagrams of flexible circuits of the signal layer, sensor wiring, and an RF energy drive circuit. FIG. **96** is a schematic diagram of one aspect of a signal layer of a flexible circuit **6550**. The flexible circuit **6550** comprises multiple layers (~4 to ~6, for example). One layer will supply the integrated circuits with power and another layer with ground. Two additional layers will carry the RF power RF1 and RF2 separately. An analog multiplexer switch **6552** has eight bidirectional translating switches that can be controlled through the I^2C bus to interface to the control circuit **210** (FIG. **14**) via the SCL-C/SDA-C interface channel. The SCL/SDA upstream pair fans out to eight downstream pairs, or channels. Any individual SCn/SDn channel or combination of channels can be selected, determined by the contents of a programmable control register. There are six down stream sensors, three on each side of the jaw member. A first side **6554*a*** comprises a first thermocouple **6556*a***, a first pressure sensor **6558*a***, and a first Hall effect sensor **6560*a***. A second side **6554*b*** comprises a second thermocouple **6556*b***, a second pressure sensor **6558*b***, and a second Hall effect sensor **6560*b***. FIG. **97** is a schematic diagram **6570** of sensor wiring for the flexible circuit **6550** shown in FIG. **96** to the switch **6552**.

FIG. **98**A illustrates an end effector **6670** comprising a jaw member **6672** and an ultrasonic blade **6674**, where the jaw member **6672** includes electrodes **6676**. The end effector **6670** can be employed in one of the surgical instruments combination ultrasonic/electrosurgical instruments **500**, **600**, **700** described herein in connection with FIGS. **30-44**, where the combination ultrasonic/electrosurgical instruments **500**, **600**, **700** are configured to switch between RF, ultrasonic, and combination RF/ultrasonic energy automatically based on a sensed/calculated measure of device parameters such as, for example, impedance, current from the motor, jaw member gap, tissue compression, temperature, among others, implying tissue thickness and/or type. Referring to FIG. **984**A, the end effector **6670** may be positioned by a physician to surround tissue **6678** prior to compression, cutting, or stapling. As shown in FIG. **98**A, no compression may be applied to the tissue while preparing to use the end effector **6670**. As shown in FIG. **98**A, the tissue **6678** is not under compression between the jaw member **6672** and the ultrasonic blade **6674**.

Referring now to FIG. **98**B, by engaging the trigger on the handle of a surgical instrument, the physician may use the end effector **6670** to compress the tissue **6678**. In one aspect, the tissue **6678** may be compressed to its maximum threshold, as shown in FIG. **98**B. As shown in FIG. **98**A, the tissue **6678** is under maximum compression between the jaw member **6672** and the ultrasonic blade **6674**.

Referring to FIG. **99**A, various forces may be applied to the tissue **6678** by the end effector **6670**. For example, vertical forces F1 and F2 may be applied by the jaw member **6672** and the ultrasonic blade **6674** of the end effector **6670** as tissue **6678** is compressed between the two. Referring now to FIG. **99**B, there is shown various diagonal and/or lateral forces also may be applied to the tissue **6678** when compressed by the end effector **6670**. For example, a force F3 may be applied. For the purposes of operating the combination ultrasonic/electrosurgical instruments **500**, **600**, **700** (FIGS. **30-44**), it may be desirable to sense or calculate the various forms of compression being applied to the tissue by the end effector. For example, knowledge of vertical or lateral compression may allow the end effector to more precisely or accurately apply a staple operation or may inform the operator of the surgical instrument such that the surgical instrument can be used more properly or safely.

In one form, a strain gauge can be used to measure the force applied to the tissue **6678** by the end effector shown in FIGS. **98**A-B, **99**A-B. A strain gauge can be coupled to the end effector **6670** to measure the force on the tissue **6678** being treated by the end effector **6670**. With reference now also to FIG. **100**, in the aspect illustrated in FIG. **100**, a system **6680** for measuring forces applied to the tissue **6678** comprises a strain gauge sensor **6682**, such as, for example, a micro-strain gauge, is configured to measure one or more parameters of the end effector **6670** such as, for example, the amplitude of the strain exerted on a jaw member of an end effector, such as the jaw member **6672** of FIGS. **99**A-B, during a clamping operation, which can be indicative of the tissue compression. The measured strain is converted to a digital signal and provided to a processor **6690** of a microcontroller **6688**. A load sensor **6684** can measure the force to operate the ultrasonic blade **6674** to cut the tissue **6678** captured between the jaw member **6672** and the ultrasonic blade **6674** of the end effector **6670**. A magnetic field sensor **6686** can be employed to measure the thickness of the captured tissue **6678**. The measurement of the magnetic field sensor **6686** also may be converted to a digital signal and provided to the processor **6690**.

Further to the above, a feedback indicator **6694** also can be configured to communicate with the microcontroller **6688**. In one aspect, the feedback indicator **6694** can be disposed in the handle of the combination ultrasonic/electrosurgical instruments **500**, **600**, **700** (FIGS. **30-44**). Alternatively, the feedback indicator **6694** can be disposed in a shaft assembly of a surgical instrument, for example. In any event, the microcontroller **6688** may employ the feedback indicator **6694** to provide feedback to an operator of the surgical instrument with regard to the adequacy of a manual input such as, for example, a selected position of a firing trigger that is used to cause the end effector to clamp down on tissue. To do so, the microcontroller **6688** may assess the selected position of the jaw member **6672** and/or firing trigger. The measurements of the tissue **6678** compression, the tissue **6678** thickness, and/or the force required to close the end effector **6670** on the tissue, as respectively measured by the sensors **6682**, **6684**, **6686**, can be used by the microcontroller **6688** to characterize the selected position of the firing trigger and/or the corresponding value of the speed of end effector. In one instance, a memory **6692** may store a technique, an equation, and/or a look-up table which can be employed by the microcontroller **6688** in the assessment.

In view of the above, in one aspect the present disclosure provides a surgical instrument comprising: a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft, the end effector comprising a first jaw and a second jaw configured for pivotal movement between a closed position and an open position; a handle assembly coupled to a proximal end of the shaft; a battery assembly coupled to the handle assembly; a radio frequency (RF) energy output powered by the battery assembly and configured to apply RF energy to a tissue; an ultrasonic energy output powered by the battery assembly and configured to apply ultrasonic energy to the tissue; and a controller configured to, based at least in part on a measured tissue characteristic, start application of RF energy by the RF energy output or application of ultrasonic energy by the ultrasonic energy output at a first time.

In one aspect, the controller may be further configured to, based at least in part on the measured tissue characteristic, switch between RF energy applied by the RF energy output and ultrasonic energy applied by the ultrasonic energy output at a second time. The controller may be further configured to, based at least in part on the measured tissue characteristic, terminate the RF energy applied by the RF energy output after a first period and apply only ultrasonic energy by the ultrasonic energy output for a second period. The controller may be further configured to, based at least in part on the measured tissue characteristic, terminate the ultrasonic energy applied by the ultrasonic energy output after a first period and apply only RF energy by the RF energy output for a second period second time. The controller may be further configured to, based at least in part on the measured tissue characteristic, control a level of RF energy applied by the RF energy output or ultrasonic energy applied by the ultrasonic energy output. The controller may be further configured to, based at least in part on the measured tissue characteristic, reduce a level of RF energy applied by the RF energy output from a therapeutic energy level to a non-therapeutic energy level suitable for measuring tissue impedance without a therapeutic effect on the tissue. The controller may be further configured to, based at least in part on the measured tissue characteristic, control a waveform of RF energy applied by the RF energy output or ultrasonic energy applied by the ultrasonic energy output. The controller may be further configured to determine the measured tissue characteristic based on behavior of impedance between the first and second jaws across the tissue.

The controller may be further configured to determine the measured tissue characteristic based on behavior of a gap distance between the first and second jaws. The controller may be further configured to determine the measured tissue characteristic based on behavior of a force applied by one or both of the first and second jaws on the tissue. The force may be measured by a current or voltage of a motor driving one or both of the first and second jaws.

The controller may be further configured to determine the measured tissue characteristic based on a time required to reach a constant force. The measured tissue characteristic may be tissue thickness. For a thick tissue, the first time may be determined to be a long delay after a force applied by one or both of the first and second jaws on the tissue has reached a constant force and for a thin tissue, the first time may be determined to be a short delay after the force has reached the constant force. For a thick tissue, RF energy may be applied before and after a first period, where impedance between the first and second jaws across the tissue may be below a first impedance threshold and ultrasonic energy is applied. For a thin tissue, only RF energy may be applied.

The controller may be further configured to adjust the first impedance threshold based on a secondary tissue characteristic different from the measured tissue characteristic. For a thick tissue, ultrasonic energy may be switched to RF energy when a force applied by one or both of the first and second jaws on the tissue falls below a first force threshold, for a thin tissue, only RF energy may be applied. The measured tissue characteristic may be tissue compressibility. The measured tissue characteristic may be tissue short circuit condition.

The controller may be further configured to determine that there is a tissue short circuit condition when impedance between the first and second jaws across the tissue is below a second impedance threshold, and a gap distance between the first and second jaws is above a gap distance threshold. The controller may be further configured to determine that there is a tissue short circuit condition when impedance between the first and second jaws across the tissue is below a third impedance threshold, and a force applied by one or both of the first and second jaws on the tissue is above a second force threshold.

In another aspect, the present disclosure provides a method for operating a surgical instrument, the surgical instrument comprising a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft, the end effector comprising a first jaw and a second jaw configured for pivotal movement between a closed position and an open position, a handle assembly coupled to a proximal end of the shaft, and a battery assembly coupled to the handle assembly, the method comprising: measuring a tissue characteristic; and starting, based at least in part on the measured tissue characteristic, application of RF energy by a RF energy output or application of ultrasonic energy by a ultrasonic energy output at a first time. The method may further comprise switching, based at least in part on the measured tissue characteristic, between RF energy applied by the RF energy output and ultrasonic energy applied by the ultrasonic energy output at a second time.

In another aspect, the present disclosure provides a surgical instrument comprising: a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft, the end effector comprising a first jaw and a second jaw configured for pivotal movement between a closed position and an open position; a handle assembly coupled to a proximal end of the shaft, the handle assembly comprising a button configured to be displaced when pressed by a user; a battery assembly coupled to the handle assembly; a sensor configured to measure a displacement or intensity of the button; a radio frequency (RF) energy output powered by the battery assembly and configured to apply RF energy to a tissue; an ultrasonic energy output powered by the battery assembly and configured to apply ultrasonic energy to the tissue; and a controller configured to: control the RF energy output to apply RF energy or control the ultrasonic energy output to apply ultrasonic energy at a low level when the displacement or intensity measured by the sensor is above a first threshold but below a second threshold higher than the first threshold, and control the RF energy output to apply RF energy or control the ultrasonic energy output to apply ultrasonic energy at a high level when the displacement or intensity measured by the sensor is above the second threshold.

The sensor may be a Hall sensor. The sensor may be configured to measure force applied to the button and the displacement or intensity of the button is the force applied to the button. The sensor may be configured to be calibrated after the surgical instrument is manufactured, such that an output of the sensor is substantially proportional to the displacement or intensity of the button. The sensor may be configured to be calibrated if the output of the sensor reaches maximum when the displacement or intensity of the button has not reached maximum. The sensor may be configured to be calibrated if the displacement or intensity of the button reaches maximum when the output of the sensor has not reached maximum.

In another aspect, the present disclosure provides a surgical instrument comprising: a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft, the end effector comprising a first jaw and a second jaw configured for pivotal movement between a closed position and an open position; a handle assembly coupled to a proximal end of the shaft, the handle assembly comprising a button configured to be displaced when pressed by a user; a battery assembly coupled to the handle assembly; a sensor configured to measure a displacement or intensity of the button; a radio frequency (RF) energy output powered by the battery assembly and configured to apply RF energy to a tissue; an ultrasonic energy output powered by the battery assembly and configured to apply ultrasonic energy to the tissue; and a controller configured to: control the RF energy output to apply RF energy or control the ultrasonic energy output to apply ultrasonic energy when the displacement or intensity measured by the sensor is above a first threshold but below a second threshold higher than the first threshold, and control the RF energy output to apply RF energy and control the ultrasonic energy output to apply ultrasonic energy when the displacement or intensity measured by the sensor is above the second threshold.

The sensor may be a Hall sensor. The sensor may be configured to be calibrated after the surgical instrument is manufactured, such that an output of the sensor is substantially proportional to the displacement or intensity of the button. The sensor may be configured to be calibrated after the surgical instrument is manufactured, such that an output of the sensor is substantially proportional to the displacement or intensity of the button. The sensor may be configured to be calibrated if the displacement or intensity of the button reaches maximum when the output of the sensor has not reached maximum.

In another aspect, the present disclosure provides a surgical instrument comprising: a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft, the end effector comprising a first jaw and a second jaw configured for pivotal movement between a closed position and an open position, wherein the first or second jaw comprising: a first sensor configured to measure a tissue characteristic at a first location, and a second sensor configured to measure the tissue characteristic at a second location; a handle assembly coupled to a proximal end of the shaft; a battery assembly coupled to the handle assembly; a radio frequency (RF) energy output powered by the battery assembly and configured to apply RF energy to a tissue; an ultrasonic energy output powered by the battery assembly and configured to apply ultrasonic energy to the tissue; and a controller configured to, based at least in part on the measured tissue characteristic at the first location and the second location, control the RF energy output to apply RF energy or control the ultrasonic energy output to apply ultrasonic energy.

The first and second sensors may be vertical pressure sensors. The controller may be further configured to: apply high energy at a location with high sensor output; and apply low energy at a location with low sensor output. The first and second sensors may be transverse pressure sensors. The first and second sensors may be temperature sensors. The first and second sensors may be configured to sense presence or absence of tissue at the first location and the second location.

The controller may be further configured to: set a first impedance threshold below which RF energy is switched to ultrasonic energy when tissue is present at both the first location and the second location; and set a second impedance threshold below which RF energy is switched to ultrasonic energy when tissue is present at only one of the first location and the second location, wherein the first and second impedance thresholds are different. The controller may be further configured to: set a first force threshold used to control application of energy when tissue is present at both the first location and the second location; and set a second force threshold used to control application of energy when tissue is present at only one of the first location and the second location, wherein the first and second force thresholds are different. The controller may be further configured to: apply energy at a location where tissue is present; and not apply energy at a location where tissue is absent.

Modular Battery Powered Handheld Surgical Instrument with Variable Motor Control Limits, Motor Control Limit Profile, and Motor Control Limits Based on Tissue Characterization

In another aspect, the present disclosure provides a modular battery powered handheld surgical instrument with variable motor control limits. Disclosed is a surgical instrument that includes a shaft assembly; a handle assembly; a battery assembly; a first driver configured to drive movement of at least a first component of the surgical instrument; a second driver configured to drive movement of at least a second component of the surgical instrument; a motor powered by the battery assembly and configured to selectively actuate the first driver or the second driver using an output of the motor; and a controller for the motor, configured to: perform a first limiting operation to the output of the motor when the motor actuates the first driver and a first limiting condition is met, and perform a second limiting operation to the output of the motor when the motor actuates the second driver and a second limiting condition is met.

In another aspect, the present disclosure provides a modular battery powered handheld surgical instrument with motor control limit profile. Disclosed is a surgical instrument that includes a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft; a handle assembly coupled to a proximal end of the shaft; a battery assembly coupled to the handle assembly; a driver configured to drive movement of at least one component of the surgical instrument; a motor powered by the battery assembly and configured to actuate the driver using an output of the motor; and a controller for the motor, configured to perform a first limiting operation to the output of the motor when a first limiting condition is met, wherein the first limiting condition defines a maximum control variable profile of the motor over a range of the movement of the at least one component.

In another aspect, the present disclosure provides a modular battery powered handheld surgical instrument with motor control limits based on tissue characterization. A surgical instrument comprises a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft; a handle assembly coupled to a proximal end of the shaft; a battery assembly coupled to the handle assembly; a driver configured to drive movement of at least one component of the surgical instrument; a motor powered by the battery assembly and configured to actuate the driver using an output of the motor; and a controller for the motor, configured to perform a limiting operation to the output of the motor when a limiting condition is met, wherein the limiting condition is determined based at least in part on a tissue characteristic.

FIGS. **101-84** illustrate various aspects of the present disclosure that is directed to motor and a motor controller where a first limiting threshold is used on the motor for the purpose of attaching a modular assembly and a second threshold associated with a second assembly step or functionality of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-70**. According to various aspects of the present disclosure, a system includes a motor **4830**, **1160**, **1432** (FIGS. **103**, **49**, and **63**, respectively), a motor controller (e.g., motor control circuits **726** shown in FIG. **36**, motor drive circuits **1165**, **4982** shown in FIGS. **50** and **80**, respectively, as well as the motor control circuit **1430** and primary processor **1302** as shown in FIG. **63**), a first controller limiting threshold associated to an attached assembly or function step of operation and a second threshold associated with another assembly or function step. The first threshold is applied when its system is in operation, and the second threshold is used as opposed to the first if a second assembly is coupled or in operation. One way to enable this functionality is by having a multi-purpose motor that actuates both transducer torqueing onto the handpiece, articulation control and clamp arm control, where the torqueing of transducer has a different limit than the articulation limit and both are different from the clamp arm limit. Different control responses can also be created when the system exceeds the motor current threshold, where this threshold can be based, in part, on the mechanism function or longitudinal position of specific components in the system.

FIG. **101** is a graph **4800** showing a limiting operation on a torque applied to an ultrasonic transducer **130**, **130'** as described herein in connection with FIGS. **4-10**B. The horizontal axis represents time (t), and the vertical axis represents torque (T). The ultrasonic transducer may be used by a surgical instrument to apply ultrasonic vibrational energy to tissue. The torque applied to the ultrasonic transducer may be measured by measuring the motor load current of the motor **4830** (FIG. **73**) driving (e.g., torqueing) the ultrasonic transducer. As shown in graph **4800**, the torque applied to the ultrasonic transducer increases over time until the torque reaches a predetermined torque limit **4804** at a time **4802**. At that time **4802**, the motor is cut off to reduce the torque to zero. Therefore, an electronic current limiter limits the torque to a known force.

FIG. **102** is a graph **4810** comparing different limiting conditions and different limiting operations when the motor is driving different moving components. The horizontal axis represents time (t), and the vertical axis represents motor current (I). Motor current may be an indication of force or torque output by the motor. In some aspects, a single motor may selectively actuate different drivers for driving torque on a transducer **4812**, user activated distal head rotation **4814**, articulation **4816**, or clamp arm movement **4818**, among others. Articulation **4816** as used herein may be articulation movement of two portions of the shaft about an articulation joint coupled therebetween. Distal head rotation **4814** as used herein may be axial rotation of the distal one of the two portions. Clamp arm movement **4818** as used herein may be pivotal movement of two jaws (clam arms) of an end effector of a surgical instrument between a closed position and an open position.

As shown in graph **4810**, torque on transducer **4812** has a first current limit **4822**, distal head rotation **4814** has a second current limit **4820**, articulation **4816** has a third current limit **4826**, clamp arm movement has a fourth current limit **4824**, and so on. When torque on transducer **4812**, distal head rotation **4814** or articulation **4816** reaches their respective current limits **4822**, **4820**, **4826**, the motor is cut off completely (e.g., safety limits). In contrast, when clamp arm movement **4818** reaches its current limit **4824**, the current is maintained at a value below or about the current limit **4824**, not completely cut off (e.g., constant force limit). It will be appreciated that in various aspects a controller for the motor may be configured to perform limiting operations to the output of the motor when a first limiting condition is met. In one aspect, the limiting condition defines a maximum control variable profile of the motor over a range of the movement of the at least one component. In one aspect, the control variable profile is a current profile of the motor. In another aspect, the variable profile is a voltage profile of the motor. In other aspects, the control variable profile of the motor may be power profile of the motor, a torque profile of the motor, or pother profiles of the motor. In other aspects, the control variable profile is position profile, inertia profile, velocity profile, or acceleration profile, either of the motor or an element coupled to the shaft of the motor.

FIG. **103** is a perspective view of a motor **4830** comprising a strain gauge **4834** used for measuring torque. The motor **4830** comprises planetary gear **4832** in its body, and comprises a pair of fixed magnets **4836**, **4838**, a coiled wire **4840** and the strain gauge **4834** (and its microchip) on the output shaft of the motor **4830**. The coiled wire **4840** rotates to power the strain gauge **4834**. Therefore, the torque applied to the transducer can be directly measured, instead of being measured using the motor current. The coiled wire **4840** generates an electric current as it rotates through the magnetic field produced by the pair of magnets **4836**, **4838**.

In another aspect of the present disclosure, the rate of change of the user trigger position is used to influence the motor control parameters. For instance, if the user is making many opposing trigger motions, implying fine dissection or precision, the system runs potentially different maximum thresholds as well as dampened acceleration and velocity direction to the motor **4830** to smooth out fine movement of the jaw members. Alternatively, a minimum trigger motion can be used to indicate a desired motion of the jaw member based on the time between larger motions, allowing the motor controller to remove hand tremor or other undesired motions to powered jaw motion.

FIG. **104** are graphs **4850** and **4860** showing smoothing out of fine trigger button movement by a motor controller. A first graph **4850** of FIG. **104** shows displacement of a trigger button over time, and a second graph **4860** of FIG. **104** shows velocity of movement over time. The trigger button, for example, may be located at a handle assembly or module and is used by a user to control jaw member opening or closing. As shown in the first graph **4850**, a first trigger button movement **4852** is manifested as a high frequency and/or low magnitude movement over a period T1, which may indicate user hand tremor. A second trigger button movement **4854**, **4864** is a low frequency and/or high magnitude movement over a period T2, which may indicate intentional movement. Therefore, when the motor controller translates the trigger button movement to velocity of jaw member opening/closing, the motor controller may perform a proportional adjustment on the first trigger button movement **4852**, thus forming a smooth portion **4862** of velocity curve. For the intentional movement **4854**, no smoothing operation is performed.

In another aspect of the present disclosure, a system includes a motor **4830** (FIG. **103**) to actuate a mechanism, a motor controller to control the motor velocity or torque, where measurement of current or a parameter related to current through the motor **4830** is used to control the motor **4830**. A non-linear threshold is used to trigger motor **4830** adjustments at different magnitudes based on position, inertia, velocity, or acceleration. For instance, upper and lower current thresholds could be employed to set a bounded pressure range for the clamp arm to provide the motor control circuit an operational range including mechanical advantage adjustments for the mechanism. Another version of this is where the I-blade radial threshold force limit is based on the system mechanical advantage and advancement stroke location of the I-blade.

Yet another version is where the slope rise or fall of current or voltage is used to provide a forecasting element to determine if they will likely exceed the threshold, and is subsequently used to vary the threshold limit. For instance, tissue compressibility used in conjunction with motor current rise could be used to provide a calculation of the maximum force expected on the components, based on the current location of closure mechanism in relation to full closure.

In still another example, position measurement of an actuated body is performed and the adjustment of the rate of change of the actuation is based on its position within its overall stroke. For instance, the clamp arm is allowed to make large acceleration and velocity changes when between 95% and 25% of its overall limits, but it must run within a much slower speed when outside of this range. At 95% of stroke (which may be considered fully open) the mechanism is approaching its max open condition, and in order to limit overstress on its mechanical stops the system begins to slow down and limit speed/torque. At 25% the jaws are in contact with tissue or approaching contact with tissue, so the system begins to limit acceleration and speed in order to soften its handling of tissue and minimize tearing. Also, each of the powered capabilities being run by the same motor **4830** could have differing limits indicating acceleration or velocity limits based on being calibrated or based on the inertia and frictional needs of the system. For instance, heavier systems have more constraining limits than lighter systems which are easier to start and stop.

FIG. **105** is a graph **4870** showing a non-linear force limit **4872** and corresponding velocity adjustment for I-blade RF motor **4830** control, where F represents force and V represents velocity. As shown in FIG. **105**, the non-linear force limit **4872** sets a lower force limit at both ends of the range of movement δ and a higher force limit in the middle of the range of movement δ. In the beginning, a relatively high velocity **4884** is applied, and the force **4874** on the I-blade increases rapidly. At a first point **4878** in the range of movement, the force **4874** exceeds the force limit **4872**, thus the velocity is reduced accordingly to a medium velocity **4886**. Therefore, the force **4874** gradually falls back below the force limit **4872**. When the force **4874** exceeds the force limit **4872** again at a second point **4880** in the range of movement, the velocity is further reduced to a relatively low velocity **4888**. Therefore, the force **4874** gradually falls back below the force limit **4872** again. A dash dot force curve **4876** shows the ideal behavior of the force on the I-blade (ideal tissue compression limit).

FIG. **106** is a graph **4890** showing force limits **4896**, **4898** and corresponding velocity adjustments for jaw motor control. As shown in FIG. **106**, a upper force limit **4896** and a lower force limit **4898** are provided for the force (pressure) applied by the jaws on the tissue. In the beginning of the range of jaw movement δ, the force (F) **4892** and the velocity (V) **4894** both increase. At a first point δ**1** in the range of movement, the force enters the region between the upper and lower limits **4896**, **4898**. Thereafter the velocity is set to be a constant velocity. At a second point δ**2** in the range of movement, the force exceeds the upper limit **4896**. Thereafter the constant velocity is lowered, and the force falls back under the upper limit **4896**. At a third point δ**3** in the range of movement, the force falls under the lower limit

4898. Thereafter the constant velocity is raised, and the force increases to above the lower limit **4898**. At a fourth point δ**4** in the range of movement, the force exceeds the upper limit **4896**. Thereafter the velocity is lowered, and the force falls back under the upper limit **4896**. Because the rate of decrease of the force is relatively high, a linearly increasing velocity instead of constant force is applied between the fourth point δ**4** and a fifth point δ**5**, where the force falls below the lower limit **4898**. Repeated description of the remaining operations is omitted. When the rate of decrease of the force is relatively high, an increasing velocity is applied. When the rate of increase of the force is relatively high, a decreasing velocity is applied. Therefore, a forecast is provided to control the velocity before the force moves outside the permitted range.

FIG. **107** are graphs **4900**, **4910** showing different velocity limits when the motor **4830** is driving different moving components. As shown in a first graph **4900** of FIG. **107**, a higher velocity limit **4902** is set for jaw clamping, and a lower velocity limit **4906** is set for articulation. A solid curve **4904** represents the opening and closing of jaw members, and its velocity does not exceed the higher velocity limit **4902**. A dashed curve **4908** represents articulation, and its velocity does not exceed the lower velocity limit **4906**.

As shown in a second graph **4910** of FIG. **107**, jaw clamping **4912** may have a different full velocity range than articulation **4920**. Jaw clamping **4912** has two limited velocity zones **4914**, **4916** at both ends of the range of movement, and a full velocity zone **4918** in the middle of the range of movement (e.g., 10% to 95% of the range of movement). Articulation **4920** has two limited velocity zones **4922**, **4924** at both ends of the range of movement, and a full velocity zone **4926** in the middle of the range of movement (e.g., 25% to 75% of the range of movement). The full velocity zone of jaw clamping **4918** has a larger range than the full velocity zone of articulation **4926**. The reason may be that articulation is a heavier system and needs a larger range to start and stop.

FIG. **108** is a graph **4950** showing velocity limits **4952**, **4954** at the beginning and the end of the range of movement. As shown in FIG. **108**, at one end of the range of movement δ (e.g., when jaws are approaching tissue), a first velocity limit **4952** may be set for the jaw movement in a range of, e.g., 10 degrees. At the other end of the range of movement δ (e.g., when jaws are approaching mechanical stop), a second velocity limit **4954** may be set for the jaw movement in a range of, e.g., 5 degrees. As shown in FIG. **108**, the force limits described with reference to FIG. **106** may be used in conjunction with the velocity limits described with reference to FIG. **108**.

FIG. **109** are diagrams **4960**, **4970** showing velocity limits at the beginning and the end of the range of movement for jaw clamping and I-beam advance. As shown in a first diagram **4960** of FIG. **109**, at both ends **4962**, **4966** of the range of movement of a jaw **4968** (acceleration and braking), velocity is limited. A full velocity zone **4964** is provided in the middle of the range of movement. As shown in a second diagram **4970** of FIG. **109**, at both ends **4972**, **4976** of the range of movement of an I-beam **4978** (acceleration and braking), velocity is limited. A full velocity zone **4974** is provided in the middle of the range of movement.

In addition, control algorithms can be created to provide functional intelligence to the articulation, head rotation, and clamp action. Methods of measuring parameters from the motor **4830** related to the force experienced by the device can also be created that does not measure current or voltage from the motor **4830**. FIG. **110** illustrates diagrams **4980**, **4982** showing measurement of out-of-phase aspect of the current through the motor **4830**. As shown in FIG. **110**, the inductor and resistor in parallel to the primary lower H-bridge may be used to determine an out of phase aspect of the current through the motor **4830**. The H-bridge **4982** is similar to the motor drive circuit **1165** described in connection with FIG. **50**.

In another aspect of the present disclosure, a system comprises a motor **4830** to actuate a mechanism, a motor controller to control the motor velocity or torque, a sensor associated with physical properties of the moving mechanism which is used to adjust a predefined threshold which triggers a change in the motor control operation. In one aspect, motor control speed changes based on the thickness and compressibility of the tissue sensed. In this example, thickness may be measured by linear encoder, and either time to force (velocity) or force reached at a predefined time used to compute the compressibility of the tissue. Tissue stiffness could also be compensated for via measurement of force at two predetermined times or two predetermined positions.

In another aspect, measurement of current and voltage in one or more locations within the kinematic portion of the motor control circuit may be employed to sense dynamic braking and motor acceleration parameters including their impacts by the bodies in rest and in motion that they are influencing. These force/inertia measurements could then be used to change the control parameters of the motor to minimize its effects on the desired motion/force of the actuation body. In addition, a pressure intensity switch can be employed that creates a response in tissue treatment energy proportionate to the intensity of the pressure on the activation button of the device.

In yet another aspect, control methods for adjusting speed, torque, and acceleration of the motor **4830** may be employed. In this aspect, motor control parameters are based on the measured current through the motor **4830**. This can be paired with force limiting on clamp arm using either a mechanical force limiting device such as a force limiting spring in line with the motor or through a current limit on the motor to control the force supplied by the motor, thus limiting the clamp arm force. Alternately, use of PWM, pulse amplitude modulation (PAM), pulse position modulation (PPM), or PDM or other signal modification can be used to control motor speed and torque while keeping the motor parameters (voltage, current, etc.) close to the ideal motor voltage/torque balance point.

FIG. **111** is a graph **4990** showing adjustment of velocity limit based on tissue type. A first curve **4992** shown in dashed line represents pre-programmed velocity behavior of a tissue. A second curve **4994** shown in solid line represents actual velocity behavior of a tissue. As shown in graph **4990**, it takes a time to for the pre-programmed curve **4992** to reach a threshold velocity vt, and a pre-programmed velocity limit v0 is provided for the pre-programmed curve **4992**. For the actual velocity curve **4994**, it takes at time t1 which is less than t0 to reach the threshold velocity vt. This may indicate that the amount of tissue between the jaws is less than a pre-programmed amount, or that the tissue is thinner than a pre-programmed thickness. The amount of tissue may be detected by two or more sensors located at the jaws and configured to detect tissue presence. Therefore, a lower revised velocity limit v1 is provided for the actual velocity curve **4994**. In other aspects of the present disclosure, determination of tissue thickness may be based on any combination of force behavior, gap distance behavior and impedance behavior.

FIG. **112** is a graph **5000** showing different wait periods and different velocity limits in I-beam motor control for tissues with different thickness. A first force curve **5002** and a first velocity curve **5008** (both shown in solid line) represent a thin tissue. A second force curve **5004** and a second velocity curve **5012** (both shown in dash-dot line) represent a medium tissue. A third force curve **5006** and a third velocity curve **5016** (both shown in dash line) represent a thick tissue.

As shown by the first force curve **5002** and the first velocity curve **5008** (thin tissue), in the beginning, the velocity increases to reach a threshold at time t1. Thereafter, the velocity is set to be zero, until it starts to increase again at time tE1. The period between time t1 and time tE1 is thin tissue wait period. When the velocity reaches a first velocity limit **5010**, it is kept constant until the force falls to zero, which may indicate the tissue has been cut through.

As shown by the second force curve **5004** and the second velocity curve **5012** (medium tissue), in the beginning, the velocity increases to reach a threshold at time t2. Thereafter, the velocity is set to be zero, until it starts to increase again at time tE2. The period between time t2 and time tE2 is medium tissue wait period. When the velocity reaches a second velocity limit **5014**, it is kept constant until the force falls to zero, which may indicate the tissue has been cut through.

As shown by the third force curve **5006** and the third velocity curve **5016** (thick tissue), in the beginning, the velocity increases to reach a threshold at time t3. Thereafter, the velocity is set to be zero, until it starts to increase again at time tE3. The period between time t3 and time tE3 is thick tissue wait period. When the velocity reaches a third velocity limit **5018**, it is kept constant until the force falls to zero, which may indicate the tissue has been cut through.

As shown in FIG. **112**, the thicker the tissue is, the longer the wait period is. Also, the thicker the tissue is, the lower the velocity limit is.

FIG. **113** are graphs **5020**, **5030** depicting a relationship between trigger button displacement and sensor output. In both graphs **5020**, **5030**, the vertical axis represents displacement of a trigger button. The trigger button, for example, may be located at a handle assembly or module and is used by a user to control application of RF and/or ultrasonic energy. In both graphs **5020** and **5030**, the horizontal axis represents output of a displacement sensor, for example a Hall sensor. As shown in the first graph **5020**, the sensor output is roughly proportional to the button displacement. A first zone **5022** may be an "OFF" zone, where button displacement is small and no energy is applied. A second zone **5024** may be an "ON" zone or normal energy zone, where button displacement is medium and normal energy is applied. A third zone **5026** may be a "HIGH" zone of increased energy zone, where button displacement is large and energy with high intensity is applied. Alternatively, the third zone **5026** may be a "HYBRID" zone, where both RF energy and ultrasonic energy are applied. Therefore, more than one active position of the trigger button is provided. In other aspects of the present disclosure, detents instead of continuous travel may be provided for the trigger button. Alternatively, as shown in the second graph **5030**, a three dimensional touch force output level may be provided, where an off zone **5032**, a normal energy zone **5034** and a high energy zone **5036** are provided.

Other aspects of the present disclosure include motor control algorithms used to create tactile feedback (not induced by the normal operation of the mechanism) to indicate to the user the status or limits of a powered

actuation. For instance, feedback to surgeon of status of mechanisms or completion of operations is provided. For example, vibration of the device is provided when the energy deliver is finished, or when the impedance curve and force output indicate completion the device is shaken. Or each of the drive systems may have a predefined off center or impact aspect to the very beginning of the strokes allowing the system to repeat moves in opposed directions, and allowing it to create a bigger vibration than the vibration the system can generate within it normal function. In one aspect, there is a specific off-center load mechanism with a dedicated clutch which allows the device to uncouple from the primary mechanism and couple to this shaker element, allowing it to have a vastly amplified and selectable vibration feedback. There are numerous different shake profiles and intensities which can be used independently or in concert with one another to create differing feedback patterns. The tactile feedback can indicate such things as end of articulation stroke, full clamping is achieved, end of coagulation, error messages such as, but not limited to, the need to reposition the device on tissue or indications that no further energy should be applied to the tissue. For instance the indication of the end of articulation can occur when the user is asking the system to articulate more in a specific direction but the system is at it full limit and is incapable of more articulation in that direction. In this case the motor moves forward and backward fractions of an inch repeatedly shaking the device, thus indicating the device cannot move while not effectively changing the motion of the joint.

Indicating that the user should not apply more energy to the tissue can be done when the device senses that the impedance of the tissue indicates that no more energy should be applied or when the tissue is too thick for energy application or even when an error condition exists such as when RF energy is requested from a modular shaft that only has capabilities for ultrasonic energy activation or when a shaft module is attached to a transducer and the two have mismatched frequencies. Additional examples of this can be found in U.S. Pat. No. 9,364,279, which is herein incorporated by reference in its entirety.

In one aspect, the present disclosure provides a surgical instrument, comprising: a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft; a handle assembly coupled to a proximal end of the shaft; a battery assembly coupled to the handle assembly; a first driver configured to drive movement of at least a first component of the surgical instrument; a second driver configured to drive movement of at least a second component of the surgical instrument; a motor powered by the battery assembly and configured to selectively actuate the first driver or the second driver using an output of the motor; and a controller for the motor, configured to: perform a first limiting operation to the output of the motor when the motor actuates the first driver and a first limiting condition is met, and perform a second limiting operation to the output of the motor when the motor actuates the second driver and a second limiting condition is met.

The end effector may comprise a pair of jaws configured for pivotal movement between a closed position and an open position; and the first driver is configured to drive the pivotal movement. The surgical instrument may further comprise a transducer configured to output an ultrasonic energy, wherein: the second driver is configured to apply a torque to the transducer. The second limiting condition may be met when the torque applied to the transducer reaches a maximum torque.

The shaft may comprise an articulation joint, a first portion of the shaft comprising the proximal end, and a second portion of the shaft comprising the distal end, the first and second portions being coupled about the articulation joint; the second portion of the shaft is configured for articulation movement about the articulation joint; and the first driver is configured to drive the articulation movement.

The second driver may be configured to drive axial rotation the second portion. The surgical instrument may further comprise a blade, wherein: the first or second driver is configured to advance the blade. The surgical instrument may further comprise a third driver configured to drive movement of at least a third component of the surgical instrument, wherein: the motor is configured to selectively actuate the first driver, the second driver or the third driver using the output; and the controller for the motor is further configured to perform a third limiting operation to the output of the motor when the motor actuates the third driver and a third limiting condition is met.

The first limiting condition may be met when the output of the motor reaches a first maximum velocity; and the second limiting condition may be met when the output of the motor reaches a second maximum velocity. The first limiting condition may be met when a current of the motor reaches a first maximum current; and the second limiting condition may be met when the current of the motor reaches a second maximum current. The first limiting operation may control the current of the motor to be substantially zero. The second limiting operation may control the current of the motor to substantially equal to the second maximum current.

The first limiting condition may define a first maximum velocity profile of the output of the motor over a range of the movement of the first component; and the second limiting condition defines a second maximum velocity profile of the output of the motor over a range of the movement of the second component.

The first maximum velocity profile may comprise a first limited velocity region near one end of the range of the movement of the first component, a second limited velocity region near the other end of the range of the movement of the first component, and a first full velocity region between the first and second limited velocity regions; and the second maximum velocity profile comprises a third limited velocity region near one end of the range of the movement of the second component, a fourth limited velocity region near the other end of the range of the movement of the second component, and a second full velocity region between the third and fourth limited velocity regions.

In another aspect, the present disclosure provides, a method for operating a surgical instrument, the surgical instrument comprising a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft, and a handle assembly coupled to a proximal end of the shaft, the method comprising: selectively actuating a first driver or a second driver using an output of a battery-powered motor; driving, with the first driver, movement of at least a first component of the surgical instrument when the motor actuates the first driver; performing a first limiting operation to the output of the motor when the motor actuates the first driver and a first limiting condition is met; driving, with the second driver, movement of at least a second component of the surgical instrument when the motor actuates second driver; and performing a second limiting operation to the output of the motor when the motor actuates the second driver and a second limiting condition is met.

The driving movement of at least a first component of the surgical instrument may comprise driving a pivotal movement of a pair of jaws of the end effector between a closed position and an open position. The driving movement of at least a second component of the surgical instrument may comprise applying a torque to a transducer configured to output an ultrasonic energy. The first limiting condition is met when a current of the motor may reach a first maximum current; and the second limiting condition may be met when the current of the motor reaches a second maximum current.

In another aspect, the present disclosure provides a surgical instrument, comprising: a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft; a handle assembly coupled to a proximal end of the shaft, the handle assembly comprising a button configured to be displaced when pressed by a user; a battery assembly coupled to the handle assembly; a driver configured to drive movement of at least one component of the surgical instrument; a motor powered by the battery assembly and configured to actuate the driver using an output of the motor; and a controller for translating displacement of the button to a velocity of the motor, the controller being configured to: detect a movement of the button having frequency above a frequency threshold and magnitude below a magnitude threshold; and perform a smoothing operation on the movement when translating the displacement to the velocity of the motor. The end effector may comprise a pair of jaws configured for pivotal movement between a closed position and an open position; and the driver is configured to drive the pivotal movement.

In another aspect, the present disclosure provides a surgical instrument, comprising: a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft; a handle assembly coupled to a proximal end of the shaft; a battery assembly coupled to the handle assembly; a driver configured to drive movement of at least one component of the surgical instrument; a motor powered by the battery assembly and configured to actuate the driver using an output of the motor; and a controller for the motor, configured to perform a first limiting operation to the output of the motor when a first limiting condition is met, wherein the first limiting condition defines a maximum control variable profile of the motor over a range of the movement of the at least one component.

The maximum control variable profile may be a non-constant control variable profile. The maximum control variable profile may be a non-linear control variable profile. The first limiting operation may comprise decreasing a velocity of the motor.

The controller may be further configured to perform a second limiting operation to the output of the motor when a second limiting condition is met, wherein the second limiting condition defines a minimum control variable profile of the motor over the range of the movement of the at least one component. The second limiting operation may comprise increasing a velocity of the motor.

The controller may be further configured to: apply a first velocity profile of the motor that decreases over time when a rate of increase of the current of the motor is above a first threshold, and apply a second velocity profile of the motor that increases over time when a rate of decrease of the current of the motor is above a second threshold.

The end effector may comprise a pair of jaws configured for pivotal movement between a closed position and an open position; and the driver is configured to drive the pivotal movement. The surgical instrument may further comprise a transducer configured to output an ultrasonic energy, wherein: the driver is configured to apply a torque to the transducer. The surgical instrument may further comprise a blade, wherein: the driver is configured to advance the blade. The control variable profile of the motor may be any one of a current profile of the motor or a voltage profile of the motor.

In another aspect the present disclosure provides, a surgical instrument, comprising: a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft; a handle assembly coupled to a proximal end of the shaft; a battery assembly coupled to the handle assembly; a driver configured to drive movement of at least one component of the surgical instrument; a motor powered by the battery assembly and configured to actuate the driver using an output of the motor; and a controller for the motor, configured to perform a limiting operation to the output of the motor when a limiting condition is met, wherein the limiting condition defines a maximum velocity profile of the motor over a range of the movement of the at least one component.

The maximum velocity profile may be a non-constant velocity profile. The maximum velocity profile may be a non-linear velocity profile. The maximum velocity profile may comprise a first limited velocity zone near one end of a range of movement of the at least one component, a second limited velocity zone near the other end of the range of movement of the at least one component, and a full velocity zone between the first and second limited velocity zones.

The end effector may comprise a pair of jaws configured for pivotal movement between a closed position and an open position; and the driver is configured to drive the pivotal movement. The surgical instrument may further comprise a transducer configured to output an ultrasonic energy, wherein: the driver is configured to apply a torque to the transducer. The surgical instrument may further comprise a blade, wherein: the driver is configured to advance the blade.

In another aspect, the present disclosure provides a method for operating a surgical instrument, the surgical instrument comprising a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft, and a handle assembly coupled to a proximal end of the shaft, the method comprising: actuating a driver using an output of a battery-powered motor; driving, with the driver, movement of at least one component of the surgical instrument; and performing a first limiting operation to the output of the motor when a first limiting condition is met, wherein the first limiting condition defines a maximum control variable profile of the motor over a range of the movement of the at least one component. The first limiting operation may comprise decreasing a velocity of the motor. The method may further comprise performing a second limiting operation to the output of the motor when a second limiting condition is met, wherein the second limiting condition defines a minimum control variable profile of the motor over the range of the movement of the at least one component. The control variable profile of the motor may be any one of a current profile of the motor or a voltage profile of the motor.

In one aspect, the present disclosure provides a surgical instrument, comprising: a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft; a handle assembly coupled to a proximal end of the shaft; a battery assembly coupled to the handle assembly; a driver configured to drive movement of at least one component of the surgical instrument; a motor powered by the battery assembly and configured to actuate the driver using an output of the motor; and a controller for the motor, configured to perform a limiting operation to the output of the motor when a limiting condition is met, wherein the limiting condition is determined based at least in part on a tissue characteristic.

The end effector may comprise at least two sensors configured to detect tissue presence, and the tissue characteristic is an amount of tissue detected by the at least two sensors. The tissue characteristic may be tissue thickness. The limiting condition may be a maximum velocity of the motor determined based at least in part on the tissue thickness.

The controller may be further configured to apply a delay between when the driver is first actuated and when the driver is actuated again, wherein a length of the delay is determined based at least in part on the tissue characteristic. The end effector may comprise a pair of jaws configured for pivotal movement between a closed position and an open position; and the driver is configured to drive the pivotal movement.

The surgical instrument may further comprise a transducer configured to output an ultrasonic energy, wherein: the driver is configured to apply a torque to the transducer. The surgical instrument may further comprise a blade, wherein: the driver is configured to advance the blade.

In another aspect, the present disclosure provides a method for operating a surgical instrument, the surgical instrument comprising a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft, and a handle assembly coupled to a proximal end of the shaft, the method comprising: actuating a driver using an output of a battery-powered motor; driving, with the driver, movement of at least one component of the surgical instrument; determining, based at least in part on a tissue characteristic, a limiting condition; and performing a limiting operation to the output of the motor when the limiting condition is met.

The tissue characteristic may be tissue thickness. The limiting condition may be a maximum velocity of the motor determined based at least in part on the tissue thickness. The method may further comprise applying a delay between when the driver is first actuated and when the driver is actuated again, wherein a length of the delay is determined based at least in part on the tissue characteristic. The driving movement of at least one component of the surgical instrument may comprise advancing a blade of the surgical instrument.

In another aspect, the present disclosure provides, a surgical instrument, comprising: a shaft assembly comprising a shaft and an end effector coupled to a distal end of the shaft; a handle assembly coupled to a proximal end of the shaft; a battery assembly coupled to the handle assembly; a driver configured to drive movement of at least one component of the surgical instrument; a motor powered by the battery assembly and configured to actuate the driver using an output of the motor; and a controller for the motor, configured to provide a tactile feedback to a user with the at least one component.

The at least one component may be dedicated for providing the tactile feedback. The at least one component may be configured to handle or treat a tissue.

The controller may be configured to provide the tactile feedback when a desired operation of the surgical instrument is competed. The controller may be configured to provide the tactile feedback when an error has occurred. The controller may be configured to provide the tactile feedback when the surgical instrument cannot provide a desired type of energy. The controller may be configured to provide the tactile feedback when a transducer configured to output ultrasonic energy is coupled to a transducer driver having a mismatched frequency.

Modular Battery Powered Handheld Surgical Instrument with Multi-Function Motor Via Shifting Gear Assembly

In another aspect, the present disclosure provides a modular battery powered handheld surgical instrument with multi-function motor via shifting gear assembly. Disclosed is a modular handheld surgical instrument that includes a modular handle assembly, a modular ultrasonic transducer assembly and a modular shaft assembly is disclosed. The modular handle assembly includes a shift gear translatable between a first position and a second positon, wherein when the shift gear is translated to the first position, the surgical instrument is configured to rotate a drive gear of a first shaft of the modular shaft assembly to perform at least one shaft actuation function and when the shift gear is translated to the second position, the surgical instrument is configured to rotate a torque gear of a second shaft of the modular shaft assembly to controllably torque an ultrasonic waveguide of the modular shaft assembly to an ultrasonic transducer of the modular ultrasonic transducer assembly to a predefined torque.

Before discussing aspects of the present disclosure in detail, a modular handheld surgical instrument is generally disclosed as using a single motor to drive not only multiple independent shaft functions (e.g., shaft rotation, jaw closure, etc.) but also torque limited attachment of an ultrasonic waveguide to an ultrasonic transducer, wherein electronic clutching/shifting may be used to switch between functions. The modular handheld surgical instrument may provide torque limited motor driven attachment of the ultrasonic waveguide via the same motor in the handle that controls shaft actuation of clamping, rotation and/or articulation. A shifting assembly in the handle may move a gear from a primary drive assembly to a spur gear to allow the motor to rotate the nozzle/shaft/waveguide. The shifting assembly may also activate the motor inducing the handle to torque the waveguide into place with a pre-desired minimum torque. For instance, the handle housing may have a transducer torqueing mechanism which shifts the motor longitudinally uncoupling the primary drive shaft spur gear and coupling the transducer torqueing gear which rotates the shaft and nozzle therefore screwing the ultrasonic waveguide into/onto the ultrasonic transducer.

Turning now to FIGS. **114-116** there is shown a sequence of events for attaching and torqueing an ultrasonic waveguide in a battery powered modular handheld surgical instrument according to an aspect of the present disclosure. With reference to FIGS. **114-116**, a system **3000** includes a modular handle assembly housing **3002** comprising a motor assembly **3620** (e.g., FIG. **125**) and a shifting assembly **3022**. The motor assembly **3620** comprises a motor **3004** and a shaft **3006** to rotate a primary gear **3008** (e.g., motor gear). The shaft **3006** of the motor **3004** extends through a first aperture defined in a distal portion of the modular handle assembly housing **3002**. The shifting assembly **3022** comprises a shifting bar **3024**, a shaft **3026**, a spring **3028**, and a shifting spur gear **3012**. The shaft **3026** of the shifting assembly **3022** extends through a second aperture defined in the distal portion of the modular handle assembly housing **3002** such that the shaft **3026** of the shifting assembly **3022** is positioned substantially parallel to the shaft **3006** of the motor **3004**. Teeth **3010** of the primary gear **3008** are meshingly engageable with teeth **3014** of the shifting spur gear **3012** to realize multiple functions of the system **3000**. System functions include shaft actuation functions such as clamping, rotation, and articulation, via a primary drive gear **3016**, and coupling functions, wherein an ultrasonic waveguide **3032** of a modular shaft assembly **3030** being attached to a modular handle assembly **3060** is coupleable, via a slip clutch gear **3020** (e.g., torque gear), to an ultrasonic transducer shaft **3044** of a modular ultrasonic transducer assembly **3040**. In one particular aspect of the present disclosure, throughout the coupling function, torque applied by the motor **3004** is limited.

FIG. **114** shows a stage in the sequence of events for attaching and torqueing an ultrasonic waveguide in the system **3000**. FIG. **114** is an elevation view of the system **3000** wherein the modular ultrasonic transducer assembly **3040** is attached to the modular handle assembly **3060**. FIG. **114** shows a modular ultrasonic transducer assembly housing **3042** installed against the modular handle assembly housing **3002**. In one example aspect of the present disclosure, the modular ultrasonic transducer assembly **3040** comprises a transducer subassembly **3050** including an ultrasonic transducer **3054** that defines a cylindrical distal portion, having teeth **3046** arranged radially about its external surface, and a ultrasonic transducer shaft **3044** that extends distally through an aperture defined in a distal portion of the modular ultrasonic transducer assembly housing **3042**. The distal end of the ultrasonic transducer shaft **3044** includes an externally threaded portion **3048**. The threaded portion **3048** extends proximally from the distal end toward the transducer **3054** a specified length. Further in view of FIG. **114**, a modular shaft assembly **3030**, comprising a cylindrical ultrasonic waveguide **3032**, is shown positioned for attachment to the modular handle assembly **3060** and the modular ultrasonic transducer assembly **3040**.

As shown in FIG. **114**, the ultrasonic waveguide **3032** extends proximally from the modular shaft assembly **3030** a specified length. In the example aspect of the present disclosure, the proximal end of the ultrasonic waveguide **3032** comprises a chamfer **3034** as well as internal threads **3106** (See FIG. **115**) configured to receive the threaded portion **3048** of the ultrasonic transducer shaft **3044**. Notably, as shown in the example aspect of FIG. **114**, when the modular shaft assembly **3030** is positioned for attachment to the modular handle assembly **3060** and the modular ultrasonic transducer assembly **3040**, a sloped face **3036** of the chamfer **3034** is positioned in interfering alignment with the distal end of the threaded portion **3048**. In one example aspect, the distal tip of the threaded portion **3048** may be configured to interact with the sloped face **3036** of the chamfer **3034** (e.g., threaded portion **3048** may also comprise a chamfer). In an alternative aspect of the present disclosure, the ultrasonic transducer shaft **3044** could comprise the chamfer and the internal threads while the ultrasonic waveguide **3032** comprises the external threaded portion and the optional chamfer.

FIG. **115** shows another stage in the sequence of events for attaching and torqueing an ultrasonic waveguide in the system **3000**. FIG. **115** is another elevation view of system **3000** shown in FIG. **114** wherein the transducer subassembly **3050** of the modular ultrasonic transducer assembly **3040** further comprises a spring **3102** positioned in a compressive state between a proximal end of transducer subassembly **3050** and a proximal portion of the modular ultrasonic transducer assembly housing **3042**. As shown in FIG. **114** the spring **3102** is configured to force the transducer subassembly **3050** distally within the modular ultrasonic transducer assembly housing **3042** such that the ultrasonic transducer shaft **3044** extends through the aperture defined in the distal portion of the modular ultrasonic transducer assembly housing **3042**.

Turning again to FIGS. **114-115**, as the modular shaft assembly **3030** is being attached to the modular handle assembly **3060** and the modular ultrasonic transducer assembly **3040**, prior to a coupling between the ultrasonic waveguide **3032** and the ultrasonic transducer shaft **3044**, the chamfer **3034** at the proximal end of the ultrasonic waveguide **3032** interferingly interfaces with the distal end of the threaded portion **3048** of the ultrasonic transducer shaft **3044**. In the example aspect, the sloped face **3036** of the chamfer **3034** interacts with the distal end of the threaded portion **3048** causing the transducer subassembly **3050** to translate proximally against (e.g., further compressing) the spring **3102**. When the modular shaft assembly **3030** is fully seated against and/or attached to the modular handle assembly **3060**, the ultrasonic waveguide **3032** is axially aligned with the ultrasonic transducer shaft **3044** and the spring **3102** naturally forces the transducer subassembly **3050** distally such that the distal end of the threaded portion **3048** of the ultrasonic transducer shaft **3044** interfaces with the internal threads **3106** at the proximal end of the ultrasonic waveguide **3032**. Such an interface provides multiple advantages.

For example, after the chamfer **3034** interacts with the threaded portion **3048**, tactile feedback (e.g., click and/or vibration) may be produced when the spring **3102** forces the threaded portion **3048** of the ultrasonic transducer shaft **3044** against the internal threads **3106** of the ultrasonic waveguide **3032**. This indicates to the user not only that the modular shaft assembly **3030** is fully seated against the modular handle assembly **3060** but also that the ultrasonic transducer shaft **3044** is axially aligned with the ultrasonic waveguide **3032** and that a coupling between the ultrasonic transducer shaft **3044** and the ultrasonic waveguide **3032** can commence. As another example, when the spring **3102** forces the threaded portion **3048** against the internal threads **3106** the applied pressure enables the threaded portion **3048** to screw into the internal threads **3106**. This is ideal since the threaded portion **3048** and the internal threads **3106** are concealed from view when the modular ultrasonic transducer assembly **3040** and the modular shaft assembly **3030** are attached to the modular handle assembly **3060**.

Referring to FIGS. **114-116**, the shifting assembly **3022** is translatable distally (e.g., FIG. **116**) and proximally (e.g., FIG. **115**) while the shifting spur gear **3012** is engaged with primary gear **3008**. For example, the shifting spur gear **3012** is translatable between at least a first position (e.g., FIG. **115**, **3104**) and a second position (e.g., FIG. **116**, **3202**). Furthermore, the primary gear **3008** comprises teeth **3010** of a longitudinal width to accommodate translation of the shifting spur gear **3012** between at least the first position **3104** and the second position **3202**. In an alternative aspect, the primary gear **3008** comprises teeth **3010** of a longitudinal width to accommodate translation of the shifting spur gear **3012** to a position where the teeth **3014** of the shifting spur gear **3012** are not meshingly engaged with the teeth **3018** of the primary drive gear **3016** or the teeth **3022** of the slip clutch gear **3020** (e.g., a neutral position). In yet another alternative aspect, it should be appreciated that, in lieu of the shifting assembly **3022**, the motor assembly **3620** may be configured to shift/translate (e.g., in a manner similar to the shifting assembly **3022**) distally and proximally within the modular handle assembly housing **3002** such that the primary gear **3008** engages the primary drive gear **3016** in the first position **3104** and engages the slip clutch gear **3020** in the second position **3202**.

Looking to FIG. **115**, in the first position **3104**, the teeth **3014** of the shifting spur gear **3012** meshingly engage the teeth **3018** of the primary drive gear **3016**. In the example aspect, the primary drive gear **3016**, when rotated by the motor **3004** via the primary gear **3008** and the shifting spur gear **3012**, is configured to perform shaft actuation functions such as clamping, rotation, and/or articulation. In other aspects, the primary drive gear **3016** may be configured to perform other desired shaft and/or end effector functions upon rotation. In one aspect, in view of FIG. **117**, the primary drive gear **3016** may rotate a primary rotary drive **3304** (FIG. **118**) including a first internal shaft **3306**, a cylindrical coupler **3308**, and a second internal shaft **3310**. The primary drive gear **3016** is fixedly attached to a proximal end of the first internal shaft **3306**. In such an aspect, the cylindrical coupler **3308** may be utilized to convert the rotational motion from the primary drive gear **3016** to translational motion (e.g., to realize the desired shaft/end effector actuation functions). Here, a proximal end of the cylindrical coupler **3308** may be fixedly attached to a distal end of the first internal shaft **3306**. A distal end of the cylindrical coupler **3308** may comprise internal threads **3320** to meshingly engage external threads **3322** on a proximal end of the second internal shaft **3310** to cause the second internal shaft **3310** to translate distally and/or proximally a defined distance (e.g., 2 mm) to perform one or more independent shaft actuation functions (e.g., jaw closure, articulation, etc.).

In one example aspect, in view of FIG. **117**, the second internal shaft **3310** may include a magnet **3324**. In such an aspect, a sensor **3326** (e.g., hall sensor) may be positioned externally to detect/monitor the distal and/or proximal translation of the second internal shaft **3310**. It should be appreciated that the second internal shaft **3310** of the primary rotary drive **3304** is restrictable from rotation distal of the threaded interface between the cylindrical coupler **3308** and the second internal shaft **3310** such that the second internal shaft **3310** is translatable distally and/or proximally. In one aspect, in view of FIG. **119**, an exterior shaft **3038** may be fixedly coupled to the ultrasonic waveguide **3032** via pins **3344**, **3346** that extend from the exterior shaft **3038** through longitudinal cutouts **3340**, **3342** (e.g., defined in the second internal shaft **3310**) to the ultrasonic waveguide **3032**.

Here, in light of FIGS. **115**, **119**, when the shifting spur gear **3012** is in the first position **3104** (e.g., when performing shaft actuation functions), the exterior shaft **3038** is selectively coupleable to a fixed external housing **3330** (e.g., housing of the modular shaft assembly **3030**) via an actuatable exterior shaft lock mechanism **3356**, **3358** (e.g., push button, lock pin, etc.). In such an aspect, the exterior shaft **3038** may comprise a detent **3352**, **3354** to accommodate the exterior shaft lock mechanism **3356**, **3358**. For example, the detent **3352**, **3354** may be configured to receive a tip of the exterior shaft lock mechanism **3356**, **3358** to selectively restrict the exterior shaft **3038** from rotating.

FIG. **119** shows the exterior shaft lock mechanism **3356** in an actuated state. When the exterior shaft **3038** is restricted from rotation by the exterior shaft lock mechanism **3356**, **3358** and detent **3352**, **3354**, the second internal shaft **3310** is also restricted from rotation via the pins **3344**, **3346**. Notably, in such an aspect, although the second internal shaft **3310** is restricted from rotation, the second internal shaft **3310** remains translatable distally and/or proximally (e.g., pins **3344**, **3346** permit distal and/or proximal translation via the longitudinal cutouts **3340**, **3342**) to perform the desired shaft actuation functions. In various aspects, the longitudinal cutouts **3340**, **3342** and the pins **3344**, **3346** may be positioned at or about an antinode of the ultrasonic waveguide **3032**.

Furthermore, it should be appreciated that distal versus proximal translation of the second internal shaft **3310** is a function of the motor **3004** rotation direction (e.g., clockwise versus counter-clockwise) and thread direction (e.g., right-handed threads versus left-handed threads) at the threaded interface between the cylindrical coupler **3308** and the second internal shaft **3310**. Further, in an alternative aspect, it should be appreciated that if rotational motion is desired to perform shaft actuation functions, the primary rotary drive **3304** may simply comprise the first internal shaft **3306**. In such an aspect, the motor **3004** would rotate the primary drive gear **3016** to rotate the first internal shaft **3306** to perform desired shaft actuation functions using clockwise and counter-clockwise rotation directions. In various aspects, the primary rotary drive **3304** (e.g., the first internal shaft **3306**, the cylindrical coupler **3308**, second internal shaft **3310**) is supported between the ultrasonic waveguide **3032** and/or the exterior shaft **3038** (e.g., via bearings, lubricious seals/o-rings/spacers, combinations thereof, etc.).

FIG. **116** shows another stage in the sequence of events for attaching and torqueing an ultrasonic waveguide in the system **100**. Turning to FIG. **116**, in the second position **3202**, the teeth **3014** of the shifting spur gear **3012** meshingly engage the teeth **3022** of the slip clutch gear **3020**. Notably, in light of FIG. **119**, when the shifting spur gear **3012** is in the second position **3202**, the exterior shaft **3038** is selectively uncoupleable from the fixed external housing **3330** (e.g., housing of the modular shaft assembly **3030**) via the exterior shaft lock mechanism **3356**, **3358**. FIG. **119** shows the exterior shaft lock mechanism **3358** in an unactuated state. In such an aspect, when the exterior shaft lock mechanism **3356**, **3358** is unactuated, the tip of the exterior shaft lock mechanism is withdrawn from the detent **3352**, **3354** in the exterior shaft **3038** to permit the exterior shaft **3038** to rotate.

As such, in the example aspect, the slip clutch gear **3020** when rotated by the motor **3004** via the primary gear **3008** and the shifting spur gear **3012**, is configured to rotate the exterior shaft **3038** and the ultrasonic waveguide **3032** coupled thereto (FIG. **119**, e.g., the exterior shaft **3038** may be coupled to the ultrasonic waveguide **3032** via the pins **3344**, **3346** that extend from the exterior shaft **3038** to the ultrasonic waveguide **3032**). In such an aspect, rotation of the exterior shaft **3038** also rotates the ultrasonic waveguide **3032** causing the internal threads **3106** (FIG. **115**) of the ultrasonic waveguide **3032** to screw onto the threaded portion **3048** of the ultrasonic transducer shaft **3044**. In light of FIGS. **115-116**, the transducer subassembly **3050** translates distally as the ultrasonic waveguide **3032** screws onto the ultrasonic transducer shaft **3044**. In such an aspect, although the second internal shaft **3310** will also rotate (e.g., via pins **3344**, **3346**), the primary drive gear **3016** is free to rotate since the shifting spur gear **3012** is at the second position **3202**.

In various aspects, the ultrasonic transducer shaft **3044**, at a proximal end of threaded portion **3048** may comprise a receiving face **3056** (FIG. **114**) configured and/or shaped to receive the sloped face **3036** of the chamfer **3034**. When the ultrasonic waveguide **3032** is fully screwed onto the threaded portion **3048**, the receiving face **3056** makes surface contact with the sloped face **3036** of the chamfer **3034** (FIG. **116**). As further described below, the motor **3004** may be torque limited when the ultrasonic waveguide **3032** is screwed onto the threaded portion **3048** to control the torque experienced at the interface between receiving face **3056** and sloped face **3036**.

In one alternative aspect of the present disclosure, in lieu of screwing the ultrasonic waveguide **3032** onto/into the ultrasonic transducer shaft **3044** as described herein, the ultrasonic transducer shaft **3044** may be screwed onto/into the ultrasonic waveguide **3032**. In view of FIGS. **10A-10B**, a shifting worm gear **206** may rotatably interface with teeth **202** of the ultrasonic transducer **130** to advance/screw/couple a threaded portion of the ultrasonic transducer shaft **156** into the ultrasonic waveguide **145**. It should be appreciated that the shifting worm gear **206** may also rotatably interface with the teeth **202** of the ultrasonic transducer **130** to withdraw/unscrew/uncouple the threaded portion of the ultrasonic transducer shaft **156** from the ultrasonic waveguide **145**.

FIGS. **120-121** depict a torque lever for use in a device according to an aspect of the present disclosure. As previously discussed, the shifting assembly **3022**, comprising the shifting bar **3024**, the shaft **3026**, the spring **3028**, and the shifting spur gear **3012**, is translatable distally and proximally within the modular handle assembly **3060**. In FIG. **120**, a top elevation view of the system **3000**, the shifting assembly **3022** further includes a torque lever **3402** and a shift linkage **3414**. The torque lever **3402** includes a first flange **3406** and a second flange **3408** and is pivotable about a lever pivot **3404**. The first flange **3406** protrudes in a direction transverse to the lever pivot **3404** and may extend at least partially outside the modular handle assembly housing **3002**. Notably, in the example aspect, the first flange **3406** is located for manual actuation by a user between a first point **3410** and a second point **3412**. In an alternative aspect, an automated mechanism (e.g., servo motor, user accessible actuation button, etc.) may be configured to actuate the first flange **3406** between the first point **3410** and the second point **3412** (e.g., inside the modular handle assembly housing **3002**). Furthermore, the first flange **3406** may comprise ribs **3434** for a user to grip and pivot the torque lever **3402** (FIG. **121**). The second flange **3408** protrudes in a direction transverse to the lever pivot **3404** and is coupled to the shifting bar **3024** and the shift linkage **3414** at a first shift pin **3416**.

In light of FIGS. **115**, **116**, **120**, and **122**, the shifting bar **3024**, in response to movement of the first flange **3406** of the torque lever **3402** between the first point **3410** toward the second point **3412**, is operable to compress the spring **3028**, positioned about the shaft **3026** between a portion of the shifting bar **3024** and a distal portion of the modular handle assembly housing **3002**, to translate the shifting spur gear **3012** (e.g., while engaged with the primary gear **3008**) from a first position **3104** (e.g., where the shifting spur gear **3012** is engaged with the primary drive gear **3016**) to a second position **3202** (e.g., where the shifting spur gear **3012** is engaged with the slip clutch gear **3020**). Notably, in light of FIG. **122**, the shifting assembly **3022** may be configured to naturally lock (e.g., to resist the force of the spring **3028**) when the first flange **3406** of the torque lever **3402** has been pivoted to the second point **3412**. In an alternative aspect, the shifting assembly **3022** may utilize a separate lock (not shown) to maintain the shifting spur gear **3012** in the second position **3202**. In one aspect, an alternative mechanism to force the shifting assembly **3022** proximally (e.g., spring coupled/attached to shifting assembly **3022** in an alternative manner) may be utilized. In light of FIGS. **120** and **122**, it should be appreciated that, in response to movement of first flange **3406** of the torque lever **3402** between the second point **3412** toward the first point **3410**, the shifting bar **3024** is operable to release compression of the spring **3028** to translate the shifting spur gear **3012** from the second position **3202** to the first position **3104**.

Furthermore, in the example aspect of FIG. **122**, a distal end of the shaft **3026** may be configured to contact/actuate a first shift switch **3504** during distal translation of the shifting assembly **3022**. The first shift switch **3504** may be normally in an unactuated "off" position. When the first flange **3406** of the torque lever **3402** is fully pivoted to the second point **3412**, the distal end of shaft **3026** may interferingly actuate the first shift switch **3504** to an "on" position (e.g., FIG. **116**). In the example aspect, the first shift switch **3504** comprises leads electrically coupled to the motor control circuit **726** (FIG. **36**) and/or the motor drive circuit **1165** (FIG. **50**). The motor control circuit **726** and/or the motor drive circuit **1165** may be configured to automatically turn the motor **3004** "on" when the first shift switch **3504** is actuated. Notably, when the first flange **3406** of the torque lever **3402** is pivoted to the second point **3412**, the shifting spur gear **3012** is engaged with the slip clutch gear **3020** (e.g., FIG. **116**). As such, when the first shift switch **3504** is actuated, the motor **3004** may be configured to automatically couple or begin coupling the ultrasonic waveguide **3032** to the ultrasonic transducer shaft **3044**. In one alternative aspect, when the first shift switch **3504** is actuated, a separate control on the modular handle assembly **3060** may be actuatable to couple or begin coupling the ultrasonic waveguide **3032** to the ultrasonic transducer shaft **3044**. In such an aspect, the motor control circuit **726** and/or the motor drive circuit **1165** is configured to turn the motor **3004** "on" when the separate control on the modular handle assembly **3060** is actuated.

In an alternative aspect, the motor control circuit **726** (FIG. **36**) and/or the motor drive circuit **1165** (FIG. **50**) may be configured to disable operation of the motor **3004** until the first shift switch **3504** or a second shift switch (not shown) is actuated. In such an aspect, the second shift switch may be positioned for actuation when the first flange **3406** of the torque lever **3402** is fully pivoted to the first point **3410** (e.g., when the shifting spur gear **3012** is engaged with the primary drive gear **3016**). The second shift switch may comprise leads electrically coupled to the motor control circuit **726** and/or the motor drive circuit **1165**. In such an aspect, the motor control circuit **726** and/or the motor drive circuit **1165** may be operable, via the first shift switch **3504** and the second shift switch, to disable the motor **3004** until one of the first shift switch **3504** or the second shift switch is actuated. A separate control on the modular handle assembly **3060** may be actuatable to selectively operate the motor **3004** when one of the first shift switch **3504** or the second shift switch is actuated. This advantageously prevents unintentional actuation of the motor **3004** as well as any potential for grinding between the shifting spur gear **3012** and the primary drive gear **3016** and/or the slip clutch gear **3020** as the shifting spur gear **3012** translates from the first position **3104** to the second position **3202**.

Further in view of FIGS. **77** and **79**, the shift linkage **3414**, in response to movement of the first flange **3406** of the torque lever **3402** between the first point **3410** and the second point **3412**, is configured to operate a transducer locking mechanism **3418** via a second shift pin **3420**. The transducer locking mechanism **3418** includes a lock bar **3422** and a pin **3424**. The pin **3424** is configured to travel within a slot **3426**, which may be defined in the modular handle assembly housing **3002**, between an unlocked position **3428** (e.g., FIG. **120**) and a locked position **3502** (e.g., FIG. **122**). In the example aspect, when the first flange **3406** of the torque lever **3402** is pivoted to the first point **3410**, the pin **3424** is in the unlocked position **3428** and when the first flange **3406** of the torque lever **3402** is pivoted to the second point **3412**, the pin **3424** is in the locked position **3502**.

Further in view of FIG. **120**, a first end of the lock bar **3422** includes the pin **3424** and a keyed surface **3430**. The keyed surface **3430** is configured to lock the ultrasonic transducer **3054**. A second end of the lock bar **3422** is coupled to the shift linkage **3414** at the second shift pin **3420** so that rotation of the torque lever **3402** about the lever pivot **3404** operates the transducer locking mechanism **3418**. In one various aspect, the shift linkage **3414** is shaped to accommodate the torque lever **3402** as it pivots about the lever pivot **3404**. For example, in view of FIG. **120**, the shift linkage **3414** comprises an arcuate shape to accommodate a round body of the torque lever **3402** to conserve space within the modular handle assembly housing **3002**.

Further in view of FIG. **120**, when the pin **3424** of the transducer locking mechanism **3418** is in the unlocked position **3428**, the keyed surface **3430** of the lock bar **3422** is disengaged from the ultrasonic transducer **3054**. In view of FIG. **122**, as the first flange **3406** of the torque lever **3402** is pivoted from the first point **3410** to the second point **3412**, the pin **3424** travels within the slot **3426** from the unlocked position **3428** toward the locked position **3502**. When the pin **3424** is in the locked position **3502**, the keyed surface **3430** engages the ultrasonic transducer **3054** to control movement of the ultrasonic transducer **3054** (e.g., via teeth **3046**) and the ultrasonic transducer shaft **3044**. In one aspect, the keyed surface **3430** may be configured to restrict rotation while permitting proximal and distal translation of the transducer subassembly **3050**. In such an aspect, the teeth **3046** of the ultrasonic transducer **3054** comprise a longitudinal width to accommodate translation of the transducer subassembly **3050** (FIGS. **114-116**, e.g., as the ultrasonic waveguide **3032** is screwed onto/into the ultrasonic transducer shaft **3044**). In one example aspect, the keyed surface **3430** comprises teeth **3432** (FIG. **121**) configured to meshingly engage the teeth **3046** of the ultrasonic transducer **3054**. Notably, locking the ultrasonic transducer **3054** from rotation, via the transducer locking mechanism **3418**, permits the system **3000** to limit torque applied by the motor **3004** when coupling the ultrasonic waveguide **3032** and ultrasonic transducer shaft **3044**.

The above aspects describe numerous structures to couple the ultrasonic waveguide **3032** of a modular shaft assembly **3030** and the ultrasonic transducer shaft **3044** of a modular ultrasonic transducer assembly **3040**. The above aspects also describe numerous structures not only to torque the ultrasonic waveguide **3032** to the ultrasonic transducer shaft **3044** but also to limit that torque in a controlled manner. In particular, when a modular ultrasonic transducer assembly **3040** and a modular shaft assembly **3030** are attached to a modular handle assembly **3060** such that the ultrasonic waveguide **3032** of the modular shaft assembly **3030** is axially aligned with the ultrasonic transducer shaft **3044** of the modular ultrasonic transducer assembly (e.g., described above, FIG. **114**), the first flange **3406** of the torque lever **3402** may be pivoted from the first point **3410** toward the second point **3412** (e.g., described above, FIGS. **120**, **129**). As the first flange **3406** of the torque lever **3402** approaches the second point **3412** movement of the ultrasonic transducer **3054** and the ultrasonic transducer shaft **3044** is controlled via the transducer locking mechanism **3418** (e.g., rotation restricted while translation permitted).

Simultaneously, while the first flange **3406** of the torque lever **3402** is pivoted from the first point **3410** toward the second point **3412**, the shaft **3026** and the shifting spur gear

3012 are translated distally so that the shifting spur gear **3012** can engage slip clutch gear **3020** and the distal end of the shaft **3026** can actuate the first shift switch **3504**. When the first flange **3406** of torque lever **3402** reaches the second point **3412**, the shifting spur gear **3012** engages the slip clutch gear **3020** and the distal end of the shaft **3026** actuates the first shift switch **3504**. Upon actuation of the first shift switch **3504**, the motor **3004** may rotate the slip clutch gear **3020** via the shifting spur gear **3012** to rotate the exterior shaft **3038** coupled to the ultrasonic waveguide **3032**. Since the ultrasonic transducer **3054** is in a locked state (e.g., rotation prevented), rotation of the exterior shaft **3038** rotates the ultrasonic waveguide **3032** relative to the ultrasonic transducer shaft **3044** such that ultrasonic waveguide **3032** screws onto the ultrasonic transducer shaft **3044** thereby coupling the ultrasonic waveguide **3032** and the ultrasonic transducer shaft **3044**. Such a coupling occurs internal to the device since the modular shaft assembly **3030** and the modular ultrasonic transducer assembly **3040** are attached to the modular handle assembly **3060**.

Notably, in the above described aspects, the motor control circuit **726** (FIG. **36**) and/or the motor drive circuit **1165** (FIG. **50**) may be configured to attach/couple the ultrasonic waveguide **3032** and the ultrasonic transducer shaft **3044** to a pre-desired torque. Such a pre-desired torque may be a minimum torque adequate for the ultrasonic waveguide **3032** and the ultrasonic transducer shaft **3044** to remain attached while the ultrasonic transducer **3054** excites the ultrasonic waveguide **3032** during operation of the device. Such a pre-desired torque would avoid damaging the ultrasonic transducer shaft **3044** and/or the ultrasonic waveguide **3032** (e.g., the threads, the chamfer, the sloped face, the receiving face, etc.). The motor control circuit **726** and/or the motor drive circuit **1165** may be configured to stop the motor **3004** from rotating the slip clutch gear **3020** (e.g., once the pre-desired torque is realized) despite the first shift switch **3504** being in an actuated state.

Upon a successful coupling of the ultrasonic waveguide **3032** and the ultrasonic transducer shaft **3044**, the first flange **3406** of the torque lever **3402** may be pivoted from the second point **3412** back to the first point **3410**. As the first flange **3406** of the torque lever **3402** is pivoted from the second point **3412** toward the first point **3410**, the transducer locking mechanism **3418** disengages from the ultrasonic transducer **3054** thereby releasing control of the ultrasonic transducer **3054** and the ultrasonic transducer shaft **3044**. Simultaneously, while the first flange **3406** of the torque lever **3402** is pivoted from the second point **3412** toward the first point **3410**, the shaft **3026** and the shifting spur gear **3012** are translated proximally so that the shifting spur gear **3012** can engage the primary drive gear **3016**. When the first flange **3406** of the torque lever **3402** reaches the first point **3410**, the shifting spur gear **3012** engages the primary drive gear **3016**. At the first point **3410**, the motor control circuit **726** (FIG. **36**) and/or the motor drive circuit **1165** (FIG. **50**) may be configured to rotate the primary drive gear **3016** via the shifting spur gear **3012** to perform shaft actuation functions such as clamping, rotation, and articulation (e.g., via a separate control(s) on the modular handle assembly **3060** as described herein). Notably, the primary drive gear **3016** may be configured to perform other desired shaft and/or end effector functions upon rotation (e.g., jaw closure).

It should be appreciated that the above aspects also describe numerous structures to uncouple/decouple the ultrasonic waveguide **3032** of the modular shaft assembly **3030** and the ultrasonic transducer shaft **3044** of the modular ultrasonic transducer assembly **3040**. This is particularly useful when wishing to detach the modular shaft assembly **3030** and/or modular ultrasonic transducer assembly **3040** from the modular handle assembly **3060** (e.g., to attach another/different modular shaft assembly and/or modular ultrasonic transducer assembly). For example, the first flange **3406** of the torque lever **3402** may be pivoted from the first point **3410** to the second point **3412** such that movement of the ultrasonic transducer **3054** and ultrasonic transducer shaft **3044** is controlled via the transducer locking mechanism **3418** (e.g., rotation restricted while translation permitted) and the ultrasonic waveguide **3032** is rotatable relative to the ultrasonic transducer shaft **3044**. At this point, the motor control circuit **726** (FIG. **36**) and/or the motor drive circuit **1165** (FIG. **50**) may be configured to rotate the ultrasonic waveguide **3032** in the opposite direction (e.g., counterclockwise if left-handed threads, clockwise if right-handed threads) such that the ultrasonic waveguide **3032** unscrews from the ultrasonic transducer shaft **3044**. The modular shaft assembly **3030** and the modular ultrasonic transducer assembly **3040** can then be detached from the modular handle assembly **3060**.

It should be appreciated that the above described aspects permit a single motor not only to perform conventional shaft and/or end effector functions (e.g., clamping, rotation, articulation, etc.) but also (i) to couple an ultrasonic waveguide **3032** of a modular shaft assembly **3030** to an ultrasonic transducer shaft **3044** of a modular ultrasonic transducer assembly **3040**, and (ii) to uncouple/decouple the ultrasonic waveguide **3032** of the modular shaft assembly **3030** from the ultrasonic transducer shaft **3044** of the modular ultrasonic transducer assembly **3040**. Such aspects permit a modular handheld surgical instrument to comprise numerous combinations of a modular handle assembly **3060**, a modular ultrasonic transducer assembly **3040** and a modular shaft assembly **3030**.

FIGS. **117**, **118** and **121** depict example torqueing mechanisms for use in a device according to an aspect of the present disclosure. As described above, when the shifting spur gear **3012** has been translated to the second position **3302** the teeth **3014** of the shifting spur gear **3012** meshingly engage the teeth **3022** of the slip clutch gear **3020**. Further, as described above, when the shifting spur gear **3012** is in the second position, the exterior shaft lock mechanism **3356**, **3358** may be withdrawn from the detent **3352**, **3354** in the exterior shaft **3038** (e.g., to uncouple the exterior shaft **3038** from the fixed external housing **3330**) to permit the exterior shaft **3038** to rotate.

In one example aspect, the slip clutch gear **3020**, when rotated by the motor **3004** via the primary gear **3008** and the shifting spur gear **3012**, is configured to rotate the exterior shaft **3038** (FIG. **118**) which may include a first shaft portion **3314**, a second shaft portion **3316**, and a third shaft portion **3318** (FIG. **117**). The slip clutch gear **3020** is coupled to a proximal end of the first shaft portion **3314**. An internal surface at a proximal end of the second shaft portion **3316** may be fixedly attached to an external surface at a distal end of the first shaft portion **3314** and an internal surface at a distal end of the second shaft portion **3316** may be fixedly attached to an external surface at a proximal end of the third shaft portion **3318**. In one example aspect, the second shaft portion **3316** may be sized (e.g., longitudinal length, internal diameter, etc.) to enable the first internal shaft **3306** and the cylindrical coupler **3308** of the primary rotary drive **3304** to translate the second internal shaft **3310** distally and/or proximally (e.g., FIG. **117**).

In line with above, the exterior shaft **3038** is coupled to the ultrasonic waveguide **3032** (e.g., via pins **3344**, **3346**) such that rotation of the exterior shaft **3038** rotates the ultrasonic waveguide **3032** relative to the ultrasonic transducer shaft **3044** (e.g., to couple the ultrasonic waveguide **3032** to the ultrasonic transducer shaft **3044** and/or to decouple the ultrasonic waveguide **3032** from the ultrasonic transducer shaft **3044**). In one aspect, such a coupling between the exterior shaft **3038** and the ultrasonic waveguide **3032** may occur distal of the second shaft portion **3316** of the exterior shaft **3038**. As previously indicated, the second internal shaft **3310** of the primary rotary drive **3304** may also rotate (e.g., via pins **3344**, **3346**) with the exterior shaft **3038** and the ultrasonic waveguide during the coupling and/or decoupling procedure. Notably however, since the shift spur gear **3012** is translated to the second position **3302** during the ultrasonic waveguide **3032** coupling and/or decoupling procedure, such rotation may simply result in negligible translation of the second internal shaft **3310** (e.g., translation permitted via the cutouts **3340**, **3342**) followed by free rotation of the cylindrical coupler **3308**, the first internal shaft **3306**, and the primary drive gear **3016**. Stated differently, since the shift spur gear **3012** is translated to the second position **3302**, the primary drive gear **3016** is able to rotate freely without damaging any device components.

It should be appreciated that the shift clutch gear **3020** may rotate the exterior shaft **3038** to rotate the ultrasonic waveguide **3032** with resect to ultrasonic transducer shaft **3044** via clockwise and counter-clockwise rotation directions. Further, it should be appreciated that the exterior shaft **3038** may simply comprise the first shaft portion **3314**. In such an aspect, the shift clutch gear **3020** would rotate the first shaft portion **3314**, in a similar manner as described above, to rotate the ultrasonic waveguide **3032** with respect to the ultrasonic transducer shaft **3044** via clockwise and counter-clockwise rotation directions.

FIGS. **117**, **118** and **121** depict an example mechanical aspect for controlling torque applied at the threaded interface between the ultrasonic waveguide **3032** and the ultrasonic transducer shaft **3044** during the ultrasonic waveguide **3032** coupling and/or decoupling procedure. In one example, an interface (FIG. **117-118**, **3350**, FIG. **121**, **3440**) coupling the slip clutch gear **3020** to the first shaft portion **3314** of the exterior shaft **3038** may comprise a slip clutch subsystem **3442** (FIG. **121**). In one aspect of the slip clutch subsystem **3442**, an external surface at the proximal end of the first shaft portion **3314** (e.g., that interfaces an internal surface of slip clutch gear **3020**) may comprise a plurality of teeth extending radially outward **3444** and the internal surface of the slip clutch gear **3020** (e.g., that interfaces with the external surface at the proximal end of the first shaft portion **3314**) may comprise a plurality of teeth extending radially inward **3446**.

Turning to FIG. **121**, the plurality of teeth **3444** associated with the first shaft portion **3314** may be configured (e.g., via selected slope of the teeth, diameter measured at tips of the teeth, size of the teeth, resilient material of the teeth, etc.) to interface with the plurality of teeth **3446** associated with the slip clutch gear **3020** and the plurality of teeth **3446** associated with the slip clutch gear **3020** may be configured (e.g., via selected slope of the teeth, diameter measured at tips of the teeth, size of the teeth, resilient material of the teeth, etc.) to interface with the plurality of teeth **3444** associated with the first shaft portion **3314** such that when screwing (e.g., coupling) the ultrasonic waveguide **3032** onto the ultrasonic transducer shaft **3044**, the slip clutch gear **3020** and the first shaft portion **3020** remain fixedly coupled until a predefined threshold (e.g., desired maximum) torque is reached at the threaded interface between the ultrasonic waveguide **3032** and the ultrasonic transducer shaft **3044**. Once the predefined threshold torque is reached, the slip clutch subsystem **3442** (e.g., via the teeth **3446** and the teeth **3444**) is configured to slip such that the slip clutch gear **3020** spins around the proximal end of the first shaft portion **3314** without further rotating the exterior shaft **3038** (e.g., or the ultrasonic waveguide **3032** coupled thereto). Larger torque thresholds may be achieved through teeth **3444/3446** comprising a steeper slope, a larger size, a greater coefficient of friction, and/or a larger amount of interference with opposing teeth and smaller torque thresholds may be achieved through teeth **3444/3446** comprising a smoother/shallower slope, a smaller size, a lesser coefficient of friction, and a smaller amount of interference with opposing teeth. It should be appreciated that combinations of such parameters may also be used to realize a desired maximum and/or limited torque threshold.

In another aspect, the plurality of teeth **3444** associated with the first shaft portion **3314** may be configured (e.g., via selected slope of the teeth, diameter measured at tips of the teeth, size of the teeth, resilient material of the teeth, etc.) to interface with the plurality of teeth **3446** associated with the slip clutch gear **3020** and the plurality of teeth **3446** associated with the slip clutch gear **3020** may be configured (e.g., via selected slope of the teeth, diameter measured at tips of the teeth, size of the teeth, resilient material of the teeth, etc.) to interface with the plurality of teeth **3444** associated with the first shaft portion **3314** such that when unscrewing (e.g., decoupling) the ultrasonic waveguide **3032** from the ultrasonic transducer shaft **3044**, the slip clutch gear **3020** and the first shaft portion **3020** remain fixedly coupled until a predefined threshold (e.g., desired maximum) torque is reached. In various aspects, the slip clutch subsystem **3442** may be configured to slip at a higher predefined threshold when unscrewing (e.g., decoupling) the ultrasonic waveguide **3032** from the ultrasonic transducer shaft **3044** than when screwing (e.g., coupling) the ultrasonic waveguide **3032** onto/into the ultrasonic transducer shaft **3044** (e.g., two-way slip). In other various aspects, the slip clutch subsystem **3442** may be configured to not slip when unscrewing (e.g., decoupling) the ultrasonic waveguide **3032** from the ultrasonic transducer shaft **3044** (e.g., teeth **3446** associated with the slip clutch gear **3020** and the teeth **3444** associated with the first shaft portion **3314** meshingly engage without slippage) but may be configured to slip at a predefined threshold when screwing (e.g., coupling) the ultrasonic waveguide **3032** onto/into the ultrasonic transducer shaft **3044** (e.g., one-way slip).

FIGS. **117** and **118** further represent alternative electrical aspects for controlling torque applied at the threaded interface between the ultrasonic waveguide **3032** and the ultrasonic transducer shaft **3044** during the ultrasonic waveguide **3032** coupling and/or decoupling procedure. In one aspect, the slip clutch gear **3020** is fixedly attached to first shaft portion **3314**. As such, in lieu of the slip clutch subsystem **3442** described above, the motor control circuit **726** (FIG. **36**) and/or the motor drive circuit **1165** (FIG. **50**) may be configured (e.g., an electronic current limiter) to alter the current supplied to the motor **3004** such that the torque produced at the threaded interface between the ultrasonic waveguide **3032** and the ultrasonic transducer shaft **3044** is limited to a predefined threshold (e.g., desired maximum) torque. In various aspects, the motor control circuit **726** and/or the motor drive circuit **1165** may be configured to alter the current supplied to the motor **3004** such that the torque produced at the threaded interface between the ultrasonic waveguide **3032** and the ultrasonic transducer shaft **3044** is a higher predefined threshold when unscrewing (e.g., decoupling) the ultrasonic waveguide **3032** from the ultrasonic transducer shaft **3044** than when screwing (e.g., coupling) the ultrasonic waveguide **3032** onto/into the ultrasonic transducer shaft **3044**. In other various aspects, the motor control circuit **726** and/or the motor drive circuit **1165** may be configured to deliver full current to the motor **3004** such that the torque produced at the threaded interface between the ultrasonic waveguide **3032** and the ultrasonic transducer shaft **3044** is not electrically limited when unscrewing (e.g., decoupling) the ultrasonic waveguide **3032** from the ultrasonic transducer shaft **3044** but may be configured to alter the current supplied to the motor **3004** such that the torque produced at the threaded interface between the ultrasonic waveguide **3032** and the ultrasonic transducer shaft **3044** is limited to a predefined threshold torque when screwing (e.g. coupling) the ultrasonic waveguide **3032** onto/into the ultrasonic transducer shaft **3044**.

Turning to FIGS. **123**-**125**, proper torque may be verified by a combination of motor torque measurements and an ultrasonic ping that returns a known good reflection profile. In view of FIG. **123**, the motor control circuit **726** and/or the motor drive circuit **1165** may be configured to cut off current **3602** supplied to the motor **3004** when a predefined torque limit **3605** is reached. In view of FIG. **124**, the motor control circuit **726** and/or the motor drive circuit **1165** may be configured to limit current supplied to the motor **3004** based on the function being performed. For example, torqueing the ultrasonic waveguide **3032** onto/into the ultrasonic transducer shaft **3044** (FIG. **124**, **3604**) may be limited to a first current limit **3606**, rotating the shaft and/or end effector (FIG. **124**, **3608**, e.g., distal roll) may be limited to a second current limit **3610**, articulating the shaft and/or end effector (FIG. **124**, **3612**) may be limited to a third current limit **3614**, and/or clamping the end effector (FIG. **124**, **3616**, e.g. constant force clamp arm) may be limited to a fourth current limit **3618**. FIG. **125** illustrates the motor assembly **3620** comprising the motor **3004**, the shaft **3006**, and the primary gear **3008** as previously discussed herein. In FIG. **125**, the motor assembly **3620** may further comprise a planetary gear box **3622**, a first magnet **3624** and a second magnet **3626** fixedly attached coaxially about the shaft **3006**, a wire coiled around the shaft **3628** positioned within the first and second magnets, and a strain gauge **3630** including a microchip (not shown). In one aspect, the coiled wire **3628** rotates with the shaft **3006** within the first and second magnets **3624**, **3626** to power the strain gauge **3630**. The strain gauge **3630** is configured to measure the torque applied by the motor **3004** via the planetary gear box **3622**. In one aspect, the motor control circuit **726** and/or the motor drive circuit **1165** may be configured to electronically limit the current supplied to the motor **3004** based on the torque measured via the strain gauge **3630**.

In an alternative aspect, a combination of mechanical aspects (e.g., slip clutch subsystem **3442**) and electrical aspects (e.g., altering the motor **3004** current via the motor control circuit **726** and/or the motor drive circuit **1165**) is contemplated for controlling torque applied at the threaded interface between the ultrasonic waveguide **3032** and the ultrasonic transducer shaft **3044** during the ultrasonic waveguide **3032** coupling and/or decoupling procedure.

In one aspect, the present disclosure provides a surgical instrument, comprising: a modular handle assembly, comprising: a motor assembly comprising a motor and a motor gear; and a shift gear translatable between a first position and a second position; a modular ultrasonic transducer assembly coupled to the modular handle assembly, wherein the modular ultrasonic transducer assembly comprises: an ultrasonic transducer; and an ultrasonic transducer shaft extending distally from the ultrasonic transducer; and a modular shaft assembly coupled to the modular ultrasonic transducer assembly and the modular handle assembly, wherein the modular shaft assembly comprises: a first rotatable shaft extending distally along a shaft axis, wherein the first rotatable shaft comprises a drive gear; an ultrasonic waveguide coaxially aligned with the first rotatable shaft about the shaft axis; and a second rotatable shaft coaxially aligned with the first rotatable shaft and the ultrasonic waveguide about the shaft axis, wherein the second rotatable shaft comprises a torque gear; wherein when the shift gear is translated to the first position, the motor gear is configured to rotate the drive gear via the shift gear, and wherein when the shift gear is translated to the second position, the motor gear is configured to rotate the torque gear via the shift gear. The shift gear may be manually or electronically translatable between the first position and the second position.

The drive gear upon rotation may be operable to perform at least one shaft actuation function comprising one or more than one of clamping, rotating, or articulating an actuation device. The torque gear upon rotation may be operable to couple the ultrasonic waveguide to the ultrasonic transducer shaft or decouple the ultrasonic waveguide from the ultrasonic transducer shaft. The torque gear upon rotation may be operable to controllably torque the ultrasonic waveguide to the ultrasonic transducer shaft. The torque gear may be coupled to the second rotatable shaft via a slip-clutch subsystem, wherein the slip-clutch subsystem is configured such that the torque gear and the second rotatable shaft remain fixedly coupled until a predefined torque is reached and such that the torque gear slips about the second rotatable shaft after the predefined torque is reached to controllably torque the ultrasonic waveguide to the ultrasonic transducer shaft. The slip-clutch subsystem may be further configured such that the torque gear slips about the second rotatable shaft at a higher predefined torque when decoupling the ultrasonic waveguide from the ultrasonic transducer shaft.

The surgical instrument may further comprise a motor circuit, wherein the motor circuit is configured to alter current supplied to the motor to controllably torque the ultrasonic waveguide to the ultrasonic transducer shaft. The surgical instrument may further comprise a strain gauge configured to measure torque applied by the motor, and wherein the motor circuit is configured to limit current supplied to the motor based on the torque measured by the strain gauge.

In another aspect, the present disclosure provides a modular handheld surgical instrument, comprising: a motor assembly comprising a motor and a motor gear; a shift gear translatable distally and proximally between a first position and a second position; a transducer assembly comprising an ultrasonic transducer shaft; and a shaft assembly, comprising: a first shaft extending distally along a shaft axis, wherein the first shaft comprises a drive gear; an ultrasonic waveguide coaxially aligned inside the first shaft about the shaft axis; and a second shaft coaxially aligned outside the first shaft and the ultrasonic waveguide about the shaft axis, wherein the second shaft comprises a torque gear coupled to the second shaft via a slip-clutch subsystem, wherein the slip-clutch subsystem is configured such that the torque gear does not slip about the second shaft until a predefined torque is reached to controllably torque the ultrasonic waveguide to the ultrasonic transducer shaft; wherein when the shift gear is translated to the first position, the motor gear is configured to rotate the drive gear via the shift gear, and wherein when the shift gear is translated to the second position, the motor gear is configured to rotate the torque gear via the shift gear.

The drive gear upon rotation may be operable to perform at least one shaft actuation function comprising one or more than one of clamping, rotating, or articulating an actuation device. The torque gear upon rotation may be operable to couple the ultrasonic waveguide to the ultrasonic transducer shaft or decouple the ultrasonic waveguide from the ultrasonic transducer shaft. The slip-clutch subsystem may be further configured such that the torque gear slips about the second shaft at a higher predefined torque when decoupling the ultrasonic waveguide from the ultrasonic transducer shaft. The slip-clutch subsystem may be further configured such that the torque gear does not slip about the second shaft when decoupling the ultrasonic waveguide from the ultrasonic transducer shaft. The shift gear may be manually or electronically translatable between the first position and the second position.

In another aspect, the present disclosure provides a modular handheld surgical instrument, comprising: a motor assembly comprising a motor and a motor gear; a shift gear translatable distally and proximally between a first position and a second position; a transducer assembly comprising an ultrasonic transducer shaft; and a shaft assembly, comprising: a first shaft extending distally along a shaft axis, wherein the first shaft comprises a drive gear; an ultrasonic waveguide coaxially aligned inside the first shaft about the shaft axis; and a second shaft coaxially aligned outside the first shaft and the ultrasonic waveguide about the shaft axis, wherein the second shaft comprises a torque gear; and a motor circuit, wherein the motor circuit is configured to alter current supplied to the motor to controllably torque the ultrasonic waveguide to the ultrasonic transducer shaft via the torque gear; wherein when the shift gear is translated to the first position, the motor gear is configured to rotate the drive gear via the shift gear, and wherein when the shift gear is translated to the second position, the motor gear is configured to rotate the torque gear via the shift gear.

The first shaft may be selectively coupleable to at least two independent actuation devices, and wherein the drive gear upon rotation is operable to actuate the at least two independent actuation devices. The modular handheld surgical instrument may further comprise a strain gauge configured to measure torque applied by the motor, and wherein the motor circuit is configured to limit current supplied to the motor based on the torque measured by the strain gauge. The motor circuit may be configured to alter current supplied to the motor such that the torque when coupling the ultrasonic waveguide to the ultrasonic transducer shaft does not exceed a predefined torque. The motor circuit may be configured to alter current supplied to the motor such that the torque when decoupling the ultrasonic waveguide from the ultrasonic transducer shaft is a higher predefined torque.

Modular Battery Powered Handheld Surgical Instrument with a Plurality of Control Programs

In another aspect, the present disclosure provides, a modular battery powered handheld surgical instrument with a plurality of control programs. A battery powered modular surgical instrument includes a control handle assembly with a processor coupled to a memory, a shaft assembly operably coupled to the control handle assembly and detachable therefrom. The shaft assembly includes circuit modules and a memory. The control programs are configured to operate the circuit modules. The control programs include computer executable instructions. The instrument also includes a transducer assembly operably coupled to the control handle assembly and detachable therefrom. The transducer assembly includes a memory and a transducer configured to convert a drive signal to mechanical vibrations. The control programs are configured to control the conversion of the drive signal to mechanical vibrations. The instrument also includes a battery assembly operably coupled to the control handle assembly and detachable therefrom. The battery assembly comprises a memory and is configured to power the surgical instrument.

FIGS. **62**, **63**, and **126-131** describe aspects of the present disclosure. In one aspect, the subject matter comprises a segmented circuit design for any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-70** to enable a plurality of control programs to operate in a plurality of different shaft assemblies **110**, **490**, **510**, **1110**, **1210**, transducer assemblies **104**, **486**, **504**, knife drive assemblies **1104**, **1152**, **1204**, and/or battery assemblies **106**, **484**, **506**, **1106**, **1206**, where the plurality of the control programs may reside in the different assemblies and are uploaded to the handle assembly **102**, **482**, **502**, **1102**, **1201** when attached thereto. FIGS. **126-131** describe aspects of the present disclosure. In one aspect, the subject matter of the present disclosure comprises controlling the operation of a battery operated modular surgical instrument **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** as described herein in connection with FIGS. **1-70** with a plurality of control programs. The battery operated modular surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** comprise components such as a handle assembly **102**, **482**, **502**, **1102**, **1202** comprising a controller, a shaft assembly **110**, **490**, **4510**, **110**, **1210**, a transducer assembly **104**, **486**, **504**, a knife drive assembly **1104**, **1152**, **1204**, and/or a battery assembly **106**, **484**, **506**, **1106**, **1206**.

In various aspects, the battery powered modular surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** as described herein in connection with FIGS. **1-70** further comprise an end effector **112**, **492**, **512**, **1000**, **1112**, **1212** and some aspects comprise a motor assembly **1160**, **1260**. The components of the modular surgical instrument are modular components that may be combined into a single modular component. For example, a proximal end of the shaft can be attached to the handle assembly such that the shaft assembly and the handle assembly are operably coupled to form a single modular component. Each of the plurality of control programs comprise machine executable instructions that may be executed by a processor of the modular surgical instrument. Although the processor is generally located either in the handle assembly or the battery assembly, in various aspects the processor may be located in any modular component such as the shaft assembly, transducer assembly, and/or the knife drive assembly. Executing a control program corresponding to a modular component controls the operation of the modular component by, for example, causing the modular component such as an ultrasonic shaft assembly to apply ultrasonic energy for a surgical application or procedure in accordance with the operation algorithm embodied in the executed control program. The modular surgical instrument is configured to treat patient tissue in surgical applications or procedures involving the application of a particular energy modality. Energy modalities can include ultrasonic energy, combination of ultrasonic and high-frequency current (e.g., RF) energy, high-frequency current energy with I-blade knife configuration, or a high-frequency current energy and opposable jaw with knife, for example.

FIG. **126** is a system schematic diagram illustrating components of a battery powered modular surgical instrument **2400**, such as the battery operated modular surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-70**, according to various aspects of the present disclosure. The components include a transducer assembly **2402**, a control handle assembly **2404**, a shaft assembly **2406**, and a battery assembly **2408**. The transducer assembly **2402** comprises a modular transducer that may be configured to implement a particular operation of the surgical instrument **2400**. For example, the modular transducer assembly **2402** may be an ultrasonic transducer operating at a 31 kHz resonant frequency or an ultrasonic transducer operating at a 55 kHz resonant frequency. Therefore, a user of the surgical instrument **2400** may select a modular variation of the transducer assembly **2402** for operation of the modular surgical instrument. A subset of a plurality of control programs embodies an algorithm, protocol or procedure corresponding to an operation or a function of the transducer assembly **2402**. For example, the subset of control programs can correspond to operation of the transducer assembly **2402** at 31 kHz or 51 kHz frequency.

In various aspects, the ultrasonic transducer component of the ultrasonic transducer assembly **2402** receives electrical power through an ultrasonic electrical signal from a generator via for example, a cable. The ultrasonic transducer converts the received electrical power into ultrasonic vibration energy. The operation of the ultrasonic transducer is described herein in connection with FIGS. **1-15**. The transducer assembly **2402** is operably coupled to the control handle assembly **2404** via a pair of connectors **2420***a*, **2420***b*, such as a male coupler and a corresponding female socket coupler. The transducer assembly **2402** comprises an ultrasonic transducer, drive circuitry, and a memory device **2410**. The memory device **2410** may be a volatile memory device or a nonvolatile memory device such as a random-access memory (RAM), dynamic RAM (DRAM), synchronous (SDRAM), read-only memory (ROM) erasable programmable ROM (EPROM), electrically EPROM (EEPROM), flash memory or other suitable memory device. In some aspects, the memory device **2414** is a plurality of nonvolatile memory devices, volatile memory device, a combination, or a sub-combination thereof. In various aspects, the memory device **2410** stores a main RTOS of the modular surgical instrument. Operation of the main RTOS is described in commonly owned U.S. Patent Application Publication No. 2016/0074038, now U.S. Pat. No. 9,730,695, which is incorporated herein by reference in its entirety.

In some aspects, the memory device **2410** stores all or a subset of control programs corresponding to the operation of the selected modular variant of the ultrasonic transducer. For example, executing an ultrasonic transducer control program by a processor can control the ultrasonic transducer by causing the ultrasonic transducer to convert an ultrasonic drive signal into a particular mode of vibration such as longitudinal, flexural, torsional and harmonics thereof. An ultrasonic transducer control program also can be configured to control the operation of the ultrasonic transducer by monitoring the rate at which the ultrasonic transducer converts a drive signal into vibrations based on monitored characteristics such as ultrasonic transducer tissue impedance. Alternatively or additionally, in various aspects, an ultrasonic transducer control program may be configured to operate a plurality of circuit modules such as software, programs, data, drivers, and/or application program interfaces (APIs). The ultrasonic transducer assembly **2402** comprises a plurality of circuit modules and the circuit modules control the operation of the ultrasonic transducer. As further described with reference to FIGS. **127-131** control programs corresponding to the ultrasonic transducer may comprise component identification, RTOS update, usage counter, energy update, 55 kHz, 31 kHz, and RF control programs, or any combination or sub-combination thereof. In some aspects, the memory device **2410** stores a plurality of control programs each corresponding to an operation or a function of the ultrasonic transducer. In other aspects, multiple control programs in conjunction correspond to an operation or a function of the ultrasonic transducer. In various aspects, the transducer assembly **2402** comprises a processor coupled to the memory device **2410**.

The control handle assembly **2404** is operably coupled via the pair of connectors **2420***a*, **2420***b* to the transducer assembly **2402**. The control handle assembly **2404** is operably coupled to the shaft assembly **2406** via a pair of connectors **2422***a*, **2422***b* such as a male coupler and a corresponding female socket coupler. The control handle assembly **2404** is operably coupled to the battery assembly **2408** via a pair of connectors **2424***a*, **2424***b* such as a male coupler and a corresponding female socket coupler. The control handle assembly **2404** is a modular control handle that may be, for example, a control handle configured to support a particular drive system of the surgical instrument such as a rotatable drive shaft assembly **2406** configured to advance an end effector of the surgical instrument such as a staple driver, cutting member or another type of end effector for other types of surgical instruments, graspers, clip appliers, access device, drug/gene therapy devices, ultrasound, RF, and or laser devices. In some aspects, the control handle assembly **2404** may comprise a closure trigger that is configured to transition between an unactuated position and an actuated position. The unactuated position corresponds to an open or unclamped configuration of the shaft assembly **2406** and the actuated portion corresponds to a closed or clamped configuration of the shaft assembly **2406**. A user of the surgical instrument may control the actuation of the closure trigger. In various aspects, the control handle assembly **2404** may comprise a plurality of handle housing segments that may be connected by for example, screws, snap features, or adhesive to form a handle grip such as a pistol grip. In some aspects, the control handle assembly **2404** comprises a motor.

The control handle assembly **2404** comprises a processor **2412** coupled to a memory device **2414**. The processor **2412** and the memory device **2414** may be integrated into a single integrated circuit (IC) or multiple ICs. The processor **2412** may be a microprocessor, a programmable gate-array (PGA), an application-specific IC (ASIC), controller, microcontroller, digital signal processor (DSP), programmable logic device (PLD) or a combination or sub-combination thereof. The memory device **2414** may be a volatile memory device or a nonvolatile memory device such as a RAM, DRAM, SDRAM, ROM, EPROM, EEPROM, flash memory, or other suitable memory device. In some aspects, the memory device **2414** is a plurality of nonvolatile memory devices, volatile memory device, a combination, or a sub-combination thereof. In various aspects, the memory device **2414** stores the main RTOS **2502** of the modular surgical instrument. In some aspects, the control handle assembly **2404** comprises one or more primary controllers. In various aspects, the control handle assembly **2404** also comprises one or more safety controllers. More generally, the modular control handle assembly **2404** operably supports a plurality of drive systems that are configured to generate and apply various control motions to corresponding portions of the modular shaft assembly **2406**. For example, the control handle assembly **2404** may comprise a handle assembly comprising an elongate body, a proximal end, a distal end, and a cavity configured to accept another component of the modular surgical instrument. Therefore, a user of the modular surgical instrument may select a modular variation of the control handle assembly **2404** to support a patient treatment operation of the modular surgical instrument. A modular variation can correspond to a particular control program of the plurality of control programs. In some aspects, the memory device **2414** stores a plurality of control programs each corresponding to an operation or a function of the control handle assembly **2404**. In other aspects, multiple control programs in conjunction correspond to an operation or a function of the control handle assembly **2404**.

Different modular variations of the control handle assembly **2404** may be configured to support the application of a particular energy modality. A subset of a plurality of control programs embodies an algorithm, protocol, or procedures corresponding to an operation or a function of the control handle assembly **2404**. For example, a control program of the plurality of control programs can correspond to operation of a control handle assembly **2404** configured to apply RF energy opposable jaw. In some aspects, the memory device **2414** stores a subset of control programs corresponding to the operation or function of the selected modular variant of the control handle assembly **2404**. For example, executing a control handle control program by the processor **2412** can control the control handle assembly **2404** by causing the control handle assembly **2404** to actuate a shaft assembly of the shaft assembly **2406**. Alternatively or additionally, in various aspects, the control handle assembly **2404** comprises a plurality of circuit modules such as software, programs, data, drivers, and/or application program interfaces (APIs) to control the operation of the control handle assembly **2404**. The plurality of circuit modules may be implemented by one or more hardware components, e.g., processors, DSPs, PLDs, ASICs, circuits, registers and/or software components, e.g., programs, subroutines, logic and/or combinations of hardware and software components. In various aspects, a control handle control program may be configured to operate the plurality of circuit modules. For example, executing a control handle control program by the processor **2412** may operate a handle motor circuit module by applying the control motions generated by a motor of the modular surgical instrument to actuate the shaft assembly **2406**. As further described with reference to FIGS. **127-129**, control programs corresponding to the control handle assembly **2404** may comprise RTOS software, motor control, switch, safety control, RTOS update, and energy update control programs, or any combination or sub-combination thereof. Control programs corresponding to the control handle assembly **2404** may further comprise the control programs corresponding to the transducer assembly **2402** and the shaft assembly **2406**. Transducer control programs include 55 kHz, 31 kHz, and RF control programs. Shaft assembly **2406** control programs include ultrasonic, RF I-blade, RF opposable jaw, and combination ultrasonic and RF control programs, for example.

The shaft assembly **2406** is operably coupled via connectors **2422***a*, **2422***b* such as a male coupler and a corresponding female socket coupler, to the control handle assembly **2404**. In various aspects, the shaft assembly **2406** is operably connected to an end effector of the modular surgical instrument to perform one or more surgical procedures. The shaft assembly **2406** may comprise an articulation joint and an articular lock that are configured to detachably hold the end effector in a particular position. Operation of the articulation joint and articular lock is described in commonly owned U.S. Patent Application Publication No. 2014/0263541, which is incorporated herein by reference in its entirety. The shaft assembly **2406** is a modular shaft comprising modular variations of shaft assemblies that may be, for example, configured to be coupled to a particular end effector of the surgical instrument such as an ultrasonic blade. In various aspects, the shaft assembly **2406** may comprise an interchangeable shaft assembly that is configured for removable attachment via for example, a latch, from a housing of the modular surgical instrument. The shaft assembly of the shaft assembly **2406** also may include a spine configured to support a shaft frame, a firing member, and a closure tube extending around the spine. The shaft assembly may support axial trial of the firing member within the spine of the shaft assembly **2406**. The shaft assembly of the shaft assembly **2406** also may comprise a slip ring assembly configured to conduct electric power between the shaft assembly **2406** and the end effector. The operation of interchangeable shaft assemblies of the shaft assembly **2406** is further described in commonly owned U.S. Patent Application Publication No. 2015/0272579, now U.S. Pat. No. 9,743,929, which is incorporated herein by reference in its entirety. Modular variations of shaft assemblies of the shaft assembly **2406** are configured to be actuated by various corresponding drive systems of the modular surgical instrument.

The shaft assembly **2406** comprises a memory device **2416**. The memory device **2416** may be a volatile memory device or a nonvolatile memory device such as a RAM, DRAM, SDRAM, ROM, EPROM, EEPROM, flash memory, or other suitable memory device. In some aspects, the memory device **2416** is a plurality of nonvolatile memory devices, volatile memory device, a combination, or a sub-combination thereof. In various aspects, the memory device **2416** stores the main RTOS of the modular surgical instrument. In various aspects, the shaft assembly **2406** comprises a processor coupled to the memory device **2416** that may be integrated into a single IC or multiple ICs. The processor may be a microprocessor, a PGA, an ASIC, controller, microcontroller, DSP, PLD or a combination or sub-combination thereof. In some aspects, the memory device **2416** stores a plurality of control programs each corresponding to a function or an operation of the control handle assembly **2404**. In other aspects, multiple control programs in conjunction correspond to an operation or a function of the control handle assembly **2404**. Modular variations of shaft assemblies of the shaft assembly **2406** may be configured to be operably coupled to modular variations of end effectors in accordance with a selected energy modality or an operation modality of the modular surgical instrument. Moreover, modular variations of drive systems of the modular surgical instrument may be configured to generate and apply at least one control motion to actuate modular variations of shaft assemblies in accordance with a selected energy modality or an operation modality of the modular surgical instrument. Therefore, a user of the modular surgical instrument may select a modular variation of the shaft assembly **2406** based on for example, shaft assemblies configured to apply other motions and forms of energy such as, for example, high-frequency current (e.g., RF) energy, ultrasonic energy and/or motion to end effector arrangements adapted for use in connection with various surgical applications and procedures.

Different modular variations of the shaft assembly **2406** can correspond to particular control programs of the plurality of control programs. A subset of a plurality of control programs embodies an algorithm, protocol or procedure corresponding to an operation of the shaft assembly **2406**. For example, a control program of the plurality of control programs can correspond to a shaft assembly configured to be operably coupled to an end effector applying a combination of ultrasonic and RF energy. In some aspects, the memory device **2416** stores all or a subset of control programs corresponding to the operation or the function of the selected modular variant of the shaft assembly **2406**. For example, executing a shaft attachment control program can enable the actuation of a latch actuator assembly to actuate a lock yoke. Alternatively or additionally, in various aspects, the shaft assembly **2406** comprises a plurality of circuit modules such as software, programs, data, drivers, and/or application program interfaces (APIs) to control the operation of the shaft assembly **2406**. The plurality of circuit modules may be implemented by one or more hardware components, e.g., processors, DSPs, PLDs, ASICs, circuits, registers and/or software components, e.g., programs, subroutines, logic and/or combinations of hardware and software components. In various aspects, a shaft control program may be configured to operate the plurality of circuit modules. For example, a shaft attachment control program may operate a shaft attachment circuit module by causing the latch actuator assembly to cooperate with the lock yoke to couple the shaft assembly **2406** to the control handle assembly **2404**. As further described with reference to FIGS. **127-129**, control programs corresponding to the shaft assembly **2406** may comprise component identification, RTOS update, usage counter, energy update, or any combination or sub-combination thereof. The component identification control program may comprise specific component identification control programs such as ultrasonic, RF I-blade, RF opposable jaw, and combination ultrasonic and RF component identification control programs.

The battery assembly **2408** is operably coupled via connectors **2424***a*, **2424***b* such as a male coupler and a corresponding female socket coupler, to the control handle assembly **2404**. The battery assembly **2408** comprises a Lithium-ion ("Li"), other suitable battery or a plurality thereof that may be connected in series. In some aspects, the battery assembly **2408** may be a battery such as the 14.4 volt nickel metal hydride (NiMH) SmartDriver Battery, available from MicroAire Surgical Instruments, Charlottesville, VA. The battery assembly **2408** is configured to provide power for the operation of the modular surgical instrument. Specifically, the battery assembly **2408** is rechargeable and applies voltage to components of the modular surgical instrument, including for example, an electric motor. In various aspects, the voltage polarity applied to the electric motor by the battery assembly **2408** is reversible between a clockwise polarity and a counterclockwise polarity. The applied voltage may operate the electric motor to drive a drive member to effectuate an end effector of the modular surgical instrument. In various aspects, the battery assembly **2408** is a component of a power assembly of the modular surgical instrument. The battery assembly **2408** comprises a memory device **2418**. In some aspects, the battery assembly **2408** further comprises a processor. The memory device **2418** may be a volatile memory device or a nonvolatile memory device such as a RAM, DRAM, SDRAM, ROM, EPROM, EEPROM, flash memory, or other suitable memory device. In some aspects, the memory device **2418** is a plurality of nonvolatile memory devices, volatile memory device, a combination, or a sub-combination thereof. In various aspects, the memory device **2418** stores the main RTOS of the modular surgical instrument. In various aspects, the battery assembly **2408** comprises a processor coupled to the memory device **2418** that may be integrated into a single IC or multiple ICs. The processor may be a microprocessor, a PGA, an ASIC, controller, microcontroller, DSP, PLD or a combination or sub-combination thereof.

Modular variations of the battery assembly **2408** may be configured to power modular variations of end effectors in accordance with a selected energy modality or an operation modality of the modular surgical instrument. For example, differences in the voltage rating of modular battery variants **2408** can correspond to different end effectors configured to achieve an effect according to a selected surgical procedure or operation, including endocutter, grasper, cutter, stapler, clip applier, access device, drug/gene therapy delivery device, and energy delivery device using an ultrasonic, RF I-blade, RF opposable jaw, combination ultrasonic or RF energy modality. Different modular variants of the battery assembly **2408** can correspond to particular control programs of the plurality of control programs. A subset of a plurality of control programs embodies an algorithm, protocol or procedure corresponding to an operation or function of the battery assembly **2408**. For example, a control program of the plurality of control programs can correspond to a battery assembly **2408** configured to provide power to a modular surgical instrument configured to apply an ultrasonic energy modality. In some aspects, the memory device **2418** stores all or a subset of control programs corresponding to the operation or the function of the selected modular variant of the battery assembly **2408**. In some aspects, the memory device **2418** stores a plurality of control programs each corresponding to an operation or a function of the control handle assembly **2404**. In other aspects, multiple control programs in conjunction correspond to an operation or a function of the control handle assembly **2404**. For example, executing a power management control program can enable a power management controller of the modular surgical instrument to modulate the power output of the battery in accordance with predetermined power requirements, such as the power requirements of an attached shaft assembly.

Alternatively or additionally, in various aspects, the battery assembly **2408** comprises a plurality of circuit modules such as software, programs, data, drivers, and/or application program interfaces (APIs) to control the operation of the battery assembly **2408**. The plurality of circuit modules may be implemented by one or more hardware components, e.g., processors, DSPs, PLDs, ASICs, circuits, registers and/or software components, e.g., programs, subroutines, logic and/or combinations of hardware and software components. In various aspects, a battery control program may be configured to operate the plurality of circuit modules. For example, a battery charge monitoring control program may operate a charge monitoring circuit module by causing a controller communicating with the memory device **2418** and a state of charge monitoring circuit to measure the state of charge of the battery assembly **2408**. The controller may further compare the measured state of charge with any charge values previously stored in the memory device **2418** and display the measured value of the state of charge on an LCD screen of the modular surgical instrument, as described, for example, in U.S. Patent Application Publication No. 2016/0106424, which is incorporated herein by reference. As further described with reference to FIGS. **127-129**, control programs corresponding to the battery assembly **2408** may comprise maximum number of uses, charge and drainage, RTOS update, usage counter, energy update, motor control, RTOS software, switch, safety control programs or any combination or sub-combination thereof. Control programs corresponding to the battery assembly **2408** may further comprise the control programs corresponding to the transducer assembly **2402** and the shaft assembly **2406**. Ultrasonic transducer control programs include 55 kHz, 31 kHz, and RF control programs. Shaft control programs include ultrasonic, RF I-Blade, RF opposable jaw, and combination ultrasonic and RF control programs.

FIGS. **127-129** describe pluralities of control programs **2500**, **2600**, **2700** distributed between the transducer assembly **2402**, control handle assembly **2404**, shaft assembly **2406**, and battery assembly **2408**. FIG. **127** describes a distribution of pluralities of control programs **2500** according to one aspect of the present disclosure in which the memory device **2414** of the control handle assembly **2404** stores a plurality of control programs **2500** comprising base operating control programs corresponding to the general operation of the modular surgical instrument, according to one aspect of the present disclosure. Moreover, the memory device **2414** stores a plurality of control programs **2500** comprising base operating control programs corresponding to transducer and shaft modalities corresponding to energy modalities of the modular surgical instrument. The memory devices **2412**, **2416**, **2418** store pluralities of control programs wherein each plurality of the pluralities of control programs corresponds to a function or operation of the respective specific component of the modular surgical instrument such as the transducer assembly **2402**, shaft assembly **2406**, and battery assembly **2408**. In some aspects, the memory device **2414** stores a basic input/output system (BIOS) program that is configured to control the communication between the processor **2412** and the modular components of the modular surgical instruments.

During operation of the modular surgical instrument, the BIOS program loads the component identification **2528** control program to the memory device **2414** for the processor **2412** to execute. Execution of the component identification **2524** control program enables the processor **2412** to select the corresponding one or more of the transducer **2509** base operating control programs **2510**, **2512**, **2514** for the BIOS to load to the memory device **2414**. The BIOS also loads the component identification **2532** control program to the memory device **2414** for the processor **2412** to execute. Execution of the component identification **2532** control program enables the processor **2412** to select the corresponding one or more of the shaft **2515** base operating control programs **2516**, **2518**, **2520**, **2522** for the BIOS to load to the memory device **2414**. The processor **2412** can then execute the control programs in the memory device **2414** to implement the selected modular variants of the modular shaft assembly **2406** and transducer assembly **2402**. If any of the modular components of the modular surgical instrument have an updated version of the RTOS software **2502** stored in their respective memory devices **2412**, **2416**, **2418**, the processor **2404** may download the updated version to the memory device **2414** via the respective RTOS update **2528**, **2536**, **2546** control programs.

The plurality of base operating control programs corresponding to general operation comprise the RTOS software **2502**, motor control **2504**, switch control **2506**, and safety control **2508** control programs. The plurality of base operating control programs corresponding to the transducer control programs **2509** and shaft control programs **2515** comprise 55 kHz, 31 kHz, or RF control programs **2510**, **2512**, **2514**, respectively, for the transducer assembly **2402** and ultrasonic **2516**, combination ultrasonic and RF **2518**, RF I-Blade **2520**, and RF opposable jaw **2522** control programs for the shaft assembly **2406**. In various aspects, the memory device **2414** comprises a nonvolatile memory device such as ROM and a volatile memory device such as RAM. The BIOS program may be stored in the nonvolatile memory device and also may contain addresses of the modular components. The RTOS software **2502** is configured to control the execution, by the processor **2412**, of the pluralities of control programs distributed between components of the modular surgical instrument. In some aspects, the RTOS software **2502** is stored in the nonvolatile memory. When the modular surgical instrument powers up, the BIOS program of the modular surgical instrument loads the RTOS software **2502** from the nonvolatile memory device to the volatile memory device.

The motor control **2504** comprises an algorithm, protocol or procedure to control operation of a motor of the modular surgical instrument, such as by controlling a direction of rotation. In some aspects, the motor control **2504** is configured to determine the direction of the motor by determining the polarity of the voltage applied by the battery assembly **2408**. The switch control **2506** comprises an algorithm, protocol or procedure to control operation of an end effector of a motor of the modular surgical instrument, such as by controlling the direction of articulating the end effector. The direction may be clockwise or counterclockwise. In various aspects, the switch control **2506** is configured to control whether a switch is in a closed or an open position, wherein a closed position may activate the motor to articulate the end effector in a particular direction. The safety control **2508** comprises an algorithm, protocol or procedure to control operation of a safety controller that is configured to perform safety critical applications such as by interrupting power to the motor if an error or fault condition is detected by the safety controller.

The 55 kHz control program **2510** comprises an algorithm, protocol or procedure to control operation of an ultrasonic transducer portion of the transducer assembly **2402** that converts received electrical power into ultrasonic vibration energy at a resonant frequency of 55 kHz. The 31 kHz control program **2512** comprises an algorithm, protocol or procedure to control operation of an ultrasonic transducer portion of the transducer assembly **2402** that converts received electrical power into ultrasonic vibration energy at a resonant frequency of 31 kHz. The RF control program **2514** comprises an algorithm, protocol or procedure to control the delivery of high-frequency current (e.g., RF) energy to an end effector to facilitate the application of RF energy by the modular surgical instrument **2400**. The ultrasonic control program **2516** comprises an algorithm, protocol or procedure to control operation of an ultrasonic waveguide located within the shaft assembly **2406** that is configured to apply ultrasonic energy for a surgical procedure or operation. The combination ultrasonic and RF control program **2518** comprises an algorithm, protocol or procedure to control operation of a shaft assembly **2406** that is configured to apply either ultrasonic or RF energy for a surgical procedure or operation. The RF I-blade control program **2520** comprises an algorithm, protocol or procedure to control operation of a shaft assembly **2406** that is configured to apply RF energy to an end effector with a cutting member such as an I-blade, for a surgical procedure or operation. The RF opposable jaw control program **2522** comprises an algorithm, protocol or procedure to control operation of a shaft assembly **2406** that is configured to apply RF energy via an end effector comprising opposable jaw members, for a surgical procedure or operation.

A plurality of control programs **2500** corresponding to the transducer assembly **2402** comprise the component identification **2524**, usage counter **2526**, RTOS update **2528**, and energy update **2530** control programs. The component identification **2524** comprises an algorithm, protocol or procedure to identify the modular variant of the transducer assembly **2402** to the BIOS and RTOS programs. For example, the component identification **2524** may identify an ultrasonic transducer as a 55 kHz transducer. The usage counter **2526** comprises an algorithm, protocol or procedure to monitor the usage of the ultrasonic transducer. For example, the usage counter **2526** can maintain a usage cycle count corresponding to the number of instances that the ultrasonic transducer is used by a user of the modular surgical instrument. In some aspects, if the usage cycle count value calculated by the usage counter **2526** exceeds a predetermined value, the processor **2412** may disable the transducer assembly **2402** or disable the entire modular surgical instrument from performing an operation or function. The RTOS update control program **2528** comprises an algorithm, protocol or procedure to identify whether the memory device **2410** stores an update to the RTOS software **2502** stored in the memory device **2414**.

In various aspects, a user of the modular surgical instrument may upload an updated version of the RTOS software **2502** via, for example, a computer coupled to the memory device **2410** through a transmission media, to the memory device **2410**. When the modular transducer assembly **2402** is attached to the modular surgical instrument, the updated version stored in the memory device **2410** is downloaded by the processor **2412** to be stored in the memory device **2414**. The updated version may overwrite the existing version of the RTOS software **2502** stored in the memory device **2414**. The energy update **2530** comprises an algorithm, protocol or procedure to identify whether the memory device **2410** stores an update to a surgical procedure energy algorithm stored in the memory device **2414**. The surgical procedure energy algorithm comprises one or more techniques to employ one or more energy modalities based on tissue parameters such as the type of tissue to be treated in the surgical procedure and tissue impedance. For example, a particular surgical procedure energy algorithm involves applying RF energy for a portion of the surgical procedure, ultrasonic energy for a second portion, and a combination of RF and ultrasonic energy for a third portion in accordance with a particular surgical procedure. An update to a surgical procedure energy algorithm comprises a change to an existing surgical energy algorithm. Such a change may be, for example, increasing the frequency at which RF energy is applied for a portion of the surgical procedure in order to perform the procedure at an improved level of precision and control. In various aspects, a user of the modular surgical instrument may upload the update to a surgical procedure energy algorithm the memory device **2410** that the processor may download and store in the memory device **2414**.

The plurality of control programs corresponding to the shaft assembly **2406** comprises the component identification **2532**, the usage counter **2534**, the RTOS update **2536**, and the energy update **2538**. The component identification **2532** comprises an algorithm, protocol or procedure to identify the modular variant of the shaft assembly **2406** to the BIOS and RTOS programs, such as a shaft assembly **2406** configured to apply ultrasonic energy. The usage counter **2534** comprises an algorithm, protocol or procedure to monitor the usage of the shaft assembly **2406**. As described previously in connection with usage counter **2526**, the usage counter **2534** can maintain a usage cycle count. As described previously in connection with RTOS update **2528**, the RTOS update **2536** comprises an algorithm, protocol or procedure to identify whether the memory device **2416** stores an updated version of the RTOS software **2502**. As described previously in connection with energy update **2530**, the energy update **2538** comprises an algorithm, protocol or procedure to identify whether the memory device **2416** stores an updated version of a surgical procedure energy algorithm.

A plurality of control programs **2500** corresponding to the battery assembly **2408** comprises usage counter **2540**, maximum number of uses **2542**, charge and drainage **2544**, RTOS update **2546**, and energy update **2548**. The usage counter **2540** comprises an algorithm, protocol or procedure to monitor the usage of the battery assembly **2408**. As described previously in connection with usage counter **2526** and usage counter **2534**, the usage counter **2540** can maintain a usage cycle count. The maximum number of uses **2542** comprises an algorithm, protocol or procedure to determine a maximum usage value. In some aspects, if the usage cycle count exceeds the maximum usage value, the processor **2412** may disable the battery assembly **2408** or disable the entire modular surgical instrument from performing an operation or function. The charge and drainage **2544** comprises an algorithm, protocol or procedure to control charging and raining a rechargeable battery assembly **2408**. In various aspects, a dedicated drainage circuit may implement the power drain function of the charge and drainage **2544** control program. A state of charge monitoring circuit may implement the recharge function of the charge and drainage **2544** control program. In some aspects, the state of charge monitoring circuit may measure the current state of charge, compare the current state with a previously stored state in the memory device **2418**, and display the measured or previously stored value on an LCD screen. As described previously in connection with RTOS update **2528** and RTOS update **2536**, the RTOS update **2546** comprises an algorithm, protocol or procedure to identify whether the memory device **2418** stores an updated version of the RTOS software **2502**. As described previously in connection with energy update **2530** and energy update **2538**, the energy update **2548** comprises an algorithm, protocol or procedure to identify whether the memory device **2418** stores an updated version of a surgical procedure energy algorithm.

FIG. **128** describes a distribution of pluralities of control programs **2600** according to one aspect of the present disclosure in which the memory device **2418** of the battery assembly **2408** stores a plurality of control programs **2500** comprising base operating control programs corresponding to the general operation of the modular surgical instrument, according too one aspect of the present disclosure. Moreover, the memory device **2418** stores a plurality of control programs **2500** comprising base operating control programs corresponding to transducer and shaft modalities corresponding to energy modalities of the modular surgical instrument. The memory device **2418** also stores a plurality of control programs **2500** corresponding to functions or operations of the battery assembly **2408**. The memory devices **2412**, **2416** store pluralities of control programs wherein each plurality of the pluralities of control programs corresponds to a function or operation of the respective specific component of the modular surgical instrument such as the transducer assembly **2402** and shaft assembly **2406**. The processor **2412** is located in the battery assembly **2408** instead of the control handle assembly **2404**. In some aspects, the processor **2412** is still located in the control handle assembly **2404** and the battery assembly **2408** comprises another processor.

In some aspects, the memory device **2418** stores the BIOS program that is configured to control the communication between the processor **2412** and the modular components of the modular surgical instruments. During operation of the modular surgical instrument, the BIOS program loads the component identification **2602** control program to the memory device **2418** for the processor **2412** to execute. Execution of the component identification **2602** control program enables the processor **2412** to select the corresponding one or more of the ultrasonic transducer base operating control programs **2510**, **2512**, **2514** for the BIOS to load to the memory device **2418**. The BIOS also loads the component identification **2532** control program to the memory device **2418** for the processor **2412** to execute. Execution of the component identification **2532** control program enables the processor to select the corresponding one or more of the shaft base operating control programs **2516**, **2518**, **2520**, **2522** for the BIOS to load to the memory device **2418**. The processor **2412** can then execute the control programs in the memory device **2418** to implement the selected modular variants of the modular shaft assembly **2406** and transducer assembly **2402**. In various aspects, the memory device **2418** comprises a nonvolatile memory device and a volatile memory device. The BIOS program may be stored in the nonvolatile memory device and also may contain addresses of the modular components. In some aspects, the RTOS software **2502** is stored in the nonvolatile memory of the memory device **2418**. The RTOS software **2502** is configured to control the execution, by the processor **2412**, of the pluralities of control programs distributed between components of the modular surgical instrument. The BIOS program of the modular surgical instrument can load the RTOS software **2502** from the nonvolatile memory device to the volatile memory device when the modular surgical instrument powers up.

As described in connection with FIG. **127**, the plurality of base operating control programs corresponding to general operation comprise the RTOS software **2502**, the motor control **2504**, the switch control **2506**, and the safety control **2508**. As described in connection with FIG. **127** base operating control programs corresponding to transducer and shaft modalities comprise the 55 kHz, the 31 kHz, and the RF control programs **2510**, **2512**, **2514**, respectively, for the transducer assembly **2402** and ultrasonic control program **2516**, combination ultrasonic and RF control program **2518**, the RF I-blade control program **2520**, and the RF opposable jaw control program **2522** control programs for the shaft assembly **2406**. As described in connection with FIG. **127**, control programs corresponding to functions or operations of the battery assembly **2408** comprise usage counter **2610**, maximum number of uses **2612**, charge and drainage **2614**, and energy update **2616**. Usage counter **2610**, maximum number of uses **2612**, charge and drainage **2614**, and energy update **2616** each comprise an algorithm, protocol or procedure that is substantially the same as usage counter **2540**, maximum number of uses **2542**, charge and drainage **2544**, and energy update **2548**.

In the distribution of pluralities of control programs **2600** according to the aspect of FIG. **128**, the plurality of control programs **2600** corresponding to the control handle assembly **2404** comprise the plurality of base operating control programs corresponding to general operation and the plurality of base operating control programs corresponding to transducer control programs **2509** and shaft control programs **2515** modalities. Therefore, the memory device **2414** does not store any of the pluralities of control programs. The plurality of control programs **2600** corresponding to a function or operation of the transducer assembly **2402** comprises the component identification **2602** and usage counter **2604**. The component identification **2602** and the usage counter **2604** each comprise an algorithm, protocol or procedure that is substantially the same as the component identification **2528** and the usage counter **2530**. The plurality of control programs **2600** corresponding to a function or operation the shaft assembly **2406** comprises the component identification **2606** and usage counter **2608**. The component identification **2606** and usage counter **2608** each comprise an algorithm, protocol or procedure that is substantially the same as the component identification **2536** and the usage counter **2538**.

FIG. **129** describes a distribution of pluralities of control programs **2700** according to one aspect of the present disclosure in which the memory device **2414** stores a plurality of control programs **2700** comprising base operating control programs corresponding to the general operation of the modular surgical instrument, according too one aspect of the present disclosure. The memory device **2410** stores a plurality of control programs **2700** comprising base operating control programs corresponding to transducer control programs **2509** modalities corresponding to energy modalities of the modular surgical instrument. The memory device **2416** stores a plurality of control programs **2700** comprising base operating control programs corresponding to shaft modalities corresponding to energy modalities of the modular surgical instrument. Therefore, the control handle assembly **2404** stores the general operation base operating control programs, while the transducer assembly **2402** and the shaft assembly **2406** store their respective energy modality base operating control programs. The memory devices **2412**, **2414**, **2416**, **2418** also store pluralities of control programs **2700** wherein each plurality of the pluralities of control programs **2700** corresponds to a function or operation of the respective specific component of the modular surgical instrument such as the transducer assembly **2402**, control handle assembly **2404**, shaft assembly **2406**, and battery assembly **2408**.

In some aspects, the memory device **2418** stores the BIOS program that is configured to control the communication between the processor **2412** and the modular components of the modular surgical instruments. During operation of the modular surgical instrument, the BIOS program loads the component identification **2704** control program to the memory device **2414** for the processor **2412** to execute. Execution of the component identification **2704** control program enables the processor **2412** to select the corresponding one or more of the transducer base operating control programs **2510**, **2512**, **2514** for the BIOS to load to the memory device **2414**. The BIOS also loads the component identification **2712** control program to the memory device **2414** for the processor **2412** to execute. Execution of the component identification **2532** control program enables the processor to select the corresponding one or more of the shaft base operating control programs **2516**, **2518**, **2520**, **2522** for the BIOS to load to the memory device **2414**. The processor **2412** can then execute the control programs in the memory device **2414** to implement the selected modular variants of the modular shaft assembly **2406** and transducer assembly **2402**. In various aspects, the memory device **2414** comprises a nonvolatile memory device and a volatile memory device. The BIOS program may be stored in the nonvolatile memory device and also may contain addresses of the modular components.

In some aspects, the RTOS software **2502** is stored in the nonvolatile memory of the memory device **2414**. The RTOS software **2502** is configured to control the execution, by the processor **2412**, of the pluralities of control programs distributed between components of the modular surgical instrument. In some aspects, one or more of the transducer assembly **2402**, shaft assembly **2406** and the battery assembly **2408** comprises one or more additional processors. As described in connection with FIGS. **127** and **128**, the plurality of base operating control programs **2600**, **2700** corresponding to general operation comprise the RTOS software **2502**, the motor control **2504**, the switch control **2506**, and the safety control **2508**. As described in connection with FIGS. **127** and **128**, base operating control programs **2600**, **2700** corresponding to transducer and shaft modalities comprise the 55 kHz **2510**, the 31 kHz **2512**, and the RF **2514** control programs for the transducer assembly **2402** and ultrasonic control programs **2516**, combination ultrasonic and the RF, the RF I-blade, and the RF opposable jaw control programs **2518**, **2520**, **2522** for the shaft assembly **2406**. In the distribution of the plurality of control programs **2700** according to the aspect of FIG. **129**, the plurality of control programs **2700** corresponding to a function or operation of the control handle assembly **2404** comprise the energy update **2702**. The energy update **2702** control program comprises an algorithm, protocol or procedure to identify whether the memory device **2414** stores an update to an existing surgical procedure energy algorithm stored in the memory device **2414**. The plurality of control programs corresponding to a function or operation of the transducer assembly **2402** comprise the component identification **2704**, RTOS update **2706**, usage counter **2708**, and energy update **2710** control programs. The component identification **2704**, RTOS update **2706**, usage counter **2708**, and energy update **2710** control programs each comprise an algorithm, protocol or procedure that is substantially the same as the component identification **2524**, RTOS update **2528**, usage counter **2526**, and energy update **2530** control programs.

The plurality of control programs **2700** corresponding to a function or operation of the shaft assembly **2406** comprise component identification **2712**, RTOS update **2714**, usage counter **2716**, and energy update **2718** control programs. The component identification **2712**, RTOS update **2714**, usage counter **2716**, and energy update **2718** control programs each comprise an algorithm, protocol or procedure that is substantially the same as the component identification **2532**, RTOS update **2536**, usage counter **2534**, and energy update **2538** control programs. The plurality of control programs corresponding to a function or operation of the battery assembly **2408** comprise usage counter **2720**, maximum number of uses **2722**, charge and drainage **2724**, RTOS update **2726**, and energy update **2728** control programs. The usage counter **2720**, maximum number of uses **2722**, charge and drainage **2724**, RTOS update **2726**, and energy update **2728** control programs each comprise an algorithm, protocol or procedure that is substantially the same as the usage counter **2540**, maximum number of uses **2542**, charge and drainage **2544**, RTOS update **2546**, and energy update **2548** control programs.

FIG. **130** is a logic diagram **2800** of a process for controlling the operation of a battery assembly operated modular surgical instrument with a plurality of control programs, according to one aspect of the present disclosure. With reference to FIGS. **126**, **127** and **131**, at the outset, the processor **2412** identifies **2802** a plurality of control programs. Each of the plurality of control programs **2500**, **2600**, **2700** (FIGS. **127-129**), for example, comprise computer executable instructions that may be executed by the processor **2412**. In various aspects, the processor **2412** comprises a primary controller and a safety controller. As previously described, in various aspects, some of the plurality of control programs may be configured to operate a plurality of circuit modules of the shaft assembly **2406**. Some of the plurality of control programs also may be configured to control the conversion of a drive signal to mechanical vibrations. The drive signal may be an electrical drive signal. The plurality of control programs may be stored in the memory devices **2410**, **2414**, **2416**, **2418** by the processor **2412** according to a predetermined distribution. For example, as described partially with reference to FIG. **127**, one example predetermined distribution may be that the memory device **2414** stores the RTOS software **2502** and motor control **2504** control programs of the plurality of control programs, the memory device **2410** stores the component identification **2524** and usage counter **2526** control programs of the plurality of control programs, the memory device **2416** stores the RTOS update **2536** and energy update **2538** control programs of the plurality of control programs, and the memory device **2418** stores the maximum number of uses **2542** and charge and drainage **2544** control programs of the plurality of control programs. In some aspects, each of the memory devices **2410**, **2414**, **2416**, **2418** may be a nonvolatile memory device or a volatile memory device. In other aspects, each of the memory devices **2410**, **2414**, **2416**, **2418** may comprise both a nonvolatile memory device and a volatile memory device.

After identifying **2802** the control programs, the processor **2412** determines **2804** the subset of the plurality of control programs necessary to operate the modular surgical instrument based on a desired operation of the modular surgical instrument. For example, a subset of the plurality of control programs may be necessary to operate the plurality of circuit modules of the shaft assembly **2406**. The processor **2412** selects **2806** at least one of the plurality of control programs to implement operation of the modular surgical instrument.

In various aspects, one of the memory devices **2410**, **2414**, **2416**, **2418** is a nonvolatile memory device storing at least one of the plurality of control programs. Selecting **2806** the at least one control program by the processor **2412** comprises the processor **2412** downloading the at least one control program from the nonvolatile memory device to a volatile memory device located in the control handle assembly **2404**, shaft assembly **2406**, transducer assembly **2402**, or battery assembly **2408**. When the selected at least one control program is stored in the volatile memory device, the processor **2412** executes **2808** the selected at least one control program. The processor **2412** may continue to select **2806** and execute **2808** selected control programs continuously during operation of the modular surgical instrument.

As previously described, the processor **2412** may store **2810** the plurality of control programs according to a predetermined distribution. In various aspects, the control handle assembly **2404**, shaft assembly **2406**, transducer assembly **2402**, or battery assembly **2408** may comprise additional processors in addition to the processor **2412**. The additional processors may be secondary processors. In some aspects, the processor **2412** may download **2812** a first, second, and third subset of the plurality of control programs stored respectively within the memory devices **2410**, **2416**, **2418** according to the predetermined distribution to the memory device **2414**. In other aspects, the predetermined distribution is the memory device **2414** storing each of the plurality of control programs. In some aspects, the predetermined distribution is the memory device **2414** storing the base operating control programs corresponding to the general operation of the modular surgical instrument, as described in FIG. **72**. The memory device **2414** may also store base operating control programs corresponding to transducer assembly and shaft assembly modalities corresponding to energy modalities of the modular surgical instrument, as described in FIG. **127**. In other aspects, the memory device **2410** stores the base operating control programs corresponding to transducer assembly modalities corresponding to energy modalities of the modular surgical instrument. The memory device **2416** stores the base operating control programs corresponding to shaft assembly modalities corresponding to energy modalities of the modular surgical instrument.

The processor **2412** identifies **2814** the modular variants of the components of the modular surgical instrument to be used by a user of the modular surgical instrument. A modular variant may be, for example, a modular shaft assembly **2406** comprising rotary shaft control or a modular transducer assembly **2402** comprising a transducer operating at 31 kHz resonant frequency. The processor **2412** determines **2816** the corresponding control programs of the plurality of control programs that correspond to the identified modular variants. The processor **2412** selects **2818** the corresponding subset of the plurality of control programs based on a look-up table. For example, the processor **2412** may select a base operating control program corresponding to a transducer assembly modality such as the 55 kHz **2510** control program if the modular variant of the modular transducer is a transducer operating at 55 kHz resonant frequency. Similarly, the processor **2412** may select a base operating control program corresponding to a shaft assembly modality such as the RF I-blade **2520** control program if the modular variant of the modular shaft is a modular shaft configured to implement a RF I-blade energy modality. The look-up table may comprise an indication that the selected subset of the plurality of control programs corresponds to at least one of the shaft assembly **2406** or the transducer assembly **2402**. In some aspects, the processor **2402** may determine **2816** the control programs corresponding to the identified modular variants based on the look-up table. After selecting **2818** the corresponding subset of control programs, the processor **2412** may execute the subset of control programs. The process of the logic diagram **2800** terminates **2820**.

FIG. **131** is a logic diagram **2900** of a process for controlling the operation of a battery assembly operated modular surgical instrument with a plurality of control programs, according to one aspect of the present disclosure. With reference to FIGS. **126**, **127** and **131**, at the outset, a user of the modular surgical instrument attaches the detachable modular components together such that the modular components of the modular surgical instrument are operably coupled together. For example, in various aspects, the user attaches **2902** a shaft assembly **2406** to the control handle assembly **2404** to operably couple the shaft assembly **2406** to the control handle assembly **2404**. A proximal end of the shaft assembly **2406** may be attached to the control handle assembly **2404**. The user attaches **2904** the transducer assembly **2402** to the control handle assembly **2404** to operably couple the transducer assembly **2402** to the control handle assembly **2404**. For example, a distal end of the transducer assembly **2402** may be attached to a proximal end of the control handle assembly **2404**. The user attaches **2906** the battery assembly **2408** to the control handle assembly **2404** to operably couple the battery assembly **2408** to the control handle assembly **2404**. As previously described, a plurality of control programs may be stored in the memory devices **2410**, **2414**, **2416**, **2418** by the processor **2412** according to a predetermined distribution. After the shaft assembly **2406**, transducer assembly **2402**, and battery assembly **2408** are each attached to the control handle assembly **2404**, the processor **2402** determines and identifies **2908** the plurality of control programs according to the predetermined distribution. In various aspects, the processor **2420** is alternatively located in one of the shaft assembly **2406**, transducer assembly **2402**, or battery assembly **2408**. The identified plurality of control programs stored according to the predetermined distribution in the memory devices **2410**, **2414**, **2416**, **2418** may be uploaded to a volatile memory device located in the control handle assembly **2404**, shaft assembly **2406**, transducer assembly **2402**, or battery assembly **2408**. In aspects, one of the memory devices **2410**, **2414**, **2416**, **2418** may comprise the volatile memory device. For example, the memory device **2414** may comprise the volatile memory device.

In some aspects, each of the memory devices **2410**, **2414**, **2416**, **2418** each stores a subset of the plurality of control programs according to the predetermined distribution. For example, the processor **2412** may determine and identify **2908** a first subset of the plurality of control programs in memory device **2414**, a second subset of the plurality of control programs in memory device **2416**, a third subset of the plurality of control programs in memory device **2410**, and a fourth subset of the plurality of control programs in memory device **2418**. In aspects, the first subset of the plurality of control programs comprises the plurality of base operating control programs corresponding to general operation such as the motor control **2504** and switch control **2506** control programs. Each of the first, second, third and fourth subsets of the plurality of control programs may be uploaded to the volatile memory device located in the control handle assembly **2404**, shaft assembly **2406**, transducer assembly **2402**, or battery assembly **2408**, as previously described. In various aspects, after the user attaches new modular variants of each of the transducer assembly **2402**, shaft assembly **2406**, and battery assembly **2408** to a previously used control handle assembly **2404**, a second plurality of control programs is stored in the memory devices corresponding to the attached new modular variants of the transducer assembly **2402**, shaft assembly **2406**, and battery assembly **2408** according to a second predetermined distribution. The new modular variants may be a new modular variant manufactured from a factory. In some aspects, the second plurality of control programs comprises different versions of the plurality of control programs.

Therefore, after the new modular variants are attached, the processor **2412** determines **2910** whether any of the second plurality of control programs are updated versions of the corresponding control program of the plurality of control programs. The updated control programs may correspond to a previous version of the updated control program stored in the memory device **2414**. For example, the memory device of a new modular transducer assembly may store an updated version of the 55 kHz control program **2510**. The memory device **2414** may have stored a previous version of the 55 kHz control program **2510**. Thus, the processor **2412** may determine **2910** that the 55 kHz control program **2510** is an updated version and upload the updated version to the memory device **2414**. In general, if the processor **2412** determines **2910** that at least one of the second plurality of control programs is an updated version corresponding to a previous version stored in the memory device **2414**, the processor **2412** uploads **2912** the updated version of the at least one control program to the memory device **2414**. The processor **2412** deletes **2914** the previous version of the at least one control program previously stored in the memory device **2414**. If the processor **2412** determines **2910** that the at least one of the second plurality of control programs is not an updated version, the processor **2412** does not upload **2912** the updated version.

The processor **2412** selects **2916** at least one of the plurality of control programs or second plurality of control programs to implement operation of the modular surgical instrument, as previously described in connection with FIG. **130**. As previously described in FIG. **130**, the processor **2412** downloads the at least one control program from the corresponding nonvolatile memory device to a volatile memory device. The processor **2412** may select **2916** based on a look-up table. As previously described, when the selected at least one control program is stored in the volatile memory device, the processor **2412** executes **2918** the selected at least one control program. The processor H**112** may continue to select H**616** and execute H**618** selected control programs continuously during operation of the modular surgical instrument. In various aspects, the processor **2412** may write usage data to the memory device of the corresponding modular component of selected control programs. For example, the processor **2412** may write **2920** time of usage data to the memory devices **2410**, **2416**, **2418** of the transducer assembly **2402**, shaft assembly **2406**, and battery assembly **2408** based on executing the usage counter **2526**, usage counter **2534**, usage counter **2540** control programs, respectively. For another example, the processor **2412** may write motor usage data to the memory device **2414** based on executing the motor control **2504** control program. In aspects, the processor **2412** may write data to the memory device **2416** such as time in use, maximum force, and shaft error data as well as control handle assembly **2404**, transducer assembly **2402**, and battery assembly **2408** serial number data. The processor **2412** also may write data to the memory device **2410** such as time in use, index of number of re-uses, error data as well as shaft assembly **2406**, control handle assembly **2404**, and battery assembly **2408** serial number data. The processor **2412** also may write data to the memory device **2418** such as number of rechargeable uses, end effector functional data, error data, and battery discharge data.

The processor **2412** determines **2922** whether any of the attached transducer assembly **2402**, shaft assembly **2406**, and battery assembly **2408**, or whether the control handle assembly **2404** has an updated version of the main RTOS or BIOS program. If the processor **2412** determines **2922** that there is an updated version of the main RTOS or BIOS program, the processor **2412** updates **2924** the main RTOS or BIOS program based on the updated version stored in the corresponding memory device **2410**, **2414**, **2416**, **2418**. If the processor **2412** determines **2922** that there is not an updated version of the main RTOS or BIOS program, the processor **2412** does not update **2924**. The process of the logic diagram **2900** terminates **2926**.

In one aspect, the present disclosure provides a battery powered modular surgical instrument, comprising: a control handle assembly comprising a processor coupled to a first memory device; a shaft assembly having a proximal end operably coupled to the control handle assembly and detachable from the control handle assembly, wherein the shaft assembly comprises a plurality of circuit modules and a second memory device, wherein a plurality of control programs is configured to operate the plurality of circuit modules, wherein each of the plurality of control programs comprises computer executable instructions; a transducer assembly operably coupled to the control handle assembly and detachable from the control handle assembly, wherein the transducer assembly comprises a third memory device and comprises a transducer that is configured to convert a drive signal to mechanical vibrations, wherein the plurality of control programs is configured to control the conversion of the drive signal to mechanical vibrations; a battery assembly operably coupled to the control handle assembly and detachable from the control handle assembly, wherein the battery assembly comprises a fourth memory device and the battery assembly is configured to power the modular surgical instrument.

The processor may comprise a primary controller and a safety controller. The plurality of control programs may be stored according to a predetermined distribution between the first memory device, second memory device, third memory device, or fourth memory device. The first memory device, second memory device, third memory device, or fourth memory device may be a nonvolatile memory device storing at least one control program of the plurality of control programs, wherein the processor is configured to download the at least one control program to a volatile memory device located in the control handle assembly, shaft assembly, transducer assembly, or battery assembly.

The processor may be configured to select at least one of the plurality of control programs based on a look-up table. The look-up table may comprise an indication that the selected at least one of the plurality of control programs corresponds to at least one of the shaft assembly or the transducer assembly. The plurality of control programs may comprise any one of a component identification program, a usage counter program, a real-time operating system (RTOS) program, an energy update program, and/or a motor control program.

The surgical instrument may further comprise an end effector coupled to a distal end of the shaft assembly; a motor positioned in the control handle assembly and configured to operate the end effector, wherein the motor control program is configured to control the operations of the motor. The processor may be a first processor, wherein one of the shaft assembly, transducer assembly or battery assembly comprises a second processor.

In another aspect, the present disclosure provides a method to operate a battery powered modular surgical instrument, comprising: a control handle assembly comprising a processor coupled to a first memory device, a shaft assembly having a proximal end operably coupled to the control handle assembly and detachable from the control handle assembly, wherein the shaft assembly comprises a plurality of circuit modules and a second memory device, wherein a plurality of control programs is configured to operate the plurality of circuit modules, wherein each of the plurality of control programs comprises computer executable instructions, a transducer assembly operably coupled to the control handle assembly and detachable from the control handle assembly, wherein the transducer assembly comprises a third memory device and comprises a transducer that is configured to convert a drive signal to mechanical vibrations, wherein at least one of the plurality of control programs is configured to control the conversion of the drive signal to mechanical vibrations, and a battery assembly operably coupled to the control handle assembly and detachable from the control handle assembly, wherein the battery assembly comprises a fourth memory device and the battery assembly is configured to power the modular surgical instrument, wherein the method comprises the steps of: identifying, by the processor, the plurality of control programs; selecting, by the processor, at least one of the plurality of control programs; executing, by the processor, at least one of the plurality of control programs.

The processor may comprise a primary controller and a safety controller, the method may further comprise executing at least one of the plurality of control programs by the primary controller and the safety controller. The method may comprise storing, by the processor, the plurality of control programs according to a predetermined distribution between the first memory device, second memory device, third memory device, or fourth memory device. The first memory device, second memory device, third memory device, or fourth memory device may be a nonvolatile memory device storing at least one control program of the plurality of control programs, the method further comprising: downloading, by the processor, the at least one control program to a volatile memory device located in the control handle assembly, shaft assembly, transducer assembly, or battery assembly. The method may further comprising: identifying, by the processor, a modular variant of one or more of the shaft assembly or transducer assembly; selecting, by the processor, the at least one control program based on determining that the at least one control program corresponds to the identified modular variant based on a look-up table.

In another aspect, the present disclosure provides a method to operate a battery powered surgical instrument, comprising: a control handle assembly comprising a processor coupled to a first memory device, a shaft assembly detached from the control handle assembly, wherein the shaft assembly comprises a plurality of circuit modules and a second memory device, wherein a plurality of control programs is configured to operate the plurality of circuit modules, wherein the plurality of control programs is stored in the first memory device, wherein each of the plurality of control programs comprises computer executable instructions, a transducer assembly detached from the control handle assembly, wherein the transducer assembly comprises a third memory device and comprises a transducer that is configured to convert a drive signal to mechanical vibrations, wherein at least one of the plurality of control programs is configured to control the conversion of the drive signal to mechanical vibrations, and a battery assembly detached from the control handle assembly, wherein the battery assembly comprises a fourth memory device and the battery assembly is configured to power the modular surgical instrument, wherein the method comprises the steps of: attaching, by a user of the surgical instrument, a proximal end of the shaft assembly to the control handle assembly to operably couple the shaft assembly to the control handle assembly; attaching, by the user of the surgical instrument, the transducer assembly to a distal end of the shaft assembly to operably couple the transducer assembly to the control handle assembly; attaching, by the user of the surgical instrument, the battery assembly to the control handle assembly to operably couple the battery assembly to the control handle assembly; selecting, by the processor, at least one of the plurality of control programs based on a look-up table; executing, by the processor, the at least one of the plurality of control programs.

The plurality of control programs may be a first plurality of control programs, the may method further comprise determining, by the processor, that a second plurality of control programs is stored according to a predetermined distribution between the second memory device, third memory device, or fourth memory device; uploading, by the processor, at least one of second plurality of control programs to the first memory device. The at least one second plurality of control programs may be an updated version of at least one of the first plurality of control programs, the method may further comprise deleting, by the processor, at least one of the first plurality of control programs from the first memory device. The plurality of control programs may be a first plurality of control programs, wherein the first plurality of control programs comprises a motor control program and a switch program, the method may further comprise identifying, by the processor, a second plurality of control programs stored in the second memory device; identifying, by the processor, a third plurality of control programs stored in the third memory device; identifying, by the processor, a fourth plurality of control programs stored in the fourth memory device.

The method may further comprise writing, by the processor, shaft assembly usage data to the second memory device based on executing, by the processor, at least one of the second plurality of control programs; writing, by the processor, transducer assembly usage data to the third memory device based on executing, by the processor, at least one of the third plurality of control programs; writing, by the processor, battery assembly usage data to the fourth memory device based on executing, by the processor, at least one of the fourth plurality of control programs. The method may further comprise writing, by the processor, usage data to the first memory device based on executing, by the processor, at least one of the second plurality of control programs.

Modular Battery Powered Handheld Surgical Instrument with Energy Conservation Techniques

In another aspect, the present disclosure provides a modular battery powered handheld surgical instrument with energy conservation techniques. Disclosed is a method of conserving energy in a surgical instrument that includes a segmented circuit having a plurality of independently operated circuit segments, a voltage control circuit, an energy source, a memory, and a processor coupled to the memory. The processor is configured to control a state of a circuit segment of the plurality of circuit segments. The state can be an energized state or a deenergized state. The method includes transmitting an energizing signal from the processor to a voltage control circuit, receiving the energizing signal by the voltage control circuit to apply a voltage to a circuit segment of the plurality of circuit segments, applying the voltage to the circuit segment by the voltage control circuit to cause the circuit segment to transition from the deenergized state to the energized state in accordance with an energization sequence that is different from a deenergization sequence.

FIGS. **62**, **63**, and **132-135** describe aspects of the present disclosure. In one aspect, the subject matter comprises a segmented circuit design for any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-70** to enable selectively powering individual circuit segments of a segmented circuit. The selective powering may be achieved in a controlled manner according to an energy conservation method. The controlled manner may include executing an energization sequence or a deenergization sequence by the segmented circuit. The energization sequence may be different from a deenergization sequence. In a non-use state of the surgical instrument, such as when there are no users using the surgical instrument, energy conservation is realized by executing an energization sequence that is different from a deenergization sequence. The selective powering technique according to the present disclosure enables verification of each of the plurality of circuit segments according to a power management mode before powering the circuit segments. The power management mode is further described below with reference to FIG. **132**. The surgical instrument may be a segmented modular mixed energy surgical instrument that enables control and customization of the surgical instrument. In various aspects, the surgical instrument may comprise a shaft, a handle, a transducer, a battery, a motor, and an end effector, as previously described in connection with FIGS. **1-70**

With reference back to FIG. **62**, there is illustrated the components of a control circuit **1300** of a surgical instrument, according to one aspect of the present disclosure. The control circuit **1300** comprises a processor **1302** coupled to a volatile memory **1304**, one or more sensors **1306**, a non-volatile memory **1308** and a battery **1310**. The surgical instrument may comprise a handle housing to house the control circuit **1300** and to contain general purpose controls to implement the power conservation mode further described in FIG. **132**. In general, the processor **1302** is electrically coupled to each of the plurality of circuit segments of the segmented circuit **1401** as illustrated in FIG. **63** to activate or deactivate the circuit segments in accordance with energization and deenergization sequences such as the example sequences described in FIGS. **132**, **133**, and **135**.

FIG. **63** is a system diagram **1400** of a segmented circuit **1401** comprising a plurality of independently operated circuit segments **1402**, **1414**, **1416**, **1420**, **1424**, **1428**, **1434**, **1440**, according to one aspect of the present disclosure. One or more sensors **1306** may include an accelerometer to verify the function or operation of each of the plurality of circuit segments, based on a safety check and a Power On Self Test (POST) further described in connection with FIG. **71**. The battery **1310** powers the surgical instrument by providing a source voltage that causes a current. The battery **1310** may comprise the motor control circuit segment block **1428** illustrated in FIG. **63**.

In the aspect shown in FIG. **133**, the plurality of circuit segments start first in the standby mode, transition second to the sleep mode, and transition third to the operational mode. However, in other aspects, the plurality of circuit segments may transition from any one of the three modes to any other one of the three modes. For example, the plurality of circuit segments may transition directly from the standby mode to the operational mode. Each individual circuit segment may be placed in a particular state by the voltage control circuit **1408** based on the execution by the processor **1302** of machine executable instructions. The states comprise a deenergized state, a low energy state, and an energized state. The deenergized state corresponds to the sleep mode, the low energy state corresponds to the standby mode, and the energized state corresponds to the operational mode. Transition to the low energy state may be achieved by, for example, the use of a potentiometer.

As further described with reference to FIG. **133**, the plurality of circuit segments may transition from the sleep mode or the standby mode to the operational mode in accordance with an energization sequence. The plurality of circuit segments also may transition from the operational mode to the standby mode or the sleep mode in accordance with a deenergization sequence. The energization sequence and the deenergization sequence may be different. In some aspects, the energization sequence comprises energizing only a subset of circuit segments of the plurality of circuit segments. In some aspects, the deenergization sequence comprises deenergizing only a subset of circuit segments of the plurality of circuit segments.

Referring back to the system diagram **1400** in FIG. **63**, the segmented circuit **1401** comprises a plurality of circuit segments comprising a transition circuit segment **1402**, a processor circuit segment **1414**, a handle circuit segment **1416**, a communication circuit segment **1420**, a display circuit segment **1424**, a motor circuit segment **1428**, an energy treatment circuit segment **1434**, and a shaft circuit segment **1440**. The transition circuit segment comprises a wake up circuit **1404**, a boost current circuit **1406**, a voltage control circuit **1408**, a safety controller **1410** and a POST controller **1412**. The transition circuit segment **1402** is configured to implement a deenergization and an energization sequence, a safety detection protocol, and a POST. The voltage control circuit **1408** may also comprise a processor and memory device, as illustrated in FIG. **134**.

FIG. **132** is a logic diagram **2000** of a process for controlling the operation of a surgical instrument according to the energy conservation method comprising the POST and operation verification, according to one aspect of the present disclosure. The logic diagram **2000** defines an energization sequence. With reference to FIGS. **62**, **63**, and **132**, at the outset, each of the plurality of circuit segments of the surgical instrument is in sleep mode, except for the transition circuit segment **1402**. The transition circuit segment **1402** is in operational mode and receives a voltage provided by the battery **1310**. Consequently, all circuits of the transition circuit segment **1402** are in an energized state. In one aspect, the logic diagram **2000** defines an energization sequence that begins after the accelerometer button sensor **1405** sends a signal to the voltage control circuit **1408**, as previously described. In other aspects, the energization sequence begins when another sensor such as a temperature sensor sends a signal to the voltage control circuit **1408**. The temperature sensor may send the signal when a temperature of the surgical instrument sensed by the temperature sensor exceeds a predetermined threshold.

After receiving the signal, the voltage control circuit **1408** applies voltage to the processor circuit segment **1414**, causing the processor **1302** and the volatile memory **1304** to transition to operational mode. The processor **1302** transmits an energizing signal to the voltage control circuit **1408** to energize **2002** the handle circuit segment **1416**, causing the voltage control circuit **1408** to apply voltage to the handle circuit segment **1416**, causing the handle control sensors **1418** to transition to operational mode. The POST controller **1412** performs **2004** a POST to verify proper operation of the handle circuit segment **1416**. If the POST is passed successfully, the handle circuit segment **1416** remains in operational mode. The POST controller **1402** also performs **2006** a POST to verify proper operation of the battery, such as by transmitting a signal to the accelerometer button sensor **1405** to sense an incremental motor pulse generated by the motor **1432**. The incremental motor pulse indicates that the battery is properly providing voltage to the motor **1432**.

The processor **1302** transmits an energizing signal to the voltage control circuit **1408** to energize **2008** the shaft circuit segment **1440**, causing the voltage control circuit **1408** to apply voltage to the shaft circuit segment **1440**, causing circuits of the shaft circuit segment **1440** to transition to operational mode. In various aspects, a detachable shaft assembly **110**, **490**, **510**, **1110**, **1210** (FIGS. **1**, **25**, **30**, **45**, **54**, respectively) comprises a memory device that stores a control program. The shaft circuit segment **1440** transmits an update signal to the processor **1302**. The update signal indicates the version of the control program stored in the memory device of the shaft assembly. When the processor **1302** receives the update signal, the processor **1302** compares **2010** the current version of the control program stored in the non-volatile memory **1308** with the version of the control program stored in the memory device of the shaft assembly. If the version is a new version, the processor **1302** updates **2012** the current version with the new version of the control program by storing the new version of the control program in the non-volatile memory **1308** and reboots **2012** the shaft assembly **110**, **490**, **510**, **1110**, **1210** or the surgical instrument **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** (FIGS. **1-70**).

To select **2014** a control program for the shaft assembly **110**, **490**, **510**, **1110**, **1210** (FIGS. **1**, **25**, **30**, **45**, **54**, respectively), the shaft module controller **1442** determines the set of shaft modules comprising the control program to be selected. The shaft module controller **1442** selects **2014** a shaft module of the set of shaft modules. The processor **1302** transmits an energizing signal to the voltage control circuit **1408** to energize **2016** the energy treatment circuit **1434**, causing the voltage control circuit **1408** to apply voltage to the energy treatment circuit **1434**, causing circuits of the energy treatment circuit **1434** to transition to operational mode. In various aspects, the detachable ultrasonic transducer **104**, **104'** (FIGS. **8** and **9**, respectively) comprises the energy treatment circuit **1434** and comprises a memory device that stores a control program. The energy treatment circuit segment **1434** transmits an update signal to the processor **1302**. The update signal indicates the version of the control program stored in the memory device of the ultrasonic transducer **104**, **104'**. When the processor **1302** receives the update signal, the processor **1302** compares **2018** the current version of the control program stored in the non-volatile memory **1308** with the control program version stored in the memory device of the ultrasonic transducer **104**, **104'** for updating purposes.

To select **2020** a control program for the ultrasonic transducer **104**, **104'** (FIGS. **8** and **9**, respectively), the processor **1302** determines the energy treatment modality corresponding to the selected shaft modality. The processor **1302** executes a control program corresponding to the determined energy treatment modality. One of or a combination of the RF amplifier and safety circuit **1436** and an ultrasonic signal generator circuit **1438** implement the determined energy treatment modality. In other aspects, a transducer controller located at the ultrasonic transducer **104**, **104'** may compare **2018** the control program versions and select **2020** a control program. As described above, the ultrasonic transducer and the shaft **110**, **490**, **510**, **1110**, **1210** (FIGS. **1**, **25**, **30**, **45**, **54**, respectively) implement a modality of the surgical instrument. The processor **1302** is configured to identify **2022** any mismatch between the ultrasonic transducer **104**, **104'** and the shaft assembly **110**, **490**, **510**, **1110**, **1210**. For example, the processor **1302** can detect a mismatch and transmit an energizing signal to the voltage control circuit **1408** to cause the voltage control circuit **1408** to apply voltage to the display circuit segment **1424**. The processor **1302** subsequently transmits a warning signal to the LCD display **1426** to cause the LCD display **1426** to generate **2024** a user warning. If the processor **1302** does not detect a mismatch, the processor **1302** transmits a match signal to the LCD display **1426** to cause the LCD display **1426** to generate **2026** an indication that the ultrasonic transducer **104**, **104'** and the shaft assembly **110**, **490**, **510**, **1110**, **1210** match. Based on the indication of the match, a user of the surgical instrument may operate the surgical instrument.

The energization sequence defined by the logic diagram **2000** of FIG. **132** differs from the deenergization sequence defined by the logic diagram **2100** of FIG. **133**. According to one aspect of the present disclosure, energizing circuits segments of the plurality of circuit segments in accordance with an energization sequence that is different from a deenergization sequence enables energy conservation through a reduction of power consumption of the surgical instrument. Specifically, energizing the segmented circuit **1401** with a different sequence from the energization sequence enables delaying or avoiding transitioning individual circuit segments with relatively high power consumption requirements to operational mode. For example, in the energization sequence defined by logic diagram **2000**, the motor circuit segment **1428** is not transitioned from sleep mode or standby mode until a user of the surgical instrument requires the motor circuit segment **1428** to be in operational mode. Non-use power drain is reduced based on this delay or avoidance in transitioning the motor circuit segment **1428** because the motor circuit segment **1428** requires relatively high power consumption in comparison to other circuit segments of the plurality of circuit segments.

Moreover, in the energization sequence defined by logic diagram **2000**, the communication circuit segment **1420** and the motor circuit segment **1428** are not transitioned into an energized state. Therefore, the energization sequence defined by logic diagram **2000** differs from a deenergization sequence that comprises deenergizing each circuit segment of the plurality of circuit segments. Energy conservation is achieved based on avoiding non-use power drain of the communication circuit segment **1420** and the motor circuit segment **1428** while the other circuit segments of the plurality of circuit segments are in operational mode and a user of the surgical instrument is not using the surgical instrument. More generally, an energization sequence may be defined to conserve energy by for example, energizing circuit segments that have low power consumption requirements before energizing circuit segments that have high power consumption requirements and skipping the energization of circuit segments. If a user of the surgical instrument desires to view diagnostic information about the surgical instrument but does not require the motor of the surgical instrument in operation, an energization sequence may be defined that comprises energizing the display circuit segment **1424** and does not comprise energizing the motor circuit segment **1428**.

In various aspects, the communication circuit segment **1420** and the motor circuit segment **1428** are energized into operational mode for the use of the user of the surgical instrument, after a certain time. The certain time in which the communication circuit segment **1420** and the motor circuit segment **1428** remain deenergized results in energy conservation. More generally, energization of only a subset of the plurality of circuit segments results in energy conservation, even if the remaining circuit segments are energized later.

FIG. **133** is a logic diagram **2100** of a process for controlling the operation of a surgical instrument according to the energy conservation method comprising a deenergization sequence and an energization sequence, according to one aspect of the present disclosure. With reference to FIGS. **62**, **63**, and **133**, the logic diagram **2100** defines a deenergization sequence and an energization sequence. At the outset, each of the plurality of circuit segments of the surgical instrument is in operational mode and one or more circuit segments are performing a function. In various aspects, a subset of the plurality of circuit segments may be in standby mode or sleep mode at the outset of performing a deenergization sequence. In the deenergization sequence defined by the logic diagram **2100**, each circuit segment of the plurality of circuit segments is transitioned from operational mode to In various aspects, the predefined standby mode transition may comprise transitioning a different subset of the plurality of circuit segments to standby mode standby mode or sleep mode. In some aspects, the deenergization sequence begins after the processor **1302** determines the surgical instrument has remained **2102** in an inactive state for a predetermined time t. An inactive state for a predetermined time t is defined by none of the plurality of circuit segments performing a function for the duration of time t. In various aspects, predetermined time t may be set based on input by a user of the surgical instrument. For example, the predefined time t may be ten seconds, thirty seconds, one minute, or some other predetermined amount of time indicated by a user through the GUI. Failure to perform a function by any individual circuit segment of the plurality of circuit segment during elapsed time t indicates an inactive state of the surgical instrument.

The processor **1302** determines the correct deenergization sequence to initiate. In some aspects, the deenergization sequence to be initiated may be stored in the nonvolatile memory **1308** of the control circuit. In other aspects, the deenergization sequence to be initiated may be determined based on, for example, user initiated motions sensed by the accelerometer button sensor **1405** or other sensor, user input through the GUI, or a deenergization sequence stored in an external device that is transferred to the nonvolatile memory **1308** through an electrical connection. The processor **1302** transmits a first series of signals to the voltage control circuit **1408** according to the determined deenergization sequence. The deenergization sequence defined by the logic diagram **2100** comprises a standby mode transition sequence and a sleep mode transition sequence. The deenergization sequence may be determined based on deenergizing the circuit segments in decreasing order, starting with the circuit segment with the highest power consumption requirement to the circuit segment with the lowest power consumption requirement. The standby mode transition sequence begins when the voltage control circuit **1408** receives the series of signals, causing the voltage control circuit **1408** to reduce voltage applied to the LCD display **1426** to dim **2104** the LCD display **1426**. The dimmed LCD display **1426** is in a low power state. After a predetermined time t_1, if the inactive state persists, the voltage control circuit **1408** deenergizes **2106** by reducing voltage applied to the display circuit segment **1424** to transition the display circuit segment **1424** to a low power state. After a predetermined time t_2, if the inactive state persists, the voltage control circuit **1408** deenergizes **2108** by reducing voltage applied to the energy treatment circuit segment **1434** to transition the energy treatment circuit segment **1434** to a low power state.

After the predetermined time t_3 elapses with the surgical instrument remaining in the inactive state, the standby mode transition sequence terminates and the sleep mode transition sequence begins when the processor **1302** transmits a second series of signals to the voltage control circuit **1408** according to the determined deenergization sequence. The voltage control circuit **1408** receives the second series of signals, causing the voltage control circuit **1408** to remove voltage from circuit segments in accordance with the sleep mode transition sequence. Over the duration of a predetermined time t_4, if the inactive state persists, the voltage control circuit **1408** deenergizes **2110**, **2114** by removing voltage applied to the motor circuit segment **1428** and shaft circuit segment **1440**, and deenergizes **2112** by reducing voltage applied to the handle circuit segment **1416**. The motor circuit segment **1428** and shaft circuit segment **1440** transition to a deenergized state. The handle circuit segment **1416** transitions to a low power state. The handle control sensors **1418** detect an actuation of one or more handle controls of the surgical instrument at a reduced sampling rate. Over the duration of a predetermined time t_5, if the inactive state persists, the voltage control circuit **1408** deenergizes **2116**, **2118**, **2120** by removing voltage applied to the communication circuit segment **1420**, the handle circuit segment **1416**, and the processor circuit segment **1414**. At the end of t_6, the sleep mode transition sequence terminates. The transition circuit segment **1402** remains in an energized state. Energy conservation is achieved based on transitioning circuit segments with higher power consumption requirements to circuit segments with lower power consumption, in the order of descending power consumption requirements. In various aspects, if the deenergization sequence is interrupted by such as, for example, a user of the surgical instrument actuating the energy modality actuator, the processor **1302** will transmit a signal to voltage control circuit **1408** indicating which circuit segments already transitioned to a low power or a deenergized state in accordance with the interrupted deenergization sequence. Consequently, when the deenergization sequence restarts after interruption, the voltage control circuit **1408** can skip the previously completed steps of the deenergization sequence to further conserve energy.

The voltage control circuit **1408** energizes **2122** by applying voltage to the processor circuit segment **1414** to transition the processor circuit segment **1414** to an energized state. The processor **1302** determines the correct energization sequence to initiate. In some aspects, the energization sequence to be initiated may be stored in the nonvolatile memory **1308** of the control circuit. In other aspects, the energization sequence to be initiated may be determined based on, for example, user initiated motions sensed by the accelerometer button sensor **1405** or other sensor, user input through the GUI, or an energization sequence stored in an external device that is transferred to the nonvolatile memory **1308** through an electrical connection. The energization sequence may be determined based on energizing the circuit segments in increasing order, starting with the circuit segment with the lowest power consumption requirement to the circuit segment with the highest power consumption requirement. The processor **1302** transmits a third series of signals to the voltage control circuit **1408** according to the determined energization sequence. The determined energization sequence begins when the voltage control circuit **1408** receives the third series of signals causing the voltage control circuit **1408** to apply voltage in accordance with the energization sequence.

The voltage control circuit **1408** energizes **2124** by applying voltage to the handle circuit segment **1416** to transition the handle circuit segment **1416** to an energized state. The voltage control circuit **1408** energizes **2126** by applying voltage to the display circuit segment **1424** to transition the display circuit segment **1424** to an energized state. The voltage control circuit **1408** energizes **2128** by applying voltage to the communication circuit segment **1420** to transition the communication circuit segment **1420** to an energized state. The voltage control circuit **1408** energizes **2130** by applying voltage to the motor circuit segment **1428** to transition the motor circuit segment **1428** to an energized state. The voltage control circuit **1408** energizes **2132** by applying voltage to the energy treatment circuit segment **1434** to transition the energy treatment circuit segment **1434** to an energized state. The voltage control circuit **1408** energizes **2134** by applying voltage to the energy treatment circuit segment **1434** to transition the energy treatment circuit segment **1434** to an energized state.

Energy conservation is achieved based on transitioning circuit segments with lower power consumption requirements to circuit segments with higher power consumption, in the order of ascending power consumption requirements. Therefore, even if all circuit segments of the plurality of circuit segments are transitioned to an energized state in accordance with ascending power consumption energization sequence, those circuit segments with relatively high power consumption requirements will have remained in a low power or deenergized state for a longer time in comparison to if those circuit segments transitioned according to an energization sequence that is not an ascending power consumption energization sequence. In various aspects, if the energization sequence is interrupted by for example, a user of the surgical instrument actuating the energy modality actuator, the processor **1302** will transmit a signal to voltage control circuit **1408** indicating which circuit segments already transitioned to an energized state in accordance with the interrupted energization sequence. Consequently, when the energization sequence restarts after interruption, the voltage control circuit **1408** can skip the previously completed steps of the energization sequence to further conserve energy. As previously described, the energization sequence may comprise energizing only a subset of circuit segments of the plurality of the circuit segments. Because the remaining circuit segments of the plurality of the circuit segments were not energized, the non-use power drain of the surgical instrument is reduced. For example, in the energized sequence defined by the logic diagram **2100**, the shaft circuit segment **1440** did not transition to an energized state. Moreover, skipping a circuit segment with relatively high power consumption requirements achieves additional energy conservation. For example, the user may only need to use the LCD display **1426**. An energization sequence comprising only the display circuit segment **1424** avoid the relatively power consumption requirements of the motor circuit segment **1428** and the energy treatment circuit segment **1434**. Therefore, combining the energy conservation methods described above can achieve more energy conservation.

FIG. **134** illustrates a control circuit **2200** for selecting a segmented circuit of a surgical instrument, according to an aspect of the present disclosure. With reference to FIGS. **62**, **63**, and **134**, the control circuit **2200** comprises a processor **2202**, a memory device **2204**, a multiplexer (MUX) **2206**, and an energy source **2208**. The processor **2202**, memory device **2204**, MUX **2206**, and energy source **2208** are coupled to each other and to the segmented circuit **1401** for electrical communication through one or more wired or wireless connection media. The processor **2202** may be, for example, an Analog Devices ADSP-21469 SHARC Digital Signal Processor, available from Analog Devices, Norwood, MA. The memory device **2204** may be a RAM, DRAM, SDRAM, ROM, EPROM, EEPROM, flash memory, or other suitable memory device. The MUX **2206** may be, for example, a power MUX such as the TPS2110 Power MUX, available from Texas Instruments, Dallas, TX or a plurality of suitable power MUXes. The energy source **2208** may be a battery such as the 14.4 volt nickel metal hydride (NiMH) SmartDriver Battery, available from MicroAire Surgical Instruments, Charlottesville, VA. In aspects, the voltage control circuit **1408** may comprise the control circuit **2200**. The MUX **2206** comprises a plurality of select lines S_0 to S_n, a plurality of inputs corresponding to circuit segments of the segmented circuit **1401**, and an output. In other aspects, the MUX **2206** may instead be a plurality of MUXes.

The energy source **2208** provides voltage to the MUX **2206**. When the MUX **2206** selects an input of the plurality of inputs to be output, the circuit segment corresponding to the selected input is energized or deenergized. The control circuit **2200** comprises a plurality of electronic switches. Each individual electronic switch of the plurality of electronic switches is configured to switch between an open and closed configuration. The plurality of electronic switches may be solid state devices such as transistors or other types of switches such as wireless switches, ultrasonic switches, accelerometers, inertial sensors, among others. The electronic switch of the selected input switches to an open configuration if the control circuit **2200** is implementing a deenergization sequence comprising deenergizing the selected circuit segment. The electronic switch of the selected input switches to a closed configuration if the control circuit **2200** is implementing an energization sequence comprising energizing the selected circuit segment.

FIG. **135** is a logic diagram **2300** of a process for controlling the operation of a surgical instrument according to the energy conservation method comprising a deenergization sequence and an energization sequence that is different from the deenergization sequence, according to one aspect of the present disclosure. With reference to FIGS. **62**, **63**, and **135**, the energy conservation method is performed with a surgical instrument comprising a segmented circuit **1401**, a memory device **1304**, a processor **1302**, a voltage control circuit **1408**, a safety controller **1410**, a POST controller **1412**, and an energy source **2208**. The processor **1302** is configured to control a state of each circuit segment of the plurality of circuit segments. The energy conservation method begins when the processor **1302** determines **2302** the existing state of the plurality of circuit segments. The existing state can be an energized state or a deenergized state. The processor **1302** determines **2304** whether to initiate an energization sequence or a deenergization sequence. As described previously, an energization or deenergization sequence may be stored in the nonvolatile memory **1308** or determined based on input of a user of the surgical instrument. After determining the energization or deenergization sequence, the processor **1302** transmits **2306** a signal to the voltage control circuit **1408**. If the determined sequence is an energization sequence, the signal is an energizing signal. If the determined sequence is a deenergization sequence, the signal is a deenergizing signal. The voltage control circuit **1408** receives **2308** the transmitted signal. If the transmitted signal is an energizing signal, receiving the transmitted signal causes the voltage control circuit **1408** to apply a voltage to a circuit segment of the plurality of circuit segments. The plurality of circuit segments may comprise a transition circuit segment **1402**, a processor circuit segment **1414**, a handle circuit segment **1416**, a communication circuit segment **1420**, a display circuit segment **1424**, a motor circuit segment **1428**, an energy treatment circuit segment **1434**, and a shaft circuit segment **1440**. If the transmitted signal is a deenergizing signal, receiving the transmitted signal causes the voltage control circuit **1408** to remove a voltage from a circuit segment of the plurality of circuit segments.

The voltage control circuit **1408** determines **2310** if a first circuit segment is the correct circuit segment. The correct circuit segment is determined **2310** by the voltage control circuit **1408** based on whether the circuit segment is in accordance with the energization or deenergization sequence determined by the processor **1302**. If the voltage control circuit **1408** determines **2310** that the first circuit segment is an incorrect circuit segment, the voltage control circuit **1408** switches **2312** to a second different circuit segment and determines **2310** whether the second circuit segment is the correct circuit segment. If the voltage control circuit **1408** also determines **2310** that the second circuit segment is an incorrect circuit segment, the voltage control circuit **1408** switches **2312** to a third different circuit segment. The voltage control circuit **1408** switches to each of the circuit segments of the plurality of circuit segments until the voltage control circuit **1408** determines the correct circuit segment. After the voltage control circuit **1408** determines **2310** the correct circuit segment, the voltage control circuit **1408** determines **2314** whether the received signal is an energizing signal or a deenergizing signal. If the received signal is determined **2314** to be an energizing signal, a POST controller **1412** performs **2316** a POST to verify **2318** proper operation of the correct circuit segment. If the correct circuit segment fails the POST, proper operation is not verified and a GUI of the surgical instrument may display **2320** a user warning. In aspects, the POST may comprise verifying the identity of the correct circuit segment with an encryption parameter. In other aspects, the POST may comprise verifying proper operation of the correct circuit segment with an accelerometer such as the accelerometer button sensor **1405**. In various aspects, the POST controller **1412** may initiate a locking protocol comprising disabling operation the correct circuit segment. A locked out circuit segment may function substantially similarly to a circuit segment in standby mode or sleep mode. The POST controller **1412** may also initiate a reduced performance mode of the correct circuit segment. In other aspects, a safety controller **1410** may also initiate the locking protocol or reduced performance mode, such as during operation of the surgical instrument. In aspects, one or more secondary processors of the surgical instrument may be configured to perform critical functions when the processor **1302** is determined to be in a malfunctioning state based on a POST of the processor circuit segment **1414**.

If the correct circuit segment passes the POST, the voltage control circuit **1408** applies **2322** voltage to the correct circuit segment to cause the correct circuit segment to transition from the deenergized state to the energized state. If the received signal is determined **2314** to be a deenergizing signal, the voltage control circuit **1408** removes **2322** voltage from the correct circuit segment to cause the correct circuit segment to transition from the energized state to the deenergized state. In some aspects, the POST controller **1412** performs a POST on the correct circuit segment after a deenergizing signal is received. The voltage control circuit **1408** determines **2324** whether the determined sequence comprises one or more circuit segments that have not transitioned in accordance with the determined sequence. If the voltage control circuit **1408** determines **2324** that there are one or more circuit segments to transition in accordance with the determined sequence, the logic diagram **2300** terminates **2326**. The processor **1302** transmits **2306** a signal to the voltage control circuit **1408**. If the voltage control circuit **1408** determines **2324** that there are further circuit segments to transition in accordance with the determined sequence, the voltage control circuit **1408** identifies **2328** a circuit segment of the one or more circuit segments. The voltage control circuit **1408** transmits **2330** a signal identifying a circuit segment of the one or more circuit segments. Based on the received signal, the processor **1302** transmits **2306** another signal to the voltage control circuit **1408**. Therefore, the process defined by the logic diagram **2300** restarts at the transmit **2306** signal step. The process defined by the logic diagram **2300** continues until the process terminates **2326**.

In one aspect, the present disclosure provides a method of conserving energy in a surgical instrument, the surgical instrument comprising a segmented circuit comprising a plurality of independently operated circuit segments, a voltage control circuit, an energy source, a memory device, and a processor coupled to the memory device, wherein the processor is configured to control a state of a circuit segment of the plurality of circuit segments, wherein the state can be an energized state or a deenergized state, the method comprising: transmitting an energizing signal from the processor to a voltage control circuit; receiving the energizing signal by the voltage control circuit to cause the voltage control circuit to apply a voltage to a circuit segment of the plurality of circuit segments; applying the voltage to the circuit segment by the voltage control circuit to cause the circuit segment to transition from the deenergized state to the energized state in accordance with an energization sequence that is different from a deenergization sequence.

The method comprises deenergizing, by the voltage control circuit, the circuit segment from the energized state to the deenergized state in accordance with the deenergization sequence. The deenergization sequence may comprise transmitting a deenergizing signal from the processor to the voltage control circuit; receiving the deenergizing signal by the voltage control circuit to cause the voltage control circuit to remove the voltage from the circuit segment; removing the voltage applied to the circuit segment, by the voltage control circuit, to cause the circuit segment to transition from the energized state to the deenergized state in accordance with the deenergization sequence that is different from the energization sequence.

Applying the voltage to the circuit segment by the voltage control circuit may comprise applying the voltage to: a handle control sensor circuit, a LCD circuit, a communication circuit, a motor circuit, an energy treatment circuit and a modular power circuit. Applying the voltage to the circuit segment by the voltage control circuit may comprise applying the voltage to: the energy treatment circuit, wherein the energy treatment circuit comprises a RF control circuit and an ultrasonic signal generator the modular power circuit, wherein the modular power circuit comprises a shaft module controller, wherein the shaft module controller is coupled to a modular control actuator, an end effector sensor, and a nonvolatile memory. Prior to applying the voltage to the circuit segment in accordance with the energization sequence the method further comprises performing a safety check, by a safety controller, and performing a Power On Self Test (POST), by a POST controller, to verify proper operation of the circuit segment.

The method may further comprise implementing a locking protocol, wherein the locking protocol comprises disabling operation of the circuit segment by the safety controller or the POST controller. The POST may comprise verifying proper operation of the circuit segment with an accelerometer. The POST may comprise verifying the identity of the circuit segment with an encryption parameter.

In another aspect, the present disclosure provides a surgical instrument comprising: a segmented circuit comprising a plurality of independently operated circuit segments, wherein a state of a circuit segment of the plurality of circuit segments can be an energized state or a deenergized state; a memory device; a processor coupled to the memory device, wherein the processor is configured to transmit an energizing signal; a voltage control circuit, wherein the voltage control circuit is configured to receive the energizing signal to be transmitted by the processor, the energizing signal to cause the voltage control circuit to apply a voltage to a circuit segment of the plurality of circuit segments, the voltage to be applied to cause the circuit segment to transition from the deenergized state to the energized state in accordance with an energization sequence that is different from a deenergization sequence; and an energy source to power the surgical instrument.

The processor may be further configured to transmit a deenergizing signal; the voltage control circuit is further configured to receive the deenergizing signal. Deenergizing the signal to cause the voltage control circuit to remove the voltage from the circuit segment, the voltage to be removed to cause the circuit segment to transition from the energized state to the deenergized state in accordance with the deenergization sequence that is different from the energization sequence.

The plurality of circuit segments may comprise a handle control sensor circuit, a LCD circuit, a communication circuit, a motor circuit, an energy treatment circuit and a modular power circuit. The energy treatment circuit comprises a RF control circuit and an ultrasonic signal generator, wherein the modular power circuit comprises a shaft module controller, wherein the shaft module controller is coupled to a modular control actuator, an end effector sensor, and a nonvolatile memory.

The surgical instrument may further comprise a safety controller configured to perform a safety check; and a POST controller configured to perform a Power On Self Test (POST), the safety check and the POST to verify proper operation of the circuit segment.

In another aspect, the present disclosure provides a surgical instrument comprising a segmented circuit comprising a plurality of independently operated circuit segments, wherein a state of a circuit segment of the plurality of circuit segments can be an energized state or a deenergized state; a memory device; a wake up circuit to determine an energization sequence based on an output of an accelerometer, wherein the energization sequence is different from a deenergization sequence; a processor coupled to the memory device, wherein the processor is configured to transmit an energizing signal in accordance with the energization sequence; a voltage control circuit, wherein the voltage control circuit is configured to receive the energizing signal to be transmitted by the processor, the energizing signal to cause the voltage control circuit to apply a voltage to a circuit segment of the plurality of circuit segments, the voltage to be applied to cause the circuit segment to transition from the deenergized state to the energized state in accordance with the energization sequence; and an energy source to power the surgical instrument.

The plurality of circuit segments may comprise a handle control sensor circuit, a LCD circuit, a communication circuit, a motor circuit, an energy treatment circuit and a modular power circuit. The energy treatment circuit may comprise a RF control circuit and an ultrasonic signal generator, wherein the modular power circuit comprises a shaft module controller, wherein the shaft module controller is coupled to a modular control actuator, an end effector sensor, and a nonvolatile memory.

The surgical instrument may further comprise a safety controller configured to perform a safety check; and a POST controller configured to perform a Power On Self Test (POST), the safety check and the POST to verify proper operation of the circuit segment. The surgical instrument may further comprise a secondary processor configured to perform a critical function when the processor is determined to be in a malfunctioning state.

Modular Battery Powered Handheld Surgical Instrument with Voltage Sag Resistant Battery Pack

In another aspect, the present disclosure provides a modular battery powered handheld surgical instrument with voltage sag resistant battery pack. Disclosed is a system including a battery assembly for use in a modular handheld surgical instrument, wherein the battery assembly includes a primary battery pack, a secondary battery pack, and a circuit configured to supply a first output voltage to a first load of the modular handheld surgical instrument via the primary battery pack, supply a second output voltage to a second load of the modular handheld surgical instrument via the secondary battery pack, and upon sensing a sagging voltage from the primary battery pack, supply a supplemental voltage to the first load, via the secondary battery pack, to prevent the battery assembly from supplying an output voltage below a predetermined threshold output voltage under the first load.

A battery pack, optionally disposable, that includes both primary and secondary cells or at least two different types of primary cells that can compensate within the battery pack for a voltage sag to prevent the battery pack output from sagging below a predetermined threshold under load is disclosed.

According to one aspect of the present disclosure, a voltage sag resistant battery pack is disclosed for use with any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-70**. The battery assembly **400**, disposable battery assembly **410**, reusable battery assembly **420**, or battery assembly **430**, described in FIGS. **16**, **17**, **18**, and **19** respectively above, may comprise a voltage sag resistant battery pack as described herein. In particular, the battery assembly **400**, disposable battery assembly **410**, reusable battery assembly **420**, or battery assembly **430** may comprise at least a switch mode power supply circuit (e.g., FIG. **22**) or a linear power supply circuit (e.g., FIG. **24**) to restore a sagging output voltage. Various structural aspects associated with each battery assembly **400**, **410**, **420**, **430** have been described above. Accordingly, for conciseness and clarity of disclosure, such structural aspects of each battery assembly **400**, **410**, **420**, **430** are incorporated herein by reference and will not be repeated here.

Turning now to FIG. **36**, there is shown a circuit **710** of the surgical instrument **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-70** according to various aspects of the present disclosure. In particular, the circuit **710** comprises a power supply **712** configured to deliver an output voltage and to restore a sagging output voltage. For example, the power supply **712** may be located within a housing of the battery assembly **400**, **410**, **420**, **430** in addition to a primary battery pack including a primary battery **715**, **717**, a secondary battery pack including a secondary battery **720**, and a switch mode power supply **727**. In one aspect, the primary battery **715**, **717** and the secondary battery **720** each may comprise a different chemical composition, such as, for example, Li-ion, NiMH, NiCd, and the like. In another aspect, the primary battery **715**, **717** and/or the secondary battery **720** may be a rechargeable battery.

In yet another aspect, the primary battery **715**, **717** and the secondary battery **720** may be non-rechargeable batteries such that the primary battery pack and the secondary battery pack are disposable. As shown in FIG. **36**, the primary battery **715**, **717** may be used to directly power the motor control circuits **726** and the energy circuits **732** and a separate secondary battery **720** may be used to drive the switch mode power supply **727** to power not only the motor control circuits **726** and the energy circuits **732** but also the handle electronic circuits **730**. As discussed above, the primary battery **715**, **717** may comprise a number of Li-ion batteries (e.g., four) and the secondary battery **720** may comprise a number of NiMH batteries (e.g., two), for example, as shown in FIG. **17**.

In this vein, FIG. **22** depicts an example switch mode power supply circuit **460** usable in power supply **712** of FIG. **36**. In particular, the switch mode power supply circuit **460** comprises a switching regulator **464** configured to produce an output voltage V_{out} from an input voltage V_{in}. With reference to FIG. **36** in view of FIG. **21**, the input voltage V_{in} in FIG. **22** corresponds to V_x in FIG. **21** delivered by the secondary battery **454** and the output voltage V_{out} in FIG. **22** corresponds to V_b in FIG. **21** delivered by the switch mode power supply circuit **460**. Details of an example switching regulator **464** are depicted in FIG. **23** described above. Notably, in line with the description of FIG. **21** above, the output voltage V_{out} of FIG. **22** may be greater than a desired output voltage V_o of the supplemental power source circuit **450** to accommodate a voltage drop across the diode **458**. In particular, as shown in FIG. **21**, the diode **458** may be placed at the output of switch mode power supply **727** in FIG. **36** to allow current to flow only from the switch mode power supply **727** to the motor control circuits **726**, the energy circuits **732**, and the handle electronic circuits **730** and to sense voltage sag at the output of the switch mode power supply **727**. Ultimately, the output voltage V_o may be a voltage required by (i) the motor control circuits **726** to drive the motor **729** and/or (ii) the energy circuits **732** to drive the ultrasonic transducer **734** and/or the high-frequency (e.g., RF) current electrode **736**.

Again in reference to FIG. **36** in view of FIG. **21**, when the switch **718**, **453** (FIGS. **36**, **21**) is closed, the primary battery **715**, **717**, **452***a*, **452***b*, **452***c*, **452***d* (FIGS. **36**, **21**) delivers the output voltage (V_o) directly to the motor control circuits **726** and the energy circuits **732**. For example, the switch **718** may be closed when the primary battery pack including primary battery **715**, **717** is inserted into the battery assembly **400**, **410**, **420**, **430** (FIGS. **16-19**). It should be understood that the switch **718** may be eliminated such that primary battery **715**, **717** is electrically connected to the circuit **710** when the power supply **712** is attached to the surgical instrument **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-70**.

In a first scenario, as the primary battery **715**, **717** maintains the output voltage V_o (e.g., no sagging V_o, V_o required by (i) the motor control circuits **726** to drive the motor **729** and/or (ii) the energy circuits **732** to drive the ultrasonic transducer **734** and/or the high-frequency (e.g., RF) current electrode **736** is supplied) the secondary battery **720** drives the switch mode power supply **727** to power handle electronics circuits **730** (e.g., to drive control electronics) when switch **723** is closed. As described above, the switch **723** may be closed when the secondary battery pack including the secondary battery **720** is inserted into the battery assembly **400**, **410**, **420**, **430**. It should be understood that the switch **723** may be eliminated such that the secondary battery **720** is electrically connected to the circuit **710** when the power supply **712** is attached to the surgical instrument **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-70**. Notably, with reference to FIG. **21**, when the output voltage V_o is not sagging, the output voltage V_o is greater than the output voltage V_b from the switch mode power supply **456**, and the diode **458** is off and does not conduct.

In a second scenario, if/when the primary battery **715**, **717** is unable to maintain the output voltage V_o (e.g., voltage sagging occurs or begins to occur, V_o required by (i) the motor control circuits **726** to drive the motor **729** and/or (ii) the energy circuits **732** to drive the ultrasonic transducer **734** and/or the high-frequency (e.g., RF) current electrode **736** is not suppliable), the secondary battery **720** drives the switch mode power supply **727** to not only deliver power to the handle electronics circuits **730** but also to deliver supplemental power to the motor control circuits **726** and/or the energy circuits **732** when switch **723** is closed. Similar to above, the switch **723** may be closed when the secondary battery pack including the secondary battery **720** is inserted into the battery assembly **400**, **410**, **420**, **430**. It should be understood that the switch **723** may be eliminated such that the secondary battery **720** is electrically connected to the circuit **710** when the power supply **712** is attached to the surgical instrument **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-70**. This supplemental power enables the primary battery to maintain the output voltage V_o demanded by the motor control circuits **726** and/or the energy circuits **732** or at least a minimum V_o required by the motor control circuits **726** and/or the energy circuits **732**. Notably, with reference to FIG. **21**, when the output voltage V_o sags below V_b by more than the diode **458** turn-on voltage (e.g., ~0.7V), the diode **458** turns-on and conducts at which point the output V_b minus ~0.7V is applied to the output V_o. This condition persists until the output voltage V_o increases above V_b to turn off the diode **458**. Accordingly, as described in FIG. **21** above, the switch mode power supply **727** in FIG. **36** is capable of supplying additional current to the motor control circuits **726** and/or the energy circuits **732** when needed.

In another aspect of the present disclosure, a linear power supply circuit **470** (FIG. **24**) may supplant the switch mode power supply circuit **456**, **460**, **727** (FIGS. **21**, **22**, **36**). In light of FIG. **36**, the power supply **712** may include a battery assembly **400**, **410**, **420**, **430** comprising a primary battery pack including a primary battery **715**, **717**, a secondary battery pack including a secondary battery **720**, and a linear power supply (e.g., replacing switch mode power supply **727** shown in FIG. **36**). In one aspect, the primary battery **715**, **717** and the secondary battery **720** each may comprise a different chemical composition (e.g., Li-ion, NiMH, NiCd, among others).

In another aspect, the primary battery **715**, **717** and/or the secondary battery **720** may be a rechargeable battery. In yet another aspect, the primary battery **715**, **717** and the secondary battery **720** may be non-rechargeable batteries such that the primary battery pack and the secondary battery pack are disposable. Similar to the description above, the circuit **710** comprising the power supply **712** is configured to deliver an output voltage and to restore a sagging output voltage. In light of FIG. **36**, the primary battery **715**, **717** may be used to directly power the motor control circuits **726** and the energy circuits **732** and a separate secondary battery **720** may be used to drive the linear power supply to power not only the motor control circuits **726** and the energy circuits **732** but also the handle electronic circuits **730**. For example, the primary battery **715**, **717** may comprise a number of Li-ion batteries (e.g., four) and the secondary battery **720** may comprise a number of NiMH batteries or NiCd batteries (e.g., two) as shown in FIG. **17**, for example. In this vein, FIG. **24** depicts an example linear power supply circuit **470** usable in the power supply **712** shown in FIG. **36**. In particular, the linear power supply circuit **470** comprises a transistor **472** configured to produce an output voltage V_{out} from an input voltage V_{in}.

With reference to FIG. **36** in view of FIG. **21**, the input voltage V_{in} shown in FIG. **24** corresponds to voltage V_x in FIG. **21** supplied by secondary battery **454** and the output voltage V_{out} shown in FIG. **24** corresponds to the voltage V_b shown in FIG. **21** supplied by the linear power supply circuit **470**. Notably, in line with the description of FIG. **21** above, the output voltage V_{out} shown in FIG. **24** may be greater than a desired output voltage V_o to accommodate a voltage drop across the diode **458**. For example, in view of FIG. **21**, the diode **458** may be placed at the output of the linear power supply to allow current to only flow from the linear power supply to the motor control circuits **726**, the energy circuits **732**, and the handle electronic circuits **730** and to sense voltage sag at the output of the linear power supply. Ultimately, the output voltage V_o may be a voltage required by (i) the motor control circuits **726** to drive the motor **729** and/or (ii) the energy circuits **732** to drive the ultrasonic transducer **734** and/or the high-frequency (e.g., RF) current electrode **736**.

Again in reference to FIG. **36** in view of FIG. **21**, when the switch **718**, **453** (FIGS. **36**, **21**) is closed, the primary battery **715**, **717**, **452***a*, **452***b*, **452***c*, **452***d* (FIGS. **36**, **21**) delivers output voltage (V_o) directly to the motor control circuits **726** and the energy circuits **732**. For example, the switch **718** may be closed when the primary battery pack including primary battery **715**, **717** is inserted into the battery assembly **400**, **410**, **420**, **430**. It should be understood that the switch **718** may be eliminated such that the primary battery **715**, **717** is electrically connected to the circuit **710** when the power supply **712** is attached to the surgical instrument **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1**-**70**.

In a first scenario, as the primary battery **715**, **717** maintains the output voltage V_o (e.g., no sagging V_o, V_o required by (i) the motor control circuits **726** to drive the motor **729** and/or (ii) the energy circuits **732** to drive the ultrasonic transducer **734** and/or the high-frequency (e.g., RF) current electrode **736** is supplied) secondary battery **720** drives the linear power supply (e.g., replacing switch mode power supply **727** shown in FIG. **36**) to power the handle electronics circuits **730** (e.g., to drive control electronics), when the switch **723** is closed. As described above, the switch **723** may be closed when the secondary battery pack including secondary battery **720** is inserted into the battery assembly **400**, **410**, **420**, **430**. It should be understood that the switch **723** may be eliminated such that the secondary battery **720** is electrically connected to the circuit **710** when the power supply **712** is attached to the surgical instrument **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1**-**70**. Notably, with reference to FIG. **21**, when the output voltage V_o is not sagging, the output voltage V_o is greater than the output voltage V_b from the linear power supply, and the diode **458** is off and does not conduct.

In a second scenario, if/when the primary battery **715**, **717** is unable to maintain the output voltage V_o (e.g., voltage sagging occurs or begins to occur, V_o required by (i) the motor control circuits **726** to the drive the motor **729** and/or (ii) the energy circuits **732** to drive the ultrasonic transducer **734** and/or the high-frequency (e.g., RF) current electrode **736** is not suppliable), the secondary battery **720** drives the linear power supply (e.g., replacing switch mode power supply **727** shown in FIG. **36**) to not only deliver power to the handle electronics circuits **730** but also deliver supplemental power to the motor control circuits **726** and/or the energy circuits **732** when switch **723** is closed. Similar to above, the switch **723** may be closed when the secondary battery pack including the secondary battery **720** is inserted into the battery assembly **400**, **410**, **420**, **430**. It should be understood that the switch **723** may be eliminated such that the secondary battery **720** is electrically connected to the circuit **710** when the power supply **712** is attached to the surgical instrument **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1**-**70**. This supplemental power enables the primary battery to maintain the output voltage V_o demanded by the motor control circuits **726** and/or the energy circuits **732** or at least a minimum V_o required by the motor control circuits **726** and/or the energy circuits **732**. Notably, with reference to FIG. **21**, when the output voltage V_o sags below V_b by more than the diode **458** turn-on voltage (e.g., ~0.7V), the diode **458** turns-on and conducts at which point the output V_b minus ~0.7V is applied to the output V_o. This condition persists until the output voltage V_o increases above V_b to turn off the diode **458**. Accordingly, the linear power supply is capable of supplying additional current to the motor control circuits **726** and/or the energy circuits **732** when needed.

Next, as previously discussed, FIG. **21** depicts a supplemental power source circuit **450** to maintain a desired output voltage (e.g., required or minimum output voltage V_o) according to one aspect of the present disclosure. In accordance with another aspect of the present disclosure, the secondary battery **454** of the supplemental power source circuit **450** may comprise a rechargeable battery. In this vein, it should be appreciated that use of the secondary rechargeable battery **454** (e.g., as described in the various first and/or second scenarios above) may deplete the charge of the secondary rechargeable battery **454** from a full charge. Since a full charge of the supplemental rechargeable battery **454** is desired (e.g., to avoid voltage sagging during continued use of the surgical instrument **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1**-**70**), the supplemental power source circuit **450** may further comprise a mechanism to recharge the secondary rechargeable battery **454**. Since, as described above, the diode **458** prevents current associated with the output voltage V_o produced by the primary batteries **452***a*, **452***b*, **452***c*, **452***d* from flowing to the secondary rechargeable battery **454** (e.g., via the switch mode power supply **456** or the linear power supply, as discussed above), the supplemental power source circuit **450** may further comprise a charger **455** to recharge the secondary rechargeable battery **454**.

In particular, the primary batteries **452***a*, **452***b*, **452***c*, **452***d* may deliver the output voltage V_o not only to the motor control circuits **726** (FIG. **36**) and the energy circuits **732** (FIG. **36**) but also to the charger **455** to return a depleted secondary rechargeable battery **454** to full charge and/or to keep the secondary rechargeable battery **454** at full charge. The charger **455** may be configured to charge the secondary rechargeable battery **454** only when the motor control circuits **726** and/or the energy circuits **732** are inactive (e.g., the motor **729**, the ultrasonic transducer **734**, and/or the high-frequency (e.g., RF) current electrode **736** are not being actuated). In one alternative aspect of the present disclosure, the charger **455** may be configured to charge the secondary rechargeable battery **454** irrespective of the state (e.g., active/inactive) of the motor control circuits **726** and/or the energy circuits **732**. In one various aspect, the charger **455** may be configured to charge the secondary rechargeable battery **454** when the switch **453** (FIG. **21**) is closed (e.g., when the primary battery pack (e.g., including the primary batteries **452***a*, **452***b*, **452***c*, **452***d*) is inserted into the battery assembly **400**, **410**, **420**, **430**). Notably, the primary battery pack (e.g., including the primary batteries **452***a*, **452***b*, **452***c*, **452***d*) of the battery assembly **400**, **410**, **420**, **430** may be replaceable to recharge the secondary rechargeable battery **454**. In one example, the primary batteries **452***a*, **452***b*, **452***c*, **452***d* are disposable non-rechargeable batteries. In another example, the primary batteries **452***a*, **452***b*, **452***c*, **452***d* are replaceable rechargeable batteries.

In one aspect, the present disclosure provides a system, comprising: a battery assembly for use in a modular handheld surgical instrument, wherein the battery assembly comprises: a primary battery pack; a secondary battery pack; and a circuit configured to: supply a first output voltage to a first load of the modular handheld surgical instrument via the primary battery pack; supply a second output voltage to a second load of the modular handheld surgical instrument via the secondary battery pack; and upon sensing a sagging voltage from the primary battery pack, supply a supplemental voltage to the first load, via the secondary battery pack, to avoid supplying the first load an output voltage below the first output voltage.

The first load may comprise at least one of a motor driven by a motor control circuit, an ultrasonic transducer driven by a first energy circuit, or a high-frequency current electrode driven by a second energy circuit. The second load may comprise control electronics driven by a handle electronics circuit and at least one of a motor driven by a motor control circuit, an ultrasonic transducer driven by a first energy circuit, or a high-frequency current electrode driven by a second energy circuit.

The circuit may comprise a power supply circuit positioned between the secondary battery pack and the first load, wherein the power supply circuit is configured to produce a power supply output voltage from an input voltage supplied by the secondary battery pack; and a diode positioned between an output of the power supply circuit and the first load, wherein the diode is configured to sense the sagging voltage from the primary battery pack. The diode may be configured to conduct the supplemental voltage when the sagging voltage drops below the power supply output voltage by more than a turn-on voltage associated with the diode. The supplemental voltage may comprise the power supply output voltage less the diode turn-on voltage. The diode may configured to conduct the supplemental voltage until the sagging voltage exceeds the power supply output voltage to turn off the diode.

The power supply circuit may comprise a switch mode power supply circuit or a linear power supply circuit. The circuit may be configured to be resistant to radiation sterilization. The secondary battery pack may comprise a plurality of rechargeable batteries, and wherein the circuit further comprises a charger configured to charge the plurality of rechargeable batteries to a full charge. The charger may be configured to charge the plurality of rechargeable batteries when the first load is inactive.

The primary battery pack may comprise a plurality of energy cells, and wherein the secondary battery pack comprises a plurality of energy cells. The plurality of energy cells of the primary battery pack may comprise a chemical composition different than the plurality of energy cells of the secondary battery pack. The plurality of energy cells of the primary battery pack and the plurality of energy cells of the secondary battery pack may be non-rechargeable such that the battery assembly is disposable.

In another aspect, the present disclosure provides a voltage sag resistant battery assembly for use in a modular handheld surgical instrument, the battery assembly comprising: a first battery pack; a second battery pack; and circuitry to: supply a first output voltage to a first load via the first battery pack; supply a second output voltage to a second load via the second battery pack; and upon sensing a sagging voltage from the first battery pack, supply a supplemental voltage to the first load, via the second battery pack, to restore the sagging voltage to the first output voltage.

The circuit may comprise a power supply circuit positioned between the second battery pack and the first load, wherein the power supply circuit is configured to produce a power supply output voltage from an input voltage supplied by the second battery pack; and a diode positioned between an output of the power supply circuit and the first load, wherein the diode is configured to sense the sagging voltage from the first battery pack. The second battery pack may comprise a plurality of rechargeable batteries, and wherein the circuit further comprises a charger configured to charge the plurality of rechargeable batteries to a full charge.

In another aspect, the present disclosure provides a system, comprising a modular handheld surgical instrument; and a battery assembly coupled to the modular handheld surgical instrument, wherein the battery assembly comprises: a primary battery pack including at least one energy cell; a secondary battery pack including at least one energy cell; and a circuit configured to: supply a first output voltage to a first load of the modular handheld surgical instrument via the primary battery pack; and upon sensing a sagging voltage from the primary battery pack, supply a compensating voltage to the first load, via the secondary battery pack, to prevent the battery assembly from supplying an output voltage below a predetermined threshold output voltage under the first load.

The circuit may comprise a power supply circuit positioned between the secondary battery pack and the first load, wherein the power supply circuit is configured to produce a power supply output voltage from an input voltage supplied by the secondary battery pack; and a diode positioned between an output of the power supply circuit and the first load, wherein the diode is configured to sense the sagging voltage from the primary battery pack, and wherein the compensating voltage comprises the power supply output voltage less a voltage drop associated with the diode. The compensating voltage may supplement the sagging voltage to meet the predetermined threshold output voltage.

Modular Battery Powered Handheld Surgical Instrument with Multistage Generator Circuits

In another aspect, the present disclosure provides a modular battery powered handheld surgical instrument with multistage generator circuits. Disclosed is a surgical instrument that includes a battery assembly, a handle assembly, and a shaft assembly where the battery assembly and the shaft assembly are configured to mechanically and electrically connect to the handle assembly. The battery assembly includes a control circuit configured to generate a digital waveform. The handle assembly includes a first stage circuit configured to receive the digital waveform, convert the digital waveform into an analog waveform, and amplify the analog waveform. The shaft assembly includes a second stage circuit coupled to the first stage circuit to receive, amplify, and apply the analog waveform to a load.

FIG. **136** illustrates a generator circuit **5500** partitioned into a first stage circuit **5504** and a second stage circuit **5506**, according to one aspect of the present disclosure. In one aspect, the surgical instruments **100**, **470**, **500**, **600**, **700**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-70** may comprise a generator circuit **5500** partitioned into multiple stages. For example, a surgical instruments **100**, **470**, **500**, **600**, **700**, **1100**, **1150**, **1200** may comprises a generator circuit **5500** partitioned into at least two circuits: a first stage circuit **5504** and a second stage circuit **5506** of amplification enabling operation of RF energy only, ultrasonic energy only, and/or a combination of RF energy and ultrasonic energy. A combination modular shaft assembly **5514** may be powered by a common first stage circuit **5504** located within the handle assembly **5512** and a modular second stage circuit **5506** integral to the modular shaft assembly **5514**. As previously discussed throughout this description in connection with the surgical instruments **100**, **470**, **500**, **600**, **700**, **1100**, **1150**, **1200**, the battery assembly **5510** and the shaft assembly **5514** are configured to mechanically and electrically connect to the handle assembly **5512**. The end effector assembly is configured to mechanically and electrically connect the shaft assembly **5514**.

Turning now to FIG. **136**, there is shown a generator circuit **5500** partitioned into multiple stages located in multiple modular assemblies of a surgical instrument, such as the surgical instruments **100**, **470**, **500**, **600**, **700**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-70**, for example. In one aspect, a control stage circuit **5502** may be located in a battery assembly **5510** of the surgical instrument. The control circuit **5502** is a control circuit **210** as described in connection with FIG. **14**. The control circuit **210** comprises a processor **214**, which includes internal memory **217** (e.g., volatile and non-volatile memory), and is electrically coupled to a battery **211**. The battery **211** supplies power to first, second, and third stage circuits **5504**, **5506**, **5508**, respectively. As previously discussed, the control circuit **210** generates a digital waveform **1800** (FIG. **67**) using circuits and techniques described in connection with FIGS. **65** and **66**. Returning to FIG. **136**, the digital waveform **1800** may be configured to drive an ultrasonic transducer, high-frequency (e.g., RF) electrodes, or a combination thereof either independently or simultaneously. If driven simultaneously, filter circuits may be provided in the corresponding first stage circuits **5504** to select either the ultrasonic waveform or the RF waveform. Such filtering techniques are described in commonly owned U.S. patent application Ser. No. 15/265,293, titled TECHNIQUES FOR CIRCUIT TOPOLOGIES FOR COMBINED GENERATOR, now U.S. Pat. No. 10,610,286, which is herein incorporated by reference in its entirety.

The first stage circuits **5504** (e.g., the first stage ultrasonic drive circuit **177**, the first stage RF drive circuit **702**, and the first stage sensor drive circuit **219**) are located in a handle assembly **5512** of the surgical instrument. As previously described, the control circuit **210** provides the ultrasonic drive signal to the first stage ultrasonic drive circuit **177** via outputs SCL-A/SDA-A of the control circuit **210**. The first stage ultrasonic drive circuit **177** is described in detail in connection with FIG. **11**. The control circuit **210** provides the RF drive signal to the first stage RF drive circuit **702** via outputs SCL-B/SDA-B of the control circuit **210**. The first stage RF drive circuit **702** is described in detail in connection with FIG. **34**. T the control circuit **210** provides the sensor drive signal to the first stage sensor drive circuit **219** via outputs SCL-C/SDA-C of the control circuit **210**. Generally, each of the first stage circuits **5504** includes a digital-to-analog (DAC) converter and a first stage amplifier section to drive the second stage circuits **5506**. The outputs of the first stage circuits **5504** are provided to the inputs of the second stage circuits **5506**.

The control circuit **210** is configured to detect which modules are plugged into the control circuit **210**. For example, the control circuit **210** is configured to detect whether the first stage ultrasonic drive circuit **177**, the first stage RF drive circuit **702**, or the first stage sensor drive circuit **219** located in the handle assembly **5504** is connected to the battery assembly **5502**. Likewise, each of the first stage circuits **5504** can detect which second stage circuits **5506** are connected thereto and that information is provided back to the control circuit **210** to determine the type of signal waveform to generate. Similarly, each of the second stage circuits **5506** can detect which third stage circuits **5508** or components are connected thereto and that information is provided back to the control circuit **210** to determine the type of signal waveform to generate.

In one aspect, the second stage circuits **5506** (e.g., the ultrasonic drive second stage circuit **166**, the RF drive second stage circuit **574**, and the sensor drive second stage circuit **221**) are located in a shaft assembly **5514** of the surgical instrument. As previously described, the first stage ultrasonic drive circuit **177** provides a signal to the second stage ultrasonic drive circuit **166** via outputs US-Left/US-Right. The second stage ultrasonic drive circuit **166** is described in detail in connection with FIGS. **12** and **13**. In addition to a transformer (FIGS. **12** and **13**), the second stage ultrasonic drive circuit **166** also may include filter, amplifier, and signal conditioning circuits. The first stage high-frequency (RF) current drive circuit **702** provides a signal to the second stage RF drive circuit **574** via outputs RF-Left/RF-Right. The second stage RF drive circuit **574** is described in detail in connection with FIG. **35**. In addition to a transformer and blocking capacitors **706**, **708** (FIG. **35**), the second stage RF drive circuit **574** also may include filter, amplifier, and signal conditioning circuits. The first stage sensor drive circuit **219** provides a signal to the second stage sensor drive circuit **221** via outputs Sensor-1/Sensor-2. The second stage sensor drive circuit **221** may include filter, amplifier, and signal conditioning circuits depending on the type of sensor. The outputs of the second stage circuits **5506** are provided to the inputs of the third stage circuits **5508**.

In one aspect, the third stage circuits **5508** (e.g., the ultrasonic transducer **130**, the RF electrodes **223***a*, **223***b*, and the sensors) may be located in various assemblies **5516** of the surgical instruments. In one aspect, the second stage ultrasonic drive circuit **166** provides a drive signal to the ultrasonic transducer **130** stack. In one aspect, the ultrasonic transducer **130** is located in the ultrasonic transducer assembly of the surgical instrument. In other aspects, however, the ultrasonic transducer **130** may be located in the handle assembly **5512**, the shaft assembly **554**, or the effector. In one aspect, the second stage RF drive circuit **574** provides a drive signal to the RF electrodes **223***a*, **223***b*, which are generally located in the end effector portion of the surgical instrument. In one aspect, the second stage sensor drive circuit **221** provides a drive signal to various sensors **225** located throughout the surgical instrument.

FIG. **137** illustrates a generator circuit **5600** partitioned into multiple stages where a first stage circuit **5604** is common to the second stage circuit **5606**, according to one aspect of the present disclosure. In one aspect, the surgical instruments **100**, **470**, **500**, **600**, **700**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-70** may comprise a generator circuit **5600** partitioned into multiple stages. For example, a surgical instruments **100**, **470**, **500**, **600**, **700**, **1100**, **1150**, **1200** may comprises a generator circuit **5600** partitioned into at least two circuits: a first stage circuit **5604** and a second stage circuit **5606** of amplification enabling operation of high-frequency (RF) energy only, ultrasonic energy only, and/or a combination of RF energy and ultrasonic energy. A combination modular shaft assembly **5614** may be powered by a common first stage circuit **5604** located within the handle assembly **5612** and a modular second stage circuit **5606** integral to the modular shaft assembly **5614**. As previously discussed throughout this description in connection with the surgical instruments **100**, **470**, **500**, **600**, **700**, **1100**, **1150**, **1200**, the battery assembly **5610** and the shaft assembly **5614** are configured to mechanically and electrically connect to the handle assembly **5612**. The end effector assembly is configured to mechanically and electrically connect the shaft assembly **5614**.

As shown in the example of FIG. **137**, the battery assembly **5610** portion of the surgical instrument comprises a first control circuit **560**, which includes the control circuit **210** previously described. The handle assembly **5612**, which connects to the battery assembly **5610**, comprises a common first stage drive circuit **177**. As previously discussed, the first stage drive circuit **177** is configured to drive ultrasonic, high-frequency (RF) current, and sensor loads. The output of the common first stage drive circuit **177** can drive any one of the second stage circuits **5606** such as the second stage ultrasonic drive circuit **166**, the second stage high-frequency (RF) current drive circuit **574**, and/or the second stage sensor drive circuit **221**. The common first stage drive circuit **177** detects which second stage circuit **5606** is located in the shaft assembly **5614** when the shaft assembly **5614** is connected to the handle assembly **5612**. Upon the shaft assembly **5614** being connected to the handle assembly **5612**, the common first stage drive circuit **177** determines which one of the second stage circuits **5606** (e.g., the second stage ultrasonic drive circuit **166**, the second stage RF drive circuit **574**, and/or the second stage sensor drive circuit **221**) is located in the shaft assembly **56144**. The information is provided to the control circuit **210** located in the handle assembly **5610** in order to supply a suitable digital waveform **1800** (FIG. **67**) to the second stage circuit **5606** to drive the appropriate load, e.g., ultrasonic, RF, or sensor. It will be appreciated that identification circuits may be included in the third stage circuits **5608** such as the ultrasonic transducer **130**, the electrodes **223***a*, **223***b*, or the sensors **225**. Thus, when a third stage circuit **5608** is connected to a second stage circuit **5606**, the second stage circuit **5606** knows the type of load that is required base don the identification information.

In one aspect, the present disclosure provides, a surgical instrument, comprising: a battery assembly, comprising a control circuit comprising a battery, a memory coupled to the battery, and a processor coupled to the memory and the battery, wherein the processor is configured to generate a digital waveform; a handle assembly comprising a first stage circuit coupled to the processor, the first stage circuit comprising a digital-to-analog (DAC) converter and a first stage amplifier circuit, wherein the DAC is configured to receive the digital waveform and convert the digital waveform into an analog waveform, wherein the first stage amplifier circuit is configured to receive and amplify the analog waveform; and a shaft assembly comprising a second stage circuit coupled to the first stage amplifier circuit to receive the analog waveform, amplify the analog waveform, and apply the analog waveform to a load; wherein the battery assembly and the shaft assembly are configured to mechanically and electrically connect to the handle assembly.

The load may comprise any one of an ultrasonic transducer, an electrode, or a sensor, or any combinations thereof. The first stage circuit may comprise a first stage ultrasonic drive circuit and a first stage high-frequency current drive circuit. The control circuit may be configured to drive the first stage ultrasonic drive circuit and the first stage high-frequency current drive circuit independently or simultaneously. The first stage ultrasonic drive circuit may be configured to couple to a second stage ultrasonic drive circuit. The second stage ultrasonic drive circuit may be configured to couple to an ultrasonic transducer. The first stage high-frequency current drive circuit may be configured to couple to a second stage high-frequency drive circuit. The second stage high-frequency drive circuit may be configured to couple to an electrode.

The first stage circuit may comprise a first stage sensor drive circuit. The first stage sensor drive circuit may be configured to a second stage sensor drive circuit. The second stage sensor drive circuit may be configured to couple to a sensor.

In another aspect, the present disclosure provides a surgical instrument, comprising: a battery assembly, comprising a control circuit comprising a battery, a memory coupled to the battery, and a processor coupled to the memory and the battery, wherein the processor is configured to generate a digital waveform; a handle assembly comprising a common first stage circuit coupled to the processor, the common first stage circuit comprising a digital-to-analog (DAC) converter and a common first stage amplifier circuit, wherein the DAC is configured to receive the digital waveform and convert the digital waveform into an analog waveform, wherein the common first stage amplifier circuit is configured to receive and amplify the analog waveform; and a shaft assembly comprising a second stage circuit coupled to the common first stage amplifier circuit to receive the analog waveform, amplify the analog waveform, and apply the analog waveform to a load; wherein the battery assembly and the shaft assembly are configured to mechanically and electrically connect to the handle assembly.

The load may comprise any one of an ultrasonic transducer, an electrode, or a sensor, or any combinations thereof. The common first stage circuit may be configured to drive ultrasonic, high-frequency current, or sensor circuits. The common first stage drive circuit may be configured to couple to a second stage ultrasonic drive circuit, a second stage high-frequency drive circuit, or a second stage sensor drive circuit. The second stage ultrasonic drive circuit may be configured to couple to an ultrasonic transducer, the second stage high-frequency drive circuit is configured to couple to an electrode, and the second stage sensor drive circuit is configured to couple to a sensor.

In another aspect, the present disclosure provides a surgical instrument, comprising a control circuit comprising a memory coupled to a processor, wherein the processor is configured to generate a digital waveform; a handle assembly comprising a common first stage circuit coupled to the processor, the common first stage circuit configured to receive the digital waveform, convert the digital waveform into an analog waveform, and amplify the analog waveform; and a shaft assembly comprising a second stage circuit coupled to the common first stage circuit to receive and amplify the analog waveform; wherein the shaft assembly is configured to mechanically and electrically connect to the handle assembly.

The common first stage circuit may be configured to drive ultrasonic, high-frequency current, or sensor circuits. The common first stage drive circuit may be configured to couple to a second stage ultrasonic drive circuit, a second stage high-frequency drive circuit, or a second stage sensor drive circuit. The second stage ultrasonic drive circuit may be configured to couple to an ultrasonic transducer, the second stage high-frequency drive circuit is configured to couple to an electrode, and the second stage sensor drive circuit is configured to couple to a sensor.

Modular Battery Powered Handheld Surgical Instrument with Multiple Magnetic Position Sensors

In another aspect, the present disclosure provides a modular battery powered handheld surgical instrument with multiple magnetic position sensors. A system comprising a surgical instrument is disclosed. The surgical instrument includes a handle, a shaft, a plurality of magnets, a plurality of sensors configured to determine a distance away from one or more of the plurality of magnets, and a processor communicatively coupled to the plurality of sensors. The processor is configured to determine a three dimensional change in position of the shaft by computing a three dimensional change in position of the one or more magnets, using the change in the distances determined by the one or more plurality of sensors.

FIGS. **138-142** illustrate various aspects of the present disclosure that is directed to multiple magnetic position sensors disposed along the length of a shaft assembly of any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-70**. The magnetic position sensors disposed along the length of the shaft assembly are paired in different configurations allowing multiple sensors to detect a single magnet coupled to a shaft motion (e.g., articulation, end effector rotation, jaw member opening/closing) in order to determine a three dimensional position of the actuating component from a stationary reference plane and simultaneously diagnosing any error from external sources.

Aspects of the present disclosure are presented for a system and surgical instrument configured to detect three dimensional position and change in position of various parts using a series of magnets and sensors. In some aspects, a surgical instrument includes a handle assembly, a shaft coupled to the handle assembly and an end effector coupled to a distal end of the shaft. Multiple magnets may be fixedly coupled to the shaft, while multiple sensors may also be coupled to the shaft and configured to determine a distance away from each of the multiple magnets. The shaft may be configured to move, such as rotate, translate, and/or articulate about an articulation joint. The magnets may be coupled to the shaft such that the magnets move along with how the shaft moves. However, the sensors may be coupled to the shaft such that the sensors remain in a stationary position while the shaft moves. For example, the shaft may include an exterior shaft and an interior shaft, such as a primary rotary driver, and the exterior shaft may be configured to move while the primary rotary driver does not. The magnets may be affixed to the exterior shaft and the sensors may be affixed to the primary rotary driver or a portion therein. The sensors may detect change in distances of three or more magnets as the shaft moves, for example, which may be processed by a processor to determine a change in three dimensional position of the shaft. The data of the sensors may be transmitted to a processor via a flexible circuit running on the inside of the primary rotary driver.

In some aspects, a portion of the series of sensors and magnets may be configured to detect translation or other movements of the end effector. For example, the end effector may include a pair of jaws, where at least one of the jaws is configured to open and close via a clap arm coupled to a portion of the primary rotary driver. The clamp arm may ultimately be communicatively coupled to a portion of the handle assembly configured to manipulate the end effector, such as a handle or a trigger. Pulling the trigger of the handle assembly may cause one or more of the jaws to open or close at the end effector. One or more of the magnets may be affixed to a portion of the clamp arm that moves when the clamp arm is manipulated. One or more of the sensors may be affixed to a portion of the shaft near the end effector that does not move when the clamp arm is manipulated. Thus, the sensors may be configured to detect a change in distance of these magnets as the clamp arm is manipulated, which may be processed by a processor to determine a change in movement of the jaws of the end effector. The data of the sensors may be transmitted to a processor via a flexible circuit running on the inside of the primary rotary driver.

In some aspects, the system and surgical instrument may also be configured to determine an error condition of the surgical instrument using the series of magnets and sensors. For example, an error condition may occur if movement of the magnets is detected to exceed known tolerances or thresholds based on the detections of one or more of the sensors. For example, the shaft of the surgical instrument may be configured to rotate only so far, and if the sensors detect a change in position of one or more magnets that exceeds the known rotation or angle, then an error condition may be generated. As another example, positions of multiple magnets may be monitored, and if one of the positions of the magnets is calculated to be inconsistent with calculated positions of the other magnets, an error condition may be generated.

Referring to FIG. **138**, illustration **5100** shows an example system of shaft control electronics interconnecting inside a portion of a handle assembly and shaft assembly of a medical device such as any one of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-70**, according to various aspects. A module **5102** shows a portion of the inner workings of a handle assembly that is coupled to a module **5104** that comprises portions of a shaft assembly, according to some aspects. The module **5102** includes a flexible circuit **5106** that runs through a portion of the handle assembly and into the shaft assembly. More specifically, in some aspects, the flexible circuit **5106** runs through a primary rotary driver **5108**. The primary rotary driver **5108** is configured to drive rotation of at least a portion of the shaft assembly. While the primary rotary driver **5108** runs contiguously through a portion of the handle assembly and the proximal end of the shaft assembly, in this illustration, a portion of the primary rotary driver is not shown to reveal the flexible circuit **5106** inside.

The proximal end of the flexible circuit **5106** may be coupled to an electric contact **5110**, which touches electrical contact points **5112** at the handle. A gear **5138** of the primary rotary driver **5108** may be coupled to a rotating gear **5140** in the handle assembly that is driven by the handle **5136**. The electric contact **5110** and contact pointe **5112** electrically couple the sensors **5126**, **5128**, **5130** to the control circuits such as control circuits **210** (FIG. **14**), **300** (FIG. **15**), **1300** (FIG. **62**), **1400** (FIG. **63**), **1500** (FIG. **64**), **1900** (FIG. **68**A), **1910** (FIG. **68**B), **1920** (FIG. **68**C), **1954** (FIG. **70**), for example.

The distal end of the shaft assembly **5104** may be coupled to an end effector, which may include one or more jaws mechanically and electrically coupled at connector **5134**. The flexible circuit **5106** may be configured to transmit electrical signals to the end effector via the connector **5134**.

Still referring to the shaft assembly **5104**, the primary rotary driver **5108** may be placed inside an exterior shaft **5114**, according to some aspects. The primary rotary driver **5108** may be configured to rotate the exterior shaft **5114** while the flexible circuit **5106** remains stationary. In some aspects, the shaft assembly **5104** may also include a distal piece of an exterior shaft **5116** that is coupled to the proximal exterior shaft **5114** via an articulation joint **5118**. While the primary rotary driver **5108** may run only through the end of the proximal exterior shaft **5114**, the flexible circuit **5106** may run through the distal exterior shaft **5116**, and may be configured to articulate and bend as the exterior shaft **5116** articulates about the articulation joint **5118**.

In some aspects, the exterior shaft portions of the shaft assembly **5104** may include one or more magnets, such as magnets **5120**, **5122**, **5124**. These magnets may be affixed to the exterior shaft portions, and thus may move or change position as the exterior shaft portions move or change position. Meanwhile, the flexible circuit **5106** may include one or more sensors configured to measure distance away from one or more magnets, such as sensors **5126**, **5128**, **5130**. A Hall effect sensor is an example type of sensor configured to measure a distance away from one or more magnets. As shown, the shaft assembly **5104** may include sets of sensor-magnet pairs in close proximity to each other, such as the sensor **5126** being in close proximity to the magnet **5120**, the sensor **5128** being in close proximity to the magnet **5122**, and the sensor **5130** being in close proximity to the magnet **5124**. This configuration may allow the sensors to measure distances to only those magnets in close proximity.

In some aspects, the shaft assembly **5104** may also include a translating shaft portion, such as the shaft portion **5132** configured to slide back and forth at the distal end of the shaft assembly **5104**. The magnet **5124** may be configured to translate or slide along with the shaft portion **5132**, while the sensor **5130** remains stationary, as it is attached to the flexible circuit **5106**.

In some aspects, one or more of the sensor-magnet pairs may be configured to measure a degree of rotation of the shaft assembly **5104**. For example, as the magnet **5120** rotates along with the exterior shaft **5114**, the change in distance away from the sensor **5126** may be measured. Based on the distance or the change in distance over time, it may be determined how much the exterior shaft **5114** has rotated.

Similarly, in some aspects, one or more of the sensor-magnet pairs may be configured to measure a degree of articulation of the shaft assembly **5104**. For example, as the magnet **5122** articulates along with the distal exterior shaft portion **5116** about the articulation joint **5118**, the sensor **5128** may be configured to measure a distance to the magnet **5122**. Due to the placement of the sensor **5128** being slightly away from the center of the articulation joint **5118**, the distance away from the magnet **5122** will vary as the distal exterior shaft **5116** articulates. Based on the distance or the change in distance over time, it may be determined how the distal exterior shaft **116** is articulating about the articulation joint **5118**.

Furthermore, in some aspects, one or more of the sensor-magnet pairs may be configured to measure an amount of translation of the shaft assembly **5104**. For example, as the magnet **5124** translates along with the translating portion of the shaft **5132**, the sensor **5130** may be configured to measure the distance away from the magnet **5124**. Based on the distance or change in distance over time, it may be determined how much the shaft assembly is translating away from the sensor **5130**.

In some aspects, the computed positions or changes in positions according to the techniques described herein may be compared to predetermined tolerances. For example, a microprocessor and memory within the handle assembly of the medical device may have a table of distances or tolerances that reflect maximum ranges of movement by various parts of the shaft assembly. If it is computed that one or more of the magnets exceeds one of the values in this table, then an error condition may be set an alarm may be activated.

In some aspects, any or all of these techniques may be included in a medical device to measure and monitor changes in position of the shaft assembly. The illustration **5100** provides merely some examples of where the magnets and sensors may be positioned, but it may be readily apparent to those of ordinary skill in the art how the magnets and the sensors may be positioned to achieve similar effects, and aspects are not so limited.

Referring to FIG. **139**, illustration **5100** provides an example of a zoomed in portion of the flexible circuit **5106**, according to some aspects. As shown, the flexible circuit may comprise a series of substrate layers **5202**. The layers may be flexible enough to bend and twist. Hall effect sensors **5204** and **5206**, or other kinds of sensors configured to measure magnet strength, may be affixed to the flexible circuit **5106**, as shown.

Referring to FIGS. **140**A and **140**B, in some aspects, a plurality of sensors may be used in combination to measure and monitor movements of the medical device, according to some aspects. For example, as shown in illustration **5160** of FIG. **140**A, a cross-sectional view of a portion of the shaft assembly shows the exterior shaft **5162** comprising one magnet **5164**. The interior shaft **5166** may include a flexible circuit **5174**. The flexible circuit **5174** may comprise three sensors **5168**, **5170**, and **5172**, positioned at various distances away from the magnet **5164**. The exterior shaft portion **5162** may be configured to rotate while the interior shaft **5166**, or at least a portion thereof, may be stationary.

Referring to FIG. **140**B, illustration **5180** shows a rotational movement of the exterior shaft. As shown, the magnet **5164** rotates along with the exterior shaft **5162**. The angle of rotation is indicated by the arrow **5182** in relation to the centerline **5176**. Because the three sensors **5168**, **5170**, and **5172**, are positioned at varying distances away from the magnet **5164**, a trilateration calculation may be conducted using the three different distances to the magnet **5164** to determine more unambiguously a position of the magnet **5164** as it is being rotated. Based on the calculated position, a degree or angle of rotation of the exterior shaft **5162** may be determined.

In general, with three or more sensors, a three-dimensional position of the magnets may be determined, assuming that the three or more sensors are positioned at uniquely different distances away from the magnet to be measured. Therefore, the principles shown in illustrations **5100** and **5180** may be used to measure other types of changes in position, such as translation distances or articulation angles.

In some aspects, instead of or in addition to multiple sensors, multiple magnets may be affixed to the moving portion of the shaft assembly, and the one or more sensors may be configured to measure the multiple distances of the multiple magnets to determine a three-dimensional position that way.

Referring to FIG. **141**, illustration **5190** provides another example of how multiple sensors can be used to determine a three-dimensional position of one or more magnets, according to some aspects. In this example, the shaft assembly includes two parts about an articulation joint **5198**. Two sensors, **5192** and **5194**, are positioned along a flexible circuit **5199** on one side of the articulation joint **5198**. On the other side is a magnet **5196** that is configured to move along with that portion of the shaft assembly as it articulates about the articulation joint **5198**. The two sensors **5192** and **5194** may be sufficient to determine a unique three-dimensional position of the magnet **5196**, because the range of motion of the magnet **5196** is limited to only one plane of motion that remains on only one side of the multiple sensors. In general, number of sensors or magnets sufficient for determining unambiguous positions or changes in positions may be based on where the sensors in magnets are positioned, as well as how many degrees of freedom are available for the various parts to move.

In some aspects, the sensors and/or magnets may be positioned with different configurations. Spacing of position sensors and magnets allows for higher resolution associated with location. For instance, sensors on opposite sides could sense the same magnet at different times, or different magnets would interact with sensors at different locations along travel. In addition, further resolution can be achieved through patterns of magnets. For instance, magnets can be spaced such that sensing two magnets in quick succession on one side, followed by another group on the other side signifies a different location than two in one on one side followed by a number of magnets on the other side. This may reduce the number of magnets or sensors required to achieve this effect.

Referring to FIG. **142**, illustration **5200** shows a variation of a shaft assembly of a medical device that includes a seal to protect sensitive electronics, according to some aspects. In this case, the outside shaft **5202** includes at least two inner tubular structures. The rotary drive **5204** is still present and is enclosed by the outside shaft. Within that is a tube **5206** with flanges **5208**, such as a rubber tube with flanges. The flanges **5208** may be configured to pressed against the inside surface of the rotary shaft **5204** to form a seal. The seal may be airtight and watertight. The tube **5206** may then house electronics and other materials sensitive to liquid or other exposure, such as the flexible circuit **5210**.

In one aspect, the present disclosure provides, a system comprising: a surgical instrument comprising: a handle assembly; a shaft coupled to the handle assembly; a plurality of magnets fixedly coupled to the shaft; a plurality of sensors coupled to the shaft and each configured to determine a distance away from one or more of the plurality of magnets; and a processor communicatively coupled to the plurality of sensors; wherein: the shaft is configured to change position relative to a stationary reference plane; one or more of the plurality of magnets are configured to change position corresponding to the changed position of the shaft; one or more of the plurality of sensors are further configured to determine a change in the distances of the one or more of the plurality of magnets relative to the one or more of the plurality of sensors as the one or more magnets change position; and the processor is configured to determine a three dimensional change in position of the shaft by computing a three dimensional change in position of the one or more magnets, using the change in the distances determined by the one or more plurality of sensors.

The shaft may comprise an articulation joint, a first component and a second component of the shaft, the first and the second components coupled about the articulation joint; the second component of the shaft is configured to articulate about the articulation joint; a first magnet of the plurality of magnets is fixedly coupled to the second component of the shaft; a first sensor of the plurality of sensors is coupled to the first component of the shaft; the first sensor is configured to measure a change in distance away from the first magnet as the second component of the shaft is articulated about the articulation joint; and the processor is configured to determine the three dimensional change in position of the articulation of the second component of the shaft using the measured change in distance of the first magnet away from the first sensor.

The second magnet of the plurality of magnets may be fixedly coupled to the second component of the shaft at a position different from where the first magnet is coupled to the second component; the first sensor is further configured to measure a change in distance away from the second magnet as the second component of the shaft is articulated about the articulation joint; and the processor is further configured to determine the three dimensional change in position of the articulation of the second component of the shaft using the measured change in distance of the first magnet away from the first sensor and the measured change in distance of the second magnet away from the first sensor.

The shaft may comprise an exterior shaft and an inner shaft positioned within the exterior shaft; the inner shaft is configured to rotate within the exterior shaft; a first magnet of the plurality of magnets is fixedly coupled to the exterior shaft; a first sensor of the plurality of sensors is fixedly coupled to the inner shaft; the first sensor is configured to measure a change in distance away from the first magnet as the exterior shaft is rotated relative to the inner shaft; and the processor is configured to determine a degree of rotation of the exterior shaft relative to the inner shaft using the measured change in distance of the first magnet away from the first sensor. A second magnet of the plurality of magnets may be fixedly coupled to the exterior shaft at a position different from where the first magnet is coupled to the exterior shaft; the first sensor is further configured to measure a change in distance away from the second magnet as the exterior shaft is rotated relative to the inner shaft; and the processor is further configured to determine the degree of rotation of the exterior shaft using the measured change in distance of the first magnet away from the first sensor and the measured change in distance of the second magnet away from the first sensor.

The surgical instrument may further comprise an end effector coupled to a distal end of the shaft, the end effector comprising a pair of surgical jaws; the shaft comprises a clamp arm driver coupled to the pair of surgical jaws that is configured to translate proximally and distally to manipulate movement of the pair of surgical jaws; a first magnet of the plurality of magnets is fixedly coupled to the clamp arm driver; a first sensor of the plurality of sensors is coupled to a portion of the shaft that does not translate when the clamp arm driver translates; the first sensor is configured to measure a change in distance away from the first magnet as the clamp arm driver translates; and the processor is configured to determine a distance of translation of the clamp arm driver using the measured change in distance of the first magnet away from the first sensor.

A second magnet of the plurality of magnets may be fixedly coupled to the clamp arm driver at a position different from where the first magnet is coupled to the exterior shaft; the first sensor is further configured to measure a change in distance away from the second magnet as the clamp arm driver translates; and the processor is further configured to determine the distance of translation of the clamp arm driver using the measured change in distance of the first magnet away from the first sensor and the measured change in distance of the second magnet away from the first sensor. The processor may be further configured to determine an error condition of the surgical instrument by measuring whether the change in the distances determined by the one or more plurality of sensors fails to satisfy at least one predetermined threshold. The surgical instrument may further comprise a flexible circuit configured to deliver an electrical connection to a distal end of the shaft and articulate about an articulation joint of the shaft. The surgical instrument may further comprise a rotary driver communicatively coupled to the handle assembly and configured to cause rotation of at least a portion of the shaft.

In another aspect, the present disclosure provides a surgical instrument, comprising a plurality of magnets; a plurality of sensors configured to determine a distance away from one or more of the plurality of magnets; and a processor communicatively coupled to the plurality of sensors, wherein the processor is configured to determine a three dimensional change in position of the surgical instrument by computing a three dimensional change in position of the one or more magnets, using the change in the distances determined by the one or more plurality of sensors.

The processor may be configured to determine the three dimensional change in position of an articulation of a component of the surgical instrument using the measured change in distance of a first magnet away from a first sensor. The processor may be configured to determine a degree of rotation of an exterior component of the surgical instrument relative to an inner component of the surgical instrument using the measured change in distance of a first magnet away from a first sensor. The processor may be configured to determine a distance of translation of a component of the surgical instrument using the measured change in distance of the first magnet away from the first sensor. The processor may be further configured to determine an error condition of the surgical instrument by measuring whether the change in the distances determined by the one or more plurality of sensors fails to satisfy at least one predetermined threshold.

The surgical instrument may further comprise a flexible circuit configured to deliver an electrical connection to a portion of the surgical instrument. The surgical instrument may further comprise a rotary driver and configured to rotate of at least a portion of the surgical instrument.

In another aspect, the present disclosure provides a method of controlling a surgical instrument comprising a plurality of magnets, a plurality of sensors configured to determine a distance away from one or more of the plurality of magnets, and a processor communicatively coupled to the plurality of sensors, the method comprising: determining, by the processor, a three dimensional change in position of the surgical instrument by computing a three dimensional change in position of the one or more magnets, using the change in the distances determined by the one or more plurality of sensors.

The method may comprise determining, by the processor, the three dimensional change in position of an articulation of a component of the surgical instrument using the measured change in distance of a first magnet away from a first sensor. The method may comprise determining, by the processor, a degree of rotation of an exterior component of the surgical instrument relative to an inner component of the surgical instrument using the measured change in distance of a first magnet away from a first sensor. The method may comprise determining, by processor, a distance of translation of a component of the surgical instrument using the measured change in distance of the first magnet away from the first sensor. The method may comprise determining, by the processor, an error condition of the surgical instrument by measuring whether the change in the distances determined by the one or more plurality of sensors fails to satisfy at least one predetermined threshold.

Modular Battery Powered Handheld Surgical Instrument Containing Elongated Multi-Layered Shaft

In another aspect, the present disclosure provides a modular battery powered handheld surgical instrument containing elongated multi-layered shaft. Disclosed is a system that includes a surgical instrument that includes a handle assembly. The handle assembly includes a handle; a shaft assembly coupled to the handle assembly; and a flexible circuit. The handle assembly comprises a proximal portion of the flexible circuit and the shaft assembly comprises a distal portion of the flexible circuit; the shaft assembly further comprising: an exterior shaft positioned around the distal portion of the flexible circuit and configured to change position while a sub-portion of the distal portion of the flexible circuit remains stationary; and shaft control electronics comprising a stationary portion and a non-stationary portion, the stationary portion coupled to the distal portion of the flexible circuit and the non-stationary portion coupled to the exterior shaft; wherein the stationary portion of the shaft control electronics is configured to measure a distance to the non-stationary portion of the shaft control electronics as the exterior shaft changes position.

In one aspect, the present disclosure provides a system comprising a surgical instrument, the surgical instrument comprising: a handle assembly comprising a handle; a shaft assembly coupled to the handle assembly; and a flexible circuit, wherein the handle assembly comprises a proximal portion of the flexible circuit and the shaft assembly comprises a distal portion of the flexible circuit; the shaft assembly further comprising: an exterior shaft positioned around the distal portion of the flexible circuit and configured to change position while at least a sub-portion of the distal portion of the flexible circuit remains stationary; and shaft control electronics comprising a stationary portion and a non-stationary portion, the stationary portion coupled to the distal portion of the flexible circuit and the non-stationary portion coupled to the exterior shaft; wherein the stationary portion of the shaft control electronics is configured to measure a distance to the non-stationary portion of the shaft control electronics as the exterior shaft changes position.

The processor may be coupled to the flexible circuit and configured to determine a three dimensional change in position of the exterior shaft by computing a three dimensional change in position of the non-stationary portion of the shaft control electronics, based on a change in distance between the stationary portion and the non-stationary portion.

The handle assembly may comprise a first module comprising the proximal portion of the flexible circuit; and a second module comprising the handle; the first module configured to be decoupled from the second module. The first module may comprise a first set of contacts communicatively coupled to the proximal portion of the flexible circuit, and the second module comprises a second set of contacts communicatively coupled to the handle. The first set of contacts may be configured to be fixably interlocked to the second set of contacts while the first module is coupled to the second module. The handle may be configured to transmit electronic signals to the flexible circuit through the first and second set of contacts.

The shaft assembly may comprise an articulation joint, a first component and a second component of the shaft assembly, the first and the second components coupled about the articulation joint; the second component of the shaft is configured to articulate about the articulation joint; the flexible circuit is configured to articulate within the articulation joint; the non-stationary portion of the shaft control electronics is fixedly coupled to the second component of the shaft; and the stationary component of the shaft control electronics is configured to measure a change in distance away from the non-stationary component as the second component of the shaft assembly is articulated about the articulation joint. The exterior shaft may be configured to rotate about the flexible circuit; and the stationary component of the shaft control electronics is configured to measure a change in distance away from the non-stationary component as the exterior shaft is rotated about the flexible circuit.

The surgical instrument may further comprise an end effector coupled to a distal end of the shaft assembly, the end effector comprising a pair of surgical jaws; the shaft assembly comprises a clamp arm driver coupled to the pair of surgical jaws that is configured to translate proximally and distally to manipulate movement of the pair of surgical jaws; the non-stationary component of the shaft control electronics is fixedly coupled to the clamp arm driver; and the stationary component of the shaft control electronics is configured to measure a change in distance away from the non-stationary component as the clamp arm driver translates. The processor may be further configured to determine an error condition of the surgical instrument by measuring whether the change in the distances determined by the stationary component of the shaft control electronics fails to satisfy at least one predetermined threshold. The surgical instrument may further comprise a rotary driver communicatively coupled to the handle assembly and configured to cause rotation of at least a portion of the shaft assembly. A computer readable medium having no transitory signals comprising instructions that, when executed by a processor, cause the processor to perform the operations described above.

Modular Battery Powered Handheld Surgical Instrument with Motor Drive

In another aspect, the present disclosure provides, modular battery powered handheld surgical instrument with motor drive. Disclosed is a surgical instrument that includes a handle assembly; a shaft assembly and an end effector. The shaft assembly includes an exterior shaft, a first clutch positioned within the exterior shaft to drive a first function of the surgical instrument when engaged. A second clutch within the exterior shaft to drive a second function when engaged. A rotary driver is positioned within the exterior shaft and configured to rotate within the exterior shaft. The rotary driver includes a first clutch engagement mechanism to engage the first clutch and a second clutch engagement mechanism configured to engage the second clutch independent of the first clutch engagement mechanism engaging the first clutch, such that, at different times both the first clutch and the second clutch are engaged simultaneously only the first clutch is engaged only the second clutch is engaged and neither the first clutch nor the second clutch is engaged simultaneously.

With reference ow to FIGS. **143-146**B, in various aspects, the present disclosure provides a modular battery powered handheld surgical instrument with motor drive comprising a primary rotary drive configured to be selectably coupleable to at least two independent actuations (first, second, both, neither) and utilizing a clutch mechanism that is located entirely in the distal modular elongated tube of the shaft assembly. The motor drive may be adapted for use with any of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-70**.

Aspects of the present disclosure are presented for a system and surgical instrument comprising a handle assembly and shaft assembly, with the shaft assembly including two or more distinct clutches configured to move various parts of the surgical instrument when engaged. The shaft assembly also includes an interior mechanism, such as a rotary drive, capable of selectively coupling to the clutches using independent actuation mechanisms. Each of the clutches may be engaged independently through the independent actuation mechanisms, such that the first clutch may be engaged only, the second clutch may be engaged only, both clutches may be engaged simultaneously, or neither of the clutches may be engaged at the same time. In some aspects, more clutches are included in the shaft assembly and may be engaged independently in similar manners. Example actions engaged by the clutches include engaging a clutch for articulating a portion of the shaft assembly, engaging another clutch to cause distal head rotation of the shaft assembly that includes an end effector, and engaging a third clutch to cause jaw closure and opening of the end effector.

In some aspects, the clutches are comprised of a micro electrical clutching design, whereby an actuation mechanism inside the shaft assembly engages the clutch by activating an electric coil that creates a magnetic field. The clutch itself may include a ferrofluid that reacts in the presence of magnetism. Thus, when the actuation mechanism creates the magnetic field in close proximity to the clutch, the actuation mechanism attaches to the clutch via the magnetic field. Rotation of the rotary drive may then cause rotation of a component within the shaft assembly coupled to the clutch. The other clutches within the shaft assembly may be engaged in a similar manner.

In some aspects, the rotary drive may be slidable within the shaft assembly, allowing for a single actuation mechanism to slide to different positions in order to engage multiple clutches at different times.

In some aspects, the surgical instrument may include one or more locking mechanisms configured to maintain the actuation mechanism in its deactivated state until it may be unambiguously determine that one or more of the clutches is desired to be engaged. The locking mechanism may be biased in a locked state and may be automatically deactivated within the first few degrees of clutch drive motion to ensure that system is placed in a specific articulated or actuated state.

Referring to FIG. **143**, illustration **5700** shows the inner workings of a portion of a shaft assembly that is part of the surgical instrument, according to some aspects. This shaft assembly includes a rotary drive **5702** that may interact with three clutch systems, **5704**, **5706**, and **5708** through actuation mechanisms **5716**, **5718**, **5720**, respectively, coupled to a rotary drive **5702**. In this case, each of the actuation mechanisms include electrically conductive coils wrapped around the rotary drive **5702**. Each of the coils may receive power through electrically conductive wires, such as wires **5710**, **5712**, and **5714**, respectively. These wires may be coupled to in electrical power source.

Each of the clutches may comprise a material or mechanism that can be activated to attach to or engage with the actuation mechanisms. For example, the clutch systems **5704**, **5706**, and **5708** each include a cylindrical cavity, **5722**, **5724**, and **5726**, respectively, positioned cylindrically around the rotary drive **5702**. These cylindrical cavities may be filled with a ferrofluid, such as a magnetorheological fluid. The ferrofluid may be resting in a deactivated state that is semi-viscous. When in the presence of magnetism, such as a magnetic field, the ferrofluid may activate, harden, and align with the magnetic field such that the ferrofluid attaches to the magnetic source. Thus, electricity flowing to each of the electrically conductive coils, causing a magnetic field, may be used to engage the clutch in its local proximity. Because each of the coils **5716**, **5718**, **5720**, may receive power independently, any and all combinations of clutches may be engaged at any particular time.

In some aspects, the shaft assembly may include other components, such as an articulation joint **5728**, and an end effector comprising jaws **5742**, for example. Certain portions of the shaft assembly may also be configured to rotate longitudinally along with rotary drive, such as rotating distal portion **5738**. Each of these components may be activated and manipulated using the clutch systems described herein.

For example, the clutch system **5704** may be configured to control movement about the actuation joint **5728**, according to some aspects. The clutch cavity **5722** that includes the ferrofluid may be coupled to a first articulation component **5734**. Towards the distal end of the articulation component **134** is a gear, that is coupled to a second gear of a second articulation component **5736**. When the clutch system **5704** is engaged, such as in the manner described above, the rotary drive **5702**, when rotated, may also cause the first articulation component **5734** to correspondingly rotate. The distal gear then causes the rotation of the second articulation component **5736** which causes articulation about the articulation joint **5728**. As shown in this example, because of the articulation joint **5728**, the rotary drive **5702** is actually broken up into two main parts, including a distal rotary drive component **5730** that is coupled to the proximal rotary drive component **5702** via a system of gears **5732** and other connectors, as shown.

As another example, the clutch system **5706** may be configured to control rotation of the distal end of the shaft assembly, according to some aspects. The clutch cavity **5724** that includes the ferrofluid may be coupled to the rotatable distal portion **5738**, as shown. Thus, when the clutch system **5706** is engaged, such as in the manner described above, the rotary drive **5702**, when rotated, may cause rotatable distal portion **5738** to correspondingly rotate.

As another example, the clutch system **5708** may be configured to control jaw movement of the jaws **5742** of the end effector, according to some aspects. The clutch cavity **5726** that includes the ferrofluid may be coupled to a connector **5740** via a series of angled threads, similar to threads of a screw. Thus, when the clutch system **5708** is engaged, rotation by the rotary driver **5702** causes the connector **5740** to move proximally and distally, along with some rotation. The longitudinal motion of the connector **5740** may pull or push a fulcrum of the jaws, causing them to open and close. In other cases, rotation of the rotary drive **5702** may cause the connector **5740** to only rotate, wherein some other mechanism may be designed to cause the jaws to open and close. Examples may include a rotating gear system that converts the rotation into translation movement, a pair of electrical contacts coupled to the distal end of the connector **5740** that activates jaw movement, and other mechanisms apparent to those with skill in the art.

The example design in illustration **5700** shows that the different clutch systems may be engaged independently, meaning any combination of them may be engaged simultaneously, or separately, as desired by a user. This is achievable by the independent activations of the coils **5716**, **5718**, **5720**, via the independent wires **5710**, **5712**, **5714**. Therefore, the example design as shown allows for the shaft assembly to simultaneously articulate, rotate, and open and close the jaws, or any sub combination thereof, and aspects are not so limited. Similarly, additional clutch systems configured to perform or manipulate different actions of the shaft assembly may be further included, and are consistent with the teachings of this disclosure.

Referring to FIGS. **144**A, **144**B, **144**C, **144**D, **144**E, and **144**F another variation for including multiple clutch systems is shown, according to some aspects. Referring to FIG. **144**A, in this case, the rotary drive **5802** includes a pair of permanent magnets **5804***a*, **5804***b*. In other cases, more than two permanent magnets may be used. The rotary drive **5802** is configured to slide back and forth, within the remaining exterior shaft portion of the shaft assembly. The rotary drive **5802** is also configured to rotate in the direction of the arrows, as shown. In some aspects, the rotary drive **5802** may be contained in a rotating primary shaft **5804** that does not slide back and forth within the exterior shaft. The rotating primary shaft **5804** may be configured to rotate when engaged with one or more clutches. The outer, exterior frame of the shaft assembly **5806** is configured to not move initially and includes a pair of cylindrical clutch cavities **5808**, **5810**. These cavities may again be filled with magnetorheological fluid that remains in a highly viscous state outside the presence of a magnetic field. The clutch cavities **5808**, **5810** may be positioned in close proximity to each other, such that the length of the magnets **5804***a*, **5804***b* may influence both ferrofluids within both cavities at the same time when the magnets are slid into a particular position.

Referring to FIG. **144**B, the rotary drive **5802** has been slid into a place such that magnets **5804***a*, **5804***b* can influence both ferrofluids found in the clutch cavities **5808**, **5810** via their magnetic fields. That is, the rotary drive **5802** is slid into a position where the magnets **5804***a*, **5804***b* are roughly both in between the clutch cavities **5808**, **5810**. This may cause the ferrofluids to align and attach to the rotating primary shaft **5804**. When this occurs, rotation of the rotary drive **5802** causes rotation of the rotating primary shaft **5804**, along with rotation of the exterior frame **5806**. The rotation of the exterior frame **5806** may cause one of the actions described in FIG. **143** to occur. In general, shown herein is another example of how a rotary drive may be configured to engage one or more clutch mechanisms of the shaft assembly. Once a clutch is engaged, rotation of the rotary drive may cause an action to occur consistent with any of the methods for connecting a clutch to an action, described herein.

Referring to FIG. **144**C, shown is a cross-sectional view of the shaft assembly when the rotary drive is not engaged with a clutch, according to some aspects. Here, the rotary drive **5802** is configured to move freely, independent of the exterior shaft portion **5806**. Because the rotary drive **5802** is not engaged with a clutch, only the rotary drive **5802** rotates, while the exterior shaft **5806** stays in place.

Referring to FIG. **144**D, shown is a cross-sectional view of the shaft assembly when the rotary drive is engaged with a clutch, according to some aspects. Here, the shaded regions **5812** signify that a clutch is engaged by the ferrofluid being aligned and attaching to the rotary drive **5802**, due to the near proximity of the magnets. Thus, when the rotary drive **5802** is rotated, a component coupled to the clutch, such as the exterior shaft **5806** in this case, will similarly rotate, as shown.

Referring to FIG. **144**E, in some aspects, the sliding design of the rotary drive with magnets may also allow for different combinations of clutch mechanisms to be engaged. For example, rather than having the rotary drive **5802** slid into a position that simultaneously engages both clutch cavities **5808**, **5810**, the rotary drive **5802** may be slid into place to engage only clutch cavity **5808**, as shown. In this position, the magnets **5804***a*, **5804***b* are able to only influence the ferrofluid in the clutch **5808**. Thus, assuming that the clutch **5808** is coupled to a separate action component that is different than what the clutch **5810** is coupled to, then rotation of the rotary drive **5802** causes only the action component of the clutch **5808** to be engaged, while the action component of the clutch **5810** remains inert.

Referring to FIG. **144**F, similarly, the rotary drive **5802** may be slid into a third position to allow engagement only of the second clutch **5810**. As shown, the magnets **5804***a*, **5804***b* are able to only influence the ferrofluid in the clutch **5810**. Thus, assuming that the clutch **5808** and the clutch **5810** have different action components or functions attached to them, then rotation of the rotary drive **5802** in this position causes only the action component of the clutch **5810** to be engaged, while the action component or function of the clutch **5808** remains inert. In general, because of the design of the shaft assembly that includes the two clutch cavities spaced near each other, and with the magnets in the rotary drive **5802** sufficiently long enough, it is possible for the rotary drive **5802** to engage both clutches simultaneously, either clutch only, or neither clutches at the same time.

Referring to FIG. **145**A, in some aspects, the shaft assembly may include a locking element to help stabilize engagement of the clutches. The locking element may be biased into a locked state to hold the clutch engagement mechanism in a non-energize state but is automatically deactivated within a first few degrees of clutch drive motion. In this example, the locking element includes a set of ridges or teeth **5902** on the outer side of an interior shaft **5906**. An outer shaft may include a set of ridges or grooves **5904** to allow the teeth **5902** to rotate into place. If these teeth **5902** and grooves **5904** are out of alignment, then the shaft assembly may be considered in an unlocked state. Furthermore, the rotational alignment of the teeth **5902** may be controlled by a spring mechanism **5908**. The spring mechanism **5908** may be manually biased such that the distal end of the locking mechanism **5910** is retracted away from the distal gear **5912**. In some aspects, when the distal end of the locking mechanism **5910** does not engaged the distal gear **5912**, the clutch mechanism may be considered in an unlocked state.

Referring to FIG. **145**B, shown is an example of when the locking element is in the locked state, according to some aspects. Here, the spring mechanism **5908** is relaxed or allowed to enter its naturally biased state. In its naturally biased state the spring mechanism **5908** causes the interior shaft **5906** to rotate, thereby causing the teeth **5902** to rotate and align with the grooves **5904**, as shown. This creates a notable space **5914** that shows the components are in alignment. In addition, the teeth at the distal end of the locking element **5910** is now pushed to engage with the gear **5912**. The distal end **5910** engages the distal gear **5912**, preventing it from moving so that the gears **5912** are unable to rotate.

Referring to FIGS. **146**A and **146**B, shown are side views of the locking element, according to some aspects. In FIG. **146**A, the spring mechanism **5952** is naturally biased to push against the other locking mechanism components. This causes the teeth **5954** of the interior shaft to align with the grooves of the outer shaft **5956**. In contrast, a force pushing out towards the spring mechanism **5952** causes the spring mechanism **5952** to compress, and allows for a rotation of the teeth of the interior shaft **5954**. This shows that the teeth **5954** are out of alignment with the grooves **5956**, representing the locking element in an unlocked state.

In one aspect, the present disclosure provides a system comprising a surgical instrument comprising: a handle assembly; a shaft assembly coupled to the handle assembly; and an end effector coupled to a distal end of the shaft assembly; the shaft assembly comprising: an exterior shaft; a first clutch positioned within the exterior shaft and configured to drive a first function of the surgical instrument when engaged; a second clutch positioned within the exterior shaft and configured to drive a second function of the surgical instrument when engaged; and a rotary driver positioned within the exterior shaft and configured to rotate within the exterior shaft, the rotary driver comprising: a first clutch engagement mechanism configured to engage the first clutch; and a second clutch engagement mechanism configured to engage the second clutch independent of the first clutch engagement mechanism engaging the first clutch, such that, at different times: both the first clutch and the second clutch are engaged simultaneously; only the first clutch is engaged; only the second clutch is engaged; and neither the first clutch nor the second clutch is engaged simultaneously.

The first clutch may comprise a cavity containing a magnetorheological fluid configured to be magnetized in the presence of an electric current. The first clutch engagement mechanism may comprise an electrical conduit configured to cause the first clutch engagement mechanism to engage the first clutch by generating an electrical current near the magnetorheological fluid that causes a magnetic attraction between the first clutch and the first clutch engagement mechanism. The second clutch may comprise a cavity containing the magnetorheological fluid configured to be magnetized in the presence of an electric current; and the second clutch engagement mechanism comprises a second electrical conduit configured to cause the second clutch engagement mechanism to engage the second clutch by generating a second electrical current that causes a magnetic attraction between the second clutch and the second clutch engagement mechanism.

The first function may comprise a rotation of a distal end of the shaft assembly. The second function may comprise a closure of a pair of jaws of the end effector. The shaft assembly may further comprise an articulation joint; a proximal shaft component coupled to the handle assembly;

and a distal shaft component coupled to the proximal component and configured to articulate about the articulation joint. The rotary shaft may further comprise a proximal rotary shaft component coupled to the handle assembly and a distal rotary shaft component coupled to the proximal rotary shaft component and configured to articulate about the articulation joint as the distal shaft component articulates. The distal rotary shaft component may comprise the first clutch engagement mechanism and the second clutch engagement mechanism.

The proximal shaft component may comprise a third clutch positioned within the exterior shaft and configured to drive a third function of the surgical instrument when engaged; and the proximal rotary shaft component comprises a third clutch engagement mechanism configured to engage the third clutch independent of the first and second clutch engagement mechanisms engaging the first clutch and the second clutch, respectively. The third function may comprise an articulation of the shaft about the articulation joint. The surgical instrument may further comprise a locking mechanism.

In another aspect, the present disclosure provides a surgical instrument comprising a handle assembly; a shaft assembly coupled to the handle assembly; and an end effector coupled to a distal end of the shaft; the shaft assembly comprising: an exterior frame; a first clutch positioned within the exterior frame and configured to drive a first function of the surgical instrument when engaged; a second clutch positioned within the exterior frame and configured to drive a second function of the surgical instrument when engaged; and a rotary driver having a portion positioned within the exterior frame and configured to rotate about a longitudinal axis and slide longitudinally within the exterior frame, the rotary driver comprising: a clutch engagement mechanism configured to: engage only the first clutch when the rotary driver is slid into the exterior frame at a first position; engage both the first and the second clutch simultaneously when the rotary driver is slid into the exterior frame at a second position; engage only the second clutch when the rotary driver is slid into the exterior frame at a third position; and engage neither the first nor the second clutch when the rotary driver is slid into the exterior frame at a fourth position.

Modular Battery Powered Handheld Surgical Instrument with Self-Diagnosing Control Switches for Reusable Handle Assembly

In another aspect, the present disclosure provides a modular battery powered handheld surgical instrument with self-diagnosing control switches for reusable handle assembly. Disclosed is a system and medical device that includes self diagnosing control switches. The control switch may be slidable within a slot in order to control activation of some function of the medical device. Due to natural wear and tear of movement of a control switch, the distances along the sliding slot that correspond to how much energy is used for the function may need to be adjusted over time in order to reflect the changing physical attributes of the actuator mechanism. The self diagnosing control switches of the present disclosures may be configured to automatically adjust for these thresholds using, for example, Hall effect sensors and magnets. In addition, in some cases, the self diagnosing control switches may be capable of indicating external influences on the controls, as well as predict a time until replacement is needed.

With reference ow to FIGS. **147**-**149**, in various aspects, the present disclosure provides self diagnosing control switches within the battery powered, modular, reusable handle containing control switches capable of adjusting their thresholds for triggering an event by detecting degradation in their function/performance to compensate for its own wear. In one aspect, the self diagnosing control switches are configured to indicate predicted time to failure/replacement or when they have exceeding their adjustability range, requiring replacement. The self diagnosing control switches may be adapted for use with any of the surgical instruments **100**, **480**, **500**, **600**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1**-**70**.

Aspects of the present disclosure are presented for a system and surgical instrument that includes self diagnosing control switches, examples of which are shown in FIGS. **150**-**152**. The system and surgical instrument may be battery-powered, modular, and/or with a reusable handle assembly that may be capable of adjusting the thresholds which trigger an event, such as when to turn on or off power to a component of the surgical instrument. In addition, in some cases, the self diagnosing control switches may be capable of indicating external influences on the controls, as well as predict a time until replacement is needed.

In some aspects, a self diagnosing control switch of the system or surgical instrument may include one or more sensors capable of measuring or detecting a signal strength of some kind, such as strength of a magnetic field. As an example, Hall-effect sensors may be located on a stationary component of the surgical instrument, such as on the inside of an outer frame of a handle assembly. One or more magnets may be placed on an actuator component, such as a sliding switch. The switch to may be configured to deliver power to a component of the surgical instrument when slid to one side, and may be configured to turn power off to the component when slid to the other side. In some aspects, the actuator component can also include a spring biased element on either side of where he can slide, or at the center of its functional range.

The slot in which the actuator mechanism slides in may offer a continuous voltage output, such that zero voltage is output when the actuator component is slid all the way on one side of the slot, and the voltage output continuously and monotonically increases to a maximum voltage output as the actuator component slides to the other side. The "on" threshold and the "off" thresholds for triggering the associated component with this switch may exist at some intermediary points along the sliding scale. However, due to natural wear and tear of movement of the switch and the electrical components connected to the actuator component, the distances along the sliding slot to represent when these thresholds occur may need to be adjusted over time in order to reflect the changing physical attributes of the actuator mechanism. The self diagnosing control switches of the present disclosures may be configured to automatically adjust for these thresholds using, for example, Hall effect sensors and magnets.

In some aspects, the self diagnosing control switch may be also configured to deliver a warning signal to signify when the self adjusting mechanisms are reaching their limits. At this point, it may be a signal to replace the modular component that includes this failing self diagnosing control switch.

Referring to FIG. **147**, graph **6201** shows an example plot of the amount of voltage output by a power switch in relation to its sliding distance within a slot of a self diagnosing control switch, according to some aspects. In this case, the

X axis **6202** represents the voltage output of the control switch, while the Y axis **6204** represents displacement along the sliding slot of the control switch, starting from one side (e.g., the “off” position). While voltage is used in this example and the following figures, other measurements of energy may be used in the self diagnosing control switches, and aspects are not so limited, including current, power, impedance, among others. Continuing with this example, the curve **6206** represents an amount of voltage output by the control switch at a given displacement along the slot within which the control switch slides. Phrased another way, the graph **6201** shows how much distance the control switch needs to be slid along the slot in order to deliver a desired output voltage. As shown, this displacement is not necessarily a one-to-one or linear relationship. For example, as shown, the initial sliding distance of the control switch is fairly insensitive to change in voltage, while the last parts of displacement of the control switch a relatively much more sensitive to change in voltage.

In some aspects, the self diagnosing control switches may be configured to provide an alert if it is determined that outside influences may be trying to tamper with or disrupt the predicted displacement versus voltage profile **6206**. First off, a voltage may be detected that is beyond intended range of the control switch. These voltages are designated by the shaded regions **6212** and **6214**. If a voltage is ever detected within these regions, an alarm can be set off and may signal that there is either a malfunction or there is a suspected tampering. In addition, the expected displacement versus voltage profile curve **6206** may be recorded, such as stored in a memory coupled to a processor of the medical instrument. The displacement along the sliding slot may be measured and compared to the voltage output is. At any given point, if it is detected that there is a drastic change in voltage that does not match with displacement, compared to the previously recorded voltage at that same displacement, then it may also be determined that there is either a malfunction or a suspected tampering, and the alarm may also be raised.

In some aspects, the control switch may be subjected to one or more thresholds that determine when an event occurs. For example, a first threshold **6208** may be in place to determine when a functional component associated with this control switch is powered on, e.g., representing an “on” threshold. In some aspects, this threshold **6208** is activated only in one direction. That is, the “on” threshold is utilized only when the control switch is being slid from the lower voltage to the higher voltage, and is not utilized when sliding the control switch from the higher voltage down to the lower voltage. Instead, in some aspects, a second threshold **6210** may be considered in these use cases. The second threshold **6210** may be used to determine when a second event occurs, such as when the functional component associated with this control switch is powered off, e.g., representing an “off” threshold. Again, the second threshold may be utilized only when the control switch is being slid from the higher voltage down to the lower voltage, and is not utilized when sliding the control switch from the lower voltage up to the higher voltage. In this way, the use of two or more thresholds may allow for a buffer zone **6212** in which no change in event occurs when the control switch is displaced into this zone. The use of such buffer zones may increase safety and prevent harmful dithering of quick oscillations between on and off states, for example. In general, other types of events may be codified with these determined thresholds, such as changing speeds or changing power levels, and other events apparent to those with skill in the art. Similarly, any number of thresholds may be utilized, and aspects are not so limited.

The self diagnosing switches may be configured to adjust these thresholds to account for changing physical conditions over time of the electrical and mechanical components that make up this power, according to some aspects. Referring to FIG. **148**, graph **6220** provides an example of how the displacement vs. voltage profile of a control switch may change over time, due to natural wear and tear of the mechanical and electrical components. While the original plot **6206** is shown in solid line, the curve **6222** showed in dashed line represents an example of the amount of displacement needed to achieve a desired voltage for a control switch, due to natural wear and tear for using the control switch over time. In this example, the initial sliding of the control switch from the off position now produces a much more drastic voltage output. In addition, the middle of the curve **6222** shows that the voltage changes very little in the middle, and towards the maximum displacement, the voltage again changes more drastically.

The changing attributes of the control switch over time may be a problem if the “on” and “off” thresholds **6224**, **6226**, respectively, remain as they were according to the initial displacement versus voltage profile. This may be because the amount of voltage being output at an intended displacement of the control switch is now considerably different at the same amount of displacement along the sliding slot. Thus, the thresholds **6224**, **6226** may not accurately reflect when a functional component associated with the control switch has enough power to be turned on, and/or no longer has enough power and is to be turned off.

To address this, in some aspects, a sensor may be coupled to the functional component or the self diagnosing control switch to measure voltage output. The self diagnosing control switch may include a system to measure displacement sliding along the slot, such as including one or more sensors, such as a hall effect sensor, and one or more tracking components, such as a magnet, to be included in or around the sliding slot. For example, a Hall effect sensor may be coupled to the stationary frame around the control switch, while the actuator component of the control switch may have a magnet coupled to it. The sensors may then deliver data to a processor that monitors the relationship between voltage output and displacement along the sliding slot. Over time, it may be determined that the voltage output has changed at a given point of displacement, based on the simultaneous readings from the different sensors. Specified thresholds for when an event (e.g., “on” or “off”) occurs may be adjusted to account either for the change in the displacement at a given voltage or the change in voltage at a given displacement. Thereafter, the processor may activate or deactivate the event in accordance with the new thresholds. In general, these thresholds may be continually adjusted based on automatic feedback of the self diagnosing control switch. In other cases, a user may manually activate a calibration or self diagnosing routine to cause the processor to perform this kind of maintenance and adjust the thresholds as needed.

Referring to FIG. **149**, in some aspects, the self diagnosing control switch of the present disclosures may be configured to transmit an alert when it is determined that automatic adjustments to account for wear and tear are no longer possible. The graph **6230** provides two examples showing abnormal endpoints of the displacement versus voltage relationship, both signaling that the full range of intended voltage and/or displacement of the control switch is no longer possible. With original curve **6206** as a reference, the curve **6232** shown in heavy dashed line is an example of a situation where the maximum sliding displacement of the control switch still fails to reach even close to the maximum voltage output. That is, even at the maximum displacement, approximately only 4V-6V can be achieved. In this case, the self diagnosing control switch may be configured to transmit an alert or alarm signifying that the desired voltage cannot be achieved, or more generally, that there is a malfunction. In general, this signal may represent that repairs are needed, or that this modular component associated with this control switch needs to be replaced.

As another example, the curve **6234** shown in light dashed line shows a situation where maximum sliding displacement of the control switch is unnecessary, as maximum voltage may already be achieved after just a small amount of sliding. This may also be a problem because it may signal that a mechanism for controlling the output voltage is malfunctioning. Also in this case, the self diagnosing control switch may be configured to transmit an alert or alarm signifying that proper voltage control is not present or malfunctioning. In general, this signal may represent that repairs are needed, or that this modular component associated with this control switch needs to be replaced. In general, these examples show how the self diagnosing control switch of the present disclosures may be used to provide alerts for end-of-life stages of the control switches.

In addition, in some aspects, the self diagnosing control switches also may be configured to anticipate end-of-life scenarios. For example, the displacement versus voltage curves may be continually recorded over time, such that the changes over time may be monitored. These changes over time may be used to create an extrapolation profile that projects when these changes, assuming they continue at the observed rate (or based on other projected factors), will result in an abnormal endpoint, similar to the example curves **6232**, **6234**. The self diagnosing control switch may then be configured to report this projected end-of-life term, or deliver a signal representing that there is a set amount of time—or number of uses—estimated until repairs are needed.

Examples of self diagnosing control switches provided on battery powered modular surgical instrument are shown in FIGS. **150-152**. The diagnostic techniques described in connection with FIGS. **147-149** may be adapted for use with the battery powered modular surgical instruments **6301**, **6502**, **6602** shown in FIGS. **150-152**. Turning first to FIG. **150**, there is shown a side elevational view of one aspect of a handle assembly **6301** of a modular surgical instrument, according to one aspect of the present disclosure. The handle assembly **6301** may be engaged with an interchangeable shaft assembly (not shown). The handle assembly **6301** includes a closure trigger **6303**, pistol grip portion **6305**, an interface **6307**, and a user display **6310**, which may also be configured to receive an input from a user. The handle assembly **6301** also includes a closure release button assembly **6312**, which can be depressed by the operator to release a closure lock thereby returning an end effector (not shown) to a desired configuration. The handle assembly **6301** is shown with a socket **6314** for a battery and the battery pack is removed.

The interface **6307** may comprise at least one user-actuated input device, such as a switch, that may be configured to provide functionality as described herein. According to aspects the interface **6307** may be utilized to implement an articulation function of an interchangeable shaft assembly. Additionally, according to the aspect shown in FIG. **150**, the interface **6307** comprises a plurality of user-actuated input devices, switches **6316**, **6318**, **6320**, which can be utilized by the operator, in part, to activate and control functions associated with an end effector of an interchangeable shaft assembly, as described herein. According to an aspect, the handle assembly **6301** may be combined with an interchangeable shaft assembly having a cutting member as described herein. In such an aspect, the switches **6316**, **6318**, **6320** may be used to control activation and speed of the cutting member.

As shown in FIG. **150**, the switch **6316** is a button style switch and switches **6318** and **6320** are part of a toggle style switch. The interface **6307** may be configured such that the switch **6316** is dedicated to activation and/or selection of the speed of a cutting member. Further, switch **6318** may be dedicated to increasing the speed of the cutting member and the switch **6320** may be dedicated to decreasing the speed of the cutting member. Other means for controlling the switches **6318**, **6320** and/or replacing functionality of the switches **6318**, **6320** with a single switch device are also contemplated within the scope of the present disclosure.

FIG. **151** is a side elevational view of another aspect of a handle assembly **6502** of a battery powered modular surgical instrument, according to one aspect of the present disclosure. According to aspects and with reference to FIG. **150**, the switches **6318**, **6320** of the handle assembly **6301** may be replaced by a single user-actuated input device. As shown in FIG. **151**, the interface **6508** of the handle assembly **6502** comprises a rotary potentiometer **6504** that may be used to adjust the speed of cutting member. For example, in one aspect, counter-clockwise rotation of the rotary potentiometer **6504** increases the speed of the cutting member whereas clockwise rotation of the rotary potentiometer **6504** decreases the speed of the cutting member. The output of the rotary potentiometer **6504** can be provided to an external A/D converter or to an A/D converter of a controller. In other aspects, the rotary potentiometer **6504** may be replaced with a rotary encoder whose output can be provided to the controller **6402**.

FIG. **152** is a side elevational view of another aspect of a handle assembly **6602** of a battery powered modular surgical instrument, according to one aspect of the present disclosure. In addition, in the aspect shown in FIG. **152**, the interface **6608** of the handle assembly **6602** comprises a slide potentiometer **6604** that may be used to adjust the speed of cutting member such that sliding the sliding bar to the left increases the speed of the cutting member whereas sliding the sliding bar to the right decreases the speed of the cutting member. The output of the slide potentiometer **6604** can be provided to an external A/D converter or to an internal A/D converter of a controller. In other aspects, the slide potentiometer **6604** may be replaced with a slide encoder whose output can be provided to the controller **6402**.

According to aspects and with reference to FIGS. **151** and **152**, the rotary potentiometer **6504** or the slide potentiometer **6604** may be used to replace the functionality associated with electrical contact switches. Further, referring to FIGS. **151** and **152**, circuits associated with the rotary potentiometer **6504** or the slide potentiometer **6504** may be combined into a single circuit. In one aspect, the rotary potentiometer **6504** or slide potentiometer **6604** may comprise a variable resistor coupled to an A/D converter. Accordingly, a controller may receive a digitized signal from either the rotary potentiometer **6504** or slide potentiometer **6604** and provide it to a motor controller **6420** for controlling the speed of a motor.

In another aspect, the present disclosure provides a system comprising: a surgical instrument comprising: a handle assembly; a shaft assembly coupled to the handle assembly; and an end effector coupled to a distal end of the shaft assembly; a self-diagnosing control switch system comprising: a control switch slidable within a slot comprising a first end and a second end of the slot, the control switch configured to: slide within the slot between the first end and the second end; and allow an amount of energy to be delivered to a functional component of the surgical instrument, the amount of energy in proportion to a degree of displacement of the control switch away from the first end of the slot; a displacement sensor configured to measure the degree of displacement of the control switch away from the first end of the slot; an energy sensor configured to measure the amount of energy delivered to the functional component; and a processor configured to: determine a threshold level of displacement of the control switch away from the first end of the slot that triggers a functional event of the functional component, based on the measured degree of displacement and the measured amount of energy delivered to the functional component.

The processor may be further configured to determine a limit to the degree of displacement. The processor may be further configured to: determine that the measured degree of displacement exceeds the determined limit; and transmit an alarm based on the determined exceeded limit. The processor may be further configured to determine a limit to the amount of energy delivered to the functional component. The processor may be further configured to: determine that the measured amount of energy exceeds the determined limit; and transmit an alarm based on the determined exceeded limit. The processor may be further configured to: record, in a memory, the threshold level of displacement of the control switch as a baseline threshold level of displacement away from the first end of the slot that triggers a functional event; receive, from the energy sensor, a second measurement of an amount of energy delivered to the functional component when the control switch is positioned at the baseline threshold level of displacement; and compare the second measurement of energy to the measurement of energy used to create the baseline threshold level of displacement.

The processor may be further configured to determine that the comparison of the second measurement of energy to the measurement of energy used to create the baseline threshold level of displacement satisfies a predetermined level of change; and determine an adjusted threshold level of displacement of the control switch away from the first end of the slot that triggers the functional event of the functional component, based on the comparison between the second measurement of energy and the measurement of energy used to create the baseline threshold of displacement. The processor may be further configured to calculate an end-of-life term of the control switch at which time the control switch is predicted to no longer function, based on an extrapolation calculation using the adjusted threshold level and the baseline threshold level. The displacement sensor may comprise a Hall effect sensor positioned next to the control switch, and the control switch comprises a magnet.

In another aspect, the present disclosure provides a method for self-diagnosing operation of a control switch in a surgical instrument system, the method comprising: receiving, from a displacement sensor, a baseline degree of displacement of the control switch in a slidable slot of the surgical instrument system; receiving, from an energy sensor, an amount of energy to be delivered to a functional component of the surgical instrument in proportion to the measured baseline degree of displacement of the control switch; determining a baseline threshold level of displacement of the control switch away from the first end of the slot that triggers a functional event of the functional component, based on the measured baseline degree of displacement and the measured amount of energy delivered to the functional component; recording, in a memory, the baseline threshold level of displacement of the control switch; receiving, from the energy sensor, a second measurement of an amount of energy delivered to the functional component when the control switch is positioned at the baseline threshold level of displacement; and comparing the second measurement of energy to the measurement of energy used to create the baseline threshold level of displacement.

The method may further comprise determining that the comparison of the second measurement of energy to the measurement of energy used to create the baseline threshold level of displacement satisfies a predetermined level of change; and determining an adjusted threshold level of displacement of the control switch away from the first end of the slot that triggers the functional event of the functional component, based on the comparison between the second measurement of energy and the measurement of energy used to create the baseline threshold of displacement.

Modular Battery Powered Handheld Surgical Instrument with Reusable Asymmetric Handle Housing

In another aspect, the present disclosure provides a modular battery powered handheld surgical instrument with reusable asymmetric handle housing. Disclosed is a surgical instrument that includes a handle assembly that includes a handle housing. The handle housing comprises two asymmetric portions, a first portion configured to support mechanical and electrical components of the surgical instrument and a second portion comprising a removable cover.

FIGS. **70** and **155**A-**155**B illustrate various configurations of reusable and serviceable handle housings for housing assemblies which are divided into two asymmetric halves such that the control circuits, wiring harness, coupling mechanisms can be supportably housed in one side of the handle housing and support all the actuation forces within that side while the other side is removably attached to cover the primary housing. The handle assemblies illustrated in FIGS. **70** and **155**A-**155**B can be used in the surgical instruments **100**, **470**, **500**, **600**, **700**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1**-**69**.

FIG. **153**A illustrates a cross sectional view of a reusable and serviceable handle assembly **600** with a removable service cover **6004** in an open position, according to one aspect of the present disclosure. FIG. **153**B illustrates a cross sectional view of the reusable and serviceable handle assembly **6000** with the service cover in a closed position, according to one aspect of the present disclosure. The handle assembly **600** comprises a housing **6002** and a perimeter elastomeric seal **6008** to prevent fluid from passing from inside to the outside of the handle assembly **6000**. The seal **6008** is disposed between the removable service cover **6004** and the housing **6002**. A removable switch **6006** is located and supported within the housing **6002**. The removable switch **6006** is accessible for service or replacement by removing the removable service cover **6004**. The removable switch **6006** is electrically coupled to a removable electrical contact **6010**, which is ultimately electrically coupled to a control circuit such as the control circuit **210** (FIG. **14**), **1300**

(FIG. **62**), **1400** (FIG. **63**), **1500** (FIG. **64**) via interface circuits such as circuits **6550**, **6570** (FIGS. **90-91**), for example.

FIG. **154**A illustrates a cross sectional view of a reusable and serviceable handle assembly **6100** with a removable service cover **6104** in an open position, according to one aspect of the present disclosure. FIG. **154**B illustrates a cross sectional view of the reusable and serviceable handle assembly **6100** with the removable service cover **6104** in a closed position, according to one aspect of the present disclosure. The handle assembly **6100** comprises a housing **6102** and a perimeter elastomeric seal **6108** to prevent fluid from passing from inside to the outside of the handle assembly **6100**. The seal **6108** is disposed between the removable service cover **6104** and the housing **6102**. The removable service cover comprises a post **6114** defining a recess **6118** to receive a first magnet **6112***a*. The housing **6102** defines an aperture **6116** to receive the post **6114**. The housing **6102** includes a second magnet **6112***b* that is aligned to magnetically couple to the first magnet **6112***a* when the service cover **6104** is closed as shown in FIG. **154**B when the removable service cover **6104** is magnetically attached to the housing **6102**.

In one aspect, the magnets **6112***a*, **6112***b* can be rare-earth magnets or other strong permanent magnets made from alloys of rare-earth elements (elements in the lanthanide series, plus scandium and yttrium). Rare-earth magnets are the strongest type of permanent magnets made, producing significantly stronger magnetic fields than other types such as ferrite or alnico magnets. The magnetic field typically produced by rare-earth magnets can exceed 1.4 tesla, whereas ferrite or ceramic magnets typically exhibit fields of 0.5 to 1 tesla, for example. Two types of rare-earth magnets that may be employed, among others, are neodymium magnets and samarium-cobalt magnets. Magnetostrictive rare-earth magnets such as Terfenol-D also may be employed. Rare-earth magnets may be plated or coated to protect them from breaking, chipping, or crumbling into powder. Magnets made of alloys of yttrium and cobalt, YCo5, that have a large magnetic anisotropy constant also may be employed.

TABLE 2 below provides a comparison of various types of materials for permanent magnets. Relevant properties used to compare permanent magnets are: remanence (Br), which measures the strength of the magnetic field; coercivity (Hci), the material's resistance to becoming demagnetized; energy product (BHmax), the density of magnetic energy; and Curie temperature (Tc), the temperature at which the material loses its magnetism. Rare earth magnets have higher remanence, much higher coercivity and energy product, but (for neodymium) lower Curie temperature than other types. The TABLE 2 below compares the magnetic performance of the two types of rare-earth magnet, neodymium (Nd2Fe14B) and samarium-cobalt (SmCo5), with other types of permanent magnets.

TABLE 2

Magnet	B_r (T)	H_{ci} (kA/m)	$(BH)_{max}$ (kJ/m^3)	T_c (° C.)
$Nd_2Fe_{14}B$ (sintered)	1.0-1.4	750-2000	200-440	310-400
$Nd_2Fe_{14}B$ (bonded)	0.6-0.7	600-1200	60-100	310-400
$SmCo_5$ (sintered)	0.8-1.1	600-2000	120-200	720
Sm(Co, Fe, Cu, Zr) (sintered)	0.9-1.15	450-1300	150-240	800
Alnico (sintered)	0.6-1.4	275	10-88	700-860
Sr-ferrite (sintered)	0.2-0.4	100-300	10-40	450

With reference now to FIGS. **70** and **154**A-**154**B, the handle assemblies **1970**, **6000**, **6100** can be used with modular battery powered handheld surgical instruments **100**, **470**, **500**, **600**, **700**, **1100**, **1150**, **1200** described herein in connection with FIGS. **1-69** comprising modular disposable shafts, control and wiring harnesses. In one aspect, each of the handle assemblies **1970**, **6000**, **6100** is configured to asymmetrically part when opened so that the switches **1980**, **6006**, wiring harnesses, and/or control electronics can be supportably housed in one side with the other side and removably attached to cover the housing **1974**, **6002**, **6102**. In the examples illustrated in FIGS. **70** and **154**A-**154**B, the handle housing **1974**, **6002**, **6102** is asymmetric with sides which allow for the nesting of life limited components like the switches **1980**, **6006** motor control board, and inner electrical connectors. The housing **1974**, **6002**, **6102** includes a small slot at the top, bottom or side of the handles. A flat bladed tool can be used to apply side load to the handle assemblies **1970**, **6000**, **6100**, creating a shear load against the fasteners. As shown in FIGS. **153**A-**153**B, in on aspect, the fasteners holding the service cover **6104** are magnets **6112***a*, **6112***b*. Accordingly, the flat blade tool needs only to overcome the magnetic attraction to remove the service cover **6104** from the housing **6102**. Once the magnets **6112***a*, **6112***b* are moved relative to one another, the service cover **6104** easily separates from the housing **6102**. The housing **6102** also can include a rotary control that includes a mechanism to push the service cover **6104** apart from the housing **6102** in a shear mode. The magnets **6112***a*, **6112***b* do not easily prevent the relative movement of the service cover **6104** and the housing **6102** in this mode and one of the magnets **6112***a*, **6112***b* is moved apart from one another, they are easily separated.

The removable switches **1980**, **6006** controls, wiring harnesses, and control boards are housed within a structural skeletal frame of the housing **1974**, **6002**, **6102**. The outer service covers **1972**, **6002**, **6102** are removable on one or both sides of the handle assemblies **1970**, **6000**, **6100** to allow servicing, maintenance, and cleaning but the skeleton protects and restrains the electronic components not the removable service covers **1972**, **6002**, **6102**. In one aspect, the service covers **1972**, **6002**, **6102** include interlocking features between them and the central skeletal system. In one aspect, as illustrated in FIGS. **153**A-**153**B, the magnets **6112***a*, **6112***b* hold the removable service cover **6104** in place but the key interlocks prevent shearing forces applied to the service cover **6104** from inadvertently dislodging the service cover **6104**.

FIGS. **155**A-**155**B illustrates one aspect of a handle assembly **6120**, according to one aspect of the present disclosure. FIG. **155**A illustrates the handle assembly **6120** in a secured fastened configuration and FIG. **155**B illustrates the handle assembly **6120** in an unlatched configuration. The handle assembly **6120** comprises a first housing **6122***a* (e.g., shroud) and a second housing **6122***b* (e.g., shroud) that can be latched and unlatched. A rotation knob **6124** is located on a distal end of the housing assembly **6120**. As previously discussed, the rotation knob **6124** is operably coupled to a shaft assembly (not shown). The handle assembly **6120** comprises a removable trigger **6126** that is operably coupled to an end effector (not shown). High-frequency (e.g., RF) energy and ultrasonic energy is activated using removable switches **6128***a*, **6128***b*. The removable switches **6128***a*, **6128***b* may be referred to as buttons, for example. An aperture **6130** is defined by the handle assembly **6120** to receive an ultrasonic transducer assembly (not shown).

In the configuration shown in FIG. **155**A, the two housings **6122***a*, **6122***b* (shrouds) of the handle assembly **6120** are secured fastened using hooked tabs **6132***a*, **6132***b* on the first shroud **6122***a*, as shown in FIG. **155**B, that snap into access slots **6134***a*, **6134***b* defined by the second shroud **6134***b*. The access slots **6134***a*, **6134***b* enable access to the hooked tabs **6132***a*, **6132***b*. The hooked tabs **6132***a*, **6132***b* can be accessed at the parting line **6136** by inserting a tool to deflect the hooked tabs **6132***a*, **6132***b* unlatching the shroud **6122***a*, **6122***b* halves.

In another aspect, the present disclosure provides a surgical instrument comprising: a handle assembly comprising a handle housing, wherein the handle housing comprises two asymmetric portions, a first portion configured to support mechanical and electrical components of the surgical instrument and a second portion comprising a removable cover.

The first portion and the removable cover each may comprise a fastener to removably connect the removable cover to the first portion of the handle housing. The fastener may comprise a permanent magnet. The fastener may comprise a hooked tab and a corresponding slot to receive the hooked tab and snap fit the removable cover to the first portion of the handle housing.

The surgical instrument may further comprise an elastomeric seal positioned about a perimeter of the first portion of the handle housing to provide a seal between the first portion of the handle housing and the removable cover. The surgical instrument may further comprise a removable switch is located within and supported by the first portion of the handle housing. The surgical instrument may further comprise a removable electrical contact located within and supported by the first portion of the handle housing. The surgical instrument may further comprise a removable motor located within and supported by the first portion of the handle housing. The surgical instrument may further comprise a removable trigger located within and supported by the first portion of the handle housing. The surgical instrument may further comprise a removable electrical contact located within and supported by the first portion of the handle housing. The first portion of the handle housing may define an aperture to receive a removable ultrasonic transducer assembly.

In another aspect, the present disclosure provides a surgical instrument comprising: a handle assembly comprising a handle housing, wherein the handle housing comprises two asymmetric portions, a first portion configured to support mechanical and electrical components of the surgical instrument and a second portion comprising a removable cover; an elastomeric seal positioned about a perimeter of the first portion of the handle housing to provide a seal between the first portion of the handle housing and the removable cover; a removable switch is located within and supported by the first portion of the handle housing; a removable electrical contact located within and supported by the first portion of the handle housing; a removable trigger located within and supported by the first portion of the handle housing; and a removable electrical contact located within and supported by the first portion of the handle housing.

The first portion of the handle housing may define an aperture to receive a removable ultrasonic transducer assembly. The first portion and the removable cover each may comprise a fastener to removably connect the removable cover to the first portion of the handle housing. The fastener may comprise a permanent magnet. The fastener may comprise a hooked tab and a corresponding slot to receive the hooked tab and snap fit the removable cover to the first portion of the handle housing. The surgical instrument may further comprise a removable motor located within and supported by the first portion of the handle housing.

In another aspect, the present disclosure provides a handle assembly, comprising a handle housing, wherein the handle housing comprises two asymmetric portions, a first portion configured to support mechanical and electrical components of the surgical instrument and a second portion comprising a removable cover, wherein the first portion and the removable cover each comprises a fastener to removably connect the removable cover to the first portion of the handle housing; an elastomeric seal positioned about a perimeter of the first portion of the handle housing to provide a seal between the first portion of the handle housing and the removable cover; a removable switch is located within and supported by the first portion of the handle housing; a removable electrical contact located within and supported by the first portion of the handle housing; a removable trigger located within and supported by the first portion of the handle housing; a removable electrical contact located within and supported by the first portion of the handle housing; and a removable motor located within and supported by the first portion of the handle housing. The fastener may comprise a permanent magnet. The fastener may comprise a hooked tab and a corresponding slot to receive the hooked tab and snap fit the removable cover to the first portion of the handle housing.

Modular Battery Powered Handheld Surgical Instrument with Curved End Effectors Having Asymmetric Engagement Between Jaw and Blade

In another aspect, the present disclosure provides a modular battery powered handheld surgical instrument with curved end effectors having asymmetric engagement between jaw and blade. Disclosed is an end effector for a surgical instrument that includes an ultrasonic blade and a jaw member including an asymmetric electrode comprising a first electrode and a second electrode. The first electrode defines a first width and the second electrode defines a second width. The first width is not equal to the second width. A first gap is defined between the first electrode and the ultrasonic blade and a second gap is defined between the second electrode and the ultrasonic blade. The first gap is not equal to the second gap.

FIG. **156** is a plan view of one aspect of an end effector **6650**. The end effector **6650** comprises a jaw member **6652** and a shaft **6654**. The jaw member **6652** pivots about pivot point **6656** and defines a pivot angle. FIG. **157** is a side view of the end effector **6650** shown in FIG. **156** with a partial cut away view to expose the underlying structure of the jaw member **6652** and an ultrasonic blade **6658**. An electrode **6660** is fixedly mounted to the jaw member **6652**. The electrode **6660** is electrically coupled to the RF drive circuit **702** (FIG. **34**) and is configured to apply RF energy to tissue located between the jaw member **6652** and the ultrasonic blade **6658**. FIG. **158** is a partial sectional view of the end effector shown in FIGS. **156**, **127** to expose the ultrasonic blade and right and left electrodes **6660***a*, **6660***b*, respectively. The jaw member **6652** and the ultrasonic blade **6658** are wider at a proximal end and are narrower at a distal end. Also, the jaw member **6652** and the ultrasonic blade **6658** define more curvature at a distal end relative to the proximal end. These features are clearly shown in the sectional views of FIGS. **159-164**.

FIG. **159** is a cross-sectional view taken at section **174-174** of the end effector **6650** shown in FIG. **156**. The end effector **6650** comprises an ultrasonic blade **6658** acoustically coupled to an ultrasonic transducer which is electrically driven by the ultrasonic drive circuit **177** (FIG. **11**). The jaw member **6652** comprises an electrode **6660** comprising asymmetric electrodes **6660***a*, **6660***b*. A first electrode **6660***a* is located on the right side and a second electrode **6660***b* is located on the left side (from the perspective of the operator) of the jaw member **6652**. The right side electrode **6660***a* defines a first width W**1** and defines a first gap G**1** between the electrode **6660***a* and the ultrasonic blade **6658**. The left side electrode **6660***b* defines a second width W**2** and defines a second gap G**2** between the electrode **6660***b* and the ultrasonic blade **6658**. In one aspect the first width W**1** is less than the second width W**2** and the first gap G**1** is less than the second gap G**2**. With reference also to FIG. **158**, an electrically insulative element is disposed between the first and second electrodes **6660***a*, **6660***b*. In one aspect, the electrically insulative element comprising a soft polymeric pad **6662** located between the ultrasonic blade **6658** and the jaw member **6652** and a high density polymeric pad **6664** located adjacent the soft polymeric pad **6662** to prevent the ultrasonic blade **6658** from shorting the electrodes **6660***a*, **6660***b*. In one aspect, the soft and high density polymeric pads **6662**, **6664** can be made of polymers known under the tradename TEFLON (polytetrafluoroethylene polymers and copolymers), for example. Accordingly, the polymeric pads **6662**, **6664** may be made of soft TEFLON and high density TEFLON, respectively.

FIG. **160** is cross-sectional view taken at section **175-175** of the end effector **6650** shown in FIG. **156**. At the plane where section **175-175** the end effector **6650** is thinner and has more curvature than section **174-174**. The right side electrode **6660***a* defines a third width W**3** and defines a third gap G**3** between the electrode **6660***a* and the ultrasonic blade **6658**. The left side electrode **6660***b* defines a fourth width W**4** and defines a fourth gap G**4** between the electrode **6660***b* and the ultrasonic blade **6658**. In one aspect the third width W**3** is less than the fourth width W**4** and the third gap G**3** is less than the fourth gap G**4**. In operation, tissue located in the gaps G**1**, G**2**, G**3**, G**4** defined between the electrodes **6660***a*, **6660***b* and the ultrasonic blade **6658** is sealed by high-frequency current transmitted through the electrodes **6660***a*, **6660***b*, the tissue, and the ultrasonic blade **6658**. The tissue located between the polymeric pad **6662** and the ultrasonic blade **6658** is cut by friction generated by the ultrasonic vibrations.

FIG. **161** is a cross-sectional view taken at a section similar to section **174-174** of the end effector **6650** shown in FIG. **156**, except that the ultrasonic blade **6658'** has a different geometric configuration. The end effector **6650'** comprises an ultrasonic blade **6658'** acoustically coupled to an ultrasonic transducer which is electrically driven by the ultrasonic drive circuit **177** (FIG. **11**). The jaw member **6652'** comprises an electrode **6660'** comprising asymmetric electrodes **6660***a*', **6660***b*'. A first electrode **6660***a*' is located on the right side and a second electrode **6660***b*' is located on the left side (from the perspective of the operator) of the jaw member **6652'**. The right side electrode **6660***a*' defines a first width W**1** and defines a first gap G**1** between the electrode **6660***a*' and the ultrasonic blade **6658'**. The left side electrode **6660***b*' defines a second width W**2** and defines a second gap G**2** between the electrode **6660***b*' and the ultrasonic blade **6658'**. In one aspect the first width W**1** is less than the second width W**2** and the first gap G**1** is less than the second gap G**2**. An electrically insulative element is disposed between the first and second electrodes **6660***a*', **6660***b*'. In one aspect, the electrically insulative element comprises a soft polymeric pad **6662'** located adjacent to a high density polymeric pad **6664'** to prevent the ultrasonic blade **6658'** from shorting the electrodes **6660***a*', **6660***b*'. In one aspect, the polymeric pads **6662'**, **6664'** can be made of polymers known under the tradename TEFLON (polytetrafluoroethylene polymers and copolymers), for example. In the aspect shown in FIGS. **176** and **177**, the ultrasonic blade **6658'** comprises features **6668'** suitable for cutting and features **6670'** suitable for coagulating tissue. The cutting features **6668'** define a smaller surface area relative to the coagulation features **6670'** to enable a more suitable cutting interface between the tissue and the ultrasonic blade **6658'**. The larger surface area of the coagulation features **6670'** are better suited to coagulation or seal tissue. Thus, in operation, tissue located between the electrodes **6660***a*', **6660***b*' and the coagulation features **6670'** is sealed and tissue located between the cutting feature **6668'** and the polymeric pad **6662'** is cut.

FIG. **162** is cross-sectional view taken at a section similar to section **175-175** of the end effector **6650** shown in FIG. **156**, except that the ultrasonic blade **6658'** has a different geometric configuration. At the plane where section **175-175** the end effector **6650'** is thinner and has more curvature than the end effector **6650'** at section **174-174**. The right side electrode **6660***a*' defines a third width W_3 and defines a third gap G_3 between the electrode **6660***a*' and the ultrasonic blade **6658'**. The left side electrode **6660***b*' defines a fourth width W_4 and defines a fourth gap G_4 between the electrode **6660***b*' and the ultrasonic blade **6658'**. In one aspect the third width W_3 is less than the fourth width W_4 and the third gap G_3 is less than the fourth gap G_4. In operation, tissue located in the gaps G_1, G_2, G_3, G_4 defined between the electrodes **6660***a*', **6660***b*' and the ultrasonic blade **6658'** is sealed by high-frequency current transmitted through the electrodes **6660***a*', **6660***b*', the tissue, and the ultrasonic blade **6658'**. The tissue located between the polymeric pad **6662'** and the ultrasonic blade **6658'** is cut by friction generated by the ultrasonic vibrations.

FIG. **163** is a cross-sectional view taken at a section similar to section **174-174** of the end effector **6650** shown in FIG. **156**, except that the ultrasonic blade **6658"** has a different geometric configuration. The end effector **6650"** comprises an ultrasonic blade **6658"** cacoustically coupled to an ultrasonic transducer which is electrically driven by the ultrasonic drive circuit **177** (FIG. **11**). The jaw member **6652"** comprises an electrode **6660"** comprising asymmetric electrodes **6660***a*". A first electrode **6652***a*" is located on the right side and a second electrode **6660***b*" is located on the left side (from the perspective of the operator) of the jaw member **6652"**. The right side electrode **6660***a*" defines a first width W_1 and defines a first gap G_1 between the electrode **6660***a*" and the ultrasonic blade **6658"**. The left side electrode **6660***b*" defines a second width W_2 and defines a second gap G_2 between the electrode **6660***b*" and the ultrasonic blade **6658"**. In one aspect the first width W_1 is less than the second width W_2 and the first gap G_1 is less than the second gap G_2. An electrically insulative element is disposed between the first and second electrodes **6660***a*", **6660***b*". In one aspect, the electrically insulative element comprises a soft polymeric pad **6662"** located adjacent to a high density polymeric pad **6664"** to prevent the ultrasonic blade **6658"** from shorting the electrodes **6660***a*", **6660***b*". In one aspect, the polymeric pads **6662"**, **6664"** can be made of polymers known under the tradename TEFLON (polytetrafluoroethylene polymers and copolymers), for example.

FIG. **164** is cross-sectional view taken at a section similar to section **175-175** of the end effector **6650** shown in FIG. **156**, except that the ultrasonic blade **6658"** has a different geometric configuration. At the plane where section **175-175** the end effector **6650"** is thinner and has more curvature than the end effector **6650"** at section **98-98**. The right side electrode **6660***a*" defines a third width W_3 and defines a third gap G_3 between the electrode **6660***a*" and the ultrasonic blade **6658"**. The left side electrode **6660***b*" defines a fourth width W_4 and defines a fourth gap G_4 between the electrode **6660***b*" and the ultrasonic blade **6658"**. In one aspect the third width W_3 is less than the fourth width W_4 and the third gap G_3 is less than the fourth gap G_4. In operation, tissue located in the gaps G_1, G_2, G_3, G_4 defined between the electrodes **6660***a*", **6660***b*" and the ultrasonic blade **6658'** is sealed by high-frequency current transmitted through the electrodes **6660***a*", **6660***b*", the tissue, and the ultrasonic blade **6658"**. The tissue located between the polymeric pad **6662"** and the ultrasonic blade **6658"** is cut by friction generated by the ultrasonic vibrations. In the aspect shown in FIGS. **178** and **179**, the ultrasonic blade **6658"** comprises features **6668"** suitable for cutting and features **6670"** suitable for coagulating tissue. The cutting features **6668"** define a smaller surface area relative to the coagulation features **6670"** to enable a more suitable cutting interface between the tissue and the ultrasonic blade **6658"**. The larger surface area of the coagulation features **6670"** are better suited to coagulation or seal tissue. Thus, in operation, tissue located between the electrodes **6660***a*", **6660***b*" and the coagulation features **6670"** is sealed and tissue located between the cutting feature **6668"** and the polymeric pad **6662"** is cut.

The surgical instruments described herein also can include features to allow the energy being delivered by the ultrasonic drive circuit **177** (FIG. **11**) and the RF drive circuit **702** (FIG. **34**) to be dynamically adjusted or switched based on the type of tissue being treated by an end effector of a surgical instrument and various characteristics of the tissue. In one aspect, the power output from the ultrasonic drive circuit **177** and the RF drive circuit **702** that is delivered to the end effector of the surgical instrument can include an input that represents the tissue type to allow the energy profile of the from the ultrasonic drive circuit **177** and the RF drive circuit **702** to be dynamically changed during the procedure based on the type of tissue being effected by the end effector of the surgical instrument. As disclosed herein, techniques for controlling the ultrasonic drive circuit **177** (FIG. **11**) and the RF drive circuit **702** (FIG. **34**) based on the tissue type may be provided. Various techniques can be used to select a power profile to allow the energy being delivered from the ultrasonic drive circuit **177** and the RF drive circuit **702** to dynamically change based on the tissue type being treated by the surgical instrument.

In one form, a strain gauge can be used to measure the force applied to the tissue by the end effector. A strain gauge can be coupled to the end effector to measure the force on the tissue being treated by the end effector. With reference now to FIG. **165**, a system **6680** for measuring forces applied to the tissue grasped by the end effector comprises a strain gauge sensor **6682**, such as, for example, a micro-strain gauge, is configured to measure one or more parameters of the end effector **6650**, **6650'**, **6650"** as shown in FIGS. **159-164**, for example. In one aspect, the strain gauge sensor **6682** can measure the amplitude or magnitude of the strain exerted on a jaw member of an end effector **6650**, **6650'**, **6650"** during a clamping operation, which can be indicative of the tissue compression. The measured strain is converted to a digital signal and provided to a processor **6690** of a microcontroller **6688**. A load sensor **6684** can measure the force to operate the ultrasonic blade **6658**, **6658'**, **6658"** as shown in FIGS. **159-164**, for example, to cut the tissue captured between the jaw member and the ultrasonic blade **6658**, **6658'**, **6658"** of the end effector **6650**, **6650'**, **6650"**. A magnetic field sensor **6686** can be employed to measure the thickness of the captured tissue **6678**. The measurement of the magnetic field sensor **6686** also may be converted to a digital signal and provided to the processor **6690**.

Further to the above, a feedback indicator **6694** also can be configured to communicate with the microcontroller **6688**. In one aspect, the feedback indicator **6694** can be disposed in the handle of the combination ultrasonic/electrosurgical instruments **500**, **600**, **700** (FIGS. **30-44**). Alternatively, the feedback indicator **6694** can be disposed in a shaft assembly of a surgical instrument, for example. In any event, the microcontroller **6688** may employ the feedback indicator **6694** to provide feedback to an operator of the surgical instrument with regard to the adequacy of a manual input such as, for example, a selected position of a firing trigger that is used to cause the end effector to clamp down on tissue. To do so, the microcontroller **6688** may assess the selected position of the jaw member **6672** and/or firing trigger. The measurements of the tissue **6678** compression, the tissue **6678** thickness, and/or the force required to close the end effector **6658**, **6658'**, **6658"** on the tissue, as respectively measured by the sensors **6682**, **6684**, **6686**, can be used by the microcontroller **6688** to characterize the selected position of the firing trigger and/or the corresponding value of the speed of end effector. In one instance, a memory **6692** may store a technique, an equation, and/or a look-up table which can be employed by the microcontroller **6688** in the assessment.

In one aspect, the present disclosure provides an end effector for a surgical instrument, the end effector comprises an ultrasonic blade; and a jaw member comprising an asymmetric electrode comprising a first electrode and a second electrode; wherein the first electrode may define a first width and the second electrode may define a second width, wherein the first width is not equal to the second width; and wherein a first gap may be defined between the first electrode and the ultrasonic blade and a second gap may be defined between the second electrode and the ultrasonic blade, wherein the first gap is not equal to the second gap.

The ultrasonic blade and the jaw member may define a curvature. The ultrasonic blade and the jaw member may define a greater curvature at a distal end relative to the curvature at a proximal end. The ultrasonic blade and the jaw member may define a greater width at a proximal end relative to a distal end. The end effector may further comprise an electrically insulative polymer pad disposed between the first and second electrodes. The electrically insulative polymer pad may comprise a first soft polymeric pad and a second and high density polymeric pad positioned adjacent to the soft polymeric pad. The ultrasonic blade may comprise features for cutting and features for coagulating tissue.

In another aspect, the present disclosure provides a jaw member of an end effector for a surgical instrument, the jaw member comprising: an asymmetric electrode comprising a first electrode and a second electrode; wherein the first electrode may define a first width and the second electrode may define a second width, wherein the first width is not equal to the second width. The jaw member may define a curvature. The jaw member may define a greater curvature at a distal end relative to the curvature at a proximal end. The jaw member may define a greater width at a proximal end relative to a distal end. The jaw member may further comprise an electrically insulative polymer pad disposed between the first and second electrodes. The electrically insulative polymer pad may comprise a first soft polymeric pad and a second and high density polymeric pad positioned adjacent to the soft polymeric pad.

In another aspect, the present disclosure provides, a surgical instrument, comprising a handle assembly; an end effector operably coupled to the hand assembly, the end effector comprising: an ultrasonic blade; and a jaw member comprising an asymmetric electrode comprising a first electrode and a second electrode; wherein the first electrode may define a first width and the second electrode may define a second width, wherein the first width is not equal to the second width; and wherein a first gap may be defined between the first electrode and the ultrasonic blade and a second gap may be defined between the second electrode and the ultrasonic blade, wherein the first gap is not equal to the second gap.

The ultrasonic blade and the jaw member may define a curvature. The ultrasonic blade and the jaw member may define a greater curvature at a distal end relative to the curvature at a proximal end. The ultrasonic blade and the jaw member may define a greater width at a proximal end relative to a distal end.

The surgical instrument may further comprise an electrically insulative polymer pad disposed between the first and second electrodes. The electrically insulative polymer pad may comprise a first soft polymeric pad and a second and high density polymeric pad positioned adjacent to the soft polymeric pad. The ultrasonic blade may comprise features for cutting and features for coagulating tissue.

Aspects of the devices disclosed herein can be designed to be disposed of after a single use, or they can be designed to be used multiple times. Various aspects may, in either or both cases, be reconditioned for reuse after at least one use. Reconditioning may include any combination of the steps of disassembly of the device, followed by cleaning or replacement of particular pieces, and subsequent reassembly. In particular, aspects of the device may be disassembled, and any number of the particular pieces or parts of the device may be selectively replaced or removed in any combination. Upon cleaning and/or replacement of particular parts, aspects of the device may be reassembled for subsequent use either at a reconditioning facility, or by a surgical team immediately prior to a surgical procedure. Those skilled in the art will appreciate that reconditioning of a device may utilize a variety of techniques for disassembly, cleaning/replacement, and reassembly. Use of such techniques, and the resulting reconditioned device, are all within the scope of the present application.

By way of example only, aspects described herein may be processed before surgery. First, a new or used instrument may be obtained and if necessary cleaned. The instrument may then be sterilized. In one sterilization technique, the instrument is placed in a closed and sealed container, such as a plastic or TYVEK bag. The container and instrument may then be placed in a field of radiation that can penetrate the container, such as gamma radiation, x-rays, or high-energy electrons. The radiation may kill bacteria on the instrument and in the container. The sterilized instrument may then be stored in the sterile container. The sealed container may keep the instrument sterile until it is opened in a medical facility. A device may also be sterilized using any other technique known in the art, including but not limited to beta or gamma radiation, ethylene oxide, or steam.

While various details have been set forth in the foregoing description, it will be appreciated that the various aspects of the techniques for operating a generator for digitally generating electrical signal waveforms and surgical instruments may be practiced without these specific details. One skilled in the art will recognize that the herein described components (e.g., operations), devices, objects, and the discussion accompanying them are used as examples for the sake of conceptual clarity and that various configuration modifications are contemplated. Consequently, as used herein, the specific exemplars set forth and the accompanying discussion are intended to be representative of their more general classes. In general, use of any specific exemplar is intended to be representative of its class, and the non-inclusion of specific components (e.g., operations), devices, and objects should not be taken limiting.

Further, while several forms have been illustrated and described, it is not the intention of the applicant to restrict or limit the scope of the appended claims to such detail. Numerous modifications, variations, changes, substitutions, combinations, and equivalents to those forms may be implemented and will occur to those skilled in the art without departing from the scope of the present disclosure. Moreover, the structure of each element associated with the described forms can be alternatively described as a technique for providing the function performed by the element. Also, where materials are disclosed for certain components, other materials may be used. It is therefore to be understood that the foregoing description and the appended claims are intended to cover all such modifications, combinations, and variations as falling within the scope of the disclosed forms. The appended claims are intended to cover all such modifications, variations, changes, substitutions, modifications, and equivalents.

For conciseness and clarity of disclosure, selected aspects of the foregoing disclosure have been shown in block diagram form rather than in detail. Some portions of the detailed descriptions provided herein may be presented in terms of instructions that operate on data that is stored in one or more computer memories or one or more data storage devices (e.g. floppy disk, hard disk drive, Compact Disc (CD), Digital Video Disk (DVD), or digital tape). Such descriptions and representations are used by those skilled in the art to describe and convey the substance of their work to others skilled in the art. In general, an algorithm refers to a self-consistent sequence of steps leading to a desired result, where a "step" refers to a manipulation of physical quantities and/or logic states which may, though need not necessarily, take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It is common usage to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like. These and similar terms may be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities and/or states.

Unless specifically stated otherwise as apparent from the foregoing disclosure, it is appreciated that, throughout the foregoing disclosure, discussions using terms such as "processing" or "computing" or "calculating" or "determining" or "displaying" or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

In a general sense, those skilled in the art will recognize that the various aspects described herein which can be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or any combination thereof can be viewed as being composed of various types of "electrical circuitry." Consequently, as used herein "electrical circuitry" includes, but is not limited to, electrical circuitry having at least one discrete electrical circuit, electrical circuitry having at least one integrated circuit, electrical circuitry having at least one application specific integrated circuit, electrical circuitry forming a general purpose computing device configured by a computer program (e.g., a general purpose computer configured by a computer program which at least partially carries out processes and/or devices described herein, or a microprocessor configured by a computer program which at least partially carries out processes and/or devices described herein), electrical circuitry forming a memory device (e.g., forms of random access memory), and/or electrical circuitry forming a communications device (e.g., a modem, communications switch, or optical-electrical equipment). Those having skill in the art will recognize that the subject matter described herein may be implemented in an analog or digital fashion or some combination thereof.

The foregoing detailed description has set forth various forms of the devices and/or processes via the use of block diagrams, flowcharts, and/or examples. Insofar as such block diagrams, flowcharts, and/or examples contain one or more functions and/or operations, it will be understood by those within the art that each function and/or operation within such block diagrams, flowcharts, and/or examples can be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or virtually any combination thereof. In one form, several portions of the subject matter described herein may be implemented via an application specific integrated circuits (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), or other integrated formats. However, those skilled in the art will recognize that some aspects of the forms disclosed herein, in whole or in part, can be equivalently implemented in integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computer systems), as one or more programs running on one or more processors (e.g., as one or more programs running on one or more microprocessors), as firmware, or as virtually any combination thereof, and that designing the circuitry and/or writing the code for the software and or firmware would be well within the skill of one of skill in the art in light of this disclosure. In addition, those skilled in the art will appreciate that the mechanisms of the subject matter described herein are capable of being distributed as one or more program products in a variety of forms, and that an illustrative form of the subject matter described herein applies regardless of the particular type of signal bearing medium used to actually carry out the distribution. Examples of a signal bearing medium include, but are not limited to, the following: a recordable type medium such as a floppy disk, a hard disk drive, a Compact Disc (CD), a Digital Video Disk (DVD), a digital tape, a computer memory, etc.; and a transmission type medium such as a digital and/or an analog communication medium (e.g., a fiber optic cable, a waveguide, a wired communications link, a wireless communication link (e.g., transmitter, receiver, transmission logic, reception logic, etc.), etc.).

In some instances, one or more elements may be described using the expression "coupled" and "connected" along with their derivatives. It should be understood that these terms are not intended as synonyms for each other. For example, some aspects may be described using the term "connected" to indicate that two or more elements are in direct physical or electrical contact with each other. In another example, some aspects may be described using the term "coupled" to indicate that two or more elements are in direct physical or electrical contact. The term "coupled," however, also may mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other. It is to be understood that depicted architectures of different components contained within, or connected with, different other components are merely examples, and that in fact many other architectures may be implemented which achieve the same functionality. In a conceptual sense, any arrangement of components to achieve the same functionality is effectively "associated" such that the desired functionality is achieved. Hence, any two components herein combined to achieve a particular functionality can be seen as "associated with" each other such that the desired functionality is achieved, irrespective of architectures or intermedial components. Likewise, any two components so associated also can be viewed as being "operably connected," or "operably coupled," to each other to achieve the desired functionality, and any two components capable of being so associated also can be viewed as being "operably couplable," to each other to achieve the desired functionality. Specific examples of operably couplable include but are not limited to physically mateable and/or physically interacting components, and/or wirelessly interactable, and/or wirelessly interacting components, and/or logically interacting, and/or logically interactable components, and/or electrically interacting components, and/or electrically interactable components, and/or optically interacting components, and/or optically interactable components.

In other instances, one or more components may be referred to herein as "configured to," "configurable to," "operable/operative to," "adapted/adaptable," "able to," "conformable/conformed to," etc. Those skilled in the art will recognize that "configured to" can generally encompass active-state components and/or inactive-state components and/or standby-state components, unless context requires otherwise.

While particular aspects of the present disclosure have been shown and described, it will be apparent to those skilled in the art that, based upon the teachings herein, changes and modifications may be made without departing from the subject matter described herein and its broader aspects and, therefore, the appended claims are to encompass within their scope all such changes and modifications as are within the true scope of the subject matter described herein. It will be understood by those within the art that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as "open" terms (e.g., the term "including" should be interpreted as "including but not limited to," the term "having" should be interpreted as "having at least," the term "includes" should be interpreted as "includes but is not limited to," etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases "at least one" and "one or more" to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles "a" or "an" limits any particular claim containing such introduced claim recitation to claims containing only one such recitation, even when the same claim includes the introductory phrases "one or more" or "at least one" and indefinite articles such as "a" or "an" (e.g., "a" and/or "an" should typically be interpreted to mean "at least one" or "one or more"); the same holds true for the use of definite articles used to introduce claim recitations.

In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of "two recitations," without other modifiers, typically technique at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to "at least one of A, B, and C, etc." is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., "a system having at least one of A, B, and C" would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to "at least one of A, B, or C, etc." is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., "a system having at least one of A, B, or C" would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that typically a disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms unless context dictates otherwise. For example, the phrase "A or B" will be typically understood to include the possibilities of "A" or "B" or "A and B."

With respect to the appended claims, those skilled in the art will appreciate that recited operations therein may generally be performed in any order. Also, although various operational flows are presented in a sequence(s), it should be understood that the various operations may be performed in other orders than those which are illustrated, or may be performed concurrently. Examples of such alternate orderings may include overlapping, interleaved, interrupted, reordered, incremental, preparatory, supplemental, simultaneous, reverse, or other variant orderings, unless context dictates otherwise. Furthermore, terms like "responsive to," "related to," or other past-tense adjectives are generally not intended to exclude such variants, unless context dictates otherwise.

It is worthy to note that any reference to "one aspect," "an aspect," "one form," or "a form" technique that a particular feature, structure, or characteristic described in connection with the aspect is included in at least one aspect. Thus, appearances of the phrases "in one aspect," "in an aspect," "in one form," or "in an form" in various places throughout the specification are not necessarily all referring to the same aspect. Furthermore, the particular features, structures or characteristics may be combined in any suitable manner in one or more aspects.

With respect to the use of substantially any plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations are not expressly set forth herein for sake of clarity.

In certain cases, use of a system or method may occur in a territory even if components are located outside the territory. For example, in a distributed computing context, use of a distributed computing system may occur in a territory even though parts of the system may be located outside of the territory (e.g., relay, server, processor, signal-bearing medium, transmitting computer, receiving computer, etc. located outside the territory).

A sale of a system or method may likewise occur in a territory even if components of the system or method are located and/or used outside the territory. Further, implementation of at least part of a system for performing a method in one territory does not preclude use of the system in another territory.

All of the above-mentioned U.S. patents, U.S. patent application publications, U.S. patent applications, foreign patents, foreign patent applications, non-patent publications referred to in this specification and/or listed in any Application Data Sheet, or any other disclosure material are incorporated herein by reference, to the extent not inconsistent herewith. As such, and to the extent necessary, the disclosure as explicitly set forth herein supersedes any conflicting material incorporated herein by reference. Any material, or portion thereof, that is said to be incorporated by reference herein, but which conflicts with existing definitions, statements, or other disclosure material set forth herein will only be incorporated to the extent that no conflict arises between that incorporated material and the existing disclosure material.

In summary, numerous benefits have been described which result from employing the concepts described herein. The foregoing description of the one or more forms has been presented for purposes of illustration and description. It is not intended to be exhaustive or limiting to the precise form disclosed. Modifications or variations are possible in light of the above teachings. The one or more forms were chosen and described in order to illustrate principles and practical application to thereby enable one of ordinary skill in the art to utilize the various forms and with various modifications as are suited to the particular use contemplated. It is intended that the claims submitted herewith define the overall scope.

The invention claimed is:

1. A surgical instrument comprising:

a handle assembly coupled to a proximal end of a shaft assembly;

the shaft assembly coupled to a distal end of the handle assembly; and

an end effector coupled to a distal end of the shaft assembly, and comprising:

a first jaw and a second jaw configured for pivotal movement between a closed position and an open position; and

a segmented flexible circuit fixedly attached to the first jaw and positioned between the first jaw and the second jaw such that the segmented flexible circuit touches tissue when the first jaw and the second jaw close onto the tissue;

the segmented flexible circuit comprising:

a plurality of segments comprising a distal segment fixedly attached to a distal end of the first jaw, and a plurality of lateral segments fixedly attached to lateral sides of the first jaw;

wherein each segment of the plurality of segments comprises an electrode, an electrical insulation, a compressive layer, a laminate layer, and a plurality of data sensors positioned intermediate the electrode and the first jaw, wherein the plurality of data sensors comprises a temperature sensor and two force sensors, wherein the electrical insulation is positioned between the electrode and the temperature sensor, wherein the temperature sensor is positioned between the electrical insulation and one of the force sensors, wherein the one of the force sensors is positioned between the temperature sensor and the compressive layer, wherein the compressive layer is positioned between the one of the force sensors and the other of the force sensors, wherein the other of the force sensors is positioned between the compressive layer and the laminate layer, wherein the laminate layer is positioned between the other of the force sensors and the first jaw;

a plurality of electrical connectors, wherein the plurality of electrical connectors electrically couples the plurality of segments to an energy source; and

wherein each segment of the plurality of segments is individually addressable to treat the tissue by providing a first power level from the energy source to a first electrode of a first segment of the plurality of segments without providing the first power level to a second electrode of a second segment of the plurality of segments, and wherein an amount of power provided to each of the plurality of segments is determined based on data provided by the plurality of data sensors corresponding to each segment of the plurality of segments.

2. The surgical instrument of claim **1**, wherein the end effector further comprises:

a Hall effect sensor positioned at a distal end of one of the first jaw or the second jaw, and

a magnet in the other of the first jaw and the second jaw;

wherein the combination of the Hall effect sensor and the magnet are configured to measure an aperture between the first jaw and the second jaw.

3. The surgical instrument of claim **1**, wherein the plurality of lateral segments comprises a first lateral segment located adjacent to the distal segment and a second lateral segment located at the proximal end of the end effector and adjacent to the first lateral segment.

4. The surgical instrument of claim **1**, wherein the second jaw comprises an ultrasonic blade.

5. The surgical instrument of claim **1**, wherein the plurality of data sensors of each of the plurality of segments are configured to detect a presence of tissue positioned between the second jaw and itself.

6. The surgical instrument of claim **5**, wherein the energy source is configured to individually provide energy to each of the plurality of segments after it is detected that tissue is positioned between the second jaw and each of the plurality of segments.

7. The surgical instrument of claim **5**, wherein the plurality of data sensors of at least one of the plurality of segments comprises a Hall effect sensor configured to detect a thickness of tissue positioned between the second jaw and itself.

8. The surgical instrument of claim **5**, wherein the force sensor is configured to detect a force applied to the tissue positioned between the second jaw and itself.

9. The surgical instrument of claim **5**, wherein the temperature sensor is configured to detect a temperature of the tissue positioned between the second jaw and itself.

10. The surgical instrument of claim **1**, wherein the first jaw comprises a left side and a right side, and the first segment is positioned on the left side of the first jaw and adjacent to the distal segment, and the second segment is positioned on the right side of the first jaw and adjacent to the distal segment.

11. The surgical instrument of claim **10**, wherein the electrode of the first segment is configured to apply RF energy from the energy source at the first power level, based on a first sensed characteristic about the tissue positioned at the plurality of data sensors of the first segment, and the second segment is configured to simultaneously apply RF energy from the energy source at a second power level that is different than the first power level, based on a second sensed characteristic about the tissue positioned at the plurality of data sensors of the second segment, wherein the second sensed characteristic is different than the first sensed characteristic.

12. An end effector of a surgical instrument comprising:

a first jaw and a second jaw configured for pivotal movement between a closed position and an open position; and

a segmented flexible circuit configured to fixedly attach to the first jaw positioned between the first jaw and the second jaw such that the segmented flexible circuit touches tissue when the first jaw and the second jaw close onto the tissue;

the segmented flexible circuit comprising:

a plurality of segments, wherein the plurality of segments comprises a distal segment configured to fixedly attach to a distal end of the first jaw, a plurality of lateral segments configured to fixedly attach to lateral sides of the first jaw;

wherein each segment of the plurality of segments comprises an electrode, an electrical insulation, a compressive layer, a laminate layer, and a plurality of data sensors positioned intermediate the electrode and the first jaw, wherein the plurality of data sensors comprises a temperature sensor and two force sensors, wherein the electrical insulation is positioned between the electrode and the temperature sensor, wherein the temperature sensor is positioned between the electrical insulation and one of the force sensors, wherein the one of the force sensors is positioned between the temperature sensor and the compressive layer, wherein the compressive layer is positioned between the one of the force sensors and the other of the force sensors, wherein the other of the force sensors is positioned between the compressive layer and the laminate layer, wherein the laminate layer is positioned between the other of the force sensors and the first jaw; and

a plurality of electrical connectors, wherein the plurality of electrical connectors electrically couples the plurality of segments to an energy source on a proximal end;

wherein the plurality of segments are each individually addressable to treat tissue by providing a first power level from the energy source to a first electrode of a first segment of the plurality of segments without providing the first power level to a second electrode of a second segment of the plurality of segments, and wherein an amount of power provided to each of the plurality of segments is determined based on data provided by the plurality of data sensors corresponding to each of the plurality of segments.

13. The end effector of claim 12, further comprising:

a Hall effect sensor positioned at a distal end of one of the first jaw or the second jaw, and

a magnet in the other of the first jaw and the second jaw;

wherein the combination of the Hall effect sensor and the magnet are configured to measure an aperture between the first jaw and the second jaw.

14. The end effector of claim 12, wherein the plurality of lateral segments comprises a first lateral segment located adjacent to the distal segment and a second lateral segment located at the proximal end of the end effector and adjacent to the first lateral segment.

15. The end effector of claim 12, wherein the plurality of data sensors of each of the plurality of segments configured to detect a presence of tissue positioned between the second jaw and itself.

16. The end effector of claim 15, wherein the energy source is configured to individually provide energy to each of the plurality of segments after it is detected that tissue is positioned between the second jaw and each of the plurality of segments.

17. The end effector of claim 12, wherein:

the first jaw comprises a left side and a right side;

the first segment is positioned on the left side of the first jaw and adjacent to the distal segment, and the second segment is positioned on the right side of the first jaw and adjacent to the distal segment;

the electrode of the first segment is configured to apply RF energy from the energy source at a first power level, based on a first sensed characteristic about the tissue positioned at the first segment; and

the second segment is configured to simultaneously apply RF energy from the energy source at a second power level that is different than the first power level, based on a second sensed characteristic about the tissue positioned at the plurality of data sensors of the second segment, wherein the second sensed characteristic is different than the first sensed characteristic.

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